Principles of Statistical Lower Bounds

Methods and Applications

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Preface

What can be learned from data, and how accurately? These questions are fundamental to statistics and machine learning. We develop procedures to estimate unknown quantities, test hypotheses, quantify uncertainty, and make predictions. But constructing a procedure and establishing its performance answer only part of the question. We also need to know whether a better procedure is possible, which features determine the difficulty of the task, and how much information the data provide.

Lower bounds are essential to these inquiries. An upper bound describes what a particular procedure can achieve; a lower bound establishes a limitation that applies to every procedure under the stated model and criterion. When the two match, they identify the optimal accuracy and explain why it cannot be improved. When they differ, the gap points to an unresolved question: either a better procedure remains to be found or the lower bound has not yet captured the full difficulty of the task.

The value of lower bounds extends beyond certifying optimality. They reveal how sample size, dimension, smoothness, sparsity, and other structural features affect statistical difficulty. They distinguish limitations inherent in the observations from shortcomings of a particular method. They also clarify when additional assumptions are necessary, when adaptation is possible, and when reliable uncertainty quantification requires more than accurate estimation. Lower bounds are thus not simply the final step of an optimality argument. They are a way to understand a statistical problem and can guide the search for effective procedures.

This book develops the principal ideas and techniques of statistical lower bounds, with an emphasis on how to use them. A central theme is the construction of alternatives that differ in what matters for the task yet remain difficult to distinguish from the observations. Turning this idea into a useful bound requires careful choices: the statistical experiment, the target, the loss or coverage criterion, the alternatives, and the measure of indistinguishability. Learning to make these choices is as important as learning the inequalities themselves.

Applications connect the general methods to questions in statistics, data science, and machine learning. They include estimation of functionals, optimization from noisy observations, classification, clustering, ranking, crowdsourcing, and signal detection and localization. Each is chosen both for its importance and for the lower-bound ideas it illustrates. Discussions of matching procedures and upper bounds place the impossibility results in context and show what is needed to establish optimality.

The statistical task matters as much as the observation model. Detecting a signal need not be as difficult as estimating it. An estimator may adapt to unknown structure even when an honest confidence interval cannot attain the corresponding class-specific length. These

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differences arise from the requirements of estimation, testing, and uncertainty quantification, not merely from the choice of proof technique.

The exposition follows a common progression: from the statistical question and decision criterion to hard alternatives, an information comparison, a lower bound, and its statistical interpretation. Each argument explains not only how the bound is established, but also what it reveals about the statistical problem.

The book is intended for graduate students and researchers with a background in probability, mathematical statistics, and basic analysis. Problems are formulated before their lower bounds are stated, and proofs emphasize the construction and calibration of difficult alternatives. Exercises offer opportunities to work through the arguments, compare methods, and extend the examples.

The aim is to make lower bounds an accessible and practical part of the reader's statistical toolkit—and to encourage the habit of asking whether a method's performance reflects what can fundamentally be learned from the available data.

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Reading pathways and notation

Organization of the book

Part I: Foundations, Information, and Hard Families. Chapters 1–3 introduce statistical decision problems and probability distances, then develop two-point, mixture, constrained-risk, hypercube, and packing arguments. The emphasis is on choosing alternatives suited to the target and loss.

Part II: Minimax Inference and Adaptation. Chapters 4–7 develop the theory of estimation risk, testing separation, honest confidence-set size, and adaptation to unknown structure. Moduli, coverage-transfer arguments, and cross-class risk comparisons explain why these tasks require different reductions, even when they share hard alternatives.

Part III: Structured Decisions and Detection. Chapter 8 studies sparse regression and structured learning, emphasizing how estimation, honest inference, prediction, and recovery determine the hard alternatives. Chapter 9 builds on Chapter 5's separation benchmarks to distinguish detection, selection, and localization, showing how rarity and observation structure change the information needed for each task.

In the application chapters, read each major lower-bound result together with its proof or proof outline and the accompanying comparison remark. The lower bound identifies a fundamental limitation; the construction explains its source; and the comparison describes a procedure and its attainable performance. These remarks show when the lower bound is sharp and identify differences in assumptions or benchmarks when the guarantees do not match. Upper bounds are attributed to the relevant source theorems or justified in the text.

Each part closes with a synthesis comparing its constructions. *Lower-Bound Techniques: A Section-by-Section Guide* (page 259) links the problems in Parts II and III to their principal tools. Appendix A collects reusable technical facts; Appendix B contains longer proofs and supporting calculations.

The core prerequisites are likelihoods, conditional expectation, elementary convexity, and Gaussian tail bounds; functional estimation also uses Hilbert-space geometry and duality.

Three routes through the book

Proof construction. Readers new to lower bounds can begin with Part I, then work through selected estimation and testing examples in Chapters 4–5 before turning to Part III.

Uncertainty quantification. Readers interested in confidence sets and adaptation can follow Part I with Chapters 5–7, consulting Chapter 4 for the corresponding estimation benchmarks.

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Applications. Readers familiar with the basic methods can start with a problem in Chapter 8 or 9. The chapter introductions and internal references identify the earlier constructions needed for each argument.

Notation and conventions

This reference collects recurring symbols and conventions. Statistical models, parameter classes, and task-specific benchmarks are developed where they are first needed; the pointers below locate their definitions. One-dimensional function problems use $[0, 1]$ as the standard domain, with $x_0 = 1/2$ for symmetric interior constructions.

Sample size and dimension. We use n for sample size or noise precision and d for ambient vector dimension, including the number of regression covariates. We generally use s for sparsity and m for a truncation level or effective dimension. Counts for separate samples, blocks, or groups and other local roles are defined in the corresponding model. The symbol p is retained for integrability exponents, densities, and probabilities.

Gaussian observations. Section 1.2 defines the Gaussian sequence and continuous white-noise models and explains their common shorthand $Y = f + \varepsilon W$. In the path formulation, B denotes Brownian motion.

Sets, vectors, and functions. Write $[m] = \{1, \dots, m\}$, $|A|$ for the cardinality of a finite set, and $A \triangle B$ for symmetric difference. The symbols $\mathbf{1}_A$, $\mathbf{1}\{A\}$, and $\mathbb{I}\{A\}$ denote indicators; for a coordinate set, $\mathbf{1}_A$ is its indicator vector, and $\mathbf{1}$ is the all-ones vector. For vectors or sequences, $\|x\|_p = (\sum_j |x_j|^p)^{1/p}$, $0 < p < \infty$, and $\|x\|_\infty = \sup_j |x_j|$; $\|x\|_0$ counts nonzero coordinates. The expression for $p < 1$ is a quasi-norm. Sequence spaces ℓ_p consist of sequences with finite corresponding expression. For functions, $\|f\|_r = (\int |f|^r)^{1/r}$, $1 \le r < \infty$, uses Lebesgue measure on the specified domain unless another measure is indicated, as in $L_r(\mu)$ or $L_2(P)$. For pointwise-defined functions, $\|f\|_\infty = \sup_x |f(x)|$. Inner products use the corresponding sum or integral.

The identity matrix is I_d, and $A^\top$ is the transpose of A. For a square matrix, $\mathrm{tr}(A)$ is its trace and $\det(A)$ its determinant; $\mathrm{diag}(a_1, \dots, a_d)$ has the displayed diagonal entries and zeros elsewhere. For label vectors, $H(u, v) = \sum_j \mathbf{1}\{u_j \ne v_j\}$ is unnormalized Hamming distance.

Probability and distributions. We use $\mathbb{P}$ for probability, $\mathbb{E}$ for expectation, and Var and Cov for variance and covariance. Subscripts specify the law or parameter: $\mathbb{P}_P(A) = P(A)$ and $\mathbb{P}_\theta(A) = P_\theta(A)$. For random variables or random vectors, $U \perp V$ denotes independence. Absolute continuity, $P \ll Q$, means that $Q(A) = 0$ implies $P(A) = 0$ for every measurable A. Write δ_x for unit point mass at x. The law $P^{\otimes n}$ describes n independent observations from P; i.i.d. means independent and identically distributed. A prior π induces the mixture law $P_\pi = \int P_\theta \, \pi(d\theta)$.

Write $N(\mu, \sigma^2)$ for the univariate normal law and $N_d(\mu, \Sigma)$ for the d-variate normal law with mean μ and covariance Σ. Write Φ and φ for the standard-normal distribution function and density, $\bar{\Phi} = 1 - \Phi$, and $z_\alpha = \Phi^{-1}(1 - \alpha)$. Other uses of ϕ or φ, including tests, are specified locally. The laws $\mathrm{Ber}(p)$, $\mathrm{Bin}(m, p)$, $\mathrm{Pois}(\lambda)$, and $\mathrm{Multinomial}(m, p)$ denote, respectively, a Bernoulli variable with success probability p, its m-trial binomial sum, a Poisson count with mean λ, and category counts with total m and probability vector p. The

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chi-square law χ^2_m is the law of $\sum_{j=1}^m Z_j^2$ for independent $Z_j \sim N(0,1)$; it is distinct from chi-square divergence.

Orders and limits. Logarithms are natural unless a base is displayed. Write $a \wedge b = \min\{a,b\}$, $a \vee b = \max\{a,b\}$, $x_+ = [x]_+ = \max\{x,0\}$, and $1/\infty = 0$. The floor $\lfloor x \rfloor$ and ceiling $\lceil x \rceil$ round down and up to integers. For nonnegative sequences, $a_n \lesssim b_n$, equivalently $b_n \gtrsim a_n$, means $a_n \le C b_n$; $\asymp$ means both inequalities. The notation $a_n = O(b_n)$ has the same upper-order meaning, using $|a_n|$ for signed sequences. Constants are independent of the varying sample size, dimension, or noise level, unless stated; they may depend on fixed class and coverage parameters. For positive sequences, $a_n \sim b_n$ means $a_n/b_n \to 1$, and $a_n = o(b_n)$ or $a_n \ll b_n$ means $a_n/b_n \to 0$; $\gg$ reverses the latter comparison. A quantity $o_P(1)$ converges to zero in P-probability; $\rightsquigarrow$ denotes convergence in distribution. Between a random variable and a law, $\sim$ specifies distribution instead.

Distances and information. Section 1.3 defines total variation TV, testing affinity η, normalized Hellinger distance H, Hellinger affinity ρ, Kullback–Leibler (KL) divergence KL, chi-square divergence χ^2, and chi-square affinity ξ. Total variation is half the L_1 distance between densities. The second law is the reference for KL and chi-square divergence; chi-square affinity has the reference law first. The comma and $\|$ notations for the two divergences are interchangeable. Definition 3.4 introduces mutual information; the proof of Lemma 3.5 defines the entropy quantities used there.

Losses. A hat denotes a data-based rule. Section 1.2 defines risk and minimax risk; symbols such as R_n and R^* retain the locally specified target and loss. Scalar targets use absolute error $|\widehat{T} - T|$ or squared error $(\widehat{T} - T)^2$. For $1 \le r < \infty$, rth-power ℓ_r loss is $\|\widehat{\theta} - \theta\|_r^r$ for vectors or sequences; rth-power L_r loss is $\|\widehat{f} - f\|_r^r$ for functions. The corresponding risk does not take an rth root. Normalized squared error for m coordinates is $m^{-1}\|\widehat{\theta} - \theta\|_2^2$; normalization is always explicit. Uniform loss is $\|\widehat{f} - f\|_\infty$, or its stated power. General norm and metric losses specify the norm or metric and the power separately.

Task-specific benchmarks. Section 5.1 defines testing errors and separation; its total-error criterion uses the sum of the two worst-case errors. Chapter 6 defines honest coverage and confidence-set size, and Chapter 7 defines adaptation benchmarks. Learning losses are specified with their problems in Chapter 8. The marginal false-discovery and missed-discovery rates, mFDR and MDR, are defined in Section 9.1.

Parameter classes and moduli. The bounded Hölder balls and their density subclasses are defined in (2.1) and (2.2). Other smoothness classes retain their local definitions; in particular, $L_p(\alpha, M)$ in (4.7) is a weighted sequence class, not a loss or an L_p function space. Chapter 4 defines the moduli used throughout; the subscript + on ω_+ denotes symmetrization, not positive part.

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Part I

Foundations, Information, and Hard Families

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1

Foundations, Distances, and Local Information

This chapter develops the language and comparisons needed to establish statistical lower bounds. The starting point is the statistical problem itself: what is observed, what is unknown, what must be learned, and how errors are measured. These choices determine what accurate inference would require the data to reveal. A lower bound identifies accuracy levels that no procedure can attain uniformly over the parameter class.

We begin with an exact minimax calculation for the normal mean, then specify the observation model, parameter class, target, loss, and admissible procedures. The main reduction turns accurate estimation into discrimination among separated targets. Total variation characterizes optimal binary testing error; Hellinger distance, Kullback–Leibler (KL) divergence, and chi-square divergence help bound it. Product formulas and data processing extend these comparisons to samples and transformed observations. Chapters 2 and 3 use this reduction with pairs, mixtures, hypercubes, and packings.

Local-information inequalities provide a complementary approach. Under their respective conditions, Hammersley–Chapman–Robbins and Cramér–Rao give variance bounds for unbiased estimation through finite and differentiable local comparisons. The van Trees inequality bounds prior-averaged squared-error risk and can yield minimax lower bounds without unbiasedness. We distinguish these guarantees as we develop them.

1.1 Why lower bounds matter

How many observations are needed to estimate a normal mean to a prescribed accuracy? An exact minimax calculation answers this question by comparing the sample mean with a lower bound that applies to every estimator.

Consider independent observations $X_1, \dots, X_n \sim N(\theta, \sigma^2)$, where the variance $\sigma^2 > 0$ is known and the mean $\theta \in \mathbb{R}$ is unknown. Under squared-error loss, the sample mean $\bar{X} = n^{-1} \sum_{i=1}^{n} X_i$ has risk $\mathbb{E}_\theta(\bar{X} - \theta)^2 = \sigma^2/n$ for every θ. Is that accuracy optimal among all estimators, including biased ones? Place a normal distribution $N(0, \tau^2)$ on the unknown mean, with $\tau > 0$. Completing the square shows that the conditional distribution of θ given the sample is normal, with variance

$$v_\tau = (n/\sigma^2 + 1/\tau^2)^{-1}.$$

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Conditional squared error is minimized by the conditional mean, so every estimator has average risk at least v_τ under this distribution. Its worst-case risk is at least its average risk. Letting $\tau \to \infty$ and comparing with $\bar{X}$ therefore gives

$$\inf_{\widehat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}_\theta (\widehat{\theta} - \theta)^2 = \frac{\sigma^2}{n}.$$

Thus worst-case root mean squared error at most $\delta > 0$ is attainable exactly when $n \geq \sigma^2/\delta^2$. The auxiliary distribution on θ is a proof device, not an assumption about the true parameter.

The normal-mean calculation contains the ingredients needed throughout the book: an observation model, a target, a loss, and a comparison that applies to every procedure under consideration. The theory of statistical experiments in Le Cam [87] and the minimax treatment in Tsybakov [103] place these ingredients in a general framework.

For more complex models, the first objective is often a rate: how the best possible error changes with sample size, dimension, or structural complexity. Matching leading constants requires a finer comparison. Table 1.1 connects the terminology of lower bounds with common machine-learning language.

Book language	Common machine-learning translation	What the lower bound is trying to isolate
Minimax risk	best achievable worst-case error	the accuracy no procedure can uniformly improve upon
Statistical experiment	data-generating model	which information is available to the procedure
Hard subfamily or packing	deliberately chosen difficult instances	a subset that retains the target difficulty while simplifying the comparison of laws
Modulus	largest target change under a constraint on observation-law distance	how much the target can move while the data remain hard to distinguish
Information bound	bound on the distinguishability of selected laws	whether the chosen separation can be turned into a risk or testing lower bound
Cost of adaptation	cost of one procedure serving several classes	which classes impose incompatible performance requirements
Decision loss	classification, ranking, or clustering error	which differences between targets matter for the task

Table 1.1 *A translation guide between statistical decision theory and learning terminology. The correspondences describe proof roles, not identities between numerical quantities.*

The hard-family arguments developed in this book follow a common progression:

1. Statistical question **2.** Decision criterion **3.** Hard alternatives
4. Information comparison **5.** Lower bound **6.** Statistical interpretation

The criterion specifies what a procedure must achieve and how the alternatives should differ. We place those alternatives inside the parameter class, quantify their separation, and control the distinguishability of their observation laws. A testing or risk comparison then yields the

lower bound. The final step is to explain the limitation it identifies: which features determine difficulty, whether a matching procedure is available, and what additional assumptions could change the conclusion. Yu [114] compares the pairwise and many-point routes; honest inference and adaptation require the coverage and cross-class comparisons developed later.

A worst-case bound does not rule out better performance at particular parameters or on smaller subclasses. Chapter 7 asks when one procedure can attain the benchmarks of several classes simultaneously.

1.2 Statistical decision problems and minimax reductions

To compare procedures, we specify the observations, inferential target, loss, and procedure class. This section formalizes these ingredients, introduces two recurring Gaussian experiments, and reduces estimation to a finite testing problem. Suppose we observe $X \sim P_\theta$ on the measurable space $(\mathcal{X}, \mathscr{X})$, with unknown parameter $\theta \in \Theta$. The family $\{P_\theta : \theta \in \Theta\}$ is the statistical experiment.

A procedure reports an action in the measurable space $(\mathcal{A}, \mathscr{A})$, while the map $\tau : \Theta \to \mathcal{T}$ specifies the inferential target in the target space $(\mathcal{T}, \mathscr{T})$. The measurable loss $L : \mathcal{A} \times \mathcal{T} \to [0, \infty]$ measures the error of an action relative to that target. The class $\mathfrak{A}$ specifies the procedures under consideration.

A nonrandomized decision rule is a measurable map $\delta : \mathcal{X} \to \mathcal{A}$; randomized rules are included by adjoining an independent random seed. Its risk at θ and the minimax risk over Θ are

$$R(\theta, \delta) = \mathbb{E}_\theta L\big(\delta(X), \tau(\theta)\big), \qquad R^*(\Theta, L; \mathfrak{A}) = \inf_{\delta \in \mathfrak{A}} \sup_{\theta \in \Theta} R(\theta, \delta).$$

Unless stated otherwise, $\mathfrak{A}$ contains all measurable procedures and is suppressed from the notation. A lower bound on R^* limits every procedure in $\mathfrak{A}$. To establish optimality, a matching upper bound must concern the same experiment, target, action space, loss, and procedure class.

For binary testing, type-I error is rejection under the null and type-II error is acceptance under the alternative; worst-case errors take suprema over the respective hypothesis classes. Minimax zero–one risk uses the larger of these two errors. Chapter 5 instead uses their sum. The sum lies between the maximum and twice the maximum, so the two minimax benchmarks agree up to a factor of two.

When indexing an experiment directly by its laws, we write $\theta(P)$ for the target attached to P; scalar or functional targets are also denoted by $T(P)$ or $T(f)$.

Gaussian sequence and white-noise experiments. The Gaussian sequence and continuous white-noise models describe noisy coefficients and noisy integrals of a signal, respectively. Expressing both as noisy linear measurements lets us use the same distance-based lower-bound arguments in either model.

With known noise level $\varepsilon > 0$, the *Gaussian sequence model* observes

$$Y_i = \theta_i + \varepsilon Z_i, \qquad Z_i \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad i \geq 1, \quad \theta \in \ell_2. \tag{1.1}$$

Its finite-dimensional version observes $i = 1, \dots, d$ with $\theta \in \mathbb{R}^d$. The *continuous white-noise model* observes the entire continuous path

$$Y(t) = \int_0^t f(s)\, ds + \varepsilon B(t), \qquad 0 \le t \le 1, \quad f \in L_2[0,1], \tag{1.2}$$

where B is standard Brownian motion with $B(0) = 0$. We also write this model as $dY(t) = f(t)\, dt + \varepsilon\, dB(t)$; this is stochastic differential notation, not a claim that the observed path is differentiable.

To unify these models, let H be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle_H$ and norm $\| \cdot \|_H$. For a signal $f \in H$, a direction $h \in H$ specifies the measurement $\langle f, h \rangle_H$. An *isonormal Gaussian process* describes the noise: it is a family $\{W(h) : h \in H\}$ for which every finite subfamily has a joint Gaussian distribution with mean zero and covariance

$$\mathbb{E}\{W(h)W(g)\} = \langle h, g \rangle_H, \qquad h, g \in H.$$

Thus $W(h)$ has variance $\|h\|_H^2$, and orthogonal directions give independent noise variables. The covariance identity also implies $W(ah + bg) = aW(h) + bW(g)$ almost surely for each fixed $a, b \in \mathbb{R}$ and $h, g \in H$. The *Gaussian shift experiment* with unknown signal $f \in H$ observes the family

$$Y(h) = \langle f, h \rangle_H + \varepsilon W(h), \qquad h \in H. \tag{1.3}$$

Each measurement $Y(h)$ has mean $\langle f, h \rangle_H$ and variance $\varepsilon^2 \|h\|_H^2$. We write $Y = f + \varepsilon W$ for this family, or $Y = f + n^{-1/2} W$ with precision $n = \varepsilon^{-2}$. In infinite dimensions this shorthand does not represent an H-valued observation or noise vector. Write P_f for the law of the observed family; procedures are measurable with respect to the sigma-field generated by its measurements, with independent randomization allowed.

The sequence model corresponds to $H = \ell_2$ and $f = \theta$. Set $W(h) = \sum_i h_i Z_i$, with convergence in mean square. For the coordinate vectors e_i, the measurements $Y(e_i) = Y_i$ recover (1.1).

The white-noise model corresponds to $H = L_2[0,1]$. Set $W(h) = \int_0^1 h(t)\, dB(t)$, the stochastic integral of the deterministic function h. Then $Y(h) = \int_0^1 h(t)\, dY(t)$ means $\int_0^1 h(t) f(t)\, dt + \varepsilon \int_0^1 h(t)\, dB(t)$. Taking $h = \mathbf{1}_{[0,t]}$ recovers the path value $Y(t)$ in (1.2). In these two models the Hilbert norm is, respectively, the sequence ℓ_2 norm and the function L_2 norm; we write both as $\| \cdot \|_2$ when the model is clear.

An orthonormal basis (φ_i) of $L_2[0,1]$ connects the two descriptions directly: the coefficients $Y_i = \int_0^1 \varphi_i(t)\, dY(t)$ follow (1.1) with $\theta_i = \langle f, \varphi_i \rangle$. The full coefficient sequence retains the white-noise information, while Parseval's identity preserves signal distances. A finite truncation need not retain that information. The density estimation and fixed-design regression models remain separate experiments; identifying either with white noise requires an additional argument.

Reduction from estimation to testing. A lower bound over a large parameter class can be proved on a finite subfamily with separated targets and observation laws that are difficult to distinguish. An accurate estimator would identify the target by selecting its nearest neighbor; a lower bound on identification error therefore limits estimation accuracy.

We now specialize to metric estimation. Let $\mathcal{P}$ be a class of observation laws and $\theta : \mathcal{P} \to (\Theta, d_\Theta)$ its target map, with Θ carrying the Borel sigma-field. Estimators take values in Θ. The following reduction applies to any finite family of targets separated by a positive distance Δ; later chapters construct such families and bound the resulting testing error.

Proposition 1.1 (Reduction from Estimation to Testing) *Let $\mathcal{P}$ be a statistical experiment with target map $\theta : \mathcal{P} \to (\Theta, d_\Theta)$, where d_Θ is a metric. Suppose a finite subexperiment $\{P_\nu : \nu \in \mathcal{V}\}$, with $|\mathcal{V}| \geq 2$, has Δ-separated targets for some $\Delta > 0$:*

$$d_\Theta\big(\theta(P_\nu), \theta(P_{\nu'})\big) \geq \Delta \qquad \text{for all } \nu \neq \nu'.$$

For a measurable estimator $\widehat{\theta}$ taking values in Θ, define the nearest-neighbor decoder with any fixed tie-breaking rule:

$$\widehat{\nu} \in \arg\min_{\nu \in \mathcal{V}} d_\Theta\big(\widehat{\theta}, \theta(P_\nu)\big).$$

Then

$$\sup_{\nu \in \mathcal{V}} \mathbb{P}_\nu\Big(d_\Theta\big(\widehat{\theta}, \theta(P_\nu)\big) \geq \Delta/2\Big) \geq \sup_{\nu \in \mathcal{V}} \mathbb{P}_\nu(\widehat{\nu} \neq \nu).$$

Proof If $d_\Theta(\widehat{\theta}, \theta(P_\nu)) < \Delta/2$, the triangle inequality places every other target at distance greater than $\Delta/2$ from $\widehat{\theta}$. Thus $\theta(P_\nu)$ is the unique nearest target and $\widehat{\nu} = \nu$. Consequently,

$$\{\widehat{\nu} \neq \nu\} \subseteq \Big\{d_\Theta\big(\widehat{\theta}, \theta(P_\nu)\big) \geq \Delta/2\Big\},$$

which yields the claim after taking probabilities and suprema. □

For loss $L(\widehat{\theta}, \theta(P)) = d_\Theta(\widehat{\theta}, \theta(P))^r$, $r \geq 1$, the loss is at least $(\Delta/2)^r$ on the error event above. Hence

$$\sup_{P \in \mathcal{P}} \mathbb{E}_P L\big(\widehat{\theta}, \theta(P)\big) \geq (\Delta/2)^r \inf_{\widehat{\nu}} \sup_{\nu \in \mathcal{V}} \mathbb{P}_\nu\{\widehat{\nu} \neq \nu\}.$$

The infimum is over measurable decoders taking values in $\mathcal{V}$. The separation Δ uses the same metric as the loss; for rth-power L_r or ℓ_r loss, take $d_\Theta(u, v) = \|u - v\|_r$.

A one-dimensional Gaussian location model makes the reduction concrete. If $X \sim N(\theta, 1)$ with $\theta \in \{-a, a\}$, $a > 0$, and the loss is squared error, the two target values are separated by $2a$. Any estimator $\widehat{\theta}(X)$ induces a decoder by thresholding at zero, with either fixed tie rule. A decoding error therefore implies $|\widehat{\theta} - \theta| \geq a$.

The likelihood-ratio test thresholds X at zero and has error $\Phi(-a)$ under either law. It minimizes the average of the two errors and, because its errors are equal, also their maximum. The reduction therefore gives worst-case squared-error risk at least $a^2\Phi(-a)$, a constant multiple of a^2 when a is bounded above by a fixed constant. Chapter 2 develops this two-point argument as Le Cam's method; Chapter 3 uses larger finite families.

For general observation laws, distances between distributions can bound testing error without explicitly finding the optimal test. We develop these comparisons next.

1.3 Probability distances used in lower bounds

Suppose an observation has law P under the null hypothesis and Q under the alternative. How accurately can a test distinguish them? Total variation gives the minimum sum of testing errors exactly. Hellinger distance, Kullback–Leibler divergence, and chi-square divergence provide bounds that are often easier to calculate. We relate these quantities to testing error, then examine independent and transformed observations.

Throughout this section, P and Q are probability measures on a common measurable space.

Definition 1.2 (Total Variation and Affinity) Let μ be a common σ-finite dominating measure for P and Q, for example $P + Q$, and write $p = dP/d\mu$ and $q = dQ/d\mu$. The total variation distance and testing affinity (also called total-variation affinity) are

$$\mathrm{TV}(P, Q) = \sup_A \left|\mathbb{P}_P(A) - \mathbb{P}_Q(A)\right| = \frac{1}{2}\int |p - q|\, d\mu, \qquad \eta(P, Q) = \int (p \wedge q)\, d\mu.$$

These quantities do not depend on the particular dominating measure, and

$$\mathrm{TV}(P, Q) = 1 - \eta(P, Q).$$

Hellinger distance compares square roots of densities. Its affinity has a simple product formula, making it useful for independent observations. The underlying integral construction originates in Hellinger [72, Section 4].

Definition 1.3 (Hellinger Distance and Affinity) With the same density notation, the squared Hellinger distance and the Hellinger affinity are

$$H^2(P, Q) = \frac{1}{2}\int (\sqrt{p} - \sqrt{q})^2\, d\mu, \qquad \rho(P, Q) = \int \sqrt{pq}\, d\mu.$$

These quantities do not depend on the choice of dominating measure. With the book's normalization,

$$H^2(P, Q) = 1 - \rho(P, Q).$$

Kullback and Leibler [84] measure discrepancy through an expected log likelihood ratio. Chi-square divergence instead uses a likelihood-ratio second moment, which is particularly useful for mixtures. Unlike the preceding distances, these divergences need not be symmetric or finite.

Definition 1.4 (Kullback–Leibler Divergence and Chi-Square Divergence) With the same density notation, the Kullback–Leibler divergence (also called relative entropy) and the chi-square divergence are

$$\mathrm{KL}(P, Q) = \int p \log \frac{p}{q}\, d\mu, \qquad \chi^2(P, Q) = \int \frac{(p - q)^2}{q}\, d\mu.$$

We use $0 \log 0 = 0$ and zero integrand where both densities vanish. Both divergences have value $+\infty$ if $P \not\ll Q$ or the displayed integral is not finite. The second law Q is the reference law in both divergences. It is often convenient to use the chi-square affinity

$$\xi(P, Q) = \int \frac{q^2}{p}\, d\mu = 1 + \chi^2(Q, P),$$

when $Q \ll P$. Here the reference law is the first argument: $\xi(P, Q) = \mathbb{E}_P(dQ/dP)^2$.

The next comparisons turn Hellinger or KL bounds into total-variation bounds. The KL comparison is Pinsker's inequality; see Tsybakov [103, Section 2.4].

Lemma 1.5 (Basic Comparisons) *Let P and Q be probability measures on a common measurable space. With the distances and affinities defined above,*

$$H^2(P, Q) \le \mathrm{TV}(P, Q) \le \sqrt{2}\, H(P, Q), \qquad \mathrm{TV}^2(P, Q) \le \frac{1}{2}\, \mathrm{KL}(P, Q),$$

allowing $\mathrm{KL}(P, Q) = +\infty$. *In particular,*

$$\eta(P, Q) \le \rho(P, Q).$$

Proof Let p and q be densities with respect to μ. Since

$$|p - q| = |\sqrt{p} - \sqrt{q}|(\sqrt{p} + \sqrt{q}),$$

Cauchy–Schwarz gives

$$\mathrm{TV}(P, Q) = \frac{1}{2} \int |p - q|\, d\mu \le \frac{1}{2} \Big(\int (\sqrt{p} - \sqrt{q})^2\, d\mu \Big)^{1/2} \Big(\int (\sqrt{p} + \sqrt{q})^2\, d\mu \Big)^{1/2}.$$

Because $\int (\sqrt{p} - \sqrt{q})^2\, d\mu = 2H^2(P, Q)$ and $\int (\sqrt{p} + \sqrt{q})^2\, d\mu = 2 + 2\rho(P, Q) \le 4$, this yields

$$\mathrm{TV}(P, Q) \le \sqrt{2}\, H(P, Q).$$

For the reverse comparison, with the ratio defined as zero when $p = q = 0$,

$$(\sqrt{p} - \sqrt{q})^2 = \frac{(p - q)^2}{(\sqrt{p} + \sqrt{q})^2} \le |p - q|,$$

and therefore

$$H^2(P, Q) = \frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2\, d\mu \le \frac{1}{2} \int |p - q|\, d\mu = \mathrm{TV}(P, Q).$$

The identity $\eta(P, Q) = 1 - \mathrm{TV}(P, Q)$ and the normalization $\rho(P, Q) = 1 - H^2(P, Q)$ then give $\eta(P, Q) \le \rho(P, Q)$.

For Pinsker's inequality, reduce the observation to the indicator of $\{p \ge q\}$. This retains the total-variation difference; we show that the corresponding binary KL divergence is no larger than $\mathrm{KL}(P, Q)$. The assertion is immediate if $\mathrm{KL}(P, Q) = +\infty$, so assume $P \ll Q$ and $\mathrm{KL}(P, Q) < \infty$. Let

$$A = \{x : p(x) \ge q(x)\}, \qquad u = \mathbb{P}_P(A), \qquad v = \mathbb{P}_Q(A).$$

Then $u - v = \mathrm{TV}(P, Q)$. For any measurable set B with $\mathbb{P}_P(B), \mathbb{P}_Q(B) > 0$, normalization on B and nonnegativity of relative entropy give

$$\int_B p \log \frac{p}{q}\, d\mu = \mathbb{P}_P(B)\, \mathrm{KL}\big(P(\,\cdot \mid B), Q(\,\cdot \mid B)\big) + \mathbb{P}_P(B) \log \frac{\mathbb{P}_P(B)}{\mathbb{P}_Q(B)} \geq \mathbb{P}_P(B) \log \frac{\mathbb{P}_P(B)}{\mathbb{P}_Q(B)}.$$

If $\mathbb{P}_P(B) = 0$, both contributions are zero; if $\mathbb{P}_Q(B) = 0$, absolute continuity also gives $\mathbb{P}_P(B) = 0$. Applying the inequality on A and A^c gives

$$\mathrm{KL}(P, Q) \geq u \log \frac{u}{v} + (1 - u) \log \frac{1 - u}{1 - v}.$$

To bound this binary divergence, fix $v \in (0, 1)$ and set, for $t \in [-v, 1 - v]$,

$$g_v(t) = (v + t) \log \frac{v + t}{v} + (1 - v - t) \log \frac{1 - v - t}{1 - v}.$$

For $-v < t < 1 - v$, direct differentiation yields $g_v(0) = g_v'(0) = 0$ and

$$g_v''(t) = \frac{1}{v + t} + \frac{1}{1 - v - t} \geq 4.$$

Taylor's formula with integral remainder gives $g_v(t) \geq 2t^2$ in the interior; continuity extends the bound to the endpoints. When $v \in \{0, 1\}$, finite $\mathrm{KL}(P, Q)$ forces $u = v$; otherwise the binary divergence is $+\infty$, so the inequality remains valid. Taking $t = u - v = \mathrm{TV}(P, Q)$ proves

$$\mathrm{KL}(P, Q) \geq 2\, \mathrm{TV}^2(P, Q),$$

which is the stated Pinsker inequality. □

Lemma 1.6 (χ^2 Controls Total Variation) *Let P and Q be probability measures on a common measurable space with $P \ll Q$ and $\chi^2(P, Q) < \infty$. Then*

$$\mathrm{TV}(P, Q) \leq \frac{1}{2} \sqrt{\chi^2(P, Q)}.$$

Proof Put $L = dP/dQ$. The likelihood-ratio form of total variation and Cauchy–Schwarz give

$$\mathrm{TV}(P, Q) = \frac{1}{2} \mathbb{E}_Q |L - 1| \leq \frac{1}{2} \{\mathbb{E}_Q (L - 1)^2\}^{1/2} = \frac{1}{2} \sqrt{\chi^2(P, Q)}.$$

□

Figure 1.1 summarizes the available comparisons. Their statistical consequence follows from the next lemma: the testing affinity $\eta(P, Q)$ is the smallest possible sum of type-I and type-II errors [103, Section 2.2]. Thus an upper bound on total variation gives a lower bound on testing error.

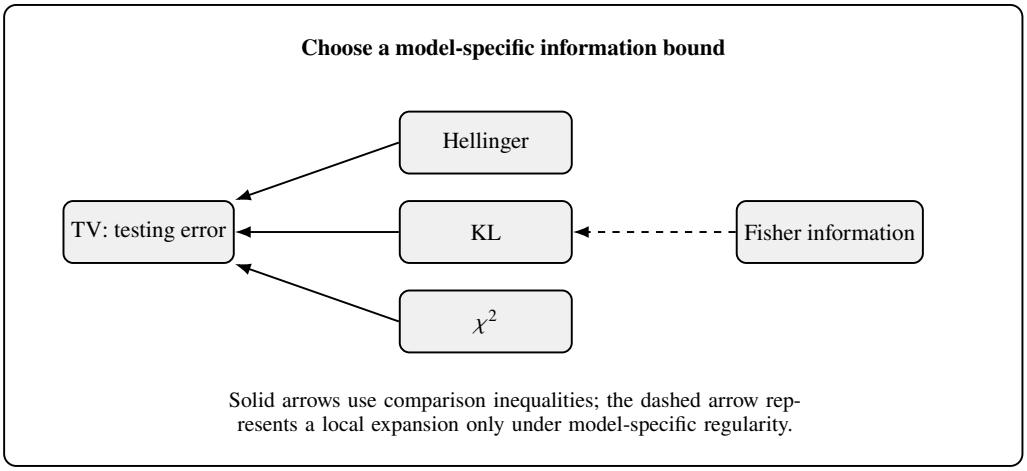


Figure 1.1 Ways to control testing error. Hellinger distance, KL, and chi-square divergence can each bound total variation under their stated conditions. Fisher information connects to divergences only through an appropriate local expansion; the diagram is not an ordering of the five quantities.

Lemma 1.7 (Testing Affinity as Minimum Total Error) *Let P, Q be probability laws on a common measurable space $(\mathcal{X}, \mathscr{X})$. For measurable tests $\psi : \mathcal{X} \to [0,1]$, where $\psi(x)$ is the probability of choosing Q over P,*

$$\inf_{0 \le \psi \le 1} \{\mathbb{E}_P \psi + \mathbb{E}_Q(1-\psi)\} = \eta(P,Q) = 1 - \mathrm{TV}(P,Q).$$

For densities p, q relative to a common dominating measure, the rule $\psi = \mathbb{I}\{q > p\}$ attains the infimum, with arbitrary values in $[0,1]$ on $\{p = q\}$.

Proof Write p and q for densities of P and Q with respect to a common dominating measure μ. For any measurable test $\psi : \mathcal{X} \to [0,1]$,

$$\mathbb{E}_P \psi + \mathbb{E}_Q(1-\psi) = \int \psi p \, d\mu + \int (1-\psi) q \, d\mu = \int q \, d\mu + \int \psi (p-q) \, d\mu.$$

Minimize pointwise: choose $\psi(x) = 0$ where $p \ge q$ and $\psi(x) = 1$ where $p < q$. The resulting likelihood-ratio test $\psi^* = \mathbb{I}\{p < q\}$ has total error

$$\int_{\{p \ge q\}} q \, d\mu + \int_{\{p < q\}} p \, d\mu = \int (p \wedge q) \, d\mu = \eta(P,Q). \qquad \square$$

Minimizing total error uses likelihood-ratio threshold one. If the type-I error must instead be at most a prescribed level, the threshold must be chosen to meet that constraint. Neyman and Pearson [92] show that the resulting rule minimizes type-II error.

Lemma 1.8 (Neyman–Pearson at a Prescribed Level) *Observe $X \sim P_i$, $i \in \{0,1\}$, and test $H_0 : P_0$ against $H_1 : P_1$. Let p_0, p_1 be densities with respect to $\mu = P_0 + P_1$. For each $0 < \alpha < 1$, there exist $c \geq 0$ and $\gamma \in [0,1]$ such that*

$$\varphi_\alpha = \mathbb{I}\{p_1 > cp_0\} + \gamma \mathbb{I}\{p_1 = cp_0\}, \qquad \mathbb{E}_{P_0}\varphi_\alpha = \alpha.$$

Every measurable test $0 \leq \varphi \leq 1$ with $\mathbb{E}_{P_0}\varphi \leq \alpha$ satisfies

$$\mathbb{E}_{P_1}(1-\varphi) \geq \mathbb{E}_{P_1}(1-\varphi_\alpha).$$

Proof Under P_0, the ratio $R = p_1/p_0$ is finite and nonnegative almost surely. Choose an upper quantile $c \geq 0$ with

$$\mathbb{P}_{P_0}(R > c) \leq \alpha \leq \mathbb{P}_{P_0}(R \geq c),$$

and choose γ so that $\mathbb{P}_{P_0}(R > c) + \gamma \mathbb{P}_{P_0}(R = c) = \alpha$. The rule also rejects on $\{p_0 = 0 < p_1\}$, which has zero probability under P_0. Thus its type-I error is exactly α. For any competing test, the threshold rule gives $(\varphi_\alpha - \varphi)(p_1 - cp_0) \geq 0$ pointwise. Integration yields

$$\mathbb{E}_{P_1}(\varphi_\alpha - \varphi) \geq c\, \mathbb{E}_{P_0}(\varphi_\alpha - \varphi) \geq 0,$$

which proves the type-II error bound. □

The prescribed-level formulation is used for Gaussian testing in Section 5.2 and for transferring confidence-interval coverage in Section 6.2.

For independent observations, information accumulates according to the following product formulas. They determine how small a one-observation perturbation must be to keep the full sample difficult to distinguish.

Remark 1.9 (Products) For every integer $n \geq 1$, if $P^{\otimes n}$ and $Q^{\otimes n}$ are product measures, then

$$\rho(P^{\otimes n}, Q^{\otimes n}) = \rho(P,Q)^n.$$

If in addition $P \ll Q$, then

$$\mathrm{KL}(P^{\otimes n}, Q^{\otimes n}) = n\,\mathrm{KL}(P,Q).$$

The same likelihood-ratio factorization gives

$$1 + \chi^2(P^{\otimes n}, Q^{\otimes n}) = \{1 + \chi^2(P,Q)\}^n.$$

Lemma A.7 proves these identities, allowing infinite divergences; recall that $\xi(P,Q) = 1 + \chi^2(Q,P)$.

Data processing. Sometimes the observed data are a statistic of a more informative experiment whose divergence is easier to calculate. Data processing justifies using that calculation as an upper bound. Let P and Q be the laws of X, with $P \ll Q$, and write $L = dP/dQ$. If only a measurable statistic $Z = Z(X)$ is retained, denote its laws by P_Z and Q_Z. The likelihood-ratio identity $(dP_Z/dQ_Z)(Z) = \mathbb{E}_Q(L \mid Z)$ holds Q-almost surely. Conditional Jensen's inequality applied to the convex function $t \log t$ gives

$$\mathrm{KL}(P_Z, Q_Z) = \mathbb{E}_Q\big[\mathbb{E}_Q(L \mid Z) \log \mathbb{E}_Q(L \mid Z)\big] \leq \mathbb{E}_Q(L \log L) = \mathrm{KL}(P,Q).$$

Thus a joint divergence calculated with latent variables included bounds the observed-data divergence from above, as in the clustering argument of Section 8.4. For missing observations, a direct calculation can be sharper; Sections 8.5 and 8.6 treat independent missingness.

Section 3.4 places these comparisons in the broader f-divergence framework of Ali and Silvey [1] and Csiszár [53]. The next section takes a complementary route through local information inequalities.

1.4 Local-information lower bounds

An unbiased estimator must track changes in its target even when the corresponding observation laws are close. How much variance does this require? The Hammersley–Chapman–Robbins (HCR) inequality [71, 50] answers through a finite likelihood-ratio comparison; the Cramér–Rao bound follows from a suitable local expansion. To obtain a risk bound without unbiasedness, the van Trees inequality [105] averages squared error over a prior. These methods also provide lower bounds through parametric submodels of larger statistical models.

For a scalar target, HCR compares the change in an estimator's expectation with the chi-square divergence between two observation laws. Under unbiasedness, this change equals the target separation and variance equals squared-error risk. Otherwise, its numerator depends on the estimator. Write $\mathrm{Var}_\theta(U) = \mathbb{E}_\theta(U - \mathbb{E}_\theta U)^2$ for the variance of a square-integrable real random variable U.

Theorem 1.10 (Hammersley–Chapman–Robbins Inequality) *Let $\{P_\theta : \theta \in \Theta \subset \mathbb{R}\}$ be a scalar-parameter statistical experiment, let $T : \Theta \to \mathbb{R}$ be the target, and let $\widehat{T} = \widehat{T}(X)$ be a real-valued estimator. Fix $\theta \in \Theta$ and a comparison point $\theta' \in \Theta$ with $\theta' \neq \theta$, $P_{\theta'} \ll P_\theta$, $0 < \chi^2(P_{\theta'}, P_\theta) < \infty$, and $\mathbb{E}_\theta \widehat{T}^2 < \infty$. Then*

$$\mathrm{Var}_\theta(\widehat{T}) \geq \frac{\left(\mathbb{E}_{\theta'}\widehat{T} - \mathbb{E}_\theta \widehat{T}\right)^2}{\chi^2(P_{\theta'}, P_\theta)}.$$

In particular, if $\widehat{T}$ is unbiased for T, meaning that $\mathbb{E}_\vartheta \widehat{T} = T(\vartheta)$ for every $\vartheta \in \Theta$, then

$$\mathrm{Var}_\theta(\widehat{T}) \geq \sup_{\theta' \neq \theta:\, P_{\theta'} \ll P_\theta,\, 0 < \chi^2(P_{\theta'}, P_\theta) < \infty} \frac{\left(T(\theta') - T(\theta)\right)^2}{\chi^2(P_{\theta'}, P_\theta)}.$$

Proof Write $L = dP_{\theta'}/dP_\theta$. We express the expectation difference as a covariance with L, then apply Cauchy–Schwarz. First, the comparison expectation exists, since $\mathbb{E}_{\theta'}|\widehat{T}| = \mathbb{E}_\theta(|\widehat{T}|L)$ and

$$\mathbb{E}_\theta(|\widehat{T}|L) \leq \{\mathbb{E}_\theta \widehat{T}^2\}^{1/2}\{\mathbb{E}_\theta L^2\}^{1/2} = \{\mathbb{E}_\theta \widehat{T}^2\}^{1/2}\{1 + \chi^2(P_{\theta'}, P_\theta)\}^{1/2}.$$

Since $\mathbb{E}_\theta(L - 1) = 0$,

$$\mathbb{E}_{\theta'}\widehat{T} - \mathbb{E}_\theta \widehat{T} = \mathbb{E}_\theta\left[(\widehat{T} - \mathbb{E}_\theta \widehat{T})(L - 1)\right].$$

The Cauchy–Schwarz inequality gives

$$\left(\mathbb{E}_{\theta'}\widehat{T} - \mathbb{E}_\theta \widehat{T}\right)^2 \leq \mathrm{Var}_\theta(\widehat{T})\, \mathbb{E}_\theta(L - 1)^2 = \mathrm{Var}_\theta(\widehat{T})\, \chi^2(P_{\theta'}, P_\theta),$$

which proves the first display after rearrangement. The unbiased case follows by substituting $\mathbb{E}_{\theta'}\widehat{T} = T(\theta')$ and $\mathbb{E}_\theta \widehat{T} = T(\theta)$ and then taking the supremum over $\theta' \neq \theta$. □

The Cramér–Rao bound follows by letting the comparison point approach the parameter of interest. This step requires the quadratic expansion of chi-square divergence stated below, not merely differentiability of the density. For a vector parameter, optimizing over directions gives the strongest bound for the scalar target.

Corollary 1.11 (Cramér–Rao from HCR) *Let $\{P_\theta : \theta \in \Theta\}$ be a dominated scalar-parameter model with densities p_θ with respect to μ, and let θ be an interior point of Θ. Assume that the score $\partial_\theta \log p_\theta$ exists P_θ-almost surely and that the local chi-square expansion*

$$\chi^2(P_{\theta+h}, P_\theta) = I(\theta)h^2 + o(h^2), \qquad I(\theta) = \int \left(\frac{\partial}{\partial\theta}\log p_\theta(x)\right)^2 p_\theta(x)\, d\mu(x),$$

holds as $h \to 0$ through values satisfying $\theta + h \in \Theta$, where $I(\theta)$ is the Fisher information and $0 < I(\theta) < \infty$. Assume also that the scalar target $T : \Theta \to \mathbb{R}$ is differentiable at θ. If $\widehat{T}$ is square-integrable under P_θ and unbiased for $T(\vartheta)$ in a neighborhood of θ, then

$$\operatorname{Var}_\theta(\widehat{T}) \geq \frac{\{T'(\theta)\}^2}{I(\theta)}.$$

In the vector-parameter case, let θ be an interior point of $\Theta \subset \mathbb{R}^d$. Let $S_\theta(x) = \nabla_\theta \log p_\theta(x)$ denote the score vector and assume that the Fisher information matrix $I(\theta) = \mathbb{E}_\theta[S_\theta(X)S_\theta(X)^\top]$ is finite and nonsingular. Assume also that for every nonzero direction u the one-dimensional submodel $t \mapsto P_{\theta+tu}$ satisfies

$$\chi^2(P_{\theta+tu}, P_\theta) = t^2 u^\top I(\theta)u + o(t^2) \qquad (t \to 0).$$

If T is differentiable at θ, then every square-integrable estimator unbiased in a neighborhood of θ satisfies

$$\operatorname{Var}_\theta(\widehat{T}) \geq \nabla T(\theta)^\top I(\theta)^{-1} \nabla T(\theta).$$

Proof Apply Theorem 1.10 with $\theta' = \theta + h$. Under the assumed local chi-square expansion and differentiability of the target,

$$\chi^2(P_{\theta+h}, P_\theta) = I(\theta)h^2 + o(h^2), \qquad T(\theta + h) - T(\theta) = T'(\theta)h + o(h).$$

Substituting these expansions into the HCR bound gives

$$\operatorname{Var}_\theta(\widehat{T}) \geq \frac{\{T'(\theta)h + o(h)\}^2}{I(\theta)h^2 + o(h^2)}.$$

Divide numerator and denominator by h^2 and let $h \to 0$ to obtain

$$\operatorname{Var}_\theta(\widehat{T}) \geq \frac{\{T'(\theta)\}^2}{I(\theta)}.$$

For the vector statement, restrict to a nonzero direction $u \in \mathbb{R}^d$. Define the target along the submodel $t \mapsto P_{\theta+tu}$ by

$$\phi_u(t) = T(\theta + tu).$$

Its derivative is $\phi_u'(0) = u^\top \nabla T(\theta)$, while the Fisher information along this submodel is the quadratic form $u^\top I(\theta)u$. Applying the scalar inequality to unbiased estimation of $\phi_u(t)$ at $t = 0$ yields

$$\mathrm{Var}_\theta(\widehat{T}) \geq \frac{\{u^\top \nabla T(\theta)\}^2}{u^\top I(\theta)u} \qquad \text{for every } u \neq 0.$$

If $\nabla T(\theta) = 0$, the asserted lower bound is immediate. Otherwise, maximize the right-hand side over u. Writing $v = I(\theta)^{1/2}u$ gives

$$\frac{\{u^\top \nabla T(\theta)\}^2}{u^\top I(\theta)u} = \frac{\{v^\top I(\theta)^{-1/2}\nabla T(\theta)\}^2}{\|v\|_2^2} \leq \nabla T(\theta)^\top I(\theta)^{-1}\nabla T(\theta)$$

by the Cauchy–Schwarz inequality, with equality for $v \propto I(\theta)^{-1/2}\nabla T(\theta)$. This proves

$$\mathrm{Var}_\theta(\widehat{T}) \geq \nabla T(\theta)^\top I(\theta)^{-1}\nabla T(\theta).$$

□

For the classical scalar information bound under differentiability and regularity assumptions, see Cramér [52, Section 32.3] and Rao [94]. HCR remains useful when a finite perturbation gives a stronger bound or the local expansion required by Corollary 1.11 is unavailable.

To remove the unbiasedness requirement, the van Trees argument averages risk over a prior on a parameter interval. Integration by parts relates estimation error to the prior average of the target's derivative; Cauchy–Schwarz then gives a denominator combining information from the observations and the prior. The boundary and integrability conditions below justify these steps. A useful prior keeps the average derivative away from zero without making its own information contribution too large.

Let $\{P_\theta : a \leq \theta \leq b\}$ be a dominated one-dimensional experiment with densities p_θ with respect to a measure μ. Let π be a continuously differentiable prior density on $[a, b]$ such that $\pi(a) = \pi(b) = 0$, $\pi(\theta) > 0$ for $a < \theta < b$, and

$$\int_a^b \frac{\{\pi'(\theta)\}^2}{\pi(\theta)}\, d\theta < \infty.$$

Let $T : [a, b] \to \mathbb{R}$ be absolutely continuous with

$$\int_a^b |T'(\theta)|\pi(\theta)\, d\theta < \infty.$$

Theorem 1.12 (van Trees Inequality) *For the experiment, prior, and target just specified, suppose there is a measurable score $S_\theta(x)$, equal to $\partial_\theta \log p_\theta(x)$ wherever $p_\theta(x) > 0$ and that derivative exists, such that*

$$\partial_\theta p_\theta(x) = p_\theta(x)S_\theta(x)$$

for almost every (x, θ) with respect to $\mu(dx)\pi(\theta)\, d\theta$. Assume also that

$$\int S_\theta(x)p_\theta(x)\, d\mu(x) = 0$$

for π-almost every θ.

Suppose $\theta \mapsto p_\theta(x)\pi(\theta)$ is absolutely continuous for μ-almost every x, with zero boundary values at a and b, and that the average Fisher information is finite:

$$\int_a^b I(\theta)\pi(\theta)\,d\theta < \infty, \qquad I(\theta) = \mathbb{E}_\theta S_\theta(X)^2.$$

Then every estimator $\widehat{T}(X)$ with finite integrated squared-error risk satisfies

$$\int_a^b \mathbb{E}_\theta(\widehat{T}(X) - T(\theta))^2\pi(\theta)\,d\theta \geq \frac{\left(\int_a^b T'(\theta)\pi(\theta)\,d\theta\right)^2}{\int_a^b I(\theta)\pi(\theta)\,d\theta + \int_a^b \frac{\{\pi'(\theta)\}^2}{\pi(\theta)}\,d\theta}.$$

For $T(\theta) = \theta$, the numerator equals one.

Proof We first justify integration by parts, which expresses the prior average of T' as an expectation involving the estimation error and joint score. Cauchy–Schwarz then gives the risk bound.

Let S_θ be the score from the statement, and define the prior score

$$s_\pi(\theta) = \frac{\pi'(\theta)}{\pi(\theta)}.$$

Write $q(x,\theta) = p_\theta(x)\pi(\theta)$ for the joint density. On the set where $q > 0$, the derivative assumptions give

$$\partial_\theta q(x,\theta) = q(x,\theta)\{S_\theta(x) + s_\pi(\theta)\}.$$

The identity also holds almost everywhere on $\{q = 0\}$, since a nonnegative absolutely continuous function has derivative zero almost everywhere on its zero set. Let R denote the finite integrated squared-error risk and put

$$J = \int_a^b I(\theta)\pi(\theta)\,d\theta + \int_a^b \frac{\{\pi'(\theta)\}^2}{\pi(\theta)}\,d\theta < \infty.$$

In fact $J > 0$: if the prior-score term vanished, positivity of π in (a,b) and continuity of π' would force $\pi' \equiv 0$, contradicting $\pi(a) = 0$ and $\int_a^b \pi = 1$. Because $(S_\theta + s_\pi)^2 \leq 2S_\theta^2 + 2s_\pi^2$, Cauchy–Schwarz gives

$$\iint |\widehat{T}(x) - T(\theta)|\,|\partial_\theta q(x,\theta)|\,d\mu(x)\,d\theta \leq \sqrt{2RJ} < \infty.$$

Also $\iint |T'(\theta)|q(x,\theta)\,d\mu(x)\,d\theta = \int_a^b |T'(\theta)|\pi(\theta)\,d\theta < \infty$. The absolute continuity of T and $q(x,\cdot)$, together with the zero boundary values of q, therefore justifies integration by parts for almost every x and then Fubini's theorem. It gives

$$\begin{aligned}
&\int_a^b \mathbb{E}_\theta\Big[(\widehat{T}(X) - T(\theta))(S_\theta(X) + s_\pi(\theta))\Big]\,\pi(\theta)\,d\theta \\
&= \iint (\widehat{T}(x) - T(\theta))\,\partial_\theta\{p_\theta(x)\pi(\theta)\}\,d\mu(x)\,d\theta \\
&= \iint T'(\theta)p_\theta(x)\pi(\theta)\,d\mu(x)\,d\theta \\
&= \int_a^b T'(\theta)\pi(\theta)\,d\theta.
\end{aligned}$$

The cross term in the joint score's second moment is integrable by Cauchy–Schwarz and vanishes by $\mathbb{E}_\theta S_\theta(X) = 0$. Hence

$$\int_a^b \mathbb{E}_\theta\{S_\theta(X) + s_\pi(\theta)\}^2\pi(\theta)\,d\theta = \int_a^b \{I(\theta) + s_\pi(\theta)^2\}\pi(\theta)\,d\theta = J.$$

Applying Cauchy–Schwarz to the preceding integration-by-parts identity gives

$$\left\{\int_a^b T'(\theta)\pi(\theta)\,d\theta\right\}^2 \le RJ.$$

Dividing by $J > 0$ proves the inequality; for $T(\theta) = \theta$, the numerator is $(\int_a^b \pi)^2 = 1$. □

Worst-case risk dominates prior-averaged risk. Thus van Trees gives a minimax lower bound over all estimators when the prior is supported in the parameter class and the stated regularity conditions hold. This is the key distinction from the unbiased-estimation bounds above. If the observations have been transformed or filtered, the information and regularity conditions must refer to the resulting observation law.

Local asymptotic minimax theory provides a complementary framework for risk over shrinking parameter neighborhoods. Under local asymptotic normality and suitable loss conditions, a Gaussian shift approximation yields asymptotic lower bounds for arbitrary, possibly biased estimators. See van der Vaart [104, Theorem 8.11] and Ibragimov and Has'minskii [73]. The finite-sample arguments developed in this book do not require that framework.

1.5 Exercises

1.1. For $p, q \in [0,1]$, calculate total variation and Hellinger affinity for $P = \mathrm{Ber}(p)$ and $Q = \mathrm{Ber}(q)$. Check the boundary values $p, q \in \{0,1\}$.

1.2. Prove $\mathrm{TV}(P,Q) \le \sqrt{1 - \rho(P,Q)^2}$ from the density formulas by Cauchy–Schwarz. Compare this inequality with Pinsker's bound for nearby Bernoulli laws.

1.3. Revisit the normal-mean calculation in Section 1.1 using the estimation-to-testing reduction under squared-error loss. Choose two means separated by a constant multiple of $\sigma/\sqrt{n}$ and obtain a minimax lower bound of the form $c\sigma^2/n$ with an explicit constant $c > 0$.

1.4. Let $X \sim N(0,1)$ under both parameter values $\theta = 0$ and $\theta = h \neq 0$, but observe $Y = X + \theta$. Calculate the KL divergence between the two latent laws and between the two observed laws. Explain why this does not contradict the data-processing argument in Section 1.3, identifying the common-map condition that fails.

1.5. Construct a case in which an estimator has very small risk at one parameter point but poor worst-case risk. Explain why neither fact contradicts the other.

1.6. Show directly that Hellinger affinity tensorizes for independent observations, including the case of a zero affinity.

1.7. Derive the centered likelihood-ratio identity used in the Hammersley–Chapman–Robbins inequality, including its integrability condition.

1.8. Compute the chi-square divergence of two Gaussian location laws and recover a finite-difference variance bound for an unbiased estimator of a linear functional.

1.9. Give a model with parameter-dependent support in which formal score differentiation omits a boundary contribution.

1.10. Under the assumptions of Theorem 1.12, verify its integration-by-parts identity using $q(x,\theta) = p_\theta(x)\pi(\theta)$. Then repeat that step allowing nonzero endpoint values of π and q, retaining the absolute-continuity and integrability conditions needed for integration by parts and assuming the boundary terms are integrable. Identify the boundary correction and explain why the uncorrected van Trees bound cannot simply be reused.

1.11. For positive-definite information matrix I and fixed gradient g, optimize $(u^\top g)^2/(u^\top I u)$ over nonzero directions u.

Selected hints. For Exercise 1.2, factor $p - q = (\sqrt{p} - \sqrt{q})(\sqrt{p} + \sqrt{q})$ and integrate the two squared factors separately. For Exercise 1.3, compare means 0 and $\sigma/\sqrt{n}$ and use the nearest-mean decision rule induced by an estimator. For Exercise 1.5, compare the estimator that is identically zero with the sample mean in the normal location model.

2

Two Points, Mixtures, and Constrained Risk

A lower bound can sometimes be established by examining only two possible parameters, even when the original model is infinite-dimensional. The difficulty is choosing a pair that captures the statistical task. Two densities may differ at a point while remaining close as sampling laws; two signals may differ in a direction that an observation operator strongly attenuates. Chapter 1 supplied the estimation-to-testing reduction. This chapter develops the constructions that make that reduction informative.

Section 2.1 develops Le Cam's two-point method: choose admissible alternatives and balance their target separation against statistical distance. A single pair can miss uncertainty about an unknown sign, support, or location. Section 2.2 compares a null law with a mixture that hides this structure while preserving target separation. Likelihood-ratio truncation controls rare extremes, anticipating the detection arguments of Chapter 9.

Sections 2.3 and 2.4 ask how accuracy at one parameter or over one class limits the same estimator's performance elsewhere. Brown and Low's constrained-risk inequality [12] links the two risks through a likelihood-ratio moment. Cai and Low [37] extend the comparison to two priors, accounting for variation of the target within a prior as well as the gap between prior target means. Moment matching provides a way to keep the observation mixtures close despite that gap.

Section 2.5 changes the error criterion, bounding the probability of missing a specified accuracy rather than only expected loss. This gives benchmarks for reliable estimation and recovery; coverage requirements turn related testing comparisons into confidence-set size bounds. The chapter's constructions recur in functional estimation, honest inference, and adaptation. Chapter 3 extends the toolkit to recovering many uncertain features or identifying one member of a large separated family.

2.1 Le Cam's two-point method

Le Cam's method bounds estimation risk by restricting the parameter class to two alternatives. Suppose $X \sim P_\vartheta$, with $\vartheta \in \Theta$, and the target is $T(\vartheta)$. Choose $\vartheta_0, \vartheta_1 \in \Theta$ whose target values are separated under the loss but whose observation laws remain difficult to distinguish. The examples show how changing support, local smoothness, and attenuation by an observation operator determine a useful pair.

The lemma allows a general loss and is stated directly in terms of two target values: set $\theta_i = T(\vartheta_i)$ and $P_i = P_{\vartheta_i}$ for $i = 0, 1$. See Yu [114] for the two-point method and Bretagnolle and Huber [11] for related testing inequalities in density estimation.

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Lemma 2.1 (Le Cam's Two-Point Lemma) *Let P_0, P_1 be probability laws on a common measurable observation space, with targets θ_0, θ_1, measurable action space $\mathcal{A}$, and nonnegative measurable loss $L : \mathcal{A} \times \{\theta_0, \theta_1\} \to [0, \infty]$. Set*

$$c(\theta_0, \theta_1) = \frac{1}{2} \inf_{t \in \mathcal{A}} \{L(t, \theta_0) + L(t, \theta_1)\}.$$

If $c(\theta_0, \theta_1) < \infty$, every measurable $\mathcal{A}$-valued estimator δ satisfies

$$\max\{\mathbb{E}_0 L(\delta, \theta_0), \mathbb{E}_1 L(\delta, \theta_1)\} \geq c(\theta_0, \theta_1)\, \eta(P_0, P_1) \geq \frac{1}{2} c(\theta_0, \theta_1) \rho(P_0, P_1)^2.$$

Proof Let $\mu = P_0 + P_1$ and $p_i = dP_i/d\mu$ for $i = 0, 1$. On the common part of the two laws, the losses must sum to at least $2c(\theta_0, \theta_1)$. Thus, for any estimator δ,

$$\begin{aligned} \max\{\mathbb{E}_0 L(\delta, \theta_0), \mathbb{E}_1 L(\delta, \theta_1)\} &\geq \frac{1}{2}\{\mathbb{E}_0 L(\delta, \theta_0) + \mathbb{E}_1 L(\delta, \theta_1)\} \\ &\geq \frac{1}{2} \int \{L(\delta, \theta_0) + L(\delta, \theta_1)\}(p_0 \wedge p_1)\, d\mu \\ &\geq c(\theta_0, \theta_1) \int (p_0 \wedge p_1)\, d\mu. \end{aligned}$$

The last expression equals $c(\theta_0, \theta_1)\eta(P_0, P_1)$. To obtain the Hellinger-affinity form, factor $|p_0 - p_1|$ and apply Cauchy–Schwarz:

$$\begin{aligned} \mathrm{TV}(P_0, P_1) &= \frac{1}{2} \int |\sqrt{p_0} - \sqrt{p_1}|(\sqrt{p_0} + \sqrt{p_1})\, d\mu \\ &\leq \frac{1}{2}\{2 - 2\rho(P_0, P_1)\}^{1/2}\{2 + 2\rho(P_0, P_1)\}^{1/2} \\ &= \sqrt{1 - \rho(P_0, P_1)^2}. \end{aligned}$$

Hence $\eta = 1 - \mathrm{TV} \geq 1 - \sqrt{1 - \rho^2}$. Since $1 - \sqrt{1 - u} \geq u/2$ for $0 \leq u \leq 1$, taking $u = \rho(P_0, P_1)^2$ gives $\eta(P_0, P_1) \geq \rho(P_0, P_1)^2/2$. □

For the losses used most often below, the loss-separation constant has a simple form.

Remark 2.2 (Useful Special Cases) When the action space is $\mathbb{R}$, absolute error $L(t, \theta) = |t - \theta|$ gives

$$c(\theta_0, \theta_1) = \frac{1}{2} |\theta_0 - \theta_1|.$$

Squared error $L(t, \theta) = (t - \theta)^2$ gives

$$c(\theta_0, \theta_1) = \frac{1}{4} (\theta_0 - \theta_1)^2.$$

When the action space is $\mathbb{R}^d$, with $d \geq 1$, squared Euclidean loss $L(t, \theta) = \|t - \theta\|_2^2$ gives

$$c(\theta_0, \theta_1) = \frac{1}{4} \|\theta_0 - \theta_1\|_2^2.$$

Example 2.3 (Uniform Upper-Endpoint Estimation) For real $u < v$, $\mathrm{Unif}(u, v)$ denotes the uniform law with density $(v-u)^{-1}\mathbf{1}_{(u,v)}$. Suppose $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}(0, \theta)$ with $\theta \in [a, b] \subset (0, \infty)$, and assume that for all sufficiently large n the interval $[a, b]$ contains both 1 and $1 + n^{-1}$ (for example, $0 < a < 1 < b$). We estimate the upper endpoint under squared-error loss. The sample maximum suggests that accuracy is governed by the small unobserved interval near that endpoint; the hard pair changes the support on just that scale. Set $\theta_0 = 1$ and $\theta_1 = 1 + n^{-1}$. For one observation, the densities are

$$p_0(y) = \mathbf{1}_{[0,1]}(y), \qquad p_1(y) = \frac{1}{1+1/n}\mathbf{1}_{[0,1+1/n]}(y).$$

Hence the Hellinger affinity is

$$\rho(P_{\theta_0}, P_{\theta_1}) = \int \sqrt{p_0 p_1}\, dy = \int_0^1 \frac{dy}{\sqrt{1+1/n}} = \left(1 + \frac{1}{n}\right)^{-1/2}.$$

Product structure gives

$$\rho(P_{\theta_0}^{\otimes n}, P_{\theta_1}^{\otimes n}) = \rho(P_{\theta_0}, P_{\theta_1})^n = \left(1 + \frac{1}{n}\right)^{-n/2} > e^{-1/2}.$$

Under squared error, Remark 2.2 gives

$$c(\theta_0, \theta_1) = \frac{1}{4}(\theta_1 - \theta_0)^2 = \frac{1}{4n^2}.$$

Lemma 2.1 therefore yields

$$\inf_{\delta} \sup_{\theta \in [a,b]} \mathbb{E}_\theta (\delta - \theta)^2 \gtrsim n^{-2}.$$

This matches the squared-error rate achieved by the sample maximum; see Lehmann and Casella [88, Example 2.1.14]. Unlike regular Gaussian location, this model has changing support. The overlap calculation reveals the relevant separation scale without a score expansion. □

For pointwise density estimation, the support stays fixed and a local perturbation changes the target. We specify the smoothness class before constructing a narrow, mean-zero bump that preserves total mass.

For $\alpha > 0$ and $M > 0$, write $\Lambda^\alpha(M)$ for the Hölder ball on $[0, 1]$ used throughout the book. Put $r = \lceil \alpha \rceil - 1$ and $\beta = \alpha - r \in (0, 1]$; at integer smoothness, $r = \alpha - 1$ and $\beta = 1$. With $C^r[0, 1]$ denoting the functions with r continuous derivatives, define

$$\begin{aligned}\Lambda^\alpha(M) = \big\{ f \in C^r[0,1] : \big\|f^{(k)}\big\|_\infty \le M, \quad 0 \le k \le r, \\ |f^{(r)}(x) - f^{(r)}(y)| \le M|x-y|^\beta, \quad x, y \in [0,1] \big\}.\end{aligned} \tag{2.1}$$

For $0 < \alpha \le 1$ this reduces to the usual condition $\|f\|_\infty \le M$ and $|f(x) - f(y)| \le M|x-y|^\alpha$. We write $[u]_{C^\beta} = \sup_{x \ne y} |u(x) - u(y)|/|x-y|^\beta$ for the Hölder seminorm on the function's domain; bumps supported on a subinterval are extended by zero when this seminorm is taken on $\mathbb{R}$. When this smoothness ball is used as a density model, we use the explicit density subclass

$$\mathcal{D}_\alpha(M) = \left\{ f \in \Lambda^\alpha(M) : f \ge 0, \ \int_0^1 f(x)\, dx = 1 \right\}. \tag{2.2}$$

The density examples assume $M > 1$, leaving room to perturb the uniform density within the sup-norm constraint. At $M = 1$, nonnegativity, unit integral, and $\|f\|_\infty \le 1$ force $f = 1$ almost everywhere.

Example 2.4 (Running Example: Pointwise Density Estimation over a Hölder Ball) Suppose we observe n i.i.d. observations $X_1, \dots, X_n$ from a density f on $[0,1]$. Write P_f for the law of one draw and $\mathbb{E}_f$ for expectation under the full sample law $P_f^{\otimes n}$. At a fixed interior point $x_0 \in (0,1)$, consider estimation of $f(x_0)$ under squared error over the density class $\mathcal{D}_\alpha(M)$ in (2.2). We use Le Cam's two-point argument to lower-bound the minimax risk

$$\inf_{\hat f(x_0)} \sup_{f \in \mathcal{D}_\alpha(M)} \mathbb{E}_f\big(\hat f(x_0) - f(x_0)\big)^2.$$

Choose a C^∞ bump g supported on $[-1,1]$ with $\int g = 0$, $g(0) \neq 0$, and $\|g\|_\infty \le 1$. For a bandwidth $0 < h < \frac{1}{2}\min\{x_0, 1-x_0\}$ define

$$f_0(x) \equiv 1, \qquad f_1(x) = 1 + c_0 h^\alpha g\Big(\frac{x - x_0}{h}\Big),$$

where $c_0 > 0$ will be chosen to ensure class membership and statistical closeness.

To check class membership, first note that for $0 \le k \le r$ the kth derivative of the perturbation is

$$c_0 h^{\alpha-k} g^{(k)}\Big(\frac{x - x_0}{h}\Big),$$

and its rth derivative has β-Hölder seminorm at most $c_0 [g^{(r)}]_{C^\beta}$. Since $M > 1$, choosing c_0 below a constant depending only on $M - 1$, α, and g controls the sup norms and the Hölder seminorm uniformly in $h \le 1$. The same choice makes $f_1 \ge 0$. Finally, the support condition keeps the bump inside $[0,1]$, and the change of variables $u = (x - x_0)/h$ gives

$$\int_0^1 (f_1 - f_0)(x)\, dx = c_0 h^{\alpha+1} \int g(u)\, du = 0.$$

The bump changes the target by

$$|f_1(x_0) - f_0(x_0)| = c_0 |g(0)| h^\alpha \asymp h^\alpha.$$

To control statistical closeness, compute the one-sample chi-square divergence relative to f_0:

$$\begin{aligned}\chi^2(P_{f_1}, P_{f_0}) &= \int_0^1 \frac{(f_1 - f_0)^2}{f_0}\, dx = c_0^2 h^{2\alpha} \int g\Big(\frac{x - x_0}{h}\Big)^2\, dx \\ &= c_0^2 h^{2\alpha+1} \int g(u)^2\, du \asymp h^{2\alpha+1}.\end{aligned}$$

Tensorization gives

$$\chi^2(P_{f_1}^{\otimes n}, P_{f_0}^{\otimes n}) = (1 + \chi^2(P_{f_1}, P_{f_0}))^n - 1 \le \exp\{C n c_0^2 h^{2\alpha+1}\} - 1,$$

for a constant C depending only on g. Take $h \asymp n^{-1/(2\alpha+1)}$ and choose the fixed constant c_0 small enough that this product chi-square divergence is bounded by a numerical constant strictly below 4. Lemma 1.6 then shows that the product total-variation distance is bounded

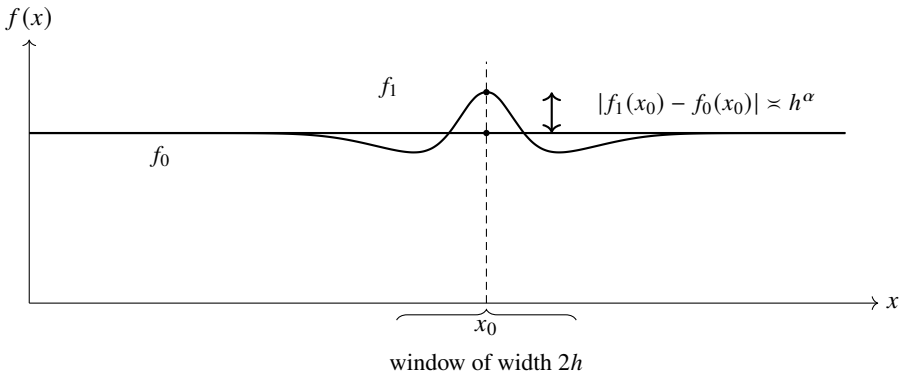


Figure 2.1 Running density example, pointwise version (schematic, not the literal normalized densities used in the proof). The admissible construction uses a compactly supported, mean-zero bump near x_0: its short support keeps the laws close while the target $f(x_0)$ moves by order h^α.

strictly below one, and hence the testing affinity is bounded away from zero. At that bandwidth the pointwise separation becomes

$$|f_1(x_0) - f_0(x_0)| \asymp n^{-\alpha/(2\alpha+1)},$$

so Lemma 2.1 yields

$$\inf_{\widehat{f}(x_0)} \sup_{f \in \mathcal{D}_\alpha(M)} \mathbb{E}_f \big(\widehat{f}(x_0) - f(x_0)\big)^2 \gtrsim n^{-2\alpha/(2\alpha+1)}.$$

At the fixed interior point x_0, a bounded kernel supported on $[-1, 1]$ and reproducing polynomials through degree $\lceil \alpha \rceil - 1$ attains the matching squared-error rate with $h \asymp n^{-1/(2\alpha+1)}$ and $h \le \min\{x_0, 1 - x_0\}$. This localizes the kernel argument of Tsybakov [103, Theorem 1.1 and Propositions 1.2–1.3] to the interior; his Definition 1.2 uses the same integer Hölder convention. Chapter 3 combines disjoint bumps under squared L_2 loss in the same density model.

Figure 2.1 summarizes the balance: smoothness limits the bump height, while its width determines how much information the sample carries about the perturbation. □

The same localized-bump construction also applies to *midpoint fixed-design Gaussian regression*. For an integer $n \ge 1$, observe

$$Y_i = f(x_i) + \varepsilon_i, \qquad x_i = \frac{i - 1/2}{n}, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, 1), \qquad i = 1, \dots, n, \tag{2.3}$$

where $f : [0, 1] \to \mathbb{R}$ is unknown and the design points x_i are deterministic. For $f \in \Lambda^\alpha(M)$, $\alpha, M > 0$, the target is $f(x_0)$ at a fixed $x_0 \in (0, 1)$, under squared-error loss. A bump of width h and height h^α affects at most $2nh + 1$ design points, so its Kullback–Leibler divergence is bounded by a constant times $(nh + 1)h^{2\alpha}$. Taking $h = n^{-1/(2\alpha+1)}$ and a sufficiently small fixed amplitude gives the squared-error minimax lower rate $n^{-2\alpha/(2\alpha+1)}$, also for randomized estimators. The design condition is essential: concentrating every observation at x_0 instead permits risk of order n^{-1}.

The density and regression examples hide a perturbation in a small region. In inverse problems, such as image deblurring, the measurement operator instead attenuates some signal directions. With isotropic Gaussian noise, singular-vector coordinates put a finite-dimensional linear inverse problem in diagonal form. A linear functional is difficult to estimate when it depends strongly on poorly observed directions.

We compare coordinatewise and directional perturbations to see what a single-coordinate pair can miss. Consider the diagonal Gaussian experiment

$$Y_j = \kappa_j \theta_j + \sigma Z_j, \qquad j = 1, \dots, d,$$

where $d \geq 1, \sigma > 0, L > 0$, the Z_j are independent standard Gaussians, $\kappa_j > 0$ are known, and $\|\theta\|_2 \leq L$. Thus we consider only nonzero singular directions. For known $c = (c_1, \dots, c_d)^\top \in \mathbb{R}^d$, estimate $T(\theta) = \sum_{j=1}^d c_j \theta_j$ under squared-error loss. Dividing Y_j by κ_j gives noise standard deviation σ/κ_j. The first pair uses opposite perturbations in one coordinate, limited by both this noise level and the parameter ball.

Proposition 2.5 (A Coordinatewise Inverse-Problem Obstruction) *In the diagonal Gaussian model above, for every coordinate j with $c_j \neq 0$, every estimator $\widehat{T}$ of $T(\theta) = c^\top \theta$ satisfies*

$$\sup_{\|\theta\|_2 \leq L} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 \geq \frac{c_j^2}{2} \min\left\{ L^2, \frac{\sigma^2}{4\kappa_j^2} \right\}.$$

Hence the right side may be maximized over j.

Proof Apply Proposition 2.6, proved below, with $D = I_d$ and $v = e_j$, where e_j is the jth standard basis vector and $K = \operatorname{diag}(\kappa_1, \dots, \kappa_d)$. Then $\|Dv\|_2 = 1$, $\|Kv\|_2 = \kappa_j$, and $c^\top v = c_j$, giving the stated bound. □

A small singular value κ_j permits a larger undetectable movement in θ_j. To capture the combined effect of many coefficients, the next proposition allows arbitrary directions and an ellipsoidal parameter constraint.

Let

$$\Theta_D(L) = \{\theta \in \mathbb{R}^d : \|D\theta\|_2 \leq L\},$$

where D is a known invertible diagonal matrix, and write $K = \operatorname{diag}(\kappa_1, \dots, \kappa_d)$ for the diagonal matrix with entries $\kappa_1, \dots, \kappa_d$. The observation is $Y = K\theta + \sigma Z$ and the target is $c^\top \theta$, still under squared-error loss. The Euclidean ball above is the case $D = I_d$. Opposite points along any direction now replace the coordinate pair. Their allowable displacement is the smaller of the ellipsoid radius in that direction and the displacement that keeps Gaussian KL small.

Proposition 2.6 (Directional Two-Point Lower Bound) *In the diagonal Gaussian model above with parameter class $\Theta_D(L)$, for every nonzero $v \in \mathbb{R}^d$ and every estimator $\widehat{T}$ of $c^\top\theta$,*

$$\sup_{\theta\in\Theta_D(L)} \mathbb{E}_\theta(\widehat{T} - c^\top\theta)^2 \geq \frac{(c^\top v)^2}{2} \min\left\{\frac{L^2}{\|Dv\|_2^2}, \frac{\sigma^2}{4\|Kv\|_2^2}\right\}.$$

The supremum of the right-hand side over nonzero v is therefore another valid lower bound.

Proof Set $t = \min\{L/\|Dv\|_2, \sigma/(2\|Kv\|_2)\}$ and use $\theta_\pm = \pm tv$. The first constraint ensures $t\|Dv\|_2 \leq L$, so both parameters lie in the ellipsoid. Their targets differ by $2t|c^\top v|$. The second constraint controls the distance between their Gaussian laws:

$$\mathrm{KL}(P_{\theta_+}, P_{\theta_-}) = \frac{\|2tKv\|_2^2}{2\sigma^2} = \frac{2t^2\|Kv\|_2^2}{\sigma^2} \leq \frac{1}{2}.$$

Pinsker's inequality bounds total variation by $1/2$. The squared-error two-point inequality with target separation $2t|c^\top v|$ gives at least $(c^\top v)^2 t^2/2$, the stated expression. □

Rescaling v leaves the bound unchanged. Choosing its direction balances target separation against the parameter constraint and the information in the observations.

For example, take $D = K = I_d$, $c = \mathbf{1}$, and $L \geq \sigma/2$. Any coordinate direction gives $\sigma^2/8$, while $v = \mathbf{1}/\sqrt{d}$ gives $d\sigma^2/8$. The estimator $\widehat{T} = \sum_j Y_j$ has risk $d\sigma^2$ uniformly in θ, matching the directional bound within a constant factor. The same information inequality thus recovers the dimension factor missed by a coordinate direction.

The examples select a pair through support overlap, local smoothness, or poorly observed directions. We next consider difficulty caused by uncertainty over a family of alternatives.

2.2 Point-versus-mixture lower bounds

An unknown sign, support, or location can make estimation difficult even when each alternative is distinguishable from a fixed null. We average the alternative laws while keeping every target separated from the null target. The affinity between the null and this mixture then bounds minimax risk.

The comparison is binary: we need not identify the particular alternative. Sign and support mixtures illustrate the second-moment calculation. Truncation and Hellinger bounds then address cases where rare observations make the full second moment uninformative.

Proposition 2.7 (Point-versus-Mixture Lower Bound) *Let $P_0, P_1, \ldots, P_m$ be probability measures on a common measurable space, with $m \geq 1$, and let $\theta_0, \theta_1, \ldots, \theta_m$ be the corresponding target values. Let δ be a measurable estimator taking values in an action space $\mathcal{A}$, and let $L : \mathcal{A} \times \{\theta_0, \ldots, \theta_m\} \to [0, \infty]$ be measurable. Define the mixture law*

$$\bar{P} = \frac{1}{m}\sum_{i=1}^{m} P_i,$$

and the loss-separation constant

$$c_{\min} = \min_{1\leq i\leq m} \frac{1}{2} \inf_{t\in\mathcal{A}} \{L(t, \theta_0) + L(t, \theta_i)\}.$$

Assume $c_{\min} < \infty$. *Then*

$$\max_{0 \le i \le m} \mathbb{E}_i L(\delta, \theta_i) \ge c_{\min}\, \eta(P_0, \bar{P}).$$

Proof For any estimator δ,

$$\max_{0 \le i \le m} \mathbb{E}_i L(\delta, \theta_i) \ge \frac{1}{2}\Big(\mathbb{E}_0 L(\delta, \theta_0) + \frac{1}{m}\sum_{i=1}^{m} \mathbb{E}_i L(\delta, \theta_i)\Big).$$

Let $p_0, p_1, \dots, p_m$ and $\bar{p} = m^{-1}\sum_i p_i$ be densities relative to a common dominating measure μ. The alternatives can have different targets, so their losses must be averaged with weights determined by the observation. For $\bar{p}(x) > 0$, define

$$\bar{L}(x) = \sum_{i=1}^{m} \frac{p_i(x)}{m\bar{p}(x)} L(\delta(x), \theta_i),$$

and set $\bar{L}(x) = 0$ when $\bar{p}(x) = 0$. Then

$$\frac{1}{m}\sum_{i=1}^{m} L(\delta(x), \theta_i) p_i(x) = \bar{L}(x)\bar{p}(x),$$

for every x. When $\bar{p}(x) > 0$, these weights are nonnegative and sum to one. Averaging the inequalities

$$\frac{1}{2}\Big(L(\delta(x), \theta_0) + L(\delta(x), \theta_i)\Big) \ge c_{\min}, \qquad i = 1, \dots, m,$$

with those coefficients yields

$$\frac{1}{2}\Big(L(\delta(x), \theta_0) + \bar{L}(x)\Big) \ge c_{\min} \qquad (\bar{p}(x) > 0).$$

By nonnegativity, for $\bar{p}(x) > 0$,

$$\frac{1}{2}\{L(\delta(x), \theta_0) p_0(x) + \bar{L}(x)\bar{p}(x)\} \ge \frac{1}{2}\{L(\delta(x), \theta_0) + \bar{L}(x)\}(p_0 \wedge \bar{p})(x) \ge c_{\min}(p_0 \wedge \bar{p})(x).$$

When $\bar{p}(x) = 0$, the same bound holds because its right-hand side is zero. Integrating over x gives

$$\max_{0 \le i \le m} \mathbb{E}_i L(\delta, \theta_i) \ge c_{\min} \int (p_0 \wedge \bar{p})\, d\mu = c_{\min}\, \eta(P_0, \bar{P}).$$

□

Only the constituent alternatives must belong to the model; their mixture need not. The bound uses their uniform loss separation $c_{\min}$, not equality of their targets. Figure 2.2 illustrates the comparison; see Tsybakov [103, Section 2.6] for related reductions.

Under 0–1 loss, the proposition also limits label recovery: identifying the label requires distinguishing the null from the alternatives. Chapter 3 addresses the additional difficulty of distinguishing among the alternative labels.

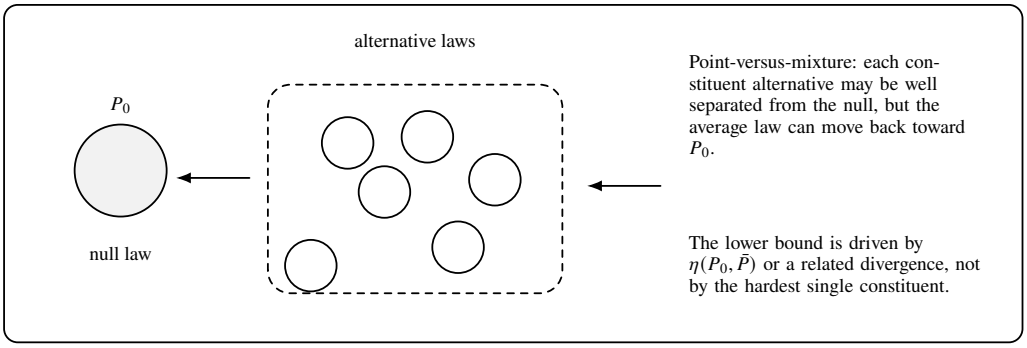


Figure 2.2 Point-versus-mixture geometry. A spread-out alternative family can average back toward the null, making the mixture much harder to distinguish than any one constituent alternative.

Corollary 2.8 (Finite-Family Classification Lower Bound) *Consider a sequence of finite classification experiments with $k \geq 1$, distinct labels $\Theta = \{\theta_0, \theta_1, \ldots, \theta_k\}$, laws P_{θ_j}, and 0–1 loss $L(a, b) = \mathbb{I}(a \neq b)$. Let a classifier T take values in Θ, and define the uniform alternative mixture*

$$\bar{P} = \frac{1}{k} \sum_{j=1}^{k} P_{\theta_j}.$$

If, along the asymptotic regime under consideration,

$$H(P_{\theta_0}, \bar{P}) = o(1),$$

then

$$\inf_T \max_{0 \leq j \leq k} \mathbb{P}_{\theta_j}(T \neq \theta_j) \geq \frac{1}{2} - o(1).$$

Proof Under 0–1 loss and distinct labels,

$$c_{\min} = \frac{1}{2} \inf_t \{\mathbb{I}(t \neq \theta_0) + \mathbb{I}(t \neq \theta_j)\} = \frac{1}{2},$$

for every $j \geq 1$. Proposition 2.7 therefore gives

$$\max_{0 \leq j \leq k} \mathbb{P}_{\theta_j}(T \neq \theta_j) \geq \frac{1}{2} \eta(P_{\theta_0}, \bar{P}) = \frac{1}{2}\big(1 - \mathrm{TV}(P_{\theta_0}, \bar{P})\big).$$

Using $\mathrm{TV}(P, Q) \leq \sqrt{2} H(P, Q)$, we obtain

$$\max_{0 \leq j \leq k} \mathbb{P}_{\theta_j}(T \neq \theta_j) \geq \frac{1}{2} - \frac{1}{\sqrt{2}} H(P_{\theta_0}, \bar{P}) = \frac{1}{2} - o(1).$$

□

The next two examples mix over signs on a fixed support (*vertical mixing*) and over supports with a fixed amplitude (*horizontal mixing*). Sign averaging preserves each alternative's squared norm; support overlap determines the second moment in the sparse example.

Example 2.9 (Vertical-Mixing Lower Bound in a Dense Gaussian Class) Consider the *finite Gaussian model*, obtained by retaining d coordinates in (1.1) and setting $\varepsilon = \sigma$:

$$Y_j = \theta_j + \sigma Z_j, \qquad Z_j \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad j = 1, \dots, d, \tag{2.4}$$

with $d \geq 1$ and $\sigma > 0$. Under squared ℓ_2 loss for the mean vector, compare zero with the sign-symmetric class

$$\Theta_{\text{vert}}(a) = \{\theta \in \mathbb{R}^d : \theta_j = \pm a\sigma d^{-1/4} \text{ for all } j\},$$

where $a > 0$ is a sufficiently small fixed constant. Let Q be the uniform mixture over this class and let $P_0 = N_d(0, \sigma^2 I_d)$. For one coordinate the mixture

$$M_b = \tfrac{1}{2}N(b, \sigma^2) + \tfrac{1}{2}N(-b, \sigma^2), \qquad b = a\sigma d^{-1/4},$$

satisfies the following Gaussian likelihood-ratio identity. Averaging the two signs produces the hyperbolic cosine, whose first nonconstant term is of order b^4/σ^4:

$$\xi\big(N(0,\sigma^2), M_b\big) = \int \left(\frac{dM_b}{dN(0,\sigma^2)}\right)^2 dN(0,\sigma^2) = \cosh(b^2/\sigma^2) \leq \exp\{b^4/(2\sigma^4)\}.$$

By product structure,

$$\xi(P_0, Q) \leq \exp(a^4/2),$$

so Lemma 1.6 gives

$$\eta(P_0, Q) = 1 - \mathrm{TV}(P_0, Q) \geq 1 - \frac{1}{2}\{\exp(a^4/2) - 1\}^{1/2}.$$

For a sufficiently small fixed a, this is bounded below by a positive numerical constant. Every constituent alternative has $\|\theta\|_2^2 = a^2\sigma^2 d^{1/2}$, and Proposition 2.7 gives

$$\inf_{\delta} \sup_{\theta \in \Theta_{\text{vert}}(a) \cup \{0\}} \mathbb{E}_\theta \|\delta - \theta\|_2^2 \gtrsim \sigma^2 d^{1/2}.$$

The zero estimator has maximum risk equal to the common squared norm, so the bound is sharp up to constants on $\Theta_{\text{vert}}(a) \cup \{0\}$. The key gain comes from sign averaging: the second moment depends on the fourth power of the coordinate amplitude. Section 4.2 calibrates this construction to smoothness constraints for quadratic-functional estimation.

□

Example 2.10 (Horizontal-Mixing Lower Bound for Sparse Normal Means) Use the finite Gaussian model (2.4), with noise standard deviation $\sigma > 0$. Under squared ℓ_2 loss for the mean vector, compare the zero mean with the sparse class

$$\Theta(s, d, a) = \{\theta \in \mathbb{R}^d : \theta_j \in \{0, a\sigma\},\ \|\theta\|_0 = s\}, \qquad 1 \leq s \leq d/2. \tag{2.5}$$

Here d, s are positive integers and $a > 0$ is an amplitude to be chosen. Let $P_0 = N_d(0, \sigma^2 I_d)$ and let $Q_{s,d}(a)$ be the uniform mixture over all supports of size s in $\{1, \dots, d\}$. If J and J' are independent uniformly random such supports and $M = |J \cap J'|$, the Gaussian likelihood-ratio calculation gives

$$\xi(P_0, Q_{s,d}(a)) = \mathbb{E} \exp(a^2 M).$$

Only coordinates shared by the two supports contribute to this second moment. Conditional on J, their number M is a hypergeometric count. Lemma A.11, with $k = s$ and $t = a^2$, bounds its exponential moment by $\{1 + (s/d)(e^{a^2} - 1)\}^s$. We use the slightly weaker form

$$\xi(P_0, Q_{s,d}(a)) \le \left(1 - \frac{2s}{d} + \frac{2s}{d} e^{a^2}\right)^s.$$

Choose $a^2 = c \log(1 + d/s^2)$ with a sufficiently small numerical constant $0 < c \le 1$. For $x = d/s^2$, concavity gives $(1 + x)^c - 1 \le cx$. Hence the last display is at most $\exp\{2s^2(e^{a^2} - 1)/d\} \le e^{2c}$ throughout $1 \le s \le d/2$. Lemma 1.6 then bounds total variation strictly below one. The mixture affinity is bounded away from zero, while each constituent has squared norm $sa^2\sigma^2$. Proposition 2.7 yields

$$\inf_{\delta} \sup_{\theta \in \Theta(s,d,a) \cup \{0\}} \mathbb{E}_\theta \|\delta - \theta\|_2^2 \gtrsim s\sigma^2 \log\left(1 + \frac{d}{s^2}\right).$$

For $\Theta(s, d, a) \cup \{0\}$ at the chosen amplitude, the zero estimator has maximum risk $sa^2\sigma^2$, so the bound is sharp up to constants for this fixed-amplitude class.

This fixed-amplitude benchmark is distinct from the minimax rate over the full sparse Gaussian mean class. Section 3.2 uses Fano's method to capture the cost of identifying a support and obtains a stronger bound in some regimes. The discussion after Example 3.6 compares the rates and explains why $s \ll \sqrt{d}$ alone does not make them comparable. □

When the full second moment is too large. Both examples control affinity through the full likelihood-ratio second moment. If rare observations dominate that moment, truncation can give a sharper comparison. For a null Q and alternative $P \ll Q$, let $L = dP/dQ$ and retain $\widetilde{L} = L\mathbf{1}_A$ on a measurable event A. Write $m_1 = \mathbb{E}_Q \widetilde{L} = \mathbb{P}_P(A)$ and $m_2 = \mathbb{E}_Q \widetilde{L}^2$. The triangle inequality and Cauchy–Schwarz give

$$\mathrm{TV}(P, Q) \le \frac{1}{2}\left\{\sqrt{m_2 - 2m_1 + 1} + 1 - m_1\right\}.$$

The first term bounds $\mathbb{E}_Q|\widetilde{L} - 1|$; the second is the discarded alternative mass $\mathbb{E}_Q(L - \widetilde{L})$. Thus $m_1 \to 1$ and $m_2 \to 1$ imply indistinguishability along a sequence of experiments. A small retained second moment alone does not establish this convergence: the discarded *alternative* probability must also vanish. Section 4.3.1 applies this argument to the maximum coordinate, and Section 9.1 to rare signals.

For a sparse mixture, one can instead bound the Hellinger integrand differently at moderate and extreme likelihood values. Let $M = (1 - \varepsilon)Q + \varepsilon G$, where Q is the null, $G \ll Q$ is the signal law, $0 < \varepsilon < 1$, and $g = dG/dQ$. Rationalizing the square root gives

$$\begin{aligned} H^2(M, Q) &= \frac{1}{2}\mathbb{E}_Q\left(\sqrt{1 + \varepsilon(g - 1)} - 1\right)^2 \\ &\le \frac{1}{2}\mathbb{E}_Q \min\{\varepsilon^2(g - 1)^2, \varepsilon|g - 1|\}. \end{aligned}$$

The quadratic bound is effective for moderate likelihood values and the linear bound for extremes. Neither region is discarded, so neither must have vanishing probability. By the Hellinger-affinity product identity of Section 1.3, $nH^2(M, Q) \to 0$ implies $\mathrm{TV}(M^{\otimes n}, Q^{\otimes n}) \to 0$. Section 9.2 develops the tail calculations.

2.3 The constrained-risk inequality

How does accuracy at one parameter restrict an estimator's performance elsewhere? The comparisons of Sections 2.1 and 2.2 bound the larger of two risks without specifying which is large. A constrained-risk inequality instead converts an upper bound on one risk into a lower bound on the other for the same estimator.

Brown and Low [12] establish this principle under squared-error loss. The version below treats finite-dimensional targets under rth-power ℓ_r loss. Hölder's inequality transfers the accuracy constraint under P_0 to a bound on how far the estimator's expectation under P_1 can move from the first target. If the targets are farther apart, the estimator must have nonzero bias under P_1.

Let P_0, P_1 be probability laws on a common measurable observation space with $P_1 \ll P_0$, and let $\theta_0, \theta_1 \in \mathbb{R}^d$ be their targets, where $d \geq 1$ is a finite integer. Fix $1 \leq r < \infty$ and its Hölder conjugate w, so $1/r + 1/w = 1$ and $w = \infty$ when $r = 1$. Define

$$\Delta_r = \|\theta_1 - \theta_0\|_r, \qquad I_w = \begin{cases} \left(\mathbb{E}_0\left[(dP_1/dP_0)^w\right]\right)^{1/w}, & 1 < w < \infty, \\ \operatorname{ess\,sup}_{P_0}(dP_1/dP_0), & w = \infty, \end{cases},$$

where the essential supremum is the smallest P_0-almost-sure upper bound on the likelihood ratio.

Theorem 2.11 (Constrained-Risk Inequality for rth-Power ℓ_r Loss) *For the laws and targets above, assume $I_w < \infty$. If a measurable $\mathbb{R}^d$-valued estimator δ and $\varepsilon \geq 0$ satisfy*

$$\mathbb{E}_0 \|\delta - \theta_0\|_r^r \leq \varepsilon^r \qquad \text{and} \qquad \Delta_r > \varepsilon I_w,$$

then

$$\|\mathbb{E}_1 \delta - \theta_1\|_r \geq \Delta_r - \varepsilon I_w,$$

and

$$\mathbb{E}_1 \|\delta - \theta_1\|_r^r \geq (\Delta_r - \varepsilon I_w)^r.$$

Proof Write $L = dP_1/dP_0$. Hölder's inequality, including its L_1–L_∞ form when $r = 1$, gives

$$\mathbb{E}_0[\|\delta - \theta_0\|_r L] \leq \left(\mathbb{E}_0 \|\delta - \theta_0\|_r^r\right)^{1/r} \|L\|_{L_w(P_0)} \leq \varepsilon I_w < \infty.$$

Thus δ is integrable under P_1, and Jensen's inequality for the norm yields

$$\|\mathbb{E}_1 \delta - \theta_0\|_r = \|\mathbb{E}_0[(\delta - \theta_0)L]\|_r \leq \mathbb{E}_0[\|\delta - \theta_0\|_r L] \leq \varepsilon I_w.$$

The triangle inequality gives $\|\mathbb{E}_1 \delta - \theta_1\|_r \geq \Delta_r - \varepsilon I_w$. Since $x \mapsto \|x\|_r^r$ is convex for $r \geq 1$, a final application of Jensen gives

$$\mathbb{E}_1 \|\delta - \theta_1\|_r^r \geq \|\mathbb{E}_1 \delta - \theta_1\|_r^r \geq (\Delta_r - \varepsilon I_w)^r,$$

using $\Delta_r > \varepsilon I_w$ in the last step. □

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A smaller risk at P_0 permits a larger likelihood-ratio norm I_w while keeping εI_w below the target separation. Unlike the preceding two-point constructions, this argument need not keep a divergence bounded by a fixed constant. The loss uses the vector ℓ_r norm, whereas I_w is the $L_w(P_0)$ norm of the likelihood ratio. For squared-error loss, $I_2^2 = \xi(P_0, P_1) = 1+\chi^2(P_1, P_0)$; the next example lets this moment grow as the risk constraint tightens.

Example 2.12 (The Cost of Superefficiency in Gaussian Location) Suppose we observe $X \sim P_\theta = N(\theta, n^{-1})$, with $\theta \in \mathbb{R}$ and integer $n \geq 1$. Thus X has the distribution of the sample mean of n independent $N(\theta, 1)$ variables. We estimate θ under squared-error loss and, for a measurable estimator $\widehat{\theta}_n$, write

$$R_n(\widehat{\theta}_n, \theta) = \mathbb{E}_\theta(\widehat{\theta}_n - \theta)^2.$$

The estimator X has risk n^{-1} at every θ. Following Brown and Low [12, Theorem 2], we examine the cost of improving on this benchmark at one point. For a fixed $\theta_0 \in \mathbb{R}$, a constant $C > 0$, and a sequence $b_n > 1$ tending to infinity, suppose

$$R_n(\widehat{\theta}_n, \theta_0) \leq \frac{C}{nb_n}$$

for all sufficiently large n.

Compare θ_0 with $\theta_1 = \theta_0 + \Delta$, where $\Delta > 0$. The likelihood ratio is

$$L = \frac{dP_{\theta_1}}{dP_{\theta_0}}(X) = \exp\left\{n\Delta(X - \theta_0) - \frac{n\Delta^2}{2}\right\}.$$

Under P_{θ_0}, $X - \theta_0 \sim N(0, n^{-1})$, so the Gaussian moment-generating formula gives

$$\mathbb{E}_{\theta_0} L^2 = e^{-n\Delta^2}\mathbb{E}_{\theta_0} e^{2n\Delta(X-\theta_0)} = e^{n\Delta^2}.$$

Thus $I_2 = e^{n\Delta^2/2}$. Theorem 2.11, with $r = 2$ and $\varepsilon = \sqrt{C/(nb_n)}$, yields

$$R_n(\widehat{\theta}_n, \theta_0 + \Delta) \geq \left(\Delta - \sqrt{\frac{C}{nb_n}}\, e^{n\Delta^2/2}\right)_+^2.$$

When the expression inside the positive part is nonpositive, this is simply the nonnegativity of risk.

Choose $\Delta_n = \sqrt{\log b_n / n}$. Then $I_2 = \sqrt{b_n}$ cancels the improvement factor $b_n^{-1/2}$ in the square root of the risk bound at θ_0. For all sufficiently large n,

$$R_n(\widehat{\theta}_n, \theta_0 + \Delta_n) \geq \frac{\left(\sqrt{\log b_n} - \sqrt{C}\right)_+^2}{n} \geq \frac{\log b_n}{4n}, \tag{2.6}$$

where the last inequality holds once $\log b_n \geq 4C$. Improvement by the factor b_n at θ_0 therefore forces deterioration by at least a constant multiple of $\log b_n$ relative to the usual n^{-1} risk.

If $\log b_n = o(n)$, these unfavorable alternatives converge to θ_0, so the deterioration occurs in every fixed neighborhood of that point. In particular, $b_n = n^a$ with fixed $a > 0$ turns the constraint $R_n(\widehat{\theta}_n, \theta_0) \leq Cn^{-1-a}$ into a lower bound of order $\log n/n$ at distance $\sqrt{a \log n/n}$. The conclusion concerns a moving sequence of alternatives, not deterioration at every fixed parameter. □

The same constrained-risk argument applies to a nonlinear scalar functional of a matrix parameter. The log determinant of a covariance matrix enters differential entropy estimation, likelihood-based testing, and discriminant analysis. Cai, Liang, and Zhou [27, Theorems 2 and 4] establish the Gaussian minimax squared-error rate d/n when $d/n \to 0$. Their sample has $n+1$ observations and an unknown mean; their Lemma 1 identifies the same covariance law as the uncentered covariance from the n zero-mean observations below.

Example 2.13 (Log-Determinant Estimation and a Local Penalty) Suppose

$$X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} N_d(0, \Sigma), \qquad \Sigma \in \mathbb{S}^d_{++},$$

where $\mathbb{S}^d_{++}$ is the set of real symmetric positive-definite $d \times d$ matrices. Write $P_\Sigma = N_d(0,\Sigma)^{\otimes n}$ for the joint law of $(X_1, \dots, X_n)$. We estimate $T(\Sigma) = \log\det(\Sigma)$ under squared-error loss, with $d/n \to 0$. For a positive sequence $a_n \to 0$ to be specified below, compare

$$\Sigma_0 = I_d, \qquad \Sigma_1 = (1 + a_n) I_d.$$

Then

$$|T(\Sigma_1) - T(\Sigma_0)| = d \log(1 + a_n) \asymp d a_n,$$

and the Gaussian product calculation gives

$$1 + \chi^2(P_{\Sigma_1}, P_{\Sigma_0}) = (1 - a_n^2)^{-nd/2}, \qquad 0 < a_n < 1.$$

If an estimator has local risk

$$\mathbb{E}_{\Sigma_0}(\widehat{T} - T(\Sigma_0))^2 = r_n \frac{d}{n}, \qquad 0 < r_n \to 0,$$

choose the perturbation scale

$$a_n = \frac{c_n}{\sqrt{nd}}, \qquad c_n = \min\{(\log(1/r_n))^{1/4}, (nd)^{1/4}\}.$$

Here $c_n \to \infty$ enlarges the target separation relative to $\sqrt{d/n}$; the two terms in its definition control the likelihood-ratio moment and ensure $a_n \to 0$. Put $\varepsilon = \sqrt{r_n d/n}$ and $\Delta_2 = |T(\Sigma_1) - T(\Sigma_0)| \asymp c_n \sqrt{d/n}$. We verify that $\varepsilon I_2 = o(\Delta_2)$, so the risk constraint contributes a negligible term in Theorem 2.11. For all sufficiently large n, $a_n^2 \le 1/2$ and

$$I_2 = (1 - a_n^2)^{-nd/4} \le \exp\{n d a_n^2/2\} = \exp(c_n^2/2).$$

Since $c_n^2 \le \{\log(1/r_n)\}^{1/2}$ and $c_n \to \infty$,

$$\frac{\varepsilon I_2}{\Delta_2} \lesssim \frac{\sqrt{r_n}\exp(c_n^2/2)}{c_n} \le \frac{\exp\{-\frac{1}{2}\log(1/r_n) + \frac{1}{2}\sqrt{\log(1/r_n)}\}}{c_n} \longrightarrow 0.$$

The constrained-risk inequality therefore implies

$$\mathbb{E}_{\Sigma_1}(\widehat{T} - T(\Sigma_1))^2 \gtrsim c_n^2 \frac{d}{n} \gg \frac{d}{n}.$$

Reducing risk at I_d to r_n times the usual benchmark therefore forces risk larger by the diverging factor c_n^2 along these alternatives.

For the ordinary minimax lower bound, instead take $c_n = c > 0$ fixed. Then $\Delta_2 \asymp \sqrt{d/n}$ and $I_2 = O(1)$. For a sufficiently small constant $c_0 > 0$, either the estimator's risk at Σ_0

exceeds $c_0 d/n$, or the constrained-risk inequality forces risk at least a constant multiple of d/n at Σ_1. Taking the infimum over estimators gives the minimax lower bound of order d/n. □

The scalar target accumulates changes in all d eigenvalues: its separation has order da_n, whereas $\log I_2$ has order nda_n^2. Chapter 7 uses the constrained-risk tradeoff for adaptive functional estimation in Theorem 7.1. We next extend the argument to two mixtures, where the target may vary within each family of alternatives.

2.4 Two-composite lower bounds

For a nonlinear functional, two families of signals can have different average target values even when their observation mixtures are close. This provides a lower-bound construction when comparison with a fixed null is insufficient, as in the nonsmooth-functional problem of Section 4.3. We place a prior on each family and separate their target averages. Unlike Section 2.2, we need not separate every target on one side from every target on the other.

The new issue is variation of the target under each prior. Even an accurate estimator need not be close to the prior target mean. Cai and Low's inequality [37, Theorem 2 and Corollary 1] accounts for this variation: an average-risk constraint forces a difference in average biases. Optimizing the resulting tradeoff gives a worst-case lower bound without an imposed accuracy constraint.

Let $\{P_\theta : \theta \in \Theta\}$ be a dominated statistical experiment on measurable parameter and observation spaces. To define mixture laws, assume it is a *probability kernel*: $\theta \mapsto P_\theta(A)$ is measurable for every measurable observation event A. Let $T : \Theta \to \mathbb{R}$ be a measurable scalar target.

Let μ_0 and μ_1 be priors supported on Θ. Assume T is integrable under both priors and square-integrable under μ_0. Define their target means m_i, for $i = 0, 1$, and the target standard deviation v_0 under μ_0 by

$$m_i = \int T(\theta)\, \mu_i(d\theta), \qquad v_0^2 = \int \big(T(\theta) - m_0\big)^2\, \mu_0(d\theta).$$

Let $F_i = \int P_\theta\, \mu_i(d\theta)$ have density f_i relative to a common dominating measure. Assume $F_1 \ll F_0$ and define

$$I = \left\{ \mathbb{E}_{F_0} \left(\frac{dF_1}{dF_0} - 1 \right)^2 \right\}^{1/2} = \sqrt{\chi^2(F_1, F_0)} < \infty.$$

For a measurable real-valued estimator $\widehat{T} = \widehat{T}(X)$, write its bias, wherever the expectation is finite, as

$$B(\theta) = \mathbb{E}_\theta \widehat{T} - T(\theta).$$

Theorem 2.14 (Mixture-versus-Mixture Constrained-Risk Inequality) *Under the preceding assumptions, every measurable real-valued estimator $\widehat{T}$ satisfies the following bounds.*

(i) If, for some $\varepsilon \geq 0$, the μ_0-average squared-error risk is at most ε^2,

$$\int \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2 \, \mu_0(d\theta) \leq \varepsilon^2,$$

then the average biases satisfy

$$\left| \int B(\theta) \, \mu_1(d\theta) - \int B(\theta) \, \mu_0(d\theta) \right| \geq |m_1 - m_0| - (\varepsilon + v_0) I.$$

(ii) If $|m_1 - m_0| > v_0 I$, then for every mixture weight $\lambda \in [0,1]$,

$$\int \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2 \, [\lambda\mu_0 + (1-\lambda)\mu_1](d\theta) \geq \frac{\lambda(1-\lambda)(|m_1 - m_0| - v_0 I)^2}{\lambda + (1-\lambda)(I+1)^2}.$$

In particular, the worst-case squared-error risk over Θ obeys

$$\sup_{\theta \in \Theta} \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2 \geq \frac{(|m_1 - m_0| - v_0 I)^2}{(I+2)^2}.$$

Proof We first control the change in expectation under the two mixture laws, then translate the resulting bias difference into a risk bound.

Step 1: integrability and change of measure. Under the risk premise of part (i), $(a+b)^2 \leq 2a^2 + 2b^2$ gives

$$\mathbb{E}_{F_0}(\widehat{T} - m_0)^2 \leq 2\varepsilon^2 + 2v_0^2 < \infty.$$

Also $f_1/f_0 \in L_2(F_0)$ because $I < \infty$ and $\mathbb{E}_{F_0}(f_1/f_0) = 1$. Cauchy–Schwarz therefore makes $\widehat{T}$ integrable under both F_0 and F_1, and the following bias integrals are finite.

Changing measure and expressing each expectation through its target mean and average bias gives

$$\begin{aligned} \mathbb{E}_{F_0}\left[(\widehat{T} - m_0)\left(\frac{f_1 - f_0}{f_0}\right)\right] &= \int \mathbb{E}_\theta(\widehat{T} - m_0) \, \mu_1(d\theta) - \int \mathbb{E}_\theta(\widehat{T} - m_0) \, \mu_0(d\theta) \\ &= m_1 + \int B(\theta) \, \mu_1(d\theta) - \left(m_0 + \int B(\theta) \, \mu_0(d\theta)\right). \end{aligned}$$

Step 2: bounding the bias shift. The estimator's distance from m_0 includes both estimation error and target variation. The L_2 triangle inequality under $\mu_0(d\theta)P_\theta(dx)$ separates them:

$$\begin{aligned} \{\mathbb{E}_{F_0}(\widehat{T} - m_0)^2\}^{1/2} &\leq \left\{\int \mathbb{E}_\theta(\widehat{T} - T(\theta))^2 \, \mu_0(d\theta)\right\}^{1/2} + \left\{\int (T(\theta) - m_0)^2 \, \mu_0(d\theta)\right\}^{1/2} \\ &\leq \varepsilon + v_0. \end{aligned}$$

Another Cauchy–Schwarz bound gives

$$\left|\mathbb{E}_{F_0}\left[(\widehat{T} - m_0)\left(\frac{f_1 - f_0}{f_0}\right)\right]\right| \leq \big(\mathbb{E}_{F_0}(\widehat{T} - m_0)^2\big)^{1/2} \left\{\mathbb{E}_{F_0}\left(\frac{f_1 - f_0}{f_0}\right)^2\right\}^{1/2} \leq (\varepsilon + v_0) I.$$

Using the reverse triangle inequality together with the identity from Step 1, we obtain

$$\left| \int B(\theta)\, \mu_1(d\theta) - \int B(\theta)\, \mu_0(d\theta) \right| \geq |m_1 - m_0| - (\varepsilon + v_0)I,$$

which proves part (i).

Step 3: converting the bias difference into Bayes risk. Let x be the square root of the μ_0-average risk:

$$x^2 = \int \mathbb{E}_\theta(\widehat{T} - T(\theta))^2\, \mu_0(d\theta).$$

For $\lambda = 0$ or 1, the claimed lower bound is zero and follows from nonnegativity. Assume $0 < \lambda < 1$. If $x = +\infty$, the mixture risk is infinite, so assume $x < \infty$. Part (i), with $\varepsilon = x$, gives

$$\left| \int B\, d\mu_1 - \int B\, d\mu_0 \right| \geq |m_1 - m_0| - (x + v_0)I.$$

The Cauchy–Schwarz inequality gives

$$\left| \int B\, d\mu_0 \right| \leq \left(\int B(\theta)^2\, \mu_0(d\theta) \right)^{1/2} \leq x,$$

so

$$\left| \int B\, d\mu_1 \right| \geq |m_1 - m_0| - v_0 I - (I + 1)x.$$

Therefore,

$$\int B(\theta)^2\, \mu_1(d\theta) \geq \Big(\max\{|m_1 - m_0| - v_0 I - (I + 1)x, 0\} \Big)^2.$$

For $(\mu_0 + \mu_1)$-almost every θ, the preceding integrability argument makes $B(\theta)$ finite, and Jensen's inequality gives

$$\mathbb{E}_\theta(\widehat{T} - T(\theta))^2 \geq B(\theta)^2.$$

It follows that for every $\lambda \in [0, 1]$,

$$\begin{aligned} &\int \mathbb{E}_\theta(\widehat{T} - T(\theta))^2\, [\lambda\mu_0 + (1 - \lambda)\mu_1](d\theta) \\ &\qquad \geq \lambda x^2 + (1 - \lambda) \Big(\max\{|m_1 - m_0| - v_0 I - (I + 1)x, 0\} \Big)^2. \end{aligned}$$

Now write $a = |m_1 - m_0| - v_0 I$ and $b = I + 1$. The function

$$J(x) = \lambda x^2 + (1 - \lambda)(a - bx)^2$$

is minimized over $x \geq 0$ at

$$x_* = \frac{ab(1 - \lambda)}{\lambda + b^2(1 - \lambda)}.$$

Since $a > 0$ and $0 < \lambda < 1$, this point lies in $0 < x_* < a/b$, where the positive part is active. For $x \geq a/b$, the positive-part expression in the risk bound reduces to the increasing function λx^2. Thus x_* also minimizes

$$\lambda x^2 + (1 - \lambda)(\max\{a - bx, 0\})^2,$$

and the minimum value is

$$\frac{\lambda(1-\lambda)a^2}{\lambda + b^2(1-\lambda)}.$$

This proves the first display in part (ii). Finally, choosing $\lambda = (I+1)/(I+2)$ gives

$$\lambda + (1-\lambda)(I+1)^2 = I+1, \qquad \lambda(1-\lambda) = \frac{I+1}{(I+2)^2},$$

so the Bayes risk under the mixture prior is at least $(|m_1 - m_0| - v_0 I)^2/(I+2)^2$. The supremum risk of the fixed estimator dominates its Bayes risk under this prior, which proves the last inequality. □

The effective separation is the prior-mean gap $|m_1 - m_0|$ minus the penalty $v_0 I$ for target variation. This guides the choice of priors: keep their observation laws close without letting target variation erase the gap. Taking μ_0 to be a point mass sets $v_0 = 0$ and gives the constrained-risk counterpart of Proposition 2.7.

Corollary 2.15 (Point-versus-Mixture Constrained-Risk Inequality) *Under the experiment and measurability assumptions of Theorem 2.14, fix $\theta_0 \in \Theta$ and a prior μ_1 with $\int |T(\theta)|\,\mu_1(d\theta) < \infty$. Let $F_1 = \int P_\theta\,\mu_1(d\theta)$, assume $F_1 \ll P_{\theta_0}$, and set $I = \sqrt{\chi^2(F_1, P_{\theta_0})} < \infty$. Define $m_1 = \int T(\theta)\,\mu_1(d\theta)$ and write $B(\theta) = \mathbb{E}_\theta \widehat{T} - T(\theta)$ wherever the expectation is finite. If, for some $\varepsilon \geq 0$, an estimator $\widehat{T}$ satisfies the pointwise risk constraint*

$$\mathbb{E}_{\theta_0}\big(\widehat{T} - T(\theta_0)\big)^2 \leq \varepsilon^2,$$

then

$$\left|\int B(\theta)\,\mu_1(d\theta) - B(\theta_0)\right| \geq |m_1 - T(\theta_0)| - \varepsilon I.$$

Without this risk constraint, if $|m_1 - T(\theta_0)| > 0$, every measurable real-valued estimator $\widehat{T}$ satisfies, for every $\lambda \in [0,1]$,

$$\int \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2\,[\lambda\delta_{\theta_0} + (1-\lambda)\mu_1](d\theta) \geq \frac{\lambda(1-\lambda)\big(|m_1 - T(\theta_0)|\big)^2}{\lambda + (1-\lambda)(I+1)^2},$$

and therefore

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2 \geq \frac{\big(|m_1 - T(\theta_0)|\big)^2}{(I+2)^2}.$$

Proof Apply Theorem 2.14 with $\mu_0 = \delta_{\theta_0}$. Its target-integrability assumption holds by hypothesis, and $m_0 = T(\theta_0)$, $v_0 = 0$, and $F_0 = P_{\theta_0}$. These substitutions give the stated bias and Bayes-risk bounds. The supremum risk dominates the Bayes risk under every prior, giving the final inequality. □

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Moment matching provides a way to construct suitable priors for nonsmooth targets. A representative example is estimation of the average absolute Gaussian mean

$$T(\theta) = \frac{1}{d}\sum_{i=1}^{d} |\theta_i|$$

from observations $Y_i = \theta_i + Z_i$, where $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$ for $i = 1, \ldots, d$, under squared-error loss. Priors can agree on low-order polynomial moments yet differ in their average absolute value. Those matched moments cancel the corresponding terms when the mixture likelihood ratios are expanded in Hermite polynomials, which are orthogonal under the standard Gaussian law. Polynomial approximation determines how large a target gap can remain. Section 4.3.2 carries out the construction and controls target dispersion, obtaining the leading minimax constant on a fixed bounded cube and the minimax rate, without a matching leading constant, on the full space.

Related constructions give lower bounds for discrete-distribution functionals. Wu and Yang [112] use moment-matched priors on small cell probabilities to separate Shannon entropy values while keeping Poisson count mixtures close. Jiao, Venkat, Han, and Weissman [80] use polynomial-approximation duality for power sums, including $\sum_i p_i^\alpha$ with $0 < \alpha < 1$, where p_i are cell probabilities. Jiao, Han, and Weissman [79] develop analogous constructions for ℓ_1 distance between probability vectors. These applications use related two-prior testing reductions rather than the constrained-risk inequality above. The recurring idea is to preserve a functional gap while keeping the observation laws close.

2.5 Probabilistic-error lower bounds

How likely is an estimator to miss a specified accuracy radius? An expected-loss lower bound does not answer this question: rare, large errors can inflate the expectation. We therefore apply the two-point method directly to the probability of an estimation error.

Suppose the observation has law P and the target $\theta(P)$ lies in a metric space (Θ, d). To bound the probability of an error of size at least $\delta > 0$, choose two targets separated by 2δ whose observation laws remain hard to distinguish. Apply the two-point lemma to the indicator of this error event.

Lemma 2.16 (Probabilistic-Error Lower Bound) *Let P_0 and P_1 be probability measures on a common observation space, and let their target values θ_0 and θ_1 lie in a metric space (Θ, d). Assume $d(\theta_0, \theta_1) > 0$ and put*

$$\delta = \frac{1}{2} d(\theta_0, \theta_1).$$

Then every measurable estimator $\widehat{\theta}$ taking values in Θ satisfies

$$\max\left\{\mathbb{P}_0\Big(d\big(\widehat{\theta}, \theta_0\big) \geq \delta\Big), \mathbb{P}_1\Big(d\big(\widehat{\theta}, \theta_1\big) \geq \delta\Big)\right\} \geq \frac{1}{2}\eta(P_0, P_1).$$

Proof Apply Lemma 2.1 to the measurable loss

$$L(t, \theta_i) = \mathbb{I}\{d(t, \theta_i) \geq \delta\}, \qquad t \in \Theta, \quad i = 0, 1.$$

Since $d(\theta_0, \theta_1) = 2\delta > 0$, the triangle inequality prevents both losses from being zero. Their sum is therefore at least one for every t, and it equals one at $t = \theta_0$. Thus the loss-separation constant in that lemma is exactly $1/2$. Its testing-affinity bound gives the stated inequality for every estimator. □

The probability bound also gives moment lower bounds. On the error event, the rth-power loss is at least δ^r; elsewhere it is nonnegative. Thus, for each $i \in \{0, 1\}$,

$$\mathbb{E}_i d(\widehat{\theta}, \theta_i)^r \geq \delta^r \mathbb{P}_i\{d(\widehat{\theta}, \theta_i) \geq \delta\}, \qquad r > 0.$$

Taking the maximum over the two models and applying Lemma 2.16 gives the moment lower bound $\delta^r \eta(P_0, P_1)/2$ for every estimator. The implication runs from probability to expected loss, not conversely.

Probability bounds also give necessary sample sizes for prescribed accuracy and failure probability; sparse regression provides an example in Section 8.1.1. For discrete targets, the error event is incorrect recovery, and the multiple-hypothesis bound in Chapter 3 applies to top-set selection in Section 8.5.1. The comparison of nearby segment boundaries in Section 9.3.3 likewise yields probability bounds for endpoint error. Section 7.2 shows why the criterion matters for adaptation: a fixed failure probability permits rare, large errors that squared-error risk penalizes.

For honest confidence sets, the same pair serves a different reduction. Chapter 6 transfers coverage at both targets to a common observation law. On the event that the set contains both targets, its diameter is at least their separation. This bounds the probability of a large set rather than an inaccurate estimate.

This chapter has used binary comparisons between individual laws or mixtures. Chapter 3 turns to packings, where accurate estimation requires identifying one of many alternatives, and hypercubes, where coordinatewise testing errors accumulate in the loss. Similar hard families can thus serve different reductions.

2.6 Exercises

2.1. Derive the loss-separation constants in the two-point inequality for absolute and squared error. Explain why one cannot transfer a constant between the two losses by changing notation alone.

2.2. In Example 2.3, compare the full n-sample laws at $\theta_0 = 1$ and $\theta_1 = 1 + n^{-1}$. Calculate their exact total-variation affinity η and Hellinger affinity ρ. Compare the minimum total testing error η with the indirect bound $\rho^2/2$, and deduce the corresponding squared-error lower bounds for estimating the endpoint over these two values.

2.3. Expand the second moment of a finite mixture likelihood ratio. Identify its diagonal and off-diagonal terms.

2.4. Recompute the symmetric Gaussian sign-mixture second moment and explain why the first nonconstant term is fourth order in the perturbation amplitude.

2.5. In a constrained-risk comparison with $P_0 = N(0,1)$ and $P_1 = N(h,1)$, calculate the second likelihood-ratio moment and quantify the penalty for exceptionally small risk at zero.

2.6. Let $P_0 = N(0,1)$ and $P_\pm = N(\pm a, 1)$ with $a > 0$, and write $L_\pm = dP_\pm/dP_0$. Select the sign after observing Y: use $L_*(Y) = L_+(Y)$ for $Y \geq 0$ and $L_-(Y)$ otherwise. Calculate $\mathbb{E}_0 L_*$ and compare it with $\mathbb{E}_0\{(L_+ + L_-)/2\}$. Explain why this data-selected expression cannot replace the likelihood ratio of the fixed prior mixture in Proposition 2.7.

2.7. Specialize Proposition 2.6 to the ellipsoid $\sum_{j=1}^{d} j^{2\beta}\theta_j^2 \leq L^2$, with $\beta > 0$. For $\theta_\pm = \pm tv$, give the exact class-membership and Gaussian-information constraints on t.

2.8. Compare the coordinatewise and directional bounds when $D = K = I_d$ and $c = \mathbf{1}$. Explain why the former is valid but loses a factor of d when $L \geq \sigma/2$.

2.9. For n i.i.d. observations from $f \in \mathcal{D}_\alpha(M)$, with $\alpha > 0$ and $M > 1$ as in (2.2), construct a two-point lower bound for estimating $f(1/2)$ under squared error using a compactly supported mean-zero bump. Check unit integral, nonnegativity, every required derivative supremum bound, and the top Hölder seminorm before computing the one-sample chi-square divergence relative to the flat density. Choose the bandwidth to keep the full-sample divergence bounded and derive the resulting lower-bound rate.

2.10. In Proposition 2.6, take $d = 2$, $D = I_2$, $K = \mathrm{diag}(1, 2^{-a})$, $c = (1,1)$, and $a > 0$. Compare the two coordinate choices with $v \propto K^{-2}c$, checking both the class and information constraints rather than presuming that this direction always attains the largest bound.

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3

Hypercubes, Packings, and Information

Recovering a function or a high-dimensional parameter may require learning many features at once. Assouad's lemma adds binary testing difficulties along hypercube edges, as in density estimation with disjoint bumps. Fano's inequality compares information with the logarithm of the number of separated possibilities, as in sparse-support estimation; it requires target separation but no coordinatewise loss decomposition.

The goal of this chapter is to learn how to construct and calibrate hypercubes and packings: place the alternatives inside the parameter class, relate their separation to the loss, and balance their number and amplitude against the information in the observations. These are central tools for nonparametric estimation, high-dimensional statistics, and machine learning. The examples show how dimension, sparsity, and smoothness guide the construction.

We then simplify information calculations with a common reference law. Two complementary extensions broaden the perspective: general f-divergence bounds and the packing-and-covering approach of Yang and Barron [113]. Later applications include global function estimation in Chapter 4 and classification, clustering, and ranking in Chapter 8. The decision criterion remains essential: Chapter 5 averages a sign cube to hide a signal rather than recover its coordinates.

3.1 Assouad's lemma

When loss accumulates across many features, one difficult pair may capture little of the total risk. Assouad's method [3] combines many binary comparisons by indexing a hard family with vertices $\gamma \in \{0,1\}^m$, where $m \geq 1$. Each edge changes one feature, as Figure 3.1 illustrates. Difficulty recovering individual bits yields a lower bound on the full estimation loss.

Every vertex must belong to the parameter class. Loss comparisons must hold for all vertex pairs; information bounds need only compare neighbors differing in one bit. The examples implement these requirements with Gaussian coordinates and disjoint density perturbations. The hypercube describes the parameter family; the lemma does not require independent observations.

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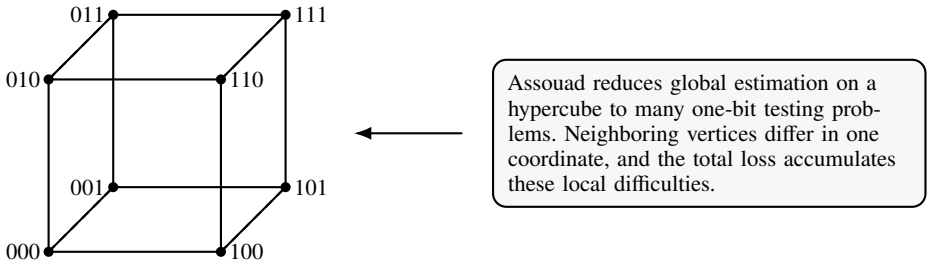


Figure 3.1 Assouad's lemma: a global estimation problem is reduced to many coordinatewise testing problems on a hypercube.

Lemma 3.1 (Assouad's Lemma) *Let $\Gamma = \{0,1\}^m$, $m \geq 1$, have Hamming distance H, and let $\{P_\gamma : \gamma \in \Gamma\}$ be an experiment on a common measurable observation space, with targets $\psi(\gamma)$ in a metric space (Θ, d). Every measurable Θ-valued estimator T satisfies, for every $r > 0$,*

$$\max_{\gamma\in\Gamma} 2^r \mathbb{E}_\gamma d^r(T, \psi(\gamma)) \geq \left[\min_{H(\gamma,\gamma')\geq 1} \frac{d^r(\psi(\gamma), \psi(\gamma'))}{H(\gamma,\gamma')}\right] \cdot \frac{m}{2} \cdot \left[\min_{H(\gamma,\gamma')=1} \eta(P_\gamma, P_{\gamma'})\right],$$

where η is the testing affinity.

Proof Let

$$\alpha = \min_{H(\gamma,\gamma')\geq 1} \frac{d^r\big(\psi(\gamma), \psi(\gamma')\big)}{H(\gamma,\gamma')}.$$

Then

$$d^r\big(\psi(\gamma), \psi(\gamma')\big) \geq \alpha H(\gamma,\gamma') \qquad \text{for all } \gamma, \gamma' \in \Gamma.$$

Define

$$\widehat{\gamma} \in \arg\min_{\gamma\in\Gamma} d\big(T, \psi(\gamma)\big),$$

with any fixed tie-breaking rule. The triangle inequality and the definition of $\widehat{\gamma}$ give

$$\alpha H(\widehat{\gamma}, \gamma) \leq d^r\big(\psi(\widehat{\gamma}), \psi(\gamma)\big) \leq \big(d(T, \psi(\widehat{\gamma})) + d(T, \psi(\gamma))\big)^r \leq \big(2d(T, \psi(\gamma))\big)^r.$$

Taking expectations under P_γ gives

$$2^r \mathbb{E}_\gamma d^r\big(T, \psi(\gamma)\big) \geq \alpha \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma).$$

It therefore suffices to lower-bound the Hamming error of every decoder $\widehat{\gamma}$.

Averaging over the hypercube gives

$$\max_{\gamma\in\Gamma} \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma) \geq \frac{1}{2^m} \sum_{\gamma\in\Gamma} \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma) = \frac{1}{2^m} \sum_{\gamma\in\Gamma} \sum_{i=1}^m \mathbb{E}_\gamma \big|\widehat{\gamma}_i - \gamma_i\big|.$$

After interchanging the sums and grouping terms according to the value of γ_i,

$$\max_{\gamma\in\Gamma} \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma) \geq \frac{1}{2} \sum_{i=1}^m \left[\frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i=0} \int \widehat{\gamma}_i \, dP_\gamma + \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i=1} \int (1 - \widehat{\gamma}_i) \, dP_\gamma\right].$$

For each coordinate i, define the two mixtures

$$P_{0,i} = \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} P_\gamma, \qquad P_{1,i} = \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 1} P_\gamma.$$

The bracketed term is the total error of the test $\widehat{\gamma}_i$ for distinguishing $P_{0,i}$ from $P_{1,i}$. The testing-affinity identity from Chapter 1 therefore gives

$$\max_{\gamma \in \Gamma} \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma) \geq \frac{1}{2} \sum_{i=1}^{m} \eta(P_{0,i}, P_{1,i}).$$

It remains to compare each mixture affinity with the worst edge affinity. Pair every vertex γ with $\gamma_i = 0$ to the unique vertex $\gamma^{(i)}$ obtained by flipping the ith bit. Then

$$P_{0,i} = \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} P_\gamma, \qquad P_{1,i} = \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} P_{\gamma^{(i)}}.$$

If p_γ and $p_{\gamma^{(i)}}$ denote densities with respect to a common dominating measure, use the elementary pointwise inequality

$$\left(\sum_j a_j \right) \wedge \left(\sum_j b_j \right) \geq \sum_j (a_j \wedge b_j),$$

for nonnegative summands with common weights. It gives

$$\begin{aligned} \eta(P_{0,i}, P_{1,i}) &= \int \left(\frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} p_\gamma \right) \wedge \left(\frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} p_{\gamma^{(i)}} \right) d\mu \\ &\geq \frac{1}{2^{m-1}} \sum_{\gamma:\, \gamma_i = 0} \int (p_\gamma \wedge p_{\gamma^{(i)}})\, d\mu. \end{aligned}$$

Every pair $(\gamma, \gamma^{(i)})$ differs in exactly one coordinate, so

$$\eta(P_{0,i}, P_{1,i}) \geq \min_{H(\gamma,\gamma')=1} \eta(P_\gamma, P_{\gamma'}).$$

Substitution into the coordinate sum yields

$$\max_{\gamma \in \Gamma} \mathbb{E}_\gamma H(\widehat{\gamma}, \gamma) \geq \frac{m}{2} \min_{H(\gamma,\gamma')=1} \eta(P_\gamma, P_{\gamma'}).$$

Combining this Hamming-error lower bound with the earlier reduction from estimation error to Hamming error proves the lemma. □

For each bit, the proof averages over the remaining bits and applies a binary testing bound. Unlike the single null-versus-mixture comparison in Section 2.2, these testing errors are summed across coordinates. The loss comparison is what makes that sum a lower bound for estimation.

Example 3.2 (Gaussian Sequence Model over an Ellipsoid) Use the Gaussian sequence model (1.1) with noise level $\varepsilon = n^{-1/2}$:

$$Y_i = \theta_i + \frac{1}{\sqrt{n}} Z_i, \qquad Z_i \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad i = 1, 2, \ldots. \tag{3.1}$$

The parameter space is the ellipsoid

$$\Theta_\alpha(M) = \left\{\theta : \sum_{i\geq 1} i^{2\alpha}\theta_i^2 \leq M\right\}, \qquad \alpha > 0, \quad M > 0. \tag{3.2}$$

We estimate the mean sequence under squared ℓ_2 loss, with α, M fixed as $n \to \infty$. Here $n \geq 1$ controls precision, not the number of observed coordinates, and M bounds the weighted squared norm, giving weighted-norm radius $\sqrt{M}$. Shifts proportional to the noise level $n^{-1/2}$ keep each bit difficult to recover. The increasing ellipsoid weights determine how many such coordinates fit in the class.

Let $r = \lfloor n^{1/(2\alpha+1)} \rfloor$ count the active coordinates in this example, and define the embedded hypercube

$$\Theta_0 = \left\{\theta : \theta_i = \gamma_i \frac{a}{\sqrt{n}},\ \gamma_i \in \{0,1\} \text{ for } 1 \leq i \leq r,\ \theta_i = 0 \text{ for } i > r\right\}, \tag{3.3}$$

where $a > 0$ is a constant chosen small enough that $\Theta_0 \subseteq \Theta_\alpha(M)$. Indeed,

$$\sum_{i\geq 1} i^{2\alpha}\theta_i^2 \leq \frac{a^2}{n}\sum_{i=1}^{r} i^{2\alpha} \lesssim \frac{a^2}{n} r^{2\alpha+1} \lesssim a^2,$$

so $a \leq c_\alpha M^{1/2}$ suffices.

Index the hypercube by $\gamma = (\gamma_1, \ldots, \gamma_r) \in \{0,1\}^r$. For two vertices,

$$\|\theta(\gamma) - \theta(\gamma')\|_2^2 = \frac{a^2}{n} H(\gamma, \gamma'),$$

so the separation factor in Lemma 3.1 equals a^2/n. Along an edge the two laws differ in exactly one Gaussian coordinate by a mean shift of size $a/\sqrt{n}$. Thresholding that standardized coordinate at the midpoint gives two error probabilities $\Phi(-a/2)$, and the likelihood-ratio rule is optimal. Hence the testing affinity is

$$\eta\big(P_\gamma, P_{\gamma'}\big) = 2\Phi(-a/2) > 0, \qquad H(\gamma, \gamma') = 1.$$

Lemma 3.1 therefore gives

$$\inf_{\widehat{\theta}} \sup_{\theta\in\Theta_0} \mathbb{E}_\theta \left\|\widehat{\theta} - \theta\right\|_2^2 \gtrsim \frac{a^2}{n} r \asymp n^{-2\alpha/(2\alpha+1)}.$$

Since $\Theta_0 \subseteq \Theta_\alpha(M)$,

$$\inf_{\widehat{\theta}} \sup_{\theta\in\Theta_\alpha(M)} \mathbb{E}_\theta \left\|\widehat{\theta} - \theta\right\|_2^2 \gtrsim n^{-2\alpha/(2\alpha+1)}.$$

The rate is attained by truncating the observed sequence or shrinking its coordinates toward zero with deterministic weights. Pinsker [93] proves asymptotic optimality of linear estimates over ellipsoids in Gaussian noise; see Johnstone [81, Theorem 5.3] for this ellipsoid with polynomially increasing weights. Here the lower bound identifies the mechanism: smoothness permits about $n^{1/(2\alpha+1)}$ difficult coordinates, each contributing risk of order $1/n$. □

Example 3.3 (Running Density Example: Global Risk via Assouad) We now estimate an entire density under squared L_2 loss. Unlike the pointwise problem of Section 2.1, this loss

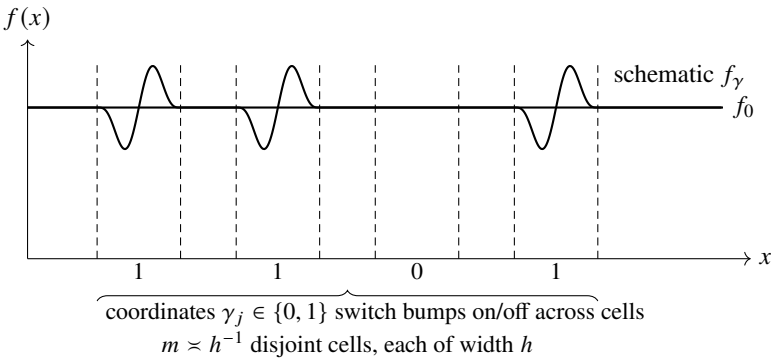


Figure 3.2 Running density example, global version. The schematic shows sign-changing, mean-zero lobes in active cells; the proof uses compactly supported C^∞ bumps. Assouad's lemma turns the one-cell loss of order $h^{2\alpha+1}$ into a global loss by summing contributions from the $m \asymp h^{-1}$ hypercube coordinates. Compare with Figure 2.1 for the local version.

accumulates error across the domain, so we replace one local perturbation with many disjoint bumps.

Suppose $X_1, \dots, X_n$ are i.i.d. from a density f on $[0,1]$. As in the pointwise example, P_f denotes the law of one observation and $\mathbb{E}_f$ expectation under $P_f^{\otimes n}$. Fix $\alpha > 0$ and $M > 1$. Over the Hölder density class $\mathcal{D}_\alpha(M)$ in (2.2), the benchmark is

$$\inf_{\widehat{f}} \sup_{f \in \mathcal{D}_\alpha(M)} \mathbb{E}_f \int_0^1 \big(\widehat{f}(x) - f(x)\big)^2 \, dx. \tag{3.4}$$

Fix an integer $R \geq 1$, choose an integer $N > 2R$, and put $h = 1/N$. Partition $[0,1]$ into N cells, leaving R cells unused at each endpoint. The remaining $m = N - 2R$ cells are $I_j = [(R + j - 1)h, (R + j)h]$, $j = 1, \dots, m$, with $m \asymp h^{-1}$ as $N \to \infty$. Choose a C^∞ bump ψ compactly supported in $(0,1)$ with $\int_0^1 \psi(u)\, du = 0$ and $\int_0^1 \psi(u)^2 \, du > 0$, and set

$$\psi_j(x) = \psi\left(\frac{x - (R + j - 1)h}{h}\right) \mathbf{1}_{I_j}(x), \qquad j = 1, \dots, m.$$

For $\gamma \in \{0,1\}^m$, define the density hypercube

$$f_\gamma(x) = 1 + c_1 h^\alpha \sum_{j=1}^m \gamma_j \psi_j(x), \tag{3.5}$$

where $c_1 > 0$ is a fixed amplitude constant to be chosen. Mean-zero bumps ensure $\int_0^1 f_\gamma = 1$. Rescaling bounds the defining derivative norms and Hölder seminorm of $f_\gamma - 1$ by a constant times c_1, uniformly in h and γ. Disjoint supports control the supremum norms, and vanishing derivatives at cell boundaries extend the within-cell Hölder bound across cells. The full class-membership calculation is given in Section 4.4.1. Thus a sufficiently small c_1, depending only on $M - 1$, α, and ψ, places every f_γ in $\mathcal{D}_\alpha(M)$ and keeps it bounded below by a positive constant.

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For the density hypercube (3.5), illustrated in Figure 3.2, disjoint supports give the following loss comparison for every pair of vertices:

$$\left\|f_\gamma - f_{\gamma'}\right\|_2^2 = c_1^2 h^{2\alpha+1} \left(\int_0^1 \psi(u)^2\, du\right) H(\gamma, \gamma').$$

If $H(\gamma, \gamma') = 1$, let j be the unique differing coordinate. The densities differ only on I_j, and

$$\left\|f_\gamma - f_{\gamma'}\right\|_2^2 = c_1^2 h^{2\alpha} \int_{I_j} \psi_j(x)^2\, dx = c_1^2 h^{2\alpha+1} \int_0^1 \psi(u)^2\, du \asymp h^{2\alpha+1}.$$

The squared target distance per differing bit is therefore of order $h^{2\alpha+1}$. For neighboring vertices, the same one-cell calculation bounds the divergence:

$$\chi^2(P_{f_\gamma}, P_{f_{\gamma'}}) = \int_0^1 \frac{(f_\gamma - f_{\gamma'})^2}{f_{\gamma'}}\, dx \lesssim h^{2\alpha+1},$$

where the denominator is bounded away from zero by the choice of c_1. For product measures,

$$\chi^2(P_{f_\gamma}^{\otimes n}, P_{f_{\gamma'}}^{\otimes n}) = (1 + O(h^{2\alpha+1}))^n - 1.$$

Choose the fixed amplitude constant c_1 sufficiently small and take $nh^{2\alpha+1} \le C$. The product chi-square divergence is then bounded by a numerical constant strictly below 4. Lemma 1.6 keeps the edge total-variation distance strictly below one and hence the edge affinity bounded below by a positive constant.

Choose $h \asymp n^{-1/(2\alpha+1)}$ and $m \asymp h^{-1}$. Lemma 3.1 yields

$$\inf_{\widehat{f}} \sup_{f \in \mathcal{D}_\alpha(M)} \mathbb{E}_f \left\|\widehat{f} - f\right\|_2^2 \ge \inf_{\widehat{f}} \sup_{\gamma \in \{0,1\}^m} \mathbb{E}_{f_\gamma} \left\|\widehat{f} - f_\gamma\right\|_2^2 \gtrsim mh^{2\alpha+1} \asymp h^{2\alpha} \asymp n^{-2\alpha/(2\alpha+1)}.$$

The exponent agrees with the pointwise Hölder benchmark, but global loss adds $m \asymp h^{-1}$ one-cell contributions of order $h^{2\alpha+1}$. Disjoint supports make these losses additive even though every decision uses the same sample. Section 4.4.1 gives an explicit finite-sample formulation and the matching kernel bound in (4.13). □

3.2 Fano's lemma and packing bounds

Fano's method reduces accurate estimation to identifying one among many separated targets: decode an estimate to its nearest target, then bound the probability of a wrong decision. Unlike Assouad's method, it requires no coordinatewise loss comparison. We seek a large separated family whose observation laws convey too little information to identify its members reliably.

The information inequality originates in Fano [65], using Shannon's entropy framework [99]. Tsybakov [103] presents its statistical packing forms, and Scarlett and Cevher [97] discuss their construction and calibration.

For $s > 0$, a finite family $\{\theta_\nu : \nu \in \mathcal{V}\}$ is a $2s$-packing when distinct targets are at distance at least $2s$. Here packing entropy refers to the logarithmic cardinality of the chosen family, without requiring it to have the largest possible cardinality. By Proposition 1.1, a decoding error entails estimation error of at least s. The strict-separation convention in Appendix A.1 is recovered by choosing any smaller radius $s' < s$.

The information calculation compares the observation laws with their uniform mixture $\bar{P}$. Their average KL divergence equals the mutual information between a uniformly chosen packing index and the observation. If this is a small fraction of the packing entropy and the packing is sufficiently large, decoding error stays bounded away from zero. The allowable information therefore grows with the logarithm of the number of alternatives, unlike the constant divergence bound typically used for two points.

Definition 3.4 (Mutual Information) For random variables X and Y with joint law P_{XY} and marginal laws P_X and P_Y, the mutual information is

$$I(X;Y) = \mathrm{KL}(P_{XY}, P_X \otimes P_Y).$$

It measures departure from independence and is zero exactly when X and Y are independent. If $P_{XY} \ll P_X \otimes P_Y$ and the log-likelihood ratio is integrable under P_{XY}, then

$$I(X;Y) = \mathbb{E}_{P_{XY}}\left[\log \frac{dP_{XY}}{d(P_X \otimes P_Y)}(X,Y)\right].$$

The KL definition also permits the value $+\infty$.

Lemma 3.5 (Fano's Lemma) *Let $\{P_\nu : \nu \in \mathcal{V}\}$ be a finite experiment on a common observation space, with $M = |\mathcal{V}| \geq 2$. Let $\theta(P_\nu)$ be the target value in a metric space (Θ, d), and assume the targets form a $2s$-packing for some $s > 0$:*

$$d\big(\theta(P_\nu), \theta(P_{\nu'})\big) \geq 2s, \qquad \nu \neq \nu'.$$

Set

$$\bar{P} = \frac{1}{M}\sum_{\nu\in\mathcal{V}} P_\nu,$$

and suppose, for some $0 < \alpha < 1$, that

$$\frac{1}{M}\sum_{\nu\in\mathcal{V}} \mathrm{KL}(P_\nu, \bar{P}) \leq \alpha \log M.$$

Every measurable Θ-valued estimator $\widehat{\theta}$ then satisfies

$$\sup_{\nu\in\mathcal{V}} \mathbb{P}_\nu\Big(d\big(\widehat{\theta}, \theta(P_\nu)\big) \geq s\Big) \geq 1 - \alpha - \frac{\log 2}{\log M}.$$

Consequently, under squared metric loss d^2,

$$\inf_{\widehat{\theta}} \sup_{\nu\in\mathcal{V}} \mathbb{E}_\nu d^2\big(\widehat{\theta}, \theta(P_\nu)\big) \geq s^2\left[1 - \alpha - \frac{\log 2}{\log M}\right]_+.$$

Proof Let V be uniform on $\mathcal{V}$ and let $X \mid (V = \nu) \sim P_\nu$. Given an estimator $\widehat{\theta}(X)$, let $\widehat{V}$ be the index of the nearest target $\theta(P_\nu)$, with a fixed rule for ties. The reduction in Proposition 1.1 shows that a decoding error requires $d(\widehat{\theta}, \theta(P_V)) \geq s$. Hence

$$\sup_{\nu\in\mathcal{V}} \mathbb{P}_\nu\{d(\widehat{\theta}, \theta(P_\nu)) \geq s\} \geq \frac{1}{M}\sum_{\nu\in\mathcal{V}} \mathbb{P}_\nu\{d(\widehat{\theta}, \theta(P_\nu)) \geq s\} \geq \mathbb{P}\{\widehat{V} \neq V\}.$$

We now bound the decoding error using entropy. Write

$$p_e = \mathbb{P}\{\widehat{V} \neq V\}.$$

For a finite-valued random variable U with probabilities p_u, write $H(U) = -\sum_u p_u \log p_u$ for its entropy, and define $H(U \mid X) = \mathbb{E}[-\sum_u p_u(X) \log p_u(X)]$, where $p_u(X) = \mathbb{P}(U = u \mid X)$ and $0 \log 0 = 0$. In this proof, H denotes entropy rather than Hamming or Hellinger distance. For mutual information $I(V; X)$, use Definition 3.4.

Because V is uniform on a set of size M, $H(V) = \log M$. Moreover, $\widehat{V}$ is a function of X, so $V \to X \to \widehat{V}$ is a Markov chain and

$$H(V \mid X) \leq H(V \mid \widehat{V}).$$

Write $h(t) = -t \log t - (1-t) \log(1-t)$, $0 \leq t \leq 1$, for binary entropy, with $0 \log 0 = 0$. Given the decoded value $\widehat{V}$, a correct decision determines V, whereas an error leaves at most $M - 1$ possibilities. The entropy of the error indicator, conditional on $\widehat{V}$, is at most its unconditional entropy $h(p_e)$. The entropy chain rule therefore gives

$$H(V \mid \widehat{V}) \leq h(p_e) + p_e \log(M - 1) \leq \log 2 + p_e \log M.$$

Therefore

$$\log M = H(V) = I(V; X) + H(V \mid X) \leq I(V; X) + \log 2 + p_e \log M,$$

which rearranges to

$$p_e \geq 1 - \frac{I(V; X) + \log 2}{\log M}.$$

It remains to identify the mutual information. Since V is uniform,

$$P_X = \bar{P} = \frac{1}{M} \sum_{\nu \in \mathcal{V}} P_\nu,$$

and the chain rule for relative entropy gives

$$I(V; X) = \frac{1}{M} \sum_{\nu \in \mathcal{V}} \mathrm{KL}(P_\nu, P_X) = \frac{1}{M} \sum_{\nu \in \mathcal{V}} \mathrm{KL}(P_\nu, \bar{P}).$$

The assumed bound makes this at most $\alpha \log M$, so

$$\mathbb{P}\{\widehat{V} \neq V\} \geq 1 - \alpha - \frac{\log 2}{\log M}.$$

This proves the probability bound. For the squared-error consequence, use

$$\mathbb{E}_\nu d^2\big(\widehat{\theta}, \theta(P_\nu)\big) \geq s^2 \mathbb{P}_\nu\Big(d\big(\widehat{\theta}, \theta(P_\nu)\big) \geq s\Big),$$

for each ν, and then take the supremum over ν and the infimum over estimators. When the bracket is nonpositive, nonnegativity of squared loss gives the displayed positive-part form. □

Birgé [8] develops related multiple-hypothesis testing bounds, and Duchi and Wainwright [62] extend distance-based Fano inequalities. Vu and Lei [109] illustrate combinatorial packing in high dimensions. Section 3.4 develops Yang and Barron's metric-entropy approach [113].

The next two examples use the same fixed-weight packing in different parameter spaces. Example 3.6 isolates the cost of an unknown sparse support. Example 3.8 embeds the packing at one resolution of a Besov sequence, where smoothness determines the amplitude, and compares Fano with Assouad on that level.

Example 3.6 (Fano Lower Bound for Sparse Gaussian Mean Estimation) For integers $d \geq 1$ and $1 \leq s \leq d$, write

$$\Theta(s, d) = \{\theta \in \mathbb{R}^d : \|\theta\|_0 \leq s\}. \tag{3.6}$$

In this example assume $1 \leq s \leq d/8$ and use the finite Gaussian model (2.4) with known $\sigma > 0$. We estimate θ under squared ℓ_2 loss over $\Theta(s, d)$. To isolate the difficulty of locating the nonzero coordinates, use the fixed-amplitude subclass $\Theta(s, d, a)$ of (2.5), which is contained in $\Theta(s, d)$, with $a > 0$ to be calibrated below.

Packing and separation. Lemma A.4 gives a family C of binary vectors of weight s such that

$$\log |C| \gtrsim s \log(ed/s), \qquad H(u, v) \geq c_1 s \quad (u \neq v),$$

where H is Hamming distance and $c_1 > 0$ is universal. To ensure $|C| \geq 8$, compare this code with a family of $\lfloor d/s \rfloor \geq 8$ disjoint supports of size s, and use whichever family has more members. Both bounds are preserved, after replacing c_1 by $\min\{c_1, 2\}$.

Set $\theta^{(v)} = a\sigma v$ for $v \in C$. For distinct $u, v \in C$,

$$\left\|\theta^{(u)} - \theta^{(v)}\right\|_2^2 = a^2\sigma^2 H(u, v) \geq c_1 a^2 \sigma^2 s,$$

so these parameters form a 2δ-packing with $\delta^2 = c_1 a^2 \sigma^2 s/4$.

Information and amplitude. Write $P_\theta = N_d(\theta, \sigma^2 I_d)$ for the observation law. Since two weight-s patterns differ in at most $2s$ coordinates,

$$\mathrm{KL}\big(P_{\theta^{(u)}} \| P_{\theta^{(v)}}\big) = \frac{\left\|\theta^{(u)} - \theta^{(v)}\right\|_2^2}{2\sigma^2} = \frac{a^2}{2} H(u, v) \leq a^2 s.$$

Balance this divergence bound with the packing entropy by choosing

$$a^2 = c \log(ed/s)$$

with a sufficiently small universal $c > 0$. For $\bar{P} = |C|^{-1} \sum_{v \in C} P_{\theta^{(v)}}$, convexity of KL divergence in its second argument gives

$$\mathrm{KL}(P_{\theta^{(u)}} \| \bar{P}) \leq \frac{1}{|C|} \sum_{v \in C} \mathrm{KL}(P_{\theta^{(u)}} \| P_{\theta^{(v)}}) \leq a^2 s \leq \tfrac{1}{4} \log |C|.$$

Risk lower bound. The information condition of Lemma 3.5 holds, and $|C| \geq 8$ makes its error factor at least $1 - 1/4 - \log 2/\log 8 = 5/12$. Since all packing points lie in $\Theta(s, d)$,

$$\inf_{\widehat{\theta}} \sup_{\theta \in \Theta(s,d)} \mathbb{E}_\theta \left\|\widehat{\theta} - \theta\right\|_2^2 \geq \frac{5}{12} \delta^2 \gtrsim s\sigma^2 \log(ed/s).$$

Thus uncertainty about the support contributes the factor $\log(ed/s)$, even when all nonzero coordinates have the same amplitude. □

Matching upper bound. For the class $\Theta(s, d)$ of vectors with at most s nonzero coordinates, Johnstone [81, Proposition 8.20] gives a nonasymptotic thresholding upper bound of order $s\sigma^2 \log(ed/s)$. Hence Example 3.6 identifies the minimax rate for $1 \le s \le d/8$, up to universal constants. Donoho and Johnstone [57, 58] develop the related shrinkage theory for sparse and inhomogeneous wavelet sequences.

Remark 3.7 The horizontal mixture in Example 2.10 and the packing above give lower bounds over the full sparse class, but they calibrate different hard families. The mixture bound and the Fano bound have respective scales

$$s\sigma^2 \log\left(1 + \frac{d}{s^2}\right) \quad \text{and} \quad s\sigma^2 \log\left(\frac{ed}{s}\right).$$

The mixture comparison averages over supports and controls a likelihood-ratio second moment. Fano retains a separated family and compares its entropy, of order $s \log(ed/s)$, with KL divergence. These different constraints permit different amplitudes.

The two scales are comparable when $s \le d^{1/2-\delta_{\mathrm{sp}}}$ for fixed $\delta_{\mathrm{sp}} \in (0, 1/2)$, with constants depending on δ_{sp}. The weaker condition $s = o(\sqrt{d})$ does not suffice. For example, take $s_n = 2^n$ and $d_n = n^2 4^n$ for integers $n \ge 2$. Then $s_n/\sqrt{d_n} = 1/n \to 0$, but the ratio of the mixture scale to the Fano scale is

$$\frac{\log(1 + n^2)}{1 + 2\log n + n \log 2} \longrightarrow 0.$$

Thus even below the $\sqrt{d}$ sparsity scale, using a multiple-hypothesis comparison of supports can give a substantially stronger bound. Section 8.1 uses a sparse packing for coefficient estimation but a nuisance-parameter mixture for honest interval length, illustrating why the inferential goal determines the comparison.

We next embed a packing in one resolution level to bound the risk of recovering an entire signal. A multiresolution representation describes broad features with a few coarse coefficients and local detail with many finer coefficients. Besov sequence balls express smoothness restrictions across and within these levels, as in the wavelet theory of Donoho and Johnstone [58].

Fix $\alpha, M > 0$, $0 < p, q \le \infty$, and a positive integer n. Use the coordinate observation law (3.1), reindexed by

$$i = 2^j + k - 1, \qquad j \ge 0, \quad 1 \le k \le 2^j. \tag{3.7}$$

Write $Y_{jk}, \theta_{jk}, Z_{jk}$ for the corresponding coordinates. Here n denotes precision, j resolution, and k position within a level. We estimate the entire sequence under squared ℓ_2 loss, $\|\widehat{\theta} - \theta\|_2^2 = \sum_{j \ge 0} \sum_{k=1}^{2^j} (\widehat{\theta}_{jk} - \theta_{jk})^2$.

The parameter space is the Besov sequence ball

$$B^{\alpha}_{p,q}(M) = \left\{ \theta : \left(\sum_{j \ge 0} \left[2^{j(\alpha + 1/2 - 1/p)} \|\theta_{j\cdot}\|_p \right]^q \right)^{1/q} \le M \right\}, \qquad \theta_{j\cdot} = (\theta_{j1}, \dots, \theta_{j,2^j}). \tag{3.8}$$

Read the constraint from the inside out: measure the coefficients at level j in ℓ_p, apply the resolution weight, then combine these weighted quantities in ℓ_q. Thus p controls concentration within a level, α controls decay across resolutions, and q controls aggregation across levels. For $p = \infty$, use the maximum coordinate magnitude and set $1/p = 0$; for $q = \infty$, use a supremum over levels. The expression is a norm when $p, q \geq 1$ and a quasi-norm otherwise; M bounds it, not its square.

This full parameter range requires a coordinate formulation of the experiment. The product $P_\theta = \bigotimes_{j\geq 0} \bigotimes_{k=1}^{2^j} N(\theta_{jk}, n^{-1})$ is defined for every real sequence θ, and the loss may be infinite. When $\theta \in \ell_2$, it agrees with the Hilbert-space experiment of Section 1.2. Some Besov balls contain sequences outside ℓ_2, but every hard alternative below has finite support.

The hard alternatives occupy one resolution level, so only one term enters the outer aggregation. When a fixed fraction of its m coefficients have a common magnitude, the within-level factor $m^{1/p}$ cancels the factor $m^{-1/p}$ in the resolution weight. The allowable magnitude is therefore of order $m^{-(\alpha+1/2)}$. Balancing it with the noise level $n^{-1/2}$ selects the hard resolution. The lower bound holds for every $\alpha > 0$; the matching upper bound below requires additional smoothness.

Example 3.8 (Besov Sequence Balls: Assouad and Fano) Choose J as the smallest integer such that

$$m = 2^J \geq \max\{8, n^{1/(2\alpha+1)}\}.$$

Then $m \asymp n^{1/(2\alpha+1)}$. At this level, put

$$\theta_{Jk} = a_J \gamma_k, \qquad a_J = c_* m^{-(\alpha+1/2)}, \qquad \gamma \in \{0,1\}^m,$$

with all other coefficients zero. Here $0 < c_* \leq M$ is a fixed constant to be chosen sufficiently small. Every vertex belongs to the Besov ball, since

$$2^{J(\alpha+1/2-1/p)} \|\theta_{J\cdot}\|_p \leq m^{\alpha+1/2-1/p} a_J m^{1/p} = c_*.$$

This includes $p = \infty$ under the convention above, and only one level contributes for any q.

Assouad: coordinatewise uncertainty. Write P_γ for the observation law. Adjacent vertices have squared target distance a_J^2 and

$$\mathrm{KL}(P_\gamma, P_{\gamma'}) = \frac{n a_J^2}{2} \leq \frac{c_*^2}{2}.$$

For small enough c_*, Pinsker's inequality keeps every edge affinity bounded away from zero. Since the squared distance between arbitrary vertices is $a_J^2 H(\gamma, \gamma')$, Lemma 3.1 gives

$$\inf_{\widehat{\theta}} \sup_{\theta \in B^\alpha_{p,q}(M)} \mathbb{E}_\theta \|\widehat{\theta} - \theta\|_2^2 \gtrsim m a_J^2 = c_*^2 m^{-2\alpha} \asymp n^{-2\alpha/(2\alpha+1)}.$$

The constants may depend on the fixed class parameters, but not on $n \geq 1$.

Fano: separated patterns. At the same level, restrict to patterns with exactly $m/8$ active coordinates. Lemma A.4 supplies a code C with $\log |C| \gtrsim m$ and pairwise Hamming distance $\gtrsim m$. As in the sparse-mean example, compare this code with a family of eight disjoint

supports of size $m/8$, and use whichever family has more members. This ensures $|C| \geq 8$ while preserving both bounds up to universal constants. For distinct codewords,

$$\|\theta^{(u)} - \theta^{(v)}\|_2^2 \gtrsim ma_J^2, \qquad \mathrm{KL}(P_u, P_v) \leq \frac{nma_J^2}{8} \leq \frac{c_*^2 m}{8}.$$

Choose c_* small enough that this KL bound is at most $\frac{1}{4}\log|C|$. Convexity in the second argument gives the same bound for divergence to the uniform mixture. Since $|C| \geq 8$, Lemma 3.5 bounds the decoding error below by $5/12$. The squared separation is of order ma_J^2, so Fano gives

$$\inf_{\widehat{\theta}} \sup_{\theta \in B_{p,q}^{\alpha}(M)} \mathbb{E}_\theta \|\widehat{\theta} - \theta\|_2^2 \gtrsim ma_J^2 = c_*^2 m^{-2\alpha} \asymp n^{-2\alpha/(2\alpha+1)}.$$

□

Both methods use $na_J^2 \leq c_*^2$ and give the same risk lower bound of order ma_J^2. Assouad adds coordinatewise costs; Fano compares information of order nma_J^2 with entropy of order m. Here a fixed fraction of the coordinates is active, so the logarithmic factor from sparse-support packing is constant.

Matching upper bound. For fixed $M > 0$, $0 < p, q \leq \infty$, and $\alpha > \max\{0, 1/p - 1/2\}$, suitably tuned level-dependent soft thresholding attains the same rate. At level j, this replaces a coefficient y by $\mathrm{sign}(y)(|y| - t_j)_+$, with threshold $t_j \geq 0$, where $\mathrm{sign}(y) = 1$ for $y \geq 0$ and -1 otherwise. A precise sequence-model statement is given by Johnstone [81, Theorems 14.2–14.4], with noise level $\varepsilon = n^{-1/2}$. His upper bound is stated for $B_{p,\infty}^{\alpha}(M)$ and therefore covers every $B_{p,q}^{\alpha}(M)$, which is a subset. Together with Example 3.8, it gives

$$\inf_{\widehat{\theta}} \sup_{\theta \in B_{p,q}^{\alpha}(M)} \mathbb{E}_\theta \|\widehat{\theta} - \theta\|_2^2 \asymp n^{-2\alpha/(2\alpha+1)} \qquad (n \to \infty).$$

The thresholds may depend on the class parameters. Outside the stated smoothness range, the lower bound is not asserted to be sharp or to imply finite minimax risk.

Adaptation asks whether one procedure can attain the class-specific rates without knowing the class parameters. Donoho and Johnstone [57, Section 3.2] choose wavelet thresholds using Stein's unbiased estimate of squared-error risk. In the Gaussian sequence model, Cai [14, Theorem 9] shrinks groups of coefficients together by factors determined by their observed squared norms, attaining the upper rate without logarithmic loss for $2 \leq p \leq \infty$ and $1 \leq q \leq \infty$. For fixed-design Gaussian regression under integrated squared loss, Cai and Zhou [46, Theorems 3–5] select block sizes and shrinkage thresholds by estimated risk, obtaining adaptive guarantees in dense and sparse regimes under their smoothness and wavelet conditions. Within these respective models and parameter ranges, adaptation need not introduce an additional logarithm. Chapter 7 contrasts this possibility with problems where adaptation necessarily incurs a cost.

Both examples bounded information by pairwise KL comparisons. Section 3.3 now shows how a common reference law can simplify that step. Appendix A.1 supplies further packing constructions.

3.3 Mutual information and information radius

Once a packing is chosen, the remaining task is to bound its mutual information. Section 3.2 expresses this as average divergence to the mixture of the packing laws, which may be difficult to evaluate. Here we replace that mixture by a convenient reference law without changing the packing or the loss reduction. The reference law need not belong to the statistical model.

Proposition 3.9 (Bounding Mutual Information by a Reference Law) *Let V be uniform on $\{1, \dots, M\}$, where $M \geq 2$, and let the conditional law of X given $V = v$ be P_v. Write $\overline{P} = M^{-1} \sum_{v=1}^{M} P_v$. For any probability measure Q with $P_v \ll Q$ for every v,*

$$\frac{1}{M} \sum_{v=1}^{M} \mathrm{KL}(P_v, Q) = I(V; X) + \mathrm{KL}(\overline{P}, Q).$$

Consequently,

$$I(V; X) \leq \frac{1}{M} \sum_{v=1}^{M} \mathrm{KL}(P_v, Q),$$

with equality when $Q = \overline{P}$.

Proof As in the proof of Lemma 3.5, $I(V; X) = M^{-1} \sum_{v=1}^{M} \mathrm{KL}(P_v, \overline{P})$. Put $p_v = dP_v/dQ$ and $\overline{p} = M^{-1} \sum_v p_v$. On $\{p_v > 0\}$,

$$\log p_v = \log \frac{p_v}{\overline{p}} + \log \overline{p}.$$

Integrating against P_v and averaging over v yields the identity, because the average of the P_v is $\overline{P}$. Since $\overline{P} \geq P_v/M$, each $\mathrm{KL}(P_v, \overline{P})$ is finite and at most $\log M$; thus the identity remains valid if $\mathrm{KL}(\overline{P}, Q) = \infty$, without subtracting infinite quantities. Nonnegativity of $\mathrm{KL}(\overline{P}, Q)$ proves the inequality. □

The mixture $\overline{P}$ minimizes the average KL divergence; this minimum is the *information radius* of the uniformly weighted family. A simpler law can be easier to use, such as the zero-mean Gaussian law for the sparse family of Example 3.6. The proposition permits any reference law dominating every alternative. Appendix A.4 gives the general-prior formulation.

The next theorem recalls the decoding inequality of Lemma 3.5 and extends its risk consequence to every positive power of a metric loss. The reference-law bound above handles the model-specific information calculation.

Theorem 3.10 (Generalized Fano via Mutual Information) *Let $\mathcal{V}$ be a finite index set of cardinality $M \geq 2$. Let V be uniform on $\mathcal{V}$, and suppose X has conditional law P_v given $V = v$. For any measurable decoder $\widehat{V} = \widehat{V}(X)$ taking values in $\mathcal{V}$,*

$$\mathbb{P}\{\widehat{V} \neq V\} \geq 1 - \frac{I(V; X) + \log 2}{\log M}.$$

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Suppose further that the parameters $\{\theta_v : v \in \mathcal{V}\}$ lie in a metric space $(\mathcal{T}, d)$ and satisfy $d(\theta_u, \theta_v) \geq 2\delta$ for all distinct u, v, where $\delta > 0$. For every $r > 0$,

$$\inf_{\widehat{\theta}} \sup_{v \in \mathcal{V}} \mathbb{E}_v d^r(\widehat{\theta}, \theta_v) \geq \delta^r \left[1 - \frac{I(V;X) + \log 2}{\log M}\right]_+,$$

where the infimum is over measurable estimators taking values in $\mathcal{T}$ and $\mathbb{E}_v$ denotes expectation under P_v.

Proof The decoding inequality is the entropy bound proved in Lemma 3.5. For the risk consequence, apply the nearest-point reduction of Proposition 1.1 with separation 2δ and a fixed tie-breaking rule. Pointwise, a decoding error entails metric error at least δ, so for every $r > 0$,

$$\sup_{v \in \mathcal{V}} \mathbb{E}_v d^r(\widehat{\theta}, \theta_v) \geq \frac{1}{M} \sum_{v \in \mathcal{V}} \mathbb{E}_v d^r(\widehat{\theta}, \theta_v) \geq \delta^r \mathbb{P}\{\widehat{V} \neq V\}.$$

Apply the decoding bound, take the infimum over estimators, and use nonnegativity for the positive-part form. Independent randomization is covered by adjoining the random seed to X, which leaves its mutual information with V unchanged. □

A reference law Q is useful when the average $\mathrm{KL}(P_v, Q)$, together with $\log 2$, is at most a fixed fraction of $\log M$; the resulting risk bound is of order δ^r. Section 3.4 places this comparison in a broader divergence framework and shows how a cover can supply the reference law.

3.4 f-divergence and global-entropy lower bounds

A packing can be easy to construct but difficult to analyze through KL divergence. We develop two ways to simplify the information comparison. An f-divergence inequality permits other divergences when bounding identification error. Yang and Barron's approach [113] constructs a reference law from a cover when comparison with a single model law is too costly. We use the latter to recover the earlier Assouad rate over a Sobolev ellipsoid, directly from Fano's inequality and the KL reference-law bound, without the general f-divergence theorem.

A unifying divergence bound. Let $F = \{\theta_1, \ldots, \theta_N\}$ consist of $N \geq 2$ distinct parameter labels with observation laws P_{θ_j}. A prior $w = (w_1, \ldots, w_N)$ assigns nonnegative weights summing to one. The Bayes error and success probabilities are

$$\bar{r}_w = \inf_T \sum_{j=1}^N w_j \mathbb{P}_{\theta_j}\{T \neq \theta_j\}, \qquad \beta_w = 1 - \bar{r}_w,$$

where T ranges over measurable classifiers taking values in F. A lower bound on $\bar{r}_w$ becomes an estimation bound when the labels index separated targets, as in Section 3.3.

For a convex function $f : [0, \infty) \to \mathbb{R}$ with $f(1) = 0$, define

$$D_f(P\|Q) = \int f\left(\frac{dP}{dQ}\right) dQ, \qquad P \ll Q.$$

The choices $f(t) = t\log t$, with $0\log 0 = 0$, and $f(t) = t^2 - 1$ give KL and chi-square divergence, respectively. Terms with zero prior weight are omitted from weighted sums.

Ali and Silvey [1] and Csiszár [53] developed this divergence family. Gushchin [70] established an equivalent uniform-prior form of the next inequality; Guntuboyina [68] gave a different convexity proof and statistical applications.

Theorem 3.11 (f-Divergence Bayes-Risk Inequality) *In the finite experiment above, let $f : [0,\infty) \to \mathbb{R}$ be convex with $f(1) = 0$. Let Q be a probability measure dominating every P_{θ_j}, and write $p_j = dP_{\theta_j}/dQ$. Let $j^*(x)$ be the smallest index maximizing $w_j p_j(x)$, and put*

$$W = \int w_{j^*(x)}\, dQ(x).$$

Then

$$\sum_{j=1}^{N} w_j D_f(P_{\theta_j}\|Q) \geq W f\left(\frac{\beta_w}{W}\right) + (1-W) f\left(\frac{1-\beta_w}{1-W}\right).$$

Use limiting values when $W = 0$ or 1. Under the uniform prior $w_j = 1/N$, write $\bar r = \bar r_w$. Then $W = 1/N$, and

$$\frac{1}{N}\sum_{j=1}^{N} D_f(P_{\theta_j}\|Q) \geq \frac{1}{N} f\big(N(1-\bar r)\big) + \frac{N-1}{N} f\left(\frac{N\bar r}{N-1}\right).$$

Proof The rule $T^*(x) = \theta_{j^*(x)}$ maximizes the integrand of the prior-average success probability. Hence

$$\beta_w = \int \max_j w_j p_j(x)\, dQ(x).$$

Introduce two joint laws on the label J and observation X:

$$\mathsf{P}(\{j\}\times dx) = w_j P_{\theta_j}(dx), \qquad \mathsf{R}(\{j\}\times dx) = w_j Q(dx).$$

Under R, the label and data are independent. Since $d\mathsf{P}/d\mathsf{R}(j,x) = p_j(x)$ on labels of positive weight,

$$D_f(\mathsf{P}\|\mathsf{R}) = \sum_{j=1}^{N} w_j D_f(P_{\theta_j}\|Q).$$

Let $A = \{(j,x) : j = j^*(x)\}$ be the event of a correct decision by T^*. Its probabilities under the two laws are

$$\mathsf{P}(A) = \beta_w, \qquad \mathsf{R}(A) = W.$$

If $0 < W < 1$, apply Jensen's inequality separately on A and A^c, normalizing the restriction of R on each event. This gives

$$\begin{aligned} D_f(\mathsf{P}\|\mathsf{R}) &= \int_A f\left(\frac{d\mathsf{P}}{d\mathsf{R}}\right) d\mathsf{R} + \int_{A^c} f\left(\frac{d\mathsf{P}}{d\mathsf{R}}\right) d\mathsf{R} \\ &\geq W f\left(\frac{\mathsf{P}(A)}{W}\right) + (1-W) f\left(\frac{\mathsf{P}(A^c)}{1-W}\right). \end{aligned}$$

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If $W = 0$ or 1, domination $\mathsf{P} \ll \mathsf{R}$ implies $\beta_w = W$; the limiting right-hand side is zero, and the inequality follows from $D_f(\mathsf{P}\|\mathsf{R}) \geq 0$. An infinite left-hand side is immediate. Finally, uniform weights give $W = 1/N$. □

The proof compares the probability of correct classification under the experiment with its reference probability W, equal to $1/N$ under uniform weights. With $f(t) = t \log t$ and $Q = N^{-1} \sum_j P_{\theta_j}$, the binary entropy bound recovers Fano's inequality in Theorem 3.10. The chi-square choice gives a different information comparison.

Corollary 3.12 (χ^2 Bound for Finite-Label Classification) *For the finite experiment above with uniform prior, the Bayes error $\bar{r}$ satisfies*

$$\bar{r} \geq 1 - \frac{1}{\sqrt{N}} \left(\inf_{Q} \frac{1}{N} \sum_{j=1}^{N} \big(\chi^2(P_{\theta_j}, Q) + 1\big) \right)^{1/2},$$

where the infimum is over probability measures Q dominating all P_{θ_j}.

Proof Taking $f(t) = t^2 - 1$ in Theorem 3.11 gives, for every admissible Q,

$$N(1 - \bar{r})^2 + \frac{N\bar{r}^2}{N - 1} \leq \frac{1}{N} \sum_{j=1}^{N} \big(\chi^2(P_{\theta_j}, Q) + 1\big).$$

Discard the nonnegative second term, take the infimum over Q, and take square roots. □

The KL form compares average divergence with $\log N$; the chi-square form compares an average likelihood-ratio second moment with N. Both concern identification among N labels, rather than the binary comparison between a null and a mixture used in Chapter 5.

Constructing a reference law from a cover. The packing supplies separated targets; the cover supplies a tractable reference law. Mixing the laws at the covering centers bounds divergence by two terms: the logarithm of the cover size and the approximation error from a nearby center. The next example balances their sum against packing entropy. Theorem A.13 gives the general formulation.

Example 3.13 (Global-Entropy Lower Bound for a Sobolev Ellipsoid) In the Gaussian sequence model (3.1), estimate θ under squared ℓ_2 loss over the ellipsoid $\Theta_\alpha(M)$ of (3.2), with fixed $\alpha, M > 0$. The weighted constraint penalizes high-frequency coefficients and represents Sobolev smoothness in an orthogonal series. We recover the lower rate $n^{-2\alpha/(2\alpha+1)}$ from Example 3.2, now through packing and covering rather than coordinatewise comparisons.

Step 1: metric entropy. Write $N(\varepsilon, \Theta_\alpha(M), \|\cdot\|_2)$ for the smallest number of ℓ_2 balls of radius ε, centered in the ellipsoid, needed to cover it; see Lemma A.1. Its logarithm is the covering entropy at scale ε. We establish

$$\log N(\varepsilon, \Theta_\alpha(M), \|\cdot\|_2) \asymp \varepsilon^{-1/\alpha} \qquad (\varepsilon \downarrow 0).$$

This is the case of an ellipsoid with polynomially decreasing semiaxes in the entropy-number framework of Carl and Stephani [47]. The following construction supplies both bounds directly.

For the entropy lower bound, construct points more than 2ε apart, so no covering ball can contain two of them. Put

$$m = \left\lfloor \left(\frac{\sqrt{M}}{8\varepsilon} \right)^{1/\alpha} \right\rfloor, \qquad a = \sqrt{M}\, m^{-\alpha-1/2}.$$

For small enough ε, $m \geq 8$ and $m \asymp \varepsilon^{-1/\alpha}$. The vectors $\theta_i^\gamma = a\gamma_i$ for $i \leq m$, with zero remaining coordinates and $\gamma \in \{0,1\}^m$, lie in the ellipsoid because their weighted squared norm is at most $a^2 m^{2\alpha+1} = M$. Lemma A.3 supplies a subfamily with log cardinality at least $(m/8)\log 2$ and squared pairwise distance at least

$$a^2 m/8 > 4\varepsilon^2.$$

The cover must therefore have at least as many balls as this separated family has points, proving the entropy lower bound.

For the entropy upper bound, truncate the tail and cover the remaining dyadic blocks separately, initially allowing centers in the ambient ℓ_2 space. Put $t = (2\sqrt{M}/\varepsilon)^{1/\alpha}$ and choose a dyadic cutoff with $t \leq 2^J \leq 4t$. This makes the omitted tail's squared norm at most $M 2^{-2\alpha J} \leq \varepsilon^2/4$. For $0 \leq j < J$, the block $2^j \leq i < 2^{j+1}$ has dimension 2^j and Euclidean radius at most

$$R_j = \sqrt{M}\, 2^{-\alpha j}.$$

Cover this ball at radius $\delta_j = \varepsilon 2^j/(4\,2^J)$. The larger blocks receive larger error allowances, while the sum of squared allowances remains small. Lemma A.2 gives at most $(1+2R_j/\delta_j)^{2^j}$ centers if $\delta_j < R_j$; otherwise, the single center zero suffices. Combine the block centers and set coordinates $i \geq 2^J$ to zero. Since errors on disjoint coordinate blocks add in squared ℓ_2 loss, the total squared approximation error is at most

$$\sum_{j<J} \delta_j^2 + M 2^{-2\alpha J} \leq \frac{\varepsilon^2}{16} + \frac{\varepsilon^2}{4} \leq \varepsilon^2.$$

To count the centers, first note that

$$1 + \frac{2R_j}{\delta_j} \leq 9\, 2^{(\alpha+1)(J-j)}.$$

The block-cover sizes multiply, so their logarithms add. Using $\sum_{j<J} 2^j \leq 2^J$ and $\sum_{j<J} 2^j(J-j) \leq 2\,2^J$, the log cardinality of the product cover is at most

$$\{\log 9 + 2(\alpha+1)\log 2\}\, 2^J \lesssim_{\alpha,M} \varepsilon^{-1/\alpha}.$$

The blockwise centers need not lie in the ellipsoid. To obtain an internal cover, start with an ambient $\varepsilon/2$-cover and replace each ball meeting the ellipsoid by one point of that intersection. The triangle inequality gives an internal ε-cover with no more centers.

Step 2: a packing and a mixture reference law. Use a finer scale for packing than for covering so that the packing entropy can dominate the information bound. For a fixed $c \in (0,1)$ to be chosen, Lemma A.1 and the entropy lower bound give a $2c\varepsilon$-packing $\{\theta^1, \ldots, \theta^N\}$ with

$$\log N \gtrsim (c\varepsilon)^{-1/\alpha}.$$

Choose an internal ε-cover $\{\vartheta^1, \dots, \vartheta^m\}$ with $\log m \lesssim \varepsilon^{-1/\alpha}$. Write $P_\theta^{(n)}$ for the law of the entire observed sequence and set

$$Q_j = P_{\vartheta^j}^{(n)}, \qquad R = \frac{1}{m}\sum_{j=1}^{m} Q_j.$$

For each packed θ^v, some center $\vartheta^{j(v)}$ satisfies $\|\theta^v - \vartheta^{j(v)}\|_2 \le \varepsilon$. The Gaussian shift identity in Lemma A.8 gives

$$\mathrm{KL}(P_{\theta^v}^{(n)}, Q_{j(v)}) = \frac{n}{2}\|\theta^v - \vartheta^{j(v)}\|_2^2 \le \frac{n}{2}\varepsilon^2.$$

Since $R \ge Q_{j(v)}/m$, comparison of likelihood ratios yields

$$\mathrm{KL}(P_{\theta^v}^{(n)}, R) \le \log m + \mathrm{KL}(P_{\theta^v}^{(n)}, Q_{j(v)}) \le \log m + \frac{n}{2}\varepsilon^2.$$

The two terms are the cost of specifying a cover center and the cost of approximating the observation law by that center.

Step 3: balance information and separation. Apply Proposition 3.9 with reference law R, followed by Theorem 3.10. Since the packing separation is $2c\varepsilon$,

$$\inf_{\widehat\theta}\sup_{\theta\in\Theta_\alpha(M)} \mathbb{E}_\theta\|\widehat\theta - \theta\|_2^2 \ge c^2\varepsilon^2\left[1 - \frac{\log m + n\varepsilon^2/2 + \log 2}{\log N}\right]_+.$$

Balance the approximation term $n\varepsilon^2$ with the covering entropy $\varepsilon^{-1/\alpha}$ by choosing $\varepsilon = n^{-\alpha/(2\alpha+1)}$, so that $n\varepsilon^2 = \varepsilon^{-1/\alpha}$. The numerator is then at most a constant times $\varepsilon^{-1/\alpha}$, while the denominator is at least a constant times $c^{-1/\alpha}\varepsilon^{-1/\alpha}$. Taking c sufficiently small, depending only on α and M, makes the bracket at least $1/2$ for all sufficiently large n. Therefore

$$\inf_{\widehat\theta}\sup_{\theta\in\Theta_\alpha(M)} \mathbb{E}_\theta\|\widehat\theta - \theta\|_2^2 \gtrsim n^{-2\alpha/(2\alpha+1)}.$$

The matching upper rate is discussed in Example 3.2. Thus the entropy argument recovers the rate, but not the sharp constant, of the Pinsker benchmark for Gaussian-noise ellipsoids [93].□

For the Sobolev ellipsoid, Assouad sums coordinatewise costs, whereas the entropy argument balances $n\varepsilon^2$ against covering entropy and uses a finer packing for separation. The latter is useful when covers are easier to construct than a coordinatewise hard family. Section 4.4 returns to bump and wavelet constructions, where density constraints and inverse operators determine class membership and the information calculation.

3.5 Exercises

3.1. Prove the Hamming form of Assouad's inequality with randomized coordinate decoders. State where decoder measurability enters.

3.2. Let $Y \sim N_m(a\gamma, (\sigma^2/n)I_m)$, where $a, \sigma > 0$ and $\gamma \in \{-1, 1\}^m$. Calculate the KL divergence between the observation laws at two vertices differing in exactly one sign.

3.3. In the setting of Lemma 3.5, let V be uniform on $M \geq 2$ labels and let p_e be a decoder's error probability. Derive $\log M \leq I(V;X) + h(p_e) + p_e \log(M-1)$, where h is the binary entropy defined in that proof. For $M = 2$ and $I(V;X) \leq (\log 2)/2$, obtain a strictly positive lower bound on p_e (an inverse binary-entropy expression suffices). Compare it with the bound obtained by replacing $h(p_e) + p_e \log(M-1)$ by $\log 2 + p_e \log M$.

3.4. Let $d \geq 8$ be an integer and $\sigma, a > 0$. Observe $Y \sim P_j = N_d(ae_j, \sigma^2 I_d)$, $j \in [d]$, where e_j is the jth standard basis vector, and estimate the mean under squared ℓ_2 loss. For a uniform index J, calculate the average KL divergences to $Q_0 = N_d(0, \sigma^2 I_d)$ and $Q_c = N_d((a/d)\mathbf{1}, \sigma^2 I_d)$. Use Proposition 3.9 to compare the resulting bounds on $I(J;Y)$. Compute the packing separation and use Theorem 3.10 to show that the minimax risk over these d mean vectors is at least $a^2/4$ when $a^2 = (\sigma^2/4)\log d$. Why are reference laws outside the finite parameter family permitted?

3.5. In the sparse-support packing of Example 3.6, let $1 \leq s \leq d/8$, $a, \sigma > 0$, $Y \sim N_d(\theta, \sigma^2 I_d)$, and $\theta^{(v)} = a\sigma v$ for binary support vectors v of weight s. Compute the squared ℓ_2 target separation and Gaussian KL divergence for two support vectors separately. Which parameters determine the support-code size, and how does the standardized amplitude a control the information cost for fixed supports?

3.6. For integers $d, n \geq 1$, let $\Theta = [0,1]^d$ and $P_\theta = N_d(\theta, n^{-1} I_d)$. For an integer $q \geq 1$, form the grid $G_q = \{0, 1/q, \ldots, 1\}^d$ and the reference mixture $R_q = |G_q|^{-1} \sum_{g \in G_q} P_g$. Calculate $|G_q|$ and show that G_q is an internal Euclidean cover of radius $\sqrt{d}/(2q)$. Following the covering argument in Example 3.13, prove $\sup_{\theta \in \Theta} \mathrm{KL}(P_\theta, R_q) \leq d \log(q+1) + nd/(8q^2)$. Evaluate this bound at $q = \lceil \sqrt{n} \rceil$ and compare it with $\sup_{\theta \in \Theta} \mathrm{KL}(P_\theta, P_0)$. Explain what additional packing and separation calculation would be needed to turn the covering bound into a risk lower bound.

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Part I Synthesis

The principal distinction among these tools is how they turn statistical indistinguishability into error. Table 3.1 compares their mechanisms and connects them to their later uses.

The local-information bounds of Section 1.4 complement these constructions. Hammersley–Chapman–Robbins uses expectation differences, which become target gaps under unbiasedness; Cramér–Rao takes a regular local limit. Van Trees averages risk over a prior and allows bias; Section 4.2 uses it for the sharp parametric quadratic-functional constant.

Tool	Essential mechanism	Principal later uses
Two-point testing	Separate targets while keeping two laws close; respect the prescribed type-I error level when needed	Linear functionals (§4.1); Gaussian testing (§5.2); intervals (§6.2)
Mixtures and moment matching	Conceal signs, supports, or target differences in averaged laws	Quadratic and nonsmooth targets (§§4.2–4.3); signal detection (§§5.4–5.6)
Constrained risk	Link small risk at a point or under a prior to error at separated alternatives	Adaptive linear and quadratic functionals (§§7.2–7.3)
Assouad	Accumulate binary errors along a hypercube under the stated loss	Global function estimation (§4.4); classification (§8.2); score estimation (§8.5.2)
Fano and reference laws	Compare packing size with information; convert decoding error to loss	Sparse regression (§8.1.1); classification and clustering (§§8.3–8.4)
Probabilistic-error bounds	Force a positive probability of error beyond a prescribed radius	Testing-to-size reduction (§6.1); confidence balls (§6.6)

Table 3.1 *General tools and their principal later applications. Each application supplies its own class, loss or coverage constraint, and calibration.*

Sharing alternatives does not mean sharing a reduction. A sign hypercube yields an estimation bound by accumulating coordinate errors, a testing bound by averaging its laws, or a confidence-radius bound by combining indistinguishability with coverage. Each use requires its own calibration.

Parts II and III develop these task-specific reductions further: coverage transfer for honest sets, cross-class comparisons for adaptation, and posterior arguments for selection. The general tools remain the foundation; the statistical requirement determines how they are used.

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Part II

Minimax Inference and Adaptation

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4

Minimax Estimation

Part I developed general methods for constructing hard alternatives and comparing their observation laws. Part II asks how these methods must be shaped by the statistical task: estimation, testing, honest uncertainty quantification, or adaptation. A signal may be detectable even when it cannot be estimated accurately. Coverage requirements and performance across parameter classes also call for arguments beyond separate minimax estimation bounds. The following four chapters develop this task-specific theory, beginning with estimation.

Minimax estimation asks how accurately an unknown quantity can be estimated when each procedure is judged by its worst-case performance over a specified parameter class. A lower bound shows that no estimator can achieve uniformly better accuracy over that class; a procedure that matches the bound is optimal under the chosen criterion. This chapter studies how the target, the measure of error, and the structure of the parameter class determine the alternatives needed for a sharp lower bound.

The Gaussian sequence and continuous white-noise models of Section 1.2 provide a common setting for observing noisy coefficients or noisy linear measurements of a function. Their common formulation lets us compare the difficulty of estimating different features of the same signal. Density estimation and fixed-design regression provide further applications, with their observation models specified where used.

The first three sections study scalar functionals under squared-error loss. Section 4.1 uses moduli to select hard pairs for linear functionals, following Donoho and Liu [59, 60], Donoho [55], and the extensions to nonconvex classes and adaptation by Cai and Low [31, 33]. Section 4.2 combines mixtures and local-information bounds to identify when quadratic-functional estimation reaches the parametric rate. Section 4.3 treats two nonsmooth targets: location mixtures hide a maximizing coordinate, whereas moment matching and polynomial approximation reveal the difficulty of estimating an average absolute mean.

Section 4.4 turns to global estimation under squared L_2 loss, using Assouad's hypercube for direct and inverse problems. Section 4.5 studies minimum values and minimizers through local two-point comparisons. The uncertainty principle of Cai, Chen, and Zhu [17] links value and location accuracy under convexity; a comparison with smoothness alone shows how shape constraints change the admissible perturbations.

Comparisons with affine and quadratic estimators distinguish intrinsic difficulty from limitations of a procedure class. Chapters 5 and 6 then adapt these constructions to testing and coverage requirements.

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4.1 Linear functionals: convex classes and unions

We study minimax squared-error estimation of a linear functional in the Gaussian models (1.1)–(1.2). Point values, regional averages, and contrasts often answer a scientific question without requiring recovery of the whole signal. Our goal is to determine how the parameter class controls their estimation accuracy, first for a convex class and then for a finite union of convex classes.

The noise level in this section is $\varepsilon = n^{-1/2}$, $n > 0$. Write P_f for the observation law, using f as a generic signal parameter and θ for its sequence representation when appropriate.

Let Θ be a specified nonempty class of signals and let T be a real-valued linear functional defined on a vector space containing Θ. The minimax squared-error benchmark is

$$R_n(\Theta, T) = \inf_{\widehat{T}} \sup_{f \in \Theta} \mathbb{E}_f \big(\widehat{T} - T(f)\big)^2, \tag{4.1}$$

where the infimum is over estimators based on the observations. Point evaluation is used on classes with specified point values, such as the continuous functions in a Hölder ball.

Throughout this chapter, constants in $\lesssim$, $\gtrsim$, and $\asymp$ may depend on fixed smoothness, radius, basis, or design parameters specified locally, but not on n, dimension, sparsity, or resolution. Moduli in theorem statements and real-valued modulus identities are assumed finite, with nonempty comparison sets.

Le Cam's two-point argument from Section 2.1 gives the lower bound; a modulus selects pairs with the largest target gap at the noise scale. For a union, we distinguish two questions: does uncertainty about the component increase minimax risk, and is a nonlinear estimator needed to attain it? The same moduli later enter interval-length and adaptation bounds under different statistical requirements.

4.1.1 Estimation over a convex class and the ordinary modulus

First suppose that the known class Θ is nonempty and convex. We seek a lower bound for $R_n(\Theta, T)$ and ask whether an affine estimator can attain its order. We begin by relating the distance between two signals to the distinguishability of their observations.

To calibrate a two-point comparison, note that in the sequence model,

$$\mathrm{KL}(P_\theta, P_{\theta'}) = \frac{n}{2} \|\theta - \theta'\|_2^2,$$

and in white noise, $\mathrm{KL}(P_f, P_g) = n\|f - g\|_2^2/2$. A distance at most $n^{-1/2}$ therefore keeps the divergence at most $1/2$. The ordinary modulus records the largest target separation at a prescribed distance.

Here the relevant distance is $d(f, g) = \|f - g\|_2$. More generally, the definition permits a semimetric on the domain of T: $d(f, g) = d(g, f) \geq 0$ and $d(f, f) = 0$, without requiring the triangle inequality or positivity for distinct signals. For $\varepsilon \geq 0$, define the *ordinary modulus*

$$\omega(\varepsilon, \Theta; T) = \sup\big\{|T(g) - T(f)| : f, g \in \Theta,\ d(f, g) \leq \varepsilon\big\}. \tag{4.2}$$

When T is fixed, write $\omega(\varepsilon, \Theta)$. The definition allows any nonempty class and real-valued target, not just a convex class and linear functional. Its statistical calibration depends on the observation model.

In the modulus approach of Donoho and Liu [59, 60] and Donoho [55], a pair approaching the modulus at the noise scale gives the following lower bound.

Theorem 4.1 (Modulus Lower Bound over a Convex Class) *In either Gaussian model* (1.1)–(1.2) *with* $\varepsilon = n^{-1/2}$, $n > 0$, *let* Θ *be nonempty and convex and let* T *be a linear functional on a vector space containing* Θ. *If* $\omega(n^{-1/2}, \Theta) < \infty$, *then*

$$R_n(\Theta, T) \geq \frac{1}{8}\,\omega^2\!\left(\frac{1}{\sqrt{n}}, \Theta\right).$$

Proof Fix $\nu \in (0,1)$. The definition of the modulus gives $f_0, f_1 \in \Theta$ such that

$$\|f_1 - f_0\|_2 \leq \frac{1}{\sqrt{n}}, \qquad |T(f_1) - T(f_0)| \geq (1-\nu)\,\omega\!\left(\frac{1}{\sqrt{n}}, \Theta\right).$$

Put $\Delta := |T(f_1) - T(f_0)|$. Then $\mathrm{KL}(P_{f_1}, P_{f_0}) = \frac{n}{2}\|f_1 - f_0\|_2^2 \leq \frac{1}{2}$, so Pinsker's inequality gives $\mathrm{TV}(P_{f_1}, P_{f_0}) \leq 1/2$ and hence $\eta(P_{f_1}, P_{f_0}) \geq 1/2$. Lemma 2.1 and Remark 2.2 therefore yield

$$\max\left\{\mathbb{E}_{f_0}(\widehat{T} - T(f_0))^2, \mathbb{E}_{f_1}(\widehat{T} - T(f_1))^2\right\} \geq \frac{1}{4}\Delta^2 \eta(P_{f_0}, P_{f_1}) \geq \frac{1}{8}\Delta^2.$$

Taking the infimum over $\widehat{T}$ and letting $\nu \downarrow 0$ proves the theorem. □

Convexity supports the affine-estimator comparison below, but is not needed for this lower bound. The next example calculates the point-evaluation modulus and shows that local averaging attains the risk order; the constant $1/8$ is not claimed to be sharp.

Example 4.2 (Ordinary Modulus for Point Evaluation on a Hölder Ball) Consider the white-noise model (1.2) with noise level $n^{-1/2}$. Fix $0 < \alpha \leq 1$ and $M > 0$. Let $\Theta = \Lambda^\alpha(M)$ be the convex Hölder ball on $[0,1]$ defined in (2.1), and let $T(f) = f(x_0)$ for a fixed interior point $x_0 \in (0,1)$. Then, as $\varepsilon \downarrow 0$,

$$\omega(\varepsilon, \Theta) \asymp \varepsilon^{\frac{2\alpha}{2\alpha+1}}.$$

We first construct a pair with a large pointwise gap, then bound that gap for every pair within L_2 distance ε. Put $\rho = \min\{x_0, 1 - x_0\} > 0$, so windows of half-width $0 < h \leq \rho$ lie in $[0,1]$.

Modulus lower bound: a localized bump. Let $\psi(u) = (1 - |u|)_+^\alpha$, $u \in \mathbb{R}$, and set

$$f_0(x) = 0, \qquad f_1(x) = Mh^\alpha \psi\!\left(\frac{x - x_0}{h}\right), \qquad 0 < h \leq \rho.$$

The map $u \mapsto (1 - |u|)_+$ is 1-Lipschitz, and $|a^\alpha - b^\alpha| \leq |a - b|^\alpha$ for $a, b \geq 0$. Thus $|f_1(x) - f_1(y)| \leq M|x - y|^\alpha$; also $\|f_1\|_\infty = Mh^\alpha \leq M$ because $h \leq \rho \leq 1/2$. Both functions therefore belong to Θ. Their target separation and L_2 distance are

$$T(f_1) - T(f_0) = Mh^\alpha, \qquad \|f_1 - f_0\|_2^2 = M^2 h^{2\alpha+1} \int_{-1}^{1} (1 - |u|)^{2\alpha}\, du = \frac{2M^2 h^{2\alpha+1}}{2\alpha + 1}.$$

Write $c_\alpha = \{2/(2\alpha+1)\}^{1/2}$. If $0 < \varepsilon \leq M c_\alpha \rho^{\alpha+1/2}$, choosing $h = (\varepsilon/(Mc_\alpha))^{2/(2\alpha+1)}$ ensures $h \leq \rho$ and makes the L_2 distance exactly ε. Consequently,

$$\omega(\varepsilon, \Theta) \geq Mh^\alpha = c_\alpha^{-2\alpha/(2\alpha+1)} M^{1/(2\alpha+1)} \varepsilon^{2\alpha/(2\alpha+1)}.$$

Modulus upper bound: local averaging. For any $f, g \in \Theta$ with $\|g - f\|_2 \le \varepsilon$, put $v = g - f$. The Hölder condition gives $|v(x_0) - v(t)| \le 2M|t - x_0|^\alpha$. For $0 < h \le \rho$, comparing $v(x_0)$ with its local average and applying Cauchy–Schwarz yields

$$\begin{aligned} |v(x_0)| &\le \frac{1}{2h}\int_{x_0-h}^{x_0+h} |v(x_0) - v(t)|\,dt + \left|\frac{1}{2h}\int_{x_0-h}^{x_0+h} v(t)\,dt\right| \\ &\le \frac{2M}{\alpha+1}h^\alpha + \frac{\varepsilon}{\sqrt{2h}}. \end{aligned}$$

The first term controls local variation; the second controls the average through the L_2 constraint. Choosing $h = (\varepsilon/M)^{2/(2\alpha+1)}$ balances their orders and is admissible when $0 < \varepsilon \le M\rho^{\alpha+1/2}$. Taking the supremum over admissible pairs gives

$$\omega(\varepsilon, \Theta) \le \left(\frac{2}{\alpha+1} + \frac{1}{\sqrt{2}}\right) M^{1/(2\alpha+1)}\varepsilon^{2\alpha/(2\alpha+1)}.$$

Together with the bump construction, this proves the modulus order for fixed α, M, and x_0.

The estimation benchmark. The bump construction alone suffices for the statistical lower bound. Substituting $\varepsilon = n^{-1/2}$ into Theorem 4.1 gives, as $n \to \infty$,

$$R_n(\Theta, T) = \inf_{\widehat{T}} \sup_{f\in\Theta} \mathbb{E}_f\big(\widehat{T} - f(x_0)\big)^2 \gtrsim n^{-\frac{2\alpha}{2\alpha+1}}.$$

The rate is attained in this white-noise model by the local average $\widehat{T}_h = (2h)^{-1}\int_{x_0-h}^{x_0+h} dY(t)$, with $0 < h \le \rho$. Its absolute bias is at most $Mh^\alpha/(\alpha+1)$ and its variance is $1/(2nh)$; choosing $h \asymp n^{-1/(2\alpha+1)}$ yields the same squared-error risk order. The Gaussian divergence calculation following (2.3) gives the corresponding pointwise lower bound in midpoint regression. Section 6.2 returns to the confidence-interval benchmark.

□

Remark 4.3 (Procedure and Upper-Bound Comparison) Affine estimators are nearly minimax over convex classes in these Gaussian models. Let $\mathcal{A}$ denote the affine estimators $a + Y(u)$, where $a \in \mathbb{R}$ and $u \in H$ are deterministic, H is the signal Hilbert space (ℓ_2 or $L_2[0,1]$), and $Y(u)$ is the measurement in (1.3). For a nonempty convex class Θ and a linear functional T satisfying $\omega(\varepsilon, \Theta) \to 0$ as $\varepsilon \downarrow 0$, Donoho and Liu [60, Theorem 2] establish the classical near-optimality bound

$$R_n^{\mathrm{aff}}(\Theta, T) := \inf_{\widehat{T}\in\mathcal{A}} \sup_{f\in\Theta} \mathbb{E}_f(\widehat{T} - T(f))^2 \le 1.25\, R_n(\Theta, T).$$

Donoho [55, Theorem 2 and Corollary 1] develops the general Gaussian optimal-recovery formulation. The modulus determines both the affine risk and the estimator's construction. The additive term in an affine estimator matters for asymmetric classes; "linear estimation" in this literature commonly includes these affine rules.

4.1.2 Unions of convex classes and cross-class moduli

We now estimate the same linear functional over a known union of convex classes, without knowing which component contains the signal. The benchmark is worst-case squared-error

risk over the union. Attaining each component's minimax rate with one estimator is the separate adaptation question of Section 7.2. A pair drawn from different components can have a larger target gap than either component permits on its own. Ordered and between-class moduli describe these comparisons.

For two nonempty classes $\mathcal{F}_1, \mathcal{F}_2$, retain the target T and semimetric d. As with the ordinary modulus, the definitions allow any real-valued target. The *ordered modulus* is

$$\omega(\varepsilon, \mathcal{F}_1, \mathcal{F}_2; T) = \sup\bigl\{T(g) - T(f) : f \in \mathcal{F}_1,\ g \in \mathcal{F}_2,\ d(f,g) \le \varepsilon\bigr\}, \tag{4.3}$$

and the *symmetrized between-class modulus* is

$$\begin{aligned}\omega_+(\varepsilon, \mathcal{F}_1, \mathcal{F}_2; T) &= \max\{\omega(\varepsilon, \mathcal{F}_1, \mathcal{F}_2; T), \omega(\varepsilon, \mathcal{F}_2, \mathcal{F}_1; T)\} \\ &= \sup\bigl\{|T(g) - T(f)| : f \in \mathcal{F}_1,\ g \in \mathcal{F}_2,\ d(f,g) \le \varepsilon\bigr\}. \end{aligned}\tag{4.4}$$

The ordered modulus retains the direction from $\mathcal{F}_1$ to $\mathcal{F}_2$; exchanging the classes can change its value. This distinction is useful for one-sided bias control. The between-class modulus takes the larger directional value, giving the absolute target gap needed for a two-point lower bound. Cai and Low use ordered moduli to construct estimators over nonconvex unions [31] and between-class comparisons to quantify adaptation costs [33].

For a single class, the symmetric cross-class comparison reduces to the ordinary modulus:

$$\omega(\varepsilon, \mathcal{F}; T) = \omega_+(\varepsilon, \mathcal{F}, \mathcal{F}; T).$$

Table 4.1 summarizes the three moduli and their roles.

Quantity	Admissible comparison	Statistical role
Ordinary $\omega(\varepsilon, \mathcal{F})$	Both endpoints in one class	Its square at the noise scale gives a minimax squared-error lower bound, up to a constant.
Ordered $\omega(\varepsilon, \mathcal{F}_1, \mathcal{F}_2)$	First endpoint in $\mathcal{F}_1$, second in $\mathcal{F}_2$	Directional bias control and construction of procedures tuned to competing classes.
Between-class $\omega_+(\varepsilon, \mathcal{F}_1, \mathcal{F}_2)$	One endpoint in each class, direction removed	Lower bounds for unions, adaptation, and honest coverage over an ambient class.

Table 4.1 *Ordinary and cross-class moduli for scalar functional estimation. All constrain the distance between two parameters, but their endpoint classes and treatment of direction differ.*

Now let $k \ge 1$ be an integer and let $\Theta = \bigcup_{j=1}^{k} \Theta_j$, where each Θ_j is nonempty and convex. Assume $\Theta_i \cap \Theta_j \ne \varnothing$ for all i, j, so the cross-class comparison sets are nonempty at every radius. In the Gaussian models, take $d(f,g) = \|f - g\|_2$. The ordinary modulus (4.2) of the union decomposes as

$$\omega(\varepsilon, \Theta) = \max_{1 \le i, j \le k} \omega_+(\varepsilon, \Theta_i, \Theta_j).$$

Thus within-component and cross-component pairs enter one ordinary modulus. Calibrating at $\varepsilon = n^{-1/2}$ gives the union lower bound.

Theorem 4.4 (Lower Bound for Linear Functionals over a Nonconvex Union) *Let Θ be a nonempty finite union of pairwise-intersecting convex classes and T a linear functional on a vector space containing Θ. In either Gaussian model* (1.1)–(1.2) *with noise level $n^{-1/2}$, $n > 0$, if $\omega(n^{-1/2}, \Theta) < \infty$, then*

$$R_n(\Theta, T) \geq \frac{1}{8}\,\omega^2\!\left(\frac{1}{\sqrt{n}}, \Theta\right).$$

Proof Put $\omega_* = \omega(n^{-1/2}, \Theta)$. The zero-modulus case is immediate. Otherwise choose $f_0, f_1 \in \Theta$ with $\|f_1 - f_0\|_2 \leq n^{-1/2}$ and $|T(f_1) - T(f_0)| \geq (1 - \nu)\omega_*$, where $0 < \nu < 1$. The two-point calculation in the proof of Theorem 4.1 does not use convexity and gives

$$\max_{i=0,1} \mathbb{E}_{f_i}(\widehat{T} - T(f_i))^2 \geq \frac{1}{8}|T(f_1) - T(f_0)|^2 \geq \frac{1}{8}(1 - \nu)^2\omega_*^2.$$

Taking the infimum over estimators and letting $\nu \downarrow 0$ proves the result. □

The next example isolates the cost of an unknown component: the union risk is larger in order than either component's minimax risk.

Example 4.5 (Point Evaluation over Two One-Sided Hölder Classes) In the white-noise model (1.2), consider point evaluation at $x_0 = 1/2$ when the smoother side of the curve is unknown. Assume $0 < \alpha_2 \leq \beta_1 < \beta_2 \leq \alpha_1 \leq 1$ and $M_1, M_2, N_1, N_2 > 0$, as in Cai and Low's example [33, Example 2]. The classes below consist of real-valued members of $L_2[0, 1]$ whose left and right restrictions are continuous at x_0 and agree with $f(x_0)$. No uniform sup-norm bound is imposed. Let

$$\mathcal{F}^R(\alpha, M) = \Big\{f : \ |f(x) - f(y)| \leq M\,|x - y|^{\alpha}\,,\ x_0 \leq x \leq y \leq 1\Big\},$$

$$\mathcal{F}^L(\alpha, M) = \Big\{f : \ |f(x) - f(y)| \leq M\,|x - y|^{\alpha}\,,\ 0 \leq x \leq y \leq x_0\Big\},$$

for $0 < \alpha \leq 1$, and define

$$\Theta(\alpha_1, M_1, \alpha_2, M_2) = \mathcal{F}^L(\alpha_1, M_1) \cap \mathcal{F}^R(\alpha_2, M_2).$$

Take the target to be point evaluation, $T(f) = f(x_0)$, and consider

$$\Theta_1 = \Theta(\alpha_1, M_1, \alpha_2, M_2), \qquad \Theta_2 = \Theta(\beta_1, N_1, \beta_2, N_2).$$

These classes are centrally symmetric, so their two ordered moduli coincide with the between-class modulus, as verified below. With all smoothness indices and radii fixed, as $\varepsilon \downarrow 0$,

$$\omega(\varepsilon, \Theta_1) \asymp \varepsilon^{2\alpha_1/(2\alpha_1+1)}, \qquad \omega(\varepsilon, \Theta_2) \asymp \varepsilon^{2\beta_2/(2\beta_2+1)},$$

and

$$\omega(\varepsilon, \Theta_1, \Theta_2) \asymp \varepsilon^{2\beta_1/(2\beta_1+1)},$$

under the displayed smoothness ordering. In this regime,

$$\frac{2\beta_1}{2\beta_1 + 1} < \min\left\{\frac{2\alpha_1}{2\alpha_1 + 1}, \frac{2\beta_2}{2\beta_2 + 1}\right\},$$

so the between-class modulus decays strictly more slowly than both within-class moduli. A cross-class two-point argument gives the corresponding squared-error lower bound over $\Theta_1 \cup \Theta_2$. Both classes are at least β_1-Hölder on the left, so a left-sided local average attains the union rate. Consequently, in the white-noise model, as $n \to \infty$,

$$\inf_{\widehat{T}} \sup_{f \in \Theta_1 \cup \Theta_2} \mathbb{E}_f \big(\widehat{T} - f(x_0)\big)^2 \asymp n^{-2\beta_1/(2\beta_1+1)}.$$

Knowing the component reveals which side of x_0 offers greater smoothness. □

Verification of Example 4.5 Following Cai and Low [33], we use one-sided perturbations. A Hölder perturbation of smoothness γ and width h changes the point value by order h^γ and has L_2 norm of order $h^{\gamma+1/2}$. Equating the norm with ε gives

$$h^{\gamma+1/2} \asymp \varepsilon, \qquad \text{so that} \qquad |T(f_1) - T(f_0)| \asymp h^\gamma \asymp \varepsilon^{2\gamma/(2\gamma+1)}.$$

The two ordered moduli agree: central symmetry sends a feasible pair $(f, g) \in \Theta_1 \times \Theta_2$ to $(-g, -f) \in \Theta_2 \times \Theta_1$ with the same distance and target increment. We first calculate this cross-class modulus, then the within-class moduli, and finally the union risk. *Ordered modulus.* For the lower bound, take a smooth cutoff ψ supported on $[-2, 2]$, equal to one on $[-1, 1]$, with $0 \le \psi \le 1$. Put $H = \tau h^{\beta_1}$, where $\tau > 0$ is fixed sufficiently small, and define

$$f_h(x) = \begin{cases} 0, & x \le x_0, \\ H\{1 - \psi((x - x_0)/h)\}, & x > x_0, \end{cases} \qquad g_h(x) = \begin{cases} H\psi((x - x_0)/h), & x \le x_0, \\ H, & x > x_0. \end{cases}$$

Both functions are smooth across x_0 because ψ is constant near zero. Put $L = \max\{1, \|\psi'\|_\infty\}$. For $0 < \gamma \le 1$, the bound $|\psi(u) - \psi(v)| \le L|u - v|^\gamma$ follows by treating $|u - v| \le 1$ and $|u - v| > 1$ separately. The right-sided α_2-Hölder constant of f_h is at most $\tau L h^{\beta_1 - \alpha_2}$, and the left-sided β_1-Hölder constant of g_h is at most τL. Their other restrictions are constant. Choosing $\tau L \le \min\{M_2, N_1\}$ therefore gives $f_h \in \Theta_1$ and $g_h \in \Theta_2$ for $0 < h \le 1/4$. Write $c_\psi^2 = \int_{-2}^{2} \psi(u)^2 \, du > 0$. Their difference is $g_h - f_h = H\psi((\cdot - x_0)/h)$, so

$$\|g_h - f_h\|_2^2 = \tau^2 c_\psi^2 h^{2\beta_1+1}, \qquad g_h(x_0) - f_h(x_0) = \tau h^{\beta_1}.$$

For sufficiently small $\varepsilon > 0$, choosing $h = (\varepsilon/(\tau c_\psi))^{2/(2\beta_1+1)} \le 1/4$ makes the distance exactly ε and gives

$$\omega(\varepsilon, \Theta_1, \Theta_2) \gtrsim \varepsilon^{2\beta_1/(2\beta_1+1)}.$$

For the upper bound, let $f \in \Theta_1$, $g \in \Theta_2$, and $\|g - f\|_2 \le \varepsilon$. Since all displayed smoothness indices are at most one, averaging over $[x_0 - h, x_0]$ and applying Cauchy–Schwarz gives

$$|g(x_0) - f(x_0)| \le h^{-1/2}\varepsilon + \frac{M_1}{\alpha_1 + 1} h^{\alpha_1} + \frac{N_1}{\beta_1 + 1} h^{\beta_1} \le C\big(\varepsilon h^{-1/2} + h^{\beta_1}\big).$$

Taking $h \asymp \varepsilon^{2/(2\beta_1+1)}$ for small ε proves the matching ordered upper bound.

Within-class moduli. For two functions in the same class, averaging on its smoother side bounds their target separation by $\varepsilon h^{-1/2} + 2Ah^\gamma/(\gamma + 1)$, with $(\gamma, A) = (\alpha_1, M_1)$ for Θ_1 and $(\gamma, A) = (\beta_2, N_2)$ for Θ_2. Balancing the terms gives the stated upper orders. For the lower bounds, compare zero with $b_h(x) = \kappa h^\gamma \psi((x - x_0)/h)$. On a side with exponent $\delta \le \gamma$, its Hölder constant is at most $\kappa L h^{\gamma - \delta} \le \kappa L$. Thus a fixed sufficiently small $\kappa > 0$ makes the

bump admissible on both sides. Its target value is κh^{γ} and its norm is $\kappa c_{\psi} h^{\gamma+1/2}$; setting the latter equal to ε gives the matching lower order.

Because the map $x \mapsto 2x/(2x+1)$ is strictly increasing on $(0,\infty)$ and the parameter ordering gives $\beta_1 < \beta_2 \le \alpha_1$, we obtain

$$\frac{2\beta_1}{2\beta_1+1} < \min\left\{\frac{2\alpha_1}{2\alpha_1+1}, \frac{2\beta_2}{2\beta_2+1}\right\}.$$

Hence the cross-class modulus decays more slowly than either fixed-class modulus and is asymptotically larger.

Minimax risk on the union. For the cross-class pair,

$$\mathrm{KL}(P_{g_h}, P_{f_h}) = \frac{n}{2}\tau^2 c_{\psi}^2 h^{2\beta_1+1}.$$

Choose $h = (n\tau^2 c_{\psi}^2)^{-1/(2\beta_1+1)}$, which is at most $1/4$ for all sufficiently large n. The divergence is $1/2$, so the two-point squared-error risk bound is at least $\tau^2 h^{2\beta_1}/8$. This yields $cn^{-2\beta_1/(2\beta_1+1)}$ for every measurable, possibly randomized estimator.

For the matching upper bound, use the left-sided white-noise average $\widehat{T}_h = h^{-1}\int_{x_0-h}^{x_0} dY(x)$, $0 < h \le 1/2$. Its absolute bias is at most $M_1 h^{\alpha_1}/(\alpha_1+1)$ on Θ_1 and $N_1 h^{\beta_1}/(\beta_1+1)$ on Θ_2. Since $\alpha_1 \ge \beta_1$ and $h \le 1$, both are at most $C_0 h^{\beta_1}$, where $C_0 = \max\{M_1/(\alpha_1+1), N_1/(\beta_1+1)\}$. Its variance is exactly $1/(nh)$, and hence

$$\sup_{f\in\Theta_1\cup\Theta_2} \mathbb{E}_f(\widehat{T}_h - f(x_0))^2 \le C_0^2 h^{2\beta_1} + \frac{1}{nh}.$$

Taking $h \asymp n^{-1/(2\beta_1+1)}$ yields the asserted union minimax rate. This is a uniform guarantee over the union, not attainment of each component's faster minimax rate. □

Remark 4.6 (Optimal Procedure and Matching Upper Bound) For a fixed finite union of closed convex classes with pairwise nonempty intersections and finite union modulus, Cai and Low's nonlinear selector attains squared-error risk of order $\omega^2(n^{-1/2}, \Theta)$ [31, Theorem 4]. To see why nonlinearity can matter, let $\mathrm{conv}(\Theta)$ denote the convex hull of Θ, the set of its finite convex combinations. In the Gaussian models above, their affine-risk identity [31, Theorems 5–6] gives, for a linear functional,

$$R_n^{\mathrm{aff}}(\Theta, T) = R_n^{\mathrm{aff}}(\mathrm{conv}(\Theta), T).$$

The convex hull can be much larger and harder than the original class. In their point-evaluation example [31, Section 5.1], the minimax risk tends to zero, whereas every affine estimator has infinite worst-case risk. Nonconvexity alone does not force such a gap: Example 4.5 admits a rate-optimal linear average.

The same moduli serve different purposes later: Section 6.4 uses coverage to bound honest interval length, while Section 7.2 uses an accuracy requirement on one class to bound risk on another.

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4.2 Quadratic functionals

To measure signal strength without cancellation, we turn from linear functionals to the sum of squared coefficients. For an orthonormal expansion of a function, this is its squared L_2 norm, a natural target in density estimation and goodness-of-fit testing [9, 85]. We observe the Gaussian sequence (1.1) with noise level $n^{-1/2}$, $n > 0$, and estimate the quadratic functional

$$Q(\theta) = \sum_{i \geq 1} \theta_i^2. \tag{4.5}$$

Write P_θ for the observation law. Over a known smoothness class $\Theta \subseteq \ell_2$, specified below, the squared-error benchmark is

$$R^*(n, \Theta) = \inf_{\widehat{Q}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \big(\widehat{Q} - Q(\theta)\big)^2, \tag{4.6}$$

where the infimum ranges over all estimators based on the sequence, allowed to use the known class parameters.

Our goal is to identify when this risk reaches the parametric order n^{-1} and whether deterministic diagonal quadratic estimators attain it. Cai and Low [32] show that a quadratic target can require a nonquadratic procedure even when the parameter class is known. We separate this restriction on estimators from the intrinsic difficulty of estimating Q.

Many individually weak coefficients can contribute substantial energy, so a single pair may miss the difficulty. A mixture can keep the observations close to pure noise while every alternative has the same positive value of Q. Smoothness limits the energy in high-index coefficients; the class constraints determine whether random signs or random supports give the stronger obstruction. We consider two families, with parameters fixed as $n \to \infty$.

For $0 < p < \infty$, $\alpha > 0$, and $M > 0$, define the weighted sequence class

$$L_p(\alpha, M) = \Big\{\theta : \Big(\sum_{i=1}^{\infty} i^{ps} |\theta_i|^p\Big)^{1/p} \leq M\Big\}, \qquad s = \alpha + \frac{1}{2} - \frac{1}{p} > 0, \tag{4.7}$$

and use the Besov ball $B^{\alpha}_{p,q}(M)$ from (3.8), here with finite $p, q > 0$ and the same restriction $s > 0$. Retain the indexing $1 \leq k \leq 2^j$, $i = 2^j + k - 1$ from (3.7), and write $\theta_j = (\theta_{j,k})_{k=1}^{2^j}$. These restrictions ensure that $Q(\theta)$ is finite; convexity is not required, and $0 < p < 1$ is allowed. Here M is the weighted (quasi-)norm radius. In particular, $L_2(\alpha, M) = \Theta_\alpha(M^2)$ by (3.2).

The smoothness index α controls decay, while p affects the allowable concentration of coefficients; in the Besov class, q governs aggregation across resolution levels. For $p \geq 2$, dense blocks of weak coefficients determine the nonparametric lower bound. For $p < 2$, uncertainty about a sparse support can be more demanding.

For the integrated squared derivative in their periodic-function model, Donoho and Nussbaum [61] construct a quadratic estimator that is asymptotically minimax among quadratic estimators and rate-optimal among all measurable estimators. Fan [64] develops further quadratic-functional benchmarks; Cai and Low [32] establish minimax rates and nonquadratic procedures needed in sparse regimes. To express the dense and sparse benchmarks together, write

$$p_* = \min\{p, 2\}, \qquad s_* = \alpha + \frac{1}{2} - \frac{1}{p_*}. \tag{4.8}$$

Two lower-bound arguments are needed. Near zero, the vertical and horizontal mixtures of Section 2.2 randomize signs and supports, respectively. Balancing their common value of Q against a likelihood-ratio second moment gives the nonparametric rates. Around a fixed nonzero signal, the van Trees inequality of Section 1.4 gives the parametric lower bound by varying the overall amplitude. The theorem states the risk order at the boundary between regimes, but asserts a sharp constant only strictly inside the parametric regime.

Theorem 4.7 (Minimax Benchmark for Quadratic-Functional Estimation) *In model* (1.1) *with* $\varepsilon = n^{-1/2}$, *let*

$$\Theta \in \left\{L_p(\alpha, M),\ B^{\alpha}_{p,q}(M)\right\}, \qquad p, q, \alpha, M > 0, \qquad s > 0,$$

with fixed class parameters and p_*, s_* *as in* (4.8).

The minimax squared-error risk for $Q(\theta)$ *satisfies, as* $n \to \infty$,

$$R^*(n, \Theta) \asymp n^{-r(\alpha,p)}, \qquad r(\alpha, p) = \begin{cases} 1, & \alpha p_* \geq \frac{1}{2}, \\ 2 - \dfrac{p_*}{1 + 2p_* s_*}, & \alpha p_* < \frac{1}{2}, \end{cases}.$$

Moreover, in the strictly parametric regime $\alpha p_* > 1/2$,

$$R^*(n, \Theta) = 4A(\Theta)n^{-1}(1 + o(1)), \qquad A(\Theta) = \sup_{\theta \in \Theta} \sum_{i \geq 1} \theta_i^2.$$

Under the stated assumptions, $0 < A(\Theta) < \infty$.

Proof of the lower bounds We first calibrate dense and sparse mixtures against the origin, then vary the amplitude of a nonzero signal. All class parameters are fixed.

Dense case: vertical mixing. Suppose $p \geq 2$ and $\alpha \leq 1/4$. Choose a block of $m \asymp n^{2/(1+4\alpha)}$ coordinates at scale m. Give each coefficient an independent Rademacher sign, equally likely to be -1 or 1, and use the common squared amplitude $a^2 = c/(n\sqrt{m})$. For the L_p ball,

$$\sum_{i \asymp m} i^{ps} |a|^p \leq C m^{ps+1} a^p = C c^{p/2} m^{p(\alpha+1/4)} n^{-p/2};$$

for a single Besov level, the corresponding Besov (quasi-)norm is bounded by $Cm^{\alpha+1/2}|a|$. Both quantities are bounded by M^p and M, respectively, after choosing c sufficiently small. Every point in the prior has target value ma^2. If L is the mixture likelihood ratio against the origin, coordinatewise Gaussian integration gives

$$\mathbb{E}_0 L^2 = \{\cosh(na^2)\}^m \leq \exp\left(\frac{1}{2} m n^2 a^4\right) = \exp(c^2/2).$$

Choosing c small enough keeps the testing affinity with the null bounded away from zero. The target gap is ma^2, so Proposition 2.7 yields

$$R^*(n, \Theta) \geq c_0 (ma^2)^2 \asymp \frac{m}{n^2} \asymp n^{-8\alpha/(1+4\alpha)}.$$

This includes the rate n^{-1} at $\alpha = 1/4$.

Sparse case: horizontal mixing. Assume $0 < p < 2$ and $\alpha p \leq 1/2$, with $s > 0$. Choose k nonzero coordinates uniformly among m possibilities and give each value $n^{-1/2}$. Every

alternative has $Q(\theta) = k/n$; the mixture second moment depends only on the overlap of two supports. Taking $k \asymp \sqrt{m}$ keeps that moment bounded, while class membership determines m. Let

$$m \asymp n^{p/(1+2ps)}, \qquad k = \lfloor \sqrt{m} \rfloor, \qquad a = n^{-1/2}.$$

For the L_p ball, let the support be a uniformly chosen k-subset of $\{1, \dots, m\}$. Then

$$\sum_i i^{ps} |\theta_i|^p \le k m^{ps} n^{-p/2} \le M^p,$$

after reducing the fixed multiplicative constant in m. For a Besov ball, choose m to be a power of two of the stated order and place the support on the corresponding level of m coefficients. Its Besov (quasi-)norm is

$$m^s k^{1/p} n^{-1/2} \le M,$$

under the same choice. Thus every point in either prior belongs to Θ and has the common target value $\Delta_n = ka^2 = k/n$. Write π for this prior, P_π for its observation mixture, and $L = dP_\pi/dP_0$, where P_0 is the law at the origin. If S, S' are two independent supports, Gaussian integration gives

$$\mathbb{E}_0 L^2 = \mathbb{E} \exp\{|S \cap S'|\}.$$

The supports are independent uniform k-subsets of m coordinates. Lemma A.11, with $t = 1$, therefore gives

$$\mathbb{E}_0 L^2 \le \left\{1 + (e-1)\frac{k}{m}\right\}^k \le \exp\{(e-1)k^2/m\} \le e^{e-1}.$$

This second-moment bound is not close to one, so the usual chi-square bound on total variation is insufficient. Log-convexity of likelihood-ratio moments instead gives a fixed positive Hellinger affinity:

$$1 = \mathbb{E}_0 L \le (\mathbb{E}_0 \sqrt{L})^{2/3} (\mathbb{E}_0 L^2)^{1/3}, \qquad \mathbb{E}_0 \sqrt{L} \ge (\mathbb{E}_0 L^2)^{-1/2}.$$

The Hellinger–total-variation comparison then gives

$$\eta(P_0, P_\pi) \ge 1 - \sqrt{1 - (\mathbb{E}_0 \sqrt{L})^2} \ge \tfrac{1}{2} (\mathbb{E}_0 \sqrt{L})^2 \ge \frac{1}{2\mathbb{E}_0 L^2} > 0.$$

Proposition 2.7 now yields

$$R^*(n, \Theta) \ge c\Delta_n^2 \asymp \frac{m}{n^2} \asymp n^{-\{2-p/(1+2ps)\}}.$$

This proves the sparse nonparametric lower bound, including its boundary.

The sharp parametric constant. Suppose $\alpha p_* > 1/2$. First, $0 < A(\Theta) < \infty$; the argument below uses only the standing class assumptions, so this assertion also holds outside the strictly parametric regime. Positivity follows from $M > 0$. For an L_p ball with $p \le 2$, removing the weights gives $\|\theta\|_2 \le \|\theta\|_p \le M$. For $p > 2$, Hölder's inequality gives

$$Q(\theta) \le M^2 \left(\sum_{i \ge 1} i^{-2sp/(p-2)} \right)^{1-2/p} < \infty,$$

because $2sp/(p-2) > 1$. For a Besov ball, $\|\theta_j\|_2 \le M2^{-js_*}$: use $\|\theta_j\|_2 \le \|\theta_j\|_p$ for $p < 2$ and $\|\theta_j\|_2 \le 2^{j(1/2-1/p)}\|\theta_j\|_p$ for $p \ge 2$. Summing $M^2 2^{-2js_*}$ proves finiteness since $s_* > 0$.

Fix $0 < \gamma < 1$ and choose $\vartheta \in \Theta$ with $Q(\vartheta) \ge (1-\gamma)A(\Theta)$. Put $\theta_0 = (1-\gamma)\vartheta$. Since $(1+\gamma)(1-\gamma) < 1$, homogeneity of both balls gives $\{t\theta_0 : 1-\gamma \le t \le 1+\gamma\} \subset \Theta$. Choose the symmetric density $\pi(t) = \gamma^{-1}\cos^2\{\pi(t-1)/(2\gamma)\}$ on this interval. It is continuously differentiable, vanishes at both endpoints, and has prior information

$$I_\pi := \int_{1-\gamma}^{1+\gamma} \frac{\{\pi'(t)\}^2}{\pi(t)}\,dt = \pi^2/\gamma^2,$$

the prior-information term in Theorem 1.12. Let $\theta_0^{(m)}$ be finite truncations of θ_0, retaining whole resolution levels for a Besov ball. Truncation preserves class membership and $Q(\theta_0^{(m)}) \to Q(\theta_0)$. The remaining observation coordinates are independent pure noise and add no information about t. In the scalar submodel $\theta = t\theta_0^{(m)}$, the Fisher information is $nQ(\theta_0^{(m)})$ and the target is $g_m(t) = t^2 Q(\theta_0^{(m)})$. Symmetry gives $\int g_m'(t)\pi(t)\,dt = 2Q(\theta_0^{(m)})$. Apply the van Trees inequality (Theorem 1.12) to each truncation, then let $m \to \infty$ at fixed n. With $g(t) = t^2 Q(\theta_0)$, this yields

$$R^*(n,\Theta) \ge \frac{\{\int g'(t)\pi(t)\,dt\}^2}{nQ(\theta_0) + I_\pi} = \frac{4Q(\theta_0)^2}{nQ(\theta_0) + I_\pi}.$$

Therefore

$$\liminf_{n\to\infty} nR^*(n,\Theta) \ge 4Q(\theta_0) \ge 4(1-\gamma)^3 A(\Theta).$$

Letting $\gamma \downarrow 0$ proves the lower constant $4A(\Theta)$. □

The transition at $\alpha p_* = 1/2$ is the *elbow phenomenon*: below it, weak coefficients force a slower rate; at and above it, the risk has order n^{-1}. The boundary conclusion concerns the rate alone, not the sharp constant. For $p \ge 2$, the elbow is $\alpha = 1/4$; for $p < 2$, it is $\alpha = 1/(2p)$ when that boundary lies in the admissible range $s > 0$.

For $p = 2$, $s_* = \alpha$ and the squared-error risk exponent below the elbow is $8\alpha/(1+4\alpha)$. Thus the risk has order $n^{-2/3}$ at $\alpha = 1/8$ and reaches n^{-1} at $\alpha = 1/4$. Section 5.6 reuses sign mixtures for testing, where the goal is to detect nonzero Q, rather than estimate its value.

Remark 4.8 (Optimal Procedure and Matching Upper Bound) A truncated sum of $Y_i^2 - n^{-1}$ attains the minimax rate in the dense regime; see Donoho and Nussbaum [61]. For sparse classes, Cai and Low [32, Sections 3.1–3.2] combine an unbiased low-frequency sum with centered coordinate thresholding. Their estimator attains the sparse-class minimax rate even where quadratic estimators are suboptimal. Strictly above the elbow, its leading risk is $4A(\Theta)/n$. For sparse Besov balls with $q > 2$, contributions from different levels can make $A(\Theta) > M^2$; Appendix B.1.2 verifies the constant.

Remark 4.9 (When Quadratic Estimators Are Suboptimal) Squaring observed coefficients and summing with deterministic weights is natural for a quadratic target. The weights and additive correction may use known class parameters, but not evidence about which coordinates carry signal. With the experiment, class, and loss unchanged, the risk gap measures the cost of this procedure restriction, not of unknown smoothness.

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More precisely, let $\mathcal{Q}$ consist of the deterministic diagonal rules

$$\widehat{Q}(Y) = \sum_{i \geq 1} a_i Y_i^2 + b, \tag{4.9}$$

where $a_i, b \in \mathbb{R}$ are deterministic and the series converges in $L_2(P_\theta)$ for every $\theta \in \Theta$. Their restricted minimax risk is

$$R_{\mathcal{Q}}^*(n, \Theta) = \inf_{\widehat{Q} \in \mathcal{Q}} \sup_{\theta \in \Theta} \mathbb{E}_\theta \big(\widehat{Q} - Q(\theta)\big)^2. \tag{4.10}$$

For the same fixed classes as in Theorem 4.7, if $0 < p < 2$ and $s = \alpha + 1/2 - 1/p > 0$, then

$$R_{\mathcal{Q}}^*(n, \Theta) \asymp n^{-r_{\mathcal{Q}}(\alpha, p)}, \qquad r_{\mathcal{Q}}(\alpha, p) = \min\left\{1, \frac{8s}{1 + 4s}\right\}.$$

Within this admissible range, $1/(2p) < \alpha < 1/p - 1/4$ permits parametric-rate estimation, although every such quadratic rule is slower. Cai and Low [32] explain this gap and construct nonquadratic estimators that exploit coefficient concentration; Figure 4.1 illustrates their comparison at $p = 1.25$. Appendix B.1.2 gives the quadratic-convex-hull proof.

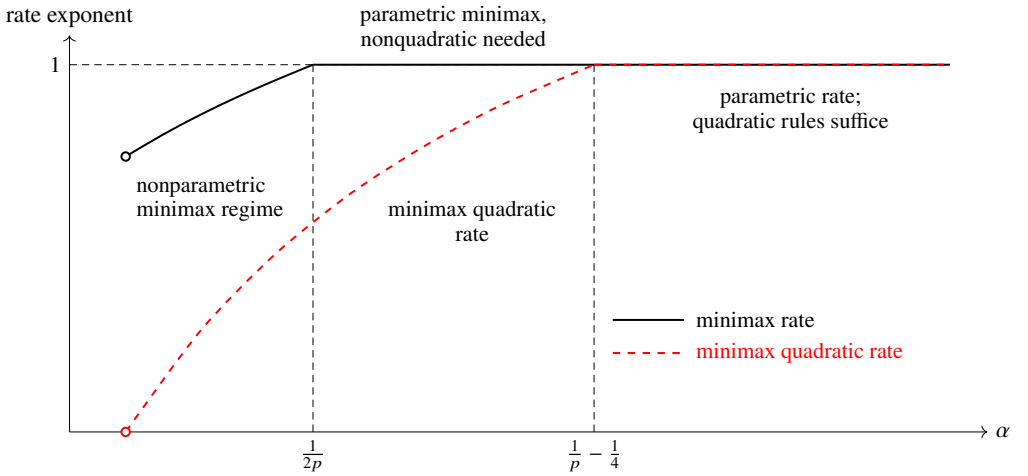


Figure 4.1 Unrestricted and quadratic-estimator rates, illustrating the comparison of Cai and Low [32]. At $p = 1.25$, the solid curve is the minimax squared-error risk exponent $r(\alpha, p)$ and the red dashed curve is its quadratic-estimator counterpart $r_{\mathcal{Q}}(\alpha, p)$. The first reaches the parametric exponent 1 at $\alpha = 1/(2p) = 0.40$; the second reaches it at $\alpha = 1/p - 1/4 = 0.55$. Between these thresholds, nonquadratic estimation is necessary to attain the minimax rate. Open circles exclude $\alpha = 0.30$, where the standing condition $s > 0$ fails.

Adaptation poses a separate question: can one estimator attain these benchmarks without knowing the class parameters? Section 7.3 recalibrates the mixtures under a risk constraint to study this cost [36, Theorems 1–2].

4.3 Nonsmooth functionals

We now study the maximum coordinate and average absolute coordinate of a Gaussian mean vector. The first measures the strongest mean response; the second aggregates magnitudes without sign cancellation. Unlike linear and quadratic functionals, these targets are nonsmooth: the maximum is nondifferentiable at ties for the largest coordinate, and absolute value has a cusp at zero. Under squared-error loss, how does their difficulty change as dimension grows? Suppose we observe

$$Y = \theta + Z, \qquad Z \sim N_d(0, I_d), \qquad \theta \in \mathbb{R}^d, \tag{4.11}$$

where $d \geq 1$ is the dimension and each coordinate is observed once with unit noise.

These targets require different lower-bound constructions. For the maximum, we mix over the location of a single elevated mean and truncate the likelihood ratio. For the average absolute mean, Cai and Low [37] use moment-matching priors and polynomial approximation to quantify the difficulty of correcting bias near zero. The maximum's minimax risk grows logarithmically with dimension; the average's decreases, but only at logarithmic rates that depend on whether the means are bounded.

4.3.1 The maximum of a Gaussian mean vector

In model (4.11), the target is

$$T(\theta) = \theta_{\max} = \max_{1 \leq j \leq d} \theta_j.$$

Our goal is the minimax squared-error risk over $\mathbb{R}^d$ as d grows, with no restrictions on the magnitudes or gaps between the means. Write

$$Y_{\max} = \max_{1 \leq j \leq d} Y_j.$$

Compare zero with alternatives that raise one uniformly chosen coordinate. Every alternative has the same maximum, so the point-versus-mixture bound of Section 2.2 converts the target gap into a risk lower bound. To obtain the sharp constant, we restrict the likelihood calculation to an event with probability tending to one under both laws. A bias correction to $Y_{\max}$ attains this constant.

Theorem 4.10 (Maximum-Coordinate Minimax Risk) *In model* (4.11)*, let* $T(\theta)$ *be the maximum coordinate. There exists a universal constant* $c > 0$ *such that*

$$\inf_{\widehat{T}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 \geq c \log d.$$

The bias-corrected maximum $\widehat{T}^* = Y_{\max} - \sqrt{2 \log d}/2$ *is asymptotically minimax:*

$$\begin{aligned} \inf_{\widehat{T}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 &\sim \frac{1}{2} \log d, \\ \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{T}^* - T(\theta)\big)^2 &\sim \frac{1}{2} \log d, \end{aligned} \qquad d \to \infty.$$

Proof of the lower bound For sufficiently large d, set

$$a_d = \sqrt{2\log d}, \qquad \delta_d = (\log d)^{-1/4}, \qquad \tau_d = (1-\delta_d)a_d.$$

Compare $P_0 = N_d(0, I_d)$ with the mixture $P_\pi = d^{-1}\sum_{j=1}^d P_j$, where $P_j = N_d(\tau_d e_j, I_d)$. Since $T(0) = 0$ and $T(\tau_d e_j) = \tau_d$, Proposition 2.7 gives

$$\inf_{\widehat{T}} \sup_{\theta\in\mathbb{R}^d} \mathbb{E}_\theta(\widehat{T} - T(\theta))^2 \geq \frac{\tau_d^2}{4}\,\eta(P_0, P_\pi).$$

It remains to prove $\eta(P_0, P_\pi) \to 1$. The untruncated second moment is $1 + (e^{\tau_d^2} - 1)/d$, which diverges at this separation. We instead control the likelihood ratio on a high-probability event and bound the omitted mass. Define

$$L_d(y) = \frac{dP_\pi}{dP_0}(y) = \frac{1}{d}\sum_{j=1}^d \exp\left(\tau_d y_j - \frac{\tau_d^2}{2}\right),$$

$$A_d = \{y_{\max} \leq a_d\}, \qquad y_{\max} = \max_{1\leq j\leq d} y_j.$$

Under P_0 and P_j, respectively,

$$\begin{aligned}
\mathbb{P}_0(A_d^c) &\leq d\,\bar{\Phi}(a_d) \to 0,\\
\mathbb{P}_j(A_d^c) &\leq (d-1)\bar{\Phi}(a_d) + \bar{\Phi}(a_d - \tau_d)\\
&= (d-1)\bar{\Phi}(a_d) + \bar{\Phi}(\delta_d a_d) \to 0.
\end{aligned}$$

Thus $\mathbb{P}_\pi(A_d^c) \to 0$ and $\mathbb{E}_0[L_d \mathbf{1}_{A_d}] = \mathbb{P}_\pi(A_d) = 1 - o(1)$.

In the truncated second moment, each off-diagonal product has expectation at most one: dropping the indicator leaves two independent likelihood ratios, each with expectation one. For each diagonal term, $A_d \subseteq \{Y_i \leq a_d\}$ and completing the Gaussian square gives

$$\mathbb{E}_0[L_d^2 \mathbf{1}_{A_d}] \leq 1 - \frac{1}{d} + \frac{1}{d}e^{\tau_d^2}\Phi(a_d - 2\tau_d).$$

Since $2\tau_d - a_d = (1-2\delta_d)a_d \to \infty$, Mills' ratio and $a_d^2 = 2\log d$ imply, for all sufficiently large d,

$$\begin{aligned}
\frac{1}{d}e^{\tau_d^2}\Phi(a_d - 2\tau_d) &\leq \frac{C}{a_d}\exp\left\{-\log d + \tau_d^2 - \frac{1}{2}(1-2\delta_d)^2 a_d^2\right\}\\
&= \frac{C}{a_d}e^{-2\delta_d^2\log d} = o(1).
\end{aligned}$$

Together with $\mathbb{E}_0[L_d \mathbf{1}_{A_d}] = 1 - o(1)$, this yields

$$0 \leq \mathbb{E}_0[(L_d\mathbf{1}_{A_d} - 1)^2] = \mathbb{E}_0[L_d^2\mathbf{1}_{A_d}] - 2\mathbb{E}_0[L_d\mathbf{1}_{A_d}] + 1 = o(1).$$

Cauchy–Schwarz and $\mathbb{E}_0[L_d\mathbf{1}_{A_d^c}] = \mathbb{P}_\pi(A_d^c) = o(1)$ now give

$$\mathbb{E}_0|L_d - 1| \leq \left\{\mathbb{E}_0[(L_d\mathbf{1}_{A_d} - 1)^2]\right\}^{1/2} + \mathbb{P}_\pi(A_d^c) = o(1).$$

Hence $\mathrm{TV}(P_0, P_\pi) \to 0$ and $\eta(P_0, P_\pi) \to 1$. Returning to the point-versus-mixture bound gives

$$\inf_{\widehat{T}} \sup_{\theta\in\mathbb{R}^d} \mathbb{E}_\theta(\widehat{T} - T(\theta))^2 \geq \frac{\tau_d^2}{4}(1 - o(1)) = \left(\frac{1}{2} + o(1)\right)\log d.$$

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This gives the universal-order lower bound for $d \geq d_0$ sufficiently large. For each remaining integer $2 \leq d < d_0$, compare $\theta = 0$ with $\theta = e_1$: their target values differ by one, their Gaussian KL divergence is $1/2$, and Pinsker's inequality gives total variation at most $1/2$. The squared-error two-point inequality therefore gives risk at least $1/8$. Choosing the universal c below both the large-d constant and $1/(8 \log d_0)$ covers these finitely many dimensions; at $d = 1$ the bound $c \log d = 0$ is automatic. □

Remark 4.11 (Optimal Procedure and Matching Upper Bound) The bias-corrected maximum $Y_{\max} - \sqrt{2 \log d}/2$ attains the leading minimax risk $\frac{1}{2} \log d$. The uncorrected maximum has the same order but a larger leading constant. The correction reduces the largest absolute bias. For $Y \sim N_d(\theta, I_d)$ and a Lipschitz function $g : \mathbb{R}^d \to \mathbb{R}$, the Gaussian Poincaré inequality states that $\mathrm{Var}_\theta\{g(Y)\} \leq \mathbb{E}_\theta \|\nabla g(Y)\|_2^2$. The maximum function has gradient of Euclidean norm one almost everywhere, so $\mathrm{Var}_\theta(Y_{\max}) \leq 1$. Appendix B.1.1 gives the complete upper-bound argument.

Although the target is scalar, uncertainty about the maximizing coordinate drives the logarithmic risk. Chapter 9 reuses this construction for detection.

Remark 4.12 (Estimating the Largest Absolute Mean) To measure the strongest effect regardless of sign, consider $S(\theta) = \max_j |\theta_j|$ in the same unit-noise Gaussian model, with squared-error loss over $\mathbb{R}^d$. Its minimax risk has the same leading constant:

$$\inf_{\widehat{S}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{S} - S(\theta)\big)^2 = \left(\frac{1}{2} + o(1)\right) \log d, \qquad d \to \infty.$$

For the lower bound, T and S agree at each parameter used in the proof of Theorem 4.10: the origin and the positive one-coordinate alternatives. Thus the same mixture argument applies.

For the upper bound, put $A(Y) = \max_j |Y_j|$ and $c_d = \sqrt{2 \log(2d)}$. The map A is 1-Lipschitz in Euclidean norm, so the Gaussian Poincaré inequality gives the variance bound below. Jensen's inequality and the triangle inequality give the bias bounds:

$$\mathrm{Var}_\theta\, A(Y) \leq 1, \qquad 0 \leq \mathbb{E}_\theta A(Y) - S(\theta) \leq \mathbb{E} \max_j |Z_j| \leq c_d.$$

The last inequality follows from $\mathbb{E} e^{t \max_j |Z_j|} \leq 2d\, e^{t^2/2}$ for $t > 0$, taking logarithms and minimizing $\log(2d)/t + t/2$. Thus $\widehat{S} = A(Y) - c_d/2$ satisfies

$$\sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{S} - S(\theta)\big)^2 \leq 1 + \frac{c_d^2}{4} = 1 + \frac{1}{2} \log(2d),$$

which matches the lower bound asymptotically. Truncating $\widehat{S}$ below at zero cannot increase squared error because $S(\theta) \geq 0$.

4.3.2 The average absolute Gaussian mean

We now average coordinate magnitudes rather than select an extreme. In model (4.11), the target is

$$T(\theta) = \frac{1}{d} \sum_{i=1}^{d} |\theta_i|.$$

This normalized ℓ_1 norm measures aggregate magnitude without cancellation. We compare minimax squared-error risk over $\mathbb{R}^d$ and the bounded cube

$$\Theta_d(M) = \{\theta \in \mathbb{R}^d : |\theta_i| \le M \text{ for } 1 \le i \le d\}, \qquad M > 0. \tag{4.12}$$

The empirical average of $|Y_i|$ reduces variability but retains bias near zero: at the zero vector its expectation is $\mathbb{E}|Z| > 0$ for $Z \sim N(0,1)$, regardless of d. Cai and Low [37] use polynomial approximation to quantify the difficulty of correcting this bias uniformly. Their two-prior lower bound, as in Section 2.4, separates the prior expectations of the target while keeping the observation mixtures close.

Polynomial approximation and moment matching. Moment matching cancels low-order terms in the Gaussian mixture expansion, making the observation laws close. It also makes the priors integrate every polynomial of the matched degree equally. Their absolute-moment gap is therefore at most twice the best uniform approximation error for $|x|$ on their common support interval. The next lemma attains this gap on $[-1,1]$. Increasing the degree improves indistinguishability but reduces target separation; the lower bound balances these effects.

For an integer $k \ge 1$, let $\mathcal{P}_k$ be the set of real polynomials of degree at most k, and define the uniform approximation error

$$\delta_k = \inf_{G \in \mathcal{P}_k} \max_{x \in [-1,1]} \big||x| - G(x)\big|.$$

Because $|x|$ is even, a best polynomial approximation may also be chosen even. The Bernstein constant is

$$\beta_* = \lim_{k \to \infty} 2k\, \delta_{2k},$$

and classical approximation theory shows that the limit exists and is positive. In particular,

$$\delta_{2k} = \frac{\beta_*}{2k}(1 + o(1)) \qquad \text{as } k \to \infty.$$

Bernstein [6, Sections 30 and 36] proves this positive-limit result. Varga and Carpenter [106, equation (1.10)] record the same normalization; their E_{2k} is our uniform approximation error δ_{2k} on $[-1,1]$.

For even $k \ge 2$, the Chebyshev alternation theorem gives at least $k+2$ ordered points where the error of a best degree-k approximation alternates between δ_k and $-\delta_k$. We weight these points so that polynomial moments cancel while the approximation errors add, producing the two priors below.

Lemma 4.13 (Moment-Matching Priors from Best Approximation) *For every even integer $k \ge 2$, there exist two probability measures ν_0 and ν_1 on $[-1,1]$, symmetric around 0, such that:*

1. $\int t^\ell\, \nu_1(dt) = \int t^\ell\, \nu_0(dt)$ *for every* $\ell = 0, 1, \dots, k$;
2. $\int |t|\, \nu_1(dt) - \int |t|\, \nu_0(dt) = 2\delta_k$.

Proof Let $f(t) = |t|$ and choose a best degree-k polynomial p. Such a polynomial exists because the polynomial space is finite dimensional and hence closed in $C([-1,1])$; a bounded minimizing sequence has a convergent subsequence in that space. Moreover, $\delta_k > 0$, since $|t|$ is not a polynomial on this interval. The alternation property stated earlier supplies points

$$-1 \le x_0 < x_1 < \cdots < x_{k+1} \le 1$$

at which the residuals $e_i := f(x_i) - p(x_i)$ have magnitude δ_k and alternate in sign.

Define the nonzero barycentric weights

$$a_i := \left\{ \prod_{j \neq i} (x_i - x_j) \right\}^{-1}, \qquad i = 0, \dots, k+1.$$

For every polynomial q of degree at most k, Lagrange interpolation on these $k+2$ nodes and comparison of the t^{k+1} coefficients give

$$q(t) = \sum_{i=0}^{k+1} q(x_i) a_i \prod_{j \neq i} (t - x_j), \qquad \sum_{i=0}^{k+1} a_i q(x_i) = 0.$$

Since a_i has sign $(-1)^{k+1-i}$, both the weights and the residuals e_i alternate in sign. Choose $\sigma \in \{-1, 1\}$ so that $\sigma a_i e_i = \delta_k |a_i|$ for every i, and put

$$S := \sum_{i=0}^{k+1} |a_i| > 0, \qquad w_i := \frac{2\sigma a_i}{S}.$$

The constant polynomial gives $\sum_i w_i = 0$, while $\sum_i |w_i| = 2$. Consequently the positive weights sum to one and the negative weights sum to minus one. Using the unit point masses δ_x, define

$$\widetilde{\nu}_1 := \sum_i (w_i)_+ \delta_{x_i}, \qquad \widetilde{\nu}_0 := \sum_i (-w_i)_+ \delta_{x_i}.$$

These are probability measures supported on $[-1, 1]$. Applying the polynomial identity to $q(t) = t^\ell$ proves moment matching for $0 \le \ell \le k$. Applying it to p gives the exact functional gap:

$$\begin{aligned} \int |t| \, \widetilde{\nu}_1(dt) - \int |t| \, \widetilde{\nu}_0(dt) &= \sum_i w_i f(x_i) = \sum_i w_i \{ f(x_i) - p(x_i) \} \\ &= \frac{2}{S} \sum_i \delta_k |a_i| = 2\delta_k. \end{aligned}$$

Finally let $R(t) = -t$ and set $\nu_j = (\widetilde{\nu}_j + R_\# \widetilde{\nu}_j)/2$, where $R_\# \widetilde{\nu}_j$ is the reflected measure. Each ν_j is a symmetric probability measure with the same support restriction. Reflection multiplies the ℓth moment by $(-1)^\ell$, so the moment equalities persist. Since $|R(t)| = |t|$, the functional gap remains $2\delta_k$. □

Minimax lower bounds. We now choose the support radius and number of matched moments. On the fixed cube, use radius M and degree of order $\log d / \log\log d$. On the full space, use radius of order $\sqrt{\log d}$ and degree of order $\log d$. These choices keep the product mixture laws

close while the target concentrates around separated prior expectations. Cai and Low [37] obtain the following lower bounds from this balance.

Theorem 4.14 (Average Absolute Normal Mean) *In model* (4.11)*, let $T(\theta)$ be the average absolute mean, with $\Theta_d(M)$ and β_* as defined above. Under squared-error loss, as $d \to \infty$:*

(i) For every fixed $M > 0$,

$$\inf_{\widehat{T}} \sup_{\theta \in \Theta_d(M)} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 \geq \beta_*^2 M^2 \left(\frac{\log\log d}{\log d}\right)^2 (1 + o(1)).$$

(ii) Over the full space $\mathbb{R}^d$,

$$\inf_{\widehat{T}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 \geq \frac{4\beta_*^2}{9e^2 \log d}(1 + o(1)).$$

Proof of part (i) We place two moment-matching measures independently on the d coordinates of the fixed cube. Polynomial approximation determines the gap between the prior expectations of the target; averaging makes fluctuations around those expectations small. Fix $M > 0$ and an even integer $k_d \geq 2$ to be chosen below. Let μ_0, μ_1 be the symmetric measures from Lemma 4.13, with their dependence on k_d suppressed. They agree on moments through order k_d and satisfy

$$\int |u|\, \mu_1(du) - \int |u|\, \mu_0(du) = 2\delta_{k_d}.$$

Scale them to $[-M, M]$ by $\nu_j(A) = \mu_j(A/M)$, where $A/M = \{u : Mu \in A\}$ for a Borel set A, and let $\Pi_j = \nu_j^{\otimes d}$ be the corresponding product priors on $\theta = (\theta_1, \ldots, \theta_d)$.

The prior means and variances satisfy

$$m_j := \mathbb{E}_{\Pi_j} T(\theta) = \int |t|\, \nu_j(dt) = M \int |u|\, \mu_j(du), \qquad j = 0, 1,$$

$$m_1 - m_0 = 2M\delta_{k_d},$$

$$v_j^2 := \mathrm{Var}_{\Pi_j}(T(\theta)) \leq \frac{M^2}{d}, \qquad j = 0, 1,$$

since $|\theta_i| \leq M$ under both product priors. For $j = 0, 1$, write the one-coordinate mixture density as

$$g_j(y) = \int \varphi(y - t)\, \nu_j(dt) = \varphi(y) r_j(y),$$

$$r_j(y) = \int e^{yt - t^2/2}\, \nu_j(dt).$$

In chi-square expressions below, a density also denotes its probability law. Symmetry and support in $[-M, M]$ give, for every $y \in \mathbb{R}$,

$$r_0(y) = \int e^{-t^2/2} \cosh(yt)\, \nu_0(dt) \geq e^{-M^2/2},$$

and therefore

$$\chi^2(g_1, g_0) = \int \frac{(g_1 - g_0)^2}{g_0} \le e^{M^2/2} \int \frac{(g_1 - g_0)^2}{\varphi} = e^{M^2/2} \int (r_1 - r_0)^2 \varphi.$$

Moment cancellation. The measures ν_0, ν_1 are supported on $[-M, M]$ and have matching moments through degree k_d. Apply Lemma A.12 with $a = M, K = k_d$, and $\pi_j = \nu_j$. Together with the denominator bound above, it gives

$$\int (r_1 - r_0)^2 \varphi \le 4 \sum_{r=k_d+1}^{\infty} \frac{M^{2r}}{r!},$$

$$\chi^2(g_1, g_0) \le 4e^{M^2/2} \sum_{r=k_d+1}^{\infty} \frac{M^{2r}}{r!}.$$

For fixed M and $k_d + 1 > eM^2$, successive terms of the series have ratio at most $M^2/(k_d + 2) < 1/e$. A geometric bound followed by Stirling's inequality gives

$$\sum_{r=k_d+1}^{\infty} \frac{M^{2r}}{r!} \le C_M \Big(\frac{eM^2}{k_d + 1} \Big)^{k_d+1},$$

$$\chi^2(g_1, g_0) \le C'_M \Big(\frac{eM^2}{k_d + 1} \Big)^{k_d+1},$$

where C_M, C'_M depend only on M.

Degree choice and risk bound. We now choose the matching degree so that the one-coordinate discrepancy remains negligible after taking products. The d-fold product mixtures satisfy

$$\zeta_d^2 := \chi^2(g_1^{\otimes d}, g_0^{\otimes d}) = \big(1 + \chi^2(g_1, g_0)\big)^d - 1.$$

Choose k_d to be the smallest even integer satisfying

$$k_d \ge \frac{\log d}{\log\log d} + \frac{\log d}{(\log\log d)^{3/2}}.$$

The extra term ensures decay while retaining $k_d = (\log d/\log\log d)(1 + o(1))$. Indeed,

$$\log\left[d \left(\frac{eM^2}{k_d + 1} \right)^{k_d+1} \right] \le -c \frac{\log d}{\sqrt{\log\log d}},$$

for all sufficiently large d and a constant $c = c(M) > 0$. Thus $d\chi^2(g_1, g_0) \to 0$, and consequently $\zeta_d \to 0$.

Since $\delta_{k_d} \asymp k_d^{-1}$ and $k_d/\sqrt{d} \to 0$, the correction $v_0\zeta_d$ is negligible relative to $m_1 - m_0 = 2M\delta_{k_d}$. Theorem 2.14, with denominator $(\zeta_d + 2)^2 \to 4$, gives

$$\inf_{\widehat{T}} \sup_{\theta \in \Theta_d(M)} \mathbb{E}_\theta \big(\widehat{T} - T(\theta)\big)^2 \ge M^2 \delta_{k_d}^2 (1 + o(1)).$$

Finally, Bernstein's approximation formula yields

$$\delta_{k_d} = \frac{\beta_*}{k_d}(1 + o(1)) = \beta_* \frac{\log\log d}{\log d}(1 + o(1)).$$

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Substituting this into the previous display proves

$$\inf_{\widehat{T}} \sup_{\theta\in\Theta_d(M)} \mathbb{E}_\theta\big(\widehat{T} - T(\theta)\big)^2 \geq \beta_*^2 M^2 \left(\frac{\log\log d}{\log d}\right)^2 (1 + o(1)),$$

which is part (i). □

For part (ii), growing prior support increases target separation and produces the slower full-space rate. Appendix B.1.3 gives the mixture calibration.

Remark 4.15 (Optimal Procedure and Matching Upper Bound) On the bounded cube, Cai and Low [37, Theorem 4] approximate the absolute-value function by a polynomial and estimate its powers unbiasedly using the probabilists' Hermite polynomials. For any prescribed $c > M^2\beta_*^2$, choosing a fixed $0 < a < 1$ sufficiently close to one and taking the least even degree above $a \log d/\log\log d$ gives worst-case squared-error risk at most $c(\log\log d/\log d)^2$ for all sufficiently large d. The leading constant can therefore be made arbitrarily close to the lower-bound constant in part (i).

Over $\mathbb{R}^d$, their hybrid estimator uses Hermite polynomials for small means and absolute observations for large means, attaining squared-error risk of order $1/\log d$ [37, Theorem 6]. This matches part (ii) in rate. Appendix B.1.4 gives an alternative Fourier construction and its risk bound.

4.4 Global function estimation under smoothness

We now estimate an entire function f on $[0, 1]$, rather than a scalar functional. The observations are independent draws from a density, noisy regression values, or a known linear transformation of a signal in Gaussian white noise. The benchmark is the smallest worst-case expected squared L_2 error over a known smoothness class. It measures accuracy across the whole domain, with loss

$$\|\widehat{f} - f\|_2^2 = \int_0^1 |\widehat{f}(t) - f(t)|^2\, dt.$$

Throughout this section, use the Hölder ball $\Lambda^\beta(M)$ from (2.1), with smoothness $\beta > 0$ and radius $M > 0$. Write $r = \lceil\beta\rceil - 1$ and $\nu_\beta = \beta - r \in (0, 1]$ for its derivative order and remaining Hölder exponent. For density estimation, use its density subclass $\mathcal{D}_\beta(M)$ from (2.2), with $M > 1$.

Our goal is to determine how smoothness and the observation model govern global accuracy. Assouad's lemma from Section 3.1 accumulates errors over disjoint bumps for density estimation and regression, and over wavelet coefficients for inverse problems. Smoothness limits their amplitude; indistinguishability determines their scale. The packing-and-covering argument of Example 3.13 measures the size of a function class globally; here explicit hypercubes show how local errors add up. Indirect observation changes the information calculation, not the loss.

Ibragimov and Has'minskii [73] provide background on nonparametric and white-noise experiments; Cavalier [49] surveys statistical inverse problems. Here we compare direct and polynomially attenuated observations without optimizing constants.

4.4.1 Global density estimation and nonparametric regression

Suppose we observe $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} f$ with $f \in \mathcal{D}_\beta(M)$. Example 3.3 introduced the density hypercube and its rate. Here we give the complete class-membership and information calculations, including integer smoothness, then adapt the construction to midpoint regression (2.3). In each experiment, write

$$R_n(\mathcal{F}) := \inf_{\widehat{f}} \sup_{f \in \mathcal{F}} \mathbb{E}_f \|\widehat{f} - f\|_2^2,$$

where the infimum ranges over measurable, possibly randomized, $L_2[0,1]$-valued estimators based on the observations.

The density class imposes nonnegativity and unit integral but no positive lower bound. We perturb the uniform density by mean-zero bumps, preserving total mass. Small amplitude keeps the alternatives between $1/2$ and $3/2$ and within the smoothness class; these pointwise bounds control divergence without restricting the full parameter space. In regression, neither positivity nor unit integral is required.

Theorem 4.16 (Global Density-Estimation Lower Bound) *Fix $\beta > 0$ and $M > 1$. For n i.i.d. observations from a density in $\mathcal{D}_\beta(M)$ of* (2.2)*, there exists $c = c(\beta, M) > 0$ such that, for every integer $n \geq 1$,*

$$R_n(\mathcal{D}_\beta(M)) \geq c\, n^{-\frac{2\beta}{2\beta+1}}.$$

Proof Choose the smooth mean-zero function ϕ from Lemma A.15: it is supported strictly inside $(0,1)$, has $\int \phi = 0$, $\int \phi^2 = 1$, and $\|\phi\|_\infty^2 \leq 32$. We place rescaled copies in disjoint cells and choose their width to keep neighboring experiments close. Errors in recovering their signs then accumulate into integrated loss.

Uniform smoothness and class membership. Retain r and ν_β from the section setup. For any integer $N \geq 1$, signs $s_k \in \{-1, 1\}$, and $\tau > 0$, define

$$C_\beta := 2 \max\left\{1, \max_{0 \leq m \leq r+1} \|\phi^{(m)}\|_{L_\infty(\mathbb{R})}\right\} < \infty,$$

$$g(x) := \tau N^{-\beta} \sum_{k=0}^{N-1} s_k \phi(Nx - k).$$

All summands, including their derivatives, have disjoint supports strictly inside their respective cells. Hence g is smooth on $\mathbb{R}$ and, for $0 \leq m \leq r+1$,

$$\|g^{(m)}\|_\infty \leq \tau N^{m-\beta} \|\phi^{(m)}\|_\infty.$$

For $d = |x - y| > 0$, the derivative bound and the mean value theorem give

$$|g^{(r)}(x) - g^{(r)}(y)| \leq \min\left\{\tau C_\beta N^{-\nu_\beta}, \frac{\tau C_\beta}{2} N^{1-\nu_\beta} d\right\} \leq \tau C_\beta d^{\nu_\beta}.$$

The last inequality uses the second bound when $Nd \leq 1$ and the first when $Nd \geq 1$; the case $d = 0$ is immediate. Also $\|g^{(m)}\|_\infty \leq \tau C_\beta$ for $0 \leq m \leq r$. These bounds hold on the closed interval, including when $\nu_\beta = 1$.

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Choose ρ and τ independently of n and, for each $\omega = (\omega_0, \dots, \omega_{N-1}) \in \{0,1\}^N$, define g_ω and f_ω as follows:

$$\rho := \min\{M - 1, 1/2\}, \qquad \tau := \min\{\rho/C_\beta, 1/\sqrt{8}\} > 0,$$
$$g_\omega(x) := \tau N^{-\beta} \sum_{k=0}^{N-1} (2\omega_k - 1)\phi(Nx - k), \qquad f_\omega := 1 + g_\omega.$$

The preceding bounds imply $g_\omega \in \Lambda^\beta(\rho)$ and $|g_\omega| \le \rho \le 1/2$, so $1/2 \le f_\omega \le 3/2$. Each summand has integral zero on $[0,1]$, so f_ω integrates to one. Its zero-order bound is at most $1 + \rho \le M$, and its positive-order derivative bounds and top Hölder seminorm are at most $\rho \le M$. Thus every f_ω belongs to $\mathcal{D}_\beta(M)$.

Neighboring experiments. Define

$$u_{Nk}(x) := \sqrt{N}\,\phi(Nx - k), \qquad A := \tau N^{-(\beta+1/2)}.$$

Disjoint support and $\int \phi^2 = 1$ make $(u_{Nk})_{k=0}^{N-1}$ orthonormal in $L_2[0,1]$. Therefore

$$f_\omega = 1 + A \sum_{k=0}^{N-1} (2\omega_k - 1)u_{Nk}, \qquad \|f_\omega - f_{\omega'}\|_2^2 = 4A^2 H(\omega, \omega'),$$

where $H(\omega, \omega')$ is the unnormalized Hamming distance between label vectors. Let P_ω be the one-observation density law. With probability laws as arguments, $H(P, Q)$ instead denotes the normalized Hellinger distance from Section 1.3. For neighboring vertices it satisfies

$$H^2(P_\omega, P_{\omega'}) = \frac{1}{2}\int_0^1 \frac{(f_\omega - f_{\omega'})^2}{(\sqrt{f_\omega} + \sqrt{f_{\omega'}})^2}\,dx \le \frac{1}{4}\|f_\omega - f_{\omega'}\|_2^2 = A^2,$$

because both densities are at least $1/2$. Choose

$$J := \left\lceil \frac{\log_2 n}{2\beta + 1} \right\rceil, \qquad N := 2^J, \qquad n^{1/(2\beta+1)} \le N < 2n^{1/(2\beta+1)}.$$

This includes $n = 1$, when $J = 0$ and $N = 1$. Hellinger tensorization and the total-variation bound give, for every edge,

$$H^2(P_\omega^{\otimes n}, P_{\omega'}^{\otimes n}) \le nA^2 = \tau^2 n N^{-(2\beta+1)} \le \frac{1}{8}, \qquad \mathrm{TV}(P_\omega^{\otimes n}, P_{\omega'}^{\otimes n}) \le \frac{1}{2}.$$

Loss conversion. For any possibly randomized L_2-valued decision $\widehat{f}$, decode $\widehat{\omega}_k = 1$ if $\langle \widehat{f} - 1, u_{Nk} \rangle \ge 0$ and $\widehat{\omega}_k = 0$ otherwise. Bessel's inequality gives, pointwise in the data and the randomization,

$$\|\widehat{f} - f_\omega\|_2^2 \ge A^2 \sum_{k=0}^{N-1} \mathbf{1}\{\widehat{\omega}_k \ne \omega_k\}.$$

For each fixed setting of the other signs, the two error probabilities at bit k sum to at least $\eta(P_\omega^{\otimes n}, P_{\omega'}^{\otimes n}) \ge 1/2$, where ω and ω' differ only at bit k. This also holds for randomized tests: total variation bounds the difference in expectations of any measurable $[0,1]$-valued

rejection probability. Averaging over all 2^N vertices therefore gives an error probability at least $1/4$ for each bit. Consequently

$$\inf_{\widehat{f}} \sup_{f \in \mathcal{D}_\beta(M)} \mathbb{E}_f \|\widehat{f} - f\|_2^2 \geq \frac{NA^2}{4} = \frac{\tau^2}{4} N^{-2\beta} \geq \frac{\tau^2}{4}\, 2^{-2\beta} n^{-2\beta/(2\beta+1)}.$$

Thus $c = \tau^2 2^{-2\beta}/4 > 0$ suffices. □

Remark 4.17 (Optimal Procedure and Matching Upper Bound) For fixed $\beta > 0$ and $M > 1$, a boundary-corrected kernel density estimator attains the minimax rate. With bandwidth $0 < h \leq 1/2$, it has the form $\widehat{f}_h(x) = (nh)^{-1} \sum_{i=1}^n K_{x,h}((X_i - x)/h)$. The kernels are supported on $[-1, 1] \cap [-x/h, (1 - x)/h]$, integrate to one, and have vanishing moments of orders $1, \ldots, r$, where $r = \lceil \beta \rceil - 1$. They can be chosen with uniformly bounded L_1 and L_2 norms. Taylor's formula bounds the bias by Ch^β, while independence bounds the variance by $C/(nh)$, uniformly in x and $f \in \mathcal{D}_\beta(M)$. Hence

$$\sup_{f \in \mathcal{D}_\beta(M)} \mathbb{E}_f \|\widehat{f}_h - f\|_2^2 \leq C \left(h^{2\beta} + \frac{1}{nh} \right), \tag{4.13}$$

where C depends only on β, M and the kernel family. Taking $h \asymp n^{-1/(2\beta+1)}$ matches the lower rate in Theorem 4.16. Boundary correction preserves the bias bound at both endpoints without requiring the density to vanish there. See Wasserman [111, Section 6.3] for kernel density estimation.

The same bumps and loss calculation apply in regression; the information bound must be recalculated at the design points.

Proposition 4.18 (Global Estimation in Fixed-Design Regression) *In the midpoint fixed-design Gaussian regression model* (2.3)*, fix $\beta, M > 0$. There exist constants $0 < c \leq C < \infty$, depending only on β, M, such that, for every integer $n \geq 1$,*

$$c\, n^{-2\beta/(2\beta+1)} \leq \inf_{\widehat{f}} \sup_{f \in \Lambda^\beta(M)} \mathbb{E}_f \|\widehat{f} - f\|_2^2 \leq C\, n^{-2\beta/(2\beta+1)}.$$

The infimum is over measurable, possibly randomized $L_2[0, 1]$-valued estimators.

Proof For the lower bound, use ϕ and C_β from the density proof, and set $\tau_{\mathrm{reg}} := \min\{M/C_\beta, 1/16\} > 0$. For $J = \lceil \log_2 n/(2\beta + 1) \rceil$, $N = 2^J$, and $\omega \in \{0, 1\}^N$, let

$$f_\omega(x) := \tau_{\mathrm{reg}} N^{-\beta} \sum_{k=0}^{N-1} (2\omega_k - 1)\phi(Nx - k).$$

The derivative bounds in the proof of Theorem 4.16 give $f_\omega \in \Lambda^\beta(M)$, including integer smoothness. If ω and ω' differ only at bit k, their difference is supported in the cell $[k/N, (k+1)/N]$, which contains at most $n/N + 1$ design points. Its height is at most $2\tau_{\mathrm{reg}} N^{-\beta} \|\phi\|_\infty$. Thus

$$\begin{aligned} \sum_{i=1}^n \{f_\omega(x_i) - f_{\omega'}(x_i)\}^2 &\leq 128 \tau_{\mathrm{reg}}^2 \{n N^{-(2\beta+1)} + N^{-2\beta}\} \\ &\leq 256 \tau_{\mathrm{reg}}^2 \leq 1. \end{aligned}$$

Let P_ω denote the joint law of the n regression responses under f_ω. The Gaussian divergence formula gives

$$\mathrm{KL}(P_\omega, P_{\omega'}) = \frac{1}{2}\sum_{i=1}^{n}\{f_\omega(x_i) - f_{\omega'}(x_i)\}^2 \le \frac{1}{2}, \qquad \mathrm{TV}(P_\omega, P_{\omega'}) \le \frac{1}{2}.$$

The integrated-loss coordinates $u_{Nk} = \sqrt{N}\,\phi(N\,\cdot - k)$ are orthonormal, independently of the design. Applying the sign-decoding calculation in the proof of Theorem 4.16 with center zero and coefficient amplitude $\tau_{\mathrm{reg}} N^{-(\beta+1/2)}$ gives

$$\inf_{\widehat{f}}\sup_{f\in\Lambda^\beta(M)} \mathbb{E}_f\|\widehat{f} - f\|_2^2 \ge \frac{\tau_{\mathrm{reg}}^2}{4}N^{-2\beta} \ge \underbrace{\frac{\tau_{\mathrm{reg}}^2}{4}2^{-2\beta}}_{c_{\mathrm{reg}}>0}\, n^{-2\beta/(2\beta+1)}.$$

The design enters only through the bound on the sampled squared differences; the loss remains the integrated squared error.

For the upper bound, use a boundary-corrected local-polynomial estimator of degree $\lceil\beta\rceil - 1$. For bandwidths $h = h_n \to 0$ with $nh_n \to \infty$, its integrated risk is at most $C\{h_n^{2\beta} + (nh_n)^{-1}\}$ for sufficiently large n. The midpoint grid satisfies the general-design conditions in Tsybakov [103, Proposition 1.13 and Corollary 1.2]. Taking $h_n \asymp n^{-1/(2\beta+1)}$ gives the upper rate. The zero estimator covers the finitely many smaller n after enlarging C.

□

4.4.2 Recovering a function from indirect observations

Blurred signals and measurements of accumulated responses reveal a transformed function rather than the function itself. The transformation can suppress fine-scale features, making reconstruction more difficult. We quantify this loss of information through attenuation across resolution levels.

Let $K : L_2[0,1] \to L_2[0,1]$ be a known compact linear operator. Suppose we observe the full path

$$dY(t) = (Kf)(t)\,dt + n^{-1/2}\,dB(t), \qquad 0 \le t \le 1, \qquad Y(0) = 0. \tag{4.14}$$

Here B is the Brownian motion of (1.2), and $n \ge 1$ is the known noise precision; the observed drift is Kf, rather than f. The unknown function belongs to the Hölder ball $\Lambda^\beta(M)$ in (2.1), with fixed $\beta, M > 0$. Our benchmark is

$$\inf_{\widehat{f}}\sup_{f\in\Lambda^\beta(M)} \mathbb{E}_f\|\widehat{f} - f\|_2^2,$$

where the infimum ranges over measurable, possibly randomized, $L_2[0,1]$-valued estimators based on Y. Thus the loss measures error in f, not in the observed transform Kf.

Propositions 2.5 and 2.6 use attenuation to make a linear functional difficult to estimate. Here the target is the whole function, so we accumulate errors across many weakly observed coefficients. The polynomial attenuation specified below describes a mildly ill-posed inverse problem [49].

Fix a compactly supported orthonormal interval-wavelet basis

$$\{\varphi_{00}\} \cup \{\psi_{\ell k} : \ell \geq 0,\ 0 \leq k < 2^{\ell}\}$$

of $L_2[0,1]$ with regularity exceeding β. Assume that K acts diagonally on its wavelet elements:

$$K\psi_{\ell k} = \kappa_\ell \widetilde{\psi}_{\ell k}, \qquad c_0 2^{-a\ell} \leq \kappa_\ell \leq C_0 2^{-a\ell}, \qquad \ell \geq 0, \tag{4.15}$$

where $a > 0$, $0 < c_0 \leq C_0 < \infty$, and $\{\widetilde{\psi}_{\ell k}\}$ is an orthonormal system in the observation space $L_2[0,1]$. All these objects are fixed independently of n. The exponent a measures how rapidly finer features are suppressed. The diagonal relation is an assumption on the operator and basis, not a consequence of compactness alone.

For a signal with zero scaling coefficient, write $f = \sum_{\ell,k} \theta_{\ell k}\psi_{\ell k}$, where $\theta_{\ell k} = \langle f, \psi_{\ell k}\rangle$. The projected observations satisfy

$$Y_{\ell k} := \int_0^1 \widetilde{\psi}_{\ell k}(t)\, dY(t) = \kappa_\ell \theta_{\ell k} + n^{-1/2} Z_{\ell k}, \qquad Z_{\ell k} \overset{\text{i.i.d.}}{\sim} N(0,1). \tag{4.16}$$

This is the diagonal experiment of Chapter 2, indexed by resolution and position. Dividing by κ_ℓ gives noise standard deviation of order $n^{-1/2}2^{a\ell}$ for each coefficient. Orthonormality leaves its contribution to squared-error loss unchanged: poorly observed coefficients still matter equally for reconstruction.

We next specify the interior wavelets used to keep the alternatives inside the Hölder ball, retaining r and ν_β from the section opening. Assume that there are fixed integers $J_0, R \geq 0$ and a compactly supported mother $\psi \in C^r(\mathbb{R})$ with zero integral and a ν_β-Hölder rth derivative such that

$$\psi_{Jk}(x) = 2^{J/2}\psi(2^J x - k) \quad \text{for } J \geq J_0, \quad R \leq k < 2^J - R.$$

Boundary elements need not have this form. Increase J_0 if necessary so that $2^{J_0} \geq 4R + 1$. At each point, only a bounded number of wavelets at a given level are nonzero. Rescaling derivatives and Hölder increments therefore gives a constant $C_{\psi,\beta}$ such that every single-level sum $g = \sum_{k=R}^{2^J - R - 1} \theta_{Jk}\psi_{Jk}$ satisfies

$$\max_{0 \leq u \leq r} \|g^{(u)}\|_\infty + \sup_{x \neq y} \frac{|g^{(r)}(x) - g^{(r)}(y)|}{|x-y|^{\nu_\beta}} \leq C_{\psi,\beta}\, 2^{J(\beta+1/2)} \max_k |\theta_{Jk}|.$$

Here the supremum is over $x, y \in [0,1]$. Consequently, for some fixed $c_\beta > 0$ depending on β, M and the basis,

$$2^{J(\beta+1/2)} \max_k |\theta_{Jk}| \leq c_\beta \quad \Longrightarrow \quad g \in \Lambda^\beta(M). \tag{4.17}$$

Three scales determine the construction: the number of active coefficients is of order 2^J, each has allowable magnitude of order $2^{-J(\beta+1/2)}$, and the operator multiplies that magnitude by a factor of order 2^{-aJ}. We choose J so that changing one sign remains difficult to detect, then sum the errors in recovering all signs.

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Theorem 4.19 (Mildly Ill-Posed Inverse-Problem Lower Bound) *In model* (4.14)*, fix* $\beta, M, a > 0$ *and* $0 < c_0 \le C_0 < \infty$*. Assume the compactness, interval-wavelet, and orthonormal observation-system conditions above, including* (4.15)*. Then, for every* $n \ge 1$*,*

$$\inf_{\widehat{f}} \sup_{f \in \Lambda^\beta(M)} \mathbb{E}_f \|\widehat{f} - f\|_2^2 \ge c\, n^{-\frac{2\beta}{2\beta+2a+1}},$$

where $c = c(\beta, a, M, c_0, C_0; \text{chosen basis}) > 0$*.*

Proof *Hard alternatives and separation.* For a level $J \ge J_0$, let $\mathcal{I}_J = \{R, \dots, 2^J - R - 1\}$ and $m = |\mathcal{I}_J| = 2^J - 2R \ge 2^{J-1}$. Fix $0 < \tau \le \min\{c_\beta, (2C_0)^{-1}\}$ and set

$$A_J = \tau 2^{-J(\beta+1/2)}, \qquad f_\omega = A_J \sum_{k \in \mathcal{I}_J} (2\omega_k - 1)\psi_{Jk}, \qquad \omega \in \{0,1\}^{\mathcal{I}_J}.$$

By (4.17), every f_ω belongs to $\Lambda^\beta(M)$. Orthonormality gives

$$\|f_\omega - f_{\omega'}\|_2^2 = 4A_J^2 H(\omega, \omega'),$$

where H is Hamming distance. The supports need not be disjoint.

Information in the full observation. Let P_ω be the full-path law in (4.14) and Q_ω the law of the finite vector $(Y_{Jk})_{k \in \mathcal{I}_J}$. On this family, the drift Kf_ω lies in the span of the corresponding $\widetilde{\psi}_{Jk}$. The orthogonal observation coordinates are pure Gaussian noise, independent of the active coordinates and with the same joint law for every ω. Thus the full path can be reconstructed from the finite vector by adding noise whose law does not depend on ω. This defines a Markov kernel T with $P_\omega = TQ_\omega$. Total variation cannot increase under this transformation, so $\mathrm{TV}(P_\omega, P_{\omega'}) \le \mathrm{TV}(Q_\omega, Q_{\omega'})$. This step ensures that the finite-coordinate calculation controls the original experiment, not merely a projection of it.

For neighboring vertices, one observed mean changes by $2\kappa_J A_J$. The Gaussian divergence is

$$\mathrm{KL}(Q_\omega, Q_{\omega'}) = \frac{n}{2}(2\kappa_J A_J)^2 \le 2C_0^2 \tau^2 n\, 2^{-J(2\beta+2a+1)}.$$

Choose

$$J = \max\left\{J_0, \left\lceil \frac{\log_2 n}{2\beta + 2a + 1} \right\rceil\right\}.$$

Then $n\, 2^{-J(2\beta+2a+1)} \le 1$, so the choice of τ makes every edge divergence at most $1/2$. Pinsker's inequality and the data-processing comparison give $\mathrm{TV}(P_\omega, P_{\omega'}) \le 1/2$ on every edge.

Accumulating coefficient errors. Set $\widehat{\omega}_k = \mathbf{1}\{\langle \widehat{f}, \psi_{Jk} \rangle \ge 0\}$. Bessel's inequality gives

$$\|\widehat{f} - f_\omega\|_2^2 \ge A_J^2 \sum_{k \in \mathcal{I}_J} \mathbf{1}\{\widehat{\omega}_k \ne \omega_k\}.$$

For each coordinate and each fixed setting of the remaining signs, the two testing errors sum to at least $\eta(P_\omega, P_{\omega'}) \ge 1/2$ for the corresponding neighboring vertices. Averaging over the cube gives error probability at least $1/4$ per coordinate. This is Assouad's argument, and it applies equally to randomized estimators. Hence

$$\inf_{\widehat{f}} \sup_{f \in \Lambda^\beta(M)} \mathbb{E}_f \|\widehat{f} - f\|_2^2 \ge \frac{mA_J^2}{4} \ge \frac{\tau^2}{8}\, 2^{-2\beta J}.$$

Finally, $2^J \le 2^{J_0+1} n^{1/(2\beta+2a+1)}$ for every $n \ge 1$. Substitution proves the stated lower bound with a constant independent of n. □

The denominator $2\beta + 2a + 1$ records the balance: smoothness contributes 2β, attenuation contributes $2a$, and the number of coefficients at a level contributes 1.

Remark 4.20 (Procedure and Upper-Bound Comparison) Spectral truncation estimates the wavelet coefficients at levels $\ell \le J$ by $Y_{\ell k}/\kappa_\ell$, estimates the scaling coefficient, and sets finer-level coefficients to zero. The cutoff limits noise amplification at fine resolutions; see Cavalier [49].

For a matching upper bound, assume that truncation in the full basis has squared approximation error at most $C2^{-2\beta J}$ uniformly over $\Lambda^\beta(M)$. Also assume that the orthogonal diagonalization extends to the scaling function with a fixed nonzero singular value. The squared L_2 risk is then at most

$$C\left\{2^{-2\beta J} + \frac{2^{(2a+1)J}}{n}\right\}.$$

The terms are squared approximation error and variance. Choosing $2^J \asymp n^{1/(2\beta+2a+1)}$ balances them and attains the lower-bound rate. The additional assumptions control the entire class; the lower-bound proof needs only one interior wavelet level.

Increasing a slows the rate relative to the direct benchmark in Section 4.4.1. Formally setting $a = 0$ recovers the direct exponent, although the identity operator is not compact and lies outside the theorem.

4.5 Noisy optimization: value and location

Rather than recovering the whole function, we now estimate its minimum value and minimizer from noisy observations. Choosing an operating condition or minimizing an expected cost requires both: the minimizer determines the decision, while the minimum value describes its performance. We first work in the Gaussian white-noise model (1.2), assuming that f is convex on $[0, 1]$ and has a unique minimizer.

We measure both errors by absolute loss. A two-point minimax benchmark fixes f and varies its competitor, revealing how accuracy depends on the shape near the minimum. Local moduli select alternatives for Le Cam's argument; raising the bottom of the graph forms a cap whose height and width describe changes in value and location.

This geometry yields the *uncertainty principle* of Cai, Chen, and Zhu [17]: the location benchmark times the squared value benchmark is comparable to the noise variance, uniformly over the convex class. A shallow minimum can permit accurate value estimation while making its location difficult to determine. The principle relates the intrinsic difficulties of the two tasks, not a tradeoff between procedures: both benchmarks are simultaneously attainable up to universal constants without knowing the local shape.

To clarify the role of convexity, we conclude with global minimax bounds in fixed-design regression under Hölder smoothness alone. Isolated wells in otherwise flat functions show

why smoothness and a unique minimizer do not ensure uniformly consistent location estimation. Section 6.3 returns to convex white-noise optimization and uses the same cap geometry to study honest confidence intervals for both targets.

4.5.1 Convex functions and local estimation benchmarks

We first quantify estimation difficulty at each convex function, rather than taking a worst case over all functions. Suppose we observe the full path in (1.2) on $[0,1]$, with known noise level $\varepsilon > 0$. Let $\mathcal{F}_{\mathrm{cvx}}$ be the class of real-valued convex functions on $[0,1]$ that attain their minimum at a unique point. The targets are

$$Z(f) = \arg\min_{t\in[0,1]} f(t), \qquad M(f) = f(Z(f)).$$

Here $M(f)$ is the minimum-value functional; no class radius M is imposed. Absolute error measures location error for $Z(f)$ and value error for $M(f)$. Convexity constrains each function's shape and is distinct from convexity of a parameter class in Section 4.1. No differentiability, positive curvature bound, or common smoothness exponent is assumed.

Minimizing risk at f alone gives zero: an estimator could simply report $T(f)$. Instead, we require accuracy at both f and a competitor g, then choose the hardest competitor. This retains dependence on f while avoiding the degenerate one-function comparison.

For a measurable estimator $\widehat{T}$ based on Y (possibly with independent randomization), its risk at h is $\mathbb{E}_h|\widehat{T} - T(h)|$. The function-specific two-point minimax benchmark is

$$R_T(\varepsilon; f) = \sup_{g\in\mathcal{F}_{\mathrm{cvx}}} \inf_{\widehat{T}} \max_{h\in\{f,g\}} \mathbb{E}_h|\widehat{T} - T(h)|, \qquad T \in \{Z, M\}. \tag{4.18}$$

For each pair $\{f,g\}$, the estimator may know both candidates but not which generated the data. The inner infimum minimizes its larger risk; the outer supremum chooses the hardest competitor. Although this allows a different estimator for each pair, the attainability result below gives one estimator for each target that works within a universal constant at every f.

The local modulus asks how much the target can change among functions close to f. Unlike the ordinary modulus of Section 4.1, it holds one function fixed, retaining information about its shape. We will show that this geometric quantity characterizes the statistical benchmark.

For a real-valued target T on a class $\mathcal{F}$, a semimetric d on $\mathcal{F}$, $f \in \mathcal{F}$, and $\varepsilon \geq 0$, define the *local modulus at* f by

$$\omega(\varepsilon, f; \mathcal{F}, T) = \sup\big\{|T(g) - T(f)| : g \in \mathcal{F},\ d(f,g) \leq \varepsilon\big\}. \tag{4.19}$$

Taking the supremum over $f \in \mathcal{F}$ recovers the ordinary modulus in (4.2). Here take $\mathcal{F} = \mathcal{F}_{\mathrm{cvx}}$ and $d(f,g) = \|f - g\|_2$, and abbreviate

$$\omega_T(\varepsilon; f) = \omega(\varepsilon, f; \mathcal{F}_{\mathrm{cvx}}, T), \qquad T \in \{Z, M\}.$$

The L_2 distance measures distinguishability in this experiment. If P_f denotes the law of the observed path, the Gaussian likelihood calculation gives

$$\mathrm{KL}(P_f, P_g) = \frac{\|f - g\|_2^2}{2\varepsilon^2}.$$

At distance at most ε, this divergence is at most $1/2$. A nearly extremal alternative in the local modulus therefore gives a two-point risk lower bound. To obtain the reverse comparison, we must also control the pairwise risk for alternatives farther from f.

To evaluate the local modulus, raise the bottom of f to a horizontal level. This changes the minimum value and creates a flat interval where a small tilt can select a new minimizer. The cap's height and width describe the target changes at an L_2 cost set by the noise level.

Formally, for $u \geq M(f)$ set $f_u(t) = \max\{f(t), u\}$. The added cap has squared L_2 cost $\|f_u - f\|_2^2$. At cost at most ε^2, define its height and maximal one-sided width by

$$\rho_M(\varepsilon; f) = \sup\{u - M(f) : u \geq M(f),\ \|f_u - f\|_2 \leq \varepsilon\},$$
$$\rho_Z(\varepsilon; f) = \sup\{|t - Z(f)| : t \in [0,1],\ f(t) \leq M(f) + \rho_M(\varepsilon; f)\}.$$

Both quantities can be calculated directly from f. The raised function itself may have a flat minimum and lie outside $\mathcal{F}_{\text{cvx}}$; the proof uses small tilts to obtain admissible competitors with unique minimizers.

Figure 4.2 shows the cap at $u_\varepsilon = M(f) + \rho_M(\varepsilon; f)$. The drawing is symmetric only for clarity: neither definition assumes symmetry.

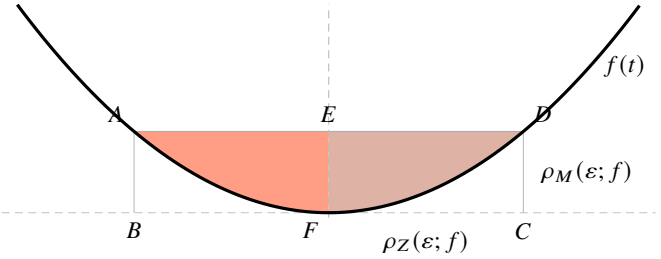


Figure 4.2 Water filling at a convex minimum, redrawn in the style of [17, Figure 1]. The cap depth is $EF = \rho_M(\varepsilon; f)$ and, in this symmetric schematic, $FC = \rho_Z(\varepsilon; f)$. Shading marks the region between the graphs of f and f_{u_ε}; the squared L_2 norm of $f_{u_\varepsilon} - f$, not the shaded area, equals ε^2.

The product involves width times *squared* height because the information cost is an integral of the squared perturbation, rather than the area of the cap. Convexity makes this comparison uniform by controlling how rapidly the cap can taper. The next theorem relates the cap dimensions to the local moduli and bounds their product.

Theorem 4.21 (Cap Geometry and Local Moduli) *For every $f \in \mathcal{F}_{\text{cvx}}$, every $\varepsilon > 0$, and each $T \in \{Z, M\}$,*

$$\rho_T(\varepsilon; f) \leq \omega_T(\varepsilon; f) \leq 3\rho_T(\varepsilon; f).$$

Moreover, the two cap dimensions obey

$$\frac{\varepsilon^2}{2} \leq \rho_Z(\varepsilon; f)\rho_M(\varepsilon; f)^2 \leq 3\varepsilon^2. \tag{4.20}$$

The modulus comparison is Proposition 2.2 of Cai, Chen, and Zhu [17]. We prove the lower comparisons by constructing admissible tilted caps, then derive the cap-product bound. Appendix B.6 proves the reverse modulus comparisons by convex secant inequalities.

Proof of the lower modulus comparisons Fix $f \in \mathcal{F}_{\mathrm{cvx}}$ and $\varepsilon > 0$. Subtracting $M(f)$ from every function preserves distances and displacements, so assume $M(f) = 0$ and write

$$z = Z(f), \quad H = \rho_M(\varepsilon; f), \quad D = \rho_Z(\varepsilon; f), \quad \int_0^1 (H - f(t))_+^2 \, dt = \varepsilon^2.$$

The last identity follows from continuity and strict increase of the cap integral as a function of its positive height. In particular, $H > 0$ and $D > 0$.

Continuous representatives. A finite convex function is continuous in the open interval, but its endpoint values may exceed its finite one-sided limits. Replacing them by these limits preserves convexity and all L_2 distances. It also preserves an attained unique minimum: strict decrease to the left of the minimizer and strict increase to its right keep the limiting value at any other endpoint above the minimum. At an endpoint minimizer, convexity forces continuity. The cap height is unchanged, as is the supremum defining its width under closure. Thus we may use continuous representatives of f and its alternatives; the continuous alternatives constructed below also belong to the original class.

Let $I = \{t : f(t) \le H\} = [a, b]$. Its maximal distance from z is D.

An admissible tilted cap. Reflect if necessary so that $b - z = D$. For sufficiently small $\lambda_{\mathrm{tilt}} > 0$, set $\ell_{\lambda_{\mathrm{tilt}}}(t) = H - \lambda_{\mathrm{tilt}}(t - a)$ and $g_{\lambda_{\mathrm{tilt}}}(t) = \max\{f(t), \ell_{\lambda_{\mathrm{tilt}}}(t)\}$. Choose the tilt so that $\ell_{\lambda_{\mathrm{tilt}}}(z) > 0$; when $z > a$, this means $\lambda_{\mathrm{tilt}} < H/(z - a)$. We check that the tilt adds no more squared distance than the original cap. If $a > 0$, then $z > a$ and $f(a) = H$, so the convex secant inequality gives

$$f(t) \ge H + \frac{H}{z - a}(a - t) \ge \ell_{\lambda_{\mathrm{tilt}}}(t), \qquad 0 \le t < a.$$

When $a = 0$ the left-hand interval is empty; for $t \ge a$, $\ell_{\lambda_{\mathrm{tilt}}}(t) \le H$. Thus

$$0 \le g_{\lambda_{\mathrm{tilt}}} - f \le (H - f)_+, \qquad \|g_{\lambda_{\mathrm{tilt}}} - f\|_2 \le \varepsilon.$$

The maximum of two convex functions is convex. It also has a unique minimizer here: to the left of z, both functions strictly decrease; to the right, f strictly increases while the affine function strictly decreases from a value above $f(z)$. Their maximum is minimized at their unique crossing, or at the right endpoint if no crossing occurs. As $\lambda_{\mathrm{tilt}} \downarrow 0$, the minimizer tends to b and the minimum to H: on every compact subinterval of (z, b) the affine function eventually exceeds f, whereas beyond b, f exceeds H. This includes $b = 1$ and $H > \max f$, when the tilted cap is eventually affine throughout the interval. Hence

$$\omega_Z(\varepsilon; f) \ge \lim_{\lambda_{\mathrm{tilt}} \downarrow 0} |Z(g_{\lambda_{\mathrm{tilt}}}) - z| = D, \qquad \omega_M(\varepsilon; f) \ge \lim_{\lambda_{\mathrm{tilt}} \downarrow 0} M(g_{\lambda_{\mathrm{tilt}}}) = H.$$

□

Proof of the cap-product bound Write $M = M(f)$, $z = Z(f)$, and $A(u) = \int_0^1 (u - f(t))_+^2 \, dt$ for $u \ge M$. The function A is continuous, $A(M) = 0$, and $A(u) \to \infty$ as $u \to \infty$. Therefore $u_\varepsilon = M + \rho_M(\varepsilon; f)$ satisfies $A(u_\varepsilon) = \varepsilon^2$. Put $H = u_\varepsilon - M$ and $D = \rho_Z(\varepsilon; f)$. The sublevel set $\{t : f(t) \le u_\varepsilon\}$ lies in $[z - D, z + D]$, and $(u_\varepsilon - f(t))_+ \le H$. Consequently $\varepsilon^2 = A(u_\varepsilon) \le 2DH^2$.

For the reverse inequality, choose a side on which points of the sublevel set approach distance D from z. If an endpoint at that distance belongs to the sublevel set, convexity bounds

f on the segment from z to that endpoint by the chord joining the two function values. The value at the endpoint is at most $M+H$. Thus, at distance $s \in [0, D]$ along this side, $(u_\varepsilon - f)_+ \geq H(1 - s/D)$ and

$$A(u_\varepsilon) \geq \int_0^D H^2(1 - s/D)^2 \, ds = DH^2/3.$$

If the extremal endpoint is not in the sublevel set, apply the same argument to shorter segments and pass to the limit. Rearranging the two inequalities gives (4.20). □

We now translate the geometry into a statistical benchmark. The following comparison of Cai, Chen, and Zhu [17, Theorem 2.1] shows that the local modulus controls the two-point minimax risk, even though the risk benchmark allows competitors at any distance from f.

Theorem 4.22 (Local Estimation Benchmarks for Convex Minima) *In the white-noise model above, for every $\varepsilon > 0$, every $f \in \mathcal{F}_{\mathrm{cvx}}$, and each $T \in \{Z, M\}$,*

$$\Phi(-1/2)\, \omega_T(\varepsilon; f) \leq R_T(\varepsilon; f) \leq \tfrac{3}{2}\, \omega_T(\varepsilon; f).$$

Proof *Lower bound.* Fix a competitor $g \in \mathcal{F}_{\mathrm{cvx}}$ and put $\Delta = |T(g) - T(f)|$ and $d = \|f - g\|_2$. The likelihood ratio between P_f and P_g depends on the one-dimensional Gaussian projection in the direction $g - f$, which gives the exact testing affinity

$$\eta(P_f, P_g) = 2\Phi\big(-d/(2\varepsilon)\big).$$

For any estimator $\widehat{T}$, the triangle inequality under the common part of the two laws yields

$$\mathbb{E}_f|\widehat{T} - T(f)| + \mathbb{E}_g|\widehat{T} - T(g)| \geq \Delta\, \eta(P_f, P_g).$$

Thus the larger risk is at least $\Delta\Phi(-d/(2\varepsilon))$. For $d \leq \varepsilon$ this is at least $\Phi(-1/2)\Delta$; taking nearly extremal g proves the lower bound.

Upper bound. First bound the minimax risk for each fixed pair; the supremum over alternatives is taken at the end. Fix f, g and put $d = \|f - g\|_2$ and $\Delta = |T(g) - T(f)|$. The likelihood-ratio test for this pair reports one of the two target values and has equal error probabilities $\Phi(-d/(2\varepsilon))$. Its maximum absolute-error risk is therefore $\Delta\Phi(-d/(2\varepsilon))$, matching the common-part lower bound. When $d = 0$, continuity on $(0, 1)$ implies that f and g agree there. For either convex function, its attained minimum equals the infimum of its interior values: convexity forbids a downward endpoint jump, and interpolation toward an attained endpoint minimum approaches that value. Thus $M(f) = M(g)$. If their attained minimizers differed, convexity and the common interior values would force every point strictly between them to have this minimum value, contradicting uniqueness. Hence also $Z(f) = Z(g)$, and the pairwise risk is zero.

To control more distant competitors, we show that the cap height and width grow at most linearly with the noise radius. To see this, replace f by $f - M(f)$, which leaves the cap dimensions unchanged, and put $H = \rho_M(\varepsilon; f)$. For $0 < c < 1$, the pointwise inequality $(cH - f)_+ \leq c(H - f)_+$ shows that $\rho_M(c\varepsilon; f) \geq cH$. Convexity gives $f(z + c(t - z)) \leq cf(t)$, where $z = Z(f)$. Applying this to the level-H sublevel set also gives $\rho_Z(c\varepsilon; f) \geq c\rho_Z(\varepsilon; f)$. Applying these inequalities at radius $x\varepsilon$ with $c = 1/x$ gives $\rho_T(x\varepsilon; f) \leq x\rho_T(\varepsilon; f)$ for $x > 1$;

equality holds at $x = 1$. If $d \le \varepsilon$, the pairwise risk is at most $\omega_T(\varepsilon; f)/2$. If $d > \varepsilon$, write $x = d/\varepsilon$. The cap comparisons in Theorem 4.21 yield

$$\Delta \le \omega_T(d; f) \le 3\rho_T(d; f) \le 3x\rho_T(\varepsilon; f) \le 3x\omega_T(\varepsilon; f).$$

To bound the risk, use $x\Phi(-x/2) \le 1/2$ for $x \ge 1$. On $[1, 3/2]$ use $\Phi(-1/2) < 1/3$, and on $[3/2, 2]$ use $\Phi(-3/4) < 1/4$. These two numerical inequalities follow from

$$\int_0^a \varphi(t)\, dt \ge \frac{a - a^3/6}{\sqrt{2\pi}}, \qquad 0 \le a \le 3/4,$$

which uses $e^{-t^2/2} \ge 1 - t^2/2$. For $x \ge 2$, Mills' inequality gives $x\Phi(-x/2) \le 2\varphi(x/2) \le 2\varphi(1) < 1/2$. Thus every pair has minimax absolute-error risk at most $3\omega_T(\varepsilon; f)/2$. Taking the supremum over g proves the upper bound. □

4.5.2 An uncertainty principle for minimum and minimizer

How are the difficulties of estimating the minimum value and its location related at the same convex function? Theorems 4.21 and 4.22 answer by converting the cap-product bound into the following relation between their local risk benchmarks, uniformly over $\mathcal{F}_{\mathrm{cvx}}$.

Theorem 4.23 (Estimation Uncertainty Principle) *In the white-noise model* (1.2)*, for every* $f \in \mathcal{F}_{\mathrm{cvx}}$ *and every* $\varepsilon > 0$*,*

$$\frac{\Phi(-1/2)^3}{2}\, \varepsilon^2 \le R_Z(\varepsilon; f) R_M(\varepsilon; f)^2 < 274\, \varepsilon^2. \tag{4.21}$$

Proof The preceding theorems give $\Phi(-1/2)\rho_T \le R_T \le (9/2)\rho_T$ for $T = Z, M$. Multiply the inequality for Z by the square of that for M, and apply (4.20). The upper multiplier is $3(9/2)^3 = 273.375 < 274$. □

At a fixed noise level, the product stays within universal multiples of ε^2: a small value benchmark forces a large location benchmark. Geometrically, a broad, shallow cap permits a large location change with little value change. Both benchmarks may nevertheless vanish as noise decreases, and their hardest competitors need not coincide.

The function-specific relation of Cai, Chen, and Zhu [17] requires no local power law. A power-shaped minimum illustrates how the two factors can vary: for $f(t) = |t - 1/2|^k$, $k \ge 1$, and $0 < u < 2^{-k}$,

$$\left\| (u - f)_+ \right\|_2^2 = 2 \int_0^{u^{1/k}} (u - s^k)^2\, ds = \frac{4k^2}{(2k+1)(k+1)} u^{(2k+1)/k}.$$

For fixed $k \ge 1$, equating this cost to ε^2 and using the geometric comparisons above gives, as $\varepsilon \downarrow 0$,

$$\omega_M(\varepsilon; f) \asymp \varepsilon^{2k/(2k+1)}, \qquad \omega_Z(\varepsilon; f) \asymp \varepsilon^{2/(2k+1)}.$$

The same orders hold for R_M and R_Z. When $k = 2$, they are $\varepsilon^{4/5}$ and $\varepsilon^{2/5}$, respectively. As k increases, the minimum becomes flatter: value estimation improves while localization worsens. In every case the product exponent is $2/(2k+1) + 4k/(2k+1) = 2$.

The following result of Cai, Chen, and Zhu [17, Theorems 3.1 and 3.3] shows that one procedure for each target can attain its local benchmark without knowing f.

Theorem 4.24 (Adaptive Estimation of Both Convex Targets) *For every known noise level $\varepsilon > 0$ in the white-noise model, there are measurable, possibly randomized estimators $\widehat{Z}_\varepsilon$ and $\widehat{M}_\varepsilon$, neither depending on the unknown f, such that simultaneously for every $f \in \mathcal{F}_{\mathrm{cvx}}$,*

$$\mathbb{E}_f|\widehat{Z}_\varepsilon - Z(f)| \le 53\,\rho_Z(\varepsilon; f), \qquad \mathbb{E}_f|\widehat{M}_\varepsilon - M(f)| \le 449\,\rho_M(\varepsilon; f).$$

Remark 4.25 (Optimal Procedure and Matching Upper Bound) Cai, Chen, and Zhu [17, Section 3.1] locate the minimizer using successively finer dyadic intervals and estimate the minimum value near that location. They combine the observed path with auxiliary Gaussian noise to obtain independent Gaussian processes for the different steps. This randomization requires no additional observations. Their supplement [18, Sections C.7 and C.9] gives the detailed risk bounds.

Theorems 4.21 and 4.22 give $\rho_T \le \omega_T \le R_T/\Phi(-1/2)$, so these procedures attain both local benchmarks up to universal constants. The same estimators work for every $f \in \mathcal{F}_{\mathrm{cvx}}$, adjusting to the local shape without knowing f. This differs from adaptation to an unknown smoothness class. Section 7.4 makes the corresponding distinction for honest confidence intervals.

4.5.3 Without convexity: minimum and minimizer

Does smoothness alone suffice to estimate a minimum and its location uniformly well? We study this question in fixed-design Gaussian regression under global minimax risk, using squared error for the value and absolute error for the location. Both the experiment and the benchmark differ from the function-specific white-noise analysis above.

Without convexity, shallow, localized wells can have widely separated minimizers and nearly indistinguishable observation laws. Inserting such a downward bump into a flat region violates convexity. The comparison shows how the shape constraint changes the permitted alternatives; it does not assert uniform localization over the entire convex class.

For an integer $n \ge 1$, suppose we observe noisy function values at a regular design,

$$Y_i = f(i/n) + \sigma\xi_i, \qquad i = 1, \dots, n,$$

where the ξ_i are independent standard normals and $\sigma > 0$ is known. Let P_f denote the joint law of these regression observations. For $0 < \beta \le 1$ and $M > 0$, use the Hölder ball $\Lambda^\beta(M)$ from (2.1). Write $m(f) = \min_{x\in[0,1]} f(x)$ and let $\Lambda^\beta_{\mathrm{uniq}}(M)$ denote the subclass with a unique minimizer x^*_f. All infima below range over measurable estimators based on the regression observations, allowing independent randomization. For value, compare a flat function with a narrow well whose depth is limited by smoothness. For location, two wells with shrinking depths and fixed, separated centers yield a nonvanishing lower bound.

Theorem 4.26 (Minimum and Minimizer under Smoothness Alone) *In the fixed-design regression model above, fix $0 < \beta \le 1$ and $M, \sigma > 0$. There are constants $c > 0$ and n_0, depending only on (β, M, σ), such that*

$$\inf_{\widehat{m}} \sup_{f \in \Lambda^\beta(M)} \mathbb{E}_f\{\widehat{m} - m(f)\}^2 \ge c\, n^{-2\beta/(2\beta+1)}, \qquad n \ge n_0.$$

For minimizer estimation,

$$\inf_{\widehat{x}} \sup_{f \in \Lambda^\beta_{\mathrm{uniq}}(M)} \mathbb{E}_f|\widehat{x} - x^*_f| \ge \frac{1}{8}, \qquad n \ge 1.$$

Proof Let $\psi(t) = (1 - |t|)_+$, and note that

$$|\psi(u) - \psi(v)| \le \min\{|u - v|, 1\} \le |u - v|^\beta.$$

For the value bound, put $h = n^{-1/(2\beta+1)}$ and take n_0 large enough that $h \le 1/8$ for $n \ge n_0$. Fix $x_0 = 1/2$ and compare $f_0 = 0$ with

$$f_1(x) = -a\psi((x - x_0)/h), \qquad a = \gamma h^\beta, \qquad \gamma = \tfrac{1}{4}\min\{M, \sigma\}.$$

Both functions belong to $\Lambda^\beta(M)$: their sup norms are at most M, and the preceding inequality gives Hölder constant $a/h^\beta = \gamma \le M$. Their minimum values are separated by a. At most $2nh + 2$ design points lie in the bump support, hence

$$\mathrm{KL}(P_{f_1}, P_{f_0}) = \frac{1}{2\sigma^2}\sum_{i=1}^n f_1(i/n)^2 \le \frac{a^2(nh+1)}{\sigma^2} = \frac{\gamma^2}{\sigma^2}(1 + h^{2\beta}) \le \frac{1}{8}.$$

Pinsker's inequality gives total variation at most $1/4$. For any estimate $\widehat{m}$, decide in favor of the closer of the two minimum values, 0 and $-a$. A wrong decision entails squared error at least $a^2/4$. The sum of the two testing-error probabilities is at least $\eta(P_{f_0}, P_{f_1}) \ge 3/4$; hence the larger squared-error risk is at least $3a^2/32$. This proves the first claim with $c = 3\gamma^2/32$.

For the location bound, set $h_0 = 1/16$, now use the centers $x_0 = 1/4$ and $x_1 = 3/4$, and define

$$g_j(x) = -b_n\psi((x - x_j)/h_0), \qquad b_n = \min\{Mh_0^\beta/2, \sigma/(4\sqrt{n})\}, \quad j = 0, 1.$$

These disjoint negative tents belong to $\Lambda^\beta(M)$: their sup norms are at most M, and each has Hölder constant $b_n/h_0^\beta \le M/2$. Their unique minimizers are x_0 and x_1. At each design point their difference has absolute value at most b_n, so

$$\mathrm{KL}(P_{g_0}, P_{g_1}) = \frac{1}{2\sigma^2}\sum_{i=1}^n\{g_0(i/n) - g_1(i/n)\}^2 \le \frac{nb_n^2}{2\sigma^2} \le \frac{1}{32}.$$

The total variation is at most $1/8$. The two minimizers are distance $1/2$ apart. The metric two-point inequality for absolute error gives maximum risk at least $(1/2)/2$ times testing affinity, hence at least $7/32 > 1/8$. □

The minimizer bound rules out uniform consistency over the fixed smoothness class: the alternatives become shallower with n, but their minimizers stay separated. Consistency at a fixed function remains possible, so this does not contradict the function-specific convex benchmarks.

Remark 4.27 (Procedure and Upper-Bound Comparison) For the minimizer, the constant estimator $\widehat{x} = 1/2$ has absolute error at most $1/2$, matching the nonvanishing lower-bound order on $[0, 1]$. For the minimum value, set $h = (\log n/n)^{1/(2\beta+1)}$ and $q = \lceil nh \rceil$. Take the smallest average over the $\lfloor n/q \rfloor$ complete consecutive blocks of q observations, omit the final incomplete block, and truncate the result to $[-M, M]$. The Hölder bias bound and the largest Gaussian fluctuation give worst-case squared-error risk of order at most $(\log n/n)^{2\beta/(2\beta+1)}$. Here a logarithmic gap remains: the fixed-location lower bound does not capture the search over locations.

Reliable localization requires quantitative separation of a minimizer from competing points. For example, a global growth condition $f(x) - m(f) \geq c_0|x - x_f^*|^q$, with $c_0, q > 0$, gives, for any selected point $\widehat{x} \in [0, 1]$,

$$|\widehat{x} - x_f^*| \leq \left\{ \frac{f(\widehat{x}) - m(f)}{c_0} \right\}^{1/q}.$$

A local growth condition also requires showing that $\widehat{x}$ lies in the relevant neighborhood. Estimating the scalar minimum alone does not provide such a point or a localization guarantee.

The target determines what must differ between hard alternatives: functional values, the whole signal, or the location of its minimum. The next chapter keeps the focus on indistinguishable observation laws but changes the goal from estimation to testing. Chapter 9 further distinguishes detection from localization.

4.6 Exercises

4.1. In the white-noise model of Example 4.2, fix $0 < \beta \leq 1$, $M > 0$, and $x_0 \in (0, 1)$. For $0 < h \leq \min\{x_0, 1 - x_0\}$, set $f_0 = 0$ and $f_h(t) = Mh^\beta(1 - |(t - x_0)/h|)_+^\beta$. Verify that both functions belong to $\Lambda^\beta(M)$, and calculate their difference in value at x_0 and their squared L_2 distance. With noise level $n^{-1/2}$, choose h to keep the Gaussian KL divergence at most $1/2$ and deduce the squared-error risk lower rate for estimating $f(x_0)$.

4.2. In the density experiment of Section 4.4.1, explain why a single-bump pointwise construction does not give the stated global squared L_2 lower rate. Explain how summing the losses from disjoint bumps in Assouad's inequality supplies the missing multiplicity.

4.3. Under the wavelet-diagonal assumptions of Section 4.4.2, compare the observation separation and loss separation of neighboring alternatives with those under direct Gaussian observation. Explain how attenuation changes the active resolution.

4.4. In Theorem 4.7, take $p = 2$ and choose two positive smoothness values on opposite sides of $\alpha = 1/4$. Calculate the unrestricted squared-error risk exponent and identify the hard-family mechanism in each regime. Distinguish this benchmark from the restriction to quadratic estimators.

4.5. Let $k \geq 2$ be even, and let ν_0, ν_1 be probability measures on $[-1, 1]$ whose moments agree through degree k. Show that the absolute difference between their expectations of $|t|$ is at most $2\delta_k$, where δ_k is the best degree-k uniform approximation error. Explain why Lemma 4.13 attains this bound. Rescale the lemma's priors to $[-M, M]$ and use them

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independently on d coordinates, as in Section 4.3.2. Calculate the target-mean gap and bound the prior variance of the average absolute mean.

4.6. For the minimum-value pair in the proof of Theorem 4.26, verify the design-point count and derive the squared-error lower bound. Explain why this pair does not establish a sharp minimax rate.

4.7. For $f(t) = |t - 1/2|^k$ with $k \geq 1$ and $0 < u < 2^{-k}$, calculate the squared L_2 cost of raising the bottom of the graph to height u and check the exponents in the estimation uncertainty product (4.21).

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5

Minimax Testing and Separation

Chapter 4 studied how accurately an unknown quantity can be estimated. We now ask a different question: can the data distinguish a specified null hypothesis from a class of alternatives? Does a regression curve exceed a reference value? Do two populations have different means? Does a high-dimensional regression contain any signal? These questions concern the presence of a departure, rather than its precise value. A signal may be detectable even when it cannot be estimated accurately.

The goal is to determine how large a departure must be to control both false rejection and missed detection uniformly over the respective hypotheses. This separation benchmark depends on what is unknown about the departure and how it is observed. A known direction, an unknown sparse support, and a smooth but otherwise unspecified signal can lead to different detection limits.

The principal tools are the two-point and mixture comparisons of Chapter 2. A pair can capture the difficulty when geometry or smoothness identifies a hard departure. When uncertainty about direction or support is essential, averaging over alternatives can conceal the signal. Every alternative must satisfy the required separation, while their mixture of observation laws remains close to a null law. Constructing and calibrating such mixtures is a central theme.

Section 5.1 defines testing errors and separation benchmarks. Sections 5.2 and 5.3 show how two-point arguments arise from different constraints. For separated convex Gaussian hypotheses, a closest pair determines the exact minimax testing error. For a regression value at a specified point, smoothness limits how narrowly a departure can be concentrated.

Sections 5.4–5.6 develop mixture arguments. Sparse Gaussian means introduce uncertainty about active coordinates and signs; the two-sample extension shows how to preserve this difficulty without revealing the mean difference through the remaining observations. Sparse regression changes the information calculation: with known noise variance, a signal also increases the marginal response variance, which a hard mixture must control. Finally, unknown-direction Gaussian testing leads to smooth-signal detection, where smoothness determines how many weak coefficients can jointly conceal a departure. Matching tests show when these lower bounds identify the separation rate.

These testing obstructions also limit the size of honest confidence sets, as Chapter 6 shows. The present chapter establishes known-class testing benchmarks; Chapter 9 extends the analysis to rare signals and distinguishes detection from selection and localization.

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5.1 Testing errors and separation

Suppose we observe $X \sim P_\theta$ in a measurable space $(\mathcal{X}, \mathscr{X})$ and must decide whether the unknown parameter $\theta \in \Theta$ belongs to a null or an alternative class. Here X denotes the full observation: a sample, a sequence, or a continuous path. We define the chapter's error and separation benchmarks, distinguishing fixed error control from errors tending to zero, and recall the pair and mixture comparisons that yield lower bounds.

For disjoint, nonempty sets $\Theta_0, \Theta_1 \subseteq \Theta$, we test $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$. A possibly randomized test is a measurable function $\varphi : \mathcal{X} \to [0,1]$, where $\varphi(x)$ is the probability of rejecting H_0 after observing $X = x$. Uniform error control uses the maximal type-I and type-II errors,

$$\alpha(\varphi) = \sup_{\theta\in\Theta_0} \mathbb{E}_\theta[\varphi(X)], \qquad \beta(\varphi) = \sup_{\theta\in\Theta_1} \mathbb{E}_\theta[1-\varphi(X)].$$

To control both errors, we use the *minimax total error*

$$\mathcal{R}(\Theta_0, \Theta_1) = \inf_\varphi \{\alpha(\varphi) + \beta(\varphi)\}. \tag{5.1}$$

Always accepting the null gives type-I error zero and type-II error one, so $0 \le \mathcal{R}(\Theta_0, \Theta_1) \le 1$. Values near zero indicate reliable discrimination; values near one indicate little improvement over a constant decision. Using the larger maximal error, as in the minimax zero–one risk of Chapter 1, changes the benchmark by at most a factor of two.

To express difficulty in terms of signal strength, index the alternative class by a minimum separation $\rho > 0$ from the null. For example, in the Gaussian experiment $X \sim N_d(\theta, n^{-1} I_d)$, with integer $d \ge 1$, known precision $n > 0$, and $\theta \in \Theta \subseteq \mathbb{R}^d$ with $0 \in \Theta$, consider

$$H_0 : \theta = 0 \qquad \text{against} \qquad H_1 : \theta \in \Theta_1(\rho), \qquad \Theta_1(\rho) = \{\theta \in \Theta : \|\theta\|_2 \ge \rho\}.$$

Here separation is measured by signal norm; Section 5.3 instead uses a function-value gap.

More generally, write $\mathcal{R}_n(\rho)$ for the minimax total error (5.1) between Θ_0 and $\Theta_1(\rho)$, with n indexing sample size or precision. As in this example, we use nested alternatives: $\Theta_1(\rho') \subseteq \Theta_1(\rho)$ for $\rho' \ge \rho$. Increasing separation then cannot make testing harder, so $\mathcal{R}_n(\rho)$ is nonincreasing wherever the alternatives are nonempty.

For a desired total error $\varepsilon_{\text{test}} \in (0,1)$, the *minimax separation radius* is

$$\rho_n^*(\varepsilon_{\text{test}}) = \inf\{\rho > 0 : \Theta_1(\rho) \ne \varnothing,\ \mathcal{R}_n(\rho) \le \varepsilon_{\text{test}}\}, \tag{5.2}$$

with the infimum of the empty set equal to $+\infty$.

A positive sequence r_n is a *fixed-error separation rate* if constants $0 < c < C < \infty$ satisfy

$$\mathcal{R}_n(cr_n) \ge 3/4, \qquad \mathcal{R}_n(Cr_n) \le 1/4$$

for all sufficiently large n, with nonempty alternatives in both comparisons. By monotonicity, these bounds give $cr_n \le \rho_n^*(1/4) \le Cr_n$. The thresholds $3/4$ and $1/4$ are convenient choices; other fixed levels require recalibrating the bounds. Unless stated otherwise, c and C may depend on fixed class parameters, but not on n or the ambient dimension.

Unlike fixed-error separation, *consistent detection* requires total error to tend to zero along the specified sequence of experiments and alternatives. Section 5.6.2 treats both criteria for smooth signals.

For a lower bound, it suffices to restrict attention to a difficult pair or mixture; a matching upper bound must control errors over the full hypotheses. Suppress n, choose $\theta_0 \in \Theta_0$, and write $P_0 = P_{\theta_0}$. For a prior π assigning probability one to Θ_1, let $P_\pi = \int P_\theta \, \pi(d\theta)$. The mixture bound of Chapter 2, written using the testing affinity of Section 1.3, gives

$$\mathcal{R}(\Theta_0, \Theta_1) \geq \eta(P_0, P_\pi) = 1 - \mathrm{TV}(P_0, P_\pi) \geq 1 - \tfrac{1}{2}\sqrt{\chi^2(P_\pi \| P_0)}. \tag{5.3}$$

For every test, the maximal null error dominates its error at θ_0, and the maximal alternative error dominates its prior average. Minimizing the sum of these two smaller errors gives the affinity; the total-variation–chi-square bound gives the last inequality. A point mass $\pi = \delta_{\theta_1}$, with $\theta_1 \in \Theta_1$, recovers the two-point bound $\mathcal{R}(\Theta_0, \Theta_1) \geq \eta(P_{\theta_0}, P_{\theta_1})$. Equality holds for simple hypotheses, but need not hold for larger classes.

When $P_\pi \ll P_0$, write $L = dP_\pi / dP_0$. Then

$$\chi^2(P_\pi \| P_0) = \mathbb{E}_0 L^2 - 1,$$

where $\mathbb{E}_0$ denotes expectation under P_0. Every alternative must satisfy the prescribed separation, but only their observation mixture needs to be close to the null.

Alternatively, fix a significance level $0 < \alpha_0 < 1$ and minimize missed detection subject to type-I error control. For fixed n and ρ, the optimal worst-case type-II error over Θ_0 and $\Theta_1(\rho)$ is

$$\beta^*_{\alpha_0}(\rho) = \inf_{\varphi:\, \alpha(\varphi) \leq \alpha_0} \beta(\varphi).$$

The best worst-case power is $1 - \beta^*_{\alpha_0}(\rho)$. Every level-$\alpha_0$ test satisfies $\alpha(\varphi) + \beta(\varphi) \geq \mathcal{R}_n(\rho)$ and hence $\beta(\varphi) \geq \mathcal{R}_n(\rho) - \alpha_0$. Taking the infimum and using nonnegativity gives

$$\beta^*_{\alpha_0}(\rho) \geq \max\{0, \mathcal{R}_n(\rho) - \alpha_0\}.$$

Thus, when $\mathcal{R}_n(\rho)$ is close to one, worst-case power cannot appreciably exceed the significance level. The next section gives exact total-error and fixed-level benchmarks for convex Gaussian hypotheses, both determined by a closest pair.

5.2 Testing separated convex Gaussian hypotheses

We first test between two convex sets of Gaussian means, representing candidate signal patterns subject to uncertainty. We seek both the minimum total error and the best worst-case power at a prescribed significance level. A closest pair determines both benchmarks: uncertainty within the sets makes testing no harder than distinguishing this pair.

Suppose we observe $X \sim N_d(\theta, \sigma^2 I_d)$, with dimension $d \geq 1$ and known noise standard deviation $\sigma > 0$. We test $H_0 : \theta \in C_0$ against $H_1 : \theta \in C_1$, where $C_0, C_1 \subset \mathbb{R}^d$ are nonempty compact convex sets with Euclidean separation

$$\Delta = \inf_{u \in C_0, v \in C_1} \|u - v\|_2 > 0.$$

Compactness ensures that a closest pair exists, giving a two-point lower bound. Convexity makes it attainable: projection along the line joining the pair separates the full classes. Whereas Section 4.1 maximizes a target gap under a distance constraint, here we minimize

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distance between hypotheses. Goldenshluger, Juditsky, and Nemirovski [67] develop convex-optimization tests for broader observation families; the Gaussian case below isolates the geometric argument.

Theorem 5.1 (Exact Testing of Separated Convex Gaussian Means) *Observe $X \sim N_d(\theta, \sigma^2 I_d)$, with integer $d \geq 1$ and known $\sigma > 0$. For testing $H_0 : \theta \in C_0$ against $H_1 : \theta \in C_1$, where $C_0, C_1 \subset \mathbb{R}^d$ are nonempty compact convex sets at Euclidean distance $\Delta > 0$, the minimax total error is*

$$\mathcal{R}(C_0, C_1) = 2\Phi\left(-\frac{\Delta}{2\sigma}\right). \tag{5.4}$$

Proof Compactness gives a closest pair $u_* \in C_0$, $v_* \in C_1$. Set $w = v_* - u_*$ and $m_* = (u_* + v_*)/2$, so $\|w\|_2 = \Delta$. For this pair, the log-likelihood ratio of the alternative to the null is $\langle w, X - m_* \rangle / \sigma^2$. The standardized projection

$$T = \frac{\langle w, X - m_* \rangle}{\sigma \Delta}$$

has variance one and means $-\Delta/(2\sigma)$ and $\Delta/(2\sigma)$ at u_* and v_*, respectively. The optimal simple-hypothesis test rejects when $T > 0$, with both errors equal to $\Phi(-\Delta/(2\sigma))$. Restriction to this pair therefore gives the stated lower bound for $\mathcal{R}(C_0, C_1)$.

To prove that the same test works uniformly over both sets, we show that projection along w places every null mean at or below u_* and every alternative mean at or above v_*. Equivalently,

$$\langle w, u - u_* \rangle \leq 0 \quad (u \in C_0), \qquad \langle w, v - v_* \rangle \geq 0 \quad (v \in C_1).$$

Indeed, convexity gives $u_* + t(u - u_*) \in C_0$ for $0 \leq t \leq 1$. Its squared distance to v_* is minimized at $t = 0$, so the right derivative $-2\langle w, u - u_* \rangle$ is nonnegative. Similarly, the squared distance from $v_* + t(v - v_*) \in C_1$ to u_* has nonnegative right derivative $2\langle w, v - v_* \rangle$, proving the second inequality.

Consequently, T has variance one and mean at most $-\Delta/(2\sigma)$ under the null and at least $\Delta/(2\sigma)$ under the alternative. Gaussian tail monotonicity gives both maximal errors $\Phi(-\Delta/(2\sigma))$, attained at the closest pair. This matches the lower bound. □

Only the separation-to-noise ratio Δ/σ enters the error formula. For example, consider nominal signal means $a, b \in \mathbb{R}^d$ subject to bounded calibration error. Take

$$C_0 = \{\theta : \|\theta - a\|_2 \leq r_0\}, \qquad C_1 = \{\theta : \|\theta - b\|_2 \leq r_1\},$$

where $r_0, r_1 \geq 0$ and $\|b - a\|_2 > r_0 + r_1$. Let $e = (b - a)/\|b - a\|_2$. The triangle inequality gives $\Delta \geq \|b - a\|_2 - r_0 - r_1$, and the points $a + r_0 e$ and $b - r_1 e$ attain equality. Consequently,

$$\mathcal{R}(C_0, C_1) = 2\Phi\left(-\frac{\|b - a\|_2 - r_0 - r_1}{2\sigma}\right).$$

Calibration uncertainty reduces separation by $r_0 + r_1$. The optimal test uses only the projection along e; orthogonal components cannot improve its minimax total error. If the balls intersect, a shared parameter forces total error at least one, attained by a constant test. Positive separation is thus necessary for minimax total error below one.

Now fix a significance level $0 < \alpha_0 < 1$ and minimize worst-case type-II error, denoted by $\beta^*_{\alpha_0}$ as in Section 5.1. The closest pair remains least favorable, but the rejection threshold changes. Retain $w = v_* - u_*$ and center the projection at the closest null mean:

$$Z = \frac{\langle w, X - u_* \rangle}{\sigma\Delta}.$$

Under every parameter, Z is normal with variance one. Its mean is at most zero on C_0 and at least Δ/σ on C_1, with equality at u_* and v_*. Rejection when $Z > z_{\alpha_0}$ therefore has maximal type-I error α_0 and maximal type-II error $\Phi(z_{\alpha_0} - \Delta/\sigma)$.

For the matching lower bound, every test of level at most α_0 over C_0 also has type-I error at most α_0 at u_*. The prescribed-level Neyman–Pearson bound of Lemma 1.8 then forces its type-II error at v_* to be at least $\Phi(z_{\alpha_0} - \Delta/\sigma)$. Hence

$$\beta^*_{\alpha_0} = \Phi\left(z_{\alpha_0} - \frac{\Delta}{\sigma}\right). \tag{5.5}$$

Thus the same projection is optimal for total error and fixed-level power, with thresholds calibrated to different criteria. If separation and noise vary with sample size, minimax total error tends to zero precisely when $\Delta/\sigma \to \infty$, and tends to one when $\Delta/\sigma \to 0$.

5.3 Testing a regression value under smoothness

We test whether a regression curve exceeds a reference value at a prespecified point, as when assessing a treatment effect at a given dose. We seek the smallest value gap permitting uniform control of both errors over a known smoothness class. The lower bound uses a localized departure: smoothness limits its width and hence the number of observations it affects.

For an integer $n \geq 1$, suppose we observe the fixed-design regression data

$$Y_i = f(i/n) + \epsilon_i, \qquad i = 1, \ldots, n, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0,1). \tag{5.6}$$

Fix $t_0 \in (0,1)$, not necessarily a design point, and use the Hölder ball $\Lambda^s(M)$ of (2.1), with known $0 < s \leq 1$ and $M > 0$. With reference value zero and $0 < \rho \leq M$, test

$$H_0 : f \in \Lambda^s(M),\ f(t_0) \leq 0, \quad \text{against} \quad H_1(\rho) : f \in \Lambda^s(M),\ f(t_0) \geq \rho.$$

Write P_f for the law of $(Y_1, \ldots, Y_n)$ and $\mathcal{R}_n(\rho)$ for the minimax total error (5.1). The restriction $\rho \leq M$ ensures that the alternative is nonempty.

Here smoothness, rather than convex geometry, selects the hard pair: compare zero with a positive bump centered at t_0. This is the modulus construction of Section 4.1, following Donoho and Liu [59], now used to bound testing error rather than estimation risk.

A bump of height ρ needs distance at least $(\rho/M)^{1/s}$ between its peak and a zero. With $h = (\rho/M)^{1/s}$, the bump below has Gaussian KL divergence at most $\rho^2(nh + 1)$. Keeping this bound small selects the scale $n^{-s/(2s+1)}$ for fixed M. The matching test averages nearby responses, balancing bias Mh^s against standard deviation of order $(nh)^{-1/2}$.

Theorem 5.2 (Pointwise Hölder Testing Rate) *For the fixed-design regression* (5.6) *and the one-sided Hölder hypotheses above, fix* $0 < s \le 1$, $M > 0$, *and* $t_0 \in (0,1)$, *and put* $r_n = n^{-s/(2s+1)}$. *There exist* $0 < c < C < \infty$ *and* n_0, *depending only on* s, M, t_0, *such that for every* $n \ge n_0$,

$$\mathcal{R}_n(cr_n) \ge \tfrac{3}{4}, \qquad \mathcal{R}_n(Cr_n) \le \tfrac{1}{4}.$$

Proof For the lower bound let $f_0 = 0$ and

$$f_1(x) = \rho\big(1 - |x - t_0|^s/h^s\big)_+, \qquad h = (\rho/M)^{1/s}.$$

For $0 < s \le 1$, the inequality $|u^s - v^s| \le |u - v|^s$ for $u, v \ge 0$ makes $x \mapsto |x - t_0|^s$ s-Hölder with constant one. The positive-part map is 1-Lipschitz, so

$$|f_1(x) - f_1(y)| \le \rho h^{-s}|x - y|^s = M|x - y|^s.$$

Together with $\|f_1\|_\infty = \rho \le M$ and $f_1(t_0) = \rho$, this verifies membership in the alternative; f_0 belongs to the null. For sufficiently small ρ, the support interval $[t_0 - h, t_0 + h]$ lies inside $[0,1]$. It contains at most $2nh + 2$ design points, so

$$\mathrm{KL}(P_{f_1}, P_{f_0}) = \tfrac{1}{2}\sum_{i=1}^{n} f_1(i/n)^2 \le \tfrac{1}{2}\rho^2(2nh + 2).$$

With $\rho = cr_n$, we have $n\rho^2 h = c^{2+1/s}M^{-1/s}$ and $\rho^2 \to 0$. Choose c small and then n_0 large so that the divergence is at most $1/8$. Pinsker's inequality and restriction to this pair give $\mathcal{R}_n(\rho) \ge \eta(P_{f_0}, P_{f_1}) \ge 3/4$.

For the upper bound, take $h = n^{-1/(2s+1)}$, $I_h = \{i : |i/n - t_0| \le h\}$, and $N_h = |I_h|$. Eventually the window lies inside $[0,1]$ and $nh \ge 1$. Its length $2h$ then gives $N_h \ge 2nh - 1 \ge nh$. The Gaussian average

$$\overline{Y}_h = \frac{1}{N_h}\sum_{i \in I_h} Y_i$$

has variance N_h^{-1} and satisfies $|\mathbb{E}_f \overline{Y}_h - f(t_0)| \le Mh^s$ uniformly over the class. Reject when $\overline{Y}_h > \rho/2$. Its mean is at most Mh^s under the null and at least $\rho - Mh^s$ under the alternative.

For $\rho = Cr_n$ with $C > 2M$, the threshold exceeds every null mean and lies below every alternative mean. In standard-deviation units, each gap is at least

$$(\rho/2 - Mh^s)\sqrt{N_h} = (C/2 - M)r_n\sqrt{N_h} \ge C/2 - M,$$

because $h^s = r_n$ and $r_n\sqrt{nh} = 1$. Taking $C = 2\{M + \Phi^{-1}(7/8)\}$ makes each error at most $1/8$, so their sum is at most $1/4$. Decrease c if necessary to ensure $c < C$, and enlarge n_0 so that $Cr_n \le M$ for all $n \ge n_0$; both alternative classes are then nonempty. □

The rate is for a prespecified location, so there is no search penalty. Dümbgen and Spokoiny [63] study multiscale testing, and Section 9.3 isolates the cost of an unknown location. We next turn to vectors with unknown active coordinates and signs, where a mixture replaces the single localized pair.

5.4 Testing sparse Gaussian means

We seek the smallest Euclidean norm at which a sparse Gaussian mean can be detected at fixed total error. This high-dimensional model describes weak effects on a small, unknown subset of features, as in genomic screening.

We begin with the finite Gaussian model (2.4): observe $Y \sim N_d(\theta, \sigma^2 I_d)$, with $d \geq 1$ and known $\sigma > 0$, and test $\theta = 0$. We then test equality of two means with an unrestricted common mean. Both lower bounds mix over supports and signs; the two-sample argument must also ensure that observations beyond the mean difference do not reveal the alternatives.

5.4.1 Testing a sparse mean vector

In this model, at most s coordinates of θ are nonzero, with unknown locations and signs. To quantify how sparsity changes the detectable norm, fix an integer $1 \leq s \leq d$ and test

$$H_0 : \theta = 0 \quad \text{against} \quad H_1(\rho) : \|\theta\|_0 \leq s, \quad \|\theta\|_2 \geq \rho, \qquad \rho > 0.$$

Write $\mathcal{R}^{\text{mean}}_{d,s,\sigma}(\rho)$ for the minimax total error (5.1). The model also represents n independent observations from $N_d(\theta, I_d)$ through their sample mean, with $\sigma^2 = 1/n$: the residuals are independent of that mean and have a parameter-free law.

Combine the mixtures of Section 2.2: choose s active coordinates uniformly and assign independent values $+a$ or $-a$, with $a > 0$ to be chosen. Every draw has squared norm sa^2. Only coordinates shared by two draws contribute to the likelihood-ratio second moment. Averaging their signs and then their support overlap determines how large a can be while the mixture remains close to the null. Collier, Comminges, and Tsybakov [51] establish the corresponding minimax benchmark.

Proposition 5.3 (Sparse Gaussian Mean Lower Bound) *In the finite Gaussian model* (2.4)*, for* $1 \leq s \leq d$*,*

$$\mathcal{R}^{\text{mean}}_{d,s,\sigma}(\rho) \geq \tfrac{3}{4} \quad \text{if} \quad 0 < \rho^2 \leq \frac{\sigma^2 s}{2} \log\left(1 + \frac{\sqrt{d}}{s}\right).$$

Proof Let P_θ be the observation law and let P_0 be its zero-mean member. For two mean vectors θ, θ', Gaussian integration gives

$$L_\theta(Y) = \frac{dP_\theta}{dP_0}(Y) = \exp\left\{\frac{\langle \theta, Y\rangle}{\sigma^2} - \frac{\|\theta\|_2^2}{2\sigma^2}\right\},$$
$$\mathbb{E}_0 L_\theta L_{\theta'} = \exp\left\{\frac{\langle \theta, \theta'\rangle}{\sigma^2}\right\}.$$

Choose a support S uniformly among the s-element subsets of $\{1, \dots, d\}$. Independently of S, draw independent uniform signs $\xi_j \in \{-1, 1\}$ and set

$$\theta_j = a\xi_j \mathbf{1}\{j \in S\}, \qquad a^2 = \frac{\sigma^2}{2} \log\left(1 + \frac{\sqrt{d}}{s}\right).$$

Every draw has s nonzero coordinates and squared norm $sa^2 \geq \rho^2$, so the prior π is supported on the alternative. Its observation mixture P_π has likelihood ratio $L_\pi = \mathbb{E}_{\theta\sim\pi} L_\theta$. For two independent draws with supports S, S', put $K = |S \cap S'|$. Each shared coordinate contributes a^2 times an independent uniform sign to the inner product. Averaging gives one factor $\cosh(a^2/\sigma^2)$ per shared coordinate, hence

$$1 + \chi^2(P_\pi \| P_0) = \mathbb{E}_0 L_\pi^2 = \mathbb{E}_{S,S'}\{\cosh(a^2/\sigma^2)^K\}. \tag{5.7}$$

It remains to average over the supports. For $z \geq 1$, the exponential-moment bound in Lemma A.11, applied to the independent uniform s-subsets S, S', gives

$$\mathbb{E} z^K \leq \left\{1 + \frac{s}{d}(z-1)\right\}^s \leq \exp\left\{\frac{s^2}{d}(z-1)\right\}. \tag{5.8}$$

Now take $z = \cosh t$, where $t = a^2/\sigma^2 = \frac{1}{2}\log(1+x)$ and $x = \sqrt{d}/s$. Since $\cosh(2t) - 1 = 2(\cosh t - 1)(\cosh t + 1)$,

$$\cosh t - 1 \leq \tfrac{1}{4}\{\cosh(2t) - 1\} = \frac{x^2}{8(1+x)} \leq \frac{x^2}{8}.$$

Because $(s^2/d)x^2 = 1$, equations (5.7) and (5.8) give $\chi^2(P_\pi \| P_0) \leq e^{1/8} - 1 < 1/4$. The mixture bound (5.3) therefore yields

$$\mathcal{R}^{\mathrm{mean}}_{d,s,\sigma}(\rho) \geq 1 - \tfrac{1}{2}\sqrt{\chi^2(P_\pi \| P_0)} > \tfrac{3}{4}.$$

□

The squared-separation scale has two regimes:

$$\sigma^2 s \log\left(1 + \frac{\sqrt{d}}{s}\right) \asymp \begin{cases} \sigma^2 s \log(1 + d/s^2), & s \leq \sqrt{d}, \\ \sigma^2\sqrt{d}, & s \geq \sqrt{d}. \end{cases}$$

For one active coordinate, this scale is of order $\sigma^2 \log(1+d)$; once $s \geq \sqrt{d}$, it reaches the unrestricted scale $\sigma^2\sqrt{d}$ of Section 5.6.1. Together with the matching test below, this shows that sparsity improves the detection rate only in the first regime. The logarithmic factor reflects uncertainty about the support, even though detection does not require recovering it.

Remark 5.4 (Optimal Procedure and Matching Upper Bound) Collier, Comminges, and Tsybakov [51, Theorem 12] construct matching tests when σ and s are known. For $s < \sqrt{d}$, the test sums squared observations whose magnitudes exceed a threshold, reducing the contribution of noise coordinates; for $s \geq \sqrt{d}$, it uses all squared observations. After centering under the null, it rejects for large values. For any prescribed fixed total error in $(0, 1)$, these tests attain that error when ρ^2 exceeds a sufficiently large constant multiple of $\sigma^2 s \log(1 + \sqrt{d}/s)$. They therefore match the lower bound up to constants depending on the error level.

5.4.2 Sparse two-sample mean testing

Comparing treatment groups across many features leads to testing equality of two means against a sparse difference. We ask how unequal sample sizes and an unrestricted common

mean affect detection. Cai, Liu, and Xia [28] study this problem under dependence. Independent Gaussian samples with known identity covariance isolate the lower-bound argument: construct a difficult mean difference while keeping the remaining observations uninformative.

Let $d, n_1, n_2 \geq 1$ be integers. Suppose we observe two independent samples

$$X_i \overset{\text{i.i.d.}}{\sim} N_d(\mu_1, I_d), \quad 1 \leq i \leq n_1, \qquad Y_i \overset{\text{i.i.d.}}{\sim} N_d(\mu_2, I_d), \quad 1 \leq i \leq n_2. \tag{5.9}$$

Write $\delta = \mu_1 - \mu_2$ and fix a known integer $1 \leq s \leq d$. Test

$$H_0 : \delta = 0 \quad \text{against} \quad H_1(\rho) : \|\delta\|_0 \leq s, \quad \|\delta\|_2 \geq \rho, \qquad \rho > 0.$$

The common mean under the null is unrestricted. Let $\mathcal{R}_{n_1,n_2,d,s}(\rho)$ denote the minimax total error (5.1) for tests of the full samples. With $\overline{X}$ and $\overline{Y}$ denoting the sample means, define the effective sample size and contrast by

$$m = \left(n_1^{-1} + n_2^{-1}\right)^{-1}, \qquad D = \overline{X} - \overline{Y} \sim N_d(\delta, m^{-1} I_d). \tag{5.10}$$

The contrast D gives the single-mean experiment with noise variance m^{-1}, so a test based on D provides an upper bound. For a lower bound, however, a test might use information discarded by D. We choose mean pairs with a fixed sample-size-weighted average. Their full-data likelihood ratios depend only on D, making the remaining observations uninformative about these alternatives.

Proposition 5.5 (Sparse Two-Sample Mean Lower Bound) *In experiment* (5.9)*, let* $1 \leq s \leq d$ *and* $m = (n_1^{-1} + n_2^{-1})^{-1}$*. Then*

$$\mathcal{R}_{n_1,n_2,d,s}(\rho) \geq \tfrac{3}{4} \qquad \text{if} \qquad 0 < \rho^2 \leq \frac{s}{2m} \log\left(1 + \frac{\sqrt{d}}{s}\right).$$

Proof For a lower bound, it suffices to restrict the null to $\mu_1 = \mu_2 = 0$ and the alternatives to

$$\mu_1 = \frac{n_2}{n_1 + n_2}\delta, \qquad \mu_2 = -\frac{n_1}{n_1 + n_2}\delta.$$

These choices preserve the difference δ and keep the pooled mean $(n_1\mu_1 + n_2\mu_2)/(n_1 + n_2)$ at zero. Let P_0 and P_δ be the corresponding full-data laws, and write $L_\delta = dP_\delta/dP_0$. The Gaussian likelihood gives

$$\begin{aligned} \log L_\delta &= n_1\langle \mu_1, \overline{X}\rangle + n_2\langle \mu_2, \overline{Y}\rangle - \tfrac{1}{2}\{n_1\|\mu_1\|_2^2 + n_2\|\mu_2\|_2^2\} \\ &= m\langle \delta, D\rangle - \frac{m}{2}\|\delta\|_2^2. \end{aligned}$$

Since $D \sim N_d(0, m^{-1} I_d)$ under P_0, its Gaussian moment-generating function yields

$$\mathbb{E}_0 L_\delta L_{\delta'} = \exp\{m\langle \delta, \delta'\rangle\}.$$

Use the prior π from Proposition 5.3, with $\sigma^2 = m^{-1}$: a uniform s-element support S and independent uniform signs ξ_j. Let P_π be its mixture of full-data laws. Thus

$$\delta_j = a\xi_j \mathbf{1}\{j \in S\}, \qquad a^2 = \frac{1}{2m}\log\left(1 + \frac{\sqrt{d}}{s}\right).$$

Every draw has s nonzero coordinates and squared norm $sa^2 \geq \rho^2$, so the prior is supported on the alternative. The cross-moment is identical to that in Proposition 5.3 with $\sigma^2 = m^{-1}$. Its support-overlap calculation therefore gives

$$\chi^2(P_\pi \| P_0) = \mathbb{E}_{S,S'}\{\cosh(ma^2)^{|S\cap S'|}\} - 1 \leq e^{1/8} - 1 < \tfrac{1}{4},$$

by (5.8) and the same amplitude calibration. Applying the mixture bound (5.3) to these full-data laws gives $\mathcal{R}_{n_1,n_2,d,s}(\rho) \geq 1 - \frac{1}{2}\sqrt{\chi^2(P_\pi \| P_0)} > 3/4$. □

Remark 5.6 (Optimal Procedure and Matching Upper Bound) The test in Remark 5.4, applied to $D = \overline{X} - \overline{Y}$ with noise variance m^{-1} and known sparsity s, attains any prescribed fixed total error in $(0, 1)$ when ρ^2 exceeds a sufficiently large constant multiple of

$$\frac{s}{m} \log\left(1 + \frac{\sqrt{d}}{s}\right).$$

This matches Proposition 5.5 up to constants depending on the error level. The common mean adds no penalty because the law of D depends only on the difference. If $n_2/n_1 \to \infty$, then $m/n_1 \to 1$: the smaller sample limits detection accuracy.

Cai, Liu, and Xia [28, Section 2.1] develop maximum-type tests for sparse alternatives under dependence. With known identity covariance, their statistic is $m \max_j D_j^2$. It targets large coordinate differences, rather than the full sparsity-dependent Euclidean benchmark above.

If the reference mean μ_2 is known, the centered sample mean $\overline{X} - \mu_2 \sim N_d(\delta, n_1^{-1} I_d)$ gives the one-sample problem of Section 5.4.1. Its noise variance is also the limit of $m^{-1} = n_1^{-1} + n_2^{-1}$ as $n_2 \to \infty$ with n_1 fixed. The next section retains sparsity but changes how the signal is observed: each response measures a random linear combination of its coordinates.

5.5 Sparse regression testing with known noise variance

We now test whether a sparse regression contains any signal, with the noise variance known. High-dimensional regression is central to statistics and machine learning, from genetic association to signal recovery from compressed measurements; testing for a signal can precede estimation or predictor selection. Our goal is to determine how small the coefficient norm can be before reliable detection becomes impossible.

Suppose we observe both $X \in \mathbb{R}^{n\times d}$ and $Y \in \mathbb{R}^n$ in the experiment

$$Y = X\beta + \sigma Z, \qquad X_{ij} \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad Z \sim N_n(0, I_n), \qquad X \perp Z, \tag{5.11}$$

where $n, d \geq 1$ are integers, $\sigma > 0$ is known, and $\beta \in \mathbb{R}^d$ is a fixed unknown vector. For a known integer $1 \leq s \leq \sqrt{d}$ and $\rho > 0$, test

$$H_0 : \beta = 0 \quad \text{against} \quad H_1(\rho) : \|\beta\|_0 \leq s, \quad \|\beta\|_2 \geq \rho.$$

Write P_β for the joint law of (X, Y), with expectation $\mathbb{E}_\beta$, and $\mathcal{R}^{\text{reg}}_{n,d,s}(\rho)$ for the minimax total error (5.1). Tests are measurable functions $\varphi(X, Y) \in [0, 1]$, and expectations average over

both design and noise. The restriction $s \le \sqrt{d}$ isolates the sparse regime. Since β is a fixed parameter, the lower-bound prior is chosen independently of the realized design.

Conditionally on X, this is a linear inverse problem, related to the diagonal model of Proposition 2.5. A singular-vector rotation generally destroys coordinate sparsity, so we retain the original coordinates and average the likelihood calculation over the random design.

The joint observations reveal information about the active coordinates, but the response alone can also reveal a signal: marginally, $Y_i \sim N(0, \sigma^2 + \|\beta\|_2^2)$. With known σ, a hard mixture must limit information from both the support and this increase in response variance.

Ingster, Tsybakov, and Verzelen [76] determine asymptotic detection boundaries; Carpentier et al. [48, Theorem 4] give non-asymptotic lower bounds for a broader class of designs. Here equal positive coefficients on a random support yield a direct Gaussian calculation. Its purpose is to make the separate roles of support overlap and response variance explicit.

Proposition 5.7 (Sparse Gaussian Regression Lower Bound) *In experiment* (5.11)*, for integers* $n, d \ge 1$ *and* $1 \le s \le \sqrt{d}$*,*

$$\mathcal{R}^{\mathrm{reg}}_{n,d,s}(\rho) \ge \tfrac{3}{4} \quad \textit{if} \quad 0 < \frac{\rho^2}{\sigma^2} \le \frac{1}{32} \min\left\{ \frac{s}{n} \log\left(1 + \frac{d}{s^2}\right), \frac{1}{\sqrt{n}} \right\}.$$

The proof averages first over response noise at a fixed design, then over the Gaussian design, and finally over two independent support draws. The design average leaves a contribution even when the supports are disjoint; the final average controls their overlap.

Proof Let S be uniform among the s-element subsets of $\{1, \dots, d\}$ and put $\beta = a\mathbf{1}_S$, where $a = \rho/\sqrt{s}$ and $\mathbf{1}_S$ is the coordinate indicator vector of S. This defines a prior π, independent of X, supported on $H_1(\rho)$. Since the design law is the same under every parameter, the likelihood ratio relative to P_0 is

$$L_\beta(X, Y) = \exp\left\{ \frac{Y^\top X\beta}{\sigma^2} - \frac{\|X\beta\|_2^2}{2\sigma^2} \right\}.$$

For an independent prior draw $\beta' = a\mathbf{1}_{S'}$, integration over Y under the null gives

$$\mathbb{E}_0[L_\beta L_{\beta'} \mid X] = \exp\left\{ \frac{\beta^\top X^\top X\beta'}{\sigma^2} \right\}. \tag{5.12}$$

Next average over the design, keeping the two supports fixed. Put $K = |S \cap S'|$, $v = \rho^2/\sigma^2$, and $t = vK/s$. Here v is the signal energy relative to noise variance, and $t = \langle \beta, \beta' \rangle / \sigma^2$ measures alignment between the two draws.

The exponent in (5.12) is a sum over independent design rows, so it suffices to calculate one row's contribution. For a row $x^\top$, let $U = x^\top\beta/\sigma$ and $V = x^\top\beta'/\sigma$. These centered Gaussian variables have variances v and covariance t. Write $\mathbb{E}_X$ for expectation over the design; the required contribution is $\mathbb{E}_X e^{UV}$.

To evaluate this moment, write UV as a difference of independent Gaussian squares. The jointly Gaussian variables $U + V$ and $U - V$ have zero covariance and variances $2(v + t)$ and $2(v - t)$, respectively. They are therefore independent, and $UV = \{(U + V)^2 - (U - V)^2\}/4$ gives

$$UV \overset{d}{=} \tfrac{1}{2}(v + t)G_1^2 - \tfrac{1}{2}(v - t)G_2^2,$$

where G_1, G_2 are independent standard Gaussians. The representation also holds at $t = v$, when $U - V = 0$. The assumed radius gives $0 \le t \le v \le 1/32$, so $v + t < 1$. Using $\mathbb{E}e^{\lambda G_1^2} = (1 - 2\lambda)^{-1/2}$ for $\lambda < 1/2$ yields

$$\mathbb{E}_X e^{UV} = \{(1 - v - t)(1 + v - t)\}^{-1/2}.$$

Multiplying over the n independent design rows gives

$$\mathbb{E}_0 L_\beta L_{\beta'} = \{(1 - t)^2 - v^2\}^{-n/2}. \tag{5.13}$$

The overlap enters through t, but the v^2 term remains even when $K = 0$. To bound these contributions separately, put $u = 2t - t^2 + v^2$. Then $0 \le u \le 2v + v^2 < 1/2$, so $-\log(1-u) \le 2u$ implies

$$\mathbb{E}_0 L_\beta L_{\beta'} \le \exp\{n(2t + v^2)\}.$$

Finally, let $P_\pi = \mathbb{E}_\pi P_\beta$ and average over the supports. The overlap bound (5.8), with dimension d and $z = e^{2nv/s}$, gives

$$\begin{aligned} 1 + \chi^2(P_\pi \| P_0) &\le e^{nv^2} \mathbb{E}_{S,S'} \exp\{2nvK/s\} \\ &\le \exp\left\{nv^2 + \frac{s^2}{d}(e^{2nv/s} - 1)\right\}. \end{aligned}$$

The two restrictions in the proposition now have separate roles: the $n^{-1/2}$ restriction controls nv^2, and the logarithmic restriction controls the support-overlap term. Write $c = 1/32$ and $q = d/s^2$. The assumed radius ensures $nv^2 \le c^2$ and $2nv/s \le 2c \log(1 + q)$. By concavity, $(1 + q)^{2c} \le 1 + 2cq$, so

$$\frac{s^2}{d}(e^{2nv/s} - 1) \le \frac{(1 + q)^{2c} - 1}{q} \le 2c.$$

Consequently,

$$\chi^2(P_\pi \| P_0) \le e^{c^2 + 2c} - 1 < e^{1/8} - 1 < \tfrac{1}{4}.$$

The mixture bound (5.3), applied to the joint observation laws, now gives $\mathcal{R}^{\mathrm{reg}}_{n,d,s}(\rho) \ge 1 - \frac{1}{2}\sqrt{\chi^2(P_\pi \| P_0)} > 3/4$. □

Remark 5.8 (Support Uncertainty and Response Variance) Proposition 5.7 has two separation restrictions. The support-overlap term gives the squared-separation scale $\sigma^2 s \log(1 + d/s^2)/n$, as in sparse Gaussian mean testing. The additional scale $\sigma^2/\sqrt{n}$ comes from information in the responses alone.

For every fixed β with $\|\beta\|_2 = \rho$, the responses are marginally independent $N(0, \sigma^2 + \rho^2)$ variables. Thus the signal increases their variance regardless of its support or signs. To see the relevant scale, let $V_n = n^{-1} \sum_{i=1}^n Y_i^2$. Gaussian moments give

$$\mathbb{E}_\beta V_n = \sigma^2 + \rho^2, \qquad \mathrm{Var}_\beta(V_n) = \frac{2(\sigma^2 + \rho^2)^2}{n}.$$

Under the null, the standard deviation of V_n is $\sqrt{2}\sigma^2/\sqrt{n}$. The squared signal norm must remain small on this scale as well as the support-overlap scale, explaining their minimum in the lower bound.

The likelihood calculation records this contribution even for disjoint supports. Although $\beta^\top \beta' = 0$, the random exponent $\beta^\top X^\top X \beta'/\sigma^2$ need not vanish. Its mean is zero, but its exponential has expectation $(1 - v^2)^{-n/2} > 1$ by (5.13), where $v = \rho^2/\sigma^2$. For small v, the logarithm of this expectation is $nv^2/2 + O(nv^4)$, again giving $\rho^2 \asymp \sigma^2/\sqrt{n}$. Replacing $X^\top X$ by nI_d inside the exponential would incorrectly discard these fluctuations.

Remark 5.9 (Procedure and Upper-Bound Comparison) Carpentier et al. [48, Theorem 3] construct a test from squared least-squares coefficient estimates, centered under the null and thresholded when $s < \sqrt{d}$. With known σ and s, it attains squared separation of order $\sigma^2 s \log(1 + d/s^2)/n$ in this joint Gaussian experiment when $n \geq 9$ and $d \leq \min\{\gamma n, n - 8\}$, for fixed $0 < \gamma < 1$. Since $s \log(1 + d/s^2) \leq \sqrt{d} < \sqrt{n}$, this matches the support term in the lower bound, with constants depending on γ and the desired total error.

The proposition's lower bound holds more broadly. Ingster, Tsybakov, and Verzelen [76, Section 4.1] give asymptotic matching results in high-dimensional regimes under their specified sparsity scaling.

Cai, Guo, and Xia [20] survey group significance and simultaneous inference. For a primary treatment of group significance with nuisance coefficients outside the tested group, see Guo et al. [69, Sections 1.1 and 1.3].

Section 8.1.2 revisits this construction when the noise variance is unknown: adjusting that variance can keep the marginal response law unchanged. The subsequent coordinate-interval argument also varies the design covariance to study honest inference.

5.6 Unknown direction and smoothness in Gaussian testing

We return to direct Gaussian observations and test a zero mean against alternatives separated in norm, now without a sparsity restriction. The finite-dimensional problem isolates uncertainty about direction; the infinite-sequence problem adds a smoothness constraint. The latter models goodness-of-fit testing when a departure from a reference curve is spread over many small coefficients. Both lower bounds use random signs, but smoothness determines how many coordinates can carry the difficult alternatives in the second problem.

5.6.1 The cost of an unknown direction

Suppose we observe $Y \sim N_d(\theta, \sigma^2 I_d)$ in the finite Gaussian model (2.4), with known $d \geq 1$ and $\sigma > 0$. To isolate the cost of an unknown direction, for $\rho > 0$ test

$$H_0 : \theta = 0 \quad \text{against} \quad H_1(\rho) : \|\theta\|_2 \geq \rho.$$

Our goal is to determine how large ρ must be to keep the total testing error below a fixed level, uniformly over all signal directions.

For comparison, suppose the alternative consists of one specified vector θ of norm ρ. The minimum total error is then $2\Phi(-\rho/(2\sigma))$, so separation of order σ suffices, independently of d. Here, however, one test must handle every direction. Unlike the convex hypotheses of Section 5.2, the alternative surrounds the null, and no single projection separates it from zero.

The sign mixture of Section 2.2 assigns every coordinate magnitude $\rho/\sqrt{d}$ and an independent random sign. Every signal has norm ρ, but averaging over the signs makes the mixture harder to distinguish from the null. This is the case $s = d$ of Section 5.4.1, with no support uncertainty.

Proposition 5.10 (Unknown-Direction Gaussian Separation) *In the finite Gaussian model* (2.4)*, test $H_0 : \theta = 0$ against $H_1(\rho) : \|\theta\|_2 \geq \rho$. There exist universal constants $0 < c < C < \infty$ such that the minimax total error is at least $3/4$ at $\rho = c\sigma d^{1/4}$ and at most $1/4$ at $\rho = C\sigma d^{1/4}$.*

Proof For the lower bound, put $\theta_j^{\xi} = \rho\xi_j/\sqrt{d}$ with independent uniform signs $\xi_j \in \{-1, 1\}$. Let π be this prior, P_π its observation mixture, and P_0 the zero-mean observation law. Every prior point belongs to $H_1(\rho)$. It remains to show that P_π is close to P_0.

Relative to P_0, the likelihood ratio for a fixed sign vector is

$$L_\xi(Y) = \exp\left\{\frac{\langle\theta^\xi, Y\rangle}{\sigma^2} - \frac{\rho^2}{2\sigma^2}\right\}.$$

Write $L = \mathbb{E}_\xi L_\xi$ for the mixture likelihood ratio. To bound its second moment, introduce an independent copy ξ' of ξ. Gaussian integration gives

$$\mathbb{E}_0 L_\xi L_{\xi'} = \exp\left\{\frac{\rho^2}{d\sigma^2}\sum_{j=1}^{d}\xi_j\xi_j'\right\}.$$

The products $\xi_j\xi_j'$ are independent uniform signs. Averaging over both sign vectors therefore gives

$$\mathbb{E}_0 L^2 = \left\{\cosh\left(\frac{\rho^2}{d\sigma^2}\right)\right\}^d \leq \exp\left(\frac{\rho^4}{2d\sigma^4}\right).$$

The inequality $\cosh t \leq e^{t^2/2}$ bounds the second moment in terms of $\rho^4/(d\sigma^4)$. Thus a sufficiently small multiple of $\sigma d^{1/4}$ keeps the mixture close to the null. Specifically, at $\rho = \frac{1}{2}\sigma d^{1/4}$,

$$\mathrm{TV}(P_0, P_\pi) \leq \tfrac{1}{2}\sqrt{\mathbb{E}_0 L^2 - 1} \leq \tfrac{1}{2}\sqrt{e^{1/32} - 1} < 1/4.$$

Equation (5.3) gives minimax total error at least $3/4$.

For the upper bound, use a statistic that does not depend on direction: $T = \|Y\|_2^2 - d\sigma^2$. Its mean and variance are

$$\mathbb{E}_\theta T = \|\theta\|_2^2, \qquad \mathrm{Var}_\theta(T) = 2d\sigma^4 + 4\sigma^2\|\theta\|_2^2.$$

Reject when $T > \rho^2/2$. Chebyshev's inequality bounds the null rejection probability by $8d\sigma^4/\rho^4$. If $u = \|\theta\|_2^2 \geq \rho^2$, the miss probability is at most

$$\frac{2d\sigma^4 + 4\sigma^2 u}{(u - \rho^2/2)^2} \leq \frac{8d\sigma^4}{\rho^4} + \frac{16\sigma^2}{\rho^2}.$$

The last inequality uses $u - \rho^2/2 \geq u/2$ and $u \geq \rho^2$. At $\rho = C\sigma d^{1/4}$, the two error bounds sum to at most $16/C^4 + 16/C^2$, since $d \geq 1$. Taking $C = 10$ makes this less than $1/4$. □

The Euclidean separation rate is $\sigma d^{1/4}$: unknown direction costs a factor $d^{1/4}$ relative to a specified alternative, although the test need not recover that direction. The lower-bound balance $\rho^4 \asymp d\sigma^4$ agrees with the upper bound's requirement that squared signal norm exceed fluctuations of order $\sigma^2\sqrt{d}$.

5.6.2 Detecting smooth signals

We now test for a departure from a known baseline curve, with a known smoothness bound but unknown shape. Signal strength is the L_2 norm of the departure, equivalently the ℓ_2 norm of its orthonormal coefficients. We seek both the fixed-error separation rate and the condition under which both errors can vanish. Unlike finite-dimensional testing, the number of coordinates available to a difficult alternative is now limited by smoothness.

After subtracting the baseline, we observe independent $Y_j \sim N(\theta_j, n^{-1})$, $j \geq 1$, as in (3.1). The known integer $n \geq 1$ indexes noise precision, not the number of observed coordinates. Fix known $s, R > 0$ and use the Sobolev ellipsoid $\Theta_s(R^2)$ from (3.2), with weighted-norm radius R. Here s denotes smoothness, not sparsity. For $0 \leq \rho \leq R$, test

$$H_0 : \theta = 0 \quad \text{against} \quad H_1(\rho) : \theta \in \Theta_s(R^2), \quad \|\theta\|_2 \geq \rho.$$

Write $\mathcal{R}_n(\rho)$ for the minimax total error (5.1). The bound $\rho \leq R$ ensures a nonempty alternative. At $\rho = 0$, extend the definition to overlapping hypotheses: the alternative includes the null, so $\mathcal{R}_n(0) = 1$.

Ingster and Suslina [75] develop Gaussian nonparametric testing theory, and Baraud [5] gives non-asymptotic separation radii for ellipsoids and other ℓ_p bodies. We derive the classical Sobolev rate by calibrating the preceding sign mixture to the ellipsoid. The lower bound must select an admissible dimension and amplitude; the matching test must also control the energy lost by truncation.

First fix an integer $m \geq 1$ and allocate the energy among the first m coordinates. Put $\theta_j = a_j\xi_j$ for $j \leq m$, where $a_j \geq 0$ and the ξ_j are independent uniform signs, and set all later coordinates to zero. The conditions

$$\sum_{j\leq m} a_j^2 = \rho^2, \qquad \sum_{j\leq m} j^{2s} a_j^2 \leq R^2$$

put every prior draw in the alternative. For the mixture likelihood ratio L relative to the zero-mean law, the coordinatewise calculation in Proposition 5.10, with noise variance $1/n$, gives

$$\mathbb{E}_0 L^2 = \prod_{j\leq m} \cosh(na_j^2) \leq \exp\left\{\frac{n^2}{2}\sum_{j\leq m} a_j^4\right\}. \tag{5.14}$$

Separation depends on the sum of squares, whereas this information bound depends on fourth powers. Cauchy–Schwarz gives $\sum_{j\leq m} a_j^4 \geq \rho^4/m$, with equality at $a_j = \rho/\sqrt{m}$. Equal amplitudes therefore minimize this second-moment upper bound at fixed m and ρ. They satisfy the smoothness constraint whenever $\rho^2 m^{2s} \leq R^2$.

Increasing m reduces the information bound but places signal in higher-frequency coefficients, increasing the smoothness cost. For small ρ, choose $m \asymp (R/\rho)^{1/s}$, the largest order

allowed by $\rho^2 m^{2s} \le R^2$. The exponent $n^2\rho^4/(2m)$ then has order $n^2\rho^{4+1/s}R^{-1/s}$. Balancing this quantity at a constant gives

$$\rho \asymp R^{1/(4s+1)} n^{-2s/(4s+1)}, \qquad m \asymp (nR^2)^{2/(4s+1)}.$$

For fixed $s, R > 0$, the following theorem absorbs the radius factor into its constants. It also shows that separation below this rate makes testing asymptotically uninformative, whereas separation above it permits both errors to vanish. Equal amplitudes suffice for these rate statements; sharp constants require a more refined profile.

Theorem 5.11 (Smooth-Signal Detection Boundary) *In experiment* (3.1)*, test* $\theta = 0$ *against* $\theta \in \Theta_s(R^2)$ *with* $\|\theta\|_2 \ge \rho$*. Fix* $s, R > 0$ *and let* $r_n = n^{-2s/(4s+1)}$*. There exist constants* $0 < c < C < \infty$ *and* n_0*, depending only on* s, R*, such that for every* $n \ge n_0$*,*

$$\mathcal{R}_n(cr_n) \ge \tfrac{3}{4}, \qquad \mathcal{R}_n(Cr_n) \le \tfrac{1}{4}. \tag{5.15}$$

Moreover, for any sequence $\rho_n \ge 0$*, if* $\rho_n/r_n \to 0$*, then* $\mathcal{R}_n(\rho_n) \to 1$*; if* $\rho_n/r_n \to \infty$ *and* $H_1(\rho_n)$ *is eventually nonempty, then* $\mathcal{R}_n(\rho_n) \to 0$*.*

Proof For the lower bound, take $m = \lfloor n^{2/(4s+1)} \rfloor$ and use the equal-amplitude sign prior above, with $a = \rho/\sqrt{m}$. Every draw has squared norm ρ^2 and

$$\sum_{j \le m} j^{2s}\theta_j^2 \le a^2 m^{2s+1} = \rho^2 m^{2s}.$$

For $\rho = cr_n$ and $c \le R$, this is at most $c^2 \le R^2$. Let P_π be the observation mixture and $L = dP_\pi/dP_0$. Equation (5.14) gives

$$\mathbb{E}_0 L^2 = \{\cosh(na^2)\}^m \le \exp\left(\frac{n^2\rho^4}{2m}\right).$$

For sufficiently large n, $m \ge \frac{1}{2}n^{2/(4s+1)}$, so the exponent is at most c^4. Taking $0 < c \le \min\{R, 1/2\}$ gives $\mathrm{TV}(P_0, P_\pi) \le \frac{1}{2}\sqrt{e^{1/16} - 1} < 1/4$, and the mixture bound (5.3) proves the first inequality.

If $\rho_n/r_n \to 0$, the same construction lies in the ellipsoid eventually and $n^2\rho_n^4/m \to 0$. The mixture bound then tends to one, as does the minimax total error, which is at most one. When $\rho_n = 0$, the alternative includes the null and the total error equals one exactly.

For the upper bound, use the energy in the first $m = \lceil n^{2/(4s+1)} \rceil$ coordinates:

$$T_m = \sum_{j \le m} (Y_j^2 - n^{-1}), \qquad S_m = \sum_{j \le m} \theta_j^2.$$

Gaussian moments and independence give $\mathbb{E}_\theta T_m = S_m$ and

$$\mathrm{Var}_\theta(T_m) = \frac{2m}{n^2} + \frac{4S_m}{n}. \tag{5.16}$$

For $j > m$, the ellipsoid weights satisfy $j^{2s} \ge (m+1)^{2s}$. Hence the sum of squared coefficients beyond m is at most $R^2(m+1)^{-2s}$, giving

$$S_m \ge \|\theta\|_2^2 - R^2(m+1)^{-2s} \ge \rho^2 - R^2 r_n^2.$$

Reject when $T_m > \tau = 4\sqrt{m}/n$. Chebyshev's inequality gives null error at most $1/8$. Since $m \le 2n^{2/(4s+1)}$, we have $\tau \le 4\sqrt{2}\, r_n^2$. For $\rho = Cr_n$ and $C^2 = R^2 + 256$, the retained energy satisfies $S_m \ge 256 r_n^2 \ge 2\tau$. Uniformly over the alternatives, the miss probability is therefore at most

$$\frac{2m/n^2 + 4S_m/n}{(S_m - \tau)^2} \le \frac{8m}{n^2 S_m^2} + \frac{16}{nS_m} \le \frac{16}{256^2} + \frac{16}{256\, nr_n^2} < \frac{1}{8},$$

where $nr_n^2 = n^{1/(4s+1)} \ge 1$. The two maximal errors sum to less than $1/4$. Enlarge n_0 so that $Cr_n \le R$ for every $n \ge n_0$; the alternatives at both cr_n and Cr_n are then nonempty, as required by the fixed-error separation benchmark.

For consistent detection, the preceding fixed-error bounds are insufficient: the null error must also tend to zero. Suppose $b_n = \rho_n/r_n \to \infty$ and the alternative is eventually nonempty. Increase the truncation level to $m_n = \lceil b_n^2 n^{2/(4s+1)} \rceil$ and reject when $T_{m_n} > \rho_n^2/2$. Relative to ρ_n^2, both the omitted squared coefficients and the null standard deviation then become negligible. Indeed,

$$\frac{R^2 m_n^{-2s}}{\rho_n^2} \le R^2 b_n^{-(4s+2)} \to 0, \qquad \frac{m_n}{n^2 \rho_n^4} \le \frac{2}{b_n^2} \to 0,$$

where the second bound uses $m_n \le 2b_n^2 n^{2/(4s+1)}$ for all sufficiently large n. Chebyshev's inequality bounds the null error by $8m_n/(n^2\rho_n^4) = o(1)$. Uniformly over the alternatives, $S_{m_n} \ge (1 - o(1))\rho_n^2$, so the miss probability is bounded by a constant times

$$\frac{m_n}{n^2 \rho_n^4} + \frac{1}{n\rho_n^2}.$$

Both terms vanish, since $n\rho_n^2 = b_n^2 n^{1/(4s+1)} \to \infty$. □

At fixed error probabilities, the hard mixture and matching test use the same order of dimension. The lower bound spreads the signal over as many coefficients as smoothness allows; the test balances truncation error against noise. Section 4.2 uses such sign mixtures to estimate the quadratic functional $Q(\theta) = \sum_j \theta_j^2$. Here the goal is only to distinguish $Q(\theta) = 0$ from $Q(\theta) \ge \rho^2$; Appendix B.2 compares the two benchmarks on this ellipsoid. Applying the sequence-model result to regression or imaging requires a justified equivalence or approximation argument.

The next chapter converts testing difficulty into confidence-set size bounds: a small honest set can induce a test, or coverage transfer can force it to contain separated targets. Section 6.6 applies these reductions to the smooth-signal construction to bound confidence-ball size.

5.7 Exercises

5.1. Prove the two closest-point projection inequalities used in Theorem 5.1. Take $C_0 = \{0\}$ and $C_1 = \{\theta \in \mathbb{R}^d : \|\theta\|_2 = a\}$ with $d \ge 2$ and $a > 0$. Which projection inequality fails, and why does the resulting linear test fail to control error uniformly over C_1?

5.2. In the convex Gaussian problem, fix $\alpha_0, \beta_0 \in (0, 1)$ with $\alpha_0 + \beta_0 < 1$, and write $z_q = \Phi^{-1}(1 - q)$. Use (5.5) to show that a test with maximal errors at most α_0 and β_0 exists exactly when $\Delta/\sigma \ge z_{\alpha_0} + z_{\beta_0}$. Express this condition in terms of the center distance and calibration radii for the two-ball example.

5.3. Construct a different compactly supported s-Hölder bump for Theorem 5.2 and check its class constant and Gaussian divergence.

5.4. In Proposition 5.3, compute the exact support-overlap distribution when $s = 1$ and when $s = d$, and recover the corresponding mixture second moments. Explain why the logarithmic support cost gives way to the unknown-direction scale as s increases.

5.5. In the two-sample experiment (5.9), let $W = (n_1\overline{X} + n_2\overline{Y})/(n_1 + n_2)$ and $D = \overline{X} - \overline{Y}$. Show that W and D are independent, and verify that the lower-bound submodel leaves the law of W unchanged. Derive the full-sample likelihood ratio and explain why the resulting lower bound applies to tests using more than D. What happens to the effective sample size when $n_2 \to \infty$ with n_1 fixed?

5.6. In Proposition 5.7, derive the likelihood cross-moment for disjoint supports. Explain why replacing $X^\top X$ by nI_d inside the exponential gives the wrong answer. Relate the remaining term to the marginal law of Y and to the restriction $n\rho^4/\sigma^4 \lesssim 1$.

5.7. Derive the exact sign-mixture second moment and energy-test variance in Proposition 5.10. Compare with projection onto a known unit vector.

5.8. For Theorem 5.11, check the tail-energy inequality and determine how the constants must change if the noise variance is σ^2/n, with fixed known $\sigma > 0$.

5.9. Derive (5.14) for unequal amplitudes. At fixed total energy and fixed support size, show that equal amplitudes minimize the fourth-power sum. Explain why this calculation alone does not optimize the profile under a weighted Sobolev constraint, even though the equal-amplitude construction gives the correct rate.

5.10. For the smooth-signal problem with $s = 1/8$, $1/4$, and $1/2$, compute the detection radius and the order of the number of coefficients in the hard mixture. Explain how these quantities change with smoothness, and distinguish the fixed-error conclusion from the consistency assertions in Theorem 5.11.

5.11. For an integer $n \geq 1$ and a fixed $\delta > 0$, observe n independent draws and test $\mathrm{Unif}(0, 1)$ against $\mathrm{Unif}(0, \vartheta)$ with $\vartheta \geq 1 + \delta$. Show that the exact minimax total error is $(1 + \delta)^{-n}$.

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6

Honest Confidence Sets

Estimation asks how accurately an unknown target can be recovered; testing asks whether specified hypotheses can be distinguished. A confidence set addresses a further question: how precisely can we describe the target while providing a reliable coverage guarantee? An interval expresses uncertainty about a treatment effect or an optimum, a band describes uncertainty across a response curve, and a ball measures uncertainty in an entire signal. The goal of this chapter is to determine how small these sets can be while retaining their prescribed coverage.

Coverage and size impose competing requirements. Reporting the entire target space guarantees coverage but provides no useful precision. A small set is informative only if its coverage is valid. We call a procedure *honest* over a parameter class when, at every parameter in that class, the reported set contains the true target with at least the nominal probability. This is a repeated-sampling guarantee that must hold at difficult parameters as well as favorable ones. For a confidence band, the guarantee concerns the entire curve simultaneously, not each location separately.

The distinction between accurate estimation and honest uncertainty quantification is especially important at simple signals. An estimator may perform well there, yet the observations may not distinguish that signal from alternatives with substantially different targets. Coverage at those alternatives can then force a large confidence set even at the simple signal. We therefore distinguish the class over which coverage is required from the class, or individual parameter, at which size is evaluated. Taking the two classes to be the same gives the minimax size benchmark; separating them reveals limitations that a worst-case comparison alone can miss.

Section 6.1 develops two lower-bound arguments. A sufficiently small honest set would yield a successful test, contradicting Chapter 5's testing bounds. Alternatively, coverage transfer between close observation laws forces a set to contain separated targets with positive probability. Both arguments use Part I's information comparisons to bound confidence-set size.

Sections 6.2–6.4 study confidence intervals through moduli over convex classes, an uncertainty principle for minimum value and minimizer, and mixture bounds over nonconvex unions. The last setting also reveals the cost of restricting the confidence interval's center to be affine.

Sections 6.5 and 6.6 treat confidence bands and balls. Bands require spatial constructions for full or relaxed coverage. For balls, we distinguish the radius required somewhere in a class from that forced even at zero. Chapter 7 then asks whether one procedure can remain honest over a collection of classes while attaining the smaller size appropriate to each.

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6.1 Honest coverage and size benchmarks

Suppose we observe $X \sim P_\theta$, with unknown $\theta \in \Theta$, and seek a confidence set for $\Psi(\theta)$ in a metric space $(\mathcal{A}, d_{\mathcal{A}})$. We formulate the smallest expected size compatible with honest coverage, distinguishing where coverage is required from where size is evaluated. Testing and coverage transfer then give general diameter and radius bounds.

A confidence procedure reports a data-dependent set $C(X) \subseteq \mathcal{A}$. Assume that $\{x : \Psi(\theta) \in C(x)\}$ is measurable for each fixed θ. Write $C = C(X)$ and define its diameter by

$$\operatorname{diam}_{d_{\mathcal{A}}}(C) = \sup_{a,b \in C} d_{\mathcal{A}}(a,b), \qquad \operatorname{diam}_{d_{\mathcal{A}}}(\varnothing) = 0. \tag{6.1}$$

When the metric is clear, write $\operatorname{diam}(C)$. Fix $0 < \alpha < 1$ and a nonempty coverage class $\mathcal{F} \subseteq \Theta$. The honest $(1-\alpha)$ procedures are

$$C_\alpha(\mathcal{F}; \Psi) = \left\{ C : \inf_{\theta \in \mathcal{F}} \mathbb{P}_\theta\{\Psi(\theta) \in C\} \geq 1 - \alpha \right\}. \tag{6.2}$$

Let $s(C) \geq 0$ measure size, for example by interval length, integrated band width, or squared ball radius, with $x \mapsto s(C(x))$ measurable. For a nonempty evaluation class $\mathcal{G} \subseteq \mathcal{F}$, define the optimal worst-case expected size by

$$S^*_\alpha(\mathcal{G}, \mathcal{F}; \Psi, s) := \inf_{C \in C_\alpha(\mathcal{F}; \Psi)} \sup_{\theta \in \mathcal{G}} \mathbb{E}_\theta s(C). \tag{6.3}$$

Taking $\mathcal{G} = \mathcal{F}$ gives the minimax benchmark; taking $\mathcal{G} = \{\theta_0\}$ evaluates size at one parameter while retaining coverage throughout $\mathcal{F}$. Section 7.1 compares this benchmark with the size attainable when the smaller class is known.

From testing to confidence-set size. For the identity target $\Psi(\theta) = \theta$, test two separated classes by rejecting whenever the confidence set contains an alternative. Under an alternative, coverage controls missed detection. Under the null, rejection requires either a coverage failure or a set large enough to reach the alternative class. A testing lower bound therefore limits how often an honest set can have small diameter.

Proposition 6.1 (An Honest Small Set Induces a Test) *Let (Θ, d) be a metric parameter space equipped with an experiment $\{P_\theta : \theta \in \Theta\}$. Let $0 \leq \alpha, \beta \leq 1$, and let nonempty $\Theta_0, \Theta_1 \subset \Theta$ have separation*

$$\Delta = \inf_{u \in \Theta_0, v \in \Theta_1} d(u,v) > 0.$$

Let $C = C(X) \subseteq \Theta$ have measurable diameter (6.1)*. Assume that $\{\theta \in C(X)\}$ is measurable for each $\theta \in \Theta_0 \cup \Theta_1$, and that $\{C(X) \cap \Theta_1 \neq \varnothing\}$ is measurable. Suppose*

$$\inf_{\theta \in \Theta_0 \cup \Theta_1} \mathbb{P}_\theta\{\theta \in C\} \geq 1 - \alpha, \qquad \inf_{\theta \in \Theta_0} \mathbb{P}_\theta\{\operatorname{diam}(C) < \Delta\} \geq 1 - \beta.$$

Then $\varphi(X) = \mathbf{1}\{C(X) \cap \Theta_1 \neq \varnothing\}$ is a test with

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta\{\varphi = 1\} \leq \alpha + \beta, \qquad \sup_{\theta \in \Theta_1} \mathbb{P}_\theta\{\varphi = 0\} \leq \alpha.$$

Proof For $\theta \in \Theta_0$, the events $\{\theta \in C\}$ and $\{\operatorname{diam}(C) < \Delta\}$ together exclude every point of Θ_1 from C. Hence $\mathbb{P}_\theta\{\varphi = 1\} \le \mathbb{P}_\theta\{\theta \notin C\} + \mathbb{P}_\theta\{\operatorname{diam}(C) \ge \Delta\} \le \alpha + \beta$. For $\theta \in \Theta_1$, $\{\theta \in C\}$ implies $\varphi = 1$, so $\mathbb{P}_\theta\{\varphi = 0\} \le \alpha$. □

To obtain a size lower bound, write $\mathcal{R} = \mathcal{R}(\Theta_0, \Theta_1)$ for the minimax total error (5.1). For an honest C, take $\beta = \sup_{\theta\in\Theta_0} \mathbb{P}_\theta\{\operatorname{diam}(C) \ge \Delta\}$ in the proposition. Its induced test gives $\mathcal{R} \le 2\alpha + \beta$. Hence

$$\sup_{\theta\in\Theta_0} \mathbb{P}_\theta\{\operatorname{diam}(C) \ge \Delta\} \ge (\mathcal{R} - 2\alpha)_+, \qquad \sup_{\theta\in\Theta_0} \mathbb{E}_\theta \operatorname{diam}(C) \ge \Delta(\mathcal{R} - 2\alpha)_+.$$

The expected-diameter bound follows from the probability bound as in Section 2.5. It is positive when $\mathcal{R} > 2\alpha$ and concerns the worst case over Θ_0, not every null parameter.

For a scalar target $T(\theta)$, the same construction rejects when the reported interval meets $T(\Theta_1)$, provided this event is measurable. Replace Δ by the positive distance between $T(\Theta_0)$ and $T(\Theta_1)$; interval diameter is its length. This reduction is useful when a testing lower bound is already available. Bull and Nickl [13] use testing conditions to study adaptive L_2 confidence sets.

A direct two-prior bound. Coverage can also be transferred directly between close observation mixtures whose target sets are separated. Cai and Guo [19, Lemma 1] use this argument for scalar targets constant on each prior's support. The metric-space version below allows separated target sets and bounds diameter and squared radius.

Let the experiment be a probability kernel on measurable parameter and observation spaces. Let $H_0, H_1 \subseteq \mathcal{F}$, let π_i be a prior supported on H_i, and let

$$P_{\pi_i} = \int P_\theta \, \pi_i(d\theta), \qquad A_i = \Psi(H_i) \subseteq \mathcal{A}, \qquad i = 0, 1.$$

Suppose the two target sets are separated by

$$\Delta := \inf\{d_{\mathcal{A}}(a_0, a_1) : a_0 \in A_0,\ a_1 \in A_1\} > 0.$$

Write $\mathbb{P}_{\pi_i}$ and $\mathbb{E}_{\pi_i}$ for probability and expectation under P_{π_i}. Choosing $H_0 \subseteq \mathcal{G}$ yields a lower bound over $\mathcal{G}$; choosing $H_0 = \{\theta_0\}$ yields one at θ_0.

Lemma 6.2 (Two-Prior Lower Bound for Confidence-Set Diameter) *For the priors and separated target sets above, let $C = C(X) \in \mathcal{C}_\alpha(\mathcal{F}; \Psi)$ have measurable diameter. Assume that the intersection events $\{C(X) \cap A_i \ne \varnothing\}$, $i = 0, 1$, are measurable. Then*

$$\mathbb{P}_{\pi_0}\{\operatorname{diam}_{d_{\mathcal{A}}}(C) \ge \Delta\} \ge \big(1 - 2\alpha - \operatorname{TV}(P_{\pi_0}, P_{\pi_1})\big)_+, \tag{6.4}$$

and consequently

$$\sup_{\theta\in H_0} \mathbb{E}_\theta \operatorname{diam}_{d_{\mathcal{A}}}(C) \ge \Delta\big(1 - 2\alpha - \operatorname{TV}(P_{\pi_0}, P_{\pi_1})\big)_+. \tag{6.5}$$

If every realized set is a metric ball with measurable radius $r(C) \ge 0$, then

$$\sup_{\theta\in H_0} \mathbb{E}_\theta r(C)^2 \ge \frac{\Delta^2}{4}\big(1 - 2\alpha - \operatorname{TV}(P_{\pi_0}, P_{\pi_1})\big)_+. \tag{6.6}$$

All three bounds remain valid with total variation replaced by $\frac{1}{2}\chi^2(P_{\pi_1}, P_{\pi_0})^{1/2}$ whenever the displayed χ^2 divergence is finite.

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Proof For $i = 0, 1$, let $E_i = \{C \cap A_i \neq \varnothing\}$. If $\theta \in H_i$, covering $\Psi(\theta)$ implies E_i, since $\Psi(\theta) \in A_i$. Honesty therefore gives $\mathbb{P}_\theta(E_i) \geq 1 - \alpha$ throughout H_i; averaging over π_i yields $\mathbb{P}_{\pi_i}(E_i) \geq 1 - \alpha$. Total variation transfers E_1 to the first mixture law:

$$\mathbb{P}_{\pi_0}(E_1) \geq \mathbb{P}_{\pi_1}(E_1) - \mathrm{TV}(P_{\pi_0}, P_{\pi_1}) \geq 1 - \alpha - \mathrm{TV}(P_{\pi_0}, P_{\pi_1}).$$

Therefore

$$\mathbb{P}_{\pi_0}(E_0 \cap E_1) \geq 1 - 2\alpha - \mathrm{TV}(P_{\pi_0}, P_{\pi_1}).$$

On $E_0 \cap E_1$, the set C contains one point from each target set, so its diameter is at least Δ. Nonnegativity of probability gives the positive part in (6.4). For a metric ball, the triangle inequality also forces radius at least $\Delta/2$. Taking expectations under P_{π_0} and using

$$\sup_{\theta \in H_0} \mathbb{E}_\theta s(C) \geq \int \mathbb{E}_\theta s(C)\, \pi_0(d\theta)$$

with $s(C) = \mathrm{diam}(C)$ or $s(C) = r(C)^2$ proves (6.5) and (6.6), respectively. The χ^2 versions follow from Lemma 1.6. □

Table 6.1 summarizes how the size criterion and lower-bound construction depend on the confidence procedure.

Object	Coverage event	Size criterion	Lower-bound construction
Interval	$T(\theta) \in [\ell, u]$	Expected length $\mathbb{E}(u - \ell)$	Pairs chosen through moduli bound length; mixtures can sharpen bounds for nonconvex and sparse classes.
Band	$f(t) \in [L(t), U(t)]$ for all t	Expected integrated width $\mathbb{E}\int(U - L)$	Localized alternatives can force width on a positive fraction of the domain. Simultaneous coverage can also introduce a logarithmic factor.
Ball	$\theta \in \{a : \|a - \delta\|_2 \leq r\}$	Expected squared radius $\mathbb{E}(r^2)$	Sign families give worst-case radius bounds; smaller-amplitude mixtures force radius even at a fixed signal.

Table 6.1 *Coverage, size, and lower-bound constructions. Here ℓ, u are interval endpoints, L, U are band envelopes, and δ, r are a Euclidean ball's center and radius.*

For scalar intervals, target separation directly bounds length; balls use the squared-radius version of the lemma. Bands require an additional spatial argument: integrated width is not sup-norm diameter, so the proof must account for where the band is forced to be wide.

6.2 Honest confidence intervals over convex classes

Point values, averages, and contrasts require uncertainty assessments as well as estimates. We seek short honest intervals for the quantities studied in Section 4.1, using the Gaussian sequence or continuous white-noise model (1.1)–(1.2), including finite-dimensional sequences. The noise level is $\varepsilon = n^{-1/2}$, with $n > 0$. Write f for the signal, X for the full observation, and $T(f)$ for the real-valued target. The norm $\|\cdot\|_2$ is ℓ_2 for sequences, $L_2[0, 1]$ for functions, and Euclidean for finite vectors.

Our goal is to distinguish the length forced at a particular signal from the optimal worst-case length over a convex class. The local modulus (4.19) fixes one signal and varies its

competitor; the ordinary modulus (4.2) varies both. In each case, coverage transfer turns a target gap between statistically close signals into a length bound. For affine targets over convex classes, optimal recovery supplies a matching worst-case upper bound.

A confidence interval is $CI(X) = [\ell(X), u(X)] \subset \mathbb{R}$, with measurable endpoints satisfying $\ell(X) \le u(X)$ and realized length $L(CI) = u(X) - \ell(X)$. Fix $0 < \alpha < 1$. For a nonempty coverage class $\mathcal{F}$, let $I_\alpha(\mathcal{F}, T)$ consist of the intervals in $C_\alpha(\mathcal{F}; T)$ from (6.2). For a nonempty evaluation class $\mathcal{G} \subseteq \mathcal{F}$, write

$$L(CI, f) = \mathbb{E}_f L(CI), \qquad L(CI, \mathcal{G}) = \sup_{f \in \mathcal{G}} L(CI, f). \tag{6.7}$$

The smallest worst-case expected length on $\mathcal{G}$ is

$$L^*_\alpha(\mathcal{G}, \mathcal{F}, T) = \inf_{CI \in I_\alpha(\mathcal{F}, T)} L(CI, \mathcal{G}). \tag{6.8}$$

Write $L^*_\alpha(\mathcal{F}, T) = L^*_\alpha(\mathcal{F}, \mathcal{F}, T)$ when length is evaluated over the full class, and $L^*_\alpha(f, \mathcal{F}, T) = L^*_\alpha(\{f\}, \mathcal{F}, T)$ when it is evaluated at f. In both cases coverage is required throughout $\mathcal{F}$.

The next two lemmas specialize Lemma 6.2 to interval length, first for mixtures and then for a pair of parameters. Neither requires a Gaussian model or a convex class. Gaussian likelihoods sharpen the comparison; convexity and an affine target enter the later attainability result.

Lemma 6.3 (Composite-Hypothesis Lower Bound) *Let H_0 and H_1 be subsets of a measurable parameter space $H = H_0 \cup H_1$, let $\theta \mapsto P_\theta$ be the probability kernel of the experiment, and let $T : H \to \mathbb{R}$ be the scalar target. Suppose that*

$$T(\theta) = \mu_0 \quad (\theta \in H_0), \qquad T(\theta) = \mu_1 \quad (\theta \in H_1).$$

Let π_i be a prior supported on H_i, with observation mixture P_{π_i}, $i = 0, 1$. Every $CI \in I_\alpha(H, T)$ satisfies

$$\sup_{\theta \in H_0} \mathbb{E}_\theta L(CI) \ge |\mu_1 - \mu_0| \big(1 - 2\alpha - \mathrm{TV}(P_{\pi_1}, P_{\pi_0})\big)_+,$$

and, by Lemma 1.6,

$$\sup_{\theta \in H_0} \mathbb{E}_\theta L(CI) \ge |\mu_1 - \mu_0| \Big(1 - 2\alpha - \tfrac{1}{2}\chi^2(P_{\pi_1}, P_{\pi_0})^{1/2}\Big)_+.$$

The same lower bound also applies to $\sup_{\theta \in H} \mathbb{E}_\theta L(CI)$.

Proof If $\mu_0 = \mu_1$, both bounds follow from nonnegative length. Otherwise apply Lemma 6.2 with $\Psi = T$, target sets $A_i = \{\mu_i\}$, and $\Delta = |\mu_1 - \mu_0|$. The intersection events reduce to $\{\mu_i \in CI\}$, which are measurable because the interval endpoints are measurable. Interval diameter equals length, so the lemma gives both inequalities; enlarging the supremum from H_0 to H gives the final assertion. □

Point-mass priors give the two-point form below. The total-variation bound holds at each parameter separately, a fact used in the local-modulus argument.

Lemma 6.4 (Two-Point Lower Bound for Confidence-Interval Length) *Let $\Theta = \{\theta_0, \theta_1\}$, let P_{θ_i} be the corresponding laws, and let the scalar target values be*

$$\mu_i = T(\theta_i), \qquad \mu_0 \le \mu_1.$$

If $CI(X)$ is a confidence interval for $T(\theta)$ with two-point coverage

$$\min_{i=0,1} \mathbb{P}_{\theta_i}\{\mu_i \in CI(X)\} \ge 1 - \alpha,$$

then, with $L(CI)$ denoting interval length,

$$\max\Big\{\mathbb{E}_{\theta_0} L(CI), \mathbb{E}_{\theta_1} L(CI)\Big\} \ge (\mu_1 - \mu_0)\big(1 - 2\alpha - \mathrm{TV}(P_{\theta_1}, P_{\theta_0})\big)_+.$$

Consequently,

$$\max\Big\{\mathbb{E}_{\theta_0} L(CI), \mathbb{E}_{\theta_1} L(CI)\Big\} \ge (\mu_1 - \mu_0)\Big(1 - 2\alpha - \tfrac{1}{2}\chi^2(P_{\theta_1}, P_{\theta_0})^{1/2}\Big)_+.$$

Proof Apply Lemma 6.3 with $H_i = \{\theta_i\}$ and point-mass priors $\pi_i = \delta_{\theta_i}$. It gives both bounds already for $\mathbb{E}_{\theta_0} L(CI)$, hence also for the displayed maximum. Interchanging the points gives the same total-variation bound for $\mathbb{E}_{\theta_1} L(CI)$. □

Gaussian likelihoods permit a sharper transfer of coverage than the generic total-variation bound. Cai and Low [30, Proposition 1] obtain the following two-point inequality by retaining the prescribed coverage level in the Neyman–Pearson comparison.

Remark 6.5 (Explicit Gaussian Two-Point Formula) Let $X \sim N(\theta, \sigma^2)$ with $\sigma > 0$, $0 < \alpha < 1$, parameter space $\Theta = \{\theta_0, \theta_1\}$, and $\theta_0 < \theta_1$. If $CI(X)$ is a $(1 - \alpha)$ confidence interval for θ, then

$$\mathbb{E}_{\theta_i} L\big(CI(X)\big) \ge (\theta_1 - \theta_0)\Big(1 - \alpha - \Phi\big((\theta_1 - \theta_0)/\sigma - z_\alpha\big)\Big)_+, \qquad i = 0, 1.$$

To verify the formula, let $A_1 = \{\theta_1 \in CI(X)\}$. Among events with $\mathbb{P}_{\theta_1}(A_1) \ge 1 - \alpha$, the Neyman–Pearson lemma (Lemma 1.8) minimizes $\mathbb{P}_{\theta_0}(A_1)$ at the upper tail $\{X \ge \theta_1 - \sigma z_\alpha\}$. Its probability under P_{θ_0} is

$$1 - \Phi\big((\theta_1 - \theta_0)/\sigma - z_\alpha\big).$$

Intersecting A_1 with $\{\theta_0 \in CI(X)\}$ under P_{θ_0} gives the stated lower bound for $\mathbb{E}_{\theta_0} L(CI)$. Reversing the two points gives the identical expression under P_{θ_1}.

For signals f_0, f_1 with $\|f_1 - f_0\|_2 > 0$ in the sequence or white-noise model, the likelihood ratio depends only on the one-dimensional Gaussian observation in the direction $f_1 - f_0$. The same calculation uses target gap $|T(f_1) - T(f_0)|$ and standardized separation $\sqrt{n}\,\|f_1 - f_0\|_2$.

We now bound interval length for $f(1/2)$ at a fixed f in the white-noise model. Choose a nearby g whose target gap nearly attains the local modulus. Transferring coverage from g to f bounds the probability of a long interval at this f, not only at a worst-case signal.

Theorem 6.6 (Local-Modulus Lower Bound for Interval Length) *In the continuous white-noise model* (1.2) *with noise level $n^{-1/2}$, let $\mathcal{F}_c$ be a convex class of pointwise-defined $L_2[0,1]$ representatives and let $T(f) = f(1/2)$. Fix $0 < \alpha < 1$, $CI \in I_\alpha(\mathcal{F}_c, T)$, and $f \in \mathcal{F}_c$. Use the local modulus* (4.19) *with this target and the L_2 semimetric, and assume it is finite at the radii below. Then for every $r > 0$,*

$$\mathbb{P}_f\left\{L(CI) \geq \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right)\right\} \geq 1 - 2\alpha - r,$$

and therefore

$$\mathbb{E}_f L(CI) \geq (1 - 2\alpha - r)\, \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right).$$

Proof Fix $\delta > 0$ and choose $g \in \mathcal{F}_c$ so that the local modulus is nearly attained:

$$|g(1/2) - f(1/2)| \geq \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right) - \delta, \qquad \|g - f\|_2 \leq \frac{r}{\sqrt{n}}.$$

The Gaussian KL formula and Pinsker's inequality imply

$$\mathrm{TV}(P_f, P_g) \leq \sqrt{n}\, \|f - g\|_2 \leq r.$$

The same bound holds after adjoining any independent randomization used to construct the interval, since a common product factor leaves total variation unchanged.

The point-mass case of (6.4), with target values $f(1/2)$ and $g(1/2)$, now gives

$$\mathbb{P}_f\left\{L(CI) \geq \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right) - \delta\right\} \geq 1 - 2\alpha - r.$$

If the target values coincide, the near-modulus choice makes the threshold nonpositive and nonnegative length gives the same inequality. As $\delta \downarrow 0$, the length events decrease to the event in the theorem. Continuity of probability therefore gives the first bound. The expected-length bound follows because

$$\mathbb{E}_f L(CI) \geq \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right) \mathbb{P}_f\left\{L(CI) \geq \omega\left(\frac{r}{\sqrt{n}}, f; \mathcal{F}_c\right)\right\}.$$

□

For $0 < r < 1 - 2\alpha$, the result gives a positive probability of a long interval at the specified f. Convexity is not needed for this lower bound: honesty and a finite local modulus suffice. Convexity becomes important when constructing intervals that attain a worst-case benchmark.

Worst-case length over a convex class. The ordinary modulus determines the optimal worst-case length. In either direct Gaussian model, let $\mathcal{F}$ be nonempty and convex, T affine, and $0 < \alpha < 1/2$. Assume $\omega(\delta, \mathcal{F})$ is finite for every $\delta > 0$ and tends to zero as $\delta \downarrow 0$. Then

$$\frac{1 - 2\alpha}{2}\, \omega\left(\frac{1 - 2\alpha}{\sqrt{n}}, \mathcal{F}\right) \leq L_\alpha^*(\mathcal{F}, T) \leq \omega\left(\frac{2z_{\alpha/2}}{\sqrt{n}}, \mathcal{F}\right).$$

Donoho's affine optimal-recovery intervals [55, Theorem 2] give the upper bound. For the lower bound, a nearly extremal pair at distance at most $(1 - 2\alpha)/\sqrt{n}$ has total variation at

most $(1-2\alpha)/2$; coverage transfer and a limiting argument suffice. Concavity of the modulus makes the bounds comparable for fixed α. Attaining the local benchmark at every f is a separate adaptation question (Section 7.4).

A fixed-design pointwise interval. For a concrete construction, return to the regression model of Theorem 5.2. Let $f \in \Lambda^s(M)$, $0 < s \le 1$, $M > 0$, and fix $t_0 \in (0,1)$. Recall that $\overline{Y}_h$ averages the N_h observations with $|i/n - t_0| \le h$. With $h = n^{-1/(2s+1)}$, this window lies in $[0,1]$ and $N_h \ge nh$ for sufficiently large n. For $0 < \alpha < 1$, a bias allowance and Gaussian quantile give

$$CI_h = \left[\overline{Y}_h - Mh^s - \frac{z_{\alpha/2}}{\sqrt{N_h}},\ \overline{Y}_h + Mh^s + \frac{z_{\alpha/2}}{\sqrt{N_h}}\right].$$

The bounds $|\mathbb{E}_f\overline{Y}_h - f(t_0)| \le Mh^s$ and $\overline{Y}_h - \mathbb{E}_f\overline{Y}_h \sim N(0, N_h^{-1})$ ensure coverage at least $1-\alpha$ uniformly over $\Lambda^s(M)$, with deterministic length of order $n^{-s/(2s+1)}$. Coverage is for the fixed point, not simultaneously over the domain.

6.3 Convex optimization: honest intervals

An interval for a minimizer describes uncertainty about the optimal decision; an interval for the minimum describes uncertainty about its performance. As in Section 4.5, we observe the full path in the Gaussian white-noise model (1.2) on $[0,1]$, with known noise level $\varepsilon > 0$. The signal $f \in \mathcal{F}_{\mathrm{cvx}}$ is convex with a unique minimizer, and the targets are its location $Z(f)$ and value $M(f)$, defined in Section 4.5.1.

Our goal is short expected length at each function, with coverage throughout $\mathcal{F}_{\mathrm{cvx}}$. Coverage transfer still applies to these nonlinear targets: the cap geometry from Chapter 4 identifies nearby functions with separated minima or minimizers. The resulting uncertainty principle explains why precise inference about the minimum value need not locate the minimizer precisely.

Fix $0 < \alpha < 0.3$. To bound the length of any globally honest interval, first require coverage only at a pair of functions. Using $I_\alpha(\{f,g\},T)$ from Section 6.2, define the function-specific two-point interval benchmark

$$L_{T,\alpha}(\varepsilon; f) = \sup_{g \in \mathcal{F}_{\mathrm{cvx}}} \inf_{C \in I_\alpha(\{f,g\},T)} \mathbb{E}_f L(C), \qquad T \in \{Z, M\}. \tag{6.9}$$

For each pair $\{f,g\}$, the interval may know both candidates, but not which generated the data. Coverage is required at both, whereas expected length is evaluated only at f; the estimation benchmark (4.18) instead uses the larger risk at f and g. Every interval honest over $\mathcal{F}_{\mathrm{cvx}}$ is admissible for every pair, so its expected length at f is at least $L_{T,\alpha}(\varepsilon; f)$. Attaining these benchmarks with one globally honest procedure is a separate requirement, addressed after the example below.

The affine-functional theory of Section 6.2 does not supply the upper bound here. Both targets are nonlinear, and convexity of each function does not make the unique-minimizer class convex: the average of $|t - 1/4|$ and $|t - 3/4|$ has an interval of minimizers.

Cai, Chen, and Zhu [17, Theorems 2.1–2.2] establish the functionwise interval benchmarks and uncertainty principle. The proof below combines two-point coverage transfer with their cap geometry; their upper comparisons give tighter constants. Recall the local moduli

$\omega_T(\cdot; f)$ and cap quantities $\rho_T(\cdot; f)$ from Section 4.5.1, and set $a_\alpha = 5/6 - 2\alpha$ and $c_\alpha = 2z_\alpha$. The range $\alpha < 0.3$ ensures $a_\alpha > 0$ and $c_\alpha > 1$, as used below; it is not a sharp restriction on the interval problem.

Theorem 6.7 (Interval Benchmarks and Uncertainty for Convex Minima) *In the Gaussian white-noise model above, let $0 < \alpha < 0.3$, $\varepsilon > 0$, and $f \in \mathcal{F}_{\mathrm{cvx}}$. For each $T \in \{Z, M\}$, the two-point honest-interval benchmark in* (6.9) *satisfies*

$$a_\alpha\, \omega_T(\varepsilon/3; f) \le L_{T,\alpha}(\varepsilon; f) \le \omega_T(c_\alpha \varepsilon; f), \tag{6.10}$$

and, more explicitly,

$$\frac{a_\alpha}{3} \rho_T(\varepsilon; f) \le L_{T,\alpha}(\varepsilon; f) \le 3c_\alpha \rho_T(\varepsilon; f).$$

Consequently the two interval benchmarks obey the functionwise uncertainty principle

$$\frac{a_\alpha^3}{54}\, \varepsilon^2 \le L_{Z,\alpha}(\varepsilon; f) L_{M,\alpha}(\varepsilon; f)^2 \le 81 c_\alpha^3\, \varepsilon^2. \tag{6.11}$$

The bounds are uniform over $\mathcal{F}_{\mathrm{cvx}}$, the class of convex functions with a unique minimizer, and make no curvature or smoothness assumption.

Proof For the lower bound, fix $T \in \{Z, M\}$ and $g \in \mathcal{F}_{\mathrm{cvx}}$, and put $\Delta = |T(g) - T(f)|$ and $d = \|g - f\|_2$. The point-mass case of Lemma 6.3 gives, for every interval honest at both f and g,

$$\mathbb{E}_f L(C) \ge \Delta\{1 - 2\alpha - \mathrm{TV}(P_f, P_g)\}_+.$$

The Gaussian white-noise KL formula and Pinsker's inequality give $\mathrm{TV}(P_f, P_g) \le d/(2\varepsilon)$. When $d \le \varepsilon/3$, the displayed lower bound is at least $a_\alpha \Delta$. Taking the infimum over C, then the supremum over such g, proves the first inequality in (6.10).

For the upper bound on the benchmark, construct an interval for each known pair. If $d < c_\alpha \varepsilon$, the deterministic interval with endpoints $T(f)$ and $T(g)$ covers both targets and has length $\Delta \le \omega_T(c_\alpha \varepsilon; f)$. If $d \ge c_\alpha \varepsilon$, the likelihood-ratio test between the two Gaussian white-noise laws has error probability $\Phi(-d/(2\varepsilon)) \le \alpha$ under each law. Report the singleton $\{T(f)\}$ or $\{T(g)\}$ according to the test. This zero-length interval has the required pairwise coverage. Taking the supremum over g proves (6.10).

It remains to express both bounds at the same noise level ε. Put $H = \rho_M(\varepsilon; f)$. For $0 < s \le 1$, $(M(f) + sH - f)_+ \le s(M(f) + H - f)_+$ pointwise. Thus a cap of height sH has L_2 norm at most $s\varepsilon$, giving $\rho_M(s\varepsilon; f) \ge s\rho_M(\varepsilon; f)$.

For the location quantity, take t with $f(t) \le M(f) + H$. Convexity gives $f(Z(f) + s(t - Z(f))) \le M(f) + sH$. Since $\rho_M(s\varepsilon; f) \ge sH$, this point lies in the cap sublevel set at noise level $s\varepsilon$, at distance $s|t - Z(f)|$ from the minimizer. Taking the supremum over t gives $\rho_Z(s\varepsilon; f) \ge s\rho_Z(\varepsilon; f)$. Apply these inequalities with $s = 1/3$, and then with $s = 1/c_\alpha$ at noise level $c_\alpha \varepsilon$, to obtain

$$\rho_T(\varepsilon/3; f) \ge \tfrac{1}{3}\rho_T(\varepsilon; f), \qquad \rho_T(c_\alpha \varepsilon; f) \le c_\alpha \rho_T(\varepsilon; f).$$

The cap–modulus comparisons of Theorem 4.21 now yield the two explicit bounds on $L_{T,\alpha}$. Multiply the bound for Z by the square of the bound for M, and use (4.20). The lower coefficient is $(a_\alpha/3)^3/2 = a_\alpha^3/54$; the upper coefficient is $3(3c_\alpha)^3 = 81c_\alpha^3$. □

Example 6.8 (Local Shape and Interval Length) The uncertainty principle fixes the product order, while local shape determines the individual length orders. Consider $f(t) = |t - 1/2|^k$, with fixed $k \geq 1$. The two-point benchmarks evaluate length at this function and range over competitors in all of $\mathcal{F}_{\mathrm{cvx}}$, not just the power family. They therefore lower-bound length at f for every globally honest interval.

The cap calculation in Section 4.5.2 gives the local moduli for both targets. Substituting them into Theorem 6.7, for fixed $0 < \alpha < 0.3$, yields

$$L_{Z,\alpha}(\varepsilon; f) \asymp \varepsilon^{2/(2k+1)}, \qquad L_{M,\alpha}(\varepsilon; f) \asymp \varepsilon^{2k/(2k+1)} \quad \text{as } \varepsilon \downarrow 0,$$

with comparison constants depending only on k and α. At the sharp minimum $k = 1$, both lengths have order $\varepsilon^{2/3}$. At the quadratic minimum $k = 2$, the location length has order $\varepsilon^{2/5}$, whereas the value length has the smaller order $\varepsilon^{4/5}$.

A flatter minimum allows statistically close alternatives to move the minimizer substantially while changing the minimum value little. Increasing k therefore slows the shrinkage of the location benchmark but accelerates that of the value benchmark. Their product, with value length squared, remains of order ε^2: a narrow interval for the optimal value need not locate the optimal decision precisely. □

Remark 6.9 (Optimal Procedure and Matching Upper Bound) Cai, Chen, and Zhu [17, Theorems 3.2 and 3.4] enlarge a data-selected dyadic interval to cover the minimizer and use local averages to form an interval for the minimum value. Each interval has coverage at least $1 - \alpha$ throughout $\mathcal{F}_{\mathrm{cvx}}$ and expected length at most an α-dependent constant times its benchmark (6.9) at every f. Both benchmarks are therefore attainable without knowing f. The uncertainty principle relates the difficulties of the two targets; it does not require sacrificing accuracy for one to improve the other.

Without convexity, smoothness alone need not permit shrinking intervals for a minimizer. To see this, return to the fixed-design Hölder regression experiment of Theorem 4.26, with $0 < \beta \leq 1$ and fixed $M, \sigma > 0$. Both the observation model and parameter class differ from the convex white-noise setting.

Apply Lemma 6.3 to that proof's hard pairs. Their observation laws have total variation at most $1/4$ for the minimum and $1/8$ for the minimizer. With $\gamma = \frac{1}{4}\min\{M, \sigma\}$ and $0 < \alpha < 1/4$, every interval honest over $\Lambda^\beta(M)$ for the minimum value has worst-case expected length at least $\gamma n^{-\beta/(2\beta+1)}/4$ for $n \geq n_0$. For the minimizer, honesty over $\Lambda^\beta_{\mathrm{uniq}}(M)$ forces worst-case expected length at least $3/16$ for every $n \geq 1$. These are lower bounds, not matching rates or a direct rate comparison with the convex white-noise experiment.

6.4 Nonconvex classes: unions, mixtures, and convex hulls

We return to linear functionals, now asking whether honest intervals can exploit structure such as sparsity without knowing the active coordinates. In the Gaussian sequence or continuous white-noise model of Section 6.2, with noise level $n^{-1/2}$, we seek an honest interval for T over a known union G of convex components. Each component represents a possible structure, such as a fixed support. The benchmark is the minimax expected length $L^*_\alpha(G, T)$, with coverage and worst-case evaluation over the union.

Following Cai and Low [30], we distinguish the difficulty of the interval problem from the limitations of affine centers. Two-point coverage transfer gives a bound through the modulus over G; affine centers face the potentially larger modulus over its convex hull. A sparse example then shows how a mixture sharpens the two-point bound and why nonlinear centers can be essential. Unlike adaptation in Section 7.4, the goal is optimal length over the union, not separate subclass benchmarks.

The convexity and intersection assumptions below permit the upper comparison; the lower-bound proof needs only a nonempty class and a finite modulus.

Theorem 6.10 (Lower Bound over a Nonconvex Union) *Consider either Gaussian model from Section 6.2, with linear functional T and the corresponding L_2 or ℓ_2 metric. Let $G = \bigcup_{j=1}^{k} \mathcal{F}_j$ be nonempty, with convex, pairwise-intersecting components. Fix $0 < \alpha < 1/2$ and assume $\omega(z_\alpha/\sqrt{n}, G) < \infty$, using the ordinary modulus (4.2) for T. With $L_\alpha^*(G) = L_\alpha^*(G, T)$ as defined in Section 6.2,*

$$L_\alpha^*(G) \geq \left(\frac{1}{2} - \alpha\right)\omega\left(\frac{z_\alpha}{\sqrt{n}}, G\right).$$

Proof Fix $\delta > 0$ and choose $f_0, f_1 \in G$ such that

$$|T(f_1) - T(f_0)| \geq \omega\left(\frac{z_\alpha}{\sqrt{n}}, G\right) - \delta, \qquad \|f_1 - f_0\|_2 \leq \frac{z_\alpha}{\sqrt{n}}.$$

Restrict any $CI \in I_\alpha(G, T)$ to the two-point subexperiment $\{f_0, f_1\}$, and put

$$d = \sqrt{n}\,\|f_1 - f_0\|_2 \leq z_\alpha.$$

For $d > 0$, apply the Gaussian likelihood-ratio calculation in Remark 6.5 to $\{T(f_1) \in CI\}$. Its probability is at least $1 - \alpha$ under P_{f_1} and hence at least $1 - \Phi(d - z_\alpha) \geq 1/2$ under P_{f_0}. The same conclusion holds when $d = 0$, since the laws coincide. Under P_{f_0}, both target values therefore belong to the interval with probability at least $1/2 - \alpha$, giving

$$\max\left\{\mathbb{E}_{f_0} L(CI), \mathbb{E}_{f_1} L(CI)\right\} \geq \left(\frac{1}{2} - \alpha\right)|T(f_1) - T(f_0)|.$$

Every interval honest over G therefore has supremum expected length at least the right-hand side. Taking the infimum over all such intervals gives

$$L_\alpha^*(G) \geq \left(\frac{1}{2} - \alpha\right)\left\{\omega\left(\frac{z_\alpha}{\sqrt{n}}, G\right) - \delta\right\}.$$

Letting $\delta \downarrow 0$ proves the claim. □

The hard pair may lie in different components, so the union modulus captures uncertainty that a within-component calculation can miss.

Remark 6.11 (Optimal Procedure and Matching Upper Bound) Cai and Low [30, Theorem 4] construct confidence intervals from affine estimators with directional bias control.

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Dividing the allowed noncoverage probability among the intervals ensures that their intersection remains honest over the union. Its worst-case expected length gives

$$L_\alpha^*(G) \le 12\,\omega\left(\frac{z_{\alpha/(2k)}}{\sqrt{n}}, G\right), \qquad 0 < \alpha < \frac{1}{2}.$$

For fixed k and α, this matches the lower-bound order. The sparse example below shows why the quantile can grow with k.

Can an interval with an affine center attain the unrestricted benchmark? Its center's worst-case absolute bias is unchanged on replacing the class by its convex hull. Cai and Low [30, Theorem 5] use this fact to show that the convex-hull modulus governs length, even with a data-dependent radius and coverage required only on the original class.

In either Gaussian model from Section 6.2, a finite-variance affine estimator has the form $a + \sum_i u_i Y_i$ with $u \in \ell_2$ in the sequence model, or $a + \int u\,dY$ with $u \in L_2$ in white noise. For such an estimator $\widehat{T}$, consider the interval

$$CI = [\widehat{T} - \gamma, \widehat{T} + \gamma]$$

with nonnegative random radius γ.

Theorem 6.12 (Affine Centers Pay the Convex-Hull Modulus) *Let $\mathcal{F}$ be a nonempty coverage class and T a linear functional on the ambient Hilbert space. Suppose the affine-centered interval above belongs to $I_\alpha(\mathcal{F}, T)$, where $0 < \alpha < 1/2$. Assume the ordinary modulus* (4.2) *on* $\mathrm{conv}(\mathcal{F})$ *is finite at $2z_{\alpha/2}/\sqrt{n}$. Then*

$$L(CI, \mathcal{F}) \ge C(\alpha)\,\omega\left(\frac{2z_{\alpha/2}}{\sqrt{n}}, \mathrm{conv}(\mathcal{F})\right),$$

where one may take

$$C(\alpha) = \min\left\{\frac{1-2\alpha}{4}, \frac{z_{0.25}(1-2\alpha)}{4z_{\alpha/2}}\right\} > 0.$$

If the interval has deterministic length, then

$$L(CI) \ge \frac{1}{2}\omega\left(\frac{2z_{\alpha/2}}{\sqrt{n}}, \mathrm{conv}(\mathcal{F})\right).$$

Proof The convex-hull modulus forces either substantial bias or substantial variance of the center. In either case, honesty forces the radius to be large with positive probability. Since the bias map $f \mapsto \mathbb{E}_f\widehat{T} - T(f)$ is affine, its absolute value at a convex combination cannot exceed the largest absolute bias among the combined signals. Thus

$$B = \sup_{f\in\mathcal{F}} |\mathbb{E}_f\widehat{T} - T(f)| = \sup_{f\in\mathrm{conv}(\mathcal{F})} |\mathbb{E}_f\widehat{T} - T(f)|.$$

Let

$$\delta = \omega\left(\frac{2z_{\alpha/2}}{\sqrt{n}}, \mathrm{conv}(\mathcal{F})\right) \quad \text{and} \quad \sigma_{\widehat{T}}^2 = \mathrm{Var}(\widehat{T}).$$

The claims are immediate if $\delta = 0$, so assume $\delta > 0$. If both $B < \delta/4$ and $\sigma_{\widehat{T}} < \delta/(4z_{\alpha/2})$ held, then for any $f, g \in \mathrm{conv}(\mathcal{F})$ with $\|f - g\|_2 \le 2z_{\alpha/2}/\sqrt{n}$ we would have

$$|T(f) - T(g)| \le |T(f) - \mathbb{E}_f\widehat{T}| + |\mathbb{E}_f\widehat{T} - \mathbb{E}_g\widehat{T}| + |\mathbb{E}_g\widehat{T} - T(g)|.$$

Now $|T(f) - \mathbb{E}_f\widehat{T}| \le B$ and $|\mathbb{E}_g\widehat{T} - T(g)| \le B$, while the affine estimator has Gaussian mean shift satisfying

$$|\mathbb{E}_f\widehat{T} - \mathbb{E}_g\widehat{T}| \le \sqrt{n}\,\sigma_{\widehat{T}}\|f - g\|_2 \le 2z_{\alpha/2}\sigma_{\widehat{T}} < \frac{\delta}{2}.$$

The common upper bound $2B + 2z_{\alpha/2}\sigma_{\widehat{T}} < \delta$ contradicts the definition of δ after taking the supremum over admissible pairs. Thus

$$B \ge \frac{\delta}{4} \qquad \text{or} \qquad \sigma_{\widehat{T}} \ge \frac{\delta}{4z_{\alpha/2}}.$$

Bias case. Suppose $B \ge \delta/4$. For any $0 < \delta_{\mathrm{app}} < \delta/4$, choose $f \in \mathcal{F}$ so that

$$B_f := |\mathbb{E}_f\widehat{T} - T(f)| \ge \delta/4 - \delta_{\mathrm{app}}.$$

This choice remains valid when $B = +\infty$. Under P_f, the statistic $\widehat{T} - T(f)$ is Gaussian with mean $\pm B_f$ and variance $\sigma^2_{\widehat{T}}$. For every $0 \le t < B_f$, the event $\{|\widehat{T} - T(f)| \le t\}$ lies strictly to one side of the Gaussian mean. By symmetry its probability is at most $1/2$; if $\sigma_{\widehat{T}} = 0$, that probability is 0. Thus

$$\mathbb{P}_f\{|\widehat{T} - T(f)| \le t\} \le \frac{1}{2} \qquad (0 \le t < B_f).$$

Coverage is impossible on $\{\gamma \le t\} \cap \{|\widehat{T} - T(f)| > t\}$, since it requires $|\widehat{T} - T(f)| \le \gamma$. Hence

$$1 - \alpha \le \mathbb{P}_f\{T(f) \in CI\} \le \mathbb{P}_f\{\gamma > t\} + \mathbb{P}_f\{|\widehat{T} - T(f)| \le t\} \le \mathbb{P}_f\{\gamma > t\} + \frac{1}{2}.$$

Thus $\mathbb{P}_f\{\gamma > t\} \ge 1/2 - \alpha$ for $0 \le t < B_f$; nonnegativity of γ then gives

$$\mathbb{E}_f L(CI) = 2\mathbb{E}_f\gamma \ge 2t\mathbb{P}_f\{\gamma > t\} \ge 2t\left(\frac{1}{2} - \alpha\right).$$

Letting $t \uparrow B_f$, taking the supremum over $f \in \mathcal{F}$, and then letting $\delta_{\mathrm{app}} \downarrow 0$ yields

$$L(CI, \mathcal{F}) \ge \left(\frac{1}{4} - \frac{\alpha}{2}\right)\delta.$$

Variance case. Suppose $\sigma_{\widehat{T}} \ge \delta/(4z_{\alpha/2})$. Honesty gives, for every $f \in \mathcal{F}$,

$$1 - \alpha \le \mathbb{P}_f\{|\widehat{T} - T(f)| \le \gamma\}.$$

Let $Z \sim N(0,1)$ and split according to whether $\gamma \le z_{0.25}\sigma_{\widehat{T}}$. For a Gaussian variable with variance $\sigma^2_{\widehat{T}}$, the probability of lying in $[-z_{0.25}\sigma_{\widehat{T}}, z_{0.25}\sigma_{\widehat{T}}]$ is largest when its mean is zero. Therefore

$$\mathbb{P}_f\{|\widehat{T} - T(f)| \le \gamma,\ \gamma \le z_{0.25}\sigma_{\widehat{T}}\} \le \mathbb{P}\{|Z| \le z_{0.25}\} = \frac{1}{2}.$$

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Consequently,

$$\mathbb{P}_f\{\gamma > z_{0.25}\sigma_{\widehat{T}}\} \geq \frac{1}{2} - \alpha,$$
$$\mathbb{E}_f L(CI) \geq 2z_{0.25}\sigma_{\widehat{T}}\Big(\frac{1}{2} - \alpha\Big) \geq \frac{z_{0.25}(1-2\alpha)}{4z_{\alpha/2}}\,\delta.$$

Taking the smaller of the constants from the bias and variance cases proves the first claim.

Deterministic length. Let γ be deterministic. In the bias case, for every sufficiently small $\delta_{\text{app}} > 0$ choose $f \in \mathcal{F}$ with $B_f \geq \delta/4 - \delta_{\text{app}}$. Coverage larger than $1/2$ forces $\gamma > t$ for every $0 \leq t < B_f$, hence $\gamma \geq B_f$. Letting $\delta_{\text{app}} \downarrow 0$ gives $L(CI) = 2\gamma \geq \delta/2$. In the variance case, honesty at any $f \in \mathcal{F}$ implies

$$1 - \alpha \leq \mathbb{P}_f\{|\widehat{T} - T(f)| \leq \gamma\} \leq \mathbb{P}\{|Z| \leq \gamma/\sigma_{\widehat{T}}\},$$

so $\gamma \geq z_{\alpha/2}\sigma_{\widehat{T}} \geq \delta/4$. In either case $L(CI) = 2\gamma \geq \delta/2$, proving the fixed-length claim. □

Remark 6.13 (Procedure and Upper-Bound Comparison) When the modulus of $\text{conv}(\mathcal{F})$ is finite and tends to zero at zero, Donoho's optimal-recovery construction [55, Theorem 2] gives a fixed-length interval centered at an affine estimator, with length of the same modulus order; see Section 6.2. Honesty on the convex hull implies honesty on $\mathcal{F}$. This matches the lower bound within the affine-centered class, not necessarily the unrestricted minimax length.

The sparse-vector ("nearly-black") example of Cai and Low [30] shows how much is lost by restricting the center to be affine. In the finite Gaussian model (2.4), take $\theta = f$, $\sigma = n^{-1/2}$, and $d = n$, and estimate

$$T(f) = \sum_{i=1}^{d} f_i.$$

For an integer sparsity level $1 \leq s_n < n^\gamma$ with fixed $0 < \gamma < 1/2$, let $G = \Theta(s_n, d)$, the finite-dimensional class in (3.6). For every subset $I \subseteq \{1, \ldots, d\}$ of size s_n, let

$$\mathcal{F}_I = \{f \in \mathbb{R}^d : \ f_j = 0 \text{ for } j \notin I\}.$$

Then $G = \bigcup_I \mathcal{F}_I$ is a union of $k = \binom{n}{s_n}$ linear subspaces. Using the ordered modulus from (4.3), we have

$$\omega(\varepsilon, \mathcal{F}_I, G) = \omega(\varepsilon, G) = \sqrt{\min\{2s_n, n\}}\,\varepsilon.$$

For all sufficiently large n, $2s_n \leq n$, giving $\sqrt{2s_n}\,\varepsilon$. A difference between two s_n-sparse vectors has at most $\min\{2s_n, n\}$ nonzero coordinates, so Cauchy–Schwarz gives the upper bound. To attain it, choose supports with minimal overlap and make the difference constant and positive on their union. The coordinates on the first support may be negative, since the class imposes no sign restriction.

The convex hull of G is $\mathbb{R}^d$, with modulus $\sqrt{n}\,\varepsilon$. At noise scale $\varepsilon \asymp n^{-1/2}$, affine centers therefore force worst-case expected length bounded away from zero. For unrestricted intervals, the two-point bound gives only $\sqrt{s_n/n}$. A mixture of positive signals on uniformly chosen supports has a common, larger target gap from zero. The support-overlap calculation of Section 5.4.1 keeps its observation law close to the null, so coverage transfer forces length even at zero.

Proposition 6.14 (Interval-Length Lower Bound for Sparse Vectors) *In the finite Gaussian model* (2.4)*, take* $\theta = f$*,* $\sigma = n^{-1/2}$*, and* $d = n$*, with target* $T(f) = \sum_{i=1}^{d} f_i$*. Fix* $0 < \gamma < 1/2$ *and* $0 < \alpha < 1/2$*. For integer* $1 \le s_n < n^{\gamma}$*, let* $G = \Theta(s_n, n)$ *and* $k = \binom{n}{s_n}$*. For all sufficiently large* n*, every* $CI(Y) \in I_\alpha(G, T)$ *satisfies*

$$\mathbb{E}_0 L(CI(Y)) \ge \left(\frac{1}{2} - \alpha\right) \frac{s_n}{\sqrt{n}} \sqrt{\frac{1}{2} \log\left(\frac{n}{s_n^2}\right)} \ge c(\alpha, \gamma)\, \omega\left(\frac{\sqrt{\log k}}{\sqrt{n}}, G\right),$$

where $c(\alpha, \gamma) > 0$ *depends only on* α, γ*, and* $\omega(\cdot, G)$ *is the ordinary modulus of* T *over* G*.*

Proof We mix over supports while keeping every target value the same. Write $s = s_n$ and let $\lambda > 0$. For each subset I of size s, define the signal

$$f_I(j) = \sqrt{\frac{\lambda}{n}} \mathbf{1}\{j \in I\}, \qquad f_0 \equiv 0,$$

and let ψ_{f_I} and ψ_{f_0} denote the corresponding joint Gaussian densities. Consider the mixture

$$\psi^* = \binom{n}{s}^{-1} \sum_I \psi_{f_I}.$$

Every mixture signal has target value $T(f_I) = s\sqrt{\lambda/n}$, while $T(f_0) = 0$. For independent uniformly chosen s-subsets I, I', put $J = |I \cap I'|$; this overlap has a hypergeometric law. A direct Gaussian calculation gives

$$\int \frac{\psi_{f_I} \psi_{f_{I'}}}{\psi_{f_0}} = e^{J\lambda}, \qquad \int \frac{(\psi^*)^2}{\psi_{f_0}} = \mathbb{E} e^{J\lambda}.$$

Take $\lambda = \frac{1}{2} \log(n/s^2) > 0$. The support-overlap bound (5.8), with dimension $d = n$ and $z = e^{\lambda}$, gives

$$1 \le \mathbb{E} e^{\lambda J} \le \exp\{s^2(e^{\lambda} - 1)/n\} = \exp\{s/\sqrt{n} - s^2/n\} = 1 + o(1),$$

because $s/\sqrt{n} \le n^{\gamma - 1/2} \to 0$, uniformly over the allowed sparsity levels. It follows that

$$\int \frac{(\psi^*)^2}{\psi_{f_0}} \to 1, \qquad \int |\psi^* - \psi_{f_0}| \to 0.$$

Honesty gives coverage of the common alternative target under the mixture. Since the mixture and null laws are close, this target must also be covered with nearly the same probability at zero. Applying Lemma 6.3 to the two priors therefore gives, for every $\varepsilon > 0$ and all sufficiently large n,

$$\mathbb{E}_{f_0} L(CI(Y)) \ge (1 - 2\alpha - \varepsilon) s \sqrt{\lambda/n}.$$

Taking $\varepsilon = 1/2 - \alpha$ proves the first displayed bound. The second follows from

$$\omega\left(\frac{\sqrt{\log k}}{\sqrt{n}}, G\right) = \sqrt{2s}\, \frac{\sqrt{\log k}}{\sqrt{n}}$$

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and $\log \binom{n}{s} \asymp s \log(n/s)$. Under $s < n^\gamma$ with $\gamma < 1/2$, $\log(n/s)$ and $\log(n/s^2)$ are comparable up to constants depending only on γ, so

$$\omega\left(\frac{\sqrt{\log k}}{\sqrt{n}}, G\right) \lesssim_\gamma \frac{s}{\sqrt{n}}\sqrt{\log(n/s^2)}.$$

□

Remark 6.15 (Optimal Procedure and Matching Upper Bound) Cai and Low [30, Section 5.2] use the finite-union construction above to obtain a nonlinear interval that attains the mixture lower-bound order under the stated sparsity restriction:

$$L^*_\alpha(G, T) \asymp_{\alpha,\gamma} \frac{s_n}{\sqrt{n}}\sqrt{\log(n/s_n^2)}.$$

This length tends to zero, whereas affine-centered intervals have worst-case expected length bounded away from zero. The mixture also shows that the unrestricted rate is forced at zero itself, not only at signals with many nonzero coordinates.

6.5 Confidence bands

An interval at one prespecified point does not describe uncertainty about a whole regression curve. For a dose–response relationship or time trend, we may wish to identify regions exceeding a reference level. A confidence band assigns an interval $[L(t), U(t)]$ to each $t \in [0, 1]$; simultaneous coverage supports conclusions at locations chosen after inspecting the data.

Suppose the curve is observed with Gaussian noise on a fixed grid and belongs to a known Hölder class. We seek the smallest expected integrated width compatible with coverage of the entire curve. Pointwise intervals do not suffice: each may cover with probability $1 - \alpha$ while failures occur at different locations. Honesty requires validity uniformly over the parameter class; simultaneity requires coverage at all locations on the same event (Figure 6.1). Both apply even when the class is known.

We also study the relaxations proposed by Cai, Low, and Ma [38]: control the proportion of the domain missed, or the size of departures from the band relative to its width. These criteria permit some failures while retaining guarantees about the curve.

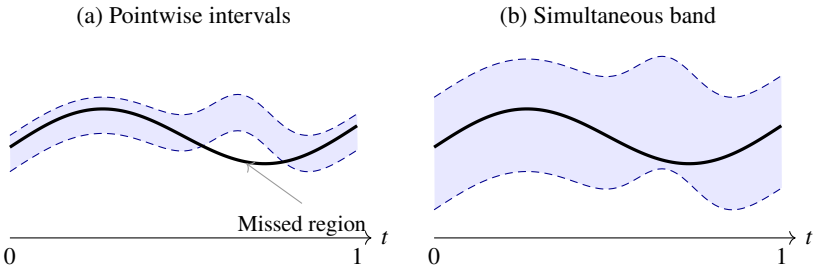


Figure 6.1 Pointwise and simultaneous coverage for the same curve f (solid black), with dashed interval boundaries. In (a), coverage fails over a short region; in (b), the band contains the entire curve. Pointwise validity controls failure at each fixed location; simultaneous validity controls any failure across the domain. Both are repeated-sampling properties, not established by one realization. The nested envelopes are illustrative, not required; shading shows realized integrated width.

For an integer $n \geq 1$ and known $\sigma > 0$, observe the fixed-design regression model

$$y_i = f(t_i) + \sigma\varepsilon_i, \qquad t_i = \frac{i}{n}, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad 1 \leq i \leq n. \tag{6.12}$$

For both full and relaxed coverage, use the Hölder class $\Lambda^\beta(M)$ from (2.1), with known $\beta, M > 0$. The fixed grid will matter: sufficiently narrow perturbations can fall between observations.

A confidence band $CB = [L, U]$ has jointly measurable random endpoints with $L(t) \leq U(t)$ for every $t \in [0,1]$ and data realization. Its full-coverage event must be measurable under each admissible f. We measure size by expected integrated width, also average width on $[0,1]$:

$$w(CB, f) = \mathbb{E}_f \int_0^1 (U(t) - L(t))\, dt, \qquad w(CB, \mathcal{F}) = \sup_{f \in \mathcal{F}} w(CB, f). \tag{6.13}$$

This criterion averages width, not coverage. A fully honest band must cover the whole curve on one event of probability at least $1 - \alpha$, even though size is not measured at the widest point.

Unlike an interval-length bound, an integrated-width bound must force uncertainty over a region, not just at one point. Disjoint bumps supply the local ambiguities; the coverage criterion determines how many the band must accommodate.

6.5.1 Full coverage and integrated width

Fix $0 < \alpha < 1/2$. We first require the band to cover the entire curve with probability at least $1 - \alpha$, uniformly over the class. The collection of honest bands is

$$\mathcal{B}_\alpha(\mathcal{F}) = \left\{ CB = [L, U] : \inf_{f \in \mathcal{F}} \mathbb{P}_f\{f(t) \in [L(t), U(t)]\ \forall t \in [0,1]\} \geq 1 - \alpha \right\}, \tag{6.14}$$

and the minimax benchmark is

$$W_\alpha(\mathcal{F}) = \inf_{CB \in \mathcal{B}_\alpha(\mathcal{F})} w(CB, \mathcal{F}). \tag{6.15}$$

We place disjoint bumps across the domain, as in Section 4.4, and convert full coverage of the curve into simultaneous coverage of their coefficients. The construction recurs, but the reduction changes: Assouad accumulates estimation errors, whereas the band must avoid missing any coefficient.

The following Gaussian problem isolates this requirement. Each unknown coordinate is zero or one, so an interval shorter than one must exclude a possibility. Under a product prior, the posterior factors across coordinates. If many remain ambiguous, reporting short intervals for too many of them makes simultaneous coverage unlikely. This forces a large total length. The worst-case bound yields the band theorem; the fixed-signal bound also enters the adaptation argument in Section 7.5.

Suppose we observe

$$X_i \sim N(c_n\theta_i, 1), \qquad i = 1, \dots, n,$$

independently, where $\theta = (\theta_1, \dots, \theta_n)$ ranges over the binary hypercube

$$\Theta_n = \{0, 1\}^n.$$

Let $C(X) = (C_1(X), \dots, C_n(X))$ be a coordinatewise interval-valued confidence set for θ, with $\theta \in C(X)$ meaning $\theta_i \in C_i(X)$ for every i. Each $C_i(X) = [\ell_i(X), u_i(X)]$ has finite measurable endpoints satisfying $\ell_i(X) \le u_i(X)$ for every data realization. Write $L(C_i(X))$ for its Lebesgue length.

Proposition 6.16 (Simultaneous Interval Length on a Gaussian Hypercube) *In this Gaussian experiment, let* $0 \le \alpha \le 1$ *and suppose* C *has simultaneous coverage*

$$\inf_{\theta\in\Theta_n} \mathbb{P}_{X|\theta}\{\theta_i \in C_i(X) \text{ for all } i\} \ge 1 - \alpha.$$

If $c_n = \sqrt{c \log n}$ *with fixed* $0 \le c < 1$*, then for every* $a < 1$ *and every* $\delta > 0$ *there exists* $N = N(\alpha, c, a, \delta)$*, uniformly over honest procedures, such that, for all* $n \ge N$*,*

$$\sup_{\theta\in\Theta_n} \mathbb{P}_{X|\theta}\Big\{\sum_{i=1}^n L(C_i(X)) \ge an\Big\} \ge 1 - \alpha - \delta,$$

and, for all sufficiently large n*,*

$$\sup_{\theta\in\Theta_n} \mathbb{E}_{X|\theta} \sum_{i=1}^n L(C_i(X)) \ge (1 - \alpha - \delta)n.$$

If, in addition, all coordinate sets have the same length, $L(C_i(X)) = L(C_1(X))$*, then for every* $\delta > 0$ *there exists* $N = N(\alpha, c, \delta)$*, uniformly over such honest procedures, such that for all* $n \ge N$ *and all* $\theta \in \Theta_n$*,*

$$\mathbb{E}_{X|\theta} \sum_{i=1}^n L(C_i(X)) \ge (1 - \alpha - \delta)n.$$

Finally, suppose $\alpha < 1/2$*. There are constants* $c_\alpha > 0$ *and* $C_\alpha > 0$ *such that, if* $0 < c_n \le c_\alpha$*, then for every* n *and every* $\theta \in \Theta_n$*,*

$$\mathbb{E}_{X|\theta} \sum_{i=1}^n L(C_i(X)) \ge C_\alpha n.$$

This fixed-signal conclusion does not require the coordinate intervals to have equal lengths. All assertions also hold when the intervals depend on an auxiliary random variable independent of X and θ, whose law does not depend on θ.

Proof of the worst-case bounds If independent parameter-free randomization R is used, condition the posterior calculations below on (X, R). The posterior of θ is then the same as that given X, and all likelihood ratios below depend only on X. Coverage is averaged jointly over X and R; no conditional coverage assumption is needed.

Worst-case length. Replace each $C_i(X)$ by the smallest member of $\{\varnothing, \{0\}, \{1\}, [0,1]\}$ containing $C_i(X) \cap \{0,1\}$. This preserves coverage at every $\theta \in \Theta_n$ and cannot increase length, so a lower bound for the reduced sets also bounds the original intervals.

Put the product prior π on Θ_n under which the coordinates θ_i are i.i.d. $\mathrm{Ber}(1/2)$, and let P denote the resulting joint law of (θ, X), with probability $\mathbb{P}$ and expectation $\mathbb{E}$. The probability assertion is immediate from nonnegative length when $a \le 0$, so fix $0 < a < 1$ and define

$$B_n(a) = \left\{ \sum_{i=1}^{n} L(C_i(X)) \le an \right\}.$$

Because $L([0,1]) = 1$ and the other three possible sets have length zero, on $B_n(a)$ at most an coordinates are left unresolved. If any coordinate set is empty, conditional simultaneous coverage is zero. Otherwise, at least $(1-a)n$ coordinates are resolved as singletons.

To bound the posterior error of a singleton report, fix a window width $H > 0$ and set

$$I_i = \mathbf{1}\{X_i \in [-H, H] \cup [c_n - H, c_n + H]\}, \qquad N_H = \sum_{i=1}^{n} I_i.$$

Under the prior predictive law, N_H is binomial. Each normal component places probability at least $2\Phi(H) - 1$ in these windows. Choose H large enough and then b so that $a < b < 2\Phi(H) - 1$. Concentration gives $\mathbb{P}\{N_H \ge bn\} \ge 1 - \delta$ for every prescribed $\delta > 0$ and all sufficiently large n.

For $X_i = x$ in either window, Bayes' formula gives

$$\frac{\mathbb{P}(\theta_i = 1 \mid X_i = x)}{\mathbb{P}(\theta_i = 0 \mid X_i = x)} = \exp\left(c_n x - \frac{c_n^2}{2} \right).$$

When $c_n = \sqrt{c \log n}$ with $c < 1$, this likelihood ratio is bounded between $n^{-c/2+o(1)}$ and $n^{c/2+o(1)}$ uniformly over the two windows. Thus both posterior probabilities are bounded below by a quantity $q_n \ge n^{-c/2+o(1)}$, and in particular $nq_n \to \infty$. On every such ambiguous coordinate, a singleton report has posterior probability at most $1 - q_n$ of being correct.

Let $A_n(b) = \{N_H \ge bn\} \cap B_n(a)$. On this event, at least bn coordinates lie in the windows, but at most an retain both possible values. Unless a coordinate set is empty, at least $(b-a)n$ such coordinates are singletons. Conditional on the data, these reports are fixed and the posterior coordinates are independent, so their simultaneous correctness probability is at most $(1 - q_n)^{(b-a)n}$. An empty set gives zero coverage; thus

$$\mathbb{E}\left[\mathbf{1}\{\theta \in C(X)\} \mathbf{1}_{A_n(b)} \right] \le (1 - q_n)^{(b-a)n}.$$

Using $\mathbb{P}(A_n(b)^c) \le \delta + \mathbb{P}(B_n(a)^c)$ and averaging honest simultaneous coverage over the prior gives

$$\begin{aligned} 1-\alpha &\le 2^{-n} \sum_{\theta\in\Theta_n} \mathbb{P}_{X|\theta}\{\theta \in C(X)\} = \mathbb{E}\,\mathbf{1}\{\theta \in C(X)\} \\ &\le (1-q_n)^{(b-a)n} + \delta + \mathbb{P}(B_n(a)^c), \\ \mathbb{P}(B_n(a)) &\le \alpha + \delta + (1-q_n)^{(b-a)n}. \end{aligned}$$

Since $nq_n \to \infty$, the last term tends to zero. The identity $\mathbb{P}(B_n(a)) = \mathbb{E}_\pi \mathbb{P}_{X|\theta}(B_n(a))$ then gives at least one θ for which $\mathbb{P}_{X|\theta}(B_n(a)) \le \alpha + \delta + o(1)$. Applying this estimate with a smaller probability slack absorbs the $o(1)$ term. We conclude that for every $a < 1$ and $\delta > 0$,

$$\sup_{\theta\in\Theta_n} \mathbb{P}_{X|\theta}\Big\{\sum_{i=1}^{n} L(C_i(X)) \ge an\Big\} \ge 1-\alpha-\delta,$$

for all sufficiently large n.

The expectation bound follows from $\mathbb{E}Z \ge an\,\mathbb{P}\{Z \ge an\}$ for $Z = \sum_i L(C_i(X))$. More explicitly, for a prescribed $\delta > 0$, first apply the probability bound with a smaller slack $\delta_1 \in (0, \delta)$ and then choose $a < 1$ sufficiently close to one that $a(1-\alpha-\delta_1) \ge 1-\alpha-\delta$. This gives the displayed coefficient $1-\alpha-\delta$ for all sufficiently large n. □

For equal-length intervals, the fixed-parameter assertion uses a mixture uniform over a binary vector and its single-coordinate flips. With sufficiently small standardized amplitude, a two-point coverage comparison instead bounds each coordinate's expected length, without requiring equal lengths. Appendix B.3 gives both proofs.

To apply the proposition, choose m bumps whose standardized amplitude is of order $\sqrt{\log m}$. Integrating the band over their supports produces coordinate intervals; their total length bounds integrated width from below after rescaling. Cai, Low, and Ma [38, Theorem 2] establish the resulting worst-case order. The proof below checks class membership and reduces the full fixed-grid observations to the Gaussian problem.

Theorem 6.17 (Expected-Width Lower Bound for Fully Honest Bands) *In regression model* (6.12)*, fix $\beta, M, \sigma > 0$ and $0 < \alpha < 1/2$. There exist constants $C_1, C_2 > 0$, depending only on (α, β, σ) and the fixed bump normalization, with the following property. For every $\varepsilon > 0$, there is $n_0 = n_0(\alpha, \beta, M, \sigma, \varepsilon)$, independent of the band, such that for every $n \ge n_0$ and every $CB = [L, U] \in \mathcal{B}_\alpha(\Lambda^\beta(M))$,*

$$\sup_{f\in\Lambda^\beta(M)} \mathbb{P}_f\left\{\int_0^1 (U-L)\,dt \ge C_1 M^{1/(2\beta+1)}\Big(\frac{\log n}{n}\Big)^{\beta/(2\beta+1)}\right\} \ge 1-\alpha-\varepsilon,$$

and hence the worst-case expected integrated width satisfies

$$\sup_{f\in\Lambda^\beta(M)} w(CB, f) \ge C_2 M^{1/(2\beta+1)}\Big(\frac{\log n}{n}\Big)^{\beta/(2\beta+1)}.$$

Proof The proof has three steps: calibrate smooth bumps, extract their Gaussian coefficients without discarding information, and turn band coverage into coefficient coverage.

Bump construction. Choose a nonnegative, nonzero C^∞ function g supported in $[1/4, 3/4]$, equal to one on $[3/8, 5/8]$, and satisfying $0 \le g \le 1$. Write $J = \int_0^1 g(t)\,dt > 0$. Rescaling the derivatives and comparing points within and across the disjoint supports gives a fixed $\tau_\beta > 0$, depending only on β and g, such that every defining bound in (2.1) is at most A/τ_β for the hypercube below. Thus $0 \le A \le (M/2)\tau_\beta$ places it in $\Lambda^\beta(M)$, with each constraint at most $M/2$.

Fix $c \in (0,1)$ and $a \in (0,1)$. Put $u = (2\beta+1)^{-1}$, choose a fixed $k > 0$ sufficiently small that $8\,2^{2\beta}\sigma^2 c\,k^{2\beta+1} \le \tau_\beta^2/4$, and, for all sufficiently large n, set

$$m = \left\lfloor k \left(\frac{M^2 n}{\log n}\right)^u \right\rfloor, \qquad p = \left\lfloor \frac{n}{8m} \right\rfloor, \qquad r = n - 8mp, \qquad h = \frac{8p}{n}.$$

There are m cells of width h, each with $8p$ assigned design points, and r points left over. Then $p \ge 1$, $0 \le r < 8m$, and $m \le h^{-1} \le 2m$. For $i = 0, \dots, m-1$ use the cell $I_i = (ih, (i+1)h)$ and the bump $g_i(t) = h^\beta g(t/h - i)$. The last r design points are unused. Every $f_\theta = A\sum_{i=0}^{m-1} \theta_i g_i$, with $\theta \in \{0,1\}^m$, belongs to $\Lambda^\beta(M)$ when $A \le (M/2)\tau_\beta$. Each block has the same positive empirical energy

$$S = h^{2\beta} \sum_{v=1}^{8p} g\left(\frac{v}{8p}\right)^2, \qquad 2ph^{2\beta} \le S \le 8ph^{2\beta}.$$

Choose $A = \sigma\sqrt{c \log m}/\sqrt{S}$. Since $S \ge nh^{2\beta+1}/4$ and $h^{-1} \le 2m$, we have $A^2 \le 8\,2^{2\beta}\sigma^2 c\, m^{2\beta+1} \log m/n$. The choice of k and $\log m \le \log n$ therefore give $0 < A \le (M/2)\tau_\beta$ for all sufficiently large n. The calibration is exact: $A\sqrt{S}/\sigma = \sqrt{c \log m}$.

Exact normal reduction. Let Q be the matrix whose ith row is the sampled g_i vector divided by $\sqrt{S}$. Disjoint blocks give $QQ^\top = I_m$. From the complete response vector define $Y = Qy/\sigma$ and $R = (I - Q^\top Q)y/\sigma$. Under f_θ, Y has independent $N(\sqrt{c \log m}\,\theta_i, 1)$ coordinates; R is independent of Y and has a law not depending on θ.

The identity $y = \sigma(Q^\top Y + R)$ recovers the full data. Hence a band using all observations induces intervals based on Y and independent, parameter-free randomization R. Proposition 6.16 applies to these intervals with probabilities taken jointly over Y and R; no conditional coverage is assumed.

Coverage and width. For a measurable ordered band $[L_y, U_y]$, put $K = Ah^\beta$ and define the truncation map $T_K(x) = \max\{-K, \min\{K, x\}\}$. With $y = \sigma(Q^\top Y + R)$, define the induced interval $C_i(Y,R) = [\ell_i(y), u_i(y)]$ by

$$\ell_i(y) = \frac{\int_{I_i} T_K(L_y(t))\,dt}{A\int_{I_i} g_i(t)\,dt}, \qquad u_i(y) = \frac{\int_{I_i} T_K(U_y(t))\,dt}{A\int_{I_i} g_i(t)\,dt}.$$

Clipping makes the endpoints finite without losing coverage of f_θ, which lies in $[0,K]$ on each occupied cell. Integrating a covered curve and dividing by $A\int_{I_i} g_i$ therefore gives $\theta_i \in$

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$C_i(Y,R)$ for every i. Clipping is measurable, monotone, and does not increase width, so, with $D = A\int_{I_i} g_i = Ah^{\beta+1}J$,

$$D\sum_{i=0}^{m-1} L(C_i(Y,R)) \le \int_0^1 \{U_y(t) - L_y(t)\}\,dt.$$

Proposition 6.16 therefore gives, for every $\epsilon > 0$ and all sufficiently large n,

$$\sup_{\theta\in\{0,1\}^m} \mathbb{P}_{f_\theta}\left\{\int_0^1 (U-L)\,dt \ge Dam\right\} \ge 1-\alpha-\epsilon.$$

Since $S \le nh^{2\beta+1}$ and $h \le 1/m$, the amplitude satisfies $Ah^\beta \ge \sigma\sqrt{cm\log m/n} \ge c_{\beta,\sigma,g}M^u(\log n/n)^{\beta/(2\beta+1)}$ for all sufficiently large n. Because $mh \ge 1/2$, $Dam \ge (aJ/2)Ah^\beta$. Thus the preceding probability bound has the displayed threshold with a positive constant independent of M, n and the band. The hypercube is a subfamily of $\Lambda^\beta(M)$, proving the first assertion. Finally choose a fixed $\epsilon_0 \in (0, 1-\alpha)$ and use $\mathbb{E}_f W \ge t\mathbb{P}_f\{W \ge t\}$ for the nonnegative integrated width W. This yields the expected-width bound, also when the expectation is infinite. □

Remark 6.18 (Procedure and Upper-Bound Comparison) Cai, Low, and Ma [38, Section 2.2] center a known-class band at a wavelet projection and balance bias against simultaneous Gaussian fluctuations. Its worst-case expected integrated width has order $M^{1/(2\beta+1)}(\sigma^2\log n/n)^{\beta/(2\beta+1)}$. This matches the lower-bound rate for periodic functions with $\beta > 1/4$, dyadic sample size, and sufficiently regular compactly supported wavelets satisfying their Assumption W. The guarantee is uniform asymptotic coverage, not finite-sample coverage over every nonperiodic class considered above.

For a known Hölder class with $\beta > 0$, honest intervals at a fixed interior point have optimal length of order $n^{-\beta/(2\beta+1)}$, with class, noise, and coverage parameters fixed. Local polynomial estimation, as developed by Tsybakov [103], attains this order when a uniform bias allowance is added to the Gaussian error margin. The band's extra logarithm is a cost of simultaneous coverage, not adaptation, even when width is integrated over the domain.

6.5.2 Relaxed coverage and local width

A band may remain informative while missing a controlled portion of the curve or allowing small departures from its boundaries. In regression model (6.12), retain coverage class $\Lambda^\beta(M)$ but evaluate expected integrated width at a specified function. We use the two criteria of Cai, Low, and Ma [38]. Define the set of noncovered points

$$N(CB,f) = \{t\in[0,1] : f(t) \notin [L(t),U(t)]\}, \tag{6.16}$$

with Lebesgue measure $\mu\{N(CB,f)\}$, and the relative excess mass

$$e_f(t) = [f(t)-U(t)]_+ + [L(t)-f(t)]_+, \qquad RE(CB,f) = \frac{\int_0^1 e_f(t)\,dt}{\int_0^1 (U(t)-L(t))\,dt}. \tag{6.17}$$

Here $e_f(t)$ is the distance from $f(t)$ to the reported interval. Uncovered measure counts locations; relative excess measures the total departure relative to integrated width. If the denominator is zero, set $RE(CB, f) = 0$ when the numerator is zero and $RE(CB, f) = +\infty$ otherwise. If the denominator is infinite, set $RE(CB, f) = 0$, including when the numerator is infinite. The theorem's relative-excess assertion assumes finite realized integrated width.

Require the chosen guarantee uniformly over $\Lambda^\beta(M)$, but evaluate width at any $h \in \Lambda^\beta(M')$, with $0 < M' < M$. The slack $M - M'$ allows perturbations around h to remain in the coverage class. The resulting lower bound applies at every such h, not just a worst-case curve.

For $\beta > 1/4$, a random-sign mixture makes many small bumps jointly difficult to detect, following Cai, Low, and Ma [38, Theorem 3]. For $\beta < 1/4$, bumps between design points give a stronger obstruction because they leave every observation unchanged. At $\beta = 1/4$, the two rates coincide. This second construction depends on the fixed grid and does not apply to continuous observation.

Theorem 6.19 (Expected-Width Lower Bound under Relaxed Coverage) *In regression model* (6.12)*, fix $\beta, M, \sigma > 0$ independently of n and let $0 < \alpha < 1/2$. Let $CB = [L, U]$ be a band with the uncovered-set and relative-excess conventions above. For the relative-excess criterion, assume $\int_0^1 (U - L) < \infty$ almost surely under every law in the coverage class. Assume either the uncovered-measure guarantee*

$$\inf_{g \in \Lambda^\beta(M)} \mathbb{P}_g \left\{ \mu\big(N(CB, g)\big) < \frac{1}{2} - \varepsilon \right\} \geq 1 - \alpha, \qquad 0 < \varepsilon < 1/2,$$

or, for some fixed $r > 0$, the relative-excess guarantee

$$\inf_{g \in \Lambda^\beta(M)} \mathbb{P}_g \{ RE(CB, g) \leq r \} \geq 1 - \alpha.$$

Then, for every $0 < M' < M$, there are constants $c > 0$ and n_0, depending only on the fixed model parameters (including σ), the coverage parameters, and the slack $M - M'$, such that, for every $h \in \Lambda^\beta(M')$ and every $n \geq n_0$,

$$w(CB, h) \geq c \begin{cases} n^{-\beta}, & 0 < \beta \leq 1/4, \\ n^{-2\beta/(1+4\beta)}, & \beta > 1/4. \end{cases}$$

The same lower bound holds for $\sup_{f \in \Lambda^\beta(M)} w(CB, f)$.

Proof outline; full proof in Appendix B.3 For $\beta > 1/4$, place independent random-sign bumps around h on $m \asymp n^{2/(1+4\beta)}$ disjoint cells, each with height $A_n \asymp m^{-\beta}$. For every fixed $\delta_{\mathrm{cov}} > 0$, there is a constant $q_{\delta_{\mathrm{cov}}} > 0$ such that every perturbation differs from h by at least $q_{\delta_{\mathrm{cov}}} A_n$ on a set of measure at least $1 - \delta_{\mathrm{cov}}$. Thus the separation extends across most of the domain, not just a few locations. Since $m\{A_n^2(n/m)\}^2 \asymp 1$, choosing the constant in the bump height sufficiently small makes the chi-square divergence of the mixture law relative to P_h arbitrarily small.

If expected width at h were too small, Markov's inequality and the guarantee at h would give a common data event on which every perturbation fails the chosen criterion. For uncovered measure, a narrow band cannot cover both curves over enough of the domain. For relative

excess, the triangle inequality bounds their integrated separation by $(2r+1)$ times the band's integrated width if both guarantees hold. The perturbations exceed this bound. Transferring the common event from P_h to the mixture makes its probability exceed α, contradicting the guarantee averaged over the alternatives.

For $\beta \le 1/4$, place bumps between consecutive design points. They vanish at every observation, so the perturbed laws equal P_h. The same coverage argument now uses zero statistical distance and gives an $n^{-\beta}$ obstruction, stronger than the mixture rate when $\beta < 1/4$ and equal in order at the boundary.

The theorem also yields bounds under expected average coverage. For fixed $0 < \alpha, \varepsilon < 1/2$, suppose that

$$\sup_{g \in \Lambda^\beta(M)} \mathbb{E}_g \mu\{N(CB, g)\} \le \alpha(1/2 - \varepsilon).$$

Markov's inequality then bounds the probability of uncovered measure at least $1/2 - \varepsilon$ by α, uniformly over the class. Thus the theorem's uncovered-measure guarantee holds. This connects the probability formulation to the average-coverage results of Cai, Low, and Ma [38].

Remark 6.20 (Procedure and Upper-Bound Comparison) Cai, Low, and Ma [38, Theorem 1] construct bands centered at a wavelet estimate with a data-selected resolution. In periodic, equispaced Gaussian regression with $n = 2^J$, the procedure adapts to $\beta \in [\beta_0, 2\beta_0]$ and $M \in [1, M_0]$, given $\beta_0 > 1/4$ and $M_0 > 1$. It uses sufficiently regular compactly supported wavelets satisfying their Assumption W. For fixed class constants, worst-case expected integrated width is $O\{(\log n/n)^{\beta/(2\beta+1)}\}$. Both the uncovered proportion and relative excess are bounded by vanishing thresholds, each with asymptotic probability at least $1 - \alpha$ uniformly over signals in each class.

The full-coverage bound concerns worst-case width; the relaxed-coverage bound holds at every function in a smaller-radius class. Permitting some failures therefore does not remove uncertainty at favorable functions, although it changes the lower-bound scale. Section 7.5 uses these local bounds to show how coverage requirements over a rough class limit width at smoother functions.

6.6 Confidence balls

A confidence ball quantifies uncertainty about an entire signal through its global norm error, rather than the location-by-location bounds of a confidence band. In the Gaussian sequence model (1.1), or the finite Gaussian model (2.4), with noise variance $1/n$, we seek the smallest expected squared radius compatible with uniform coverage over a specified class. For orthonormal function coefficients, this norm is the function's L_2 norm.

Two questions must be distinguished: how large must the radius be somewhere in the class, and how small can it be at zero? Following Cai and Low [35], we use equal-magnitude coordinates with unknown signs at two amplitudes. In the first argument, accumulated sign errors force a large radius at some signal. In the second, smaller amplitudes make the mixture close to pure noise. Coverage transfer then forces the ball to contain zero and a separated alternative with positive probability under the zero-signal law.

Thus estimation and testing lead to different radius bounds, first over a known Besov body, then over $\mathbb{R}^d$; compare Sections 4.4 and 5.6. Robins and van der Vaart [96] develop the relation between confidence-region diameter, squared-loss estimation, and testing under asymptotic honesty.

Here CB denotes a confidence ball rather than a band: a measurable random ℓ_2 ball

$$CB(\delta, r) = \{\theta : \ \|\theta - \delta\|_2 \le r\},$$

where $\delta = \delta(Y)$ is an ℓ_2-valued center and $r = r(Y) \ge 0$ is the random radius. In the finite-dimensional experiment, the center and parameter lie in $\mathbb{R}^d$. We evaluate $\mathbb{E}_\theta r^2$ at a fixed signal and, over a class Θ, use the worst-case criterion

$$R(CB, \Theta) = \sup_{\theta \in \Theta} \mathbb{E}_\theta r^2. \tag{6.18}$$

For $0 < \alpha < 1/2$, write

$$C_\alpha(\Theta) = \left\{ CB(\delta, r) : \ \inf_{\theta \in \Theta} \mathbb{P}_\theta\{\theta \in CB(\delta, r)\} \ge 1 - \alpha \right\}, \tag{6.19}$$

for the ball-restricted version of $C_\alpha(\Theta; \Psi)$ in (6.2), with identity target $\Psi(\theta) = \theta$, and define the corresponding minimax benchmark by

$$R^*_\alpha(\Theta) = \inf_{CB \in C_\alpha(\Theta)} R(CB, \Theta). \tag{6.20}$$

6.6.1 Coverage over a Besov body

For the known Besov ball $B^\beta_{p,q}(M)$ from (3.8), we seek both a worst-case radius bound and a bound at favorable signals such as zero. Assume $\beta, M > 0$, $2 \le p \le \infty$, and $1 \le q \le \infty$; these restrictions ensure square summability. Smoothness limits the amplitude at each resolution level. We first construct alternatives at one level, then choose the level giving the strongest bound.

For an integer $J \ge 1$, the truncated experiment retains levels $0 \le j < J$. Let E_J extend such a vector by zero on all levels $j \ge J$. The finite-dimensional restriction is

$$B^\beta_{p,q}(M)\big|_{j<J} = \left\{ u \in \mathbb{R}^{2^J - 1} : E_J u \in B^\beta_{p,q}(M) \right\}. \tag{6.21}$$

In truncated experiments, $B^\beta_{p,q}(M)$ abbreviates this restriction of the infinite sequence ball; otherwise it retains its original meaning. Write $\theta_j = \theta_{j\cdot}$ for the level vector from (3.8). The dimension $d = 2^J - 1$ and precision n are separate parameters unless a relation is imposed.

At level j, the alternatives have $m = 2^j$ equal-magnitude coefficients with unknown signs; all others vanish. The following bounds concern worst-case expected squared radius and its value at zero, respectively, with honesty over the whole Besov ball in both cases.

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Theorem 6.21 (Single-Level Lower Bounds for Confidence-Ball Squared Radius) *In the Gaussian sequence model* (1.1) *with* $\varepsilon = n^{-1/2}$ *and indexing* (3.7)*, let* $CB(\delta, r_\alpha) \in C_\alpha(B^\beta_{p,q}(M))$*, where* $\beta, M > 0$*,* $2 \le p \le \infty$*,* $1 \le q \le \infty$*, and* $0 < \alpha < 1/2$*. Fix an observed level* j*. The bounds also hold when only levels* $0 \le j < J$ *are observed and the parameter lies in* (6.21)*. Then for every* $0 < \tau < \frac{1}{2}(\frac{1}{2} - \alpha)$*,*

$$\sup_{\theta \in B^\beta_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \ge \frac{\tau^2}{1-\alpha-\tau} \min\left(M^2 2^{-2\beta j},\ z^2_{\alpha+2\tau} 2^j n^{-1}\right).$$

Moreover, for every $0 < \tau < 1/2 - \alpha$*, the expected squared radius at the origin satisfies*

$$\mathbb{E}_0 r_\alpha^2 \ge \frac{1}{4}(1 - 2\alpha - 2\tau) \min\left(M^2 2^{-2\beta j},\ \sqrt{\log(1+\tau^2)}\, 2^{j/2} n^{-1}\right).$$

Proof Write $m = 2^j$ and identify the selected level with $\mathbb{R}^m$. Projection of the center onto this level cannot increase its distance from a parameter supported there, so it suffices to consider the induced ball

$$\{u \in \mathbb{R}^m : \|u - \delta_j(Y)\|_2 \le r_\alpha(Y)\}.$$

Under parameters supported on this level, all off-level observations are independent, parameter-free Gaussian noise. Conditioning on them leaves the same block experiment, so the Bayes lower bound below applies even when the projected center uses the full data or additional randomization.

For $a > 0$, let

$$\mathcal{H}_m(a) = \{u \in \mathbb{R}^m : u_k \in \{-a, a\},\ 1 \le k \le m\}.$$

A vector supported on level j with block $u \in \mathcal{H}_m(a)$ has Besov contribution

$$2^{j(\beta+1/2-1/p)} \|u\|_p = 2^{j(\beta+1/2)} a.$$

Thus every such vector lies in $B^\beta_{p,q}(M)$ whenever

$$a \le M 2^{-j(\beta+1/2)}. \tag{6.22}$$

Worst-case radius. Place the independent uniform prior on the signs of $u \in \mathcal{H}_m(a)$ in the Gaussian experiment below, with loss $\mathcal{L}$:

$$Y_k = u_k + n^{-1/2} Z_k, \qquad Z_k \overset{\text{i.i.d.}}{\sim} N(0,1), \quad 1 \le k \le m,$$
$$\mathcal{L}(\widehat{u}, u) = \sum_{k=1}^m \mathbf{1}\{|\widehat{u}_k - u_k| \ge a\}.$$

An arbitrary real estimate cannot be within distance strictly less than a of both $-a$ and a. Its conditional loss is therefore minimized by choosing the more probable sign. The posterior factors across coordinates. Given $Y_k = y_k$, the posterior odds of $u_k = a$ against $u_k = -a$ equal $\exp(2nay_k)$, so a Bayes rule chooses a when $y_k \ge 0$ and $-a$ otherwise. Each coordinate has error probability $\Phi(-a\sqrt{n})$. The total risk $m\Phi(-a\sqrt{n})$ is constant over the hypercube and equals the Bayes risk, hence also the minimax risk:

$$\inf_{\widehat{u}} \sup_{u \in \mathcal{H}_m(a)} \mathbb{E}_u \mathcal{L}(\widehat{u}, u) = m\Phi(-a\sqrt{n}). \tag{6.23}$$

Choose the amplitude to satisfy the Besov constraint and keep each sign-error probability at least $\alpha + 2\tau$:

$$a = \min\left\{M2^{-j(\beta+1/2)},\ z_{\alpha+2\tau}n^{-1/2}\right\}. \tag{6.24}$$

The restriction on τ implies $\alpha+2\tau < 1/2$ and hence $z_{\alpha+2\tau} > 0$. Equations (6.23)–(6.24) give, for the projected center δ_j,

$$\sup_{u\in\mathcal{H}_m(a)} \mathbb{E}_u X_u \geq (\alpha+2\tau)m, \qquad X_u = \sum_{k=1}^{m} \mathbf{1}\{|\delta_{j,k} - u_k| \geq a\}. \tag{6.25}$$

To combine the expected-error bound with coverage, we need a probability of large error exceeding the allowed noncoverage probability α. Set $\gamma = \tau/(1-\alpha-\tau)$. Since $0 \leq X_u \leq m$,

$$\mathbb{E}_u X_u \leq \gamma m + (1-\gamma)m\mathbb{P}_u\{X_u \geq \gamma m\}.$$

Combining this inequality with (6.25) and the identity $(\alpha+2\tau-\gamma)/(1-\gamma) = \alpha+\tau$ shows that some $u \in \mathcal{H}_m(a)$ satisfies

$$\mathbb{P}_u\{X_u \geq \gamma m\} \geq \alpha + \tau. \tag{6.26}$$

Each coordinate counted by X_u contributes at least a^2 to squared error. Therefore

$$\mathbb{P}_u\{\|\delta_j - u\|_2^2 \geq \gamma m a^2\} \geq \alpha + \tau.$$

Honesty at this same u gives $\mathbb{P}_u\{\|\delta_j - u\|_2^2 \leq r_\alpha^2\} \geq 1-\alpha$. The two events intersect with probability at least τ, and on their intersection $r_\alpha^2 \geq \gamma m a^2$. Consequently,

$$\sup_{\theta\in B^\beta_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \geq \tau\gamma m a^2. \tag{6.27}$$

Finally, (6.24) gives

$$ma^2 = \min\left\{M^2 2^{-2\beta j},\ z_{\alpha+2\tau}^2\, 2^j n^{-1}\right\}.$$

Substituting $\tau\gamma = \tau^2/(1-\alpha-\tau)$ in (6.27) proves the first assertion.

Radius at the origin. We now choose the amplitude to make the mixture of observation laws close to the law at zero. This permits coverage at the alternatives to constrain radius under the zero-signal law. Set

$$\Gamma = \log(1+\tau^2), \qquad a = \min\left\{M2^{-j(\beta+1/2)},\ \Gamma^{1/4}m^{-1/4}n^{-1/2}\right\}. \tag{6.28}$$

Let P_π be the uniform mixture of the laws indexed by $\mathcal{H}_m(a)$ and let P_0 be the law at the origin. Since $\sqrt{n}Y_k \sim N(0,1)$ under P_0, the one-coordinate mixture likelihood ratio and its second moment are

$$\ell(Y_k) = e^{-na^2/2}\cosh(naY_k), \qquad \mathbb{E}_0\ell(Y_k)^2 = \cosh(na^2).$$

Independence and $\log\cosh x \leq x^2/2 \leq x^2$ yield

$$1 + \chi^2(P_\pi, P_0) = \{\cosh(na^2)\}^m \leq \exp(mn^2a^4) \leq e^\Gamma = 1+\tau^2. \tag{6.29}$$

Thus, for every data event A,

$$|\mathbb{P}_\pi(A) - \mathbb{P}_0(A)| \leq \mathrm{TV}(P_\pi, P_0) \leq \frac{1}{2}\sqrt{\chi^2(P_\pi, P_0)} \leq \frac{\tau}{2} \leq \tau. \tag{6.30}$$

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The event to transfer must depend only on the data, not on the unknown sign vector. Define

$$A_0 = \{0 \in CB(\delta, r_\alpha)\}, \qquad A_1 = \{CB(\delta, r_\alpha) \cap \mathcal{H}_m(a) \neq \varnothing\},$$

where each hypercube point is understood to be zero off level j. Honesty at the origin gives

$$\mathbb{P}_0(A_0) \geq 1 - \alpha. \tag{6.31}$$

For each $u \in \mathcal{H}_m(a)$, coverage at u implies $\mathbb{P}_u(A_1) \geq 1 - \alpha$; averaging over the signs gives

$$\mathbb{P}_\pi(A_1) \geq 1 - \alpha. \tag{6.32}$$

Applying (6.30) to A_1 gives $\mathbb{P}_0(A_1) \geq 1 - \alpha - \tau$. Both events are now evaluated under P_0. The union bound for their complements and (6.31) yield

$$\mathbb{P}_0(A_0 \cap A_1) \geq 1 - 2\alpha - \tau \geq 1 - 2\alpha - 2\tau. \tag{6.33}$$

On $A_0 \cap A_1$, the ball contains the origin and at least one $u \in \mathcal{H}_m(a)$. Their distance is $\|u\|_2 = \sqrt{m}\, a$, so the triangle inequality forces $r_\alpha \geq \sqrt{m}\, a/2$. By (6.33),

$$\mathbb{E}_0 r_\alpha^2 \geq \frac{1}{4}(1 - 2\alpha - 2\tau) m a^2. \tag{6.34}$$

By (6.28),

$$ma^2 = \min\left\{M^2 2^{-2\beta j},\ \sqrt{\log(1+\tau^2)}\, 2^{j/2} n^{-1}\right\}.$$

Substitution in (6.34) proves the second assertion. □

The different powers of m come from the amplitude choices. Before imposing the Besov constraint, $a^2 \asymp 1/n$ gives accumulated squared error of order m/n. Choosing mn^2a^4 to be a fixed constant instead makes the mixture close to zero while every alternative has squared norm of order $\sqrt{m}/n$. The constraint $ma^2 \leq M^2 m^{-2\beta}$ limits both constructions. The bound at zero is forced by coverage at other signals, not by difficulty estimating zero alone.

We next optimize over levels. Translating the mixture to any $\theta^0 \in B^\beta_{p,q}(M')$ also extends the bound at zero to every signal in a smaller-radius body: the slack $M - M' > 0$ keeps the alternatives in the coverage class. Each minimum below reflects the available radius, the finite observation dimension, and the balance between smoothness and noise.

Theorem 6.22 (Global Lower Bounds over Besov Bodies) *Work in the finite Gaussian model* (2.4) *with* $\sigma = n^{-1/2}$ *and ambient dimension* $d = 2^J - 1$*, where* $J \geq 1$*. Index its coordinates by* (3.7)*, restricted to* $0 \leq j < J$*. Let* $B^\beta_{p,q}(M)$ *denote the finite-dimensional restriction* (6.21)*, with* $\beta, M > 0$*,* $2 \leq p \leq \infty$*, and* $1 \leq q \leq \infty$*. Let* $CB(\delta, r_\alpha) \in C_\alpha(B^\beta_{p,q}(M))$ *with* $0 < \alpha < 1/2$*. Then*

$$\sup_{\theta \in B^\beta_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{M^2,\ dn^{-1},\ M^{2/(1+2\beta)} n^{-2\beta/(1+2\beta)}\right\}.$$

In addition, for every smaller Besov body $B^\beta_{p,q}(M')$ *with* $0 < M' < M$*,*

$$\inf_{\theta \in B^\beta_{p,q}(M')} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{(M - M')^2,\ d^{1/2} n^{-1},\ (M - M')^{2/(1+4\beta)} n^{-4\beta/(1+4\beta)}\right\}.$$

The constants implicit in $\gtrsim$ depend only on α, β, p, and q, not on n, J, M, or M'. In the standard asymptotic formulation with fixed M and M', the first terms inside the two minima are eventually inactive, and the displays reduce to the familiar two-regime bounds.

Proof *Worst-case radius.* Put $m_j = 2^j$. A vector supported on level j with all m_j coordinates equal in magnitude to a has Besov norm exactly

$$2^{j(\beta+1/2-1/p)} m_j^{1/p} a = m_j^{\beta+1/2} a. \tag{6.35}$$

Fix $0 < \tau < \frac{1}{2}(\frac{1}{2} - \alpha)$. The first assertion of Theorem 6.21, applied at level j, and (6.35) therefore give

$$\sup_{\theta \in B^{\beta}_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \geq c_\alpha \min\left\{ M^2 m_j^{-2\beta}, \frac{m_j}{n} \right\}, \qquad 0 \leq j < J, \tag{6.36}$$

where $c_\alpha > 0$ depends on α through the fixed choice of τ but is independent of j, n, J, and M.

To optimize (6.36) over the available levels, let

$$x = (M^2 n)^{1/(1+2\beta)}, \qquad \rho = M^{2/(1+2\beta)} n^{-2\beta/(1+2\beta)} = \frac{x}{n},$$

and let $m_{\max} = 2^{J-1} = (d+1)/2$. The value x balances the smoothness bound $M^2 m^{-2\beta}$ against the noise term m/n, before restriction to the available dyadic dimensions.

If $x \leq 1$, choose $m_j = 1$; then the right-hand side of (6.36) is $c_\alpha M^2$.

If $1 < x \leq m_{\max}$, choose a dyadic m_j with $x/2 < m_j \leq x$. The two terms in (6.36) are then at least constant multiples of ρ.

Finally, if $x > m_{\max}$, choose $m_j = m_{\max}$. The inequality $m_{\max} < x$ implies $M^2 m_{\max}^{-2\beta} > m_{\max}/n$, so the right-hand side of (6.36) is of order $m_{\max}/n \asymp d/n$. These three cases prove

$$\sup_{\theta \in B^{\beta}_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{ M^2, \frac{d}{n}, M^{2/(1+2\beta)} n^{-2\beta/(1+2\beta)} \right\}. \tag{6.37}$$

Radius on a smaller Besov body. Translate the alternatives, not the procedure, and continue to require coverage over $B^{\beta}_{p,q}(M)$. Fix $\theta^0 \in B^{\beta}_{p,q}(M')$ and write $D = M - M' > 0$. At a fixed level with $m = 2^j$, let $\mathcal{V}_m(a)$ contain the 2^m vectors supported on that level whose coordinates each equal $-a$ or a. By (6.35) and the triangle inequality for the $B^{\beta}_{p,q}$ norm,

$$\theta^0 + v \in B^{\beta}_{p,q}(M) \quad \text{for every } v \in \mathcal{V}_m(a) \quad \text{whenever} \quad a \leq D m^{-(\beta+1/2)}. \tag{6.38}$$

The triangle inequality applies because $p, q \geq 1$.

Choose $0 < \delta_{\mathrm{TV}} < 1/2 - \alpha$ depending only on α, put $\Gamma = \log(1 + \delta_{\mathrm{TV}}^2)$, and set

$$a = \min\left\{ D m^{-(\beta+1/2)}, \Gamma^{1/4} m^{-1/4} n^{-1/2} \right\}. \tag{6.39}$$

Let P_{θ^0} be the law at θ^0 and let P_π be the uniform mixture of the laws at $\theta^0 + v$, $v \in \mathcal{V}_m(a)$. Centering the observations at θ^0 gives the same calculation as at zero: the coordinate likelihood ratio is $e^{-na^2/2} \cosh\{na(Y_{j,k} - \theta^0_{j,k})\}$. Consequently,

$$1 + \chi^2(P_\pi, P_{\theta^0}) = \{\cosh(na^2)\}^m \leq \exp(mn^2 a^4) \leq 1 + \delta_{\mathrm{TV}}^2, \qquad \mathrm{TV}(P_\pi, P_{\theta^0}) \leq \delta_{\mathrm{TV}}. \tag{6.40}$$

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Apply the radius bound in Lemma 6.2 to the identity target with the ℓ_2 metric, taking $H_0 = \{\theta^0\}$ and $H_1 = \theta^0 + \mathcal{V}_m(a)$. The priors are the point mass at θ^0 and the uniform distribution on H_1. Both sets lie in the coverage class by (6.38), and their separation is $\Delta = \sqrt{m}\, a$. The intersection events are measurable because these sets are finite and the ball has measurable center and radius. Thus (6.40) gives

$$\mathbb{E}_{\theta^0} r_\alpha^2 \geq \frac{ma^2}{4}(1 - 2\alpha - \delta_{\mathrm{TV}}). \tag{6.41}$$

Substituting (6.39) and using the smaller positive coefficient $1 - 2\alpha - 2\delta_{\mathrm{TV}}$ yields

$$\mathbb{E}_{\theta^0} r_\alpha^2 \geq \frac{1}{4}(1 - 2\alpha - 2\delta_{\mathrm{TV}}) \min\left\{ D^2 m^{-2\beta}, \sqrt{\Gamma}\, \frac{\sqrt{m}}{n} \right\}. \tag{6.42}$$

The bound is uniform over $\theta^0 \in B^\beta_{p,q}(M')$ and holds at every available dyadic m. Unlike the first assertion, it bounds radius at every signal in this smaller body, not just at one difficult signal.

To optimize (6.42), absorb constants depending on the fixed δ_{TV} into $\gtrsim$ and set

$$x_D = (D^2 n)^{2/(1+4\beta)}, \qquad \rho_D = D^{2/(1+4\beta)} n^{-4\beta/(1+4\beta)} = \frac{\sqrt{x_D}}{n}.$$

The same three cases apply, now with noise term $\sqrt{m}/n$ rather than m/n. If $x_D \leq 1$, choose $m = 1$ and obtain the order D^2. If $1 < x_D \leq m_{\max}$, choose a dyadic m with $x_D/2 < m \leq x_D$ and obtain the order ρ_D. If $x_D > m_{\max}$, choose $m = m_{\max}$; the defining inequality for this case gives $D^2 m_{\max}^{-2\beta} > \sqrt{m_{\max}}/n$, so the bound is of order $\sqrt{d}/n$. Since θ^0 was arbitrary, this proves the second assertion:

$$\inf_{\theta \in B^\beta_{p,q}(M')} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{ D^2, \frac{\sqrt{d}}{n}, D^{2/(1+4\beta)} n^{-4\beta/(1+4\beta)} \right\}.$$

□

Theorem 6.22 bounds the known-class minimax benchmark from below:

$$R^*_\alpha\{B^\beta_{p,q}(M)\} \gtrsim \min\left\{ M^2, \frac{d}{n}, M^{2/(1+2\beta)} n^{-2\beta/(1+2\beta)} \right\}. \tag{6.43}$$

Remark 6.23 (Optimal Procedure and Matching Upper Bound) A ball centered at a truncated estimator attains the known-class minimax rate. Its radius accounts for noise in the retained coordinates and signal in the omitted ones. For $0 \leq K \leq J$, retain the first m_K coordinates:

$$\delta^{(K)}_{j,k} = Y_{j,k} \mathbf{1}\{j < K\}, \qquad m_K = 2^K - 1.$$

For $\theta \in B^\beta_{p,q}(M)$ and $K < J$, the Besov constraint and $p \geq 2$ give

$$\|\theta_j\|_2 \leq 2^{j(1/2-1/p)} \|\theta_j\|_p \leq M 2^{-j\beta},$$
$$\sum_{j=K}^{J-1} \|\theta_j\|_2^2 \leq C_\beta M^2 2^{-2K\beta}. \tag{6.44}$$

Here $C_\beta > 0$ depends only on β; the tail vanishes for $K = J$. With $q_{m,1-\alpha}$ the $(1-\alpha)$ quantile of χ^2_m and $q_{0,1-\alpha} = 0$, choose squared radius

$$r_K^2 = \frac{2q_{m_K,1-\alpha}}{n} + 2C_\beta M^2 2^{-2K\beta}\mathbf{1}\{K < J\}.$$

The retained noise has squared norm $n^{-1}\chi^2_{m_K}$ in distribution. Its quantile and the omitted-signal bound (6.44) together ensure honesty over $B^\beta_{p,q}(M)$. For fixed α, $q_{m,1-\alpha} \le C_\alpha m$ for $m \ge 1$.

Put $x = (M^2 n)^{1/(1+2\beta)}$. If $x \le 1$, use the zero-centered ball of radius $C_\beta^{1/2} M$. For $1 < x \le d/2$, choose $2^K \asymp x$ to balance noise and the omitted signal. For $x > \max\{1, d/2\}$, retain all coordinates: d/n is then comparable to $\min\{M^2, d/n, x/n\}$. Combined with the lower bound,

$$R^*_\alpha\{B^\beta_{p,q}(M)\} \asymp \min\left\{M^2, \frac{d}{n}, M^{2/(1+2\beta)} n^{-2\beta/(1+2\beta)}\right\}. \tag{6.45}$$

This cutoff uses the known β and M and attains the worst-case benchmark; it need not attain the smaller bound at individual signals. Cai and Low [35] develop adaptive constructions, discussed in Section 7.6.

6.6.2 Coverage over the full parameter space

Without a smoothness assumption, the ball must remain honest over $\mathbb{R}^d$ in the finite Gaussian model with noise variance $1/n$. We ask how small its expected squared radius can be even at zero. A sign mixture can now use all d coordinates: as in Section 5.6.1, it remains close to pure noise at squared separation of order $\sqrt{d}/n$. Coverage transfer forces this order of expected squared radius for any center.

Theorem 6.24 (Honest Confidence Balls over $\mathbb{R}^d$) *In the finite Gaussian model* (2.4) *with* $\sigma = n^{-1/2}$, *let* $CB(\delta, r_\alpha) \in C_\alpha(\mathbb{R}^d)$, $0 < \alpha < 1/2$, *with arbitrary measurable center and radius as in* (6.19). *Then the expected squared radius at the origin satisfies*

$$\mathbb{E}_0 r_\alpha^2 \ge \frac{1}{4}(1 - 2\alpha - 2\varepsilon)\sqrt{\log(1+\varepsilon^2)}\, d^{1/2} n^{-1} \qquad \text{for every } 0 < \varepsilon < \frac{1}{2} - \alpha.$$

In particular, if $d = n$, *every honest confidence ball over* $\mathbb{R}^d$ *has expected squared radius at the origin at least of order* $n^{-1/2}$.

Proof Fix $0 < \varepsilon < 1/2 - \alpha$ and put

$$\Gamma = \log(1+\varepsilon^2), \qquad a = \Gamma^{1/4} d^{-1/4} n^{-1/2},$$
$$\mathcal{H}_d(a) = \{u \in \mathbb{R}^d : u_i \in \{-a, a\},\ 1 \le i \le d\}.$$

Let P_π be the uniform mixture of the Gaussian laws indexed by $\mathcal{H}_d(a)$ and let P_0 be the law at the origin. Under P_0, Lemmas A.9 and A.10 give the sign-mixture second moment below: the independent signs factor the Gaussian cross-moment into d factors $\cosh(na^2)$. Using $\log\cosh x \le x^2$,

$$1 + \chi^2(P_\pi, P_0) = \{\cosh(na^2)\}^d \le \exp(dn^2a^4) = e^\Gamma = 1 + \varepsilon^2. \tag{6.46}$$

It follows that

$$\mathrm{TV}(P_\pi, P_0) \le \frac{1}{2}\sqrt{\chi^2(P_\pi, P_0)} \le \frac{\varepsilon}{2} \le \varepsilon. \tag{6.47}$$

Apply (6.4) with identity target, $H_0 = \{0\}$, $H_1 = \mathcal{H}_d(a)$, and separation $\Delta = \sqrt{d}\,a$. Since the diameter of a Euclidean ball is at most twice its radius, (6.47) gives

$$\mathbb{P}_0\{r_\alpha \ge \sqrt{d}\,a/2\} \ge 1 - 2\alpha - \varepsilon/2 \ge 1 - 2\alpha - 2\varepsilon. \tag{6.48}$$

Taking expectations, or applying (6.6) directly, yields

$$\mathbb{E}_0 r_\alpha^2 \ge \frac{1}{4}(1 - 2\alpha - 2\varepsilon)da^2 = \frac{1}{4}(1 - 2\alpha - 2\varepsilon)\sqrt{\log(1+\varepsilon^2)}\, d^{1/2}n^{-1},$$

proving the finite-sample bound and, when $d = n$, the order $n^{-1/2}$. □

Remark 6.25 (Procedure and Upper-Bound Comparison) Cai and Low [35, Theorem 5] construct a ball centered at a block-thresholded estimator, with radius based on its estimated squared error. For $d = 2^J - 1 \le n^2$, coverage is at least $1-\alpha-2/\log n$ throughout $\mathbb{R}^d$, without smoothness assumptions. Smoothness controls the expected squared radius: for $\tau > 0$, $p \ge 2$, $q \ge 1$, and $M \ge 1$,

$$\sup_{\theta \in B^\tau_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \le A_\alpha \frac{\sqrt{d}}{n} + C_\tau \min\left\{\frac{d}{n}, M^{2/(1+2\tau)} n^{-2\tau/(1+2\tau)}\right\}.$$

The first term reflects honesty over $\mathbb{R}^d$; the second is the Besov estimation rate. Using nominal error $\alpha/2$ ensures coverage at least $1 - \alpha$ for $n \ge \exp(4/\alpha)$, changing only the coverage-dependent constant. For fixed M, $d = n$, and $\tau \ge 1/2$, the upper bound has order $n^{-1/2}$. Since zero belongs to these Besov bodies, this also matches the lower bound there. The dimension restriction is needed only for the upper bound.

The coverage class determines which alternatives the radius must accommodate. Smoothness limits their size; honesty over $\mathbb{R}^d$ leaves a dimension-dependent obstruction even at zero. Section 7.6 compares these bounds with smoother-class benchmarks to determine the possible range of adaptation.

6.7 Exercises

6.1. In the fixed-design Hölder experiment of Theorem 5.2, fix $0 < \alpha < 1/2$. Verify the uniform coverage and deterministic length of the interval centered at $\overline{Y}_h$ in Section 6.2. Choose the constant in the lower-bound bump so that its total variation distance from the zero-function law is at most $(1 - 2\alpha)/2$. Apply Lemma 6.2 to obtain a positive expected-length lower bound of order $n^{-s/(2s+1)}$ at zero for every interval honest over $\Lambda^s(M)$, for all sufficiently large n. Explain why these single-location conclusions alone give neither simultaneous-band coverage nor an integrated-width bound.

6.2. Prove Proposition 6.1. Contrast inversion of level-α tests indexed by the complete parameter with inversion of separate coordinatewise tests. Show that the former gives an honest confidence set, whereas the latter need not give simultaneous band coverage. Give a two-coordinate Gaussian example.

6.3. In the Gaussian experiment of Theorem 6.24, fix $0 < \alpha < 1/2$. Let $\mathcal{H}_d(a) = \{u \in \mathbb{R}^d : u_i \in \{-a, a\}\}$ and let Q_a be the uniform mixture of its observation laws. Calculate $1 + \chi^2(Q_a, P_0)$ and, using $\log\cosh t \le t^2$, choose $a > 0$ so that $\mathrm{TV}(Q_a, P_0) \le (1 - 2\alpha)/2$. For any confidence ball $CB(\delta, r)$ honest over $\mathbb{R}^d$ at level $1 - \alpha$, use $A_0 = \{0 \in CB\}$ and $A_1 = \{CB \cap \mathcal{H}_d(a) \neq \varnothing\}$ to derive $\mathbb{E}_0 r^2 \ge c_\alpha \sqrt{d}/n$ with an explicit $c_\alpha > 0$. Specify the law under which each coverage event is initially controlled before transferring A_1 to P_0.

6.4. In Lemma 6.2, take the identity target, $H_0 = \{0\}$ and $H_1 = \{1\}$ in $\mathbb{R}$, and identical observation laws $P_0 = P_1$. The deterministic ball $CB = \{0\}$ has radius zero and coverage one on H_0. Explain why separation and zero total variation do not force a positive radius when coverage is required only on H_0. Identify the missing bound on the event $\{CB \cap H_1 \neq \varnothing\}$. Then state the coverage and prior-support assumptions needed for the lemma's squared-radius bound, and specify which prior must be supported on an evaluation class $\mathcal{G}$ to lower-bound $\sup_{\theta \in \mathcal{G}} \mathbb{E}_\theta r^2$.

6.5. In the fixed-design Hölder regression model of Theorem 4.26, fix $0 < \alpha < 1/4$. For every $(1 - \alpha)$ confidence interval honest over $\Lambda^\beta(M)$ for $m(f) = \min_{x \in [0,1]} f(x)$, use the theorem's minimum-value pair to prove that its expected length at $f = 0$ is at least $\gamma n^{-\beta/(2\beta+1)}/4$ for all sufficiently large n, where $\gamma = \frac{1}{4}\min\{M, \sigma\}$. Explain why this conclusion does not establish a matching upper bound.

6.6. In the convex white-noise optimization model, derive the lower inequality in (6.10) from the Gaussian KL formula and pairwise honesty. Explain why the corresponding estimation lower bound does not by itself prove the interval claim.

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7

The Cost of Adaptation

Chapters 4–6 established benchmarks for estimation, testing, and honest confidence sets over a given parameter class. In practice, however, the smoothness or sparsity defining the class is often unknown. A common goal in statistics and machine learning is *adaptive minimaxity*: attaining the class-specific minimax rates simultaneously with a single procedure that does not know which class contains the signal. This goal is not always attainable. This chapter studies when adaptation is possible and uses lower bounds to quantify its unavoidable cost when it is not.

The procedure may use the collection of candidate classes, but must meet the benchmark for each. Minimax optimality over their union is not enough: for nested classes with a largest member, the union benchmark is governed by that class and need not reward the greater accuracy available on smaller subclasses. Requiring that accuracy can force logarithmic or larger penalties elsewhere.

The lower-bound argument must link the performance of the same procedure across classes. For estimation, Brown and Low's constrained-risk inequality [12] converts small risk on one class into a lower bound on risk elsewhere. Cai and Low's two-mixture extension [37, Theorem 2 and Corollary 1], introduced in Chapter 2, compares families through their observation mixtures. It supports moment-matching bounds for nonsmooth functionals, and its point-versus-mixture form gives the risk-transfer argument for quadratic-functional adaptation. Between-class moduli [31, 33] select target gaps; sign and support mixtures conceal weak signals. The essential step is to calibrate these constructions against the imposed risk constraint, revealing costs that separate minimax bounds cannot detect.

For honest confidence sets, adaptation concerns size while preserving coverage. An interval should exploit a favorable subclass without losing validity over the larger collection. Statistically close alternatives outside that subclass can force substantial length within it, regardless of how well an estimator performs there. Cai and Low's confidence interval theory [30] quantifies this obstruction through between-class moduli and coverage transfer. Unlike constrained-risk estimation, the argument requires no small-risk assumption: coverage itself forces the size bound. Adaptive estimation therefore does not automatically provide an adaptive measure of uncertainty.

Section 7.1 defines the benchmarks. Section 7.2 uses between-class moduli to distinguish no-cost adaptation from logarithmic and larger penalties, and examines how the error criterion changes the conclusion. Section 7.3 treats quadratic functionals through mixtures. Sections 7.4–7.6 turn to intervals, bands, and balls, where shape, coverage, and dimension govern adaptation. Matching procedures identify when the lower bounds are sharp.

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This chapter completes Part II by separating the difficulty of a statistical task from the cost of unknown structure. Part III brings these distinctions together within structured learning problems, beginning with sparse regression, where estimation, testing, honest inference, and adaptation can be compared in a common model.

7.1 Adaptive benchmarks and local penalties

Suppose we observe $X \sim P_f^{(n)}$, where $f \in \mathcal{F}$ is unknown. We evaluate one procedure over a given collection $\{\mathcal{F}_\tau : \tau \in \mathcal{T}\}$ of parameter classes contained in $\mathcal{F}$. The index τ may specify smoothness, sparsity, shape, or several structural features; the classes need not be nested. The procedure may use the collection but not the unknown class index.

We measure the cost of unknown class membership relative to the known-class benchmarks: risk for estimation and expected length for honest intervals. Let $T(f)$ be the estimation target and $\ell(\widehat{T}, T(f))$ the loss. For each fixed class define

$$r_n(\tau) = \inf_{\widehat{T}} \sup_{f \in \mathcal{F}_\tau} \mathbb{E}_f \ell\{\widehat{T}, T(f)\}. \tag{7.1}$$

Assume $0 < r_n(\tau) < \infty$ for the indices under consideration. The estimator in (7.1) may depend on τ. Adaptive estimation instead seeks one estimator $\widehat{T}$, not depending on τ, whose classwise risks

$$R_n(\widehat{T}, \mathcal{F}_\tau) = \sup_{f \in \mathcal{F}_\tau} \mathbb{E}_f \ell\{\widehat{T}, T(f)\}$$

attain the orders $r_n(\tau)$ simultaneously. The smallest worst-class risk ratio is

$$A_n(\mathcal{T}) = \inf_{\widehat{T}} \sup_{\tau \in \mathcal{T}} \frac{R_n(\widehat{T}, \mathcal{F}_\tau)}{r_n(\tau)}. \tag{7.2}$$

The infimum uses one estimator for all τ. *Uniform rate adaptation* requires a common eventual bound on the risk ratios over $\mathcal{T}$, equivalently $A_n(\mathcal{T}) = O(1)$ as $n \to \infty$. *Classwise rate adaptation* also uses one estimator sequence, but allows the bound and starting sample size to depend on each fixed class.

Unlike the union benchmark of Section 4.1.2, this ratio compares risk with each class's own difficulty, including the faster rates available on easier classes.

Testing has its own adaptation penalties, for example under unknown smoothness [101] or an unknown ellipsoid radius at fixed smoothness [78]. This chapter focuses on estimation risk and the size of honest confidence sets.

For a real-valued target T, retain the expected-length benchmark L_α^* from Section 6.2. An adaptive interval should be short on each class while remaining honest over their union. Fix $0 < \alpha < 1/2$ and write this coverage class as

$$\mathcal{F}^{\mathrm{H}} = \bigcup_{\tau \in \mathcal{T}} \mathcal{F}_\tau.$$

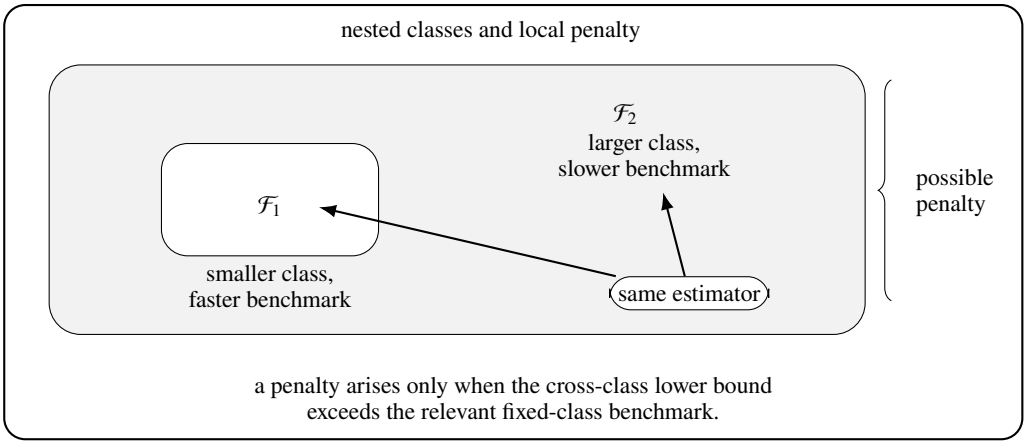


Figure 7.1 A possible obstruction to adaptation over nested classes $\mathcal{F}_1 \subset \mathcal{F}_2$. Good performance on $\mathcal{F}_1$ supplies a risk constraint at a point in that class for comparisons with nearby alternatives in $\mathcal{F}_2$. These comparisons force an adaptation penalty only if their lower bound exceeds the corresponding fixed-class benchmark. Nesting and unequal class-specific rates alone do not imply a penalty.

Both $L^*_\alpha(\mathcal{F}_\tau, \mathcal{F}_\tau, T)$ and $L^*_\alpha(\mathcal{F}_\tau, \mathcal{F}^{\mathrm{H}}, T)$ evaluate worst-case expected length on $\mathcal{F}_\tau$. The first requires coverage only there; the second requires coverage over the union. When the denominator is finite and strictly positive, the ratio

$$\frac{L^*_\alpha(\mathcal{F}_\tau, \mathcal{F}^{\mathrm{H}}, T)}{L^*_\alpha(\mathcal{F}_\tau, \mathcal{F}_\tau, T)} \tag{7.3}$$

measures the cost of requiring honesty over the union when evaluating length on $\mathcal{F}_\tau$. It can diverge even when estimation admits simultaneous class-specific rates. Each length infimum is taken separately; attaining these benchmarks with one interval across the collection requires a further construction.

The coverage-transfer bounds of Chapter 6 give length lower bounds at a reference parameter in $\mathcal{F}_\tau$. Comparison with the known-class benchmark then bounds (7.3), without any small-risk assumption on an estimator.

For estimation, Example 2.12 shows how small risk at one point forces larger risk along suitable alternatives. A classwise risk requirement supplies that constraint at every point in the class. Section 7.2 uses between-class moduli to choose the comparison points and quantify the resulting cost.

7.2 Adaptive functional estimation and between-class modulus

We estimate a linear functional $T(f)$ in the Gaussian shift model (1.3), $Y = f + n^{-1/2}W$, with known precision $n > 0$. The signal lies in a real separable Hilbert space H with norm $\|\cdot\|_2$; Section 1.2 gives the sequence, white-noise, and finite-dimensional interpretations. The estimator knows the candidate classes but not which contains f. For point evaluation, this missing information affects which observations to average and at what scale.

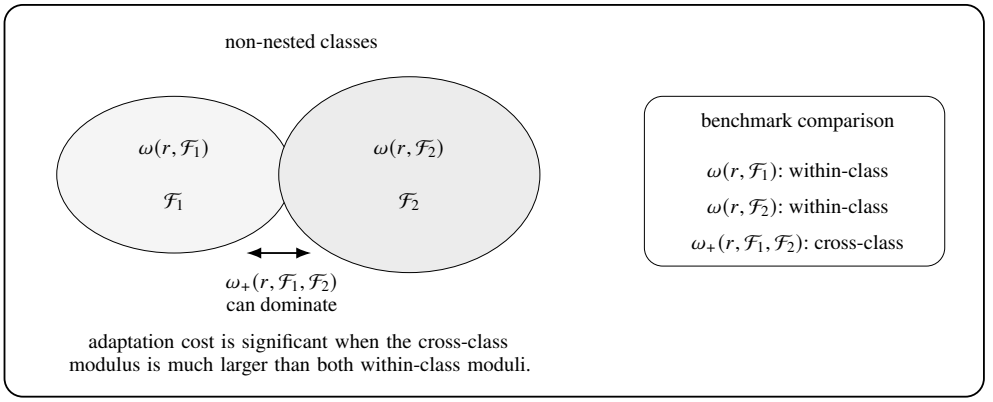


Figure 7.2 For nonnested classes, statistically close signals from different classes can have much larger target separation than pairs within either class. The between-class modulus captures this gap, which can produce an adaptation cost exceeding a logarithmic factor.

The goal is to preserve the known-class accuracy on one class while determining the best attainable risk on the other. The between-class modulus characterizes this tradeoff. We begin with squared-error loss over intersecting closed convex classes $\mathcal{F}_1, \mathcal{F}_2 \subset H$, with T linear on a vector space containing their union. White-noise examples then show logarithmic and larger costs, and explain how shape restrictions can eliminate the rate penalty. Finally, we replace squared-error risk by a fixed-probability error criterion.

7.2.1 The two-class risk tradeoff

Suppose an estimator attains the known-class risk order on $\mathcal{F}_1$. What does this require of its risk on $\mathcal{F}_2$? Separate minimax bounds cannot answer, because they allow different estimators. Cai and Low [33] link the two risks through the moduli of Section 4.1. Ordinary moduli supply the known-class benchmarks; the between-class modulus selects target gaps across the classes; ordered moduli control directional bias in the matching procedure.

To apply the constrained-risk argument of Example 2.12, take $f_1 \in \mathcal{F}_1$ and $f_2 \in \mathcal{F}_2$ with $\|f_2 - f_1\|_2 \le r$. If the root mean squared error at f_1 is at most b, the Gaussian likelihood-ratio identity and Theorem 2.11 give

$$\left\{\mathbb{E}_{f_2}(\widehat{T} - T(f_2))^2\right\}^{1/2} \ge \left(|T(f_2) - T(f_1)| - b\exp(nr^2/2)\right)_+ .$$

A useful pair has a target gap larger than $b\exp(nr^2/2)$. Small risk at f_1 can therefore allow a radius larger than $n^{-1/2}$. The between-class modulus selects the largest target gap at that radius; the risk constraint determines how far the comparison can go.

The following formulation of Cai and Low's theory [33, Theorems 1 and 3] gives matching risk orders. Retain the classwise squared-error risk $R_n(\widehat{T}, \mathcal{F}_i)$ from Section 7.1. Order the classes by their ordinary moduli at the noise radius, not by inclusion, and write

$$a_i = \omega(n^{-1/2}, \mathcal{F}_i), \qquad i = 1, 2, \qquad a_{12} = \omega_+(n^{-1/2}, \mathcal{F}_1, \mathcal{F}_2).$$

When $a_1 > 0$, define the comparison factors

$$\gamma_n^+ = \max\left\{e, \frac{a_{12}}{a_1}\right\}, \qquad \gamma_{n,0} = \max\left\{e, \frac{a_{12}}{\sqrt{C_0}\, a_1}\right\},$$

where $C_0 > 0$ is the fixed constant in the first-class risk constraint below. These ratios compare the cross-class target gap with the first-class root mean squared error scale; a large ratio permits an enlarged comparison radius.

Theorem 7.1 (Adaptive Estimation over Two Convex Classes) *In the Gaussian shift* (1.3) *on a real separable Hilbert space H, with noise level $n^{-1/2}$ and $n > 0$, let $\mathcal{F}_1, \mathcal{F}_2 \subset H$ be closed convex classes with nonempty intersection, and let T be real-valued and linear on a vector space containing their union. With the notation above, assume that all moduli in the definitions and bounds are finite and $0 < a_1 \le a_2 < \infty$.*

(i) ***Lower bound.*** *For each fixed $C_0 > 0$, every estimator $\widehat{T} = \widehat{T}(Y)$ satisfying*

$$R_n(\widehat{T}, \mathcal{F}_1) \le C_0 a_1^2$$

obeys

$$R_n(\widehat{T}, \mathcal{F}_2) \ge c(C_0)\left\{\omega_+^2\left(\sqrt{\frac{\log \gamma_{n,0}}{n}}, \mathcal{F}_1, \mathcal{F}_2\right) + a_2^2\right\}.$$

(ii) ***Upper bound.*** *There exists an estimator $\widehat{T}^* = \widehat{T}^*(Y)$ satisfying both*

$$R_n(\widehat{T}^*, \mathcal{F}_1) \le C a_1^2,$$
$$R_n(\widehat{T}^*, \mathcal{F}_2) \le C\left\{\omega_+^2\left(\sqrt{\frac{\log \gamma_n^+}{n}}, \mathcal{F}_1, \mathcal{F}_2\right) + a_2^2\right\}.$$

Here $C > 0$ is universal, and $c(C_0) > 0$ depends only on C_0.

Proof For part (i), put

$$r_{n,0} = \sqrt{\frac{\log \gamma_{n,0}}{n}}.$$

By definition, $\gamma_{n,0} \ge e$, so the following two cases are exhaustive. If $\gamma_{n,0} > e$, choose, for arbitrary $\delta_{\text{app}} \in (0,1)$, a reference parameter $f_1 \in \mathcal{F}_1$ and an alternative $f_2 \in \mathcal{F}_2$ such that $\|f_2 - f_1\|_2 \le r_{n,0}$ and

$$|T(f_2) - T(f_1)| \ge (1 - \delta_{\text{app}})\omega_+(r_{n,0}, \mathcal{F}_1, \mathcal{F}_2).$$

Either ordered modulus can supply the pair: use its signal in $\mathcal{F}_1$ as the reference, since only the absolute target difference matters. For $L = dP_{f_2}/dP_{f_1}$, the Gaussian likelihood-ratio identity gives

$$\|L\|_{L_2(P_{f_1})} = \exp\left\{\frac{n}{2}\|f_2 - f_1\|_2^2\right\} \le \sqrt{\gamma_{n,0}}.$$

The root mean squared error at f_1 is at most $\sqrt{C_0}a_1$. Since $\gamma_{n,0} > e$ implies $a_{12} = \sqrt{C_0}a_1\gamma_{n,0}$ and since the between-class modulus is nondecreasing,

$$\sqrt{C_0}a_1\|L\|_{L_2(P_{f_1})} \le \frac{a_{12}}{\sqrt{\gamma_{n,0}}} \le e^{-1/2}\omega_+(r_{n,0}, \mathcal{F}_1, \mathcal{F}_2).$$

Theorem 2.11 therefore bounds risk at the selected f_2 from below by the squared modulus times $(1 - \delta_{\text{app}} - e^{-1/2})_+^2$. The selected signal may depend on δ_{app}, so take the supremum over $\mathcal{F}_2$ before letting $\delta_{\text{app}} \downarrow 0$:

$$\sup_{f\in\mathcal{F}_2} \mathbb{E}_f(\widehat{T} - T(f))^2 \ge (1 - e^{-1/2})^2\omega_+^2(r_{n,0}, \mathcal{F}_1, \mathcal{F}_2).$$

If $\gamma_{n,0} = e$, then $r_{n,0} = n^{-1/2}$ and

$$\omega_+(r_{n,0}, \mathcal{F}_1, \mathcal{F}_2) = a_{12} \le e\sqrt{C_0}a_1 \le e\sqrt{C_0}a_2.$$

In this case the ordinary convex-class lower bound over $\mathcal{F}_2$ already controls both terms. In both cases Theorem 4.1 also gives

$$\sup_{f\in\mathcal{F}_2} \mathbb{E}_f(\widehat{T} - T(f))^2 \ge c_1 a_2^2.$$

The maximum of the cross-class and within-class lower bounds is at least one half their sum, which proves part (i).

For part (ii), let $G = \mathcal{F}_1 \cup \mathcal{F}_2$. Start with an affine estimator tuned to $\mathcal{F}_1$ and use ordered-modulus affine estimates to check both bias directions. If these comparisons reject it, use an estimator designed for G. Its fourth-moment bound controls error on the data-dependent rejection event. Define the directional comparison factors

$$\gamma_{ij} = \max\left\{e, \frac{\omega(n^{-1/2}, \mathcal{F}_i, \mathcal{F}_j)}{a_1}\right\}, \qquad r_{ij} = \sqrt{\frac{\log\gamma_{ij}}{n}}, \qquad i \ne j.$$

Appendix B.4.1 gives the bias–variance construction, fourth-moment bound, and two-sided test. It produces the first-class and union estimators $\widehat{T}_1, \widehat{T}_G$ and an acceptance indicator I_n such that

$$\widehat{T}^* = I_n\widehat{T}_1 + (1 - I_n)\widehat{T}_G, \qquad \sup_{f\in\mathcal{F}_1} \mathbb{E}_f(\widehat{T}^* - T(f))^2 \le Ca_1^2,$$

and

$$\sup_{f\in\mathcal{F}_2} \mathbb{E}_f(\widehat{T}^* - T(f))^2 \le C\left\{\max_{i\ne j} \omega^2(r_{ij}, \mathcal{F}_i, \mathcal{F}_j) + a_2^2\right\}.$$

Since $\gamma_{ij} \le \gamma_n^+$ and the ordered moduli are nondecreasing,

$$\max_{i\ne j} \omega(r_{ij}, \mathcal{F}_i, \mathcal{F}_j) \le \omega_+\left(\sqrt{\frac{\log\gamma_n^+}{n}}, \mathcal{F}_1, \mathcal{F}_2\right).$$

This proves the upper benchmark and completes the proof. □

Remark 7.2 (Optimal Procedure and Matching Upper Bound) Cai and Low [33, Theorem 3] construct an estimator that uses an estimate tuned to $\mathcal{F}_1$ when bias-adjusted comparisons accept it, and otherwise uses one designed for $\mathcal{F}_1 \cup \mathcal{F}_2$. For sufficiently large fixed C_0, it satisfies the risk constraint on $\mathcal{F}_1$ and attains the lower-bound order on $\mathcal{F}_2$. The between-class modulus therefore determines the cost of preserving accuracy on $\mathcal{F}_1$, relative to the known-class benchmark a_2^2 on $\mathcal{F}_2$.

The cross-class term determines an adaptation cost only when it exceeds the known-class risk on $\mathcal{F}_2$. The next proposition compares these terms when the moduli follow power laws.

Proposition 7.3 (Benchmark Classification via Modulus Exponents) *In the Gaussian shift and two-class setting of Theorem 7.1, suppose that, as $r \downarrow 0$, the moduli satisfy*

$$\omega(r, \mathcal{F}_i) \asymp r^{q_i}, \qquad i = 1, 2,$$

and

$$\omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \asymp r^{q_{12}},$$

for fixed $0 < q_1, q_2, q_{12} \leq 1$. The benchmark ordering implies $q_1 \geq q_2$. As $n \to \infty$, among estimators attaining the $\mathcal{F}_1$ benchmark with a suitable fixed constant, the optimal squared-error risk order over $\mathcal{F}_2$ is

$$\begin{cases} n^{-q_{12}} + n^{-q_2}, & q_{12} \geq q_1, \\ (\log n/n)^{q_{12}} + n^{-q_2}, & q_{12} < q_1 \end{cases}.$$

Consequently:

(i) If $q_{12} > q_2$, or if $q_{12} = q_1 = q_2$, then the risk over $\mathcal{F}_2$ has its known-class order n^{-q_2} and full rate adaptation is possible.

(ii) If $q_{12} = q_2 < q_1$, then the optimal risk over $\mathcal{F}_2$ has order $(\log n/n)^{q_2}$, so adaptation costs the factor $(\log n)^{q_2}$ relative to n^{-q_2}.

(iii) If $q_{12} < q_2 \leq q_1$, then the optimal risk over $\mathcal{F}_2$ has order $(\log n/n)^{q_{12}}$, which is polynomially larger than its known-class benchmark.

Proof At the statistical radius,

$$\omega(n^{-1/2}, \mathcal{F}_i) \asymp n^{-q_i/2},$$

and hence

$$\gamma_n^+ \asymp \max\left\{e, n^{(q_1 - q_{12})/2}\right\}.$$

The factor $\gamma_{n,0}$ in the lower bound has the same order because C_0 is fixed. If $q_{12} \geq q_1$, then $\log \gamma_n^+ \asymp 1$, and the squared cross-class term has order $n^{-q_{12}}$. If $q_{12} < q_1$, then $\log \gamma_n^+ \asymp \log n$, and that term has order

$$\left(\frac{\log n}{n}\right)^{q_{12}}.$$

Adding n^{-q_2} gives the displayed benchmark. Relative to this known-class risk, the cross-class term is negligible when $q_{12} > q_2$ and polynomially larger when $q_{12} < q_2$. When $q_{12} = q_2$, it

incurs a logarithmic factor precisely if $q_1 > q_2$; otherwise γ_n^+ stays bounded. This proves the classification. □

7.2.2 A logarithmic cost under unknown smoothness

Suppose we observe the white-noise model (1.2), with noise level $n^{-1/2}$, and estimate $f(x_0)$ at a fixed interior point under squared-error loss. Smoothness determines the appropriate averaging scale. To quantify the cost of choosing that scale without knowing smoothness, take the Hölder balls of (2.1):

$$\mathcal{F}_i = \Lambda^{\alpha_i}(M), \qquad 0 < \alpha_2 < \alpha_1, \quad M > 0.$$

The parameters are fixed, and these closed convex classes contain zero, with $\mathcal{F}_1 \subset \mathcal{F}_2$. Write $q_i = 2\alpha_i/(2\alpha_i + 1)$. The known-class squared-risk orders are n^{-q_i}, as the following modulus calculation verifies. We ask what risk on the rougher class is unavoidable if the estimator attains the faster rate on the smoother class.

The ordinary-modulus calculation in Example 4.2 extends to every fixed $\alpha > 0$. To see this, choose a smooth function ψ, supported on $[-1, 1]$, with $\psi(0) = 1$. For sufficiently small fixed $c > 0$, the bump

$$f_h(x) = ch^\alpha \psi\big((x - x_0)/h\big)$$

belongs to $\Lambda^\alpha(M)$ when h is small. Its derivatives and highest Hölder seminorm satisfy the defining bounds by rescaling. Its point value is ch^α, whereas $\|f_h\|_2 = c\|\psi\|_2 h^{\alpha+1/2}$. Comparing it with zero gives the required modulus lower bound.

For the upper bound at smoothness $\alpha > 1$, replace simple averaging by polynomial weights. Put $k = \lceil \alpha \rceil - 1$ and choose a polynomial K on $[-1, 1]$ with $\int K = 1$ and $\int u^j K(u)\, du = 0$ for $1 \le j \le k$. Such weights exist because the moment matrix of $1, u, \dots, u^k$ is positive definite. Taylor's formula and Cauchy–Schwarz give, for $f, g \in \Lambda^\alpha(M)$ with $\|f - g\|_2 \le r$,

$$|f(x_0) - g(x_0)| \le C\{Mh^\alpha + rh^{-1/2}\}, \qquad 0 < h \le \min(x_0, 1 - x_0).$$

For $0 < \alpha \le 1$, the same inequality follows with $K = 1/2$. Balancing the two terms proves the ordinary-modulus order for all $\alpha > 0$. Using zero from $\mathcal{F}_1$ and the rougher bump from $\mathcal{F}_2$ gives the between-class lower bound; nesting gives its upper bound. Thus,

$$\omega(r, \mathcal{F}_i) \asymp r^{q_i}, \qquad \omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \asymp r^{q_2}, \qquad r \downarrow 0.$$

Proposition 7.3 therefore gives optimal constrained risk of order $(\log n/n)^{q_2}$ on $\mathcal{F}_2$, rather than its known-class order n^{-q_2}. This is the logarithmic cost of pointwise adaptation studied by Lepski [89] and recovered by the between-class theory of Cai and Low [33, Section 3].

The construction is the localized bump from Example 4.2, but its calibration changes. The smoother-class risk constraint permits L_2 distance of order $\sqrt{\log n/n}$ rather than $n^{-1/2}$. The bump's point value increases to order $(\log n/n)^{q_2/2}$, giving the squared-risk penalty.

7.2.3 Costs beyond a logarithmic factor

The logarithmic penalty above comes from an unknown smoothness level. A larger cost can arise when the useful observations themselves are unknown. Continue to estimate $f(x_0)$

at $x_0 = 1/2$ under squared-error loss in the same white-noise experiment, but now let the smoother side be unknown. We first obtain a polynomial adaptation cost, then exhibit a union on which uniform consistency is impossible.

For $0 < a \le 1$ and $M > 0$, recall from Example 4.5 that $\mathcal{F}^L(a, M)$ imposes $|f(x)-f(y)| \le M|x-y|^a$ for $x, y \in [0, x_0]$, while $\mathcal{F}^R(a, M)$ imposes the same condition on $[x_0, 1]$. These classes have no uniform height bound. Their intersection is

$$\Theta(a_1, M_1, a_2, M_2) = \mathcal{F}^L(a_1, M_1) \cap \mathcal{F}^R(a_2, M_2).$$

For fixed $0 < b < a \le 1$, take

$$\mathcal{F}_1 = \Theta(a, 1, b, 1), \qquad \mathcal{F}_2 = \Theta(b, 1, a, 1).$$

The smoother side of the evaluation point is left in the first class and right in the second. Reflection about $1/2$ interchanges the classes, so their ordinary moduli agree. Writing $q_a = 2a/(2a+1)$ and $q_b = 2b/(2b+1)$, the calculations in that example give

$$\omega(r, \mathcal{F}_i) \asymp r^{q_a}, \qquad \omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \asymp r^{q_b}.$$

Thus each known-class risk has order n^{-q_a}, but attaining this rate on $\mathcal{F}_1$ forces risk of order at least $(\log n/n)^{q_b}$ on $\mathcal{F}_2$. Theorem 7.1 attains that order with a suitable fixed first-class constant. Relative to the second class's own benchmark, the cost is

$$n^{q_a - q_b} (\log n)^{q_b}.$$

For example, $a = 1$ and $b = 1/4$ change the risk order from $n^{-2/3}$ to $(\log n/n)^{1/3}$. Cai and Low [33, Example 2] use this geometry to exhibit a polynomial cost of adaptation.

In Chapter 4, the same alternatives gave union minimax risk of order n^{-q_b}. Here, preserving one component's faster rate forces a further logarithmic cost on the other. The construction is unchanged; the constrained-risk requirement changes its calibration.

The contrast is sharper if the opposite side is only continuous, with no uniform smoothness bound. Fix $0 < a \le 1$ and $M > 0$, and consider

$$\mathcal{G}_L = \{f \in C[0,1] \cap \mathcal{F}^L(a, M) : \|f\|_\infty \le M\},$$
$$\mathcal{G}_R = \{f \in C[0,1] \cap \mathcal{F}^R(a, M) : \|f\|_\infty \le M\}.$$

If the class is known, a one-sided local average has squared bias at most $M^2 h^{2a}$ and variance $(nh)^{-1}$. The balance $h \asymp n^{-1/(2a+1)}$, together with the localized-bump lower bound above, gives minimax risk of order $n^{-2a/(2a+1)}$ on either class. Without knowledge of the smooth side, however, no estimator is uniformly consistent over their union.

Here is a direct two-point proof. Take a smooth cutoff ψ, supported on $[-1, 1]$, with $0 \le \psi \le 1$ and $\psi(0) = 1$. For $0 < h < 1/2$, set

$$f_h(x) = \begin{cases} 0, & x \le x_0, \\ M\{1 - \psi((x - x_0)/h)\}, & x > x_0, \end{cases} \qquad g_h(x) = \begin{cases} M\psi((x - x_0)/h), & x < x_0, \\ M, & x \ge x_0. \end{cases}$$

Both functions are continuous and bounded by M. The first is constant on the left and the second on the right, so $f_h \in \mathcal{G}_L$ and $g_h \in \mathcal{G}_R$. Nevertheless,

$$g_h(x_0) - f_h(x_0) = M, \qquad \|g_h - f_h\|_2^2 = M^2 h \int_{-1}^{1} \psi(u)^2 \, du.$$

Thus $\omega_+(r, \mathcal{G}_L, \mathcal{G}_R) \geq M$ for every $r > 0$: the between-class modulus does not tend to zero. For each n, choose h so small that $n\|g_h - f_h\|_2^2 \leq 1$. The white-noise KL divergence is then at most $1/2$, and the two-point bound gives

$$\inf_{\widehat{T}} \sup_{f \in \mathcal{G}_L \cup \mathcal{G}_R} \mathbb{E}_f\{\widehat{T} - f(x_0)\}^2 \geq cM^2$$

with a universal $c > 0$. Thus the difficulty is not merely a penalty for preserving one class's faster rate: no procedure is uniformly consistent over the union. The nonvanishing between-class modulus reveals what the ordinary moduli miss.

7.2.4 Shape constraints and no-cost adaptation

Shape information can instead reduce the cross-class target gaps. Suppose the signal is known to be nonincreasing, as in a decreasing response curve, but its smoothness is unknown. In the same white-noise experiment, we ask whether one estimator can attain each class's minimax rate.

Retain squared-error loss for $T(f) = f(x_0)$, where $x_0 = 1/2$. Using the Hölder seminorm defined after (2.1), let $F_D(\alpha, M)$ consist of the nonincreasing continuous functions on $[0, 1]$ satisfying $[f]_{C^\alpha} \leq M$, where $0 < \alpha \leq 1$ and $M > 0$. Unlike the ball $\Lambda^\alpha(M)$, this class imposes no uniform sup-norm bound.

Cai and Low [33, Example 1] show that monotonicity changes the between-class modulus to the smoother-class order without changing the known-class rates. A pointwise gap must persist on one side, where the smoother function controls how quickly it can close. The next result derives this geometric fact and its consequence: adaptation without a rate penalty.

Corollary 7.4 (Shape-Constrained Point Estimation: Free Adaptation) *In the continuous white-noise model* (1.2) *with noise level* $n^{-1/2}$ *and target* $T(f) = f(1/2)$, *let* $\mathcal{F}_i = F_D(\alpha_i, M_i)$, *where* $0 < \alpha_2 < \alpha_1 \leq 1$ *and* $M_1, M_2 > 0$ *are fixed. As* $r \downarrow 0$,

$$\omega(r, \mathcal{F}_i) \asymp r^{2\alpha_i/(2\alpha_i+1)}, \quad i = 1, 2, \qquad \omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \asymp r^{2\alpha_1/(2\alpha_1+1)}.$$

Consequently there is one estimator sequence, not depending on which class contains f, *such that*

$$\sup_{f \in \mathcal{F}_i} \mathbb{E}_f\big(\widehat{T}_n - f(1/2)\big)^2 \leq Cn^{-2\alpha_i/(2\alpha_i+1)}, \qquad i = 1, 2,$$

for all sufficiently large n. These are the respective minimax risk orders; constants may depend on the fixed class parameters.

Proof *Lower bounds.* Take $M_* = \min(M_1, M_2)$ and choose $\delta > 0$ sufficiently small that $h = (\delta/M_*)^{1/\alpha_1} \leq 1/2$. Define

$$f_\delta(x) = \begin{cases} \delta, & x \leq x_0, \\ (\delta - M_*(x - x_0)^{\alpha_1})_+, & x > x_0, \end{cases} \qquad g_\delta(x) = \begin{cases} \min\{\delta, M_*(x_0 - x)^{\alpha_1}\}, & x < x_0, \\ 0, & x \geq x_0. \end{cases}.$$

Both functions are nonincreasing and α_1-Hölder with constant M_*. Since the interval has length one and $\alpha_1 > \alpha_2$, they are also α_2-Hölder with constant $M_* \le M_2$. Thus $f_\delta, g_\delta \in \mathcal{F}_1 \cap \mathcal{F}_2$, with $f_\delta(x_0) - g_\delta(x_0) = \delta$ and, integrating in the distance u from x_0,

$$\begin{aligned}\|f_\delta - g_\delta\|_2^2 &= 2\int_0^h (\delta - M_* u^{\alpha_1})^2\, du \\ &= c_{\alpha_1} M_*^{-1/\alpha_1} \delta^{2+1/\alpha_1}, \qquad c_{\alpha_1} = 2\int_0^1 (1-u^{\alpha_1})^2\, du.\end{aligned}$$

Choosing δ so that the last display equals r^2 proves the lower bounds

$$\omega(r, \mathcal{F}_1) \gtrsim r^{2\alpha_1/(2\alpha_1+1)}, \qquad \omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \gtrsim r^{2\alpha_1/(2\alpha_1+1)}.$$

Upper bounds. Take $f \in \mathcal{F}_1$, $g \in \mathcal{F}_2$, and set $\delta = |f(x_0) - g(x_0)|$. If $f(x_0) \ge g(x_0)$, monotonicity of g and smoothness of f give $f(x) - g(x) \ge \delta - M_1(x - x_0)^{\alpha_1}$ for $x_0 \le x \le 1$. If $g(x_0) > f(x_0)$, then $g(x) - f(x) \ge \delta - M_1(x_0 - x)^{\alpha_1}$ for $0 \le x \le x_0$. Therefore, with $h = (\delta/(2M_1))^{1/\alpha_1}$ and $h \le 1/2$, integration over $[x_0, x_0 + h]$ or $[x_0 - h, x_0]$, respectively, gives

$$\|f - g\|_2^2 \ge \frac{\delta^2 h}{4} = c M_1^{-1/\alpha_1} \delta^{2+1/\alpha_1}.$$

If $h > 1/2$, the same one-sided inequality gives

$$\|f - g\|_2^2 \ge \frac{\delta^2}{8}, \qquad \delta > 2M_1(1/2)^{\alpha_1}.$$

Thus $h > 1/2$ is impossible when $\|f - g\|_2 \le r$ and r is sufficiently small. The preceding bound then gives

$$\delta \le C M_1^{1/(2\alpha_1+1)} r^{2\alpha_1/(2\alpha_1+1)}.$$

The same argument with both endpoints in $\mathcal{F}_1$ proves the ordinary-modulus upper bound. We have thus shown

$$\omega(r, \mathcal{F}_1) \asymp r^{2\alpha_1/(2\alpha_1+1)}, \tag{7.4}$$

$$\omega_+(r, \mathcal{F}_1, \mathcal{F}_2) \asymp r^{2\alpha_1/(2\alpha_1+1)}. \tag{7.5}$$

Repeating the ordinary calculation with (α_2, M_2) gives $\omega(r, \mathcal{F}_2) \asymp r^{2\alpha_2/(2\alpha_2+1)}$.

From (7.4) and (7.5) we obtain

$$q_1 = q_{12} = \frac{2\alpha_1}{2\alpha_1 + 1}, \qquad q_2 = \frac{2\alpha_2}{2\alpha_2 + 1}.$$

Since $\alpha_1 > \alpha_2$, we have $q_{12} = q_1 > q_2$, the free-adaptation case of Proposition 7.3. The cross-class component is no larger than the smoother-class rate, and the total risk over $\mathcal{F}_2$ remains of order n^{-q_2}. □

Remark 7.5 (Optimal Procedure and Matching Upper Bound) For the corollary's monotone classes, the estimator in Theorem 7.1 attains each class's minimax rate without knowing which class contains the signal. For convex functions in the same white-noise model, Cai and Low [15, Theorems 1 and 3] use differences of local averages to assess bias and select

an averaging scale. Their estimator attains the function-specific two-point minimax mean squared error, up to constants, characterized by the squared local modulus at the noise scale. Here the target is a point value, not the minimum or minimizer studied under absolute loss in Section 4.5.1.

Their estimator also attains the minimax rates simultaneously over convex Hölder seminorm classes with $1 \le \alpha \le 2$, $M \ge 1$, and $n > 1$, without bounds on the function or lower derivatives [15, Theorem 4]. This smoothness range cannot be extended without qualification: faster convergence above smoothness two can force a logarithmic loss at smoothness two [15, Proposition 3].

7.2.5 Adaptation under probabilistic error

Adaptation also depends on the loss criterion. Retain the Gaussian shift (1.3), with noise level $n^{-1/2}$, and the target $T(f)$. Instead of controlling mean squared error, require the absolute estimation error to exceed a specified radius with probability at most a fixed $0 < \alpha < 1/2$. Can one estimator attain the class-specific radii without knowing the class?

Let $\mathcal{F}_1, \dots, \mathcal{F}_k \subset H$ be a fixed finite collection of convex classes with pairwise nonempty intersections, and let T be linear on a vector space containing them. The adaptation problem is posed over the successive unions $\mathcal{G}_j = \bigcup_{i \le j} \mathcal{F}_i$, a nested family even when the components are not nested. When the relevant ordinary and ordered moduli are finite, Cai and Low [34, Theorems 1–2] identify $\omega(n^{-1/2}, \mathcal{G}_j)$ as the minimax error-radius order and construct one estimator satisfying

$$\sup_{f \in \mathcal{G}_j} \mathbb{P}_f \left\{ |\widehat{T} - T(f)| > C\omega(n^{-1/2}, \mathcal{G}_j) \right\} \le \alpha, \qquad j = 1, \dots, k.$$

Randomized estimators are allowed.[1] The constant C depends on k and α, not on n, so fixed k requires no logarithmic enlargement. For increasing nested classes, $\mathcal{G}_j = \mathcal{F}_j$, and one estimator attains every component's benchmark. For nonnested classes, the guarantee concerns the successive unions, not the individual components. Their larger minimax radii can still reflect the cross-class obstructions above.

For nested classes, the contrast with Section 7.2.2 comes from rare, large errors: a probability bound controls their frequency, whereas squared-error risk also accounts for their magnitude. Nor does this guarantee provide an adaptive honest interval: the displayed radius depends on the unknown class, whereas interval endpoints must be data-determined (Section 7.4).

7.3 Adaptive estimation of quadratic functionals

We return to the quadratic functional $Q(\theta) = \sum_{i \ge 1} \theta_i^2$ in the Gaussian sequence model (1.1), with known noise variance n^{-1} and squared-error loss. Estimating this total signal energy requires deciding which weak coefficients to retain. In Section 4.2, truncation and thresholding could use the known parameter class. We now ask whether one estimator can attain those

[1] The strict and nonstrict error conventions give the same infimum of admissible radii, although a zero infimum need not be attained under the nonstrict convention.

benchmarks without knowing the smoothness, sparsity, or radius. All estimators are allowed: the issue is unknown structure, not the restriction to quadratic procedures studied there.

Let Θ be a weighted sequence class $L_p(\alpha, M)$ or a Besov ball $B^{\alpha}_{p,q}(M)$ from (4.7) and (3.8), with the single-sequence indexing of Section 4.2. The parameters p, α, M are fixed, finite, and positive, with $s = \alpha + 1/2 - 1/p > 0$; for Besov balls, q is also fixed, finite, and positive. No convexity assumption is needed. We seek classwise adaptation: comparison constants may depend on these parameters, but the estimator may not.

Retain p_*, s_* from (4.8), and $r(\alpha, p)$ and $A(\Theta) \in (0, \infty)$ from Theorem 4.7. The known-class squared-error risk has order $n^{-r(\alpha,p)}$: it is parametric, of order n^{-1}, when $\alpha p_* \geq 1/2$, and slower below this elbow. The quantity $A(\Theta) = \sup_{\theta\in\Theta} Q(\theta)$ determines the sharp constant strictly above it.

As in Section 7.2, the lower bound links risks across classes. Here mixtures replace the modulus-selected pair: random signs for $p \geq 2$ and random supports for $p < 2$ control the mixture likelihood-ratio moment while leaving Q fixed. The constrained-risk argument of Section 2.4 uses accuracy at zero, which belongs to every class. This risk constraint allows the second moment to grow polynomially in n, rather than remain bounded as in the fixed-class proof. The resulting larger target gap yields the adaptation penalty.

Cai and Low [36, Theorem 2] construct a nonquadratic estimator that is efficient above the minimax elbow and has a logarithmically inflated risk bound at or below it. The following theorem pairs this benchmark with a constrained-risk lower bound. Laurent and Massart [86] provide related model-selection methods.

Theorem 7.6 (Adaptive Benchmark for Quadratic-Functional Estimation) *In the Gaussian sequence model* (1.1) *with $\varepsilon = n^{-1/2}$ and the class restrictions above, the following assertions hold as $n \to \infty$. All estimators may depend on n.*

(i) ***Adaptive upper bound.*** *There exists a single estimator $\widehat{Q} = \widehat{Q}(Y)$, independent of (p, q, α, M), such that, for every fixed class Θ as above,*

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta\big(\widehat{Q} - Q(\theta)\big)^2 \leq \begin{cases} 4A(\Theta)n^{-1}(1 + o(1)), & \alpha p_* > \frac{1}{2}, \\ Cn^{-r(\alpha,p)}(\log n)^{2p_*s_*/(1+2p_*s_*)}, & \alpha p_* \leq \frac{1}{2}, \end{cases}$$

where C may depend on the fixed class parameters.

(ii) ***Constrained-risk lower bound.*** *Suppose $\alpha p_* \leq 1/2$. If an estimator $\widetilde{Q} = \widetilde{Q}(Y)$ satisfies, for all sufficiently large n,*

$$\mathbb{E}_0\big(\widetilde{Q} - Q(0)\big)^2 \leq C_0 n^{-\gamma} \qquad \text{for fixed } C_0 > 0 \text{ and } \gamma > r(\alpha, p),$$

then there is $C' > 0$, depending only on C_0, γ and the fixed class parameters, such that its maximum squared bias and mean squared error satisfy, for all sufficiently large n,

$$\sup_{\theta\in\Theta}\big\{\mathbb{E}_\theta\widetilde{Q} - Q(\theta)\big\}^2 \geq C' n^{-r(\alpha,p)}(\log n)^{2p_*s_*/(1+2p_*s_*)},$$
$$\sup_{\theta\in\Theta} \mathbb{E}_\theta\big(\widetilde{Q} - Q(\theta)\big)^2 \geq C' n^{-r(\alpha,p)}(\log n)^{2p_*s_*/(1+2p_*s_*)}.$$

Strictly above the elbow, part (i) matches the known-class minimax risk, including its leading constant. Below it, the displayed risk exceeds the known-class rate by a logarithmic factor.

Part (ii) makes that factor unavoidable if the same estimator attains $O(n^{-1})$ risk on a smoother class containing zero: then $\gamma = 1 > r(\alpha, p)$, and the lower bound matches part (i), already through squared bias. Any faster power rate with $\gamma > r(\alpha, p)$ gives the same obstruction.

At the elbow, $r(\alpha, p) = 1$, the lower bound instead requires $\gamma > 1$; an $O(n^{-1})$ guarantee on a smoother class does not establish the displayed boundary penalty. The restriction $\alpha p_* \le 1/2$ is also essential to the lower-bound formulation here, although it is not displayed in Cai and Low [36, Theorem 1].

Proof of the constrained-risk lower bound Write $\ell_n = \log n$, let P_0 be the observation law at zero, and take n sufficiently large that the blocks below are nonempty. Since $\alpha p_* \le 1/2$, we have $r(\alpha, p) = 2 - p_*/(1+2p_*s_*)$, including at the elbow. Each construction places a prior on Θ with constant target $\Delta_m > 0$ and observation mixture P_π satisfying $\mathbb{E}_0(dP_\pi/dP_0)^2 \le n^\kappa$. We choose $\kappa < \gamma - r(\alpha, p)$ so that Cauchy–Schwarz keeps the mixture expectation of $\widetilde{Q}$ small relative to Δ_m. This forces squared bias of order Δ_m^2.

Dense alternatives: random signs ($p \ge 2$). Here $p_* = 2$, $s_* = \alpha$, and $\alpha \le 1/4$. Put $D = 1 + 2p_*s_* = 1 + 4\alpha$ and use the block $I_m = \{m, \dots, 2m-1\}$, one Besov level for dyadic m. Smoothness permits amplitude of order $m^{-\alpha-1/2}$. The log second moment will be bounded by $mn^2a^4/2$, so keeping it of order $\log n$ requires $n^2m^{-D} \asymp \ell_n$. Accordingly, choose dyadic m and amplitude a with

$$m \asymp (n^2/\ell_n)^{1/D}, \qquad a = c_0 m^{-\alpha-1/2}.$$

The constant $c_0 > 0$ will be chosen small. Relative to the fixed-class construction in Section 4.2, the smaller block permits larger coefficients; the price is a growing second moment. Let π assign independent Rademacher signs to the vector with coordinates $\theta_i = a$ in absolute value on I_m and zero elsewhere. For the weighted sequence class $L_p(\alpha, M)$,

$$\sum_{i \in I_m} i^{ps} |\theta_i|^p \le C m^{ps+1} a^p = C c_0^p;$$

for the Besov ball, the only nonzero level contributes $m^s(m^{1/p}a) = c_0$. Thus the prior lies in Θ when c_0 is small. Every alternative has

$$Q(\theta) = \Delta_m = ma^2 = c_0^2 m^{-2\alpha}, \qquad \Delta_m^2 \asymp n^{-r(\alpha,p)} \ell_n^{2p_*s_*/D}. \tag{7.6}$$

For two independent draws from the prior, each product of signs is again Rademacher. The Gaussian likelihood cross-moment therefore contributes $\cosh(na^2)$ per active coordinate. Independence and $\cosh x \le e^{x^2/2}$ give

$$1 + \chi^2(P_\pi, P_0) = \{\cosh(na^2)\}^m \le \exp\{mn^2a^4/2\} \le n^{\kappa(c_0)},$$

where $\kappa(c_0) \downarrow 0$ as $c_0 \downarrow 0$.

Sparse alternatives: random support ($0 < p < 2$). Here $p_* = p$, $s_* = s$, and the assumptions $s > 0$ and $\alpha p \le 1/2$ imply $p > 1$. Put $D = 1 + 2ps$. With active amplitude $n^{-1/2}$, the log second moment is controlled by k^2/m. Allowing this quantity to have order $\log m$ suggests $k \asymp \sqrt{m \log m}$. The class constraint $m^s k^{1/p} n^{-1/2} \lesssim 1$ then requires $m^D \log m \lesssim n^p$. For a sufficiently small $b > 0$, choose

$$m \asymp n^{p/D} \ell_n^{-1/D}, \qquad k = \left\lfloor \sqrt{b\, m \log m} \right\rfloor.$$

Take m dyadic and round k as displayed. For large n, $1 \le k \le m$. Choose S uniformly among the k-subsets of I_m and set $\theta_i = n^{-1/2}\mathbf{1}\{i \in S\}$, with all other coordinates zero. Unlike the fixed-class support size $\sqrt{m}$, the larger support increases $Q(\theta) = k/n$. To check class membership, the one-level Besov norm equals $m^s k^{1/p} n^{-1/2}$, and the weighted L_p norm is at most 2^s times this quantity. Since $\log m \asymp \ell_n$,

$$\begin{aligned} m^s k^{1/p} n^{-1/2} &\le b^{1/(2p)} m^{D/(2p)} (\log m)^{1/(2p)} n^{-1/2} \\ &\le C b^{1/(2p)}, \end{aligned}$$

so the prior is supported on Θ when b is sufficiently small. Every alternative has

$$Q(\theta) = \Delta_m = k/n, \qquad \Delta_m^2 \asymp n^{-r(\alpha,p)} \ell_n^{2ps/D}. \tag{7.7}$$

Let S' be an independent support drawn from the same prior. In the Gaussian likelihood cross-moment, each coordinate in $S \cap S'$ contributes $n(n^{-1/2})^2 = 1$ to the exponent; all other coordinates contribute zero. Hence

$$1 + \chi^2(P_\pi, P_0) = \mathbb{E} e^{|S \cap S'|}.$$

The overlap bound in Lemma A.11, with ambient dimension m and $t = 1$, gives

$$\mathbb{E} e^{|S \cap S'|} \le \exp\{(e-1)k^2/m\} \le m^{(e-1)b}.$$

Thus in this case also $1 + \chi^2(P_\pi, P_0) \le n^{\kappa(b)}$ with $\kappa(b) \downarrow 0$ as $b \downarrow 0$.

Risk transfer: use accuracy at zero. Both priors have constant target value Δ_m. Write $\mathbb{E}_{P_\pi}$ for expectation under the data mixture and $\mathbb{E}_{\theta \sim \pi}$ for expectation over the prior. Let $L = dP_\pi / dP_0$. Since $Q(0) = 0$, the risk assumption at zero and Cauchy–Schwarz give

$$|\mathbb{E}_{P_\pi} \widetilde{Q}| = |\mathbb{E}_0(L\widetilde{Q})| \le (\mathbb{E}_0 \widetilde{Q}^2)^{1/2} (\mathbb{E}_0 L^2)^{1/2} \le C_0^{1/2} n^{-(\gamma-\kappa)/2}.$$

Choose c_0 in the dense construction or b in the sparse construction so that $\gamma - \kappa > r(\alpha, p)$. The squared expectation is then $O(n^{-(\gamma-\kappa)})$, smaller in order than the squared target gap in (7.6) and (7.7). Thus $|\mathbb{E}_{P_\pi} \widetilde{Q}| = o(\Delta_m)$ and is at most $\Delta_m/2$ for all sufficiently large n. Every alternative has target Δ_m, so the average bias must be large. Jensen's inequality yields

$$\begin{aligned} \sup_{\theta \in \Theta} \{\mathbb{E}_\theta \widetilde{Q} - Q(\theta)\}^2 &\ge \left| \mathbb{E}_{\theta \sim \pi} \{\mathbb{E}_\theta \widetilde{Q} - Q(\theta)\} \right|^2 \\ &\ge \frac{1}{4} \Delta_m^2 \ge C' n^{-r(\alpha,p)} \ell_n^{2p_* s_*/(1+2p_* s_*)}. \end{aligned}$$

Mean squared error dominates squared bias pointwise, giving the risk lower bound. □

For a concrete comparison, take $\Theta = L_2(1/8, M)$ with fixed $M > 0$. Its known-class minimax risk has order $n^{-2/3}$. If the same estimator has risk $O(n^{-1})$ on a smoother class containing zero, part (ii) applies with $\gamma = 1$. The dense construction uses

$$m \asymp n^{4/3} (\log n)^{-2/3}, \qquad \Delta_m^2 \asymp n^{-2/3} (\log n)^{1/3}.$$

Choosing the amplitude constant so that $\kappa < 1/3$ makes the risk-transfer term negligible. The optimal risk under the smoother-class requirement is therefore $n^{-2/3}(\log n)^{1/3}$, by parts (i)–(ii). The sign mixture is the same as in the fixed-class argument; accuracy at zero changes its calibration.

Remark 7.7 (Adaptive Procedure and Upper-Bound Comparison) Cai and Low [36, Theorem 2] construct an adaptive estimator combining an unbiased low-frequency term, penalized selection among block and coordinate-thresholding estimates, and a thresholded tail. In the nonparametric region $\alpha p_* < 1/2$, its squared-error risk has order

$$n^{-r(\alpha,p)}(\log n)^{2p_*s_*/(1+2p_*s_*)}.$$

This matches part (ii) under the smoother-class risk requirement discussed above. In the parametric region $\alpha p_* > 1/2$, the estimator attains the sharp asymptotic risk $4A(\Theta)/n$, with no first-order adaptation cost. Appendix B.4.3 proves the upper bound.

7.4 Adaptive confidence intervals

Adaptive estimation does not by itself provide an honest assessment of accuracy. Suppose we observe $X \sim P_f$ and report an interval for a scalar target $T(f)$, with coverage at least $1-\alpha$ throughout $\mathcal{F}$, where $0 < \alpha < 1/2$. How small can its expected length be on $\mathcal{F}_1 \subseteq \mathcal{F}$, or at an individual function? For a function value, we ask whether unknown smoothness or favorable local shape permits shorter intervals without weakening coverage elsewhere.

Retain the interval and expected-length notation of Section 6.2. The ratio (7.3) compares this length with the known-class benchmark. Cai and Low [30] characterize the comparison through between-class moduli. We use coverage transfer to contrast smoothness and shape restrictions: honesty replaces the small-risk constraint used for estimation in Sections 7.2–7.3.

7.4.1 Coverage transfer and between-class moduli

To bound length on $\mathcal{F}_1$, compare $f_0 \in \mathcal{F}_1$ with $f_1 \in \mathcal{F}$ whose target differs but whose observation law is sufficiently close. Coverage of $T(f_1)$ then transfers to P_{f_0}, forcing the interval to contain both targets with positive probability. We recall the pair and mixture bounds from Chapter 6, then use a Gaussian between-class modulus to select the target gap.

For f_0, f_1 in the coverage class and $\Delta = |T(f_1) - T(f_0)|$, the point-mass case of Lemma 6.3 gives

$$\mathbb{E}_{f_0} L(CI) \geq \Delta\{1 - 2\alpha - \mathrm{TV}(P_{f_0}, P_{f_1})\}_+. \tag{7.8}$$

Length is evaluated at f_0; the alternative f_1 may lie anywhere coverage is required.

For mixtures, retain the assumptions of Lemma 6.3: the target equals μ_i on H_i, and a prior π_i supported on H_i induces the observation law P_{π_i}, $i = 0, 1$. For $CI \in I_\alpha(H_0 \cup H_1, T)$, the lemma gives

$$L(CI, H_0) \geq |\mu_1 - \mu_0|\big(1 - 2\alpha - \mathrm{TV}(P_{\pi_1}, P_{\pi_0})\big)_+. \tag{7.9}$$

Taking H_0 to be a singleton gives a length bound at one parameter.

Corollary 7.8 (Point-Null Specialization) *Under the assumptions of Lemma 6.3, take* $0 < \alpha < 1/2$, $H_0 = \{\theta_0\}$, *and* $\pi_0 = \delta_{\theta_0}$. *Every* $CI \in I_\alpha(\{\theta_0\} \cup H_1, T)$ *then satisfies*

$$\mathbb{E}_{\theta_0} L(CI) \geq |\mu_1 - T(\theta_0)| \big(1 - 2\alpha - \mathrm{TV}(P_{\pi_1}, P_{\theta_0})\big)_+.$$

Proof Substitute $\mu_0 = T(\theta_0)$, $P_{\pi_0} = P_{\theta_0}$, and $L(CI, H_0) = \mathbb{E}_{\theta_0} L(CI)$ into (7.9). □

Section 8.1.4 uses this bound to force interval length at the zero regression vector. As in Section 7.3, mixing separates targets while limiting information, but the calibration differs. Coverage transfer requires close observation laws; constrained risk can allow a growing likelihood-ratio moment because it also uses accuracy at zero.

We now specialize to the Gaussian shift (1.3), covering the sequence and white-noise models, with noise level $n^{-1/2}$ and signal norm $\|\cdot\|_2 = \|\cdot\|_H$. The symmetrized between-class modulus (4.4) selects a target gap subject to a bound on this distance. For a calibration constant $0 < \varepsilon < \sqrt{2}(1 - 2\alpha)$, the next theorem forces length comparable to the modulus with positive probability. The expected-length obstruction therefore cannot be confined to rare, very long intervals.

Theorem 7.9 (Probabilistic Between-Class Lower Bound) *In the Gaussian shift* (1.3) *with noise level* $n^{-1/2}$, $n > 0$, *let* $\mathcal{F}_1, \mathcal{F}_2$ *be nonempty parameter classes and* $T : \mathcal{F}_1 \cup \mathcal{F}_2 \to \mathbb{R}$. *Set* $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$ *and use the between-class modulus with Hilbert distance* $\|\cdot\|_2$. *Fix* $0 < \alpha < 1/2$ *and a calibration constant* $\varepsilon > 0$. *Assume that the comparison set defining the modulus at* $\varepsilon/\sqrt{n}$ *is nonempty and that*

$$0 \leq \omega_+(\varepsilon/\sqrt{n}, \mathcal{F}_1, \mathcal{F}_2) < \infty.$$

Define

$$q_{\alpha,\varepsilon} = \left(1 - 2\alpha - \frac{\varepsilon}{\sqrt{2}}\right)_+.$$

Then, for every $0 < \delta_{\mathrm{app}} < 1$ *and every* $CI \in I_\alpha(\mathcal{F}, T)$,

$$\sup_{f \in \mathcal{F}_1} \mathbb{P}_f \left\{ L(CI) \geq (1 - \delta_{\mathrm{app}}) \omega_+\left(\frac{\varepsilon}{\sqrt{n}}, \mathcal{F}_1, \mathcal{F}_2\right) \right\} \geq q_{\alpha,\varepsilon}.$$

Consequently,

$$L(CI, \mathcal{F}_1) \geq q_{\alpha,\varepsilon} \omega_+\left(\frac{\varepsilon}{\sqrt{n}}, \mathcal{F}_1, \mathcal{F}_2\right).$$

The same conclusion holds with the roles of $\mathcal{F}_1$ *and* $\mathcal{F}_2$ *interchanged.*

Proof Fix $0 < \delta_{\mathrm{app}} < 1$ and write

$$\omega_n = \omega_+\left(\frac{\varepsilon}{\sqrt{n}}, \mathcal{F}_1, \mathcal{F}_2\right).$$

The nonempty comparison set and the definition of the supremum supply signals $f \in \mathcal{F}_1$ and $g \in \mathcal{F}_2$ with

$$\|g - f\|_2 \leq \frac{\varepsilon}{\sqrt{n}}, \qquad |T(g) - T(f)| \geq (1 - \delta_{\mathrm{app}}) \omega_n.$$

Write $\Delta = |T(g) - T(f)|$. Honesty at both endpoints and the point-mass case of (6.4) give

$$\mathbb{P}_f\{L(CI) \geq \Delta\} \geq 1 - 2\alpha - \mathrm{TV}(P_f, P_g).$$

When $\Delta = 0$, nonnegative length gives the same inequality. The Gaussian likelihood identity and Pinsker's inequality give

$$\mathrm{KL}(P_f, P_g) = \frac{n}{2}\|f - g\|_2^2 \leq \frac{\varepsilon^2}{2},$$
$$\mathrm{TV}(P_f, P_g) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(P_f, P_g)} \leq \frac{\varepsilon}{2} \leq \frac{\varepsilon}{\sqrt{2}}.$$

Thus $\mathbb{P}_f\{L(CI) \geq \Delta\} \geq q_{\alpha,\varepsilon}$. Since $\Delta \geq (1 - \delta_{\mathrm{app}})\omega_n$, this proves

$$\sup_{f \in \mathcal{F}_1} \mathbb{P}_f\big\{L(CI) \geq (1 - \delta_{\mathrm{app}})\omega_n\big\} \geq q_{\alpha,\varepsilon}.$$

For expected length, $L(CI) \geq \Delta \mathbf{1}\{L(CI) \geq \Delta\}$ yields

$$L(CI, \mathcal{F}_1) \geq \mathbb{E}_f L(CI) \geq \Delta \mathbb{P}_f\{L(CI) \geq \Delta\} \geq q_{\alpha,\varepsilon}(1 - \delta_{\mathrm{app}})\omega_n.$$

Letting $\delta_{\mathrm{app}} \downarrow 0$ proves the expected-length bound without requiring attainment of the modulus. Interchanging $\mathcal{F}_1$ and $\mathcal{F}_2$ gives the symmetric version. □

Unlike the constrained-risk argument in Section 7.2, this comparison uses the ordinary noise radius $n^{-1/2}$, not an enlarged radius calibrated to an accuracy requirement. Honesty alone forces length on the evaluation class. Neither convexity of the classes nor linearity of the target is needed. The following specialization connects the bound to the convex-class benchmarks of Chapter 6.

For this and the later functionwise bound, fix $\zeta_\alpha > 0$ such that $\zeta_\alpha < \sqrt{2}(1 - 2\alpha)$. Then $2\Phi(-\zeta_\alpha/2) \geq 1 - \zeta_\alpha/2 > 2\alpha$, so both coverage-transfer constants are positive. Here ζ_α is a calibration radius, not the normal quantile z_α.

Corollary 7.10 (Two-Class Lower Bound for Honest Confidence Intervals over Convex Spaces) *In the Gaussian experiment of Theorem 7.9, let $\mathcal{F}_1, \mathcal{F}_2$ be nonempty convex classes, let T be linear, and set $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$. For $0 < \alpha < 1/2$, assume the displayed moduli are finite and use the calibration radius ζ_α chosen above. Then, with $c_\alpha = 1 - 2\alpha - \zeta_\alpha/\sqrt{2} > 0$,*

$$L^*_\alpha(\mathcal{F}_j, \mathcal{F}, T) \geq c_\alpha\, \omega_+\left(\frac{\zeta_\alpha}{\sqrt{n}}, \mathcal{F}_j, \mathcal{F}\right), \qquad j = 1, 2.$$

Proof Apply Theorem 7.9 to $(\mathcal{F}_j, \mathcal{F})$ with $\varepsilon = \zeta_\alpha$, then take the infimum over $CI \in I_\alpha(\mathcal{F}, T)$. The comparison set is nonempty because $\mathcal{F}_j \subseteq \mathcal{F}$. □

Cai and Low [30, Theorems 2–3] construct confidence intervals from affine estimators with directional bias control. For finite nested families, their selection procedure attains the between-class expected-length benchmarks. For pairwise-intersecting convex families, their intersection procedure incurs a factor depending on the number of classes. Both maintain honest coverage over the union.

7.4.2 Smoothness and the cost of honesty

Suppose we observe white noise (1.2), with noise level $n^{-1/2}$, and seek an interval for $f(t_0)$ at a fixed interior point. The function belongs to one of two Hölder balls with fixed smoothness indices $0 < \alpha < \beta$ and positive radii; write $\alpha_0 \in (0, 1/2)$ for the noncoverage level. We require honesty on their union but evaluate length on the smoother ball, whose known-class optimal length has order $n^{-\beta/(2\beta+1)}$.

To test whether that shorter length is possible without knowing smoothness, compare zero, which belongs to both balls, with a narrow bump in the rougher ball. Its height separates the targets while its width keeps the observation laws close. Coverage at the bump then forces a longer interval even at zero. The between-class modulus quantifies this obstruction.

Corollary 7.11 (Smoothness Classes: The Between-Class Benchmark Governs) *Work in the continuous white-noise model* (1.2) *with noise level* $n^{-1/2}$ *and estimate* $T(f) = f(t_0)$ *at a fixed* $t_0 \in (0, 1)$. *Let* $\mathcal{F}_1 = \Lambda^\beta(M_\beta)$ *and* $\mathcal{F}_2 = \Lambda^\alpha(M_\alpha)$ *be the Hölder balls from* (2.1)*, with* $M_\alpha, M_\beta > 0$ *and* $0 < \alpha < \beta$, *and set* $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$. *As* $\varepsilon \downarrow 0$, *the ordinary and between-class moduli satisfy*

$$\omega(\varepsilon, \mathcal{F}_1) \asymp \varepsilon^{2\beta/(2\beta+1)}, \qquad \omega_+(\varepsilon, \mathcal{F}_1, \mathcal{F}) \asymp \varepsilon^{2\alpha/(2\alpha+1)}.$$

Consequently, for every $0 < \alpha_0 < 1/2$, *the benchmark under coverage probability at least* $1 - \alpha_0$ *over* $\mathcal{F}$ *satisfies*

$$L^*_{\alpha_0}(\mathcal{F}_1, \mathcal{F}, T) \gtrsim n^{-\alpha/(2\alpha+1)}.$$

Thus the smoother class $\Lambda^\beta(M_\beta)$ *cannot enjoy its own smaller pointwise expected-length benchmark under honesty on the union.*

Proof The within-class calculation in Example 4.2 extends to every $\beta > 0$. Choose an infinitely differentiable ψ compactly supported in $(-1, 1)$, with $\psi(0) = 1$. For sufficiently small fixed $c > 0$ and $0 < h \le \min\{t_0, 1 - t_0\}$, both zero and $b_h(t) = ch^\beta\psi((t - t_0)/h)$ belong to $\Lambda^\beta(M_\beta)$. Their target gap is ch^β and their L_2 distance is $ch^{\beta+1/2}\|\psi\|_2$. Setting the latter equal to ε gives the lower bound.

For the upper bound, put $r = \lceil\beta\rceil - 1$ and choose a bounded kernel K supported on $[-1, 1]$ with $\int K = 1$ and $\int v^j K(v)\, dv = 0$ for $1 \le j \le r$. Such polynomial weights exist by the moment-matrix argument in Section 7.2.2 and may be signed. For $u = g - f$ with $f, g \in \Lambda^\beta(M_\beta)$ and $\|u\|_2 \le \varepsilon$, Taylor's formula with Hölder remainder and Cauchy–Schwarz give $|u(t_0)| \le Ch^\beta + \|K\|_2\varepsilon h^{-1/2}$. Taking $h \asymp \varepsilon^{2/(2\beta+1)}$, small enough to stay inside the domain, proves the equivalence, including at integer smoothness:

$$\omega(\varepsilon, \Lambda^\beta(M_\beta)) \asymp \varepsilon^{2\beta/(2\beta+1)}.$$

For the cross-class lower bound, take $f = 0 \in \Lambda^\beta(M_\beta)$ and $g(t) = ch^\alpha\psi((t - t_0)/h)$, with $c > 0$ small enough that $g \in \Lambda^\alpha(M_\alpha)$. Their L_2 distance is of order $h^{\alpha+1/2}$ and their target gap is of order h^α. Choosing $h \asymp \varepsilon^{2/(2\alpha+1)}$ with a sufficiently small fixed factor gives

$$\omega_+(\varepsilon, \Lambda^\beta(M_\beta), \Lambda^\alpha(M_\alpha)) \gtrsim \varepsilon^{2\alpha/(2\alpha+1)}.$$

For the cross-class upper bound, the smoother ball is contained in a Hölder ball of smoothness α with a possibly larger fixed radius. Thus $u = g - f$ satisfies a rougher Hölder bound whether

g belongs to the smoother or rougher ball. The kernel argument gives $|u(t_0)| \lesssim h^\alpha + \varepsilon h^{-1/2}$ when $\|u\|_2 \le \varepsilon$. Optimizing in h proves the between-class modulus order for the union. Corollary 7.10 then gives

$$L^*_{\alpha_0}(\mathcal{F}_1, \mathcal{F}, T) \gtrsim \omega_+\big(\zeta_{\alpha_0}/\sqrt{n}, \mathcal{F}_1, \mathcal{F}\big) \asymp n^{-\alpha/(2\alpha+1)}.$$

□

Donoho's fixed-length confidence interval based on an affine estimator [55, Theorem 2] can be constructed for a Hölder ball of smoothness α and radius large enough to contain both classes; see Section 6.2. It has honest coverage over the union and length of order $n^{-\alpha/(2\alpha+1)}$, matching the preceding lower bound. If the smoother class were known, the optimal length would instead have order $n^{-\beta/(2\beta+1)}$. Requiring coverage over both classes therefore increases the optimal expected length on the smoother class by a factor of order $n^{\beta/(2\beta+1)-\alpha/(2\alpha+1)}$.

In Section 7.2.2, an estimator can retain the smoother-class risk rate while paying a logarithmic penalty on the rougher class. Honesty instead forces the rougher interval length even on the smoother class. Thus accurate estimation and an honest assessment of that accuracy have different adaptation costs.

7.4.3 Shape restrictions and functionwise adaptation

Shape restrictions can make adaptation possible by limiting how nearby functions differ at the target. In the same white-noise model, seek an honest interval for $f(1/2)$ when f is nonincreasing, as in a decreasing response curve, with unknown smoothness. The goal is one interval that attains each class's known-smoothness length order while remaining honest over their union.

Use the classes $F_D(\beta, M)$ from Section 7.2.4, with common fixed radius $M > 0$ and unknown $0 < \beta \le 1$. Monotonicity forces a target gap to persist on one side over a width controlled by the smoother function. The between-class modulus therefore has the known-class order. We establish this using the construction from Corollary 7.4.

Corollary 7.12 (Shape-Constrained Confidence Intervals: No Additional Lower-Bound Cost) *Work in the continuous white-noise model* (1.2) *with noise level* $n^{-1/2}$ *and estimate* $T(f) = f(x_0)$*, where* $x_0 = 1/2$*. For* $M > 0$ *and* $0 < \beta \le 1$*, use the monotone Hölder class* $F_D(\beta, M)$ *defined before Corollary 7.4, with its specified continuous representatives. Set*

$$\mathcal{F} = \bigcup_{0<\beta\le 1} F_D(\beta, M).$$

The classes are nested as β *decreases, and* $\mathcal{F}$ *is convex. Then, for each fixed* $0 < \beta \le 1$ *and as* $\varepsilon \downarrow 0$*,*

$$\omega_+\big(\varepsilon, F_D(\beta, M), \mathcal{F}\big) \asymp \omega\big(\varepsilon, F_D(\beta, M)\big) \asymp \varepsilon^{2\beta/(2\beta+1)}.$$

Consequently, for every $0 < \alpha < 1/2$*, the optimal expected-length benchmark under honesty over* $\mathcal{F}$ *satisfies*

$$L^*_\alpha(F_D(\beta, M), \mathcal{F}, T) \gtrsim n^{-\beta/(2\beta+1)}.$$

Proof Use the opposing caps from the proof of Corollary 7.4, with exponent β and radius M. They belong to $F_D(\beta, M)$ and give the ordinary-modulus lower bound. For the upper bound, take $f \in F_D(\beta, M)$ and $g \in \mathcal{F}$, and write $d = |g(x_0) - f(x_0)|$. If $g(x_0) > f(x_0)$, monotonicity keeps g above $g(x_0)$ to the left, while smoothness limits the increase of f there. The gap remains at least $d/2$ over length $(d/(2M))^{1/\beta}$. For the opposite sign, use the right side. As in the earlier proof, sufficiently small $\|f - g\|_2 \le \varepsilon$ excludes an interval extending beyond $[0, 1]$. Therefore

$$\|f - g\|_2^2 \ge cM^{-1/\beta} d^{2+1/\beta}, \qquad \omega_+(\varepsilon, F_D(\beta, M), \mathcal{F}) \asymp \omega(\varepsilon, F_D(\beta, M)) \asymp \varepsilon^{2\beta/(2\beta+1)}.$$

Applying Corollary 7.10 with $\mathcal{F}_1 = F_D(\beta, M)$ and coverage class $\mathcal{F}$ yields

$$L_\alpha^*(F_D(\beta, M), \mathcal{F}, T) \gtrsim \omega_+\left(\frac{\zeta_\alpha}{\sqrt{n}}, F_D(\beta, M), \mathcal{F}\right) \asymp n^{-\beta/(2\beta+1)}.$$

□

Remark 7.13 (Optimal Procedure and Matching Upper Bound) Cai, Low, and Xia [39, Theorems 2–3] select a local averaging scale and form interval endpoints from one-sided averages, using monotonicity to control bias. The construction in Appendix B.4.2 is honest over the entire monotone class and attains the lower-bound order on every fixed $F_D(\beta, M)$. Shape restrictions thus permit adaptation of interval length, not just estimation.

A finer goal is to adapt to each function's local shape. We now evaluate expected length at a fixed monotone or convex f, retaining honesty over the entire shape class.

Let $\mathcal{F}$ be either the class of real-valued nonincreasing functions in $L_2[0, 1]$, represented by their right-continuous versions on the interior, or the class of real-valued convex functions in that space, represented by their unique continuous versions on the interior. The target is $T(f) = f(1/2)$; endpoint values affect neither the observation law nor the target. Use the singleton expected-length benchmark $L_\alpha^*(f, \mathcal{F}, T)$ from Chapter 6 and the local modulus (4.19) with L_2 distance.

The local modulus fixes f and varies its nearby alternative. Cai, Low, and Xia [39] use it to characterize the shortest expected length under full-class honesty. The following theorem gives the lower bound; one data-driven interval attains its order at every function, as discussed afterward.

Theorem 7.14 (Functionwise Lower Bound under Shape Constraints) *In the continuous white-noise model* (1.2) *with noise level $n^{-1/2}$, let $\mathcal{F}$ be either shape class defined above. For every $f \in \mathcal{F}$ and $0 < \alpha < 1/2$ for which the displayed local modulus is finite,*

$$L_\alpha^*(f, \mathcal{F}, T) \ge c_\alpha\, \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f; \mathcal{F}\right),$$

where ζ_α is the calibration radius chosen earlier in this section and one may take

$$c_\alpha = 2\Phi(-\zeta_\alpha/2) - 2\alpha > 0.$$

Proof Fix $f \in \mathcal{F}$ and let $CI = [L, U]$ be any interval with coverage at least $1-\alpha$ uniformly over $\mathcal{F}$. Set $\varepsilon_n = \zeta_\alpha/\sqrt{n}$. For any $0 < \delta_{\text{app}} < 1$, choose $g \in \mathcal{F}$ such that

$$\|g - f\|_2 \le \varepsilon_n, \qquad |T(g) - T(f)| \ge (1 - \delta_{\text{app}})\, \omega(\varepsilon_n, f; \mathcal{F}).$$

Write $\Delta = |T(g) - T(f)|$. The point-mass case of Lemma 6.3 gives

$$\mathbb{E}_f L(CI) \ge \Delta\{1 - 2\alpha - \mathrm{TV}(P_f, P_g)\}_+.$$

In the Gaussian white-noise model, if $\sqrt{n}\,\|f - g\|_2 \le \zeta_\alpha$, then

$$\mathrm{TV}(P_f, P_g) \le 1 - 2\Phi\left(-\frac{\zeta_\alpha}{2}\right).$$

Thus the calibration of ζ_α gives $c_\alpha = 2\Phi(-\zeta_\alpha/2) - 2\alpha > 0$, and

$$\mathbb{E}_f L(CI) \ge c_\alpha\, \Delta \ge c_\alpha (1 - \delta_{\text{app}})\, \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f; \mathcal{F}\right).$$

Letting $\delta_{\text{app}} \downarrow 0$ and taking the infimum over honest intervals gives

$$L^*_\alpha(f, \mathcal{F}, T) \ge c_\alpha\, \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f; \mathcal{F}\right).$$

□

Remark 7.15 (Optimal Procedure and Matching Upper Bound) Cai, Low, and Xia [39, Theorems 2–5] construct intervals for monotone and convex functions when $0 < \alpha \le 0.2$. In each shape class, one data-driven interval is honest throughout the class and attains every function-specific length benchmark up to a constant depending only on α.

Appendix B.4.2 gives constructions for these classes for every $0 < \alpha < 1/2$. Outer-interval statistics select the scale, and inner-interval averages determine the endpoints. At each candidate scale, the inner averages are independent of the selection statistics used so far, preserving coverage after scale selection.

This adaptation to local shape also occurs for nonlinear targets. For the minimum value and minimizer of a convex function, the estimation and honest-interval procedures in Sections 4.5.2 and 6.3 attain their local benchmark orders without knowing the minimum's flatness, subject to the uncertainty principles established there.

7.5 Adaptive confidence bands

An interval for $f(t_0)$ addresses one prespecified location. To identify where a response curve exceeds a reference level, we need uncertainty statements across locations. Can a band become narrower for smoother functions while remaining valid over a rougher class? We first require simultaneous coverage of the entire curve, then allow a controlled amount of noncoverage. These requirements lead to different restrictions on adaptation.

Suppose we observe the full path in the Gaussian white-noise model (1.2), with noise level $n^{-1/2}$. Use the Hölder classes $\Lambda^\beta(M)$ from (2.1). Fix $0 < \alpha < 1/2$ as the noncoverage level; write β for a single smoothness index and α_0, α_1 when comparing classes. Unlike fixed-grid

regression in Section 6.5, white noise does not leave perturbations between design points unobserved: their L_2 size determines distinguishability.

Retain the band $CB = [L, U]$ and measurability assumptions of Section 6.5, with endpoints now based on the white-noise path. Size is expected integrated width $w(CB, f)$ from (6.13), not width at a single location. As before, $\mathcal{B}_\alpha(\mathcal{F})$ denotes fully honest bands over $\mathcal{F}$, and $W_\alpha(\mathcal{F})$ is their optimal worst-case expected integrated width, now in this experiment.

Unlike the scalar target gaps in Section 7.4, the alternatives must force uncertainty over a region to bound integrated width. Under full coverage, many local ambiguities accumulate into a width lower bound. Under the relaxed criteria of Cai, Low, and Ma [38], a random-sign mixture distributes departures across the domain. In each case, we compare the width forced at zero with an achievable smoother-class benchmark. The cited regression procedures provide a separate upper comparison.

7.5.1 Full coverage and the known-smoothness benchmark

We first construct a band for known smoothness, then show why a band honest over two smoothness classes cannot attain this width on the smoother class. For fixed β and M, piecewise-polynomial approximation and simultaneous Gaussian noise control give worst-case expected integrated width at most a constant multiple of

$$\left(\frac{\log n}{n}\right)^{\beta/(2\beta+1)}.$$

The logarithm reflects simultaneous coverage, not adaptation. A forced width of larger order than this achievable benchmark suffices to rule out adaptation; a sharp known-class lower bound is unnecessary for that comparison.

A known-class fully honest band in white noise. Fix $\beta, M > 0$ and $0 < \alpha < 1/2$ in this white-noise experiment. Put $r = \lceil \beta \rceil - 1$. For an integer $J \geq 1$, partition $[0, 1]$ into J cells I_j of length $h = 1/J$, with left endpoints $a_j = (j-1)h$; take cells half-open except for the last, which includes 1. For $0 \leq k \leq r$, let ℓ_k be the degree-k polynomial obtained by orthonormalizing $1, t, \ldots, t^r$ in $L_2[0, 1]$, with positive leading coefficient. These are the shifted Legendre polynomials, with $\|\ell_k\|_\infty = \sqrt{2k+1}$. Define

$$A_r = \sum_{k=0}^{r} \sqrt{2k+1}, \qquad e_{jk}(t) = h^{-1/2}\ell_k((t-a_j)/h)\mathbf{1}_{I_j}(t),$$

$$\widehat{f}_J = \sum_{j=1}^{J}\sum_{k=0}^{r}\left(\int e_{jk}\, dY\right) e_{jk}.$$

Set $u_{J,\alpha} = \sqrt{2\log\{2J(r+1)/\alpha\}}$. We use the constant half-width

$$R_{n,J} = (1 + A_r)Mh^\beta + \frac{A_r u_{J,\alpha}}{\sqrt{nh}}, \qquad CB_J = [\widehat{f}_J - R_{n,J}, \widehat{f}_J + R_{n,J}].$$

The two terms in $R_{n,J}$ bound approximation bias and stochastic error, respectively. To check the bias, let $\Pi_J f = \mathbb{E}_f \widehat{f}_J$. The cellwise Taylor argument in Appendix B.1.5 gives a degree-r polynomial p_j with $\|f - p_j\|_{\infty,I_j} \le Mh^\beta$, including integer smoothness and both endpoints. Write Π_{I_j} for the local $L_2(I_j)$ projection onto polynomials of degree at most r. For any bounded function g on that cell,

$$\|\Pi_{I_j} g\|_{\infty,I_j} \le \sum_{k=0}^{r} \|\ell_k\|_\infty \|\ell_k\|_1 \|g\|_{\infty,I_j} \le A_r \|g\|_{\infty,I_j}.$$

Because $\Pi_{I_j} p_j = p_j$, this proves $\|f - \Pi_J f\|_\infty \le (1 + A_r)Mh^\beta$. Orthonormality gives independent standard normal coefficient errors $Z_{jk} = \sqrt{n} \int e_{jk}(dY - f\,dt)$. A union bound over the $J(r+1)$ coefficients and the Gaussian tail bound give

$$\mathbb{P}_f\{\max_{j,k} |Z_{jk}| > u_{J,\alpha}\} \le 2J(r+1)e^{-u_{J,\alpha}^2/2} = \alpha.$$

For $t \in I_j$, only the $r+1$ basis functions supported on that cell contribute to the stochastic error. Hence

$$\begin{aligned} |\widehat{f}_J(t) - \Pi_J f(t)| &= \frac{1}{\sqrt{nh}} \left| \sum_{k=0}^{r} Z_{jk} \ell_k((t - a_j)/h) \right| \\ &\le \frac{A_r}{\sqrt{nh}} \max_{j,k} |Z_{jk}|. \end{aligned}$$

On the event $\max_{j,k} |Z_{jk}| \le u_{J,\alpha}$, which has probability at least $1 - \alpha$, combining this bound with the bias bound gives

$$\|\widehat{f}_J - f\|_\infty \le (1 + A_r)Mh^\beta + \frac{A_r u_{J,\alpha}}{\sqrt{nh}} = R_{n,J}.$$

The bound holds at every $t \in [0,1]$ simultaneously, so CB_J is fully honest over $\Lambda^\beta(M)$.

The measurable-integral construction in Appendix A supplies versions of the coefficients chosen independently of f and n. The finite expansion makes the band endpoints jointly measurable in path and location. Coverage is measurable by testing a countable dense subset of each cell and its one-sided limits.

The width is deterministically $2R_{n,J}$. For $n \ge 3$, take $J = \lceil (n/\log n)^{1/(2\beta+1)} \rceil$. Then, for fixed β, M, α,

$$W_\alpha(\Lambda^\beta(M)) \le w(CB_J, \Lambda^\beta(M)) \le C_{\beta,M,\alpha}(\log n/n)^{\beta/(2\beta+1)}.$$

Figure 7.3 summarizes the two coverage comparisons.

To show why adaptation fails under full coverage, allow a smooth bump to be present or absent in each of many disjoint cells. Choose their width so that the corresponding Gaussian projections have small mean separations. A fully honest band induces simultaneous intervals for the binary bump coefficients. The sum of their lengths forces integrated width even at zero, which belongs to the smoother class.

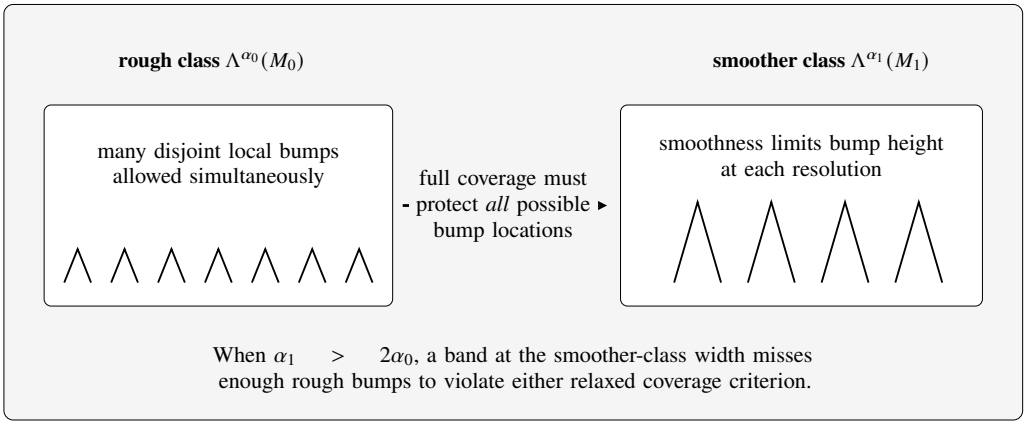


Figure 7.3 Full and relaxed coverage in band adaptation. Local perturbations in the rougher class force a lower bound on expected integrated width even at the zero function. That bound rules out the smoother-class full-coverage benchmark, although it does not recover the logarithmic factor in the rougher-class benchmark. Under the relaxed criteria, a random-sign mixture yields a different, factor-of-two smoothness obstruction.

Theorem 7.16 (Full-Coverage Bands Cannot Adapt across Two Hölder Classes) *Work in the continuous white-noise model* (1.2) *with noise level* $n^{-1/2}$ *and expected integrated width* w *defined above. Fix* $0 < \alpha_0 < \alpha_1$, $M_0, M_1 > 0$, *and* $0 < \alpha < 1/2$. *There exist* $c > 0$ *and* n_0, *depending only on these parameters and the Hölder norm in* (2.1), *such that for every* $n \geq n_0$ *and every band* CB *fully honest over* $\Lambda^{\alpha_0}(M_0) \cup \Lambda^{\alpha_1}(M_1)$,

$$w(CB, \Lambda^{\alpha_1}(M_1)) \geq w(CB, 0) \geq cM_0^{1/(2\alpha_0+1)} n^{-\alpha_0/(2\alpha_0+1)}.$$

Consequently,

$$\frac{w(CB, \Lambda^{\alpha_1}(M_1))}{(\log n/n)^{\alpha_1/(2\alpha_1+1)}} \longrightarrow \infty,$$

so no fully honest band can attain this known-class achievable width order on both classes.

Proof *Bump family.* Write $\beta = \alpha_0$. Choose a nonnegative C^∞ function g supported on $[0, 1]$, positive on $(0, 1)$, with every derivative vanishing at the endpoints. Put

$$\gamma_1 = \int_0^1 g(u)\, du > 0, \qquad \gamma_2 = \left\{ \int_0^1 g^2(u)\, du \right\}^{1/2} > 0.$$

For an integer m, let $I_i = [(i-1)/m, i/m]$, $x_i = (i-1)/m$, and

$$b_i(t) = m^{-\beta} g\{m(t - x_i)\}, \qquad 1 \leq i \leq m.$$

The derivatives of b_i through order $s = \lceil \beta \rceil - 1$ vanish at the cell boundaries. Direct rescaling of the constraints in (2.1) gives

$$\sum_{i=1}^m \theta_i b_i \in \Lambda^\beta(C_g), \qquad \theta \in \{0, 1\}^m, \tag{7.10}$$

where C_g is independent of m and θ. Indeed, the jth derivative on a cell has supremum at most $m^{j-\beta}\|g^{(j)}\|_\infty \le \|g^{(j)}\|_\infty$ for $j \le s$, and the order-$\nu_\beta = \beta - s$ seminorm of the sth derivative is bounded by a constant depending only on g. The vanishing boundary derivatives give the same bound across adjacent cells. The argument also holds for coefficients in $\{-1, 1\}$, after enlarging C_g if necessary.

Set $A = M_0/(2C_g)$ and define

$$f_\theta = A \sum_{i=1}^{m} \theta_i b_i, \qquad \theta \in \{0,1\}^m.$$

By (7.10), every f_θ belongs to $\Lambda^\beta(M_0)$, while $f_0 = 0$ belongs also to $\Lambda^{\alpha_1}(M_1)$.

Gaussian coordinates and induced intervals. Let $\phi_i = b_i/\|b_i\|_2$. The disjoint supports make these functions orthonormal, so under f_θ the coordinates

$$X_i = \sqrt{n}\int_0^1 \phi_i(t)\, dY(t) \sim N(c_n\theta_i, 1), \qquad c_n = A\gamma_2\sqrt{n}\, m^{-\beta-1/2},$$

are independent. Choose a fixed $0 < c_* < c_\alpha$, where c_α is the constant in the fixed-signal part of Proposition 6.16, and set

$$m = \left\lceil \left(\frac{A^2\gamma_2^2 n}{c_*^2}\right)^{1/(2\beta+1)} \right\rceil.$$

Then $0 < c_n \le c_*$ and $m \asymp (A^2 n)^{1/(2\beta+1)}$. The white-noise coordinates orthogonal to the span of the ϕ_i have a law that does not depend on θ and are independent of the displayed coordinates. They therefore provide parameter-free randomization; the proof of Proposition 6.16 is unchanged when such ancillary randomness is included.

For each cell define the data-dependent interval

$$C_i = \{a \in \mathbb{R} : Aab_i(t) \in [L(t), U(t)] \text{ for almost every } t \in I_i\}.$$

This essential intersection can be empty or unbounded. Its endpoints are the essential supremum of $L(t)/(Ab_i(t))$ and the essential infimum of $U(t)/(Ab_i(t))$ over the cell interior, where $b_i > 0$. Joint measurability of the band makes the events $\{0 \in C_i\}$ and $\{1 \in C_i\}$ measurable. Define instead the nonempty finite interval

$$D_i = \begin{cases} \operatorname{conv}(C_i \cap \{0,1\}), & C_i \cap \{0,1\} \neq \varnothing, \\ \{0\}, & C_i \cap \{0,1\} = \varnothing. \end{cases}$$

Whenever the band covers f_θ on all of $[0,1]$, every binary coordinate θ_i belongs to C_i and hence to D_i. Thus full honesty implies the simultaneous coverage hypothesis of Proposition 6.16 for these nonempty finite measurable intervals, with dimension m and mean separation c_n. Its fixed-signal conclusion at the zero vertex gives

$$\mathbb{E}_0 \sum_{i=1}^{m} L(D_i) \ge C_\alpha m. \tag{7.11}$$

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Band area and rate comparison. If $L(D_i) = 1$, both 0 and 1 belong to C_i, so $Ab_i(t) \le U(t) - L(t)$ for almost every $t \in I_i$. If $L(D_i) = 0$, the corresponding area inequality is automatic. Since $\int_{I_i} b_i(t)\,dt = \gamma_1 m^{-\beta-1}$, in either case

$$\int_{I_i} \{U(t) - L(t)\}\,dt \ge A\gamma_1 m^{-\beta-1} L(D_i).$$

After summing over the cells and taking expectation under $f_0 = 0$, (7.11) implies

$$w(CB, 0) \ge C_\alpha A \gamma_1 m^{-\beta} \ge c M_0^{1/(2\beta+1)} n^{-\beta/(2\beta+1)}. \tag{7.12}$$

Because $0 \in \Lambda^{\alpha_1}(M_1)$, this proves the first conclusion.

Finally put $p_j = \alpha_j/(2\alpha_j + 1)$ for $j = 0, 1$. A direct subtraction gives

$$p_1 - p_0 = \frac{\alpha_1 - \alpha_0}{(2\alpha_1 + 1)(2\alpha_0 + 1)} > 0.$$

Dividing (7.12) by $(\log n/n)^{p_1}$ produces a constant multiple of $n^{p_1 - p_0}/(\log n)^{p_1}$, which diverges and rules out adaptation. □

The obstruction does not require nesting: zero belongs to both classes, and coverage over the rougher class forces width there. The power difference alone rules out the smoother-class achievable rate.

7.5.2 Relaxed coverage and distributed alternatives

Full coverage treats any missed location as failure. If a controlled amount of noncoverage is acceptable, can the band adapt over a wider smoothness range? Retain the white-noise experiment and expected integrated width, but use either criterion from Section 6.5.2, uniformly over the union of two Hölder classes. The uncovered measure $\mu\{N(CB, f)\}$ is the fraction of $[0, 1]$ where the band misses f, with μ denoting Lebesgue measure. The relative excess mass $RE(CB, f)$ measures the integrated distance outside the band relative to its area. These criteria control how much of the curve is missed, rather than merely lowering its simultaneous coverage probability.

For either criterion, one missed bump may be harmless. We therefore place random-sign bumps across many cells around a function in a smaller-radius ball of the rougher class. The radius slack keeps every perturbed function in the coverage class. Averaging over signs keeps the observation mixture close to the law at the reference function, while the departures across the domain prevent a narrow band from meeting the chosen guarantee at both the reference function and the alternatives. This construction gives the factor-of-two restriction below.

Use the zero- and infinite-width definitions of $RE(CB, f)$ in (6.17). If expected integrated width at the reference function is infinite, the lower bound is immediate. Otherwise integrated width is finite almost surely under its law and, by equivalence of white-noise laws with L_2 drifts, under every law in the parameter class.

Theorem 7.17 (Relaxed-Coverage Bands Cannot Adapt beyond the Factor-of-Two Window) *In the continuous white-noise model* (1.2) *with noise level* $n^{-1/2}$*, let* $CB = [L, U]$ *be a band for* $\mathcal{F} = \Lambda^{\alpha_0}(M_0) \cup \Lambda^{\alpha_1}(M_1)$*, where* $0 < \alpha_0 < \alpha_1$*,* $M_0, M_1 > 0$*, and* $0 < \alpha < 1/2$*. Use the uncovered-set and relative-excess-mass definitions above. Suppose that at least one of the following conditions holds:*

$$\inf_{f \in \mathcal{F}} \mathbb{P}_f \left\{ \mu\big(N(CB, f)\big) < \frac{1}{2} - \varepsilon \right\} \geq 1 - \alpha \quad \text{for some fixed } 0 < \varepsilon < \tfrac{1}{2} \text{ independent of } n, \tag{7.13}$$

$$\inf_{f \in \mathcal{F}} \mathbb{P}_f\{RE(CB, f) \leq r\} \geq 1 - \alpha \quad \text{for some fixed } r > 0 \text{ independent of } n. \tag{7.14}$$

Then, for every fixed $0 < M_0' < M_0$ *and* $h \in \Lambda^{\alpha_0}(M_0')$*, there exist* $c_h > 0$ *and* n_0 *such that, for every* $n \geq n_0$*,*

$$w(CB, h) \geq c_h n^{-2\alpha_0/(1+4\alpha_0)}.$$

The constants may depend on h*, the fixed class parameters,* α*, and the fixed* ε *or* r *in the applicable condition, but not on the eligible band. In particular, since* 0 *belongs to both classes,*

$$w(CB, \Lambda^{\alpha_1}(M_1)) \geq c_0 n^{-2\alpha_0/(1+4\alpha_0)}.$$

If $\alpha_1 > 2\alpha_0$*, then no sequence of bands satisfying either condition can achieve the known-class upper width scale on the smoother class*

$$w(CB, \Lambda^{\alpha_1}(M_1)) = O\left[\left(\frac{\log n}{n} \right)^{\alpha_1/(2\alpha_1+1)} \right].$$

Thus $\alpha_1 \leq 2\alpha_0$ *is necessary for rate adaptation under either relaxed criterion.*

Proof outline; full proof in Appendix B.4.4 Mix over random-sign bumps. Around $h \in \Lambda^{\alpha_0}(M_0')$, place random-sign bumps in $m \asymp n^{2/(1+4\alpha_0)}$ cells, with height of order $m^{-\alpha_0}$. Broad bump plateaus separate every alternative from h on almost all the domain. Yet the sign mixture is close to P_h: its log second moment is bounded by a small constant times $n^2 m^{-(1+4\alpha_0)}$.

Combine small width with coverage at h. If expected integrated width at h were much smaller than the bump height, Markov's inequality would give high probability to small band area, hence narrow width outside a small set. Intersect this event with the relaxed-coverage event at h. For uncovered measure, the band contains h on more than half the domain, with a fixed margin. After excluding the small regions where the band is wide or the perturbation is small, it must miss each alternative on more than half the domain. For relative excess, the triangle inequality bounds integrated separation by band area plus the two excess integrals. If both relative excesses were at most r, this separation would be at most $(2r + 1)$ times the area, contradicting the perturbation size.

Transfer the contradiction. In either case, the event depends only on the data and h, not on the signs. This is essential: the same event forces failure for every alternative. Total variation transfers its probability to the mixture with a small loss; the constants make the average failure probability exceed α, contradicting uniform validity. Appendix B.3 gives the two coverage-to-width arguments used in the full proof.

The factor of two follows by comparing the forced-width exponent $2\alpha_0/(1 + 4\alpha_0)$ with the smoother-class exponent $\alpha_1/(1 + 2\alpha_1)$: they agree at $\alpha_1 = 2\alpha_0$. Beyond this value the polynomial gap dominates the logarithm in the benchmark. Full coverage rules out adaptation for every $\alpha_1 > \alpha_0$; under the relaxed criteria, attainability within $\alpha_0 < \alpha_1 \le 2\alpha_0$ remains a separate question.

Remark 7.18 (Procedure and Upper-Bound Comparison) Cai, Low, and Ma [38, Theorem 1] construct adaptive wavelet bands in periodic fixed-design regression with expected integrated width $O\{(\log n/n)^{\beta/(2\beta+1)}\}$ for $\beta \in [\alpha_0, 2\alpha_0]$, $\alpha_0 > 1/4$. The wavelet, radius, and uniform asymptotic relaxed-coverage conditions are given in Section 6.5. Transferring this guarantee to white noise requires comparing the experiments and coverage conditions; the present theorem establishes only a necessary restriction on adaptation.

7.6 Adaptive confidence balls

An adaptive confidence ball should quantify global uncertainty while becoming smaller for smoother signals. Unlike a confidence band, it controls an aggregate norm rather than error at each location. Suppose we observe $Y \sim N_d(\theta, n^{-1}I_d)$ in the finite Gaussian model (2.4), with known precision $n \ge 1$. For orthonormal signal coefficients, with all remaining coefficients zero, its Euclidean radius also measures global L_2 uncertainty.

Fix $0 < \alpha < 1/2$. Following Section 6.6, write $CB(\delta, r_\alpha)$ for a ball with data-dependent center δ and radius r_α, and use the expected-squared-radius benchmark (6.20). An adaptive center alone is insufficient: the radius must remain valid for rough signals that the observations cannot distinguish from smoother ones.

Take an integer $J \ge 1$, set $d = 2^J - 1$, and consider the truncated Besov body (6.21), with $\beta, M > 0$, $2 \le p \le \infty$, and $1 \le q \le \infty$. These bodies are convex, and $B^\tau_{p,q}(M) \subseteq B^\beta_{p,q}(M)$ whenever $\tau \ge \beta$. The parameters β, p, q, M are fixed, while d may grow with n; adaptation concerns the unknown smoothness τ. We seek one ball honest over $B^\beta_{p,q}(M)$ whose maximal expected squared radius on each smoother body satisfies

$$\sup_{\theta \in B^\tau_{p,q}(M)} \mathbb{E}_\theta r_\alpha^2 \le C_\tau R^*_\alpha\{B^\tau_{p,q}(M)\},$$

where C_τ is independent of n and d. Thus the same ball must attain the known-class benchmark without knowing τ.

The random-sign mixture in Theorem 6.22 bounds expected squared radius at every signal in a smaller-radius body, including zero. Since zero belongs to every smoother class, this bound restricts adaptation to all of them. Smoothness limits the squared norm of the undetectable block, while dimension limits its number of coordinates. Comparing the resulting bound with the known-class benchmark gives the necessary adaptation ranges.

When $d \asymp n^\kappa$ with $0 < \kappa < 2$, the critical smoothness separating these regimes is

$$\beta_c(\kappa) = \frac{1}{2\kappa} - \frac{1}{4}.$$

Theorem 7.19 (Range Obstruction for Adaptive Confidence Balls) *In the finite Gaussian model and truncated Besov setting above, let $CB(\delta, r_\alpha) \in C_\alpha(B^\beta_{p,q}(M))$, with $0 < \alpha < 1/2$, fixed $\beta, M > 0$, $2 \le p \le \infty$, and $1 \le q \le \infty$. Then, for every fixed $0 < M' < M$ and all sufficiently large n,*

$$\inf_{\theta \in B^\beta_{p,q}(M')} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{d^{1/2} n^{-1},\, n^{-4\beta/(1+4\beta)}\right\}. \tag{7.15}$$

Consequently, the possible range of rate adaptation is limited as follows.

(i) If $d \ge n^2$, rate adaptation is possible at most over

$$\beta \le \tau \le 2\beta.$$

In particular, it is impossible on $B^\tau_{p,q}(M)$ for any $\tau > 2\beta$.

(ii) If $d \asymp n^\kappa$ with $0 < \kappa < 2$ and

$$\beta \ge \beta_c(\kappa),$$

rate adaptation is again possible at most over

$$\beta \le \tau \le 2\beta.$$

(iii) If $d \asymp n^\kappa$ with $0 < \kappa < 2$ and

$$0 < \beta \le \beta_c(\kappa),$$

rate adaptation is possible at most over

$$\beta \le \tau \le \frac{1}{\kappa} - \frac{1}{2}.$$

At the boundary $\beta = \beta_c(\kappa)$, the endpoints in (ii) and (iii) coincide because $2\beta_c(\kappa) = 1/\kappa - 1/2$.

For any fixed τ above the stated upper endpoint, every sequence of balls honest over $B^\beta_{p,q}(M)$ satisfies

$$\frac{\mathbb{E}_0 r_\alpha^2}{R^*_\alpha\{B^\tau_{p,q}(M)\}} \longrightarrow \infty.$$

Above the stated endpoint, coverage over the rougher class forces an expected squared radius larger in order than the smoother-class benchmark, even at the zero signal. The theorem gives an impossibility result; attainability within the ranges is addressed in the comparison remark.

Proof We first obtain the radius forced by honesty, then compare it with the smoother-class benchmark. The dimension cases differ only in which term of the forced-radius bound is smaller.

Forced expected squared radius. Fix $0 < M' < M$ and put $D = M - M'$. The inner-body conclusion of Theorem 6.22 states that every ball honest over $B^\beta_{p,q}(M)$ satisfies

$$\inf_{\theta \in B^\beta_{p,q}(M')} \mathbb{E}_\theta r_\alpha^2 \gtrsim \min\left\{D^2, \frac{\sqrt{d}}{n}, D^{2/(1+4\beta)} n^{-4\beta/(1+4\beta)}\right\}. \tag{7.16}$$

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Since D is fixed and the last term tends to zero, the first term is eventually inactive. Absorbing the fixed powers of D into the constant proves (7.15), including at the origin because $0 \in B^{\beta}_{p,q}(M')$.

Fixed-class benchmark. Applying (6.45) with smoothness τ gives

$$R^*_\alpha\{B^\tau_{p,q}(M)\} \asymp \min\left\{M^2, \frac{d}{n}, M^{2/(1+2\tau)} n^{-2\tau/(1+2\tau)}\right\}.$$

This is the expected-squared-radius benchmark when τ is known. Honesty over the rougher body imposes the additional lower bound above.

Exponent comparison. Since the class radii are fixed, define

$$a_\beta = \frac{4\beta}{1+4\beta}, \qquad g(\tau) = \frac{2\tau}{1+2\tau}.$$

Here a_β is the smoothness exponent in the forced-radius bound, while $g(\tau)$ is the nonparametric exponent in the known-class benchmark. The map g is strictly increasing and $a_\beta = g(2\beta)$. If a ball attained the known-class rate on $B^\tau_{p,q}(M)$, then, because the origin belongs to that body,

$$\mathbb{E}_0 r_\alpha^2 \le C_\tau R^*_\alpha\{B^\tau_{p,q}(M)\}. \tag{7.17}$$

Large dimension: case (i). If $d \ge n^2$, then $\sqrt{d}/n \ge 1$, so (7.16) is of order n^{-a_β}, while (6.45) is of order $n^{-g(\tau)}$. For $\tau > 2\beta$,

$$\frac{n^{-a_\beta}}{n^{-g(\tau)}} = n^{g(\tau)-a_\beta} \longrightarrow \infty,$$

contradicting (7.17). This proves (i).

Polynomial dimension: cases (ii) and (iii). Suppose $d \asymp n^\kappa$ with $0 < \kappa < 2$ and put $b_\kappa = 1 - \kappa/2$, retaining the critical smoothness $\beta_c(\kappa)$ defined above. Then $\sqrt{d}/n \asymp n^{-b_\kappa}$, and the lower bound for the expected squared radius has order

$$\min\{n^{-a_\beta}, n^{-b_\kappa}\} = n^{-\max(a_\beta, b_\kappa)}. \tag{7.18}$$

The minimum selects the faster-decaying term, hence the larger exponent. The transition occurs where $a_\beta = b_\kappa$; subtracting gives

$$a_\beta - b_\kappa = \frac{2\kappa\{\beta - \beta_c(\kappa)\}}{1+4\beta}. \tag{7.19}$$

For both cases, suppose $g(\tau) > \max\{a_\beta, b_\kappa\}$. Then

$$g(\tau) > b_\kappa = 1 - \frac{\kappa}{2} > 1 - \kappa,$$

so $n^{-g(\tau)} = o(d/n)$; it also tends to zero and is eventually smaller than the fixed M^2 term. The known-class benchmark therefore has order $n^{-g(\tau)}$, and

$$\frac{\mathbb{E}_0 r_\alpha^2}{R^*_\alpha\{B^\tau_{p,q}(M)\}} \gtrsim n^{g(\tau)-\max\{a_\beta, b_\kappa\}} \longrightarrow \infty.$$

It remains to translate this exponent condition into a range of smoothness indices.

If $\beta \ge \beta_c(\kappa)$, smoothness determines the bound: $a_\beta \ge b_\kappa$. Since $a_\beta = g(2\beta)$ and g is strictly increasing, every $\tau > 2\beta$ gives the required divergence. This proves (ii).

If $0 < \beta \le \beta_c(\kappa)$, dimension determines the bound: (7.18) is n^{-b_κ}. Set $\tau_\kappa = 1/\kappa - 1/2 > 0$. Direct substitution gives

$$g(\tau_\kappa) = \frac{2(1/\kappa - 1/2)}{1 + 2(1/\kappa - 1/2)} = 1 - \frac{\kappa}{2} = b_\kappa. \tag{7.20}$$

Hence every $\tau > \tau_\kappa$ gives $g(\tau) > b_\kappa \ge a_\beta$ and the required divergence, proving (iii). At equality, $2\beta_c(\kappa) = \tau_\kappa$, so the two descriptions agree exactly. □

The obstruction concerns the radius, not the center's estimation accuracy. A smoothness-limited block yields the endpoint 2β; a dimension-limited block yields the smaller lower bound $\sqrt{d}/n$ and can permit a wider range. Unlike the scalar target gaps of Section 7.4, the separation here is the squared norm of many weak coordinates. The factor of two arises from this balance, not from a universal rule for confidence sets.

For example, take $d \asymp n$, so $\kappa = 1$ and $\beta_c(1) = 1/4$. If $\beta \ge 1/4$, the obstruction gives $\tau \le 2\beta$. If $0 < \beta < 1/4$, it instead gives $\tau \le 1/2$, a wider range than $[\beta, 2\beta]$. In the latter case, the finite dimension prevents the mixture from using as many coordinates as the smoothness constraint alone would allow.

Remark 7.20 (Procedure and Upper-Bound Comparison) Cai and Low [35, Theorems 4–5] center balls at block-thresholded estimators and choose radii from estimates of squared error. Their Theorem 4 treats cases (i)–(ii): given the coverage class's β and M, the procedure attains the fixed-class order for every fixed $\beta \le \tau \le 2\beta$ under uniform asymptotic coverage. It adapts to smoothness within this range, not to an arbitrary unknown radius.

For case (iii), their Theorem 5 gives coverage at least $1 - \alpha - 2/\log n$ over $\mathbb{R}^d$ when $d \le n^2$, with radius guarantees for $M \ge 1$. A Besov ball with fixed $M < 1$ lies in the radius-one ball, yielding the same rate with M-dependent constants, but not uniformly as $M \downarrow 0$. When $d \asymp n^\kappa$, $0 < \kappa < 2$, the dimension restriction holds eventually. The additional term $\sqrt{d}/n$ does not exceed the fixed-class order precisely when $\beta \le \tau \le 1/\kappa - 1/2$, as follows by comparing it with d/n and using $2\tau/(1 + 2\tau) \le 1 - \kappa/2$.

The lower bound also holds under uniform asymptotic honesty, since coverage eventually exceeds $1 - \alpha'$ for every fixed $\alpha' \in (\alpha, 1/2)$. Exact honesty follows by using nominal error $\alpha/2$ for sufficiently large n. For the remaining bounded values of n, the zero-centered ball with squared radius $C_\beta M^2$, $C_\beta = (1 - 2^{-2\beta})^{-1}$, contains the coverage class by (6.44). This dimension-independent modification preserves the classwise rates.

7.7 Exercises

7.1. Construct two nonnested convex classes for which the between-class modulus exceeds both within-class moduli at a chosen observation radius. Explain why the requested strict comparison is impossible for nested classes.

7.2. In the Gaussian white-noise experiment, fix two Hölder classes $\mathcal{F}_i = \Lambda^{\beta_i}(M_i)$, $i = 1, 2$, with $0 < \beta_1 < \beta_2$, $M_i > 0$, fixed $0 < \alpha < 1/2$, and target $T(f) = f(1/2)$. Expand the fixed-class squared-error risk and union-honest expected-length benchmarks from Section 7.1, distinguishing evaluation from coverage. Which separate infima may use i? Formulate the requirement that one interval attain both union-honest classwise length optima up to constants independent of n, and explain why the separate infima do not establish its existence.

7.3. In the Gaussian shift experiment with noise level $n^{-1/2}$, let an interval for $T(f)$ be honest over $\mathcal{F}_1 \cup \mathcal{F}_2$ at level $1-\alpha$, where $0 < \alpha < 1/2$. Suppose the comparison set defining $\omega_+(\rho, \mathcal{F}_1, \mathcal{F}_2)$ is nonempty and this modulus is finite. Starting from coverage at two points, derive a lower bound for the interval's maximal expected length on $\mathcal{F}_1$ in terms of this modulus. Check the coefficient of total variation in the coverage-transfer step.

7.4. For $\Theta = L_2(\alpha, M)$ in Theorem 7.6, with fixed $M, C > 0$, suppose $\mathbb{E}_0 \widehat{Q}^2 \le C n^{-1}$. At $\alpha = 1/8$, compute the fixed-class exponent and logarithmic factor and deduce the theorem's lower risk order. Give a sufficient restriction on κ for the risk-transfer step in the theorem's proof if $1 + \chi^2(P_\pi, P_0) \le n^\kappa$. Repeat the premise check at $\alpha = 1/4$ and state what the theorem no longer implies.

7.5. Under the respective assumptions of Theorems 7.16 and 7.17, compare the lower orders for width at zero with the smoother-class achievable order $(\log n/n)^{\alpha_1/(2\alpha_1+1)}$. Compute both ratios and determine when they diverge. What does the relaxed comparison say at $\alpha_1 = 2\alpha_0$? Explain why failure to obtain a divergent ratio is not a white-noise achievability theorem.

7.6. In Theorem 7.19, fix $\beta = 1/8$ and the other class and coverage parameters. Compare dyadic dimensions $d = 2^J - 1$ with $d \asymp n$ and with $d \ge n^2$. In each regime identify the active origin-radius lower bound, compare it with the fixed-class expected-squared-radius benchmark at smoothness $\tau \ge \beta$, and derive the necessary upper endpoint for τ. Explain why these calculations do not construct an adaptive ball.

7.7. Assume the setting of Theorem 7.1, with $\omega(\varepsilon, \mathcal{F}_1) \asymp \varepsilon^{2/3}$, $\omega(\varepsilon, \mathcal{F}_2) \asymp \varepsilon^{1/2}$, and $\omega_+(\varepsilon, \mathcal{F}_1, \mathcal{F}_2) \asymp \varepsilon^{1/2}$ as $\varepsilon \downarrow 0$. Compute the orders of $a_1, a_2, a_{12}, \gamma_{n,0}$ and the comparison radius $\sqrt{\log \gamma_{n,0}/n}$ for fixed $C_0 > 0$. Under the theorem's first-class risk constraint, derive the resulting lower bound on risk over $\mathcal{F}_2$ and its inflation relative to the second-class benchmark a_2^2. Explain why evaluating the between-class modulus only at $n^{-1/2}$ misses this penalty.

Selected hints. For Exercise 7.1, in $\mathbb{R}^2$ take $T(x, y) = y$ and the line segments $\mathcal{F}_\pm = \{(t, \pm\lambda_{\text{slope}} t) : |t| \le 1\}$, with $0 < \lambda_{\text{slope}} < 1$. At radius $0 < \varepsilon < \lambda_{\text{slope}}$, compare opposite signs at $t = \varepsilon/(2\lambda_{\text{slope}})$ with pairs on a single segment. If $\mathcal{F}_1 \subset \mathcal{F}_2$, every cross-class pair is already a pair in $\mathcal{F}_2$. For Exercise 7.3, transfer the coverage event at the second parameter to the first law before intersecting the two coverage events.

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Part II Synthesis

Chapters 4–6 ask what accuracy is possible for estimation, testing, and uncertainty quantification on specified classes. Chapter 7 asks whether one procedure can attain the class-specific benchmarks without knowing the class. Together, they distinguish the difficulty of the task from the cost of unknown structure.

Different tasks, different reductions. In Chapter 4, the *target and loss* determine which differences matter: a functional gap, accumulated coordinate errors, or changes in the minimum value and minimizer. Moduli, mixtures, and hypercubes expose these differences. Convex optimization further shows how shape constraints link value and location accuracy through function-specific benchmarks.

In Chapter 5, the *hypotheses and observation mechanism* determine what must be concealed. Testing need not identify the alternative: averaging over signs or supports can hide departures that each satisfy the prescribed separation. A hypercube used to accumulate estimation errors can therefore serve a different purpose as a binary testing mixture.

The following comparisons show why rates alone do not make the tasks equivalent. Table 7.1 fixes a point target; Table 7.2 compares vector and quadratic-functional targets in one sequence experiment.

Criterion	Fixed-design Gaussian regression: point target $f(t_0)$
Estimation	MSE $n^{-2s/(2s+1)}$; RMSE $n^{-s/(2s+1)}$.
Testing	Gap in $f(t_0)$ between composite one-sided hypotheses: $n^{-s/(2s+1)}$.
Honest interval	Minimax expected length of order $n^{-s/(2s+1)}$, with coverage at least $1-\alpha$ uniformly over $\Lambda^s(M)$, for fixed $0 < \alpha < 1/2$.

Table 7.1 *Fixed-design regression $Y_i = f(i/n) + Z_i$, $i = 1, \dots, n$, with independent $Z_i \sim N(0,1)$, $f \in \Lambda^s(M)$ as defined in* (2.1)*, fixed $0 < s \leq 1$ and $M > 0$, and a prespecified interior point t_0; see Section 5.3 and the interval construction in Section 6.2. Entries give minimax rate orders; testing uses fixed error probabilities. MSE denotes mean squared error; RMSE is its square root. Interval constants may depend on α.*

For point evaluation, RMSE, testing separation, and honest interval length have the same order. The reductions differ: a target gap forces estimation error, indistinguishability limits testing power, and coverage at separated targets forces interval length. Agreement of rates does not make an estimator, a test, and an honest interval interchangeable.

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Target	Root mean squared error	Testing gap from zero
Vector θ	ℓ_2 error $n^{-s/(2s+1)}$	$\|\theta\|_2$ gap $n^{-2s/(4s+1)}$
Quadratic $Q(\theta) = \|\theta\|_2^2$	$\max\{n^{-4s/(4s+1)}, n^{-1/2}\}$	Q-gap $n^{-4s/(4s+1)}$

Table 7.2 *Gaussian sequence observations $Y_j = \theta_j + n^{-1/2}Z_j$, $j \geq 1$, with independent $Z_j \sim N(0,1)$ and $\theta \in \Theta_s(R^2)$ for fixed $s, R > 0$; see Theorem 5.11 and* (B.14)*. Entries give rate orders; testing rates use fixed error probabilities.*

In the sequence model, detection is easier than vector estimation. For the quadratic functional $Q(\theta) = \|\theta\|_2^2$, the testing gap and estimation RMSE have the same order when $s \leq 1/4$; for $s > 1/4$, estimation incurs the larger parametric term $n^{-1/2}$. Changing the target can therefore change the relation between estimation and detection, even within one experiment.

Honest uncertainty is a distinct optimization problem. Chapter 6 minimizes size subject to coverage, rather than minimizing estimation loss. Two complementary reductions give lower bounds. A sufficiently small honest set induces a test. Direct coverage transfer instead forces the set to contain separated targets under one observation law. The latter can force substantial size at a particular parameter, not just somewhere in the class.

A scalar gap forces interval length; distributed ambiguities force integrated band width; random-sign mixtures force squared ball radius. Thus moduli pass from estimation in Section 4.1 to interval length in Section 6.2, while the mixtures of Sections 4.2 and 5.6 yield radius lower bounds in Section 6.6. The coverage class need not equal the size-evaluation class, as Cai and Low's interval theory [30] makes explicit. Cai, Chen, and Zhu [17] further link functionwise precision in value and location through an uncertainty principle.

Adaptation requires a cross-class comparison. For estimation, *constrained risk* links good performance on one class to error on another. A small risk on one class permits the likelihood-ratio moment of a pair or mixture to grow, allowing greater separation than an ordinary indistinguishability argument. Between-class moduli and mixtures over signs or supports calibrate this balance in Sections 7.2 and 7.3. The resulting lower bounds, together with matching constructions, distinguish free adaptation, logarithmic penalties, and more severe incompatibility.

For confidence procedures, *honesty over the ambient class* replaces the small-risk constraint. Coverage transfer can force large sets even on a favorable subclass. Between-class moduli at the ordinary Gaussian noise scale reveal this obstruction in Section 7.4, whereas constrained-risk estimation may require an enlarged comparison radius. Sections 7.5 and 7.6 derive necessary adaptation-range restrictions for bands and balls; matching ball constructions establish attainability under the stated coverage, radius, and dimension conditions.

Part III brings estimation, testing, honest inference, and adaptation together in sparse regression. Unknown nuisance parameters can conceal a signal or a coordinate change, obstructing detection and adaptive interval length. The remaining applications show how latent labels, incomplete observations, and structured departures alter the information available for learning and inference.

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Part III

Structured Decisions and Detection

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8

Lower Bounds for Structured Learning

Part II organized lower-bound theory around four statistical tasks: estimation, testing, honest confidence sets, and adaptation. Part III brings these tasks together within structured problems in statistics and machine learning. The organization now follows the learning problem rather than the inferential task. This allows us to compare different goals within a common model and to examine how sparsity, latent labels, nuisance parameters, and incomplete observations change the construction of hard alternatives.

This chapter studies sparse regression, classification, clustering, ranking, and crowdsourcing. These problems arise when assessing many potential predictors, predicting a diagnosis, identifying unlabeled subgroups, selecting promising alternatives, or combining assessments from different workers. Each has its own target and criterion of success. Our goal is to understand the resulting statistical limitations and to show how the methods developed earlier must be adapted to establish them.

Sparse regression provides the most direct connection to Part II. Section 8.1 treats coefficient estimation, signal detection, honest intervals, and adaptation within one random-design model. A support packing captures the difficulty of locating active predictors. Mixtures that vary the noise or design covariance reveal further obstacles: nuisance parameters can conceal a signal or a change in one coefficient. A shared information calculation then supports distinct testing and coverage arguments, and the interval-length bound also limits adaptation to unknown sparsity.

Sections 8.2–8.4 turn to prediction and grouping. In nonparametric classification, disagreement with the Bayes rule is weighted by how far the conditional class probability lies from $1/2$; the lower-bound construction must respect this weighting. Sparse linear discriminant analysis instead uses a packing of decision directions and converts disagreement into excess prediction risk. Clustering retains this geometric idea but removes the observed labels. The loss must allow a common exchange of group names, and data processing bounds the information remaining after labels are removed.

Sections 8.5–8.6 consider comparisons and heterogeneous responses. In ranking, identifying the best items and estimating their average-win scores require different constructions, even though the observations are the same: Fano isolates one uncertain boundary item, whereas Assouad accumulates small score errors. In crowdsourcing, paired task perturbations preserve identifiability, while worker effects and missing responses determine which contrasts are available. These examples show why information must be calculated for the actual observation scheme.

Chapter 9 continues this problem-based approach, distinguishing detection of a signal from selection of affected observations and localization of its boundaries.

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8.1 Sparse linear regression: estimation, testing, and honest inference

Sparse linear regression models a response through a small subset of many potential predictors. In genomic association studies, for example, the coefficients quantify adjusted associations with the response. Testing asks whether any association is present; a coordinate interval measures uncertainty about one association. Sparsity can make estimation feasible even with more predictors than observations, but unknown design covariance and noise limit detection and inference.

Suppose we observe n independent pairs (X_i, Y_i) satisfying

$$Y_i = X_i^\top \beta + \varepsilon_i, \qquad X_i \sim N_d(0, \Sigma), \qquad \varepsilon_i \sim N(0, \sigma^2), \qquad X_i \perp \varepsilon_i. \tag{8.1}$$

The coefficient vector $\beta \in \mathbb{R}^d$, positive-definite design covariance Σ, and noise standard deviation $\sigma > 0$ are unknown. Write $\theta = (\beta, \Sigma, \sigma)$ and P_θ for the law of the full sample. Let $\lambda_{\min}(\Sigma)$ and $\lambda_{\max}(\Sigma)$ denote the smallest and largest eigenvalues of Σ. Fix constants $K_\Sigma > 1$ and $K_\sigma > 1$. Using the sparse-vector class $\Theta(s, d)$ of (3.6), define

$$\mathcal{G}(s) = \left\{(\beta, \Sigma, \sigma) : \beta \in \Theta(s, d), K_\Sigma^{-1} \le \lambda_{\min}(\Sigma) \le \lambda_{\max}(\Sigma) \le K_\Sigma,\ K_\sigma^{-1} \le \sigma \le K_\sigma\right\}. \tag{8.2}$$

Here $n, d \ge 1$ and $1 \le s \le d$ are integers, and s bounds the number of nonzero coefficients. The spectral and noise bounds control conditioning and scale. The fixed constants K_Σ, K_σ, suppressed in $\mathcal{G}(s)$, affect constants but not rate orders. Neither Σ nor its inverse is assumed sparse. Unlike Section 5.5, which assumes $\Sigma = I_d$ and known σ in (5.11), both are unknown here. Unknown noise changes the testing construction, while covariance uncertainty further limits coordinate inference.

The four subsections compare coefficient-estimation error, testing separation, honest interval length, and the cost of adaptation in this common model. Sparse packings isolate support uncertainty; mixtures reveal the effect of unknown noise and covariance. Cai and Guo's covariance construction [19, Section 7.2] forces long honest intervals even at zero, showing why accurate estimation need not provide equally adaptive uncertainty quantification.

8.1.1 Estimating the coefficient vector

We estimate β under squared ℓ_2 loss, with minimax risk

$$\inf_{\widehat{\beta}} \sup_{\theta \in \mathcal{G}(s)} \mathbb{E}_\theta \|\widehat{\beta} - \beta\|_2^2.$$

The loss measures aggregate coefficient error, rather than support recovery or prediction error on the observed design. The proposition gives a probability lower bound and its expected-risk consequence; the matching upper comparison concerns probability. A fixed-weight packing from Chapter 3 isolates the cost of an unknown support, with amplitudes calibrated through the joint design–response law.

For fixed Σ and σ, the design distribution is common to two coefficient vectors. Conditional Gaussian divergence and averaging over the design give

$$\mathrm{KL}(P_{\beta,\Sigma,\sigma} \| P_{\beta',\Sigma,\sigma}) = \frac{1}{2\sigma^2} \mathbb{E} \sum_{i=1}^n \{X_i^\top (\beta - \beta')\}^2 = \frac{n}{2\sigma^2} (\beta - \beta')^\top \Sigma (\beta - \beta'). \tag{8.3}$$

For $\Sigma = I_d$ and $\sigma = 1$, this pairwise KL divergence equals that of sparse Gaussian mean estimation with noise variance $1/n$, so the packing and Fano argument of Example 3.6 apply. Testing mixtures require a different information calculation: the joint likelihood moments in Section 5.5 also reflect response variance. Raskutti, Wainwright, and Yu [95, Theorem 1(b)] develop the corresponding fixed-design theory; here we use the joint random-design laws.

Proposition 8.1 (Sparse Regression Estimation Lower Bound) *In* (8.1)*, if* $1 \le s \le d/8$*, there is a universal* $c > 0$ *such that, for every* $n \ge 1$*,*

$$\inf_{\widehat{\beta}} \sup_{\theta \in \mathcal{G}(s)} \mathbb{P}_\theta \left\{ \|\widehat{\beta} - \beta\|_2^2 \ge c \frac{s \log(ed/s)}{n} \right\} \ge \frac{1}{4}.$$

Consequently, the minimax squared-error risk is at least $cs \log(ed/s)/(4n)$.

Proof Restrict to $\Sigma = I_d$ and $\sigma = 1$. Lemma A.4 supplies a code of weight-s binary vectors with the entropy and separation bounds below. Since $d \ge 8s$, there also exist eight disjoint supports of size s. As in Example 3.6, take the larger of the code and this disjoint-support family. After decreasing the universal separation constant if necessary, the resulting family $\mathcal{V}$ satisfies

$$|\mathcal{V}| \ge 8, \qquad \log |\mathcal{V}| \ge c_0 s \log(ed/s), \qquad H(u, v) \ge c_1 s \quad (u \ne v),$$

where H is Hamming distance and $c_0, c_1 > 0$ are universal. Set $\beta_v = av$, where $a^2 = c_0 \log(ed/s)/(8n)$. The squared separation is at least $c_1 s a^2$, while (8.3) gives

$$\mathrm{KL}(P_{\beta_u, I_d, 1} \| P_{\beta_v, I_d, 1}) = \frac{na^2}{2} H(u, v) \le na^2 s \le \frac{1}{8} \log |\mathcal{V}|.$$

Write $P_v = P_{\beta_v, I_d, 1}$ and $\bar{P} = |\mathcal{V}|^{-1} \sum_v P_v$. Convexity of KL in its second argument gives

$$\frac{1}{|\mathcal{V}|} \sum_u \mathrm{KL}(P_u \| \bar{P}) \le \frac{1}{|\mathcal{V}|^2} \sum_{u,v} \mathrm{KL}(P_u \| P_v) \le \frac{1}{8} \log |\mathcal{V}|.$$

Fano's inequality in Lemma 3.5 therefore gives average decoding error at least $1 - 1/8 - \log 2/\log 8 = 13/24$. For an arbitrary estimator, choose a nearest packing point $\beta_{\widehat{v}}$, resolving ties in a fixed order. If $\widehat{v} \ne v$, then

$$\sqrt{c_1 s a^2} \le \|\beta_{\widehat{v}} - \beta_v\|_2 \le \|\beta_{\widehat{v}} - \widehat{\beta}\|_2 + \|\widehat{\beta} - \beta_v\|_2 \le 2\|\widehat{\beta} - \beta_v\|_2.$$

Thus the average probability of squared error at least $c_1 s a^2/4$ is at least $13/24$. Since the supremum over $\mathcal{G}(s)$ dominates this average, the desired maximal error probability is at least $1/4$ for every estimator. This proves the proposition with $c = c_0 c_1/32$. Finally, $\mathbb{E}Z \ge t\mathbb{P}\{Z \ge t\}$ for $Z \ge 0$ gives the stated risk bound. □

The factor s counts active coefficients, while $\log(ed/s)$ is the cost of locating them among d candidates. With a known support and enough observations for a well-conditioned fit, the corresponding squared-error scale is s/n. Thus the packing isolates the additional difficulty of an unknown support. It already occurs in the spherical, unit-noise submodel and cannot be removed by observing the design.

Remark 8.2 (Procedure and Upper-Bound Comparison) Raskutti, Wainwright, and Yu [95, Theorem 2(b)] show that least squares constrained to s-sparse vectors attains squared error of order $\sigma^2 s \log(d/s)/n$ with high probability. This matches the proposition's probability lower bound, not its expected-risk consequence, in the common range: $s \le d/8$, s above a universal constant, and the design conditions below. The source requires $d > 4s$; here $\log(d/s) \asymp \log(ed/s)$. Computation requires combinatorial subset selection, generally infeasible in high dimensions; exhaustive search considers $\binom{d}{s}$ supports.

The design matrix X, with rows $X_i^\top$, must have fixed positive lower and upper bounds on $\|Xv\|_2^2/(n\|v\|_2^2)$ for all nonzero $2s$-sparse v; the upper bound also controls column norms. Gaussian designs in (8.2) satisfy these conditions with high probability when $n \ge Cs \log d$ for sufficiently large C. The lower design inequality follows from their Appendix A, equation (42); Gaussian concentration over sparse unit vectors gives the upper one.

Computationally practical alternatives are the Lasso, which fits least squares with an ℓ_1 penalty and noise-dependent tuning [7, Theorem 7.2], and the scaled Lasso, which jointly estimates the coefficients and noise level [102, Theorem 2 and Corollary 1]. Under their respective design and tuning conditions, normalized columns, the assumed noise bounds, and sufficiently small $s \log d/n$, both attain squared coefficient error of order $s \log d/n$ in probability without knowing s. This matches the lower-bound order when $\log(ed/s) \asymp \log d$.

8.1.2 Detecting a signal with unknown noise variance

Section 5.5 studies sparse regression testing with known noise level and $\Sigma = I_d$. Here both σ and Σ are unknown. Increased response variability may therefore reflect noise rather than a regression effect. We seek a lower bound on the signal size needed for reliable detection. In model (8.1), we test

$$H_0 : \theta \in \mathcal{G}_0(s) = \{\theta \in \mathcal{G}(s) : \beta = 0\},$$
$$H_1(\rho) : \theta \in \mathcal{G}_1(s, \rho) = \{\theta \in \mathcal{G}(s) : \|\beta\|_2 \ge \rho\}.$$

Here $\rho > 0$ measures separation in coefficient norm. Following Section 5.1, define the minimax total error

$$\mathcal{R}^{\mathrm{unk}}_{n,d,s}(\rho) = \inf_{0 \le \varphi \le 1} \left\{ \sup_{\theta \in \mathcal{G}_0(s)} \mathbb{E}_\theta \varphi + \sup_{\theta \in \mathcal{G}_1(s,\rho)} \mathbb{E}_\theta (1 - \varphi) \right\}, \tag{8.4}$$

where φ is a measurable function of the full sample. Both errors must be controlled uniformly over the unknown covariance and noise variance.

We reuse the random-support construction of Section 5.5, reducing the noise variance as the signal increases to keep the response marginal unchanged. The alternatives then differ from the null only through response–predictor dependence. Ingster, Tsybakov, and Verzelen [76, Section 4.2] develop the corresponding asymptotic detection theory.

A shared covariance-mixture calculation. We first evaluate a Gaussian likelihood cross-moment, then average over random supports. The resulting bound depends only on support overlap, as in Section 5.4.1. It will control both testing separation and interval length, with different covariance perturbations and different reductions.

For positive-definite $m \times m$ matrices V, W, let ℓ_V, ℓ_W be the density ratios of $N_m(0, V), N_m(0, W)$ relative to $N_m(0, I_m)$. Since

$$\ell_V(z) = (\det V)^{-1/2} \exp\{-\tfrac{1}{2} z^\top (V^{-1} - I_m) z\},$$

multiplying the two ratios by the standard Gaussian density gives an integrand with exponent $-z^\top B z/2$, where $B = V^{-1} + W^{-1} - I_m$. When B is positive definite, Gaussian integration gives

$$\mathbb{E}_{N_m(0,I_m)} \ell_V \ell_W = (\det V \det W \det B)^{-1/2} = \det\{I_m - (V - I_m)(W - I_m)\}^{-1/2}. \tag{8.5}$$

The last equality follows from $VBW = V + W - VW = I_m - (V - I_m)(W - I_m)$; the square root is positive. The product of covariance perturbations, rather than either inverse separately, is therefore enough to evaluate the moment.

We apply this identity to a covariance that keeps both marginal laws fixed: one coordinate is standard normal, and the remaining m coordinates jointly have law $N_m(0, I_m)$. Only their correlations with the first coordinate vary. For $v \in \mathbb{R}^m$ with $\|v\|_2 \le 1/2$, define

$$V(v) = \begin{pmatrix} 1 & v^\top \\ v & I_m \end{pmatrix}. \tag{8.6}$$

Write P_* for the law of n independent $N_{m+1}(0, I_{m+1})$ observations, with expectation $\mathbb{E}_*$, and L_v for the likelihood ratio of n independent $N_{m+1}(0, V(v))$ observations relative to P_*.

The identity requires a finite Gaussian integral. The eigenvalues of $V(v)$ are $1 \pm \|v\|_2$, with all remaining eigenvalues equal to one. For two such vectors v, w, both spectra lie in $[1/2, 3/2]$. Each inverse has smallest eigenvalue at least $2/3$, so $V(v)^{-1} + V(w)^{-1} - I_{m+1}$ has smallest eigenvalue at least $1/3$, as required.

We can now apply (8.5) without computing the inverses. With $t = v^\top w$,

$$(V(v) - I_{m+1})(V(w) - I_{m+1}) = \begin{pmatrix} t & 0 \\ 0 & v w^\top \end{pmatrix}.$$

Subtracting this product from the identity gives two determinant factors: $1 - t$ from the first coordinate and $\det(I_m - v w^\top) = 1 - w^\top v = 1 - t$ from the rank-one perturbation. Their product is $(1 - t)^2$. Since $|t| \le 1/4$, taking the positive square root and multiplying over the n independent observations gives

$$\mathbb{E}_* L_v L_w = (1 - v^\top w)^{-n}. \tag{8.7}$$

Now average over the unknown locations of the correlations. Let $1 \le k \le m$ be integers, $\tau > 0$, and $k\tau^2 \le 1/4$. Draw a uniform k-subset S of $\{1, \dots, m\}$, set $v_S = \tau \mathbf{1}_S$, and let Q_* be the resulting mixture of the full-sample laws. The support is drawn once for all n rows. Squaring the mixture likelihood ratio introduces two independent prior draws S, S'. Their inner product is $v_S^\top v_{S'} = \tau^2 R$, where $R = |S \cap S'|$: only shared coordinates contribute. Equation (8.7), $-\log(1 - x) \le 2x$ for $0 \le x \le 1/4$, and (5.8) give

$$\begin{aligned} 1 + \chi^2(Q_* \| P_*) &= \mathbb{E}_{S,S'} \mathbb{E}_* L_{v_S} L_{v_{S'}} = \mathbb{E}(1 - \tau^2 R)^{-n} \\ &\le \mathbb{E} e^{2n\tau^2 R} \le \exp\left\{ \frac{k^2}{m} (e^{2n\tau^2} - 1) \right\}. \end{aligned} \tag{8.8}$$

To keep both constructions inside (8.2), fix

$$c_{\mathrm{reg}} = \min\left\{\frac{1}{32},\ (1-K_\Sigma^{-1})^2,\ \frac{1-K_\sigma^{-2}}{2}\right\} > 0. \tag{8.9}$$

This choice keeps the covariance and noise perturbations within the parameter class. We now apply the moment bound to global testing, where the distinguished coordinate is the response and the correlation vector is the regression coefficient.

Verzelen [108, Theorem 4.3] gives nonasymptotic testing lower bounds for $\Sigma = I_d$ and unknown noise variance. The following bound uses the noise-adjusted support mixture of Ingster, Tsybakov, and Verzelen [76, Section 5.5], explicitly keeping the noise level within $\mathcal{G}(s)$. Thus the obstruction arises from unknown noise alone and remains valid over our larger covariance class.

Proposition 8.3 (Sparse Regression Testing with Unknown Variance) *In* (8.1)*, let* $d \ge 3$*,* $1 \le s \le \sqrt{d}$*, and* $n \ge s \log d$*. Then*

$$\mathcal{R}^{\mathrm{unk}}_{n,d,s}(\rho) \ge \tfrac{3}{4} \quad \text{if} \quad 0 < \rho^2 \le \frac{c_{\mathrm{reg}} s}{n} \log\left(1 + \frac{d}{s^2}\right).$$

Proof Take $\theta_\circ = (0, I_d, 1)$ as the null parameter, with sample law $P_\circ$ and expectation $\mathbb{E}_\circ$. Draw S uniformly from the s-subsets of $\{1, \dots, d\}$, independently of the design. Let $\mathbf{1}_S \in \{0,1\}^d$ be its incidence vector, and set

$$a = \frac{\rho}{\sqrt{s}}, \qquad \beta_S = a\mathbf{1}_S, \qquad \Sigma_S = I_d, \qquad \sigma_S^2 = 1 - \rho^2.$$

The radius condition and $n \ge s \log d$ imply $\rho^2 \le c_{\mathrm{reg}} \log(1+d)/\log d < 2c_{\mathrm{reg}} \le 1/16$. Also $2c_{\mathrm{reg}} \le 1 - K_\sigma^{-2}$, so $K_\sigma^{-2} \le \sigma_S^2 \le 1$. Thus $\theta_S = (\beta_S, \Sigma_S, \sigma_S)$ belongs to $\mathcal{G}_1(s, \rho)$. For one observation, the joint law of (Y, X) is centered Gaussian with covariance

$$V_S = I_{d+1} + \begin{pmatrix} 0 & \beta_S^\top \\ \beta_S & 0 \end{pmatrix}. \tag{8.10}$$

In particular, $Y \sim N(0,1)$ and $X \sim N_d(0, I_d)$ under both the null and every alternative. Only their dependence changes.

Apply the shared moment bound (8.8) with $m = d$, $k = s$, and $\tau = a$. Its norm condition holds because $sa^2 = \rho^2 < 1/16$. Writing $Q = \mathbb{E}_S P_{\theta_S}$ and $L_S = dP_{\theta_S}/dP_\circ$, we obtain

$$\begin{aligned} \mathbb{E}_\circ L_S L_{S'} &= (1 - a^2|S \cap S'|)^{-n}, \\ 1 + \chi^2(Q\|P_\circ) &\le \exp\left\{\frac{s^2}{d}(e^{2na^2} - 1)\right\}. \end{aligned} \tag{8.11}$$

The radius condition and $c_{\mathrm{reg}} \le 1/32$ give $2na^2 \le \frac{1}{16}\log(1 + d/s^2)$. By concavity, $(1+x)^{1/16} - 1 \le x/16$ for $x \ge 0$, so

$$\chi^2(Q\|P_\circ) \le e^{1/16} - 1, \qquad \mathrm{TV}(Q, P_\circ) \le \tfrac{1}{2}\sqrt{e^{1/16} - 1} < \tfrac{1}{4}.$$

For every test, its maximal null error dominates $\mathbb{E}_\circ \varphi$ and its maximal alternative error dominates $\mathbb{E}_Q(1 - \varphi)$. The mixture-testing reduction (5.3) now gives $\mathcal{R}^{\mathrm{unk}}_{n,d,s}(\rho) \ge \eta(P_\circ, Q) > 3/4$. □

For disjoint supports, the cross-moment (8.11) equals one. With known noise variance, the response-variance contribution in (5.13) instead imposes the additional squared-separation restriction of order $n^{-1/2}$ in Proposition 5.7. Adjusting the unknown noise variance removes this restriction: detection must rely on response–predictor dependence, even with $\Sigma = I_d$.

Remark 8.4 (Procedure and Upper-Bound Comparison) Verzelen [108, Theorem 4.3] combines subset F-tests with a Bonferroni adjustment. The test permits arbitrary Gaussian design covariance and unknown noise variance, requiring neither Σ, σ, nor s. For $d \geq n$, $1 \leq s \leq n/2$, and sufficiently large n, both errors are at most $1/8$ whenever

$$\frac{\beta^\top \Sigma \beta}{\sigma^2} \geq C_1 \frac{s \log(ed/s)}{n} \exp\left\{ C_2 \frac{s \log(ed/s)}{n} \right\},$$

where $C_1, C_2 > 0$ are universal for these error levels. On $\mathcal{G}(s)$, the left side is at least $\|\beta\|_2^2/(K_\Sigma K_\sigma^2)$. Hence squared separation of order $s \log d/n$ suffices uniformly over $\mathcal{G}(s)$ when $s \leq d^\kappa$ for fixed $0 < \kappa < 1/2$, $d \geq n \to \infty$, and $s \log d = o(n)$. This matches Proposition 8.3 in that regime, without establishing sharp constants or optimality throughout $s \leq \sqrt{d}$. The subset search is computationally expensive.

For $\Sigma = I_d$, Ingster, Tsybakov, and Verzelen [76, Theorem 4.5 and Proposition 4.6] attain asymptotic squared separation of order $\sigma^2 s \log d/n$ when $s = \lfloor d^\kappa \rfloor$, $0 < \kappa < 1/2$, and $s \log d = o(n)$. Their higher-criticism test normalizes design–response inner products by the observed response norm and does not require σ. The orders again match because $\log(1 + d/s^2) \asymp \log d$ and σ is bounded above and away from zero.

8.1.3 An honest interval for one coefficient

Detecting a regression signal does not quantify an individual predictor's contribution. We seek an honest interval for $T(\theta) = \beta_1$; any fixed coordinate can be treated by relabeling. The interval should be short while maintaining coverage uniformly over the other coefficients, covariance, and noise. In the notation of Section 6.2, $CI \in I_\alpha(\mathcal{G}(s), T)$ has coverage at least $1 - \alpha$ throughout $\mathcal{G}(s)$, and $L(CI)$ denotes its length. The benchmark $L_\alpha^*(\mathcal{G}(s), T)$ is the smallest worst-case expected length among these honest intervals.

We bound expected length at the null parameter $\theta_\circ = (0, I_d, 1)$, retaining its full-sample law $P_\circ$ and expectation $\mathbb{E}_\circ$ from the testing problem. Since this point belongs to every sparsity class, the bound will also limit adaptation. A direct shift of β_1 gives the $n^{-1/2}$ term; the sparsity term uses Cai and Guo's joint perturbation of nuisance parameters and target [19, Theorem 3]. Write (X, Y) for a generic observation and X_j for its jth predictor.

Theorem 8.5 (Interval Length at the Zero Regression Vector) *Let $d \geq 3$, $2 \leq s \leq d$, $n \geq s \log d$, and $0 < \alpha < 1/4$. Put $k = s - 1$. In* (8.1)*, every $CI \in I_\alpha(\mathcal{G}(s), T)$ satisfies*

$$\mathbb{E}_\circ L(CI) \geq \frac{c_{\mathrm{reg}}}{8} \left\{ \frac{1}{\sqrt{n}} + \frac{k}{n} \log\left(1 + \frac{d-1}{k^2} \right) \right\}. \tag{8.12}$$

For the sparsity term, we keep the marginal law of (Y, X_1) fixed while moving the conditional coefficient of X_1 away from zero. A rotation reduces the likelihood calculation to (8.8); uniform coverage then forces the interval to accommodate the target shift even at the null.

Proof Constructing alternatives with a common target. Draw S uniformly from the k-subsets of $\{2, \dots, d\}$, and let $u_S \in \{0,1\}^{d-1}$ be its incidence vector. Set

$$a^2 = \frac{c_{\mathrm{reg}}}{n} \log\!\left(1 + \frac{d-1}{k^2}\right), \qquad a > 0, \qquad b = ka^2.$$

Because $\log(1 + (d-1)/k^2) \le \log d$ and $n \ge s \log d$, we have $b \le c_{\mathrm{reg}} \le 1/32$. For each row, specify a centered joint Gaussian law for $(Y, X_1, X_2, \dots, X_d)$ with covariance

$$V_S = \begin{pmatrix} 1 & 0 & au_S^\top \\ 0 & 1 & au_S^\top \\ au_S & au_S & I_{d-1} \end{pmatrix}. \tag{8.13}$$

The leading identity block preserves the null marginal law of (Y, X_1), while correlations with the other predictors change the conditional coefficient of X_1. To check admissibility, consider the span of $(1, 1, 0_{d-1})/\sqrt{2}$ and $(0, 0, u_S)/\sqrt{k}$, where 0_{d-1} is the zero vector in $\mathbb{R}^{d-1}$. On this span the covariance has diagonal entries one and off-diagonal entries $\sqrt{2b}$; it is the identity on the orthogonal complement. Its two nonunit eigenvalues are therefore $1 \pm \sqrt{2b}$. The design block Σ_S similarly has eigenvalues $1 \pm \sqrt{b}$, with the rest equal to one.

Let β_S be the conditional regression coefficient under this joint Gaussian law, and write $\beta_{S,-1}$ for its last $d-1$ entries. The conditional-mean equation $\Sigma_S \beta_S = \mathrm{Cov}(X, Y)$ is

$$(\beta_S)_1 + au_S^\top \beta_{S,-1} = 0, \qquad \beta_{S,-1} + au_S (\beta_S)_1 = au_S.$$

Substitution gives $(1-b)(\beta_S)_1 = -b$ and $\beta_{S,-1} = au_S/(1-b)$. The residual variance is $1 - au_S^\top \beta_{S,-1} = 1 - b/(1-b)$. Hence

$$(\beta_S)_1 = -\frac{b}{1-b}, \qquad (\beta_S)_j = \frac{a}{1-b} \mathbf{1}\{j \in S\} \ (j \ge 2), \qquad \sigma_S^2 = \frac{1-2b}{1-b}. \tag{8.14}$$

Thus Y and X_1 remain marginally independent, but both are correlated with the predictors in S. Adjusting for those predictors makes the coefficient of X_1 negative. This is why preserving the pair's marginal law does not preserve the regression target.

The Gaussian residual is independent of X, and the support has size $k + 1 = s$. Since $\sqrt{b} \le 1 - K_\Sigma^{-1}$, the design eigenvalues lie in $[K_\Sigma^{-1}, K_\Sigma]$, using $2 - K_\Sigma^{-1} \le K_\Sigma$. Also

$$\sigma_S^2 = 1 - \frac{b}{1-b} \ge 1 - 2b \ge 1 - 2c_{\mathrm{reg}} \ge K_\sigma^{-2}, \qquad \sigma_S^2 \le 1.$$

Thus $\theta_S = (\beta_S, \Sigma_S, \sigma_S)$ lies in $\mathcal{G}(s)$. Every alternative has target $-\Delta$, where $\Delta = b/(1-b)$.

Indistinguishability of the full sample. To see why a rotation helps, note that Y and X_1 have identical correlations with $X_2, \dots, X_d$. Taking their difference cancels these correlations; taking their sum combines them. Rotate the first two coordinates within each row to $(Y + X_1)/\sqrt{2}$ and $(Y - X_1)/\sqrt{2}$. The latter is standard normal and independent of the remaining coordinates under every constructed law. On the remaining d coordinates the covariance is

$$\Gamma_S = V(\sqrt{2} au_S) = \begin{pmatrix} 1 & \sqrt{2} au_S^\top \\ \sqrt{2} au_S & I_{d-1} \end{pmatrix}.$$

This is exactly the family (8.6), with $m = d - 1$, support size $k = s - 1$, and $\tau = \sqrt{2}a$. Its norm condition holds because $\|\sqrt{2} au_S\|_2^2 = 2b \le 1/16$. The common rotation and removal

of an independent coordinate with the same law under every parameter preserve likelihood moments. Thus (8.7) gives, for independent support draws S, S', $R = |S \cap S'|$, and $L_S = dP_{\theta_S}/dP_\circ$,

$$\mathbb{E}_\circ L_S L_{S'} = (1 - 2a^2 R)^{-n}. \tag{8.15}$$

Let Q be the uniform mixture of these full-sample laws. Applying (8.8) directly gives

$$1 + \chi^2(Q\|P_\circ) \le \exp\left\{\frac{k^2}{d-1}(e^{4na^2} - 1)\right\} \le e^{1/8}.$$

The last step uses $4c_{\mathrm{reg}} \le 1/8$ and $(1+x)^{1/8} - 1 \le x/8$ for $x \ge 0$. Hence $\mathrm{TV}(Q, P_\circ) \le \frac{1}{2}\sqrt{e^{1/8} - 1} < 1/4$.

From indistinguishability to length. Apply Lemma 6.3 with $H_0 = \{\theta_\circ\}$, $H_1 = \{\theta_S : |S| = k\}$, a point-mass first prior, and the uniform support prior inducing Q. The target is zero on H_0 and $-\Delta$ on H_1, and honesty on $\mathcal{G}(s)$ implies honesty on their union. Since $\alpha < 1/4$ and $\mathrm{TV}(Q, P_\circ) < 1/4$, the lemma gives

$$\mathbb{E}_\circ L(CI) \ge \Delta\{1 - 2\alpha - \mathrm{TV}(Q, P_\circ)\} \ge \frac{b}{4} = \frac{c_{\mathrm{reg}} k}{4n} \log\left(1 + \frac{d-1}{k^2}\right).$$

For the parametric term, compare $\theta_\circ$ with $(he_1, I_d, 1)$, where e_1 is the first coordinate vector and $h = 1/(4\sqrt{n})$. Equation (8.3) gives KL divergence $nh^2/2 = 1/32$, so Pinsker's inequality gives total variation at most $1/8$. The same coverage-transfer bound yields

$$\mathbb{E}_\circ L(CI) \ge h(1 - 2\alpha - 1/8) \ge \frac{3}{32\sqrt{n}}.$$

If a nonnegative quantity is at least each of two bounds, it is at least half their sum. Since $3/64 \ge c_{\mathrm{reg}}/8$, the two bounds imply (8.12). □

Remark 8.6 (One Mixture, Two Inferential Tasks) The nuisance perturbations used for the interval lower bound, following Cai and Guo [19], preserve the marginal law of (Y, X_1) but change the conditional regression coefficient. Their mixture supports both testing and coverage reductions. Its alternatives satisfy

$$|(\beta_S)_1| = \Delta = \frac{b}{1-b}, \qquad \|\beta_S\|_2^2 = \frac{b + b^2}{(1-b)^2}.$$

Since $\mathrm{TV}(Q, P_\circ) < 1/4$, every test obeys $\mathbb{E}_\circ \varphi + \mathbb{E}_Q(1 - \varphi) \ge \eta(P_\circ, Q) > 3/4$. Thus the same family obstructs testing $\beta_1 = 0$ against $\beta_1 = -\Delta$, or testing $\beta = 0$ against vectors with the displayed squared norm, within $\mathcal{G}(s)$. The separations have different orders: b for the coordinate and $\sqrt{b}$ for the vector. Coverage converts the same indistinguishability into an expected-length lower bound proportional to b. The mixture, rather than the length inequality alone, supplies the testing bound.

Conversely, an honest interval for β_1 yields a level-α test by rejecting when $0 \notin CI$. Power requires additional control, such as interval length under alternatives, and the test may miss signals in other coordinates. Proposition 8.3 addresses global testing, including $s = 1$, and establishes the obstruction from unknown noise even with $\Sigma = I_d$.

Remark 8.7 (Procedure and Upper-Bound Comparison) Cai and Guo [19, Theorem 1 and Proposition 1] center an interval at a bias-corrected scaled Lasso estimator and account for noise and remaining bias in its radius. Its expected length has order $n^{-1/2} + s \log d/n$ for fixed $0 < \gamma < 1/2$ and $2 \le s \le c_* \min\{d^\gamma, n/\log d\}$, with sufficiently small $c_* > 0$. Their class contains (8.2) upon taking $M_1 = K_\Sigma$ and $M_2 = K_\sigma$; they require only $0 < \sigma \le M_2$.

The lower bound holds more broadly, for $2 \le s \le d$ and $n \ge s \log d$. In the comparison regime, $s/2 \le k \le s$ and $1 + (d-1)/k^2 \ge d/k^2 \ge d^{1-2\gamma}$, so

$$(1 - 2\gamma) \log d \le \log\left(1 + \frac{d-1}{k^2}\right) \le \log d.$$

The second lower-bound term therefore has order $s \log d/n$. The upper guarantee uses uniform asymptotic coverage [19, Proposition 1]. For fixed $0 < \alpha < 1/4$, nominal error $\alpha/2$ gives coverage at least $1 - \alpha$ for all sufficiently large n, d along admissible sequences, yielding the matching rate with exact honesty in this sparse regime.

8.1.4 The cost of adapting interval length

The sparsity bound used to calibrate an interval may be conservative: the model allows up to s active predictors when only $s_0 \le s$ are present. Can an interval retain coverage over $\mathcal{G}(s)$ while attaining the shorter length appropriate to $\mathcal{G}(s_0)$? The two-class benchmark $L^*_\alpha(\mathcal{G}(s_0), \mathcal{G}(s), T)$ in (6.8) evaluates length on the smaller class and requires coverage on the larger one. Since $\theta_\circ$ belongs to every smaller class, the preceding pointwise bound already gives the obstruction. No new hard family is needed.

Corollary 8.8 (Sparsity Adaptation for Coordinate Intervals) *Fix $0 < \alpha < 1/4$ and $0 < \gamma < 1/2$. If $d \ge 3$, $2 \le s \le d^\gamma$, $n \ge s \log d$, and $1 \le s_0 \le s$, then*

$$L^*_\alpha(\mathcal{G}(s_0), \mathcal{G}(s), T) \ge c_\gamma \left(\frac{1}{\sqrt{n}} + \frac{s \log d}{n}\right)$$

for a constant $c_\gamma > 0$ depending only on $\gamma, K_\Sigma, K_\sigma$.

Proof The point $\theta_\circ = (0, I_d, 1)$ belongs to $\mathcal{G}(s_0)$. Hence every interval honest over $\mathcal{G}(s)$ satisfies

$$\sup_{\theta \in \mathcal{G}(s_0)} \mathbb{E}_\theta L(CI) \ge \mathbb{E}_\circ L(CI) \ge \frac{c_{\mathrm{reg}}}{8} \left\{\frac{1}{\sqrt{n}} + \frac{(1-2\gamma) s \log d}{2n}\right\},$$

by Theorem 8.5, $k \ge s/2$, and the logarithmic comparison above. Taking the infimum over these intervals proves the claim with $c_\gamma = c_{\mathrm{reg}}(1 - 2\gamma)/16$. □

Remark 8.9 (Procedure and Upper-Bound Comparison) In the matching regime above, Cai and Guo's interval has expected length at most a constant times $r(s) = n^{-1/2} + s \log d/n$ throughout $\mathcal{G}(s)$ while remaining honest there. For $2 \le s_0 \le s$, the corollary gives every such honest interval a worst-case expected-length lower bound of order $r(s)$ on $\mathcal{G}(s_0)$, even when the known-class benchmark $r(s_0)$ is smaller. The adaptation penalty vanishes when these orders agree; $s_0 = s$ recovers the fixed-class result.

Rate-optimal adaptation is thus impossible when $r(s)/r(s_0) \to \infty$, for example if $s_0 = o(s)$ and $s \log d \gg \sqrt{n}$. This specializes Cai and Guo's nonadaptivity theory [19, Theorems 2–3] to a fixed coordinate at zero. Conversely, when $s \log d \lesssim \sqrt{n}$ in the same regime, the interval for the largest admissible sparsity attains all benchmarks of order $n^{-1/2}$. As in Chapter 7, the obstacle is coverage over the larger class, not estimation at zero.

Remark 8.10 (Other Information Settings and Targets) The unknown nuisance parameters matter. For known $\Sigma = I_d$ and known fixed $\sigma > 0$, Cai and Guo [19, Theorem 7] give the parametric coordinate-length order $\sigma/\sqrt{n}$ when $s \le c \min\{d^\gamma, n/\log d\}$, with fixed $0 < \gamma < 1/2$ and sufficiently small $c > 0$; their constructed intervals are asymptotically valid. The covariance perturbations in (8.13) are then unavailable. Cai and Guo [19, Sections 3–5] also treat general linear functionals, whose loading vectors introduce further distinctions. The coordinate target studied here already exhibits the effect of unknown nuisance parameters on interval length and sparsity adaptation.

Unknown support limits coefficient estimation even with known covariance and noise. Unknown nuisance parameters create further testing and coverage obstructions, including the interval-adaptation penalty above. We next turn to predicting a future label, where separation must be measured in classification loss rather than coefficient distance.

8.2 Nonparametric classification

In medical screening and document labeling, the goal is to predict a future label from labeled examples. Accurate prediction need not require accurate estimation of every conditional label probability. We study how smoothness and the frequency of nearly tied class probabilities limit classification, using minimax expected excess error over the Bayes rule as the benchmark. An Assouad construction will turn uncertainty about local labels into a lower bound for this decision problem.

Suppose we observe i.i.d. labeled pairs $(X_i, Y_i)_{i=1}^n$ with unknown joint law P, where $X \in [0,1]^d$ and $Y \in \{0,1\}$. Write P_X for the covariate law and $m_P(x) = \mathbb{P}_P(Y = 1 \mid X = x)$ for the regression function, here the conditional probability of label 1. A classifier $g : [0,1]^d \to \{0,1\}$ predicts Y from X in an independent test pair. Its risk is $R_P(g) = \mathbb{P}_P\{g(X) \neq Y\}$, minimized by the Bayes rule $g_P^*(x) = \mathbf{1}\{m_P(x) \ge 1/2\}$, with either label admissible at a tie. Conditioning on X gives

$$R_P(g) - R_P(g_P^*) = \int |2m_P(x) - 1|\, \mathbf{1}\{g(x) \neq g_P^*(x)\}\, dP_X(x). \tag{8.16}$$

For a fitted rule, the identity holds conditional on the training sample and any independent randomization; averaging gives its expected excess risk. Disagreement with the Bayes rule costs $|2m_P(x) - 1|$ at x. Thus errors near a tie have little excess cost, although the more likely label is harder to learn there.

We impose three conditions, with all constants independent of n:

(i) *Smoothness.* For $0 < \beta \le 1$ and $L > 0$, the regression function satisfies $|m_P(x) - m_P(x')| \le L\|x - x'\|_2^\beta$ for all $x, x' \in [0,1]^d$.

(ii) *Design regularity.* The covariate law P_X has a density between positive constants $\mu_{\min}$ and $\mu_{\max}$ on its compact support $A \subseteq [0,1]^d$. For fixed $c_0, r_0 > 0$, it satisfies $\lambda(A \cap B(x,r)) \geq c_0 \lambda(B(x,r))$ whenever $x \in A$ and $0 < r \leq r_0$, where λ is Lebesgue measure and $B(x,r)$ is a Euclidean ball. This is the support-thickness condition.

(iii) *Margin.* For $\alpha, C_{\mathrm{m}} > 0$,

$$\mathbb{P}_{P_X}\{0 < |m_P(X) - 1/2| \leq t\} \leq C_{\mathrm{m}} t^{\alpha}, \qquad t > 0. \tag{8.17}$$

Write $\mathcal{P}_{\mathrm{cl}}$ for the class of joint laws satisfying these conditions for a given choice of their constants.

The margin condition limits the probability of nearly tied labels: larger α allows less mass near, but not at, $m_P = 1/2$. Exact ties contribute no excess loss and are excluded from the margin event, as in Audibert and Tsybakov [4]. The following theorem adapts their lower bound [4, Theorem 3.5]. Its proof makes the support-regularity check explicit and shows how Assouad's method accommodates classification loss. Cai and Wei [42] develop related Assouad constructions for transfer learning.

To make Bayes labels hard to learn, place m active regions on a grid of resolution q^{-1}, give each probability mass $w \asymp q^{-d}$, and vary its regression value by $b \asymp q^{-\beta}$ according to a sign bit. Smoothness limits b; the margin condition imposes $mw \lesssim b^{\alpha}$, hence $m \lesssim q^{d-\alpha\beta}$. The assumption $\alpha\beta \leq d$ leaves at least one active region as the grid is refined. Reversing one sign costs order nwb^2 in training-sample divergence but order wb in excess risk. Taking $q \asymp n^{1/(2\beta+d)}$ keeps neighboring laws close and yields the excess-risk scale

$$n^{-\beta(1+\alpha)/(2\beta+d)}. \tag{8.18}$$

The accumulated loss is $mwb \asymp q^{-\beta(1+\alpha)}$. In contrast, the integrated squared-error loss of Chapter 4 weights each mistaken sign by order wb^2. The hard family may have the same geometry, but its contribution to risk depends on the statistical task.

The construction uses separated support balls. On active balls the regression function lies above or below $1/2$; on background balls it equals $1/2$. Gaps permit a smooth extension without observations in the transition regions. The support varies with n, but its density and thickness constants remain fixed, so all hard families belong to one class. Connected support and positive density throughout the cube are not required.

Theorem 8.11 (Classification under a Regular Bounded-Density Design) *Fix $d \geq 1$, $0 < \beta \leq 1$, $\alpha > 0$ with $\alpha\beta \leq d$, and $L, C_{\mathrm{m}} > 0$. There exist positive constants $c, c_0, r_0, \mu_{\min}, \mu_{\max}$, depending only on these quantities, such that for every sufficiently large n a finite family $\mathcal{H}_n \subset \mathcal{P}_{\mathrm{cl}}$ with a common covariate law satisfies*

$$\sup_{P \in \mathcal{H}_n} \mathbb{E}_P\{R_P(\widehat{g}) - R_P(g_P^*)\} \geq c\, n^{-\beta(1+\alpha)/(2\beta+d)}.$$

The bound holds for every classifier $\widehat{g}$ based on n i.i.d. labeled observations, including randomized classifiers.

Proof We first construct a design whose density and thickness bounds are uniform in n, then assign regression functions satisfying the smoothness and margin conditions. Pairing neighboring sign vectors will convert testing error into excess risk. Put $q = \lceil n^{1/(2\beta+d)} \rceil$, $a = (q+1)^{-1}$, and $m = \lfloor q^{d-\alpha\beta} \rfloor$. For $n \geq 2$ we have $q \geq 2$ and $1 \leq m < q^d$; the endpoint $\alpha\beta = d$ gives $m = 1$. Enumerate the q^d sites $z_k = a(k_1 + 1, \ldots, k_d + 1)$, with $k_i \in \{0, \ldots, q-1\}$, and write $B_k = \overline{B}(z_k, a/4)$ for the closed Euclidean ball with center z_k and radius $a/4$. The balls are disjoint and contained in $[0,1]^d$. We mark m of them active and take $A_n = \bigcup_k B_k$, including both active and background balls.

Choose fixed positive c_b, c_w, sufficiently small as specified below, and set $b = c_b a^\beta$, $w = c_w a^d$. Give each active ball mass w and distribute the remaining mass equally among the $q^d - m$ background balls. Conditional on its ball, X is uniform. If $v_d = \lambda(B(0,1))$, the density equals $4^d c_w / v_d$ on an active ball and

$$\frac{4^d(1 - mw)}{v_d a^d (q^d - m)}$$

on a background ball. We choose $c_w \leq 1/4$, so $mw \leq 1/4$. Moreover, $m/q^d \leq 2^{-\alpha\beta} < 1$ and $q/(q+1) \geq 2/3$. These inequalities bound both displayed density values above and away from zero by constants depending only on the fixed parameters.

We next check that the gaps do not violate support thickness. At radii comparable to the grid spacing, the ball containing x supplies enough volume; at larger radii, many support balls contribute. For $x \in A_n$ and $r \leq a/4$, the intersection of $B(x,r)$ with the ball containing x contains a ball of radius $r/2$. For $a/4 < r \leq 2(d+1)a$, the same intersection contains a ball of radius $a/8$, and its volume is therefore at least a dimension-only multiple of r^d.

For the larger radii, use the overlapping side-$2a$ cells $C_k = \prod_i [k_i a, (k_i + 2)a]$. They cover $[0,1]^d$; their central balls are disjoint, and each occupies the fraction $v_d/8^d$ of its cell's volume. A cell that meets $B(x, r-(d+1)a) \cap [0,1]^d$ has its central ball inside $B(x,r)$: every point of the cell is at distance at most da from its center, and the extra a covers the ball radius $a/4$.

The cells meeting $B(x, r - (d+1)a) \cap [0,1]^d$ cover that intersection. Their number is therefore at least its volume divided by $(2a)^d$, even though they overlap. Their central support balls are disjoint and lie in $B(x,r)$, so their total volume is at least $v_d/8^d$ times the volume of the intersection.

For $r \geq 2(d+1)a$, the smaller radius is at least $r/2$. Since $x \in A_n \subset [0,1]^d$ and $r \leq 1/2$, a fixed dimension-only fraction of the smaller ball lies in the unit cube. The intersection, and hence the support inside $B(x,r)$, thus has volume at least a dimension-only multiple of r^d. Consequently, for some $c_0 = c_0(d) > 0$, $\lambda(A_n \cap B(x,r)) \geq c_0 v_d r^d$ for every $x \in A_n$ and $0 < r \leq r_0 = 1/2$. The constants are uniform in q, m, and n.

With the design fixed, we now assign the hard labels. For each sign vector $\omega \in \{-1,1\}^m$, prescribe regression value $(1 + b\omega_j)/2$ on active ball j and $1/2$ on every background ball. Values on distinct balls differ by at most b, while their distance is at least $a/2$. Choose c_b so that $b \leq L(a/2)^\beta$ and $b \leq 1$. The prescribed function is then β-Hölder with constant at most L on A_n. Denote it by m^0_ω and define

$$\widetilde{m}_\omega(x) = \inf_{z \in A_n} \{m^0_\omega(z) + L\|x - z\|_2^\beta\}, \qquad m_\omega(x) = \min\{1, \max\{0, \widetilde{m}_\omega(x)\}\}.$$

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For $0 < \beta \le 1$, $\|x-y\|_2^\beta$ is a metric. Its triangle inequality gives $|\widetilde{m}_\omega(x)-\widetilde{m}_\omega(y)| \le L\|x-y\|_2^\beta$. The prescribed Hölder bound and the choice $z = x$ show that $\widetilde{m}_\omega = m_\omega^0$ on A_n. Clipping preserves these values and the Hölder bound. On the observed support,

$$\mathbb{P}_{P_X}\{0 < |m_\omega(X) - 1/2| \le t\} = mw\,\mathbf{1}\{t \ge b/2\}.$$

Since $m \le (q+1)^{d-\alpha\beta}$, we have $mw = c_w m a^d \le c_w a^{\alpha\beta}$. Thus $c_w \le C_{\mathrm{m}}(c_b/2)^\alpha$ gives $mw \le C_{\mathrm{m}}(b/2)^\alpha$, which verifies the margin bound for every $t > 0$.

It remains to show that these labels cannot be learned reliably. If bit j is flipped, the one-observation Hellinger affinity is $1-w+w\sqrt{1-b^2}$: only the label law on active ball j changes. The n-observation affinity is therefore at least $1-nwb^2$. The constants may be chosen so that $nwb^2 \le 1/4$ for every $n \ge 2$; hence this affinity is at least $3/4$. The Hellinger–total-variation inequality gives edge total variation at most $\sqrt{1-(3/4)^2} \le 3/4$. On active ball j, predicting the wrong Bayes label incurs excess weight b. Let P_ω and $P_{\omega'}$ be the one-observation laws for sign vectors differing only in bit j. For each fixed point in that ball, pairing the training laws gives an average error probability at least $\eta(P_\omega^{\otimes n}, P_{\omega'}^{\otimes n})/2 \ge 1/8$. Integrating over the ball, whose design mass is w, and summing over the m bits gives maximal expected excess risk at least $mwb/8$. An independent random seed does not change the pairing bound. Finally, $m \ge \frac{1}{2}q^{d-\alpha\beta}$ and $q+1 \le 3n^{1/(2\beta+d)}$, so this bound is at least a positive constant times $n^{-\beta(1+\alpha)/(2\beta+d)}$. All calibration constants were selected before n, as required. □

At $\alpha\beta = d$, one active region suffices; for $\alpha\beta < d$, their number grows with resolution. The lower bound therefore reflects both the difficulty of each local decision and how many such decisions the margin condition permits. Smoothness is imposed on the regression function, not on the piecewise-constant design density.

Remark 8.12 (Optimal Procedure and Matching Upper Bound) The nearest-neighbor classifier $\widehat{g}_k$ predicts 1 when the average label among the k nearest training covariates is at least $1/2$, and 0 otherwise. Covariate regularity controls neighbor distances, smoothness bounds bias, and Bernoulli concentration controls label noise. The margin condition then gives constants $C, c', c_1 > 0$, depending only on the fixed class constants, such that for every $P \in \mathcal{P}_{\mathrm{cl}}$ and integer $1 \le k \le c_1 n$,

$$\mathbb{E}\{R_P(\widehat{g}_k) - R_P(g_P^*)\} \le C\left\{(k/n)^{\beta(1+\alpha)/d} + k^{-(1+\alpha)/2} + e^{-c'k}\right\}. \tag{8.19}$$

Appendix B.5 proves this bound using the concentration-to-risk argument of Audibert and Tsybakov [4, Lemma 3.1]; their Theorem 3.3 treats a local-polynomial plug-in classifier. Taking $k \asymp n^{2\beta/(2\beta+d)}$ balances the first two terms and attains (8.18), matching Theorem 8.11 over $\mathcal{P}_{\mathrm{cl}}$ with its stated class constants. This choice uses β and does not establish adaptation to unknown smoothness. For transfer learning, Cai and Wei [42] develop weighted nearest-neighbor and adaptive classifiers that combine source and target samples, and establish the corresponding minimax rates.

8.3 High-dimensional sparse LDA

High-dimensional classification often involves many measured features but relatively few labeled observations. A sparse discriminant can make accurate prediction possible, but its

relevant coordinates must be learned from the data. We ask how this uncertainty limits classification accuracy. Unlike Section 8.2, which imposes smoothness and margin conditions, we study linear discriminant analysis (LDA) for two Gaussian populations with a common covariance matrix and a sparse discriminant. The benchmark remains excess classification error relative to the optimal rule.

Suppose we observe independent labeled samples $X_i^{(j)} \sim N_d(\mu_j, \Sigma)$ for classes $j = 1, 2$ and $i = 1, \dots, n_j$, with unknown means and a common unknown positive-definite covariance matrix Σ. An independent test pair (Z, Y) has equal class probabilities and $Z \mid Y = j \sim N_d(\mu_j, \Sigma)$. Put $\delta = \mu_2 - \mu_1$, $\Omega = \Sigma^{-1}$, $\beta = \Omega\delta$, and $\Delta = (\delta^\top \Omega \delta)^{1/2}$. The Bayes rule, Fisher's linear discriminant with known parameters, assigns class 2 when $\beta^\top\{Z - (\mu_1 + \mu_2)/2\} \geq 0$ and has error $\Phi(-\Delta/2)$. We measure a fitted classifier by its excess over this error, averaged over the training sample.

Sparsity concerns β, not necessarily δ or Ω: correlations can turn a dense mean difference into a sparse discriminant, or conversely. Classification requires learning the discriminant direction and its threshold, not estimating every entry of the mean vectors and precision matrix accurately.

Cai and Zhang [45] develop the adaptive LDA procedure AdaLDA and establish minimax rates for sparse discriminants, with applications to lung-cancer and leukemia classification. We use a spherical Gaussian subfamily with bounded, nonvanishing signal strength to isolate the cost of unknown active coordinates. The key step is to show that small excess classification error would identify the corresponding Bayes rule in a packing. Fano's inequality then gives the lower bound; the comparison afterward relates this restricted family to the broader AdaLDA theory.

For the lower bound, encode class 1 by -1 and class 2 by 1, and restrict to $\mu_1 = -\mu$, $\mu_2 = \mu$, and $\Sigma = I_d$, with $\mu \neq 0$. We first use random class labels: suppose we observe n i.i.d. labeled pairs (Z_i, Y_i) from

$$\mathbb{P}(Y = y) = \tfrac{1}{2}, \qquad Z \mid Y = y \sim N_d(y\mu, I_d), \qquad y \in \{-1, 1\}. \tag{8.20}$$

An independent test pair follows the same law. For $v \neq 0$, the linear rule $C_v(z) = \operatorname{sign}(v^\top z)$ has excess error

$$\Phi(-\lambda \cos\vartheta) - \Phi(-\lambda), \tag{8.21}$$

where $\lambda = \|\mu\|_2$ and ϑ is the angle between v and μ. For λ bounded above and away from zero, this is of order ϑ^2 as $\vartheta \to 0$.

To vary the direction through its active coordinates, let $d \geq 3$, $q = d - 1$, $1 \leq k \leq q/2$, and $B = k \log(eq/k)$. Set $\mu_u = (1, hu)$ for $u \in \{0, 1\}^q$ with exactly k ones, choosing $h^2 = aB/(nk)$ for a sufficiently small universal constant $a \in (0, 1]$, specified in the proof. When $n \geq B$, we have $1 \leq \|\mu_u\|_2 \leq \sqrt{2}$, and the discriminant $2\mu_u$ has sparsity $k + 1$.

Let P_u be the one-pair law with $\mu = \mu_u$, and write $C_u(z) = \operatorname{sign}(\mu_u^\top z)$ and $R_u^* = \Phi(-\|\mu_u\|_2)$ for its Bayes rule and error. A fitted classifier $C : \mathbb{R}^d \to \{-1, 1\}$ has conditional excess risk $\mathcal{E}_u^{\mathrm{LDA}}(C) = \mathbb{P}_u\{C(Z) \neq Y\} - R_u^*$ on the independent test pair. The theorem averages over training and any independent randomization.

The transformed observations $Y_i Z_i \sim N_d(\mu_u, I_d)$ are independent, also for fixed class counts totaling n. Thus the same argument covers both sampling schemes. Their pairwise training KL is of order at most nkh^2, while the fixed-weight packing of Section 8.1.1 has

logarithmic cardinality of order B. Balancing these quantities gives $kh^2 \asymp B/n$. The proof must also turn small excess risk into accurate decoding for arbitrary classifiers, not only linear rules.

Theorem 8.13 (Sparse LDA Lower Bound) *For the family $\{\mu_u : \|u\|_0 = k\}$ defined above in model* (8.20)*, there are universal constants $B_0, c > 0$ such that, whenever $B \ge B_0$ and $n \ge B$,*

$$\inf_{\widehat{C}} \sup_{u \in \{0,1\}^q : \|u\|_0 = k} \mathbb{E}_u \mathcal{E}_u^{\mathrm{LDA}}(\widehat{C}) \ge c \frac{k \log(eq/k)}{n}.$$

The infimum includes randomized rules.

Proof The fixed-weight coding lemma (Lemma A.4) supplies $\mathcal{U}$ of weight-k vectors with $\log|\mathcal{U}| \ge c_0 B$ and pairwise Hamming distance at least $c_1 k$. Increase B_0 so $\log|\mathcal{U}| \ge 4\log 2$ and choose $a \le \min\{1, c_0/16\}$. All μ_u have norm $\lambda = (1 + kh^2)^{1/2} \in [1, \sqrt{2}]$.

We first separate the packed Bayes rules by their probability of disagreement. For $G \sim N_d(0, I_d)$ define the pseudometric $\mathfrak{d}(C, D) = \mathbb{P}\{C(G) \ne D(G)\}$. Gaussian rotational symmetry and the chord–angle inequality give, for distinct packed u, v,

$$\mathfrak{d}(C_u, C_v) = \pi^{-1} \arccos(\mu_u^\top \mu_v / \lambda^2) \ge \frac{h\sqrt{H(u,v)}}{\pi\lambda} \ge c_2 h\sqrt{k} =: \mathfrak{d}_0.$$

Next we relate disagreement to excess risk. Write f_ν for the $N_d(\nu, I_d)$ density and $A_u(C) = \{z : C(z) \ne C_u(z)\}$. With $\sinh t = (e^t - e^{-t})/2$, the Bayes identity yields

$$\mathcal{E}_u^{\mathrm{LDA}}(C) = \frac{1}{2} \int_{A_u(C)} |f_{\mu_u} - f_{-\mu_u}|\, dz = e^{-\lambda^2/2} \mathbb{E}_G[|\sinh(\mu_u^\top G)| \mathbf{1}_{A_u(C)}].$$

For any event A of G-probability b, let $V = \mu_u^\top G \sim N(0, \lambda^2)$. Since $\lambda \ge 1$, the normal density bound gives $\mathbb{P}(|V| \le t) \le \sqrt{2/\pi}\, t/\lambda \le t$ for $t \ge 0$. Taking $t = b/2$ yields

$$\mathbb{E}[|V| \mathbf{1}_A] \ge t\mathbb{P}(A \cap \{|V| > t\}) \ge t\{b - \mathbb{P}(|V| \le t)\} \ge b^2/4.$$

Because $|\sinh V| \ge |V|$ and $\lambda^2 \le 2$, it follows that

$$\mathcal{E}_u^{\mathrm{LDA}}(C) \ge (4e)^{-1} \mathfrak{d}(C, C_u)^2. \tag{8.22}$$

Decode a fitted rule by choosing the nearest packed Bayes rule in $\mathfrak{d}$. By the triangle inequality, a decoding error entails $\mathfrak{d}(C, C_u) \ge \mathfrak{d}_0/2$, and hence excess risk at least $c_3 h^2 k$.

Finally, the labels have a parameter-independent law, and conditioning on them gives a Gaussian mean-shift calculation:

$$\mathrm{KL}(P_u^{\otimes n}, P_v^{\otimes n}) = \frac{nh^2}{2} H(u, v) \le nh^2 k = aB \le \tfrac{1}{16} \log|\mathcal{U}|.$$

The information-radius bound and Fano's inequality (Lemma 3.5) give average decoding error at least $1 - 1/16 - (\log 2)/\log|\mathcal{U}| \ge 11/16$. Multiplying this probability bound by the excess-risk lower bound on a decoding error proves the assertion. Conditioning on an independent random seed covers randomized rules. □

Remark 8.14 (Procedure and Upper-Bound Comparison) For the family $\mu_u = (1, hu)$ in Theorem 8.13, the fixed rule $C_0(z) = \operatorname{sign}(z_1)$ has excess error $\Phi(-1) - \Phi(-\sqrt{1+kh^2}) \le Ckh^2 = CB/n$, matching the lower bound. This rule uses the common first coordinate; the general sparse LDA problem requires learning both the discriminant direction and its intercept.

Cai and Zhang [45, Theorems 2–3(a)] propose AdaLDA, which substitutes estimates of the sparse discriminant and class midpoint into Fisher's rule. With balanced class counts and $n_{\min} = \min(n_1, n_2)$, its conditional excess risk is at most $Cs \log d/n_{\min}$ with probability at least $1 - 8/d$. This requires $\|\beta\|_0 \le s$, covariance eigenvalues and signal strength bounded above and away from zero, and $s \log d/n_{\min} = o(1)$.

Our family satisfies the sparsity, covariance, and signal bounds with $s = k + 1$, $\Sigma = I_d$, and $2 \le \Delta \le 2\sqrt{2}$. When the additional condition $s \log d/n_{\min} = o(1)$ holds, bounded excess risk converts the probability guarantee into

$$\sup_u \mathbb{E}_u \mathcal{E}_u^{\mathrm{LDA}}(\widehat{C}_{\mathrm{AdaLDA}}) \le C \frac{s \log d}{n_{\min}} + \frac{8}{d}.$$

The lower bound also holds for balanced fixed class counts totaling n, with $n_{\min} \asymp n$. Thus expected-risk orders match when $\log(e(d-1)/k) \asymp \log d$ and $d^{-1} = O(s \log d/n_{\min})$. Cai and Zhang [45, Theorems 3(b) and 7] treat growing signals and broader classes under additional conditions.

8.4 Sparse Gaussian-mixture clustering

When diagnostic labels are unavailable, clustering can help identify previously unrecognized biological subtypes. Unlike the classification problem of Section 8.3, training observations carry no labels, and the two group names are interchangeable. We ask how uncertainty about the active coordinates of a sparse discriminant limits the accuracy of assigning a future observation to a group. The benchmark is excess error over the optimal rule, allowing one global exchange of group names.

Retain $d \ge 3$, $q = d - 1$, $1 \le k \le q/2$, $B = k \log(eq/k)$, and the family $\mu_u = (1, hu)$ from Section 8.3, where $u \in \{0,1\}^q$ has k ones. Choose $h^2 = aB/(nk)$ for a sufficiently small universal constant $a \in (0, 1]$, specified in the proof; then $n \ge B$ ensures $1 \le \|\mu_u\|_2 \le \sqrt{2}$. Suppose we observe $Z_1, \dots, Z_n$ independently from $\frac{1}{2}N_d(\mu_u, I_d) + \frac{1}{2}N_d(-\mu_u, I_d)$. An independent future pair has a uniform latent label $Y \in \{-1, 1\}$ and $Z \mid Y = y \sim N_d(y\mu_u, I_d)$. The labels are unobserved. For this orientation, the Bayes rule is $C_u(z) = \operatorname{sign}(\mu_u^\top z)$, with error $\Phi(-\|\mu_u\|_2)$.

Replacing μ_u by $-\mu_u$ exchanges component names without changing the observation law. For a fitted rule $C : \mathbb{R}^d \to \{-1, 1\}$, define excess misclustering error by

$$\mathcal{E}_u^{\mathrm{CL}}(C) = \min_{\zeta \in \{-1,1\}} \mathbb{P}_u\{C(Z) \ne \zeta Y\} - \Phi(-\|\mu_u\|_2).$$

The probability is conditional on the fitted rule. The minimum permits one label swap for the whole rule, not a separate choice for each observation. Subtracting the Bayes error measures the cost of learning the partition. We average this excess error over training and independent randomization, assessing future assignments rather than recovery of training labels.

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Cai, Ma, and Zhang [40] study the broader model $(1-\omega)N_d(\mu_1,\Sigma)+\omega N_d(\mu_2,\Sigma)$, with mixing weight bounded away from 0 and 1, bounded covariance spectrum, and sparse discriminant $\beta=\Sigma^{-1}(\mu_1-\mu_2)$. Its sign reverses when the component names are exchanged, which leaves our loss unchanged. Their clustering procedure, called CHIME, alternates estimates of component-membership probabilities with sparse discriminant updates. They apply it to glioblastoma expression profiles.

The construction follows the fixed-first-coordinate subfamily in Cai, Ma, and Zhang [40, proof of Theorem 3.3]. To adapt the LDA proof, we must separate partitions even after label swapping. We then bound the information in unlabeled observations by the labeled Gaussian KL through data processing. Fano's inequality balances nkh^2 against the packing entropy B.

Theorem 8.15 (Sparse Gaussian-Mixture Clustering Lower Bound) *For the balanced Gaussian-mixture family defined above, there are universal constants $B_0, c>0$ such that, whenever $B\ge B_0$ and $n\ge B$,*

$$\inf_{\widehat{C}}\sup_{u\in\{0,1\}^q:\,\|u\|_0=k}\mathbb{E}_u\mathcal{E}_u^{\mathrm{CL}}(\widehat{C})\ge c\frac{k\log(eq/k)}{n}.$$

The infimum ranges over randomized rules based only on the unlabeled training vectors.

Proof By Lemma A.4, choose a weight-k code $\mathcal{U}$ with $\log|\mathcal{U}|\ge c_0B$ and minimum Hamming distance at least c_1k. Choose B_0 so $\log|\mathcal{U}|\ge 4\log 2$ and $a\le\min\{1,c_0/16\}$. Every packed μ_u has norm $\lambda=(1+kh^2)^{1/2}\in[1,\sqrt{2}]$.

First compare rules up to a global label swap. For the Gaussian decision pseudometric $\mathfrak{d}$ from the proof of Theorem 8.13, set $\mathfrak{d}_\pm(C,D)=\min\{\mathfrak{d}(C,D),\mathfrak{d}(C,-D)\}$. This measures disagreement after the better of the two labelings. Simultaneous sign reversal leaves $\mathfrak{d}$ unchanged, so minimizing over signs preserves the triangle inequality. For distinct packed u,v, $\mu_u^\top\mu_v=1+h^2u^\top v>0$, so their directions meet at an angle less than $\pi/2$; reversing one increases their disagreement. The chord–angle inequality gives

$$\mathfrak{d}_\pm(C_u,C_v)=\mathfrak{d}(C_u,C_v)=\pi^{-1}\arccos(\mu_u^\top\mu_v/\lambda^2)\ge\frac{h\sqrt{H(u,v)}}{\pi\lambda}\ge c_2h\sqrt{k}=:\mathfrak{d}_0.$$

Next translate this separation into excess misclustering error. Apply the Bayes-risk bound (8.22) to C and $-C$, then minimize:

$$\mathcal{E}_u^{\mathrm{CL}}(C)\ge(4e)^{-1}\mathfrak{d}_\pm(C,C_u)^2.$$

Decode by choosing the nearest packed Bayes rule in $\mathfrak{d}_\pm$. If the codeword is wrong, the triangle inequality gives $\mathfrak{d}_\pm(C,C_u)\ge\mathfrak{d}_0/2$, so the excess error is at least c_3h^2k.

Finally, write $P_u^{\mathrm{unlab}}=\{\frac12N_d(\mu_u,I_d)+\frac12N_d(-\mu_u,I_d)\}^{\otimes n}$ for the training law. If the latent training labels Y_i were supplied, the transformation $(Z_i,Y_i)\mapsto(Y_iZ_i,Y_i)$ would give independent $N_d(\mu_u,I_d)$ vectors and uniform labels whose law does not depend on u. The labeled KL is therefore the Gaussian mean-shift KL. Deleting labels cannot increase it, so

$$\mathrm{KL}(P_u^{\mathrm{unlab}},P_v^{\mathrm{unlab}})\le\frac{nh^2}{2}H(u,v)\le nh^2k=aB\le\tfrac{1}{16}\log|\mathcal{U}|.$$

The information-radius inequality and Fano's lemma yield average decoding error at least $1-1/16-(\log 2)/\log|\mathcal{U}|\ge 11/16$. Multiplying by the excess error on a decoding failure proves

the bound. The argument also covers randomized rules by conditioning on their independent seed. □

The shared first coordinate keeps the packed directions away from their negatives. Thus label swapping cannot merge two packed partitions, and small excess error still permits decoding. This is the change in the reduction from LDA, where the observed class labels fix the orientation.

Remark 8.16 (Procedure and Upper-Bound Comparison) For the family $\mu_u = (1, hu)$ in Theorem 8.15, the fixed rule $C_0(z) = \operatorname{sign}(z_1)$ has excess error $\Phi(-1) - \Phi(-\sqrt{1 + kh^2}) \le Ckh^2 = CB/n$, as in the LDA comparison. It matches the lower-bound order on this restricted family without using training labels.

Cai, Ma, and Zhang [40, Theorems 3.1–3.2] show that CHIME attains excess misclustering error of order $s \log d/n$ with probability $1 - o(1)$. They assume $\|\beta\|_0 \le s$, bounded $\|\beta\|_1$, covariance eigenvalues bounded above and away from zero, and a mixing weight bounded away from zero and one. The guarantee also requires $s = o(\sqrt{n}/\log d)$, initialization condition (C1) with $\sqrt{s \log d/n} = o(r)$ for its radius r, separation condition (C2), and prescribed tuning and iteration count.

For our family, $(\mu_1, \mu_2) = (\mu_u, -\mu_u)$ gives $\beta = 2\mu_u$ and $s = k + 1 \ge 2$. Where the source conditions hold and the logarithmic factors are comparable, the rate agrees with $(s - 1) \log(e(d-1)/(s-1))/n$. An expected-risk comparison additionally needs a uniform failure probability $\varepsilon_n = O(s \log d/n)$: bounded loss contributes at most ε_n on failure, so $o(1)$ alone does not suffice. The lower-bound construction includes a fixed active coordinate and does not cover one-sparse means separated from zero.

The observation density and moments give a complementary view of what remains identifiable without labels. Recall the $N_d(\nu, I_d)$ density f_ν from the LDA proof. For any nonzero mean μ,

$$g_\mu(z) = \tfrac{1}{2} f_\mu(z) + \tfrac{1}{2} f_{-\mu}(z) = f_0(z) e^{-\|\mu\|_2^2/2} \cosh(\mu^\top z).$$

Thus $g_\mu = g_{-\mu}$. Writing $Z = Y\mu + G$ with independent $Y \in \{-1, 1\}$ uniform and $G \sim N_d(0, I_d)$, the labeled cross-moment is $\mathbb{E}(YZ) = \mu$, whereas $\mathbb{E}Z = 0$. Without labels, direction is instead visible through the second moment:

$$\mathbb{E}(ZZ^\top) = I_d + \mu\mu^\top.$$

Its leading eigenvector is $\pm\mu/\|\mu\|_2$, and the gap between its two largest eigenvalues is $\|\mu\|_2^2$. Hence it identifies the line spanned by μ, not its orientation. This population identity alone does not make ordinary principal-component clustering rate optimal: in high dimension, sample covariance fluctuations can require explicit use of sparsity.

Table 8.1 summarizes the comparison. The shared lower-bound order does not establish equal minimax risks in the broader labeled and unlabeled models.

Question	Sparse LDA	Gaussian-mixture clustering
Training law	Labeled Gaussian pairs (Z_i, Y_i)	Unlabeled Gaussian-mixture draws Z_i
Identifiability	Class names orient the discriminant	Component names are exchangeable; μ and $-\mu$ induce the same observation law
Scored action	Label an independent future draw	Assign an independent future draw to a group, modulo one global label swap
Local information bound	Gaussian KL for $Y_i Z_i$	Unlabeled KL bounded by the labeled KL through data processing
Source procedure	Sparse discriminant fit and Fisher plug-in	Soft component assignments and sparse discriminant updates from a suitable initialization

Table 8.1 *Support uncertainty in two different experiments. The paired constructions share a packing, but label removal changes identifiability, the loss, and the information calculation. The source procedures apply under their respective parameter-class and initialization conditions.*

8.5 Ranking from pairwise comparisons

Pairwise comparisons support recommendation and ranking when numerical ratings are unavailable. With noisy comparisons observed for only some pairs, we study two goals: selecting the best items and estimating their average-win scores. We seek the score gap needed for reliable exact selection and the minimax mean squared error of score estimation. These benchmarks explain why accurate score estimates need not identify the best items: small errors can reverse an ordering near a tie.

Suppose there are d items and at most one comparison per unordered pair of distinct items. Each pair $\{i, j\}$ is observed independently with known probability $q \in (0, 1]$. Given that it is observed, i wins with probability ρ_{ij}, independently across pairs. The probability matrix is reciprocal: $\rho_{ij} + \rho_{ji} = 1$, with $\rho_{ii} = 1/2$. No parametric relation among the comparison probabilities is assumed. Define the population average-win score $\tau_i = d^{-1} \sum_{j=1}^{d} \rho_{ij}$ and order the scores as $\tau_{(1)} \geq \cdots \geq \tau_{(d)}$. When $\tau_{(1)} > \tau_{(2)}$, write i_* for the unique maximizer. We observe both the comparison graph and its outcomes; all risks average over both sources of randomness.

The scores define the ranking even when pairwise preferences are cyclic. For three items, $\rho_{12} = \rho_{23} = \rho_{31} = 0.7$ and reverse probabilities 0.3 give $\tau_1 = \tau_2 = \tau_3 = 1/2$: every pair favors one item, but none is uniquely best by average score. A positive gap makes the target set unique; score estimation remains meaningful at ties. Shah and Wainwright [98] establish the win-count benchmark for top-set recovery. Cai, Chakraborty, and Wang [16] study its private extension; here we impose no privacy constraint.

8.5.1 Exact top-set recovery

For $1 \leq k < d$, exact top-set recovery asks for precisely the k highest-scoring items, without requiring their internal ordering. We determine how large the boundary gap $\tau_{(k)} - \tau_{(k+1)}$ must be to make the worst-case error small. The case $k = 1$, with loss $\mathbf{1}\{\widehat{i} \neq i_*\}$, isolates the key construction: give one item a small advantage and vary its identity. Fano's inequality

compares the information in its incident comparisons with the number of possible winners. For general k, the same idea leaves one boundary item uncertain.

Both lower-bound constructions use the same information calculation. For one edge with win probabilities $p, p' \in (0,1)$ under two hypotheses, the recorded outcome is missing, a win, or a loss, with laws $R_{p,q} = (1-q, qp, q(1-p))$ and $R_{p',q} = (1-q, qp', q(1-p'))$. Direct calculation gives

$$\mathrm{KL}(R_{p,q}, R_{p',q}) = qp \log \frac{p}{p'} + q(1-p) \log \frac{1-p}{1-p'} = q\,\mathrm{KL}(\mathrm{Ber}(p), \mathrm{Ber}(p')). \tag{8.23}$$

Conditional on observation, the edge contributes $\mathrm{KL}(\mathrm{Ber}(p), \mathrm{Ber}(p'))$; averaging over its presence gives the factor q. The missing outcome contributes zero. Independence then lets us sum edge divergences for the full data.

Theorem 8.17 (Top-Item Separation Lower Bound) *There are numerical constants $c_0, c_1 > 0$ such that, for $d \geq 16$, $dq \geq \log d$, and $\Delta = c_0\sqrt{\log d/(dq)}$, every possibly randomized estimator $\widehat{i}$ satisfies*

$$\sup_{\rho:\, \tau_{(1)} - \tau_{(2)} = \Delta} \mathbb{P}_\rho\{\widehat{i} \neq i_*\} \geq c_1.$$

The supremum ranges over reciprocal probability matrices in the experiment just defined.

Proof Choose c_0 small enough that $\Delta \leq 1/4$. For each $u \in [d]$, set $\rho^{(u)}_{uj} = 1/2 + \Delta$ and $\rho^{(u)}_{ju} = 1/2 - \Delta$ for $j \neq u$; all other off-diagonal probabilities are $1/2$. Then $\tau_u = 1/2 + (d-1)\Delta/d$, while $\tau_j = 1/2 - \Delta/d$ for $j \neq u$. Thus the unique best item is u and its gap is exactly Δ.

Let P_u be the joint law of the graph and outcomes under $\rho^{(u)}$, and let P_0 be the law with every win probability equal to $1/2$. Applying (8.23) to the $d-1$ changed edges and using $\mathrm{KL}(\mathrm{Ber}(1/2 \pm t), \mathrm{Ber}(1/2)) \leq 4t^2$ for $|t| \leq 1/4$ yields

$$\mathrm{KL}(P_u, P_0) \leq 4q(d-1)\Delta^2 \leq 4c_0^2 \log d.$$

If U is uniform on $[d]$ and D denotes all observed edges and outcomes, the reference-law decomposition gives $I(U; D) \leq d^{-1} \sum_u \mathrm{KL}(P_u, P_0)$. Fano's inequality therefore bounds the average error below by $1 - 4c_0^2 - \log 2/\log d$. Taking, for example, $c_0 \leq 1/8$ makes this at least a positive numerical constant for $d \geq 16$. Maximum error dominates average error, proving the claim. □

For general k, let $S_k(\rho)$ be the set of items with the k largest scores when $\tau_{(k)} > \tau_{(k+1)}$, with loss $\mathbf{1}\{\widehat{S} \neq S_k(\rho)\}$. Fixing all but one selected or excluded item leaves at least $d/2$ candidates and preserves the $\log d$ difficulty.

Theorem 8.18 (Top-k Separation in the Random-Comparison Experiment) *There are numerical constants $c_0, c_1 > 0$ such that, for every $d \geq 16$, $1 \leq k < d$, and $q \in (0,1]$ with $dq \geq \log d$, if $\Delta = c_0\sqrt{\log d/(dq)}$, then every possibly randomized top-k selector $\widehat{S}$ satisfies*

$$\sup_{\rho:\, \tau_{(k)} - \tau_{(k+1)} = \Delta} \mathbb{P}_\rho\{\widehat{S} \neq S_k(\rho)\} \geq c_1.$$

The supremum ranges over reciprocal comparison matrices; probabilities are taken under the random-edge observation law defined above.

Proof For a vector $u \in \mathbb{R}^d$ with $|u_i - u_j| \le 1/4$, set $\rho_{ij} = 1/2 + u_i - u_j$ for $i \ne j$ and $\rho_{ii} = 1/2$. Then ρ is reciprocal, all edge probabilities lie in $[1/4, 3/4]$, and direct summation gives $\tau_i = 1/2 + u_i - \bar{u}$, where $\bar{u} = d^{-1}\sum_j u_j$. Consequently every score gap equals the corresponding gap between two entries of u.

First suppose $k \le d/2$. Fix $k-1$ items at $u_i = 2\Delta$, leaving $M = d-k+1 \ge d/2$ candidates. Under hypothesis h, set candidate h to Δ and the others to zero. The top-k set consists of the fixed items and h, with boundary gap Δ. Let P_h be the resulting joint observation law and P_{ref} the law with every candidate set to zero. Only the $d-1$ edges incident to h change, each by Δ in win probability. The Bernoulli bound $\mathrm{KL}(\mathrm{Ber}(p), \mathrm{Ber}(p+t)) \le Ct^2$ for $p, p+t \in [1/4, 3/4]$ and (8.23) give joint divergence at most $Cqd\Delta^2$.

If $k > d/2$, fix $d-k-1$ items at -2Δ, leaving $M = k+1 \ge d/2$ candidates. Set candidate h to $-\Delta$ and the others to zero. The top-k set excludes the fixed low items and h, again with gap Δ. Define P_h and P_{ref} as above, setting all candidates to zero for the reference law. The same KL bound holds. In either case, identifying the top set identifies h, and the average KL to the reference is at most $Cc_0^2 \log d$.

Since $M \ge d/2$ and $d \ge 16$, we have $\log M \ge \frac{3}{4}\log d$. Choose $c_0 \le 1/8$ small enough that $Cc_0^2 \log d \le \frac{1}{16}\log M$. The first restriction ensures $2\Delta \le 1/4$, keeping all win probabilities in $[1/4, 3/4]$. Let D denote the observed graph and outcomes. For H uniform on the M hypotheses, the information-radius inequality gives $I(H; D) \le M^{-1}\sum_h \mathrm{KL}(P_h, P_{\mathrm{ref}}) \le \frac{1}{16}\log M$. An independent random seed does not change this bound. Fano's inequality yields average decoding error at least $1 - 1/16 - (\log 2)/\log M$, a positive numerical constant since $M \ge 8$. Correct top-set selection identifies H, so the worst-case selection error is at least this large. □

The $\log d$ cost comes from one uncertain boundary item; a packing of all $\binom{d}{k}$ sets is unnecessary. The additive probabilities restrict only the hard subfamily. As in Section 3.3, the reference law is used only to bound information and need not have a unique top set.

Remark 8.19 (Optimal Procedure and Matching Upper Bound) Let W_i be the number of observed comparisons won by item i. Select the k items with the largest W_i, breaking ties by a fixed rule. This rule requires neither comparison probabilities nor a parametric preference model. Since $\mathbb{E}_\rho W_i = q(d\tau_i - 1/2)$, expected win counts preserve the score ordering. Uniform concentration controls selection error despite dependence among the counts.

Shah and Wainwright [98, Theorem 1] show that, with one potential comparison per pair, a boundary gap at least $8\sqrt{\log d/(dq)}$ gives error probability at most d^{-14} uniformly over reciprocal comparison matrices. Together with Theorem 8.18, this identifies the optimal gap order $\sqrt{\log d/(dq)}$ for $d \ge 16$ and $dq \ge \log d$. A small constant forces nonvanishing worst-case error; the stated upper bound permits exact recovery with probability tending to one as $d \to \infty$, when its gap requirement is feasible. This is an order comparison, not a sharp-constant result.

The guarantee concerns ranking by average-win score, not recovering a latent ordering. It assumes independent Bernoulli observation of pairs, rather than a fixed-budget, repeated, or adaptive design.

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8.5.2 Average-win score estimation

Exact top-set recovery treats any inclusion or omission error as a failure. Score estimation instead measures accuracy across all items through $d^{-1}\|\widehat{\tau}-\tau\|_2^2$, where $\tau=(\tau_1,\dots,\tau_d)$. We take worst-case expected loss over the same reciprocal matrices, without imposing a score gap, and determine whether win counts attain the minimax risk.

Instead of varying one boundary item, we pair the items and perturb their scores in opposite directions. Each sign change affects only incident comparisons. Amplitude $(dq)^{-1/2}$ keeps neighboring observation laws close, while Assouad's argument adds the squared-error costs across pairs.

Proposition 8.20 (Average-Win Score Estimation) *In the same random-edge comparison experiment, suppose $d \geq 4$ is even and $dq \geq 1$. There is a universal $c > 0$ such that*

$$\frac{c}{dq} \leq \inf_{\widehat{\tau}} \sup_{\rho} \frac{1}{d}\mathbb{E}_\rho\|\widehat{\tau}-\tau(\rho)\|_2^2 \leq \frac{1}{dq},$$

where the supremum is over reciprocal win-probability matrices and the infimum includes randomized estimators.

Proof The win-count estimator in the following remark proves the upper bound. For the lower bound, put $h=d/2$ and $t=c_0/\sqrt{dq}$ for a sufficiently small numerical $c_0>0$. For each $\omega\in\{-1,1\}^h$, set $u_{2r-1}=t\omega_r$, $u_{2r}=-t\omega_r$, and $\rho_{ij}^{(\omega)}=1/2+u_i-u_j$ for $i\neq j$, with $\rho_{ii}=1/2$. Since $\sum_i u_i=0$, the average-win scores are exactly $\tau_i^{(\omega)}=1/2+u_i$. All win probabilities remain within $[1/4,3/4]$ when c_0 is small.

Flipping bit r changes at most $2d$ edges incident to its two items, each by at most $4t$ in win probability. The Bernoulli KL bound on $[1/4,3/4]$ and (8.23) give neighboring KL at most $Cqdt^2=Cc_0^2$. Choose c_0 so this is at most $1/16$; Pinsker's inequality then bounds neighboring total variation by $1/4$.

Decode bit r from the sign of $\widehat{\tau}_{2r-1}-\widehat{\tau}_{2r}$, assigning $+1$ at a tie. A wrong sign incurs squared error at least $2t^2$: the true pair $(1/2+t\omega_r,1/2-t\omega_r)$ lies squared distance $2t^2$ from the diagonal where the two estimated scores agree. On each cube edge, the two-point bound gives sum of decoding errors at least $1-\mathrm{TV}\geq 3/4$. Averaging over the uniform prior gives expected number of wrong bits at least $3h/8$. The prior-averaged normalized risk is therefore at least $(2t^2/d)(3h/8)=3t^2/8$, a constant multiple of $1/(dq)$. Maximum risk dominates this average; the same argument applies with an independent random seed included in the observations. □

Remark 8.21 (Optimal Procedure and Matching Upper Bound) The win-count estimator $\widehat{\tau}_i=W_i/(qd)+1/(2d)$ is unbiased because $\mathbb{E}W_i=q(d\tau_i-1/2)$. Each count sums $d-1$ independent indicators with variance at most q, giving

$$\sup_{\rho}\frac{1}{d}\mathbb{E}_\rho\|\widehat{\tau}-\tau\|_2^2 \leq \frac{d-1}{qd^2} \leq \frac{1}{dq}.$$

The two tasks use the same edge information differently. Exact selection compares possible identities for one boundary item, allowing KL of order $\log d$. Score estimation keeps

neighboring KL bounded and adds errors across pairs, giving risk of order $1/(dq)$ without the logarithm. Its paired perturbations preserve the mean score $1/2$ required by reciprocity. Thus exact selection and average score accuracy require different calibrations of the same observation law.

Estimating the full comparison matrix is a further target. Cai, Li, and Xiang [25] study it under additional shape and ordering restrictions, with an application to football rankings. Their structured-matrix rates do not determine the row-average estimation or top-set recovery rates considered here.

8.6 Crowdsourcing: task effects and worker heterogeneity

In crowdsourced annotation, a task may receive more positive assessments because of its own characteristics or because its assessors tend to respond positively. A two-way logistic model separates these task and worker effects. We seek to estimate relative task effects and determine how unknown worker effects and incomplete responses limit accuracy. Here worker effects describe response tendencies, not necessarily expertise or accuracy.

Suppose n workers assess $m \geq 2$ tasks. Under full observation, we observe independent responses $Y_{ij} \in \{0, 1\}$, with workers indexing rows and tasks indexing columns. Using the logistic function σ, the model is

$$\mathbb{P}_{x,a}(Y_{ij} = 1) = \sigma(x_j + a_i), \qquad \sigma(t) = (1 + e^{-t})^{-1},$$

where x_j is task j's effect and a_i is worker i's nuisance effect. Adding a constant to all task effects and subtracting it from all worker effects leaves the law unchanged. We therefore impose the identifiability constraint $\sum_j x_j = 0$ and, for fixed $M_0, M_1 > 0$, the parameter class

$$\mathcal{P}_{m,n}(M_0, M_1) = \{(x, a) \in \mathbb{R}^m \times \mathbb{R}^n : \textstyle\sum_j x_j = 0,\ \|x\|_\infty \leq M_0,\ |x_j + a_i| \leq M_1 \text{ for all } i, j\}.$$

We measure error by $m^{-1}\|\widehat{x} - x\|_2^2$ and seek its minimax expected value over both task and worker effects. As in score estimation in Section 8.5.2, paired perturbations preserve a fixed sum and allow Assouad's argument to accumulate errors. The new issue is worker heterogeneity: attainable accuracy depends on within-worker contrasts that remove these nuisance effects.

Li and Cai's unpublished manuscript [91] studies inference in this two-way logistic model under worker-level privacy constraints and provides the conditional-likelihood comparison below. Here we impose no privacy constraint and give a separate full-observation rule. In contrast to models with latent correct labels and worker-specific error probabilities [54], this likelihood assumes no latent correct answer: the target is a relative task effect, not a ground-truth label.

8.6.1 Full observation: estimating task contrasts

Under full observation, each task has n responses, suggesting normalized squared-error risk of order $1/n$. Does having n unknown worker effects prevent this rate? We show that it does not: the lower bound holds even with known worker effects, and a matching procedure removes them through within-worker contrasts.

Fix all worker effects at zero and assign equal and opposite task effects in disjoint pairs. Reversing one pair changes two response columns and costs KL of order n times squared amplitude; amplitude $n^{-1/2}$ keeps neighboring laws close.

Theorem 8.22 (Crowdsourcing Task-Effect Lower Bound) *In the full-observation logistic experiment above, fix $M_0, M_1 > 0$. There are constants $c > 0$ and n_0 depending only on these bounds such that, for $m \geq 2$ and $n \geq n_0$,*

$$\inf_{\widehat{x}} \sup_{(x,a)\in\mathcal{P}_{m,n}(M_0,M_1)} \frac{1}{m}\mathbb{E}_{x,a}\|\widehat{x} - x\|_2^2 \geq \frac{c}{n}.$$

Proof It suffices to restrict the nuisance effects to $a_i = 0$. Put $h = \lfloor m/2 \rfloor$ and choose $t = 1/(4\sqrt{n})$, with n_0 large enough that $t \leq \min(M_0, M_1)$. For each $\omega \in \{-1, 1\}^h$, set $(x_{2r-1}, x_{2r}) = (t\omega_r, -t\omega_r)$ for $r \leq h$, and set the final coordinate to zero if m is odd. Every vector is centered and belongs to the stated class.

Neighboring vertices differ in two task columns. For one worker, the two changed columns contribute

$$\begin{aligned}&\mathrm{KL}(\mathrm{Ber}(\sigma(t)), \mathrm{Ber}(\sigma(-t))) + \mathrm{KL}(\mathrm{Ber}(\sigma(-t)), \mathrm{Ber}(\sigma(t)))\\ &\quad = 2t\{2\sigma(t) - 1\} \leq t^2.\end{aligned}$$

The inequality follows from $\sigma(0) = 1/2$ and $\sigma'(u) = \sigma(u)\{1 - \sigma(u)\} \leq 1/4$. Independence across workers gives neighboring KL at most $nt^2 = 1/16$, and Pinsker's inequality gives total variation at most $1/4$.

Decode each bit from the sign of $\widehat{x}_{2r-1} - \widehat{x}_{2r}$, assigning $+1$ at a tie. A wrong sign incurs squared error at least $2t^2$: each true pair lies squared distance $2t^2$ from the diagonal where the two estimated effects agree. The two-point testing bound on each cube edge gives average bit-error probability at least $(1 - 1/4)/2 = 3/8$. Thus the uniform-prior normalized risk is at least $2t^2 m^{-1} \cdot 3h/8 \geq c/n$, since $h/m \geq 1/3$. The supremum risk dominates this Bayes risk. □

Normalization by m cancels the number of perturbed pairs, leaving the $1/n$ lower bound. To attain this order, we estimate task contrasts $x_j - x_\ell$ without estimating the worker effects. A pair of responses from one worker is *discordant* if one is positive and the other negative. Conditional on discordance, the probability that task j receives the positive response is $\sigma(x_j - x_\ell)$, which does not depend on the worker effect. The following row-wise calculation explains this cancellation more generally.

Conditional likelihood removes a nuisance parameter by conditioning on a sufficient statistic for it [2]. For worker i, the response-dependent likelihood factor is $\exp(\sum_j Y_{ij}x_j + a_i \sum_j Y_{ij})$. Among responses with the same total $R_i = \sum_j Y_{ij}$, the factor involving a_i is constant and cancels. Writing $Y_i = (Y_{i1}, \ldots, Y_{im})$, for $u \in \{0, 1\}^m$ with total r we obtain

$$\mathbb{P}_{x,a}(Y_i = u \mid R_i = r) = \frac{\exp(u^\top x)}{\sum_{v\in\{0,1\}^m:\, \sum_j v_j = r} \exp(v^\top x)}, \quad \sum_j u_j = r.$$

The denominator sums over binary arrangements with the observed total. For $1 \leq r \leq m - 1$, the relative probabilities of two arrangements that exchange a positive response between

tasks depend on their effect contrast, not on a_i. Totals zero and m provide no conditional information.

Remark 8.23 (Optimal Procedure and Matching Upper Bound) Choose task 1 as a baseline and estimate each difference $x_j - x_1$ by comparing responses from the same workers. For each $j \geq 2$, retain workers whose responses to tasks j and 1 differ. Let B_j be the fraction giving task j the positive response, with $B_j = 1/2$ if none are retained. Conditional on the retained workers, these responses are independent Bernoulli variables with common probability $\sigma(x_j - x_1)$, regardless of their worker effects.

Project B_j onto $[\sigma(-2M_0), \sigma(2M_0)]$ and apply $\operatorname{logit}(p) = \log\{p/(1-p)\}$ to estimate the difference by $\widehat{d}_j$. Set $\widehat{d}_1 = 0$ and center these estimates:

$$\widehat{x}_j = \widehat{d}_j - \frac{1}{m}\sum_{\ell=1}^{m} \widehat{d}_\ell.$$

Because $\sum_j x_j = 0$, centering the true differences recovers x. The class bounds keep retention probabilities away from zero and the logit transformation Lipschitz on the projection interval. Each estimated difference therefore has mean squared error at most C/n, and centering cannot increase the total squared error. Thus the normalized risk is at most C/n uniformly over $\mathcal{P}_{m,n}(M_0, M_1)$ for every $m \geq 2$, with C depending only on M_0, M_1. This matches Theorem 8.22 for sufficiently large n, without requiring independence among the estimated differences.

Li and Cai [91, Proposition 2.2] obtain the same order for conditional maximum likelihood under full observation and sufficiently large m. The estimator above also covers small m, including two tasks.

8.6.2 Incomplete responses and available contrasts

When each worker assesses only some tasks, the number of responses and the availability of within-worker contrasts become distinct considerations. Retain the parameter class and loss, and let $O_{ij} \overset{\text{i.i.d.}}{\sim} \operatorname{Ber}(q)$, with $q \in (0, 1]$ common to all parameters and independent of the potential responses. We observe the entire mask and Y_{ij} whenever $O_{ij} = 1$.

Each task now has nq responses on average. The masked-divergence identity (8.23) replaces n by nq in the paired-perturbation lower bound. Attaining this order is a separate question: procedures that eliminate worker effects need responses to different tasks from the same worker, not just many individual responses.

Theorem 8.24 (Sparse-Observation Crowdsourcing Lower Bound) *Fix $M_0, M_1 > 0$. There are constants $c > 0$ and $n_0 > 0$, depending only on these bounds, such that, for $m \geq 2$, $q \in (0, 1]$, and $nq \geq n_0$, the minimax risk in the Bernoulli-q observation experiment satisfies*

$$\inf_{\widehat{x}} \sup_{(x,a)\in\mathcal{P}_{m,n}(M_0,M_1)} \frac{1}{m}\mathbb{E}_{x,a}\|\widehat{x} - x\|_2^2 \geq \frac{c}{nq}.$$

The risk averages over both the response and observation-mask randomness.

Proof Restrict to $a_i = 0$ and use the paired-task cube from Theorem 8.22, now with $t = 1/(4\sqrt{nq})$. Increase n_0 so $t \le \min(M_0, M_1)$, ensuring admissibility. The mask law is the same at every vertex, so (8.23) multiplies each changed response's KL by q. Neighboring vertices differ in two task columns; their full-data KL is at most $nqt^2 = 1/16$, and Pinsker's inequality bounds their total variation by $1/4$.

The decoding and loss conversion in the proof of Theorem 8.22 are unchanged. They give normalized Bayes risk at least $3\lfloor m/2 \rfloor t^2/(4m) \ge t^2/4 = 1/(64nq)$, and hence the stated minimax bound, also for randomized estimators. □

To examine attainable accuracy, apply the same conditioning argument to the observed responses. For a realized mask, write $S_i = \{j : O_{ij} = 1\}$ and let $R_i = \sum_{j \in S_i} Y_{ij}$ now denote the observed row total. Write $Y_{i,S_i} = (Y_{ij})_{j \in S_i}$. For $u \in \{0,1\}^{S_i}$ with total r,

$$\mathbb{P}_{x,a}(Y_{i,S_i} = u \mid S_i, R_i = r) = \frac{\exp(\sum_{j \in S_i} u_j x_j)}{\sum_{v \in \{0,1\}^{S_i} : \sum_{j \in S_i} v_j = r} \exp(\sum_{j \in S_i} v_j x_j)}, \qquad \sum_{j \in S_i} u_j = r.$$

A row with only one observed task, or with observed responses all equal to zero or all equal to one, supplies no contrast after conditioning on its observed total.

Which task contrasts can these observations identify? Consider two distinct tasks j, ℓ observed for worker i. Put $D_{i,j\ell} = \mathbf{1}\{Y_{ij} \ne Y_{i\ell}\}$ and $d_{j\ell} = x_j - x_\ell$. Given $D_{i,j\ell} = 1$, the indicator $B_{i,j\ell} = \mathbf{1}\{Y_{ij} = 1, Y_{i\ell} = 0\}$ records which task received the positive response. It is Bernoulli with probability $\sigma(d_{j\ell})$, which does not depend on a_i. Its conditional log likelihood is

$$B_{i,j\ell} d_{j\ell} - \log(1 + e^{d_{j\ell}}).$$

Let e_j denote the jth coordinate vector in $\mathbb{R}^m$. The negative Hessian of this log likelihood with respect to x is

$$\sigma(d_{j\ell})\{1 - \sigma(d_{j\ell})\}(e_j - e_\ell)(e_j - e_\ell)^\top.$$

For a perturbation $v \in \mathbb{R}^m$, its quadratic form is $(v_j - v_\ell)^2$ times a positive scalar: the information concerns a difference, not an absolute level. Regard the tasks as vertices and each available discordant pair as a weighted edge. The displayed matrix is the edge's Laplacian; summing these matrices gives the Laplacian of the task-comparison graph.

The scalar factor is bounded away from zero on the parameter class. Each edge matrix annihilates constant shifts, which centering removes. If the graph is disconnected, these contrasts also leave separate component shifts undetermined. For example, with three tasks and only an edge between tasks 1 and 2, the centered perturbation $(1, 1, -2)$ changes no available contrast. A connected positive-weight graph controls every centered direction, though weak connections can make that control poor.

This graph describes identifiable contrasts, not independent observations. Pairs from one row share responses and must not be treated as independent. The full conditional row likelihood counts each worker once. Its negative Hessian is the conditional covariance matrix J_i below; for a perturbation $v \in \mathbb{R}^{S_i}$,

$$J_i = \mathrm{Cov}_{x,a}(Y_{i,S_i} \mid S_i, R_i), \qquad v^\top J_i v = \mathrm{Var}_{x,a}\left(\sum_{j \in S_i} v_j Y_{ij} \,\middle|\, S_i, R_i \right).$$

This identity follows by differentiating the log normalizing sum in the conditional likelihood. If $0 < R_i < |S_i|$, every binary configuration with that total has positive probability. Exchanging a one and a zero shows that the variance vanishes exactly when v is constant on S_i. Thus an informative row distinguishes all contrasts among its observed tasks without treating overlapping pairs as independent.

Remark 8.25 (Procedure and Upper-Bound Comparison) For two tasks, write $x = (\theta, -\theta)$ and retain workers only when both responses are observed and differ. Among these workers, a positive response to task 1 has probability $\sigma(2\theta)$, regardless of the worker effect. Estimate this probability by the observed fraction, using $1/2$ if no workers remain. Project the fraction onto $[\sigma(-2M_0), \sigma(2M_0)]$ and take half its logit to obtain $\widehat{\theta}$. Each worker is retained with probability at least $2q^2\sigma(-M_1)^2$. Consequently, $\widehat{x} = (\widehat{\theta}, -\widehat{\theta})$ has normalized expected squared error at most $C/(nq^2)$ uniformly over the two-task class, where C depends only on M_0, M_1.

Compared with the lower bound $c/(nq)$, this guarantee has an extra factor q^{-1}. The lower-bound construction fixes worker effects at zero and uses individual responses, observed with probability q. The estimator removes unknown worker effects using two responses from the same worker, both observed with probability q^2. These bounds do not determine whether the extra factor is unavoidable as q decreases.

For many tasks under Bernoulli sampling, Li and Cai [91, Theorem 2.7] obtain a sampling contribution of order $1/(nq)$ for their differentially private estimator. This requires $nq \gtrsim \log m$, $mq \gtrsim \log n$, bounded logits, a public tuning value comparable to q, and the algorithmic tuning in their equation (15). These assumptions exclude fixed $m = 2$ as n grows. The result neither closes the two-task gap nor establishes this rate for ordinary conditional maximum likelihood.

What the inferential goal changes

The constructions of Chapter 3 recur, but their statistical meaning changes. A sparse packing separates regression coefficients, prediction rules, or partitions; each requires its own conversion from decoding error to loss. One uncertain boundary item obstructs exact ranking, whereas paired perturbations accumulate estimation errors in ranking and crowdsourcing. The regression covariance mixture supplies a complementary lesson from Chapters 5–7: a common information bound can constrain testing separation, honest interval length, and adaptation, through different reductions.

Labels provide a useful bridge to Chapter 9. Classification predicts a future label; clustering learns a partition up to a global exchange of group names; sparse-mixture discovery selects signal members among the observed data. Their losses and information requirements differ. Chapter 9 places this distinction at the center of the analysis: evidence sufficient to detect a signal may not identify its members or locate its boundaries.

8.7 Exercises

8.1. Derive (8.3) by conditioning on the design. For $\Sigma = I_d$ and $\sigma = 1$, compare the packing calibration in Proposition 8.1 with Example 3.6. Why does replacing the joint experiment by a conditional one require a separate design argument?

8.2. Compare (5.13) and (8.11) for disjoint supports. Explain why changing the noise variance removes information from the response marginal, and why this construction is unavailable in Section 5.5. Does it require unknown design covariance?

8.3. For (8.13), solve for the conditional mean and variance of Y given X. Verify that the marginal law of (Y, X_1) is unchanged even though $(\beta_S)_1 \neq 0$. Rotate these two coordinates to recover the covariance family (8.6), and use the shared mixture calculation to derive (8.15). Explain why the testing alternatives in Proposition 8.3 do not provide the same common coordinate shift.

8.4. In model (8.1), fix $K_\Sigma, K_\sigma > 1$, $0 < \alpha < 1/4$, and $0 < \gamma < 1/2$. Work along sequences of integers $n, d \to \infty$ with $d \geq 3$, $n \geq s \log d$, and $2 \leq s_0 \leq s \leq c_* \min\{d^\gamma, n/\log d\}$, where $0 < c_* \leq 1$ is sufficiently small for the matching interval-length comparison preceding Section 8.1.4. Put $r(s) = n^{-1/2} + s \log d/n$. Give sequences in this regime for which $r(s)/r(s_0)$ diverges and sequences for which it stays bounded. Explain why Corollary 8.8, together with the matching known-class benchmarks, rules out rate-optimal interval adaptation in the first case but not in the second.

8.5. Derive the excess-risk identity (8.16) by conditioning on X. Explain why ties $m_P(X) = 1/2$ have zero excess cost, yet including them in the margin event (8.17) excludes the displayed hard family when $\alpha > 0$.

8.6. For the hard family in Theorem 8.11, calculate the total active design mass, the training-sample KL between neighboring sign vectors, and the excess-risk cost of a wrong sign. Use these quantities to recover the rate, and compare the KL calculation with the proof's Hellinger-affinity argument.

8.7. For Theorem 8.13, verify the Gaussian angular disagreement formula and the quadratic conversion from disagreement probability to excess risk.

8.8. Let g_μ be the density of $\frac{1}{2}N_d(\mu, I_d) + \frac{1}{2}N_d(-\mu, I_d)$. Verify that $g_\mu = g_{-\mu}$. In Theorem 8.15, explain why deleting labels contracts KL and why the loss-to-decoding argument must also allow a global sign change. Why does KL contraction alone not establish equality of labeled and unlabeled minimax risks?

8.9. In Theorem 8.17, verify the gap between the best item and every competitor and the factor q in the observed-edge KL.

8.10. Prove that $\tau_i = 1/2 + u_i - \bar{u}$ in the additive comparison subfamily used for top-k recovery. Explain why fixing $k - 1$ high items and varying one pivotal item gives at least $d/2$ hypotheses when $k \leq d/2$, and construct the symmetric low-item family when $k > d/2$.

8.11. In Proposition 8.20, compute the squared distance of the true score pair from the separating diagonal, and deduce the minimum squared error when its bit is decoded incorrectly. Why does the score-vector risk average avoid the $\log d$ factor needed for exact top-k selection?

8.12. Using the conditional row law following Theorem 8.22, verify that conditioning on a worker's row sum removes the worker effect. For a total $1 \leq r \leq m - 1$, compare two

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configurations that exchange a positive response between tasks j and ℓ to recover the contrast $x_j - x_\ell$. Explain the role of centering and why totals 0 and m contribute no conditional information.

8.13. Recompute the cube-edge KL in Theorem 8.24 with the Bernoulli observation mask included. Explain why averaging over the mask introduces a factor q in the information bound, and recover the full-observation calculation when $q = 1$.

8.14. For two tasks under Bernoulli-q observation, calculate the probability of a co-observed discordant pair. Derive the $C/(nq^2)$ risk upper bound for the clipped orientation-fraction rule using only such pairs. Explain why this upper bound and the general $c/(nq)$ lower bound do not determine the sparse-design minimax rate.

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9

Detection, Selection, and Localization

Chapter 8 brought estimation, testing, and honest inference together within structured learning problems. This chapter shifts the emphasis to the information required for detection, selection, and localization. Evidence that a signal exists need not reveal which observations carry it or where its boundaries lie. A collection of weak effects may be detectable even when individual findings cannot be selected reliably; an anomalous interval may be detectable without permitting accurate endpoint estimation. Our goal is to understand these distinctions and construct lower bounds suited to each decision.

These questions arise in large-scale screening, where selected candidates receive further investigation; watermark analysis, where evidence of a statistical signature must be distinguished from evidence at particular positions; and monitoring, where both the presence and extent of an anomaly matter. Categorical data introduce a related challenge: departures may be spread across many rarely observed categories, and the reference distribution may itself be unknown.

Section 9.1 compares detection, discovery, classification, and screening in sparse Gaussian mixtures. Truncation and Hellinger affinity control rare likelihood extremes, while posterior signal probabilities bound reliable selection. Rarity, signal strength, and variance govern the resulting boundaries.

Section 9.2 extends detection beyond Gaussian scores through product affinity and a quadratic–linear likelihood bound, which remains useful when the full second moment is infinite. An exponential example and independent watermark segments illustrate the calculation; token-level watermark models show how dependence and the information available to a selector affect its scope.

Section 9.3 first changes the decision for a single segment: location mixtures obstruct detection, Fano's inequality limits block identification, and nearby interval pairs bound endpoint accuracy. It then holds detection fixed and varies the signal family through overlapping locations, unknown signs, and multiple active blocks.

Finally, Section 9.4 studies heterogeneity among multinomial populations through a simplified model of Cai, Ke, and Turner [24]. Paired signs preserve probability mass and the average distribution, while random topic contrasts and document assignments obstruct detection. The comparison with DELVE shows how the same signal scale governs attainable power.

We begin with sparse Gaussian mixtures, where detection and selection admit explicit comparisons.

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9.1 Sparse Gaussian signals: detection, discovery, and classification

Large-scale studies, such as searches for weak associations across genetic markers, ask both whether signals are present and which findings merit further investigation. In one independent Gaussian-score model, we compare the signal strength needed for four goals: detecting a signal population, making reliable discoveries, classifying nearly all signals, and screening out many nulls while retaining nearly all signals. The model isolates the effect of these different error requirements; dependence among markers requires additional analysis.

For $i = 1, \dots, n$, let the unobserved indicators θ_i be independent $\mathrm{Ber}(\varepsilon_n)$ variables and, conditional on them, observe independent scores

$$X_i \mid \theta_i = 0 \sim N(0,1), \qquad X_i \mid \theta_i = 1 \sim N(\mu_n, \sigma^2),$$

where $0 < \varepsilon_n < 1$, $\mu_n > 0$, and the nonnull standard deviation $\sigma > 0$ is fixed. Under the global null all $\theta_i = 0$; under the alternative they follow the stated mixture law. Only $X_1, \dots, X_n$ are observed. Integrating out the indicators gives independent observations from $(1-\varepsilon_n)N(0,1) + \varepsilon_n N(\mu_n, \sigma^2)$ under the alternative. This is a specified mixture experiment, not a union over fixed unknown supports.

Detection uses the total-error criterion (5.1). For a null law P_0 and the specified mixture alternative P_1, the optimal error sum is the testing affinity $\eta(P_0, P_1)$. This global decision does not require recovering the latent indicators.

Throughout this chapter, we write a test as its $\{0,1\}$-valued decision, adjoining an independent random seed when randomization is used and averaging error probabilities over that seed. This is equivalent to the $[0,1]$-valued rejection-probability formulation in Section 5.1.

For selection under the alternative, write $I = \{i : \theta_i = 1\}$ and $N = \{i : \theta_i = 0\}$. A score-based, possibly randomized rule returns $S \subseteq \{1, \dots, n\}$; unsubscripted probabilities and expectations in the selection results refer to the joint alternative law of labels and scores, including any independent random seed. We assess false selections and missed signals through the marginal false-discovery rate (mFDR) and the missed-discovery rate (MDR),

$$\mathrm{mFDR}(S) = \frac{\mathbb{E}|S \cap N|}{\mathbb{E}|S|}, \qquad \mathrm{MDR}(S) = \frac{\mathbb{E}|I \setminus S|}{\mathbb{E}|I|},$$

with $\mathrm{mFDR}(S) = 0$ if $\mathbb{E}|S| = 0$. The marginal false-discovery rate is a ratio of expectations, not the expected realized false-discovery proportion.

The selection goals impose different requirements:

- Reliable *discovery* requires $\mathrm{mFDR}(S) \to 0$ and $\mathbb{P}(|S| \geq 1) \to 1$: report a nonempty set with high probability while making expected false selections negligible relative to expected total selections.
- Near-complete *classification* requires $\mathrm{mFDR}(S) \to 0$ and $\mathrm{MDR}(S) \to 0$: retain almost all signals as well.
- *Screening*, treated last, retains nearly all signals while limiting the fraction of nulls retained; the selected set need not be dominated by signals.

Discovery and classification do not require the probability of any false selection to vanish. Classification labels the observed scores, rather than predicting a new label as in Chapter 8. After an elementary bounded-shift calculation, we study the rare/weak calibration $\varepsilon_n = n^{-\beta}$, $\mu_n = \sqrt{2r \log n}$, with fixed $1/2 < \beta < 1$ and $r > 0$.

We begin with detection under equal variances, $\sigma = 1$. Suppressing the n subscripts, write $P_0 = N(0,1)^{\otimes n}$ and $P_{\varepsilon,\mu} = \{(1-\varepsilon)N(0,1) + \varepsilon N(\mu,1)\}^{\otimes n}$. The exact likelihood second moment gives an elementary benchmark for bounded shifts and explains why growing shifts require a more careful argument.

Theorem 9.1 (Gaussian Contamination Second Moment) *For the experiment just specified, every test obeys*

$$\mathbb{P}_0(\phi = 1) + \mathbb{P}_{\varepsilon,\mu}(\phi = 0) \geq 1 - \frac{1}{2}\sqrt{\{1 + \varepsilon^2(e^{\mu^2} - 1)\}^n - 1}.$$

In particular, if $0 < \mu \leq 1$ and $n\varepsilon^2\mu^2$ is bounded by a sufficiently small numerical constant, the sum of errors is bounded away from zero. Conversely, for $0 < \mu \leq 1$, a test based on $\sum_i X_i$ has both errors tending to zero whenever $n\varepsilon^2\mu^2 \to \infty$.

Proof For one null observation $X \sim N(0,1)$, the mixture-to-null likelihood ratio is $\ell(x) = 1 - \varepsilon + \varepsilon e^{\mu x - \mu^2/2}$. The Gaussian moment-generating identity yields $\mathbb{E}_0\ell = 1$ and

$$\mathbb{E}_0\ell^2 = 1 + \varepsilon^2\{\mathbb{E}_0 e^{2\mu X - \mu^2} - 1\} = 1 + \varepsilon^2(e^{\mu^2} - 1).$$

Independence gives $\chi^2(P_{\varepsilon,\mu}\|P_0) = \{1 + \varepsilon^2(e^{\mu^2} - 1)\}^n - 1$, the expression inside the square root in the theorem. The optimal sum of errors is $\eta(P_0, P_{\varepsilon,\mu}) = 1 - \mathrm{TV}(P_0, P_{\varepsilon,\mu})$, and Cauchy–Schwarz gives $\mathrm{TV} \leq \frac{1}{2}\sqrt{\chi^2}$, proving the display. For $\mu \leq 1$, $e^{\mu^2} - 1 \leq (e-1)\mu^2$; if $n\varepsilon^2\mu^2$ is a sufficiently small constant, the displayed lower bound is positive.

For the converse, under the alternative $\mathbb{E}\sum_i X_i = n\varepsilon\mu$ and $\mathrm{Var}(\sum_i X_i) = n\{1 + \varepsilon(1-\varepsilon)\mu^2\} \leq 2n$; under the null the mean is zero and variance is n. Reject when $\sum_i X_i > n\varepsilon\mu/2$. Chebyshev's inequality gives each error at most a constant multiple of $(n\varepsilon^2\mu^2)^{-1}$, proving the assertion. □

For $\mu \leq 1$, $n\varepsilon^2\mu^2$ is the squared shift of the sum in units of its null variance. The theorem therefore identifies when aggregate mean information suffices. Growing shifts require a different calibration: $e^{\mu^2} - 1$ is no longer comparable to μ^2, and rare observations can dominate the second moment.

9.1.1 Rare/weak Gaussian detection

Can increasingly strong signals remain undetectable because they are too rare? To locate this balance, fix $\beta \in (1/2, 1)$ and $r > 0$ and, for each $n \geq 2$, let

$$\begin{aligned} \varepsilon_n &= n^{-\beta}, & \mu_n &= \sqrt{2r\log n}, \\ P_{0,n} &= N(0,1)^{\otimes n}, & P_{1,n} &= \big\{(1-\varepsilon_n)N(0,1) + \varepsilon_n N(\mu_n,1)\big\}^{\otimes n}. \end{aligned} \tag{9.1}$$

The exponent β measures rarity and r measures logarithmic signal strength, both fixed as $n \to \infty$. Although $\mathbb{E}|I| = n^{1-\beta} \to \infty$, the contamination fraction is $o(n^{-1/2})$; tail counts can succeed where a sample-mean test does not.

Section 5.4.1 mixes fixed-size supports and signs to make a mean vector with prescribed norm difficult to detect, with information governed by support overlap. Here independent Bernoulli indicators and positive shifts define an alternative with random support size. To

locate the sharp boundary in (β, r), we track the informative score scales and truncate the second moment where rare extremes make it too large.

The classical boundary in this experiment is

$$\rho^*(\beta) = \begin{cases} \beta - \frac{1}{2}, & \frac{1}{2} < \beta \le \frac{3}{4}, \\ \left(1 - \sqrt{1-\beta}\right)^2, & \frac{3}{4} < \beta < 1. \end{cases} \tag{9.2}$$

The two formulas agree at $\beta = 3/4$. We prove the strict interior statements; behavior at equality requires separate analysis of lower-order terms. The boundary comes from Ingster's sparse-signal work [74]; Donoho and Jin [56] developed adaptive higher-criticism tests that scan standardized deviations of tail counts from their null expectations. We emphasize why truncation is needed to prove impossibility throughout the region below the boundary.

Why truncation is needed. The second-moment bound proves indistinguishability when $r < \beta - 1/2$. For $\beta \le 3/4$, this reaches the boundary. For $\beta > 3/4$, it leaves a gap: at $\beta = 0.9$ it reaches only $r < 0.4$, whereas the boundary is approximately 0.468. Extremely large likelihood values on rare observations can dominate the moment. Its divergence in the intervening range therefore does not establish detectability.

Truncate at $\sqrt{2\log n}$, the leading null maximum scale, using Section 2.2. Two checks suffice: the retained region has alternative probability tending to one, and the truncated likelihood converges to one in null mean square. Gaussian tail bounds will verify both.

Theorem 9.2 (Rare/Weak Gaussian-Mixture Detection Boundary) *In the experiment* (9.1)*, with fixed $\beta \in (1/2, 1)$ and $r > 0$, the following hold.*

(i) *If $0 < r < \rho^*(\beta)$, then $\mathrm{TV}(P_{0,n}, P_{1,n}) \to 0$. Thus the error sum of every sequence of tests is at least $1 - o(1)$.*

(ii) *If $r > \rho^*(\beta)$, there is a sequence of tests, knowing β and r, for which both error probabilities tend to zero.*

For part (ii), a single exceedance-count threshold suffices in the moderately sparse part of the detectable region; a maximum test suffices in the remaining part.

Proof For part (i), the ordinary second moment suffices when $\beta \le 3/4$; otherwise we truncate to control rare extremes. The one-coordinate likelihood ratio under $P_{0,n}$ is

$$\ell_n(x) = 1 - \varepsilon_n + \varepsilon_n \exp(\mu_n x - \mu_n^2/2).$$

All asymptotic comparisons below use fixed β, r and $n \to \infty$. For a fixed $a > 0$, the elementary normal-tail bounds give

$$\bar{\Phi}\left(\sqrt{2a\log n}\right) \asymp \frac{n^{-a}}{\sqrt{\log n}}, \tag{9.3}$$

where the constants may depend on a. This follows, for example, by integrating φ by parts and bounding $\bar{\Phi}(t)$ between $t\varphi(t)/(1+t^2)$ and $\varphi(t)/t$.

Lower bound when $\beta \le 3/4$. The second-moment identity in Theorem 9.1 holds for every positive shift. Applying it with $\varepsilon = \varepsilon_n$ and $\mu = \mu_n$ gives

$$1 + \chi^2(P_{1,n} \| P_{0,n}) = \left\{1 + \varepsilon_n^2(e^{\mu_n^2} - 1)\right\}^n \le \exp\left(n^{1-2\beta+2r}\right).$$

For $r < \beta - 1/2$, the exponent tends to zero. Hence $\chi^2 \to 0$ and $\mathrm{TV} \le \frac{1}{2}\sqrt{\chi^2} \to 0$. This proof also covers $\beta = 3/4$ below its boundary.

Lower bound when $\beta > 3/4$. Assume $r < \rho^*(\beta) < 1$, set $t_n = \sqrt{2\log n}$, and truncate the *whole* likelihood ratio $L_n = \prod_{i=1}^n \ell_n(X_i)$ on $A_n = \{\max_i X_i \le t_n\}$:

$$\widetilde{L}_n = L_n \mathbf{1}_{A_n} = \prod_{i=1}^n \{\ell_n(X_i)\mathbf{1}_{\{X_i \le t_n\}}\}.$$

Discarded probability. First check that truncation loses negligible alternative probability. The one-coordinate probability of crossing t_n is

$$(1-\varepsilon_n)\bar{\Phi}(t_n) + \varepsilon_n \bar{\Phi}(t_n - \mu_n).$$

The first term multiplied by n vanishes by (9.3). Since $r < 1$, the second term multiplied by n is at most a constant times

$$n^{1-\beta-(1-\sqrt{r})^2}/\sqrt{\log n} = o(1),$$

because $r < \rho^*(\beta)$ is equivalent to $(1-\sqrt{r})^2 > 1-\beta$. A union bound therefore yields $\mathbb{P}_{1,n}(A_n^c) = o(1)$.

Retained first moment. For $X \sim N(0,1)$, let $m_{1,n} = \mathbb{E}_0[\ell_n(X)\mathbf{1}_{\{X \le t_n\}}]$. The change-of-measure identity gives $m_{1,n} = 1 - (1-\varepsilon_n)\bar{\Phi}(t_n) - \varepsilon_n \bar{\Phi}(t_n - \mu_n)$, so $\mathbb{E}_0 \widetilde{L}_n = m_{1,n}^n \to 1$.

Retained second moment. Put $m_{2,n} = \mathbb{E}_0[\ell_n(X)^2 \mathbf{1}_{\{X \le t_n\}}]$. Expanding the square and completing the Gaussian square gives

$$\begin{aligned} m_{2,n} &= (1-\varepsilon_n)^2 \Phi(t_n) + 2\varepsilon_n(1-\varepsilon_n)\Phi(t_n - \mu_n) \\ &\quad + \varepsilon_n^2 e^{\mu_n^2}\Phi(t_n - 2\mu_n) \le 1 + \varepsilon_n^2 e^{\mu_n^2}\Phi(t_n - 2\mu_n). \end{aligned}$$

If $r \le 1/4$, the last contribution multiplied by n is at most $n^{1-2\beta+2r} = o(1)$ because $\beta > 3/4$. If $1/4 < r < \rho^*(\beta)$, then $2\mu_n - t_n = \sqrt{2\log n}(2\sqrt{r} - 1) > 0$; the Gaussian upper-tail bound makes that contribution at most a constant times

$$n^{1-2\beta+2r-(2\sqrt{r}-1)^2} = n^{2(2\sqrt{r}-r-\beta)} = o(1).$$

The final exponent is negative precisely because $r < \rho^*(\beta)$ and $r < 1$. The preceding bound gives $m_{2,n} \le 1 + o(n^{-1})$, while Cauchy–Schwarz gives $m_{2,n} \ge m_{1,n}^2$. Since $m_{1,n}^n \to 1$, these bounds imply $m_{2,n}^n \to 1$. Independence now gives

$$\mathbb{E}_0(\widetilde{L}_n - 1)^2 = m_{2,n}^n - 2m_{1,n}^n + 1 \longrightarrow 0.$$

Finally $\mathbb{E}_0|L_n - \widetilde{L}_n| = \mathbb{P}_{1,n}(A_n^c) \to 0$, so $\mathbb{E}_0|L_n - 1| \to 0$. Since $\mathrm{TV}(P_{0,n}, P_{1,n}) = \frac{1}{2}\mathbb{E}_0|L_n - 1|$, this proves (i).

Upper bound. First suppose $r < 1/4$ and $r > \rho^*(\beta)$. Necessarily $\beta < 3/4$ and $r > \beta - 1/2$. Set $q = 4r < 1$, $u_n = \sqrt{2q\log n}$, and $N_n = \sum_i \mathbf{1}_{\{X_i > u_n\}}$. Put $p_{0,n} = \bar{\Phi}(u_n)$, $p_{1,n} = \bar{\Phi}(u_n - \mu_n)$, and $\Delta_n = n\varepsilon_n(p_{1,n} - p_{0,n})$. Then $\mathbb{E}_0 N_n = np_{0,n}$, $\mathbb{E}_1 N_n = np_{0,n} + \Delta_n$, and (9.3) yields

$$p_{0,n} \asymp n^{-4r}/\sqrt{\log n}, \qquad p_{1,n} \asymp n^{-r}/\sqrt{\log n}, \qquad \Delta_n \asymp n^{1-\beta-r}/\sqrt{\log n}.$$

Both $\Delta_n^2/(np_{0,n}) \to \infty$ and $\Delta_n^2/(n\varepsilon_n p_{1,n}) \to \infty$: their polynomial exponents are respectively $1 - 2\beta + 2r > 0$ and $1 - \beta - r > 0$. Under the null and alternative, respectively, $\mathrm{Var}_0(N_n) \le$

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$np_{0,n}$ and $\mathrm{Var}_1(N_n) \le np_{0,n} + n\varepsilon_n p_{1,n}$. Therefore Chebyshev's inequality shows that the test rejecting at $np_{0,n} + \Delta_n/2$ has both errors tending to zero.

In every other case with $r > \rho^*(\beta)$ we have $r \ge 1/4$. Reject when $\max_i X_i > t_n = \sqrt{2\log n}$. Its null error is at most $n\bar{\Phi}(t_n) = o(1)$. Its alternative error is at most

$$\{1 - \varepsilon_n \bar{\Phi}(t_n - \mu_n)\}^n \le \exp\{-n\varepsilon_n \bar{\Phi}(t_n - \mu_n)\}.$$

If $1/4 \le r < 1$, the exponent's magnitude grows polynomially because $1 - \beta - (1 - \sqrt{r})^2 > 0$; at $\beta < 3/4, r = 1/4$ this follows from $3/4 - \beta > 0$, and at $\beta \ge 3/4$ it follows from $r > \rho^*(\beta)$. If $r \ge 1$, the exceedance probability under a contaminated observation is at least $1/2$, and $n\varepsilon_n \to \infty$. The alternative error therefore vanishes, completing (ii). □

The two branches of ρ^* correspond to different threshold choices. For $0 < r < 1$, consider counts above $\sqrt{2q\log n}$, $q \in (0, 1]$. When $q > r$, the null and signal exceedance probabilities have polynomial orders n^{-q} and $n^{-(\sqrt{q}-\sqrt{r})^2}$. Dividing the count's mean shift by its null standard deviation gives exponent

$$\frac{1}{2} - \beta + \frac{q}{2} - (\sqrt{q} - \sqrt{r})^2. \tag{9.4}$$

For $q \le r$ the standardized shift increases with q; maximizing over all $q \le 1$ therefore selects $q = 4r$ when $4r \le 1$ and $q = 1$ otherwise. The former gives $r = \beta - 1/2$, where an excess of moderately large scores suffices; the latter gives $r = (1 - \sqrt{1-\beta})^2$, where scores near the null maximum matter.

Our proof uses parameter-dependent thresholds. Donoho and Jin's rank-trimmed higher-criticism statistic scans standardized excesses of ordered one-sided Gaussian p-values. Their Theorems 1.1–1.2 [56], with critical values calibrated at significance levels tending to zero sufficiently slowly, give adaptivity for fixed $1/2 < \beta < 1$ and $\rho^*(\beta) < r < 1$, without knowing β, r. The same statistic also covers $r \ge 1$ by coordinatewise monotonicity and coupling with a fixed smaller detectable shift; this extension is separate from the range stated in the cited theorem.

9.1.2 Variance heterogeneity

A signal population may differ from the null in variability as well as mean: a larger nonnull variance can produce detectable extremes even without a large mean shift. Our goal is to determine how this variance changes the signal strength needed for detection. Retain the global null, rare/weak calibration, and total-error criterion above, but let a signal score have law $N(\mu_n, \sigma^2)$ for a fixed $\sigma > 0$. Cai, Jeng, and Jin [21] establish the heteroscedastic detection boundary; in the sparse regime considered here it is

$$\rho^*(\beta;\sigma) = \begin{cases} (2-\sigma^2)(\beta - \frac{1}{2}), & 0 < \sigma < \sqrt{2},\ \frac{1}{2} < \beta \le 1 - \sigma^2/4, \\ (1 - \sigma\sqrt{1-\beta})^2, & 0 < \sigma < \sqrt{2},\ 1 - \sigma^2/4 < \beta < 1, \\ 0, & \sigma \ge \sqrt{2},\ \frac{1}{2} < \beta \le 1 - 1/\sigma^2, \\ (1 - \sigma\sqrt{1-\beta})^2, & \sigma \ge \sqrt{2},\ 1 - 1/\sigma^2 < \beta < 1. \end{cases} \tag{9.5}$$

At $\sigma = 1$, this reduces to (9.2). Increased variance can make every fixed $r > 0$ detectable when $\sigma \ge \sqrt{2}$ and $1/2 < \beta \le 1 - 1/\sigma^2$. This zero boundary does not include $r = 0$.

The theorem below treats the strict sparse-regime boundary, extending the source's range $0 < r < 1$ to larger detectable shifts; equality and the differently calibrated dense regime remain outside its scope.

For the lower bound, Hellinger affinity controls likelihood extremes without requiring a finite full second moment. Let a_n be the affinity between the one-observation null and mixture laws. By Section 1.3, their squared Hellinger distance is $1 - a_n$ and the product affinity is a_n^n. Thus $1 - a_n = o(n^{-1})$ makes the full-sample laws indistinguishable: the discrepancy in one score is too small to accumulate across n observations.

Split the affinity integral at the null extreme scale. A quadratic likelihood bound controls the center, and a linear bound controls the tails without requiring a small full second moment. The central contribution peaks either inside the retained interval or at its endpoint, producing the linear and curved boundary branches, respectively. Combining this calculation with the tail bound proves the lower bound; optimizing an exceedance threshold gives the attainable boundary.

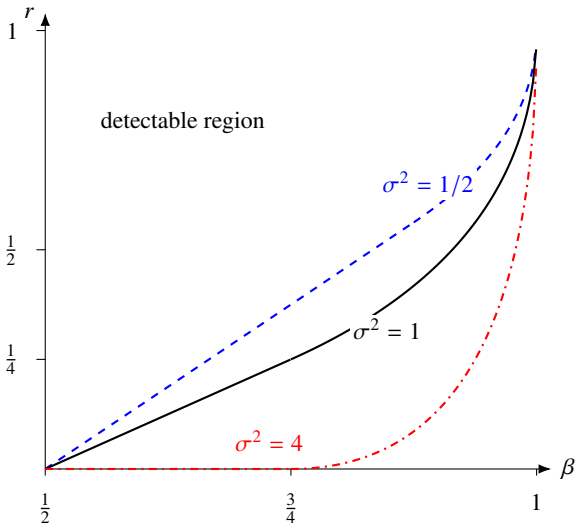


Figure 9.1 Sparse-mixture detection boundaries from [21]. The dashed, solid, and dash-dotted curves correspond to $\sigma^2 = 1/2$, 1, and 4, respectively. Above a curve reliable testing is possible; below it the two product laws merge. The solid curve is the boundary proved in Theorem 9.2. For $\sigma^2 = 4$, the boundary is zero on $1/2 < \beta \le 3/4$; the flat segment is intentional.

Theorem 9.3 (Heteroscedastic Gaussian-Mixture Detection Boundary) *Fix $1/2 < \beta < 1$, $r > 0$, and $\sigma > 0$. Let $\varepsilon_n = n^{-\beta}$, $\mu_n = \sqrt{2r \log n}$, and test $P_{0,n} = N(0,1)^{\otimes n}$ against*

$$P_{1,n} = \left\{(1 - \varepsilon_n)N(0,1) + \varepsilon_n N(\mu_n, \sigma^2)\right\}^{\otimes n}.$$

With $\rho^*(\beta;\sigma)$ as in (9.5)*, the following strict-boundary statements hold:*

(i) *If $r < \rho^*(\beta;\sigma)$, then $\mathrm{TV}(P_{0,n}, P_{1,n}) \to 0$, so every test has total error at least $1 - o(1)$.*

(ii) *If $r > \rho^*(\beta;\sigma)$, a one-sided threshold-count scan with an absolute standardized deviation, calibrated without (β, r, σ), has both errors tending to zero. The likelihood-ratio test for the specified simple hypotheses is also asymptotically reliable.*

No conclusion at $r = \rho^*(\beta;\sigma)$ is asserted.

Proof Let $g_n(x) = \varphi((x-\mu_n)/\sigma)/\{\sigma\varphi(x)\}$ be the nonnull-to-null likelihood ratio, and put $t_n = \sqrt{2\log n}$. For a null observation $X \sim N(0,1)$, split the Hellinger integral over $A_n = \{|X| \le t_n\}$ and its complement. To prove (i), we bound the deficit of the one-observation affinity

$$a_n = \mathbb{E}_0\sqrt{1-\varepsilon_n+\varepsilon_n g_n(X)}.$$

The exact identity $1-a_n = \frac{1}{2}\mathbb{E}_0[\{\sqrt{1+\varepsilon_n(g_n-1)}-1\}^2]$ permits a direct bound on these two regions. On A_n, rationalizing the square root bounds the squared term by $C\varepsilon_n^2(g_n-1)^2$; on A_n^c it is at most $\varepsilon_n|g_n-1|$. Since $\mathbb{P}_0(A_n^c) \le Cn^{-1}$ and $\varepsilon_n\mathbb{P}_0(A_n^c) \le C\varepsilon_n^2$ for $\beta < 1$, these two regions give

$$1-a_n \le \frac{\varepsilon_n}{2}\mathbb{P}_{N(\mu_n,\sigma^2)}(A_n^c) + C\varepsilon_n^2\{1+\mathbb{E}_0[g_n(X)^2\mathbf{1}_{A_n}]\}. \tag{9.6}$$

The term $\varepsilon_n^2 = o(n^{-1})$ because $\beta > 1/2$. The nonnull right tail at t_n dominates its left tail; Gaussian tail bounds give, up to a fixed power of $\log n$,

$$\varepsilon_n\mathbb{P}_{N(\mu_n,\sigma^2)}(A_n^c) \le (\log n)^C n^{-\beta-(1-\sqrt{r})_+^2/\sigma^2}. \tag{9.7}$$

This is $o(n^{-1})$ if $r < 1$ and $\beta + (1-\sqrt{r})^2/\sigma^2 > 1$; the branch check below verifies these conditions throughout the subcritical region.

Central contribution: interior peak or endpoint. It remains to control the retained second moment. The exponent in $g_n(x)^2\varphi(x)$ is

$$\frac{x^2}{2} - \frac{(x-\mu_n)^2}{\sigma^2}.$$

If $\sigma^2 < 2$, completing the square puts its maximum at $x_* = 2\mu_n/(2-\sigma^2)$, with value $\mu_n^2/(2-\sigma^2)$. The center lies within A_n exactly when $r \le (2-\sigma^2)^2/4$. Integrating the resulting Gaussian over A_n, and using $\mu_n^2 = 2r\log n$, gives

$$\varepsilon_n^2\mathbb{E}_0[g_n(X)^2\mathbf{1}_{A_n}] \le C(\log n)^C n^{-2\beta+2r/(2-\sigma^2)}. \tag{9.8}$$

When $x_* > t_n$, the exponent is increasing throughout the retained interval and its value at t_n is $\{1-2(1-\sqrt{r})^2/\sigma^2\}\log n$. Bounding the interval length by $2t_n$ gives

$$\varepsilon_n^2\mathbb{E}_0[g_n(X)^2\mathbf{1}_{A_n}] \le C(\log n)^C n^{1-2\{\beta+(1-\sqrt{r})^2/\sigma^2\}}. \tag{9.9}$$

For $\sigma^2 \ge 2$, the exponent is convex on A_n; its linear coefficient is positive, so its larger endpoint value is again the one at t_n.

Checking the boundary branches. The tail bound and the applicable central bound must both be $o(n^{-1})$. We check this against each branch of the stated boundary.

- If $\sigma^2 < 2$ and $\beta \le 1-\sigma^2/4$, subcriticality gives $r < (2-\sigma^2)(\beta-1/2) \le (2-\sigma^2)^2/4$. The peak is interior, and the exponent in (9.8) is less than -1.

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For the tail, note that $1-\sigma\sqrt{1-\beta}>0$ and $(1-\sigma\sqrt{1-\beta})^2 \geq (2-\sigma^2)(\beta-1/2)$: their difference is $2(\sqrt{1-\beta}-\sigma/2)^2$. Thus the tail exponent in (9.7) is also less than -1.

- In the remaining subcritical cases, the curved boundary gives $\beta+(1-\sqrt{r})^2/\sigma^2>1$. This controls the tail and the endpoint bound (9.9). If $\sigma^2<2$ and the peak is still interior, then $\beta>1-\sigma^2/4$ and $r\leq(2-\sigma^2)^2/4$ make the central exponent less than -1 as well.

The zero-boundary branch has no subcritical $r>0$ to check.

Together, the tail and second-moment bounds make the right-hand side of (9.6) $o(n^{-1})$. Hence $a_n^n\to 1$ and $\mathrm{TV}(P_{0,n},P_{1,n})\leq\sqrt{1-a_n^{2n}}\to 0$.

Finding a useful threshold. For (ii), fix $q\in(0,1)$ and count exceedances of $t_n(q)=\sqrt{2q\log n}$. Under the null their one-coordinate probability is $n^{-q+o(1)}$; under the nonnull Gaussian law it is $n^{-A(q)+o(1)}$, where

$$A(q)=\frac{(\sqrt{q}-\sqrt{r})_+^2}{\sigma^2}.$$

When $A(q)<q$, the signal exceedance probability dominates the null probability. The count's mean shift, divided by its null standard deviation, then has polynomial exponent

$$\delta(q)=\frac{1}{2}-\beta+\frac{q}{2}-A(q). \tag{9.10}$$

We need only thresholds with $\delta(q)>0$. They automatically satisfy $A(q)<q/2+1/2-\beta<q$, so the preceding comparison applies.

To maximize δ, note first that it increases on $q\leq r$. For $q\geq r$, write $u=\sqrt{q}$; apart from $1/2-\beta$, the objective is $u^2/2-(u-\sqrt{r})^2/\sigma^2$. When $\sigma^2<2$, its unconstrained maximizer is $u_*=2\sqrt{r}/(2-\sigma^2)$. If $u_*\leq 1$, the supremum is $1/2-\beta+r/(2-\sigma^2)$; otherwise it is the limiting endpoint value $1-\beta-(1-\sqrt{r})_+^2/\sigma^2$. For $\sigma^2\geq 2$ the limiting endpoint value governs. The supremum is positive exactly when $r>\rho^*(\beta;\sigma)$.

Scanning without knowing the parameters. Above the boundary, continuity of δ gives an interval of useful thresholds. Fix a sequence $(q_i)_{i\geq 0}$ listing all rationals in $(0,1)$ and, at sample size n, scan its first n entries. Let $N_n(q)$ be the count, $p_{0,n}(q)=\bar{\Phi}(t_n(q))$, and

$$S_n(q)=\frac{N_n(q)-np_{0,n}(q)}{\sqrt{np_{0,n}(q)(1-p_{0,n}(q))}}.$$

Reject when $|S_n(q_i)|>2^{i/2}\log(n+1)$ for at least one $0\leq i<n$. Under the null, each standardized count has variance one. Chebyshev's inequality and the union bound therefore give a false-alarm probability at most $\sum_{i\geq 0}2^{-i}/\log^2(n+1)=2/\log^2(n+1)\to 0$; no uniform lower bound on the null expected counts is needed. Above the boundary, continuity ensures that $\delta(q_i)>0$ for some fixed rational q_i, whose index is included for every sufficiently large n. At this threshold $\mathbb{E}_1 S_n(q_i)=n^{\delta(q_i)+o(1)}$, which exceeds the index-dependent logarithmic cutoff. Writing $p_{1,n}(q_i)=\bar{\Phi}((t_n(q_i)-\mu_n)/\sigma)=n^{-A(q_i)+o(1)}$ for the nonnull exceedance probability, the alternative variance divided by the square of this mean is bounded by a constant times

$$n^{-2\delta(q_i)+o(1)}+n^{-\{1-\beta-A(q_i)\}+o(1)}\longrightarrow 0,$$

since $1 - \beta - A(q_i) = \delta(q_i) + (1 - q_i)/2 > 0$. Chebyshev's inequality makes the miss probability vanish. Finally, the likelihood-ratio test minimizes total error for the two specified product laws, so it succeeds whenever this scan does. □

Remark 9.4 (Optimal Procedure and Matching Upper Bound) The threshold scan in the proof attains the strict detection boundary without knowing (β, r, σ): both errors vanish for every fixed $r > \rho^*(\beta; \sigma)$. For $0 < r < 1$, Cai, Jeng, and Jin [21, Theorem 7] attain this boundary by higher criticism, which maximizes absolute standardized deviations of empirical tail counts. The comparison concerns the boundary, not behavior at equality.

9.1.3 Discovery and classification of sparse Gaussian signals

Detecting a rare signal population does not identify which observations can reliably be reported as findings. In the alternative model of Theorem 9.3, discovery requires a nonempty selection with probability tending to one and vanishing mFDR; classification also requires vanishing MDR. We ask how much signal strength is needed for each goal beyond global detection.

Write $P_{\text{sig}} = N(\mu_n, \sigma^2)$, so $\mathbb{E}|I| = n^{1-\beta} \to \infty$ and $\text{MDR}(S) = 1 - \mathbb{E}|S \cap I|/(n\varepsilon_n)$. To derive the boundaries of Cai and Sun [41], we bound what posterior signal probabilities allow any selection rule to achieve.

We first give the conditional-expectation argument for a general selection model, without assuming Gaussian scores or independence across coordinates. For $i = 1, \dots, m$, with integer $m \geq 1$, let $\theta_i \in \{0, 1\}$ indicate a signal, let X denote all observed data, and set $q_i = \mathbb{P}(\theta_i = 1 \mid X)$. A selection rule reports indicators $D_i \in \{0, 1\}$ measurable with respect to X and, optionally, a random seed independent of the data and latent indicators. All expectations include that randomization. Write $R = \sum_i D_i$ for the number selected and $T = \sum_i \theta_i D_i$ for the number of true selections.

Each selected observation contributes its posterior signal probability to the expected number of true selections. Splitting at any $t \in (0, 1)$ gives

$$\mathbb{E}T = \mathbb{E} \sum_i D_i q_i \leq t\mathbb{E}R + \sum_i \mathbb{P}\{\theta_i = 1,\ q_i > t\}.$$

Below the cutoff, $D_i q_i \leq t D_i$; above it, bound D_i by one and use $\mathbb{E}[q_i \mathbf{1}_{\{q_i > t\}}] = \mathbb{P}\{\theta_i = 1, q_i > t\}$. If $q_i \leq b$ almost surely for every i, with $b \in [0, 1]$, then $\mathbb{E}T \leq b\mathbb{E}R$ and $\text{mFDR}(S) \geq 1 - b$ whenever $\mathbb{E}R > 0$, where $S = \{i : D_i = 1\}$. These statements allow joint selectors and require no coordinate independence.

When the data are $X = (X_1, \dots, X_m)$ and the pairs (θ_i, X_i) are independent, each posterior depends only on X_i. With common signal probability $\varepsilon \in (0, 1)$ and conditional densities f_0, f_1, Bayes' formula gives, almost surely,

$$q_i = \frac{\varepsilon f_1(X_i)}{(1 - \varepsilon) f_0(X_i) + \varepsilon f_1(X_i)}.$$

A selector may still use all observations jointly. The distinction is how often strong posterior evidence occurs: discovery needs enough such observations to make reliable selections, whereas classification needs them for nearly all signals.

Returning to the Gaussian mixture, the prior odds $\varepsilon_n/(1-\varepsilon_n)$ have order $n^{-\beta}$. A convincing signal score must overcome this rarity penalty, but must also occur often enough among the $n^{1-\beta}$ signals. Balancing them yields the discovery boundary

$$\rho_{\mathrm{dis}}(\beta;\sigma)=\begin{cases}(1-\sigma^2)\beta, & 0<\sigma<1, \quad \beta\le 1-\sigma^2,\\ (1-\sigma\sqrt{1-\beta})^2, & 0<\sigma<1, \quad \beta>1-\sigma^2,\\ (1-\sqrt{1-\beta})^2, & \sigma=1,\\ 0, & \sigma>1, \quad \beta\le 1-\sigma^{-2},\\ (1-\sigma\sqrt{1-\beta})^2, & \sigma>1, \quad \beta>1-\sigma^{-2}.\end{cases} \tag{9.11}$$

The branches meet continuously at their junctions; only their portions in $1/2<\beta<1$ apply here. Cai and Sun [41, Theorem 3] establish this boundary. We derive it from the likelihood-ratio tail in the sparse regime.

Figure 9.2 fixes $\sigma=1$ and places detection, discovery, and classification in one (β,r) plane. The dense portion $\beta\le 1/2$ preserves the source figure's visual context; the theorems here concern $1/2<\beta<1$. The inequalities among the boundaries are interpreted after the classification theorem.

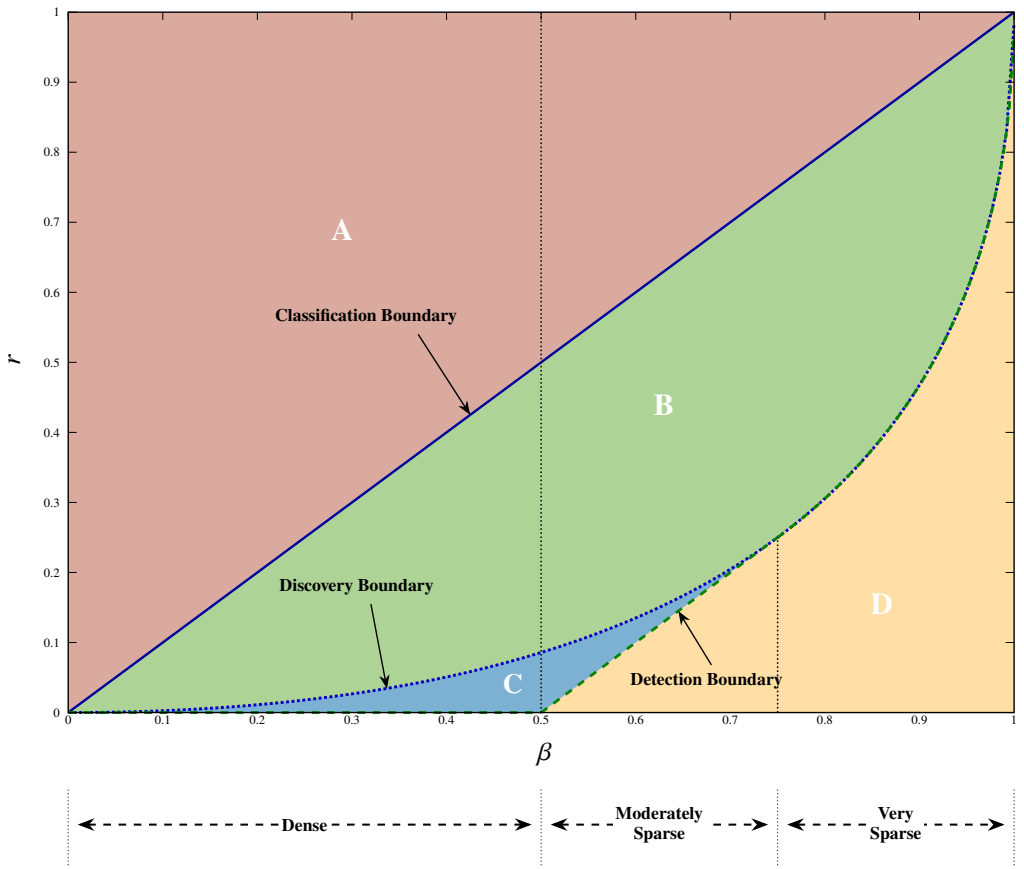


Figure 9.2 Phase diagram for the homoscedastic Gaussian mixture, closely redrawn from [41, Figure 1]. The classification line $r = \beta$, dotted discovery curve $\rho_{\text{dis}}(\beta; 1)$, and dashed detection curve $\rho^*(\beta)$ partition the plane into (A) permitting near-complete classification, (B) permitting discovery but not near-complete classification, (C) detectable but not reliably discoverable, and (D) undetectable regions. The dense range $\beta \le 1/2$ is contextual; the strict-boundary theorems here address $1/2 < \beta < 1$.

Theorem 9.5 (Discovery Boundary for Sparse Gaussian Signals) *Fix $1/2 < \beta < 1$, $r > 0$, and $\sigma > 0$ in the independent latent-mixture model with the rare/weak calibration and selection criteria of Section 9.1. Use the boundary ρ_{dis} in* (9.11)*. Allow S to depend jointly on all observations and on an independent random seed.*

(i) *If $r > \rho_{\text{dis}}(\beta; \sigma)$, there are selection rules, allowed to know (β, r, σ), with* $\text{mFDR}(S) \to 0$ *and* $\mathbb{P}(|S| \ge 1) \to 1$.

(ii) *If $r < \rho_{\text{dis}}(\beta; \sigma)$, every sequence of rules with $\mathbb{P}(|S| \ge 1) \to 1$ has* $\text{mFDR}(S) \to 1$.

No assertion is made at equality. In particular, for $\sigma = 1$ the boundary is $(1 - \sqrt{1 - \beta})^2$.

The converse applies to every joint selection rule, not just to threshold rules. The proof has two steps: identify scores with strong posterior evidence, then count how often signals produce them. Appendix B.7 evaluates the resulting optimization for each variance regime.

Proof Posterior evidence. Apply the independent-pair specialization in the posterior-selection calculation with $m = n$, $\varepsilon = \varepsilon_n$, and f_0, f_1 the $N(0,1)$ and $N(\mu_n, \sigma^2)$ densities. Retain its posterior probabilities q_i. Independence makes each q_i depend only on X_i, even though the selection decision may depend on every score. Put $D_i = \mathbf{1}_{\{i \in S\}}$ and condition on the data and the rule's independent random seed. Then

$$\mathbb{E}|S \cap I| = \mathbb{E}\sum_i D_i q_i, \qquad \mathbb{E}|S \cap N| = \mathbb{E}\sum_i D_i(1 - q_i). \tag{9.12}$$

For a score x, write $q(x)$ for this posterior probability with $X_i = x$. Its exact log-odds are

$$\log \frac{q(x)}{1 - q(x)} = -\beta \log n - \log\{\sigma(1 - \varepsilon_n)\} + \frac{x^2}{2} - \frac{(x - \mu_n)^2}{2\sigma^2}.$$

Writing $x = u\sqrt{2 \log n}$ with $u \geq 0$ therefore gives

$$\frac{\log\{q(x)/(1 - q(x))\}}{\log n} = G_\sigma(u; r) - \beta + O(1/\log n), \qquad G_\sigma(u; r) = u^2 - \frac{(u - \sqrt{r})^2}{\sigma^2}. \tag{9.13}$$

For fixed u, the posterior probability tends to one if $G_\sigma(u; r) > \beta$ and to zero if $G_\sigma(u; r) < \beta$. This identifies convincing scores. The next step asks how often they occur under the signal law.

The number of convincing signals. At $x = -u\sqrt{2 \log n}$, the second square becomes $(u + \sqrt{r})^2$. Reflection to $+u\sqrt{2 \log n}$ increases both the posterior odds and the signal density, so the negative half-line cannot improve the leading signal-count exponent. Under P_{sig}, for fixed $u > \sqrt{r}$ the normal tail obeys

$$\mathbb{P}_{\text{sig}}\{X \geq u\sqrt{2 \log n}\} = n^{-(u - \sqrt{r})^2/\sigma^2 + o(1)}.$$

The constraint $G_\sigma(u; r) \geq \beta$ retains scores whose likelihood evidence offsets the prior rarity penalty. Under the signal law, $X/\sqrt{2 \log n}$ concentrates at $\sqrt{r}$. The feasible score nearest this value has the smallest Gaussian probability exponent. Define

$$J_\sigma(r, \beta) = \inf_{\substack{u \geq 0 \\ G_\sigma(u; r) \geq \beta}} \frac{(u - \sqrt{r})^2}{\sigma^2}, \tag{9.14}$$

where the infimum is $+\infty$ if the feasible set is empty. The exponent J_σ measures how rare convincing scores are under the signal law. There are $n\varepsilon_n = n^{1-\beta}$ signals on average, so whenever the feasible set has nonempty interior, a fixed posterior cutoff $0 < d < 1$ satisfies

$$n\varepsilon_n \mathbb{P}_{\text{sig}}\{q(X) > d\} = n^{1 - \beta - J_\sigma(r, \beta) + o(1)}.$$

Indeed, $q(x) > d$ shifts the constraint $G_\sigma(u; r) \geq \beta$ by only $O(1/\log n)$; bounding the normal probability on either side of a minimizing feasible point gives the exponent.

For example, when $\sigma = 1$ and $0 < r < \beta$, the evidence constraint is $2u\sqrt{r} - r \geq \beta$. Its nearest feasible point to the typical signal scale is $u_* = (\beta + r)/(2\sqrt{r}) > \sqrt{r}$, giving

$$J_1(r, \beta) = \frac{(\beta - r)^2}{4r}.$$

The expected number of convincing signals grows polynomially when this exponent is smaller than $1 - \beta$. The smaller root of the equality is $(1 - \sqrt{1 - \beta})^2$, the discovery boundary. If $r \geq \beta$, then $J_1 = 0$: convincing scores are no longer polynomially rare under the signal law.

The two proof directions use different cutoffs. For impossibility, signals must rarely exceed any fixed posterior cutoff $d > 0$. For attainability, use $q(x) > 1 - 1/\log n$, making selected posterior probabilities tend to one. This shifts the evidence constraint by $O(\log\log n/\log n)$ and leaves the exponent unchanged when the feasible set has interior.

Attainability therefore requires a score scale with both strong posterior evidence and a diverging expected signal count:

$$G_\sigma(u; r) > \beta, \qquad \frac{(u - \sqrt{r})^2}{\sigma^2} < 1 - \beta.$$

Appendix B.7 solves these two inequalities and obtains exactly (9.11). Above that boundary, $J_\sigma < 1 - \beta$ and the feasible region has positive width. Below it, every fixed posterior cutoff has expected signal count $o(1)$. The appendix also checks the possible singleton feasible set when $\sigma < 1$, for which only a one-sided probability bound is needed.

Attainability and the converse. Above the boundary, for all sufficiently large n, select indices for which $q_i > 1 - 1/\log n$. The preceding exponent calculation gives $\mathbb{E}|S \cap I| \to \infty$. Under this coordinatewise rule the selected true-signal indicators are independent, and hence

$$\mathbb{P}(|S \cap I| = 0) = \{1 - \varepsilon_n \mathbb{P}_{\text{sig}}(q(X) > 1 - 1/\log n)\}^n \leq e^{-\mathbb{E}|S\cap I|} \longrightarrow 0.$$

Thus $\mathbb{P}(|S| \geq 1) \to 1$. By (9.12), $\mathrm{mFDR}(S) \leq 1/\log n \to 0$.

For the converse, we must allow joint rules, not just the coordinatewise rule above. Below the boundary, for every fixed $d \in (0, 1)$ the same feasible-set and Gaussian-tail bounds give $n\varepsilon_n \mathbb{P}_{\text{sig}}(q(X) > d) = o(1)$. For an arbitrary joint rule,

$$\mathbb{E}|S \cap I| = \mathbb{E}\sum_i D_i q_i \leq d\mathbb{E}|S| + n\varepsilon_n \mathbb{P}_{\text{sig}}(q(X) > d) = d\mathbb{E}|S| + o(1).$$

If $\mathbb{P}(|S| \geq 1) \to 1$, then $\mathbb{E}|S| \geq 1 - o(1)$, so $\mathbb{E}|S \cap I|/\mathbb{E}|S| \leq d + o(1)$. Since d is arbitrary, the ratio of expected true selections to expected total selections tends to zero and $\mathrm{mFDR}(S) \to 1$, proving (ii). □

For classification, convincing scores must be typical among signals, rather than merely occur somewhere in the sample. We require a vanishing expected missed fraction, not exact recovery of every label. Under the signal law, the leading posterior log-odds are $(r - \beta)\log n$, with smaller-order fluctuations for fixed σ. This gives Cai and Sun's variance-independent boundary [41, Supplementary Proposition 3] without the score-scale optimization needed for discovery.

Theorem 9.6 (Classification Boundary for Sparse Gaussian Signals) *Under the assumptions of Theorem 9.5, simultaneous* $\mathrm{mFDR}(S) \to 0$ *and* $\mathrm{MDR}(S) \to 0$ *are attainable by rules allowed to know* (β, r, σ) *if* $r > \beta$. *If* $r < \beta$, *every sequence with* $\mathrm{mFDR}(S) \to 0$ *has* $\mathrm{MDR}(S) \to 1$. *Thus the strict classification boundary is* $r = \beta$ *for every fixed* $\sigma > 0$; *the equality case is not asserted.*

Proof Under the signal law write $X_i = \mu_n + \sigma Z_i$, where $Z_i \sim N(0,1)$. The exact posterior log-odds calculation used above now becomes

$$\log \frac{q_i}{1-q_i} = (r-\beta)\log n + \mu_n \sigma Z_i + \frac{\sigma^2-1}{2} Z_i^2 - \log\{\sigma(1-\varepsilon_n)\} = (r-\beta)\log n + O_{P_{\mathrm{sig}}}(\sqrt{\log n}).$$

Here $O_{P_{\mathrm{sig}}}(\sqrt{\log n})$ means that the remainder divided by $\sqrt{\log n}$ is bounded in probability under the signal law. If $r > \beta$, then for all sufficiently large n select the indices for which $q_i > 1 - 1/\log n$. The cutoff has log-odds $\log(\log n - 1) = o(\log n)$, so its probability under P_{sig} tends to one and $\mathrm{MDR}(S) \to 0$; the conditional-expectation identity (9.12) gives $\mathrm{mFDR}(S) \le 1/\log n$.

If $r < \beta$, then for every fixed $d \in (0,1)$, $\mathbb{P}_{\mathrm{sig}}(q_i > d) \to 0$. For any rule, the same split as above yields

$$\mathbb{E}|S \cap I| \le d\mathbb{E}|S| + n\varepsilon_n \mathbb{P}_{\mathrm{sig}}(q_i > d).$$

Suppose $\mathrm{mFDR}(S) \to 0$. Since $\mathbb{E}|S \cap I| = (1 - \mathrm{mFDR}(S))\mathbb{E}|S|$, division by $n\varepsilon_n$ gives

$$\frac{\mathbb{E}|S \cap I|}{n\varepsilon_n} \le \frac{d}{1 - \mathrm{mFDR}(S)} \frac{\mathbb{E}|S \cap I|}{n\varepsilon_n} + o(1).$$

Taking $d < 1$ forces $\mathbb{E}|S \cap I|/(n\varepsilon_n) \to 0$, or $\mathrm{MDR}(S) \to 1$. This argument also permits joint and randomized selection. □

The three strict boundaries can now be compared. For $\sigma = 1$, the detection boundary $\rho^*(\beta)$ of Theorem 9.2 satisfies

$$\rho^*(\beta) \le \rho_{\mathrm{dis}}(\beta; 1) < \beta.$$

The first inequality is strict for $1/2 < \beta < 3/4$ and holds with equality for $3/4 \le \beta < 1$. Thus moderately sparse signals can be detectable without any reliable discovery; even where detection and discovery coincide, near-complete classification has the higher strict boundary $r = \beta$. All three comparisons use the same Gaussian law but different decisions.

Variance heterogeneity changes the evidence needed for these decisions. For $\sigma < 1$, a narrower signal density can make the likelihood ratio largest away from the extreme right tail, so ranking by X_i need not agree with ranking by posterior probability as it does at $\sigma = 1$ [41, Theorem 5]. For $\sigma > \sqrt{2}$ and $1/2 < \beta \le 1 - \sigma^{-2}$, both detection (9.5) and discovery (9.11) have zero boundaries, while the strict classification boundary remains $r = \beta$. These are comparisons within each fixed experiment; Figure 9.2 displays only $\sigma = 1$.

9.1.4 Screening: retaining nearly every signal

Screening is a first-stage decision: retain almost all signals while reducing the pool of nulls, without requiring that signals dominate the selected set. This is appropriate when subsequent

measurements are expensive and a smaller candidate pool can be examined more carefully. Unlike the expected missed-fraction criterion above, the screening requirements below must hold with probability tending to one.

In the same independent Gaussian latent-mixture model, fix $1/2 < \beta < 1$, $\sigma > 0$, and $0 < \kappa < 1/2$. The exponent κ specifies the desired reduction in null candidates. Require, for every fixed $\delta_{\mathrm{scr}} > 0$,

$$\mathbb{P}\{|S \cap I| \geq (1 - \delta_{\mathrm{scr}})|I|\} \longrightarrow 1, \qquad \mathbb{P}\{|S \cap N| \leq n^{-\kappa}|N|\} \longrightarrow 1.$$

Cai and Sun [41, Theorem 1] identify the strict boundary $r = \kappa$: screening is attainable above it and impossible below it, with no assertion here at equality. The required reduction in null candidates determines this boundary. Unlike discovery, screening controls the fraction of *all nulls* retained, not the composition of S; a useful screening set may contain many more nulls than signals.

The lower bound must convert a posterior bound on expected selections into a failure of screening with high probability. To do so, replace any oversized selected set by the empty set. A successful screening rule would exceed this size cap only with vanishing probability. Below the boundary, however, the capped rule captures a vanishing expected fraction of signals, which contradicts successful screening by Markov's inequality.

For the lower bound, allow also $r = 0$ by setting $\mu_n = 0$. Suppose $0 \leq r < \kappa$, put $m_n = n\varepsilon_n = n^{1-\beta}$ and $B_n = n^{1-\kappa}$, and choose a with $r < a < \kappa$. Under the signal law, the log-odds calculation above gives

$$\frac{\log\{q(X)/(1 - q(X))\}}{\log n} \longrightarrow r - \beta \quad \text{in probability.}$$

Since $q \leq q/(1-q)$, it follows that $\mathbb{P}_{\mathrm{sig}}\{q(X) > n^{a-\beta}\} \to 0$. For any joint selector, including an independent random seed, define $\widetilde{S} = S$ when $|S| \leq 2B_n$ and $\widetilde{S} = \varnothing$ otherwise. Splitting the conditional-expectation identity (9.12) at $n^{a-\beta}$ yields

$$\frac{\mathbb{E}|\widetilde{S} \cap I|}{m_n} \leq \frac{n^{a-\beta}\mathbb{E}|\widetilde{S}|}{m_n} + \mathbb{P}_{\mathrm{sig}}\{q(X) > n^{a-\beta}\} \leq 2n^{a-\kappa} + o(1) \longrightarrow 0.$$

The capped rule captures a vanishing expected fraction of signals. To finish the contradiction, we show that a successful screening rule would be left unchanged by the cap with high probability.

The latent signal count is binomial with mean m_n and variance at most m_n, so $\mathbb{P}\{\big||I| - m_n\big| \geq m_n/2\} \leq 4/m_n$. Eventually $3m_n/2 \leq B_n$, since $\kappa < \beta$. On the complementary count event, if $|S \cap I| \geq |I|/2$ and $|S \cap N| \leq n^{-\kappa}|N|$, then $|S| \leq |I| + B_n \leq 2B_n$ and $|\widetilde{S} \cap I| \geq m_n/4$. Consequently, Markov's inequality gives

$$\mathbb{P}\{|S \cap I| \geq |I|/2,\ |S \cap N| \leq n^{-\kappa}|N|\} \leq \frac{4}{m_n} + \frac{4\mathbb{E}|\widetilde{S} \cap I|}{m_n} \longrightarrow 0.$$

Successful screening would make this joint probability tend to one, a contradiction. The cap avoids needing a convergence rate for the original rule's failure probability.

Remark 9.7 (Procedure and Upper-Bound Comparison) Retain index i when $X_i > \sqrt{(r+\kappa)\log n}$. For fixed $r > \kappa$ and $\sigma > 0$, Gaussian tail bounds give both screening requirements with probability tending to one. Thus this rule attains the strict boundary of Cai and Sun [41, Theorem 1], although its threshold requires knowing r. Cai and Sun [41, Section 2] also develop data-driven cutoffs under additional estimation conditions.

Discovery and classification control false selections relative to the selected set, whereas screening controls the fraction of all nulls retained. The posterior rules above use known (β, r, σ); neither their guarantees nor the screening comparison establishes adaptation to unknown mixture parameters.

9.2 General sparse-mixture detection beyond Gaussian scores

Count data, heavy-tailed measurements, and watermark scores need not have Gaussian tails. A rare contamination may then be detectable even when the likelihood ratio has an infinite second moment. Our goal is to determine when the full observation laws permit vanishing total testing error. For n independent observations, consider

$$H_{0,n} : Y_i \overset{\text{i.i.d.}}{\sim} Q_n, \qquad H_{1,n} : Y_i \overset{\text{i.i.d.}}{\sim} M_n := (1-\varepsilon_n)Q_n + \varepsilon_n G_n, \qquad \varepsilon_n = n^{-\beta}, \quad \tfrac{1}{2} < \beta < 1. \tag{9.15}$$

Here Q_n is the specified null law and G_n the specified nonnull law; both may vary with n. The observed alternative is $M_n^{\otimes n}$ after its latent contamination indicators are integrated out. These are two simple hypotheses; testing with an unknown common null distribution is a different problem, taken up below for categorical data.

Cai and Wu [43] characterize detection boundaries through likelihood-ratio tails. We first extract the product-affinity argument used for heteroscedastic Gaussian detection, then apply it to an exponential example. Watermark models illustrate how the observed score and dependence assumptions affect the calculation. Across these models, moderate and extreme likelihood ratios require different bounds.

9.2.1 The product-affinity criterion

Independence reduces the testing problem to the Hellinger affinity of one observation. Its deficit from one has critical scale $1/n$: a deficit of $o(n^{-1})$ cannot accumulate into detectable evidence, whereas one asymptotically larger than $1/n$ permits vanishing total error. The distance comparisons of Section 1.3 make this precise for any pair of laws, not only sparse mixtures.

Proposition 9.8 (Product-Affinity Detection Criterion) *For each n, let P_n and Q_n be probability laws on one observation, both dominated by a common measure, and set $a_n = \int \sqrt{dP_n\, dQ_n} \in [0,1]$. For the simple hypotheses $P_n^{\otimes n}$ versus $Q_n^{\otimes n}$,*

$$1 - a_n^n \le \mathrm{TV}(P_n^{\otimes n}, Q_n^{\otimes n}) \le \sqrt{1 - a_n^{2n}}.$$

Consequently, if $n(1-a_n) \to 0$, every test has error sum at least $1 - o(1)$; if $n(1-a_n) \to \infty$, a likelihood-ratio test has error sum $o(1)$.

Proof Independence makes the product Hellinger affinity a_n^n. Lemma 1.5 gives $1 - a \le \mathrm{TV}$ for Hellinger affinity a; retaining the factor $2(1 + a)$ in its Cauchy–Schwarz proof gives $\mathrm{TV}^2 \le (1 - a)(1 + a)$. Apply both bounds with $a = a_n^n$.

For the limiting conclusions, if $n(1 - a_n) \to 0$, Bernoulli's inequality gives $a_n^n \ge 1 - n(1 - a_n) \to 1$; if $n(1 - a_n) \to \infty$, then $a_n^n \le e^{-n(1-a_n)} \to 0$. The minimum total error is $\eta(P_n^{\otimes n}, Q_n^{\otimes n}) = 1 - \mathrm{TV}(P_n^{\otimes n}, Q_n^{\otimes n})$ by Lemma 1.7. □

For the mixture (9.15), put $a_n = \rho(M_n, Q_n)$. If $n(1 - a_n)$ stays bounded above and away from zero, the proposition keeps the optimal error away from both zero and one but does not identify its limit.

It remains to compute the one-observation discrepancy. Suppose $G_n \ll Q_n$ and put $r_n = dG_n/dQ_n$, the nonnull-to-null likelihood ratio. Since $dM_n/dQ_n = 1 + \varepsilon_n(r_n - 1)$ and both measures have total mass one,

$$1 - a_n = \frac{1}{2}\mathbb{E}_{Q_n}\left[\left\{\sqrt{1 + \varepsilon_n(r_n - 1)} - 1\right\}^2\right]. \tag{9.16}$$

For $x \ge -1/2$, rationalizing the square root shows that $(\sqrt{1+x} - 1)^2 = x^2/(\sqrt{1+x} + 1)^2$ is, up to universal factors, $\min\{x^2, |x|\}$. In our sparse setting $x = \varepsilon_n(r_n - 1) \ge -\varepsilon_n$, so for all sufficiently large n,

$$1 - a_n \asymp \mathbb{E}_{Q_n} \min\left\{\varepsilon_n^2(r_n - 1)^2,\ \varepsilon_n|r_n - 1|\right\}.$$

The quadratic bound is useful for small likelihood deviations and the linear bound for large ones, with the transition at $\varepsilon_n|r_n - 1|$ of order one. Both bounds hold throughout the integration region, so we may choose a convenient splitting point. This controls the full Hellinger integral without requiring a finite likelihood-ratio second moment, extending the truncation idea of Section 2.2.

The exponential law $\mathrm{Exp}(\lambda)$ has density $\lambda e^{-\lambda x}\mathbf{1}_{\{x \ge 0\}}$ for rate $\lambda > 0$. For example, take $Q = \mathrm{Exp}(1)$ and $G = \mathrm{Exp}(\lambda)$ with $0 < \lambda < 1/2$. Let $\varepsilon_n = n^{-\beta}$ with fixed $\beta > 0$. The signal has a heavier right tail than the null. The strict detection boundary is $\beta = 1 - \lambda$, obtained by balancing signal rarity against this tail advantage.

For the lower bound, let $Y \sim Q$ and write $r(y) = \lambda e^{(1-\lambda)y}$. Its second moment is infinite, but its contribution below $t_n = \log n$ is finite. This threshold is the leading order of the largest null observation. On $\{Y \le t_n\}$ use the quadratic bound with $(r-1)^2 \le 2(r^2+1)$; on $\{Y > t_n\}$ use the linear bound with $|r - 1| \le r + 1$. Since

$$\mathbb{E}_Q[r(Y)^2\mathbf{1}_{\{Y \le t_n\}}] = \lambda^2 \int_0^{t_n} e^{(1-2\lambda)x}\,dx \le C_\lambda n^{1-2\lambda}, \qquad \mathbb{E}_Q[r(Y)\mathbf{1}_{\{Y > t_n\}}] = n^{-\lambda},$$

and $\mathbb{P}_Q(Y > t_n) = n^{-1}$, equation (9.16) gives

$$n(1 - a_n) \le C_\lambda\left\{n^{1-2\beta} + n^{2-2\lambda-2\beta} + n^{1-\beta-\lambda} + n^{-\beta}\right\} \longrightarrow 0 \quad \text{if } \beta > 1 - \lambda.$$

Proposition 9.8 proves indistinguishability. This is a split of a one-observation integral, not removal of all samples with an observation above $\log n$: that event has nonvanishing null probability. The calculation illustrates why a non-Gaussian tail changes the boundary.

Remark 9.9 (Optimal Procedure and Matching Upper Bound) For the exponential mixture, reject when the largest observation exceeds $(1+\zeta)\log n$, with fixed $\zeta > 0$. The expected null exceedance count is $n^{-\zeta}$, while the expected number of signals exceeding the threshold is $n^{1-\beta-\lambda(1+\zeta)}$. A union bound controls false rejection; independence makes missed detection vanish when the latter expectation diverges. For $\beta < 1-\lambda$, choose $0 < \zeta < (1-\beta-\lambda)/\lambda$. Both errors then vanish, matching the strict lower-bound boundary.

A general likelihood-tail boundary. The preceding example evaluates affinity directly. Cai and Wu [43, Theorem 1] express the same balance for a Gaussian null as an optimization over score scales. This connects the affinity criterion to the Gaussian boundaries of Section 9.1. Take $Q_n = N(0,1)$ and suppose G_n has Lebesgue density $f_{1,n}$. Write $r_n(x) = f_{1,n}(x)/\varphi(x)$ and $h_n(x) = \log r_n(x)$.

The variable u indexes scores of the form $u\sqrt{2\log n}$. The boundary balances their likelihood evidence against their rarity under the null. Under the uniform limiting condition

$$\frac{h_n(u\sqrt{2\log n})}{\log n} \longrightarrow \alpha(u) \quad \text{uniformly in } u \in \mathbb{R},$$

for a measurable function $\alpha : \mathbb{R} \to \mathbb{R}$ that is positive on a set of positive Lebesgue measure, the critical sparsity exponent is given below. The essential supremum is taken with respect to Lebesgue measure, so values on sets of measure zero do not affect it:

$$\beta^\sharp = \frac{1}{2} + \max\left\{0, \operatorname*{ess\,sup}_{u\in\mathbb{R}}\left[\alpha(u) - u^2 + \frac{u^2 \wedge 1}{2}\right]\right\}. \tag{9.17}$$

Here $\alpha(u)$ measures the log-likelihood ratio on the $\log n$ scale, while $-u^2$ records the decay of the null density. The correction $(u^2 \wedge 1)/2$ changes form at $|u| = 1$, the leading scale of the largest absolute observation in a null sample. Maximizing over u identifies the score scales that permit the sparsest detectable contamination.

The cited theorem gives reliable testing for $\beta < \beta^\sharp$ and total variation tending to zero between the joint null and alternative sample laws for $\beta > \beta^\sharp$. It makes no assertion at equality.

For fixed $r > 0$ and $G_n = N(\sqrt{2r\log n}, 1)$, one obtains $\alpha(u) = 2u\sqrt{r} - r$; the maximizing u is $2\sqrt{r}$ for $r \le 1/4$ and 1 for $1/4 \le r \le 1$, recovering (9.2). For $r \ge 1$, the maximizer is $u = \sqrt{r}$ and $\beta^\sharp = 1$. With fixed $\sigma > 0$ and $G_n = N(\sqrt{2r\log n}, \sigma^2)$, $\alpha(u) = u^2 - (u-\sqrt{r})^2/\sigma^2$ instead yields (9.5). These substitutions recover the Gaussian boundaries; a different score law requires its own likelihood-tail calculation.

9.2.2 Connection to detecting sparse language-model watermarks

An algorithmic watermark couples generated token choices with keyed pseudorandomness, giving a detector with the key a statistical signature to look for. A token is a vocabulary element, such as a word or part of a word; it is not visibly labeled as watermarked. Some schemes preserve its marginal distribution after averaging over the randomization. Editing or embedding generated passages in a larger document can leave only sparse evidence. Detection asks whether that evidence is present; discovery asks which passages or positions carry it.

We use total testing error for detection and the nonempty-selection and mFDR criteria of Section 9.1 for discovery. The models below differ in both the observations and the permitted selection rules:

Model	Observations	Scope of the comparison
Independent segments	Green-token counts	Detection and discovery; the one-bit discovery bound allows joint rules.
Growing vocabulary	Keyed token scores	Source detection boundary and a discovery boundary for local rules; joint procedures have separate guarantees.
Fixed vocabulary	Keyed token scores	A local-rule discovery bound, allowing dependence across positions.

The segment and fixed-vocabulary arguments are self-contained. The intervening source comparison explains why vocabulary growth and the joint score law matter.

Independent segments. In the green-list watermark of Kirchenbauer et al. [82], a secret-keyed pseudorandom partition identifies a fraction γ of vocabulary tokens as green at each position. Generation favors these tokens by increasing their logits, the scores whose exponentials are normalized into token probabilities. A green token can also occur without watermarking; the evidence is an excess of green choices, not the occurrence of a particular word. Their detector counts green tokens and standardizes the count against a null green fraction. Their later study considers short watermarked spans embedded in longer documents [83]. We replace the keyed-text mechanism by an explicit Bernoulli model below. In particular, its independent indicators and null probability γ are assumptions of the idealization, not consequences of a vocabulary fraction alone.

Consider n independent segments, each containing T binary green-token indicators, where $T \geq 1$ is an integer. Under the null, all indicators are independent with success probability $\gamma \in (0, 1)$. Under the alternative, each segment is independently marked with probability $\varepsilon \in (0, 1)$. Conditional on its status, the indicators within a segment are independent, with success probability $\gamma + \delta \in (0, 1)$ if marked and γ otherwise. Both T and the parameters may depend on n. The observed indicators in segment j can be reduced to their sufficient count S_j, $j = 1, \ldots, n$. Thus the count laws are

$$\begin{aligned} H_0 : &\quad S_j \sim \mathrm{Bin}(T, \gamma), \\ H_1 : &\quad S_j \sim (1 - \varepsilon)\,\mathrm{Bin}(T, \gamma) + \varepsilon\,\mathrm{Bin}(T, \gamma + \delta). \end{aligned}$$

This is (9.15) with a segment count as one observation, leaving ε and T general. The latent status can make indicators within a segment dependent after averaging over that status: the count law is a mixture of binomials, not generally $\mathrm{Bin}(T, \gamma + \varepsilon\delta)$.

Accordingly, the second-moment calculation has three steps: combine indicator likelihoods within a marked segment, average over its unknown status, and take the product over the n independent segments.

Proposition 9.10 (Exact Information Bound for an Idealized Sparse Watermark) *In the independent-segment model just specified, let W_0 and W_1 be the joint null and mixture laws. Then*

$$1 + \chi^2(W_1 \| W_0) = \left[1 + \varepsilon^2 \left\{\left(1 + \frac{\delta^2}{\gamma(1-\gamma)}\right)^T - 1\right\}\right]^n.$$

In particular, if $n\varepsilon^2\{(1+\delta^2/[\gamma(1-\gamma)])^T - 1\} \to 0$, every test has error sum at least $1 - o(1)$.

Proof For one Bernoulli observation G, the marked-to-null likelihood ratio equals $a(G) = [(\gamma+\delta)/\gamma]^G[(1-\gamma-\delta)/(1-\gamma)]^{1-G}$. Direct expansion gives

$$\mathbb{E}_0 a(G) = 1, \qquad \mathbb{E}_0 a(G)^2 = \frac{(\gamma+\delta)^2}{\gamma} + \frac{(1-\gamma-\delta)^2}{1-\gamma} = 1 + \frac{\delta^2}{\gamma(1-\gamma)}.$$

Write $G_1, \ldots, G_T$ for the indicators in one segment and $A = \prod_{t=1}^T a(G_t)$ for its marked-to-null likelihood ratio. Its second moment is therefore the Tth power of the preceding expression. Mixing the marked segment with null weight $1 - \varepsilon$ gives a one-segment likelihood ratio $1 - \varepsilon + \varepsilon A$, with second moment $1 + \varepsilon^2(\mathbb{E}_0 A^2 - 1)$. Independence across n segments proves the displayed identity. Under the stated condition its right side tends to one, so the χ^2–total-variation inequality yields the testing claim. □

Remark 9.11 (Procedure and Upper-Bound Comparison) For specified $(T, \gamma, \delta, \varepsilon)$, the likelihood-ratio test rejects when the product of segment-level mixture-to-null likelihood ratios exceeds one. It minimizes total error, and Proposition 9.8 gives its consistency criterion through one-segment affinity. The preceding second-moment bound is only a sufficient impossibility condition; its failure does not establish detectability.

For a positive watermark shift, a simpler test rejects when the total green count $U = \sum_{j=1}^n S_j$ exceeds the midpoint of its null and alternative means. Conditioning on segment status gives

$$\begin{aligned}\mathrm{Var}_0(U) &= nT\gamma(1-\gamma),\\ \mathrm{Var}_1(U) &= n\big\{(1-\varepsilon)T\gamma(1-\gamma) + \varepsilon T(\gamma+\delta)(1-\gamma-\delta)\\ &\qquad + \varepsilon(1-\varepsilon)T^2\delta^2\big\}.\end{aligned}$$

The last term accounts for the common status of indicators within a segment. Since the mean shift is $n\varepsilon T\delta$, Chebyshev's inequality gives vanishing errors when both variances are $o(n^2\varepsilon^2T^2\delta^2)$. This is a sufficient condition for the pooled test, not a matching detection boundary: pooling can miss a few strongly marked segments.

For comparison with the Gaussian model, the standardized count $Z_j = (S_j - T\gamma)/\sqrt{T\gamma(1-\gamma)}$ has null mean zero and marked mean $\delta\sqrt{T/[\gamma(1-\gamma)]}$. For $\varepsilon = n^{-\beta}$, this suggests $T\delta^2/[\gamma(1-\gamma)] \approx 2r\log n$. A sharp boundary still requires a discrete tail calculation, since the marked variance differs and a Gaussian approximation need not hold uniformly at $\sqrt{\log n}$ [43].

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Detection without reliable discovery. Within the same independent-segment model, set $T = 1$ and take $\varepsilon_n = n^{-b}$ with $0 < b < 1/2$, a less sparse regime than the earlier rare/weak calibration. An excess of green indicators can reveal watermarking in aggregate, although neither bit value reliably identifies a marked segment. We compare a pooled mean test with a posterior bound that applies even to selectors using all observed indicators jointly.

Proposition 9.12 (A Detectable Mixture with No Reliable Discoveries) *Fix $0 < \gamma < \gamma+\delta < 1$ and $b \in (0, 1/2)$. For each n, let $\varepsilon_n = n^{-b}$ and $\Theta_j \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(\varepsilon_n)$. Under the alternative, conditionally on these indicators, observe independent $G_j \sim \mathrm{Ber}(\gamma + \delta\Theta_j)$, $j = 1, \dots, n$; under the null, the G_j are independent $\mathrm{Ber}(\gamma)$. There are tests with total error tending to zero as $n \to \infty$.*

Let $D = (D_1, \dots, D_n) \in \{0,1\}^n$ be any selector measurable with respect to $(G_1, \dots, G_n)$ and, optionally, an independent random seed. Define $\mathrm{mFDR}(D)$ as in Section 9.1, with selected set $\{j : D_j = 1\}$, signal set $\{j : \Theta_j = 1\}$, and expectations under the joint alternative law. Whenever the selector reports a position with positive probability,

$$\mathrm{mFDR}(D) \geq 1 - C\varepsilon_n,$$

for a constant C depending only on γ, δ. Thus no sequence of selection rules can both select a position with probability tending to one and make $\mathrm{mFDR}(D) \to 0$.

Proof Under the mixture alternative the G_j are independent Bernoulli variables with success probability $\gamma + \varepsilon_n\delta$. The sum $U = \sum_j G_j$ has mean shift $n\varepsilon_n\delta$ from the null, while its variance under either law is at most $n/4$. Rejecting above the midpoint of the two means and applying Chebyshev's inequality makes each error at most $1/(n\varepsilon_n^2\delta^2) = \delta^{-2}n^{2b-1} \to 0$.

For discovery, independence of the coordinate pairs (Θ_j, G_j) implies $\mathbb{P}(\Theta_j = 1 \mid G_1, \dots, G_n) = \mathbb{P}(\Theta_j = 1 \mid G_j)$. Bayes' formula gives the exact bounds

$$\mathbb{P}(\Theta_j = 1 \mid G_j = 1) = \frac{\varepsilon_n(\gamma+\delta)}{\gamma + \varepsilon_n\delta} \leq \frac{\gamma+\delta}{\gamma}\,\varepsilon_n,$$
$$\mathbb{P}(\Theta_j = 1 \mid G_j = 0) = \frac{\varepsilon_n(1-\gamma-\delta)}{1-\gamma-\varepsilon_n\delta} \leq \varepsilon_n.$$

Thus even after observing all the data, each position has posterior signal probability at most $C\varepsilon_n$, where $C = (\gamma+\delta)/\gamma$. Conditioning on these data and an independent random seed gives $\mathbb{E}\sum_j \Theta_j D_j \leq C\varepsilon_n \mathbb{E}\sum_j D_j$. Subtracting the expected true selections from the expected total selections proves the displayed lower bound whenever its denominator is positive. A rule that makes no selections instead fails the discovery requirement directly. □

The bounded one-bit likelihood ratio explains how aggregate detection can succeed without reliable selection. Extending this conclusion to text generation requires a model for the joint law of tokens and keyed randomization; the independent-segment example alone does not provide such a boundary.

Comparison with a growing-vocabulary model. Token-level watermark detection uses scores that may depend on the preceding text, rather than independent segment counts. A keyed pivot combines an observed token with the watermark's keyed randomization to give

a score in $(0, 1)$ with a known null distribution. It is an observation, not the latent watermark status or its posterior probability. Li et al. [90] develop this framework for evaluating detection efficiency and constructing optimal detection rules.

Blanchet, Cai, Li, Liu, Long, and Su [10, Assumption 4.1 and the proof of Lemma B.3] study the keyed pivot $Y_t = U_{t,w_t}$, where w_t is the observed token and U_t is a fresh vector of keyed independent uniform values. Their fresh-randomness idealization gives the joint null law $\mathrm{Unif}(0,1)^{\otimes n}$. Here n counts token positions, and the preceding text determines the next-token probability vector and hence the alternative pivot law. The detection experiment observes $Y_{1:n}$.

Vocabulary and sparsity. Three exponents describe the difficulty: b measures sparsity, q concentration on a dominant token, and α growth in the number of light tokens. Using arXiv version 2, write $b \in [0, 1]$ for the source's exponent p, so the surviving fraction is $\varepsilon_n \asymp n^{-b}$. At position t, the next-token probability vector P_t assigns mass $1 - \Delta_n$ to a dominant token $w^*_{t,n}$, where $\Delta_n \asymp n^{-q}$ and $q \in [0, 1]$.

The remaining vocabulary consists of a light set $\mathcal{L}_{t,n}$ and a smaller core $C_{t,n}$. Their sizes and masses matter because rare pivot values can supply the evidence missing from the dominant token. For $\alpha \in (0, 1)$ and a fixed core exponent $\kappa_c < \alpha$, the source assumes, uniformly over positions,

$$|\mathcal{L}_{t,n}| \asymp n^{\alpha}, \qquad P_{t,w} \asymp n^{-(\alpha+q)} \quad (w \in \mathcal{L}_{t,n}),$$
$$|C_{t,n}| \asymp n^{\kappa_c}, \qquad c_C n^{-(\alpha+q)} \le P_{t,w} \le C_C n^{-(\kappa_c+q)} \quad (w \in C_{t,n}),$$

with fixed $0 < c_C \le C_C < \infty$. The source additionally assumes that the dominant-token probability is uniformly bounded away from zero when $q = 0$, and that ε_n is bounded away from one when $b = 0$.

Global detection. Under these vocabulary conditions and the source's generation assumptions, detection is possible when

$$\max\{b + q,\ 2b + q - \alpha\} < 1,$$

and impossible when the maximum is greater than one [10, Theorem 4.1]. This converse concerns the observed pivots $Y_{1:n}$, not the full token–randomization pairs; the equality boundary is not covered.

Greater sparsity or concentration makes detection harder; vocabulary-tail growth relaxes the second constraint.

Local discovery. To identify positions reliably, the source additionally assumes geometric mixing, meaning that dependence between well-separated positions decays geometrically, and considers deterministic local rules $D_t = \phi_{t,n}(Y_t)$. Each decision uses only the current pivot. Within this class, nonempty selection with probability tending to one and vanishing mFDR are attainable if $b < \alpha$ and $b + q < 1$, and impossible if $b \ge \alpha$ or $b + q > 1$ [10, Theorem 4.2]. The impossibility includes $b = \alpha > 0$, but does not cover joint selectors or the boundary $b < \alpha$, $b + q = 1$.

Joint selection. The same authors' threshold-selection procedure uses the full pivot sample to scan null-calibrated tails and estimate a proxy for their marginal false-discovery level [10, Algorithm 1 and Theorem 4.5]. Its achievability guarantee is separate from the local-rule boundary. It requires their Assumptions 4.1–4.3, the specified grid, target and slack scales, a

scan interval containing $b + q$ strictly inside $(0, 1)$, and the stated accuracy of the surviving-fraction estimator.

Repeated contexts. Reusing keyed pseudorandom values changes the joint law. Cai, Li, Long, Su, and Wen [26, Assumptions 3.1–3.2 and Proposition 3.1] use a product law conditional on the repetition partition. Applying the product calculations above requires that full conditional factorization and its component laws; pairwise independence or independence before conditioning on a data-dependent partition is insufficient.

Fixed vocabulary: a local discovery bound. Can a rule based only on the current pivot make reliable discoveries when the vocabulary is fixed? We show that its mFDR stays bounded away from zero if null positions retain positive probability. The argument compares selection probabilities under the signal and null laws of one pivot. It requires no independence across positions, but applies only to local rules, including independently randomized ones.

Let $\theta_t \in \{0, 1\}$ indicate watermark survival and $D_t \in \{0, 1\}$ selection by a local rule using only the current pivot Y_t and independent randomization. Conditional on survival, assume the pivot is a mixture of the laws defined below, indexed by probability vectors supported on at most a fixed d tokens; no core/light decomposition is required. Assume that, for some $\pi_{\min} > 0$, $\mathbb{P}(\theta_t = 0) \geq \pi_{\min}$ for all positions and all sufficiently large n. Conditional on $\theta_t = 0$, the pivot is uniform on $(0, 1)$ after averaging over contexts; uniformity conditional on the past is not required. For each such probability vector P, the signal pivot has density

$$f_{1,P}(y) = \sum_{w:P_w>0} y^{1/P_w - 1} \leq d, \qquad 0 < y < 1.$$

Indeed, each exponent is nonnegative, so every summand is at most one, and the density integrates to $\sum_w P_w = 1$. Mixtures over context-dependent next-token laws preserve the bound. Let a_t be the rule's selection probability under a uniform pivot, including its independent randomization. The rule may vary with t, but cannot use other pivots or their history. Integrating its selection probability against the signal density gives a conditional selection probability at most da_t; the marginal null probability gives an unconditional false-selection probability at least $\pi_{\min} a_t$.

Let the true-positive and false-positive counts be $\mathrm{TP} = \sum_{t=1}^{n} \theta_t D_t$ and $\mathrm{FP} = \sum_{t=1}^{n} (1-\theta_t) D_t$. The preceding bounds give $\mathbb{E}\mathrm{TP} \leq d \sum_t a_t$ and $\mathbb{E}\mathrm{FP} \geq \pi_{\min} \sum_t a_t$. Consequently, every such rule with a positive expected number of selections satisfies

$$\mathrm{mFDR} = \frac{\mathbb{E}\mathrm{FP}}{\mathbb{E}(\mathrm{TP} + \mathrm{FP})} \geq \frac{\pi_{\min}}{\pi_{\min} + d} > 0.$$

Thus bounded single-pivot evidence prevents reliable discovery for local rules, even with dependence across positions; the argument does not cover joint selectors.

Positive null mass is essential: when $b = 0$, the assumed order of the survival probability alone does not guarantee it. The result also cannot be obtained by setting $\alpha = 0$ in the growing-vocabulary theorem, whose nonempty core and restriction $\kappa_c < \alpha$ require $\alpha > 0$.

We next replace scattered contamination by a contiguous signal region, making its location and boundaries additional targets.

9.3 Segment detection, identification, and localization

A localized change in a sequence can indicate a genomic alteration or a sustained departure in a monitored process. Detecting such a change establishes its presence; identifying the affected region guides follow-up investigation; locating its boundaries determines its extent. Our goal is to understand the information required for each task. In Gaussian observations, we measure detection by total testing error, block identification by the probability of selecting the wrong candidate, and localization by expected endpoint distance. Figure 9.3 distinguishes these decisions.

Unlike the arbitrary coordinate supports in Chapter 5, a segment has contiguous active coordinates. For a single constant shift, we use a location mixture for detection, Fano's inequality for block identification, and a nearby pair for endpoint localization. We then hold the detection task fixed and vary the alternatives: candidate locations may overlap, the shift may have either sign, and several blocks may be active. This separates two issues: how the required decision determines the reduction, and how signal structure determines the information calculation.

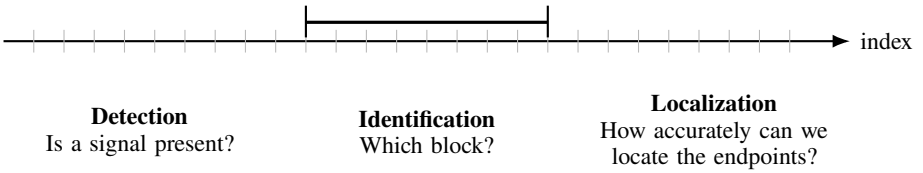


Figure 9.3 Three decision targets for a segment signal. Presence, candidate identity, and endpoint accuracy require different losses and often different least-favorable comparisons, even when the Gaussian observation model is unchanged.

9.3.1 Detection of a single segment

A sustained shift can be detectable even when its effect at each observation is small. In genomic screening or process monitoring, however, its location is usually unknown: a large block sum might reflect a signal or one of many noise fluctuations. We begin with specified disjoint candidate blocks to quantify this search cost without the additional dependence caused by overlapping candidates.

Fix integers $n, k \geq 1$ and $m \geq 2$ with $mk \leq n$. Let $B_1, \ldots, B_m \subseteq \{1, \ldots, n\}$ be known disjoint blocks, each consisting of k consecutive indices. Suppose we observe $Y \sim N_n(\theta, \sigma^2 I_n)$, with known $\sigma > 0$, and test $\theta = 0$ against $\theta = a\mathbf{1}_{B_j}$ for a fixed amplitude $a \in \mathbb{R}$ and an unknown j. The benchmark is the minimax total error: null rejection probability plus the largest miss probability over the candidate blocks. A uniform mixture over locations gives a lower bound by the comparison of Chapter 2.

The normalized block sums reduce this experiment to the Gaussian means of Section 5.4:

$$T_j = \frac{1}{\sigma\sqrt{k}} \sum_{i \in B_j} Y_i, \qquad \lambda = \frac{a\sqrt{k}}{\sigma}, \qquad j = 1, \ldots, m. \tag{9.18}$$

The scores are independent with variance one. Their means are zero under the null; under an alternative, only the active block has mean λ. Within-block residuals and observations

outside the candidate blocks are independent of the scores and have the same law under every hypothesis. Thus the full-data likelihood ratios depend only on the scores.

Theorem 9.13 (Disjoint-Block Mixture Bound) *Fix integers $n \geq 1$, $k \geq 1$, and $m \geq 2$ with $mk \leq n$, a noise level $\sigma > 0$, and an amplitude $a \in \mathbb{R}$. Let $B_1, \dots, B_m \subseteq \{1, \dots, n\}$ be disjoint blocks of size k, let $P_0 = N_n(0, \sigma^2 I_n)$ and $P_j = N_n(a\mathbf{1}_{B_j}, \sigma^2 I_n)$, and set $Q = m^{-1} \sum_{j=1}^m P_j$. Then*

$$\chi^2(Q\|P_0) = \frac{\exp(ka^2/\sigma^2) - 1}{m}.$$

Consequently every test ϕ obeys

$$\mathbb{P}_0\{\phi = 1\} + \frac{1}{m}\sum_{j=1}^m \mathbb{P}_j\{\phi = 0\} \geq 1 - \frac{1}{2}\sqrt{\frac{\exp(ka^2/\sigma^2) - 1}{m}}.$$

Proof By (9.18), the full-data likelihood ratio is $L_j = \exp(\lambda T_j - \lambda^2/2)$. Lemma A.10, applied to the normalized scores with covariance I_m, gives $\mathbb{E}_0 L_j L_\ell = 1$ for $j \neq \ell$ and $\mathbb{E}_0 L_j^2 = e^{\lambda^2}$. Hence

$$\mathbb{E}_0\left(\frac{1}{m}\sum_j L_j\right)^2 - 1 = \frac{me^{\lambda^2} + m(m-1)}{m^2} - 1 = \frac{e^{\lambda^2} - 1}{m}.$$

Substitute $\lambda^2 = ka^2/\sigma^2$ and apply the mixture testing bound (5.3). □

If $ka^2/\sigma^2 \leq \log(1 + m/4)$, the error bound is at least $3/4$. The worst-case miss probability is at least the mixture average, so this also bounds minimax total error. Persistence increases the squared signal-to-noise ratio through k, but searching among m locations requires logarithmic order in m. The same obstruction applies to any larger segment class containing these blocks.

Remark 9.14 (Optimal Procedure and Matching Upper Bound) For $a > 0$, reject when the largest normalized block sum exceeds $t_m = \sqrt{2(1+\zeta)\log m}$, with fixed $\zeta > 0$. The false-rejection and miss probabilities are at most $m^{-\zeta}$ and $\Phi(t_m - \lambda)$, respectively. Both vanish as $m \to \infty$ if $\lambda - t_m \to \infty$; in particular, $ka^2/\sigma^2 = C \log m$ suffices for fixed $C > 2(1+\zeta)$. This matches the logarithmic lower-bound order, not its sharp constant.

Under the same condition, selecting the block with the largest T_j identifies the active block with probability tending to one. Jeng, Cai, and Li [77] and Cai, Jeng, and Li [22] develop broader segment detection and identification methods. Guarantees for overlapping segments and multiple change points require separate calibration; see also Wang, Yu, and Rinaldo [110].

Varying signal profiles and sampling. The block sum is appropriate for a constant shift, but a varying signal can cancel within the segment. For a profile $f : [0,1] \to \mathbb{R}$ and a segment starting at ℓ, let the mean at index $\ell + i - 1$ be $f(i/k)$, $i = 1, \dots, k$, and zero outside. Cai and Yuan [44] study how knowledge of this profile changes detection in independent Gaussian noise. Their segment length k diverges but remains at most n^ξ for a fixed $\xi < 1$; the amplitude benchmark depends on the profile class.

For $f = Af_0$ with amplitude $A > 0$ and a fixed known shape f_0 satisfying $\int_0^1 f_0^2 = 1$ and $k^{-1}\sum_{i=1}^k f_0(i/k)^2 \to 1$, a scan of linear statistics weighted by the shape has detection rate $A \asymp \sqrt{\log n/k}$ [44, Theorem 1]. Here and below, the known noise standard deviation is fixed and suppressed in the constants.

For arbitrary profiles, use the sampled amplitude $A_k = \{k^{-1}\sum_{i=1}^k f(i/k)^2\}^{1/2}$. A scan based on sums of squared observations attains the rate $\sqrt{\log n/k} + (\log n/k)^{1/4}$ [44, Theorem 2]. Under the preceding length assumptions, this is the square-root rate when $k = O(\log n)$ and the fourth-root rate when $k/\log n \to \infty$.

For smooth profiles, Cai and Yuan [44, Section 4] also study an integral-amplitude benchmark. Using it for a sampled experiment requires control of discretization error at the claimed detection scale. Hölder smoothness alone does not ensure comparable sampled and integral energies. For fixed $c > 0$ and $0 < \nu \le 1$, consider $f_k(t) = ck^{-\nu}\sin(2\pi kt)$ on $[0,1]$, with integer $k \ge 1$. These functions have uniformly bounded ν-Hölder norms, vanish at every sampling point i/k, and yet have $\|f_k\|_2 = ck^{-\nu}/\sqrt{2}$. Indeed, for $t, u \in [0,1]$,

$$|f_k(t) - f_k(u)| \le ck^{-\nu}\min\{2, 2\pi k|t-u|\} \le 2\pi c|t-u|^\nu.$$

We therefore use sampled amplitude unless its comparability with integral amplitude is explicitly assumed.

9.3.2 Identifying the active block

An alarm alone does not tell an investigator where to look. In copy-number analysis, identifying a candidate genomic region directs subsequent biological investigation [77]. We now assume that one candidate block carries a signal and ask whether its identity can be recovered. This differs both from testing whether any signal exists and from estimating boundaries between candidate locations.

Retain the known disjoint blocks and fixed amplitude a and noise level $\sigma > 0$. Suppose we observe $Y \sim P_j = N_n(a\mathbf{1}_{B_j}, \sigma^2 I_n)$ for an unknown $j \in \{1,\dots,m\}$. A block estimator $\widehat{j} \in \{1,\dots,m\}$ is judged by its worst-case error probability $\max_j \mathbb{P}_j\{\widehat{j} \ne j\}$.

This maximum dominates the average error under a uniform block index. Fano's inequality therefore gives a lower bound by comparing the information in Y with the $\log m$ uncertainty about that index. The infimum below allows all measurable estimators, including independent randomization:

$$\inf_{\widehat{j}} \frac{1}{m}\sum_{j=1}^m \mathbb{P}_j\{\widehat{j} \ne j\} \ge 1 - \frac{ka^2/(2\sigma^2) + \log 2}{\log m}, \qquad m \ge 2. \tag{9.19}$$

Indeed, let J be uniform on $\{1,\dots,m\}$ and let $Y \mid J = j \sim P_j$. Its marginal law is the mixture Q in Theorem 9.13, and

$$I(J;Y) = \frac{1}{m}\sum_j \mathrm{KL}(P_j, Q) \le \frac{1}{m}\sum_j \mathrm{KL}(P_j, P_0) = \frac{ka^2}{2\sigma^2}.$$

The inequality is Proposition 3.9, and the last equality is Lemma A.8. For a randomized decoder, adjoin its independent seed U to the data. Since $I(J;Y,U) = I(J;Y)$, the decoding

bound in Theorem 3.10 gives (9.19). For large m, identification therefore requires information of order $\log m$. For positive shifts, a sufficiently large signal of this order permits the maximum-score rule in Remark 9.14 to identify the block; for negative shifts, reverse the scores. Detection and identification have the same logarithmic order in this model, but not the same reduction or necessarily the same sharp constant. A test distinguishes the null from a mixture, whereas a block estimator must distinguish its individual components.

For finer endpoint accuracy, nearby intervals become the relevant alternatives. Wang, Yu, and Rinaldo [110] study the related dependence of change-point localization on spacing, jump size, and noise.

9.3.3 Localizing the endpoints

Even after a candidate region is identified, its precise boundaries can matter: in copy-number analysis, they determine which genes lie within an altered region. A signal may be readily detectable because it extends over many observations, while its endpoints remain uncertain because moving a boundary changes only a few observations.

We isolate this uncertainty in a Gaussian model with one elevated segment of known length. A two-point argument bounds the expected maximum absolute endpoint error by comparing nearby shifts. Only observations entering or leaving the segment distinguish them. Thus endpoint uncertainty persists even when the segment's length is known.

Fix integers $n \geq 1$ and $1 \leq k \leq n$, a positive amplitude a, and a noise level $\sigma > 0$. Suppose we observe $Y \sim P_\ell = N_n(a\mathbf{1}_{I_\ell}, \sigma^2 I_n)$, where $I_\ell = \{\ell, \dots, \ell + k - 1\}$ and the start $\ell \in \{1, \dots, n - k + 1\}$ is unknown. An endpoint estimator reports a pair $(\widehat{L}, \widehat{R})$; it need not be constrained to have length k. We measure its error by

$$D\big((\widehat{L}, \widehat{R}), (\ell, \ell + k - 1)\big) = \max\{|\widehat{L} - \ell|, |\widehat{R} - (\ell + k - 1)|\}.$$

Compare starts ℓ and $\ell + r$, where the positive integer $r \leq k$ allows both intervals to fit in the observation range. Their endpoint pairs are distance r apart. Choosing r so that their KL divergence stays small forces expected error of that order.

Theorem 9.15 (A Finite-Sample Endpoint Obstruction) *In the Gaussian length-k interval experiment above, with endpoint loss D as defined there, let r and ℓ be integers with $1 \leq r \leq k$ and $1 \leq \ell \leq n - k + 1 - r$. Suppose $ra^2/\sigma^2 \leq 1/2$. Then every endpoint estimator satisfies*

$$\max_{u \in \{\ell, \ell + r\}} \mathbb{E}_{P_u} D\big((\widehat{L}, \widehat{R}), (u, u + k - 1)\big) \geq \frac{r}{4}.$$

Consequently the same lower bound holds for the minimax expected endpoint error over all length-k intervals whenever such an ℓ and r exist.

Proof The two mean vectors differ on precisely $2r$ coordinates: r leave the segment at the left and r enter at the right. The Gaussian shift identity gives

$$\mathrm{KL}(P_\ell, P_{\ell+r}) = \frac{\|a\mathbf{1}_{I_\ell} - a\mathbf{1}_{I_{\ell+r}}\|_2^2}{2\sigma^2} = \frac{ra^2}{\sigma^2} \leq \frac{1}{2}.$$

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Pinsker's inequality yields $\mathrm{TV}(P_\ell, P_{\ell+r}) \le 1/2$. Write $\vartheta_u = (u, u+k-1)$ and $\widehat{\vartheta} = (\widehat{L}, \widehat{R})$. The two true endpoint pairs have distance r under D. For every reported pair x, the triangle inequality gives $D(x, \vartheta_\ell) + D(x, \vartheta_{\ell+r}) \ge r$, with equality at $x = \vartheta_\ell$. Thus the loss-separation constant in Lemma 2.1 is $r/2$. Applying that lemma gives

$$\max_{u \in \{\ell, \ell+r\}} \mathbb{E}_{P_u} D(\widehat{\vartheta}, \vartheta_u) \ge \frac{r}{2}\, \eta(P_\ell, P_{\ell+r}) \ge \frac{r}{4}.$$

An independent random seed preserves total variation, so the argument also covers randomized estimators. □

The KL divergence depends on displacement r, not the full segment length k: observations in the common interior do not distinguish the endpoints. When an admissible displacement has order σ^2/a^2, the expected-error lower bound has that order. A long segment can therefore be detectable while its boundaries remain uncertain.

The finite observation range and integer-valued endpoints constrain this conclusion. We need $1 \le r \le \min\{k, n-k, \sigma^2/(2a^2)\}$ and $1 \le \ell \le n-k+1-r$. Short intervals or limited room for displacement can reduce the bound; no such integer exists when $\sigma^2/a^2 < 2$. In that stronger-signal regime an exponential testing calculation can give a different bound. The theorem is therefore a local obstruction, not a uniform minimax rate over all parameter regimes.

Remark 9.16 (Procedure and Upper-Bound Comparison) In this known-length, positive-signal experiment, the scan selects the interval with the largest sum:

$$\widehat{\ell} \in \operatorname*{arg\,max}_{1 \le j \le n-k+1} \sum_{i \in I_j} Y_i.$$

With ties broken by a fixed rule, it reports $(\widehat{\ell}, \widehat{\ell}+k-1)$ and has endpoint error $|\widehat{\ell} - \ell|$. The scan uses the known length and positive sign, but not the amplitude.

With $\rho = a^2/\sigma^2$ and $M = n-k+1$, a Gaussian comparison gives, uniformly over the true start ℓ and integers $1 \le r \le k$,

$$\mathbb{P}_\ell\{|\widehat{\ell} - \ell| \ge r\} \le 2 \sum_{h=r}^{k-1} e^{-\rho h/4} + M e^{-\rho k/4},$$

where an empty sum is zero. The first term controls overlapping competitors; the second controls up to M intervals disjoint from the true segment. Thus both boundary uncertainty and the global search affect accuracy.

In particular, the endpoint error is smaller than $r = r_n \le k$ with probability tending to one if

$$M e^{-\rho k/4} \longrightarrow 0, \qquad \frac{e^{-\rho r/4}}{1 - e^{-\rho/4}} \longrightarrow 0.$$

These conditions ensure localization in probability, not a matching bound for expected endpoint loss. The union bound can also be conservative because candidate intervals overlap.

9.3.4 Overlapping locations and unknown signs

Real shifts need not align with a preassigned partition, and monitoring must often detect decreases as well as increases. We return to total-error testing in the same Gaussian observation model and examine these uncertainties separately. First, allowing all length-k intervals introduces overlap between candidate signals. Second, on disjoint blocks, allowing either sign changes the mixture even though the locations stay fixed. Neither test is required to identify the active block or estimate its endpoints.

Overlapping locations. In the interval model of Section 9.3.3, suppose we observe $Y \sim N_n(\theta, \sigma^2 I_n)$ and test $\theta = 0$ against $\theta = a\mathbf{1}_{I_\ell}$ over all admissible starts ℓ. Under a uniform location prior, shared signal coordinates increase likelihood cross-moments above the disjoint-block value of one. The aim is to identify this additional contribution explicitly.

Proposition 9.17 (The Overlap Kernel) *In the Gaussian interval experiment of Section 9.3.3, put $M = n-k+1$ and $\rho = a^2/\sigma^2$, let $P_0 = N_n(0, \sigma^2 I_n)$, and set $Q = M^{-1}\sum_{\ell=1}^{M} P_\ell$. Then*

$$\chi^2(Q\|P_0) = \frac{1}{M^2}\left[M(e^{\rho k}-1) + 2\sum_{h=1}^{\min(k-1,M-1)} (M-h)\big(e^{\rho(k-h)}-1\big)\right].$$

Proof With $L_\ell = dP_\ell/dP_0$, Lemma A.10, applied with covariance $\sigma^2 I_n$, gives

$$\mathbb{E}_0 L_\ell L_j = \exp\left(\frac{a^2}{\sigma^2}|I_\ell \cap I_j|\right).$$

The overlap is $(k - |\ell - j|)_+$. There are M ordered pairs with distance zero and $2(M-h)$ ordered pairs with distance $h > 0$. Expand $\mathbb{E}_0(M^{-1}\sum_\ell L_\ell)^2 - 1$ and collect these terms. □

At $k = 1$ this recovers the disjoint formula. For longer intervals, nearby candidates overlap in their shifted coordinates, so their likelihood cross-moments exceed one. Adding candidate locations therefore increases both the mixture size and these overlap contributions. Restricting to disjoint intervals removes the latter at the cost of fewer alternatives.

Unknown signs. To isolate uncertainty about the direction of change, return to the disjoint blocks and observe $Y \sim N_n(\theta, \sigma^2 I_n)$ under the null $\theta = 0$ or alternatives $\theta = \pm a\mathbf{1}_{B_j}$, with $a > 0$. The normalized scores (9.18) have one nonzero mean equal to $+\lambda$ or $-\lambda$, so the signed sparse-mean calculation in Proposition 5.3 applies with support size one. The exact identity shows what averaging over signs adds to the location-only construction.

Proposition 9.18 (Signed Disjoint-Block Mixture) *Let n, k, m be positive integers with $mk \le n$, let $B_1, \dots, B_m \subseteq \{1, \dots, n\}$ be disjoint blocks of size k, and let $a > 0$ and $\sigma > 0$. Put $P_0 = N_n(0, \sigma^2 I_n)$, $P_{j,s} = N_n(sa\mathbf{1}_{B_j}, \sigma^2 I_n)$ for $j = 1, \dots, m$ and $s \in \{-1, 1\}$, and $Q_\pm = (2m)^{-1}\sum_{j,s} P_{j,s}$. Then*

$$\chi^2(Q_\pm\|P_0) = \frac{\cosh(ka^2/\sigma^2) - 1}{m}.$$

Every test between P_0 and this signed mixture has type-I error plus mixture-averaged type-II error at least

$$1 - \frac{1}{2}\sqrt{\frac{\cosh(ka^2/\sigma^2) - 1}{m}}.$$

Proof Apply the sign-averaging identity (5.7), which holds for any amplitude, to the normalized scores with dimension m, support size one, noise variance one, and amplitude $\lambda = a\sqrt{k}/\sigma$. Two independent block choices coincide with probability $1/m$. Thus

$$1 + \chi^2(Q_\pm \| P_0) = 1 - \frac{1}{m} + \frac{1}{m}\cosh(\lambda^2).$$

The score reduction preserves the full-data likelihood ratios. Subtract one and apply (5.3). □

Remark 9.19 (Optimal Procedure and Matching Upper Bound) For an unknown sign, use $\max_j |T_j|$ in the test of Remark 9.14, with the same threshold $t_m = \sqrt{2(1+\zeta)\log m}$ and fixed $\zeta > 0$. The type-I error is at most $2m^{-\zeta}$ and the miss probability at most $\Phi(t_m - |\lambda|)$, uniformly over signs and blocks. As $m \to \infty$, both vanish if $|\lambda| - t_m \to \infty$. Thus sufficiently large and small constant multiples of $\log m$ for $\lambda^2 = ka^2/\sigma^2$ separate detectability from indistinguishability. Unknown sign leaves this rate unchanged; the bounds do not determine its sharp constant.

For small $x = ka^2/\sigma^2$, the signed numerator $\cosh(x) - 1 = x^2/2 + O(x^4)$ is smaller than the unsigned numerator $e^x - 1 = x + O(x^2)$. At logarithmic signal scales both grow exponentially. Sign randomization can therefore improve a local information bound without changing the detection-rate order.

For block identification, positive and negative shifts at the same location have the same target label and cannot be counted as two distinct Fano targets. Detection instead averages over both as alternatives to the null. We retain this detection goal in the final extension, which allows several blocks to carry signal simultaneously.

9.3.5 Detecting several active blocks

Several genomic regions may be altered, or a monitored process may undergo repeated shifts. Evidence can accumulate even when no individual region is easy to identify. Retain the m disjoint blocks of size k and the observation model $Y \sim N_n(\theta, \sigma^2 I_n)$. We test $\theta = 0$ against exactly s active blocks, each with a common known shift a, where $1 \le s \le m$. The goal is detection under total testing error, not recovery of the active blocks.

The normalized scores (9.18) have s means equal to $\lambda = a\sqrt{k}/\sigma$ and all other means zero. Thus blocks replace individual coordinates in a sparse Gaussian mean problem.

Under a uniform support prior, the likelihood cross-moment depends on how many blocks two independent supports share. The common specified sign gives an exponential factor rather than the hyperbolic cosine arising from sign averaging. We can therefore use the support-overlap bound (5.8), with squared signal-to-noise ratio $\lambda^2 = ka^2/\sigma^2$ per block rather than $\rho = a^2/\sigma^2$ per coordinate.

Theorem 9.20 (A Sparse Multi-Block Detection Bound) *Let n, k, m be positive integers with $mk \le n$, let $B_1, \dots, B_m \subseteq \{1, \dots, n\}$ be disjoint blocks of k coordinates, let s be an integer with $1 \le s \le m$, and let $a \in \mathbb{R}$ and $\sigma > 0$. For $S \subseteq \{1, \dots, m\}$ with $|S| = s$, define*

$$P_S = N_n\left(a \sum_{j \in S} \mathbf{1}_{B_j}, \sigma^2 I_n\right),$$

with $P_0 = N_n(0, \sigma^2 I_n)$. Let Q be the uniform mixture of P_S over all s-subsets, and put $\lambda = a\sqrt{k}/\sigma$, so $\lambda^2 = ka^2/\sigma^2$. If S, T are independent uniform s-subsets of $\{1, \dots, m\}$, then

$$1 + \chi^2(Q \| P_0) = \mathbb{E}_{S,T} e^{\lambda^2 |S \cap T|} \le \exp\left\{\frac{s^2}{m}(e^{\lambda^2} - 1)\right\}.$$

Hence every test has null type-I error plus support-averaged type-II error at least

$$1 - \frac{1}{2}\sqrt{\exp\left\{\frac{s^2}{m}(e^{\lambda^2} - 1)\right\} - 1}.$$

Proof Write $L_S = dP_S/dP_0$. The likelihood ratio L_S equals that of the normalized block scores, whose mean vector is $\lambda \mathbf{1}_S$. Lemma A.10, with covariance I_m, gives

$$\mathbb{E}_0 L_S L_T = \exp\{\lambda^2 |S \cap T|\}.$$

Averaging over S, T proves the equality. Apply (5.8) with dimension m and $z = e^{\lambda^2}$ to obtain

$$\mathbb{E} e^{\lambda^2 |S \cap T|} \le \left\{1 + \frac{s}{m}(e^{\lambda^2} - 1)\right\}^s \le \exp\left\{\frac{s^2}{m}(e^{\lambda^2} - 1)\right\}.$$

The risk bound follows from (5.3). □

The factor s^2/m is the expected overlap of two randomly chosen supports. When $(s^2/m)(e^{\lambda^2} - 1) \to 0$, even the combined signal cannot be detected reliably. This transfers the support-mixture technique of Chapter 5 to block signals. Exact support recovery would instead require distinguishing $\binom{m}{s}$ possibilities, as in Chapter 3.

Remark 9.21 (Procedure and Upper-Bound Comparison) For known s and a, the mixture likelihood-ratio test is minimax: it minimizes average error under the uniform support prior, and permutation symmetry makes its miss probability the same for every support. The preceding chi-square bound need not be sharp in every regime.

For $a > 0$, the standardized sum $m^{-1/2} \sum_j T_j$ has variance one and means zero and $s\lambda/\sqrt{m}$ under the null and every alternative. A midpoint threshold gives vanishing total error when $s\lambda/\sqrt{m} \to \infty$. For $\lambda \to 0$, this matches the transition scale of the lower bound, which tends to one when $s^2\lambda^2/m \to 0$. For stronger individual signals, the maximum test has vanishing total error when $m \to \infty$ and $\lambda - \sqrt{2(1+\zeta)\log m} \to \infty$, with fixed $\zeta > 0$.

These simple comparisons do not settle the intermediate regimes. For related higher-criticism methods and sharp boundaries, see Donoho and Jin [56, Theorems 1.1–1.2]; their independent sparse-mixture model differs from the fixed number s of active blocks here.

The segment constructions use Gaussian shifts. We now return to detection with categorical observations, where perturbations must preserve probability mass and the common null distribution is unknown.

9.4 Detecting heterogeneity in multinomial populations

Observed word frequencies differ from document to document, even when all documents share the same underlying vocabulary distribution. Before estimating topics or clustering documents, it is therefore natural to ask whether the differences exceed sampling variation. Cai, Ke, and Turner [24] study this question within a general theory of testing equality of group-wise mean multinomial distributions. We focus on their special case in which each document forms its own group, with equal document lengths.

The testing goal resembles Gaussian signal detection, but the observation model changes the construction. Category probabilities must be nonnegative and sum to one; count coordinates are dependent because their total is fixed; and their variances depend on the unknown probabilities. The null distribution is common but unknown, rather than a specified zero-mean Gaussian law. We construct alternatives that differ across documents while leaving the average distribution unchanged. Randomizing both the category-probability contrast and its allocation across documents makes this heterogeneity difficult to detect.

9.4.1 The homogeneity problem and its benchmark

A collection of short documents can provide evidence of heterogeneity even when each document's empirical frequencies are noisy. To study when detection is possible, suppose we observe independent word-count vectors $X_i = (X_{i1}, \dots, X_{id})$ with

$$X_i \sim \text{Multinomial}(N, \Omega_i), \qquad i = 1, \dots, m, \tag{9.20}$$

where $m, d \geq 2$ and $N \geq 1$ are integers. Here X_{ij} counts occurrences of category j in document i, and $\Omega_i = (\Omega_{i1}, \dots, \Omega_{id})$ is its unknown probability vector. Each document contains N tokens. The multinomial model assumes independent token draws within each document; it does not model linguistic dependence. Put

$$\Delta_d = \{p \in [0,1]^d : \textstyle\sum_{j=1}^d p_j = 1\}, \qquad \mathcal{P}_{\text{flat}}(d) = \{p \in \Delta_d : \max_{1 \leq j \leq d} p_j \leq 2/d\}.$$

We restrict each Ω_i to this class so that no category dominates. Let $\mathbf{\Omega} = (\Omega_1, \dots, \Omega_m)$ and define

$$\bar{\Omega} = \frac{1}{m} \sum_{i=1}^m \Omega_i, \qquad D(\mathbf{\Omega}) = N \sum_{i=1}^m \|\Omega_i - \bar{\Omega}\|_2^2. \tag{9.21}$$

The discrepancy D is zero precisely when all document distributions agree. It weights between-document squared variation by the common document length, giving the equal-length form of Cai, Ke, and Turner's signal measure [24].

For $0 < \varepsilon < 1$, test

$$H_0 : \Omega_1 = \cdots = \Omega_m = p, \quad p \in \mathcal{P}_{\text{flat}}(d),$$
$$H_1(\varepsilon) : \Omega_i \in \mathcal{P}_{\text{flat}}(d) \text{ for all } i, \quad D(\mathbf{\Omega}) \geq \frac{mN\varepsilon^2}{d}.$$

Thus ε^2 is a lower bound on the normalized average squared departure $(d/m)\sum_i \|\Omega_i - \bar{\Omega}\|_2^2$. Write $\mathbb{P}_{\mathbf{\Omega}}$ for probability under the joint law in (9.20) and use the total-error benchmark

$$\mathcal{R}_{m,N,d}(\varepsilon) = \inf_{\phi} \left\{ \sup_{\mathbf{\Omega}\in H_0} \mathbb{P}_{\mathbf{\Omega}}\{\phi = 1\} + \sup_{\mathbf{\Omega}\in H_1(\varepsilon)} \mathbb{P}_{\mathbf{\Omega}}\{\phi = 0\} \right\}. \tag{9.22}$$

The infimum ranges over $\{0,1\}$-valued tests ϕ based on the count vectors, with independent randomization allowed; $\phi = 1$ means rejection. We seek the separation scale below which no test can make both worst-case errors small.

Unlike the independent Gaussian coordinates in Section 5.4, the empirical frequencies satisfy

$$\mathrm{Cov}_{\mathbf{\Omega}}(X_i/N) = \frac{1}{N}\{\mathrm{diag}(\Omega_i) - \Omega_i\Omega_i^{\top}\}.$$

Both the variances and the off-diagonal covariances depend on the unknown probabilities. Replacing the counts by independent Gaussian coordinates with a fixed noise level would discard these features. We instead use the exact multinomial likelihood. The lower bound needs only one admissible null, whereas a matching test must be valid for every common distribution allowed under H_0.

9.4.2 A two-topic mixture and the detection lower bound

We now construct documents with two distinct vocabulary distributions whose average is uniform. The test must detect their presence without knowing the vocabulary contrast or which distribution each document follows. This is a balanced version of the two-layer sign construction used for the lower bound in Cai, Ke, and Turner [24, Theorem 5]; their topic-model application shows that a two-topic family already creates the obstruction [24, Corollary 2].

Let m and d be even. Choose independent uniform vectors $u \in \{-1,1\}^{d/2}$ and $v \in \{-1,1\}^{m/2}$. Pair each coordinate with its negative to form balanced vectors $b \in \{-1,1\}^d$ and $z \in \{-1,1\}^m$:

$$b_{2j-1} = u_j, \quad b_{2j} = -u_j, \qquad z_{2h-1} = v_h, \quad z_{2h} = -v_h.$$

Define the document distributions by

$$\Omega_{ij}^{(u,v)} = \frac{1 + \varepsilon z_i b_j}{d}, \qquad i = 1,\dots,m, \quad j = 1,\dots,d. \tag{9.23}$$

Pairing categories preserves each row sum; pairing documents preserves the average distribution $p_0 = (1/d,\dots,1/d)$. Every entry lies between $(1-\varepsilon)/d$ and $(1+\varepsilon)/d$, so each row belongs to $\mathcal{P}_{\mathrm{flat}}(d)$, and

$$\bar{\mathbf{\Omega}} = p_0, \qquad D(\mathbf{\Omega}^{(u,v)}) = \frac{mN\varepsilon^2}{d}.$$

There are just two distinct row distributions, $A_{\pm} = (\mathbf{1} \pm \varepsilon b)/d$, each used by half the documents. In topic-model terminology, these are pure-topic documents: each draws all its tokens from one of the two topic distributions. The test is not given the topic assignments.

The shared vector b keeps the construction within a two-topic family; independent category patterns for different document pairs could require more topics. We average over both

b and the unknown assignments z. Unlike the Gaussian mean calculation, the likelihood cross-moment therefore involves two overlaps, one for the category patterns and one for the assignments.

Theorem 9.22 (A Multinomial Heterogeneity Lower Bound) *In* (9.20)–(9.22)*, let* $m, d \geq 2$ *be even,* $N \geq 1$*, and* $0 < \varepsilon < 1$*. Put* $b_0 = mN^2\varepsilon^4/d$*. If* $b_0 < 1/4$*, then*

$$\mathcal{R}_{m,N,d}(\varepsilon) \geq 1 - \frac{1}{2}\sqrt{(1-4b_0)^{-1/2} - 1}.$$

In particular, $b_0 \leq 1/16$ *implies* $\mathcal{R}_{m,N,d}(\varepsilon) \geq 3/4$*. The same lower bounds hold when the alternative is restricted to the two-topic family* (9.23)*.*

Proof Let P_0 be the law with $\Omega_i = p_0$ for every document, let $P_{u,v}$ be the law under (9.23), and let $Q = \mathbb{E}_{u,v} P_{u,v}$. The construction already verifies class membership and separation. We bound the mixture likelihood's second moment by averaging first over document signs and then over category signs.

For two independent prior draws (u, v) and (u', v'), write

$$U = \sum_{j=1}^{d/2} u_j u'_j, \qquad V = \sum_{h=1}^{m/2} v_h v'_h.$$

For one token in document i, the likelihood cross-moment is

$$\sum_{j=1}^{d} \frac{\Omega_{ij}^{(u,v)} \Omega_{ij}^{(u',v')}}{(p_0)_j} = 1 + \frac{2\varepsilon^2}{d} z_i z'_i U.$$

The linear terms vanish because both category-sign vectors sum to zero; their inner product is $2U$. The two documents in pair h have the same product $z_i z'_i = v_h v'_h$. For fixed prior draws, independence of tokens and documents therefore gives, with $L_{u,v} = dP_{u,v}/dP_0$,

$$\mathbb{E}_0 L_{u,v} L_{u',v'} = \prod_{h=1}^{m/2} \left(1 + \frac{2\varepsilon^2}{d} v_h v'_h U\right)^{2N} \leq \exp(\kappa U V), \qquad \kappa = \frac{4N\varepsilon^2}{d}.$$

Every base is at least $1 - \varepsilon^2 > 0$, so $1 + x \leq e^x$ applies. The likelihood ratio depends only on category counts; calculating with ordered tokens thus gives the same cross-moment as the multinomial experiment.

Now average over the document signs. Conditional on U, the products $v_h v'_h$ are independent uniform signs. Hence

$$1 + \chi^2(Q \| P_0) \leq \mathbb{E}_U \{\cosh(\kappa U)\}^{m/2} \leq \mathbb{E}_U e^{tU^2}, \qquad t = \frac{m\kappa^2}{4} = \frac{4mN^2\varepsilon^4}{d^2},$$

where $\log\cosh x \leq x^2/2$. This first average removes V but leaves a quadratic function of the category overlap U.

To average over the category signs, introduce an independent $G \sim N(0,1)$. This turns the square in the exponent into a linear function of U, allowing the expectation to factor over the

independent signs. The Gaussian moment identity $e^{tU^2} = \mathbb{E}_G e^{\sqrt{2t}\,GU}$ and the independence of the $d/2$ signs in U yield

$$\mathbb{E}_U e^{tU^2} = \mathbb{E}_G\{\cosh(\sqrt{2t}\,G)\}^{d/2} \le \mathbb{E}_G e^{dtG^2/2} = (1-dt)^{-1/2} = (1-4b_0)^{-1/2}.$$

The last integral is finite because $dt = 4b_0 < 1$. This auxiliary Gaussian variable evaluates a moment bound; it does not approximate the multinomial observations.

Every component of Q lies in the two-topic alternative and P_0 lies in the null. The mixture testing bound (5.3) now gives

$$\mathcal{R}_{m,N,d}(\varepsilon) \ge 1 - \tfrac{1}{2}\sqrt{\chi^2(Q\|P_0)}.$$

Substitution proves the result, including the restricted-alternative statement. If $b_0 \le 1/16$, then $(1-4b_0)^{-1/2} - 1 \le 2/\sqrt{3} - 1 < 1/4$, giving the asserted $3/4$ bound. □

The lower bound tends to one when $N\varepsilon^2\sqrt{m/d} \to 0$. With N fixed, more documents increase this signal scale; with m fixed, longer documents do so. A larger vocabulary spreads the contrast across more categories. At fixed total token count $B = mN$, however, $b_0 = B^2\varepsilon^4/(md)$. Dividing the same total among more documents decreases b_0 and raises the displayed error lower bound: each document supplies fewer observations about its unknown topic assignment.

This is a detection result, not a bound for learning the topics or assigning documents to them. Even two topics can be statistically indistinguishable from a common distribution at the stated scale. The proof combines the mixture method of Chapter 2 with a structured alternative: one set of signs describes the topic contrast and another describes its allocation across documents.

Remark 9.23 (Procedure and Upper-Bound Comparison) Cai, Ke, and Turner's DELVE test [24, Theorem 7] subtracts multinomial sampling bias from empirical between-document squared variation and standardizes the result. On the flat class, assume $m, d \to \infty$, $N \ge 2$, and $mN^2/d \to \infty$; these conditions imply the paper's Condition 2. For the variance estimator, use the normalization justified in their supplement [23, Propositions D.17–D.18], which for equal lengths is

$$V_* = \frac{2(1-1/m)^2}{(N-1)^2}\sum_{i=1}^{m}\sum_{j=1}^{d} X_{ij}(X_{ij}-1).$$

For fixed $0 < \alpha < 1$, the test has asymptotic level α and power tending to one when

$$\frac{D(\mathbf{\Omega})}{(\sum_i \|\Omega_i\|_2^2)^{1/2}} \longrightarrow \infty.$$

Since $m/d \le \sum_i \|\Omega_i\|_2^2 \le 2m/d$, the condition $N\varepsilon^2\sqrt{m/d} \to \infty$ suffices over $H_1(\varepsilon)$. The signal scale thus matches the lower bound under these regularity conditions. The comparison concerns fixed asymptotic level and power tending to one, not vanishing total error or sharp constants. The cited sources give the statistic and its calibration.

Cai, Ke, and Turner [24] also allow unequal document lengths and heterogeneous documents within known groups; the null then compares length-weighted group means.

Chapter perspective: choosing the lower-bound comparison

The tools of Parts I and II serve distinct decisions here. Mixture bounds obstruct detection without requiring identification of an alternative. Fano's inequality makes that identity the target; nearby pairs instead limit endpoint accuracy. The discovery arguments use posterior probabilities to bound reliable selections, rather than the distance between global testing laws. The required output determines the reduction, even when the alternatives are shared.

The constructions also suggest statistics: hidden signs lead to quadratic statistics, unknown locations to scans, and rare likelihood evidence to tail counts. In categorical homogeneity testing, calibration must hold throughout the unknown common-law null, although a single uniform null suffices for the lower bound.

9.5 Exercises

9.1. In Theorem 9.2, calculate the exponent of the alternative mean of the exceedance count after centering by its null mean and dividing by its null standard deviation, at a threshold $\sqrt{2q \log n}$ with fixed $0 < q \le 1$. Treat $q \le r$ and $q > r$ separately. Why does the optimizing threshold change at $r = 1/4$?

9.2. In Theorem 9.3, complete the square in $g_n^2 \varphi$ and recover the transition $r = (2-\sigma^2)^2/4$ when $\sigma^2 < 2$. Explain when the quadratic exponent is maximized inside the truncation interval and when it is maximized at its endpoint. Verify that the resulting exponents agree at this transition.

9.3. In the general posterior-selection setup, put $R = \sum_i D_i$, $T = \sum_i \theta_i D_i$, and $q_i = \mathbb{P}(\theta_i = 1 \mid X)$. Show $\mathbb{E}T \le t\mathbb{E}R + \sum_i \mathbb{P}\{\theta_i = 1,\ q_i > t\}$ for fixed $0 < t < 1$. If $M = \mathbb{E}\sum_i \theta_i > 0$ and this sum is $o(M)$ along a sequence of problems, prove that mFDR $\to 0$ implies MDR $= 1 - \mathbb{E}T/M \to 1$. Allow randomized joint selectors and handle $\mathbb{E}R = 0$ explicitly. Is coordinate independence needed?

9.4. For $Q = \mathrm{Exp}(1)$ and $G = \mathrm{Exp}(\lambda)$, $0 < \lambda < 1/2$, calculate the truncated likelihood-ratio second moment at $\log n$. Use (9.16) to explain why extreme likelihood ratios contribute at first rather than second order. Recover the detection boundary and maximum-test comparison in Section 9.2.

9.5. Derive the one-bit likelihood-ratio second moment in Proposition 9.10 and explain why dependence created by a context-keyed green list prevents a direct application of its product formula.

9.6. In Proposition 9.12, verify the posterior upper bound for both observed bit values. Why can no sequence of rules with positive expected numbers of selections have vanishing marginal false-discovery rate, even when each rule examines all n bits?

9.7. For the unsigned mixture of m blocks of size k in Theorem 9.13, replace disjointness by $|B_j \cap B_\ell| = r$ for every $j \ne \ell$. Assuming such blocks exist, recalculate $\chi^2(Q\|P_0)$. Explain why a lower bound obtained from a specified family of blocks also applies to a larger alternative class containing that family.

9.8. Derive (9.19) from the information-radius and Fano inequalities of Chapter 3. Why does this bound concern block identification rather than detection?

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9.9. In the Gaussian segment experiment of Section 9.3, let $I \subset J$ share their left endpoint, with right endpoints separated by r coordinates. For $P_I = N_n(a\mathbf{1}_I, \sigma^2 I_n)$ and $P_J = N_n(a\mathbf{1}_J, \sigma^2 I_n)$, compute $\mathrm{KL}(P_I, P_J)$. Contrast this with shifting an interval of known length k by $r \le k$ coordinates. Explain why the two calculations differ.

9.10. For the known-length segment scan, compute the mean and variance of the score difference for two overlapping intervals. Why do distant false intervals create the separate term involving $M = n - k + 1$ in the tail bound?

9.11. Derive the chi-square identity in Proposition 9.18 and compare its small-signal expansion with the unsigned identity.

9.12. In Theorem 9.20, set $s = m$ and calculate $1 + \chi^2(Q\|P_0)$. Compare it separately with $\{1 + (s/m)(e^{\lambda^2} - 1)\}^s$ and with $\exp\{(s^2/m)(e^{\lambda^2} - 1)\}$ from the proof. Which bound is exact in this case, and which inequality introduces slack?

9.13. Verify that the alternatives (9.23) belong to $\mathcal{P}_{\mathrm{flat}}(d)$, have average p_0, and satisfy $D(\mathbf{\Omega}) = mN\varepsilon^2/d$. Show that they use exactly two distinct document distributions. What can fail if category patterns are chosen independently for different document pairs, or if document signs are not balanced? For the full flat class, prove $m/d \le \sum_i \|\Omega_i\|_2^2 \le 2m/d$ and explain how this bound relates the DELVE signal ratio to the lower-bound scale.

9.14. Reproduce the likelihood cross-moment for two independent prior draws in Theorem 9.22. Explain why it depends on both the category-sign overlap U and the document-sign overlap V. Derive the bound on $\mathbb{E}\exp(\kappa UV)$ used in the proof, identifying where independence of token draws, documents, and prior signs enters. Why must the prior signs be held fixed before using independence of the observations?

Selected hints. For Exercise 9.7, separate diagonal likelihood cross-moments from off-diagonal cross-moments before averaging over blocks. For Exercise 9.13, use the two pairings and $\|p\|_2^2 \le \max_j p_j$, together with Cauchy–Schwarz. For Exercise 9.3, use $\mathbb{E}T = (1 - \mathrm{mFDR})\mathbb{E}R$ when $\mathbb{E}R > 0$, and treat $\mathbb{E}R = 0$ separately.

Part III Synthesis

Part III connects structural restrictions with inferential goals. Chapter 8 examines learning and inference with sparsity, latent labels, nuisance parameters, and incomplete observations. Chapter 9 emphasizes the different information needed for detection, selection, and localization. Table 9.1 compares the resulting constructions.

Problem	Hidden information and lower-bound design	Decision and scope distinction
Sparse regression	Fano support packing; noise and covariance mixtures	Vector error, signal detection, honest intervals, and sparsity adaptation
Classification and clustering	Assouad under a margin condition; Fano on sparse-support packings	Excess risk; clustering separation is modulo label exchange
Ranking and crowdsourcing	Fano for top-set recovery; Assouad on centered pairs; KL under missing observations	Ranking: exact top-set recovery or score error; crowdsourcing: task-effect error
Sparse mixtures and watermark scores	Truncation or product affinity; posterior selection bounds	Detection versus discovery; product arguments require independence
Segments	Location mixtures, Fano, and local two-point comparisons	Presence, block identity, and endpoint accuracy
Multinomial heterogeneity	Mixtures over category patterns and document assignments	Detecting heterogeneity while preserving probability mass and the average distribution

Table 9.1 *Lower-bound constructions in Part III and the decisions they address.*

Mixtures play two roles. Averaging over hard alternatives is a proof device; in a latent-label model, mixing is part of the observation mechanism. The testing mixtures of Chapter 5 extend to both settings, but detecting a mixture need not identify its signal observations. Section 9.1.3 therefore uses posterior probabilities to bound selection errors.

Sparse regression brings the four tasks of Part II into one model. Different nuisance-parameter constructions share a covariance-mixture calculation: testing requires separation from the zero vector, whereas the interval bound requires a common shift in one coefficient. Coverage transfer from Chapter 6 forces interval length even at zero. Since zero belongs to every sparsity class, this also gives the cross-class obstruction studied in Chapter 7.

Part I supplies information inequalities and hard-family constructions. Part II shows how estimation loss, testing error, honest coverage, and adaptation turn indistinguishability into different limitations. Part III brings these ideas together: parameter constraints determine the admissible alternatives, the observation scheme determines their distinguishability, and

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the inferential goal determines the lower bound. Recurring tools serve genuinely different statistical questions.

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Lower-Bound Techniques: A Section-by-Section Guide

This guide follows the substantive sections of Parts II and III. Each entry identifies the principal tool and the feature that determines its use. The linked sections state the models, assumptions, and scope of the results.

Section and problem	Principal tool	What changes in the application
Chapter 4: Minimax estimation		
§4.1 Linear functionals	Two-point bounds; moduli	Target gaps are optimized over a convex class or across components of a union.
§4.2 Quadratic functionals	Sign and support mixtures; van Trees	Weak-signal mixtures determine the nonparametric regimes; a radial submodel gives the parametric obstruction.
§4.3 Nonsmooth functionals	Location mixtures; moment-matched priors	An unknown extreme coordinate differs from accumulated approximation error near a nonsmooth point.
§4.4 Global function estimation	Assouad hypercubes	Local errors add under squared norm loss; an inverse operator changes the information per coordinate.
§4.5 Noisy optimization	Two-point bounds; local moduli and convex caps	Convexity links local value and location accuracy; smoothness alone permits localized wells in a separate regression benchmark.
Chapter 5: Testing and separation		
§5.1 Testing benchmarks	Pair and mixture reductions	Total error, prescribed-level power, and separation specify the testing obstruction.
§5.2 Convex Gaussian hypotheses	Closest pair; Neyman–Pearson	Convex geometry makes one pair determine the composite testing benchmark.
§5.3 A regression value	Two-point bump construction	Smoothness limits how narrowly a pointwise departure can be localized.
§5.4 Sparse Gaussian means	Support and sign mixtures	Overlap controls the second moment; the two-sample construction also preserves the nuisance mean.
§5.5 Regression testing: known noise	Support mixtures; joint likelihood moments	The construction must limit information from both the active support and the signal's increase in response variance.
§5.6 Unknown direction and smoothness	Random-sign mixtures	The dimension, fixed or constrained by smoothness, determines how much signal can remain hidden.

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Section and problem	Principal tool	What changes in the application
Chapter 6: Honest confidence sets		
§6.1 Coverage and size	Testing-to-size reduction; two-prior coverage transfer	Coverage is required on one class, while size may be evaluated on a smaller class or at one parameter.
§6.2 Intervals over convex classes	Modulus-selected pairs; Gaussian coverage transfer	A target gap forces interval length, rather than estimation error; prescribed coverage affects constants.
§6.3 Intervals for convex minima	Convex cap geometry; two-point coverage transfer	The value–location uncertainty principle now concerns honest interval lengths at a function.
§6.4 Nonconvex classes	Pairs and mixtures; convex-hull comparison	Unknown components affect length; affine-center restrictions introduce a separate obstruction.
§6.5 Confidence bands	Binary Gaussian families; spatial coverage arguments	Integrated width requires ambiguity across locations, with different demands under full and relaxed coverage.
§6.6 Confidence balls	Sign hypercubes; random-sign mixtures	Estimation bounds force worst-case expected squared radius; mixture comparisons force it even at zero.
Chapter 7: The cost of adaptation		
§7.1 Adaptive benchmarks	Cross-class comparison; constrained risk	One procedure must serve several classes; separate minimax bounds do not quantify the added cost.
§7.2 Linear functionals	Between-class moduli; constrained two-point bounds	Small risk on one class can permit a larger comparison radius and force an adaptation penalty.
§7.3 Quadratic functionals	Constrained risk against a mixture	The risk constraint at zero can offset growth of the mixture likelihood-ratio moment.
§7.4 Adaptive intervals	Between-class coverage transfer	Length is evaluated on a favorable subclass while coverage protects the larger class.
§7.5 Adaptive bands	Bump families; distributed sign mixtures	The coverage criterion determines how spatial ambiguity forces width on a smoother subclass.
§7.6 Adaptive balls	Random-sign mixture bound	Comparing forced expected squared radius with smoother-class benchmarks limits adaptation as dimension varies.

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Section and problem	Principal tool	What changes in the application
Chapter 8: Structured learning		
§8.1 Sparse regression inference	Fano; covariance mixtures; coverage transfer	Nuisance adjustments hide signal presence or a coordinate shift; one mixture calculation supports testing, interval, and adaptation bounds.
§8.2 Nonparametric classification	Assouad under a margin condition	Smoothness and the margin condition constrain the hard family; design mass and the conditional probability gap weight each wrong Bayes label.
§8.3 Sparse discriminant analysis	Fano; fixed-weight support packing	Small excess classification risk must distinguish separated Bayes rules, even for nonlinear classifiers.
§8.4 Sparse clustering	Fano; data processing	Separation is between partitions up to label exchange; training labels are unobserved.
§8.5 Pairwise ranking	Fano for top-set recovery; Assouad for scores	Identifying boundary items differs from controlling average score error; missing comparisons reduce information.
§8.6 Crowdsourcing	Centered hypercubes; KL under missing observations	Task identifiability and worker effects constrain the alternatives and the information in observed responses.
Chapter 9: Detection, selection, and localization		
§9.1 Sparse Gaussian signals	Second moments and truncation; posterior bounds	Detecting a rare population, discovering members, and retaining nearly every signal impose different requirements.
§9.2 General sparse mixtures	Product Hellinger affinity; likelihood-tail bounds	Non-Gaussian tails can defeat a second-moment bound; watermark scores also separate detection from discovery.
§9.3 Segments	Location mixtures; Fano; local pairs	Presence, block identity, and endpoint accuracy require different reductions on related interval families.
§9.4 Multinomial heterogeneity	Two-layer sign mixtures	Shared topic contrasts and document assignments preserve probability mass and the average distribution.

Appendix A

Technical Lemmas and Information Bounds

This appendix collects reusable ingredients for the main-text constructions: packing bounds for target separation; Gaussian, overlap, and moment-matching bounds for indistinguishability; and a smooth mean-zero bump for function estimation. It also gives the general packing–covering template and information-radius identity.

A.1 Packing and information lemmas

Packing arguments require a large separated family and a tractable comparison of its observation laws. The first lemmas supply the geometric constructions and cardinality bounds. Beginning with Lemma A.7, the focus shifts to the information calculations used repeatedly in the book: product experiments, Gaussian likelihoods, and mixture second moments.

Lemma A.1 (Packing and Covering) *Let (T, d) be a totally bounded metric space. A δ-packing is a subset whose distinct points have distance greater than δ, and a δ-cover is a collection of closed radius-δ balls with centers in T whose union contains T. If $M(\delta)$ is the largest cardinality of a δ-packing and $N(\delta)$ is the smallest cardinality of a δ-cover, then, for every $\delta > 0$,*

$$M(2\delta) \le N(\delta) \le M(\delta).$$

Proof Every radius-δ covering ball contains at most one point of a 2δ-packing, because two points in the same ball have distance at most 2δ. Hence $M(2\delta) \le N(\delta)$. Total boundedness makes $M(\delta)$ finite. Choose a δ-packing of cardinality $M(\delta)$. If the closed radius-δ balls centered at the packing points did not cover T, a point at distance greater than δ from every packing point could be added, contradicting maximal cardinality. Thus $N(\delta) \le M(\delta)$. □

Lemma A.2 (Volumetric Bound for Norm Balls) *Let $B = \{x \in \mathbb{R}^d : \|x\| \le 1\}$ be the unit ball of a norm on $\mathbb{R}^d$, and write $N(\delta; B, \|\cdot\|)$ for its covering number with centers in B. Then for every $\delta \in (0, 1)$,*

$$\delta^{-d} \le N(\delta; B, \|\cdot\|) \le \left(1 + \frac{2}{\delta}\right)^d.$$

Consequently B contains a δ-packing with at least δ^{-d} points.

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Proof Write vol for d-dimensional Lebesgue measure. A cover by N translates of δB satisfies $\operatorname{vol}(B) \le N\delta^d \operatorname{vol}(B)$, so $N \ge \delta^{-d}$. For the upper bound, choose a maximal δ-packing $\{x_1, \dots, x_M\}$. The balls $x_j + (\delta/2)B$ are disjoint and lie in $(1 + \delta/2)B$, hence

$$M \cdot (\delta/2)^d \operatorname{vol}(B) \le (1 + \delta/2)^d \operatorname{vol}(B), \qquad M \le (1 + 2/\delta)^d.$$

Lemma A.1 transfers this upper bound to the covering number and gives the asserted packing lower bound. □

The Varshamov–Gilbert existence bound gives exponentially large binary codes with minimum Hamming distance proportional to the block length; see Gilbert [66] and Varshamov [107].

Lemma A.3 (Varshamov–Gilbert Coding Lemma) *For every integer $m \ge 8$ there exists a subset $C \subset \{0,1\}^m$ such that*

$$\log |C| \ge \frac{m}{8} \log 2 \qquad \text{and} \qquad H(u,v) \ge \frac{m}{8} \quad \text{for all } u \ne v \in C.$$

Proof Choose a maximal subset $C \subset \{0,1\}^m$ with pairwise Hamming distance greater than $m/8$. Since distances are integer valued, the balls of radius $r = \lfloor m/8 \rfloor$ around its points cover the cube: otherwise another point could be added. Use the binary-entropy function defined in the proof of Lemma 3.5. By Lemma A.6 and monotonicity of h on $[0, 1/2]$, each ball contains at most $\exp\{mh(r/m)\} \le \exp\{mh(1/8)\}$ points. Since $h(1/8) = -(1/8)\log(1/8) - (7/8)\log(7/8) < (7/8)\log 2$,

$$2^m \le |C| \exp\{mh(1/8)\} < |C|\, 2^{7m/8}.$$

Thus $|C| > 2^{m/8}$, and the chosen separation is stronger than required. □

The fixed-weight version below restricts every codeword to the same number of nonzero coordinates. It supplies the sparse-support packings used in Chapters 3 and 8. See Tsybakov [103] for the general packing method; the stated form is proved here.

Lemma A.4 (Fixed-Weight Varshamov–Gilbert Lemma) *Let m, s be integers with $1 \le s \le m/2$. Then there exist universal constants $c_0, c_1 > 0$ and a family*

$$C \subset \{u \in \{0,1\}^m : \|u\|_0 = s\}$$

such that

$$\log |C| \ge c_0 s \log\Big(\frac{em}{s}\Big) \qquad \text{and} \qquad H(u,v) \ge c_1 s \quad \text{for all } u \ne v \in C.$$

Proof We first construct well-separated supports by random selection, then extend the construction to larger support sizes. Suppose $1 \le s \le m/4$. Let S, T be independent uniform s-subsets of $[m]$ and put $t = \lceil 3s/4 \rceil$. A union bound over the t-subsets of S gives

$$\mathbb{P}\{|S \cap T| \ge t\} \le \binom{s}{t} \frac{(s)_t}{(m)_t} \le 2^s (s/m)^t \le \exp\{-cs \log(em/s)\},$$

for a universal $c > 0$. Here $(a)_t = a(a-1)\cdots(a-t+1)$ denotes the falling factorial, with $(a)_0 = 1$, and $(s-i)/(m-i) \le s/m$ for $0 \le i < t$. The final inequality follows because $(3/4)\log(m/s) - \log 2$ is bounded below by a positive universal multiple of $1 + \log(m/s)$ when $m/s \ge 4$.

Draw $L = \lfloor \exp\{(c/4)s\log(em/s)\} \rfloor$ independent uniform s-subsets. The expected number of pairs with intersection at least $3s/4$ is at most $\binom{L}{2}\exp\{-cs\log(em/s)\} < 1$, so some realization has no such pair. Its incidence vectors satisfy $H(\mathbf{1}_S, \mathbf{1}_T) = 2(s - |S\cap T|) > s/2$. When the floor is at least two, decreasing a universal constant gives $\log L \ge c_0 s\log(em/s)$. The finitely many remaining cases admit two disjoint s-subsets; decreasing c_0, c_1 covers them.

For larger supports, keep some coordinates fixed and apply the first construction to the rest. If $m/4 < s \le m/2$ and m is sufficiently large, set $r = \lfloor (m-s)/4 \rfloor$ and $m' = m - s + r$. Then $r \le m'/4$ and $r \ge c_2 s$ for a universal $c_2 > 0$. Fix a set $A \subset [m]$ of size $s - r$ and apply the first part to r-subsets T of the remaining m' coordinates. The sets $A \cup T$ have size s, and their Hamming distances equal those of their variable parts. Moreover,

$$r\log(em'/r) \ge c_3 s\log(em/s),$$

because m/s and m'/r lie between universal positive constants. The resulting family has the required entropy and distance after adjusting c_0, c_1. The finitely many remaining values of m are again covered by two disjoint s-subsets and smaller universal constants. □

Lemma A.5 (Binomial-Coefficient Bounds) *Let m, s be integers with $1 \le s \le m$. Then*

$$\left(\frac{m}{s}\right)^s \le \binom{m}{s} \le \left(\frac{em}{s}\right)^s.$$

Consequently,

$$s\log\left(\frac{m}{s}\right) \le \log\binom{m}{s} \le s\log\left(\frac{em}{s}\right).$$

For fixed $c \in (0,1)$, $\log\binom{m}{s} \asymp s\log(em/s)$ uniformly for $1 \le s \le cm$, with constants depending only on c.

Proof Use $s! \ge (s/e)^s$ for the upper bound and $(m-s+j)/j \ge m/s$ for the lower bound. The latter holds for $1 \le j \le s$ because it is equivalent to $(s-j)(m-s) \ge 0$. Thus

$$\binom{m}{s} \le \frac{m^s}{s!} \le (em/s)^s, \qquad \binom{m}{s} = \prod_{j=1}^{s} \frac{m-s+j}{j} \ge (m/s)^s.$$

Taking logarithms gives the displayed bounds. If $s \le cm$, then $\log(m/s) \ge \log(1/c) > 0$, so $\log(m/s)$ and $\log(em/s)$ are comparable with constants depending only on c. □

The following volume estimates are useful for greedy-covering proofs of coding lemmas; the fixed-weight construction above instead uses random selection.

Lemma A.6 (Hamming-Ball Volume Bounds) *Let $m \geq 1$ be an integer and let $0 < \alpha \leq 1/2$.*

(a) *The Hamming ball in $\{0,1\}^m$ of radius $\lfloor \alpha m \rfloor$ satisfies*

$$\sum_{j=0}^{\lfloor \alpha m \rfloor} \binom{m}{j} \leq \exp\big(m h(\alpha)\big),$$

where h is the binary-entropy function defined in the proof of Lemma 3.5.

(b) *For integer $1 \leq s \leq m/2$, equip the size-s supports in $\{1, \dots, m\}$ with Johnson distance $d_J(S,T) := |S \setminus T| = \frac{1}{2} H(\mathbf{1}_S, \mathbf{1}_T)$. The ball $\{T : |T| = s,\ d_J(S,T) \leq t\}$ has cardinality one for $t = 0$ and, for integer $1 \leq t \leq s/2$, satisfies*

$$\sum_{j=0}^{t} \binom{s}{j}\binom{m-s}{j} \leq \exp\Big\{ C\, t \log\Big(\frac{em}{t}\Big)\Big\},$$

for a universal constant $C > 0$.

Proof For *(a)*, put $r = \lfloor \alpha m \rfloor$; the case $r = 0$ is immediate. For $r \geq 1$ and $0 < u \leq 1$, the binomial theorem gives

$$(1+u)^m \geq \sum_{j=0}^{r} \binom{m}{j} u^j \geq u^r \sum_{j=0}^{r} \binom{m}{j}.$$

To minimize $u^{-r}(1+u)^m$, differentiate its logarithm: $m/(1+u) - r/u = 0$ at $u_* = r/(m-r) \in (0,1]$. Substitution yields

$$\sum_{j=0}^{r} \binom{m}{j} \leq \Big(\frac{m}{r}\Big)^r \Big(\frac{m}{m-r}\Big)^{m-r} = \exp\{m h(r/m)\} \leq \exp\{m h(\alpha)\},$$

where the last inequality uses $r/m \leq \alpha \leq 1/2$ and monotonicity of h.

For *(b)*, a support at Johnson distance j from S removes j coordinates of S and adds j from its complement. Thus the ball has cardinality $\sum_{j=0}^{t} \binom{s}{j}\binom{m-s}{j}$; for $t = 0$ it contains only S. For $1 \leq j \leq t \leq s/2$, Lemma A.5 gives

$$\binom{s}{j}\binom{m-s}{j} \leq (es/j)^j \{e(m-s)/j\}^j \leq a(j), \qquad a(x) = (e^2 sm/x^2)^x.$$

On $1 \leq x \leq t$, $(\log a)'(x) = \log(sm/x^2) \geq \log 2 > 0$, using $t \leq s/2$ and $s \leq m/2$. Every summand, including the $j = 0$ term, is therefore at most $a(t)$. Consequently,

$$\sum_{j=0}^{t} \binom{s}{j}\binom{m-s}{j} \leq (t+1)(e^2 sm/t^2)^t \leq (t+1)(em/t)^{2t} \leq \exp\{Ct \log(em/t)\}.$$

The last two inequalities use $s \leq m$ and $\log(t+1) \leq t \log 2 \leq t \log(em/t)$. □

For the information calculations, recall from Section 1.3 that $H^2 = 1 - \rho$ and $\xi(P,Q) = 1 + \chi^2(Q,P)$, with the reference law first in ξ.

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Lemma A.7 (Tensorization for Product Experiments) *Let P, Q be probability measures on a common measurable space $(\mathcal{X}, \mathscr{X})$. Then for every integer $n \geq 1$,*

$$\rho(P^{\otimes n}, Q^{\otimes n}) = \rho(P, Q)^n, \qquad \mathrm{KL}(P^{\otimes n}, Q^{\otimes n}) = n\ \mathrm{KL}(P, Q).$$

If moreover $Q \ll P$, then

$$\xi(P^{\otimes n}, Q^{\otimes n}) = \xi(P, Q)^n,$$

where ξ is the χ^2-affinity defined in Section 1.3.

Proof Choose densities p, q with respect to a common dominating measure μ. The product densities are $p_n(x) = \prod_i p(x_i)$ and $q_n(x) = \prod_i q(x_i)$. Tonelli's theorem gives

$$\rho(P^{\otimes n}, Q^{\otimes n}) = \int \prod_{i=1}^{n} \sqrt{p(x_i) q(x_i)}\, d\mu^{\otimes n} = \left(\int \sqrt{pq}\, d\mu \right)^n.$$

For KL, if $P \not\ll Q$, a set A with $\mathbb{P}_P(A) > 0 = \mathbb{P}_Q(A)$ makes $A \times \mathcal{X}^{n-1}$ a witness to $P^{\otimes n} \not\ll Q^{\otimes n}$, so both sides are infinite. If $P \ll Q$, put $L = dP/dQ$ and $\ell = \log L$. Its negative part $\ell^- = \max\{-\ell, 0\}$ is integrable under P because $\mathbb{E}_P \ell^- = \int_{\{0<L<1\}} L(-\log L)\, dQ \leq e^{-1}$. The product log likelihood ratio is $\sum_i \ell(X_i)$. Each summand has an integrable negative part, and Tonelli's theorem applies to the sum of their positive parts. Hence

$$\mathrm{KL}(P^{\otimes n}, Q^{\otimes n}) = \mathbb{E}_{P^{\otimes n}} \sum_{i=1}^{n} \ell(X_i) = \sum_{i=1}^{n} \mathbb{E}_P \ell(X_i) = n\, \mathrm{KL}(P, Q),$$

including infinite values. When $Q \ll P$, define $q^2/p = 0$ where $p = q = 0$. Tonelli similarly gives

$$\xi(P^{\otimes n}, Q^{\otimes n}) = \int \prod_{i=1}^{n} \frac{q(x_i)^2}{p(x_i)}\, d\mu^{\otimes n} = \left(\int \frac{q^2}{p}\, d\mu \right)^n.$$

□

At $n = 0$, the product laws coincide on the singleton empty-product space. Their Hellinger and chi-square affinities are 1, and their KL divergence is 0; the KL identity extends to this case with $0 \cdot (+\infty) = 0$.

For independent observations with possibly different laws $P_i \ll Q_i$, $1 \leq i \leq m$, the same likelihood-ratio factorization gives

$$1 + \chi^2\left(\bigotimes_i P_i \middle\| \bigotimes_i Q_i \right) = \prod_i \{1 + \chi^2(P_i \| Q_i)\},$$

$$\mathrm{KL}\left(\bigotimes_i P_i \middle\| \bigotimes_i Q_i \right) = \sum_i \mathrm{KL}(P_i \| Q_i),$$

with extended values allowed.

Lemma A.8 (Gaussian Divergence Formulas with Common Covariance) *Let* $P = N_d(\mu_0, \Sigma)$ *and* $Q = N_d(\mu_1, \Sigma)$ *on* $\mathbb{R}^d$, *where* Σ *is positive definite, and put* $D = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$. *Then*

$$\mathrm{KL}(P, Q) = D/2, \qquad \rho(P, Q) = e^{-D/8},$$
$$H^2(P, Q) = 1 - e^{-D/8}, \qquad \xi(P, Q) = e^{D}.$$

For every integer $n \geq 1$, *the product laws satisfy*

$$\mathrm{KL}(P^{\otimes n}, Q^{\otimes n}) = nD/2, \qquad \rho(P^{\otimes n}, Q^{\otimes n}) = e^{-nD/8}.$$

The same formulas hold for $P = P_f$ *and* $Q = P_g$ *in the Gaussian shift experiment* (1.3), *with* $f, g \in H$, $\varepsilon > 0$, *and* $D = \|g - f\|_H^2/\varepsilon^2$.

Proof Put $\Delta = \mu_1 - \mu_0$ and $L = dQ/dP$. Expanding the Gaussian log likelihood ratio gives

$$\log L(X) = \Delta^\top \Sigma^{-1}(X - \mu_0) - D/2.$$

Under P, the first term is a centered Gaussian variable of variance D. Its moment-generating formula therefore yields, for every real t,

$$\mathbb{E}_P L^t = \exp\{(t^2 - t)D/2\}.$$

Consequently $\mathrm{KL}(P, Q) = -\mathbb{E}_P \log L = D/2$, $\rho(P, Q) = \mathbb{E}_P L^{1/2} = e^{-D/8}$, and $\xi(P, Q) = \mathbb{E}_P L^2 = e^D$. The identity $H^2 = 1 - \rho$ gives the Hellinger formula, and Lemma A.7 gives the product formulas.

For the Gaussian shift, we first identify the likelihood ratio; the same one-dimensional Gaussian calculation will then apply. Put $h = g - f$ and define

$$L = \exp\left\{\frac{Y(h) - \langle f, h\rangle_H}{\varepsilon^2} - \frac{\|h\|_H^2}{2\varepsilon^2}\right\}.$$

Under P_f, the random term in the exponent is centered Gaussian with variance D, so $\mathbb{E}_f L = 1$ and L defines a probability law. To identify it, take any finite linear combination $u = \sum_j t_j u_j$ of measurement directions. The joint Gaussian exponential formula gives

$$\mathbb{E}_f\left[L \exp\left\{i \sum_j t_j Y(u_j)\right\}\right] = \exp\left\{i\langle g, u\rangle_H - \frac{\varepsilon^2}{2}\|u\|_H^2\right\}.$$

This is the joint characteristic function of those measurements under P_g. Cylinder events generate the observation sigma-field, so $L = dP_g/dP_f$. Since $L > 0$, the laws are mutually absolutely continuous. The same Gaussian moment calculation as above now proves all four divergence identities and their product versions. □

Measurable white-noise integrals. Let $C[0, 1]$ be the space of continuous real-valued functions on $[0, 1]$, equipped with the Borel sigma-field of the uniform norm. In the white-noise model (1.2), each deterministic $h \in L_2[0, 1]$ admits a Borel measurable map $I_h : C[0, 1] \to \mathbb{R}$, chosen independently of the drift f and noise level ε, such that

$$I_h(Y) = \langle h, f\rangle + \varepsilon \int_0^1 h(t)\, dB(t) \quad \text{almost surely}$$

for every $f \in L_2[0,1]$ and $\varepsilon > 0$.

To construct this version, choose deterministic interval-step functions $s_m \to h$ in $L_2[0,1]$. Write these functions and their corresponding increment sums as

$$s_m = \sum_{\ell} c_{m\ell} \mathbf{1}_{(t_{m\ell}, t_{m,\ell+1}]},$$
$$S_m(y) = \sum_{\ell} c_{m\ell} \{y(t_{m,\ell+1}) - y(t_{m\ell})\}.$$

Each S_m is measurable. By the stochastic-integral isometry, $S_m(B) \to \int h\, dB$ in L_2; fix a subsequence along which convergence also holds almost surely. Define $I_h(y)$ as the finite limit of these sums along that subsequence where it exists, and as zero otherwise. This defines a measurable map without reference to f or ε. For each such drift and noise level,

$$S_m(Y) = \langle s_m, f \rangle + \varepsilon S_m(B), \qquad \langle s_m, f \rangle \longrightarrow \langle h, f \rangle.$$

Taking the subsequential limit proves the assertion. For any countable family of integrands, these identities hold simultaneously almost surely under each fixed observation law.

Lemma A.9 (χ^2 Mixture Second-Moment Identity) *Let $(\Omega, \mathcal{A}, \Pi)$ be a probability space, P a probability measure on the measurable observation space $(\mathcal{X}, \mathscr{X})$, and $\omega \mapsto Q_\omega$ a probability kernel dominated by P with a jointly measurable likelihood-ratio version $L(\omega, x) = dQ_\omega/dP(x)$. For the mixture $\bar{Q} = \int Q_\omega\, d\Pi(\omega)$,*

$$1 + \chi^2(\bar{Q}, P) = \int \left(\frac{d\bar{Q}}{dP}\right)^2 dP = \iint \left[\int \frac{dQ_\omega}{dP} \frac{dQ_{\omega'}}{dP}\, dP\right] d\Pi(\omega) d\Pi(\omega').$$

Equivalently, for independent $\omega, \omega' \sim \Pi$ and $L_\omega = dQ_\omega/dP$,

$$1 + \chi^2(\bar{Q}, P) = \mathbb{E}_{\omega, \omega' \sim \Pi} \int L_\omega L_{\omega'}\, dP.$$

Proof By Tonelli's theorem, the mixture has density $\bar{L}(x) = \int L(\omega, x)\, d\Pi(\omega)$ with respect to P. Nonnegativity and joint measurability permit a second application of Tonelli, without any finiteness assumption:

$$\begin{aligned} 1 + \chi^2(\bar{Q}, P) &= \int \left(\int L_\omega\, d\Pi(\omega)\right)\left(\int L_{\omega'}\, d\Pi(\omega')\right) dP \\ &= \iint \left(\int L_\omega L_{\omega'}\, dP\right) d\Pi(\omega) d\Pi(\omega'). \end{aligned}$$

For independent $\omega, \omega' \sim \Pi$, this is the asserted expectation. □

A.2 Overlap and moment-matching estimates

Mixture lower bounds depend on likelihood cross-moments. Support randomization controls shared active coordinates, as in sparse-signal detection; moment matching cancels low-degree terms, as in nonsmooth functional estimation. The following estimates combine with

Lemma A.9 to control observed mixture laws. Target separation and concentration under the priors must be checked separately.

Lemma A.10 (Gaussian Covariance and Mixture Formulas) *For positive-definite Σ_0, Σ_1 and $P = N_d(0, \Sigma_1)$, $Q = N_d(0, \Sigma_0)$ in dimension d,*

$$\mathrm{KL}(P\|Q) = \frac{1}{2}\{\mathrm{tr}(\Sigma_0^{-1}\Sigma_1 - I_d) - \log\det(\Sigma_0^{-1}\Sigma_1)\}.$$

For positive-definite $\Sigma \in \mathbb{R}^{d\times d}$, let $P_0 = N_d(0, \Sigma)$ and define $L_\theta = dN_d(\theta, \Sigma)/dP_0$ for $\theta \in \mathbb{R}^d$. Then, for every $\theta, \theta' \in \mathbb{R}^d$,

$$\mathbb{E}_0(L_\theta L_{\theta'}) = \exp\{\theta^\top \Sigma^{-1}\theta'\}.$$

Proof For the covariance formula, the log density ratio at x equals

$$\frac{1}{2}\log\frac{\det\Sigma_0}{\det\Sigma_1} - \frac{1}{2}x^\top(\Sigma_1^{-1} - \Sigma_0^{-1})x.$$

Under P, $\mathbb{E}X^\top AX = \mathrm{tr}(A\Sigma_1)$, which gives the covariance formula. For the cross-moment, write

$$L_\theta(X) = \exp\{\theta^\top\Sigma^{-1}X - \tfrac{1}{2}\theta^\top\Sigma^{-1}\theta\}$$

and apply the Gaussian moment-generating formula to $L_\theta L_{\theta'}$. The two individual quadratic terms cancel, leaving $\theta^\top\Sigma^{-1}\theta'$ in the exponent. □

Lemma A.11 (Hypergeometric Overlap Control) *Let S, S' be independent uniformly distributed k-subsets of $\{1, \dots, d\}$, and let $R = |S \cap S'|$. If $1 \le k \le d$, then, for every $t \ge 0$,*

$$\mathbb{E}\exp(tR) \le \left\{1 + \frac{k}{d}(e^t - 1)\right\}^k \le \exp\left\{\frac{k^2}{d}(e^t - 1)\right\}.$$

Proof Condition on S and put $u = e^t - 1 \ge 0$. Expand in binomial coefficients so that each term counts shared subsets. Since $e^{tR} = (1+u)^R = \sum_{j=0}^R \binom{R}{j}u^j$, it is enough to calculate the factorial moments of the hypergeometric variable. For $0 \le j \le k$,

$$\mathbb{E}\left[\binom{R}{j}\,\middle|\, S\right] = \binom{k}{j}\frac{(k)_j}{(d)_j} \le \binom{k}{j}\left(\frac{k}{d}\right)^j,$$

because a fixed j-subset of S is contained in S' with probability $(k)_j/(d)_j$, and $(k-\ell)/(d-\ell) \le k/d$ for $0 \le \ell < j$. Therefore

$$\mathbb{E}(e^{tR} \mid S) \le \sum_{j=0}^k \binom{k}{j}\left(\frac{k}{d}u\right)^j = \left(1 + \frac{k}{d}u\right)^k \le \exp\left(\frac{k^2}{d}u\right).$$

The bound is independent of S, and $u = e^t - 1$ proves both claims. □

Lemma A.12 (Moment-Matching Normal-Mixture Estimate) *Let $a \ge 0$ and let $K \ge 0$ be an integer. Let π_0 and π_1 be probability measures supported on $[-a, a]$ whose moments agree through degree K. Let p_i be the density of $P_i = \int N(\theta, 1)\, d\pi_i(\theta)$ and let φ be the standard normal density. Then*

$$\int_{\mathbb{R}} \frac{(p_1(x) - p_0(x))^2}{\varphi(x)}\, dx \le 4 \sum_{j>K} \frac{a^{2j}}{j!} \le \frac{4e^{a^2} a^{2(K+1)}}{(K+1)!}.$$

If also $p_0 \ge c_0 \varphi$ pointwise for some $c_0 > 0$, then, for every integer $d \ge 1$,

$$\chi^2(P_1^{\otimes d} \| P_0^{\otimes d}) \le \exp\left\{ \frac{4d\, e^{a^2} a^{2(K+1)}}{c_0 (K+1)!} \right\} - 1.$$

For one observation, the sharper bound is $\chi^2(P_1 \| P_0) \le 4c_0^{-1} \sum_{j>K} a^{2j}/j!$.

Proof The proof reduces the distance between mixture densities to the kernel $e^{\theta t}$. Matching moments cancels its polynomial terms; bounded support controls the remainder. Set $L_\theta(x) = e^{\theta x - \theta^2/2}$ and $L_i(x) = \int L_\theta(x)\, d\pi_i(\theta) = p_i(x)/\varphi(x)$. Completing the Gaussian square gives $\int_{\mathbb{R}} L_\theta(x) L_t(x) \varphi(x)\, dx = e^{\theta t}$. Thus Tonelli identifies $C_{ij} := \iint e^{\theta t}\, d\pi_i(\theta) d\pi_j(t)$ with $\int_{\mathbb{R}} L_i(x) L_j(x) \varphi(x)\, dx$. Since $|\theta t| \le a^2$, $0 < C_{ij} \le e^{a^2}$, so $L_i \in L_2(\varphi(x)\, dx)$ and we may expand the square:

$$A := \int_{\mathbb{R}} \frac{(p_1(x) - p_0(x))^2}{\varphi(x)}\, dx = C_{11} - C_{10} - C_{01} + C_{00}.$$

Write $m_{i,j} = \int \theta^j\, d\pi_i(\theta)$ and put

$$T_K(u) = \sum_{j=0}^{K} \frac{u^j}{j!}, \qquad R_K(u) = e^u - T_K(u), \qquad B_K = \sum_{j>K} \frac{a^{2j}}{j!}.$$

Absolute convergence gives $|R_K(u)| \le B_K$ for $|u| \le a^2$. The contribution of $T_K(\theta t)$ to the four-term expression is

$$\sum_{j=0}^{K} \frac{m_{1,j}^2 - 2m_{1,j} m_{0,j} + m_{0,j}^2}{j!} = \sum_{j=0}^{K} \frac{(m_{1,j} - m_{0,j})^2}{j!} = 0.$$

Bounded support justifies integration of the finite polynomial. Each of the four remainder integrals is at most B_K in absolute value, so $A \le 4B_K$. For $x \ge 0$,

$$\sum_{j>K} \frac{x^j}{j!} = \sum_{r \ge 0} \frac{x^{K+1+r}}{(K+1+r)!} \le \frac{x^{K+1}}{(K+1)!} \sum_{r \ge 0} \frac{x^r}{r!} = \frac{e^x x^{K+1}}{(K+1)!},$$

using $(K+1+r)! \ge (K+1)!\, r!$. Taking $x = a^2$ gives the second inequality in the lemma's first display. If $p_0 \ge c_0 \varphi$, then $\chi^2(P_1 \| P_0) \le c_0^{-1} A$, and tensorization gives

$$1 + \chi^2(P_1^{\otimes d} \| P_0^{\otimes d}) = \{1 + \chi^2(P_1 \| P_0)\}^d \le \exp\{d \chi^2(P_1 \| P_0)\}.$$

Substituting the one-observation bound gives the asserted product bound. □

A.3 Yang–Barron coverings and nonparametric minimax lower bounds

For infinite-dimensional classes, Yang and Barron [113] separate parameters by a packing and control information by a KL cover of their observation laws. A uniform mixture of the covering laws provides a reference when comparison with a single law is too costly. The template below assumes the loss-to-testing reduction separately; its packing and covering sizes refer to the selected finite families, which need not have optimal cardinalities.

Theorem A.13 (Yang–Barron Packing–Covering Lower-Bound Template) *Let $\{P_\theta^{(n)} : \theta \in \Theta\}$ be a statistical experiment indexed by $n > 0$, representing sample size or noise precision, let d be a semimetric on Θ, let L be a nonnegative measurable loss, and let $\Psi : [0,\infty) \to [0,\infty)$ be nondecreasing. Fix $\delta_{\mathrm{pack}} > 0$ and $\varepsilon \geq 0$. Suppose every finite $2\delta_{\mathrm{pack}}$-packing $\{\theta_v : v \in V\} \subset \Theta$ yields*

$$\inf_{\widehat{\theta}} \sup_{v \in V} \mathbb{E}_v L(\widehat{\theta}, \theta_v) \geq \Psi(\delta_{\mathrm{pack}}) \inf_{\widehat{v}} \sup_{v \in V} \mathbb{P}_v\{\widehat{v} \neq v\},$$

where $P_v = P_{\theta_v}^{(n)}$. Assume further that:

(a) *there exists a finite $2\delta_{\mathrm{pack}}$-packing $\{\theta_v : v \in V\} \subset \Theta$; write its cardinality as $N_{\mathrm{pack}}(2\delta_{\mathrm{pack}}) := |V| \geq 2$;*

(b) *there exists a finite family of probability measures $\{Q_j : 1 \leq j \leq M_{\mathrm{KL}}(\varepsilon)\}$ such that for every $v \in V$,*

$$\min_{1 \leq j \leq M_{\mathrm{KL}}(\varepsilon)} \mathrm{KL}(P_v, Q_j) \leq n\varepsilon^2.$$

Then

$$\inf_{\widehat{\theta}} \sup_{\theta \in \Theta} \mathbb{E}_\theta L(\widehat{\theta}, \theta) \geq \Psi(\delta_{\mathrm{pack}}) \left[1 - \frac{\log M_{\mathrm{KL}}(\varepsilon) + n\varepsilon^2 + \log 2}{\log N_{\mathrm{pack}}(2\delta_{\mathrm{pack}})}\right]_+.$$

Proof Write $N = N_{\mathrm{pack}}(2\delta_{\mathrm{pack}}) \geq 2$ and $M = M_{\mathrm{KL}}(\varepsilon) \geq 1$. Choose $j(v)$ with $\mathrm{KL}(P_v, Q_{j(v)}) \leq n\varepsilon^2$ and set $R = M^{-1}\sum_{j=1}^M Q_j$ and $\kappa = \log M + n\varepsilon^2$. For densities p_v, q_j, r relative to $\mu = \sum_v P_v + \sum_j Q_j$, we have $r \geq q_j/M$ and $P_v \ll Q_{j(v)} \ll R$. Thus

$$\log(p_v/r) \leq \log(p_v/q_{j(v)}) + \log M \quad P_v\text{-almost surely.}$$

To integrate this inequality, note that the negative part of a log density ratio has finite expectation: here it is bounded by r/p_v on $\{p_v > 0\}$, with P_v-integral at most one. The finite KL covering bound makes the right side integrable and therefore controls the positive part of the left side. Hence

$$\mathrm{KL}(P_v, R) \leq \mathrm{KL}(P_v, Q_{j(v)}) + \log M \leq \kappa.$$

For a uniform random index U on V, Proposition 3.9 now gives

$$I(U;X) \leq \frac{1}{N}\sum_{v \in V} \mathrm{KL}(P_v, R) \leq \kappa.$$

The decoding bound in Theorem 3.10 implies that every decoder has average, and hence worst-case, error at least $[1 - (\kappa + \log 2)/\log N]_+$. The assumed loss-to-testing reduction multiplies this bound by $\Psi(\delta_{\mathrm{pack}})$, and the full minimax risk is no smaller than the risk on the packing. Substitution of $\kappa = \log M + n\varepsilon^2$ proves the result. □

The packing supplies separation; the covering controls information and may use a different scale. With N and M as in the proof, choose these scales so that $\log M + n\varepsilon^2 + \log 2 < \log N$. No triangle inequality is required of the semimetric here because the loss-to-testing reduction is assumed separately. In contrast, Lemma A.1 compares packings and coverings in a metric space.

A.4 Information radius for general priors

Lower bounds based on a general prior require information comparisons beyond finite families. We extend Proposition 3.9 to this setting. When almost every observation law is dominated by the prior predictive law, that predictive law minimizes average KL divergence. A reference measure dominating the whole family also gives a mutual-information bound uniform over priors, without this additional condition.

Proposition A.14 (Mutual Information as an Information Radius) *Let $\theta \mapsto P_\theta$ be a measurable probability kernel from a standard Borel parameter space Θ to a standard Borel sample space, let Π be a prior on Θ, and write $P_X = \int P_\theta \, d\Pi(\theta)$ for its prior predictive law. Let $\vartheta \sim \Pi$ be the random parameter and $X \mid \vartheta = \theta \sim P_\theta$. If $P_\theta \ll P_X$ for Π-almost every θ, then*

$$I(\vartheta; X) = \inf_Q \int \mathrm{KL}(P_\theta, Q)\, d\Pi(\theta),$$

where the infimum ranges over probability measures Q that dominate P_θ for Π-almost every θ, and it is attained at P_X. Even without the assumption $P_\theta \ll P_X$ for Π-almost every θ,

$$I(\vartheta; X) \le \inf_Q \sup_{\theta \in \Theta} \mathrm{KL}(P_\theta, Q),$$

where the latter infimum ranges over probability measures Q that dominate the whole family $\{P_\theta : \theta \in \Theta\}$. Hence

$$\sup_\Pi I(\vartheta; X) \le \inf_Q \sup_{\theta \in \Theta} \mathrm{KL}(P_\theta, Q).$$

Proof Let $P_{\vartheta X}$ be the joint law induced by Π and the experiment. By definition $I(\vartheta; X) = \mathrm{KL}(P_{\vartheta X}, \Pi \otimes P_X)$. We compare this quantity with the average divergence to any admissible reference law. For any probability measure Q dominating P_θ for Π-almost every θ, a Q-null set is also P_X-null, so $P_X \ll Q$. The extended-valued relative-entropy chain rule gives

$$\int \mathrm{KL}(P_\theta, Q)\, d\Pi(\theta) = \mathrm{KL}(P_{\vartheta X}, \Pi \otimes Q) = I(\vartheta; X) + \mathrm{KL}(P_X, Q).$$

The chain rule is an identity in $[0, \infty]$, so infinite KL terms cause no subtraction of infinities. Since $\mathrm{KL}(P_X, Q) \ge 0$, the integral is at least $I(\vartheta; X)$. Under predictive domination, $Q = P_X$ is admissible and attains equality, proving the infimum formula.

For any Q dominating the whole family, the same chain rule gives, without requiring predictive domination,

$$I(\vartheta; X) \le \int \mathrm{KL}(P_\theta, Q)\, d\Pi(\theta) \le \sup_{\theta \in \Theta} \mathrm{KL}(P_\theta, Q).$$

Taking the infimum over such Q proves the second assertion; taking the supremum over priors proves the last. □

A related interpretation uses rate–distortion. For a nonnegative distortion function L and a prior Π, define $R_\Pi(D)$, $D \ge 0$, as the infimum of $I(\vartheta;\widetilde{\vartheta})$ over joint laws of a parameter $\vartheta \sim \Pi$ and a reconstruction $\widetilde{\vartheta}$ satisfying $\mathbb{E}L(\widetilde{\vartheta},\vartheta) \le D$. The reconstruction is the first argument of the loss. If a reconstruction based on X, possibly with independent randomization, meets this distortion constraint, data processing gives $R_\Pi(D) \le I(\vartheta;X)$. Shannon's fidelity-criterion formulation [100] motivates this comparison. The applications in this book use the finite-family bounds directly and do not require a rate–distortion formula.

A.5 Smooth perturbations for function estimation

Section 4.4.1 constructs density and regression alternatives from disjoint copies of a smooth mean-zero function. Zero integral preserves density normalization; compact support prevents boundary discontinuities. Here $C_c^\infty((0,1))$ denotes infinitely differentiable functions compactly supported in $(0,1)$, extended by zero to $\mathbb{R}$. The following construction gives the required normalization and height bound.

Lemma A.15 (Smooth Mean-Zero Bump) *There exists $\phi \in C_c^\infty((0,1))$ such that*

$$\int_{\mathbb{R}} \phi(x)\,dx = 0, \qquad \int_{\mathbb{R}} \phi(x)^2\,dx = 1, \qquad \|\phi\|_\infty^2 \le 32.$$

Proof We construct two disjoint translates of a smooth nonnegative bump, subtract them to obtain zero integral, and normalize in L_2. Define

$$q(t) := \begin{cases} \exp(-1/t), & t > 0, \\ 0, & t \le 0, \end{cases} \qquad S(t) := \frac{q(t)}{q(t)+q(1-t)}, \qquad b(t) := S(2-16|t|).$$

Every derivative of q on $(0,\infty)$ is $\exp(-1/t)$ times a polynomial in $1/t$ and tends to zero as $t \downarrow 0$. Thus q is smooth. The denominator defining S is everywhere positive, so S is also smooth, equals zero on $(-\infty,0]$, and equals one on $[1,\infty)$, with $0 \le S \le 1$. It follows that $b = 1$ on $[-1/16, 1/16]$, $b = 0$ outside $(-1/8, 1/8)$, and $0 \le b \le 1$. The absolute value does not affect smoothness because b is constant near zero.

Put

$$v(x) := b(8x-2) - b(8x-5/2), \qquad E := \int_{\mathbb{R}} v(x)^2\,dx, \qquad \phi(x) := E^{-1/2} v(x).$$

The two terms in v have disjoint supports, contained respectively in $[15/64, 17/64]$ and $[19/64, 21/64]$, and equal integrals. Their squared norms add. Since b equals one on an interval of length $1/8$ and is bounded by one on a support of length $1/4$, a change of variables gives

$$\int_{\mathbb{R}} v = 0, \qquad E = \frac{1}{4}\int_{\mathbb{R}} b(t)^2\,dt, \qquad \frac{1}{32} \le E \le \frac{1}{16}.$$

In particular, $\phi \in C_c^\infty((0,1))$, $\int \phi = 0$, and $\int \phi^2 = 1$. Since $|v| \le 1$ and $v(1/4) = 1$, we have $\|\phi\|_\infty^2 = E^{-1} \le 32$. □

Appendix B

Proof Compendium

This appendix collects derivations deferred from the main text; each split proof refers to its chapter argument. Reusable divergence and overlap lemmas are in Appendix A.

B.1 Proofs for Chapter 4: minimax estimation

B.1.1 The matching upper bound for the maximum coordinate

We prove the upper bound in Theorem 4.10; its truncated-mixture lower bound is proved in Section 4.3.1. Retain the unit-noise Gaussian model, write $Y_{\max} = \max_j Y_j$, and set $a_d = \sqrt{2\log d}$.

Let $\widehat{T}^* = Y_{\max} - a_d/2$. Fix $\theta \in \mathbb{R}^d$ and write $m = T(\theta) = \theta_{\max}$. Then

$$\widehat{T}^* - T(\theta) = R_\theta - \frac{a_d}{2}, \qquad R_\theta := Y_{\max} - m = \max_{1\le j\le d}\{Z_j - (m-\theta_j)\}.$$

Since $m - \theta_j \ge 0$ for every j, any coordinate j_* attaining m satisfies

$$Z_{j_*} \le R_\theta \le M_d := \max_{1\le j\le d} Z_j, \qquad 0 \le \mathbb{E}_\theta R_\theta \le \mathbb{E}M_d.$$

It remains to control the variance of R_θ and the mean of M_d. The Gaussian Poincaré inequality gives $\mathrm{Var}_\theta(R_\theta) \le 1$ because R_θ is 1-Lipschitz in Z. We derive the needed inequality first in one dimension, then for the smooth maximum. Let $f : \mathbb{R} \to \mathbb{R}$ be continuously differentiable with bounded derivative. The derivative bound implies at most linear growth and hence square integrability under the Gaussian law. The independent-copy variance identity, the fundamental theorem of calculus, and Cauchy–Schwarz give

$$\begin{aligned}\mathrm{Var}(f(Z)) &= \int_{x<y} (f(y)-f(x))^2\varphi(x)\varphi(y)\,dx\,dy \\ &\le \int_{\mathbb{R}} f'(t)^2 \left\{\int_{x\le t<y} (y-x)\varphi(x)\varphi(y)\,dx\,dy\right\} dt = \mathbb{E}f'(Z)^2.\end{aligned}$$

Tonelli's theorem justifies the interchange because the integrand is nonnegative. To evaluate the inner integral, use $\int_t^\infty y\varphi(y)\,dy = \varphi(t)$ and $\int_{-\infty}^t x\varphi(x)\,dx = -\varphi(t)$; it equals $\Phi(t)\varphi(t) + (1-\Phi(t))\varphi(t) = \varphi(t)$.

The corresponding product inequality follows from variance tensorization. Indeed, for independent random elements X, Y and square-integrable $G(X,Y)$, the conditional variance identity gives

$$\mathrm{Var}(G) = \mathbb{E}_X \mathrm{Var}_Y(G) + \mathrm{Var}_X(\mathbb{E}_Y G) \le \mathbb{E}_X \mathrm{Var}_Y(G) + \mathbb{E}_Y \mathrm{Var}_X(G).$$

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For the inequality, introduce an independent copy X' and apply Jensen's inequality to $\{\mathbb{E}_Y[G(X,Y)-G(X',Y)]\}^2$ in the independent-copy formula for the second term. Induction over the coordinates yields $\mathrm{Var}(G(Z)) \le \sum_j \mathbb{E}\,\mathrm{Var}_{Z_j}(G(Z))$, where each variance on the right is taken with respect to one coordinate, holding the others fixed.

Apply this inequality to the smooth maximum

$$F_\beta(z) = \beta^{-1}\log\left(\sum_{j=1}^{d} e^{\beta(\theta_j+z_j)}\right) - m, \qquad \beta > 0.$$

Its partial derivatives are the weights $w_j(z) = e^{\beta(\theta_j+z_j)}/\sum_k e^{\beta(\theta_k+z_k)}$. They are continuous, nonnegative, and sum to one, so each is bounded by one and $\sum_j w_j(z)^2 \le 1$. The scalar inequality applies to every coordinate section, giving $\mathrm{Var}(F_\beta(Z)) \le \mathbb{E}\sum_j w_j(Z)^2 \le 1$. Finally, $0 \le F_\beta(z) - R_\theta(z) \le \log(d)/\beta$. Both functions are square-integrable under the Gaussian law, and this uniform bound implies L_2 convergence as $\beta \to \infty$. Their variances therefore converge, proving $\mathrm{Var}_\theta(R_\theta) \le 1$.

The variance bound and the earlier bounds on $\mathbb{E}_\theta R_\theta$ now give

$$\mathbb{E}_\theta(\widehat{T}^* - T(\theta))^2 = \mathrm{Var}_\theta(R_\theta) + \left(\mathbb{E}_\theta R_\theta - \frac{a_d}{2}\right)^2 \le 1 + \max_{0\le u\le \mathbb{E}M_d}\left(u - \frac{a_d}{2}\right)^2.$$

For every $\lambda > 0$, Jensen's inequality and the Gaussian moment-generating function give

$$e^{\lambda \mathbb{E}M_d} \le \mathbb{E}e^{\lambda M_d} \le \sum_{j=1}^{d} \mathbb{E}e^{\lambda Z_j} = de^{\lambda^2/2}.$$

Thus $\mathbb{E}M_d \le (\log d)/\lambda + \lambda/2$. For $d \ge 2$, take $\lambda = a_d = \sqrt{2\log d}$ to obtain $\mathbb{E}M_d \le a_d$. Since $0 \le \mathbb{E}_\theta R_\theta \le a_d$, the preceding variance bound yields

$$\sup_{\theta\in\mathbb{R}^d} \mathbb{E}_\theta(\widehat{T}^* - T(\theta))^2 \le 1 + \frac{a_d^2}{4} = 1 + \frac{1}{2}\log d = \left(\frac{1}{2} + o(1)\right)\log d.$$

For $d = 1$, $a_1 = 0$ and $\widehat{T}^* = Y_1$, whose risk is exactly one.

Together with the lower bound in Section 4.3.1, this proves Theorem 4.10. □

B.1.2 Quadratic-functional benchmark derivations

We determine when restricting estimation of a quadratic functional to quadratic rules changes the attainable risk. Retain the Gaussian sequence model, target, and fixed classes of Section 4.2: n is precision, the noise variance is n^{-1}, and $L_p(\alpha, M)$ is defined in (4.7). The theorem below makes Remark 4.9's restricted benchmark precise. We also prove the upper bounds in Theorem 4.7; its unrestricted lower bounds are proved in the main text.

Use the admissible diagonal rules $\mathcal{Q}$ and restricted risk $R^*_{\mathcal{Q}}(n,\Theta)$ defined in (4.9)–(4.10). For a fixed diagonal quadratic rule, risk depends on the signal through its squared coordinates. A class Θ is *orthosymmetric* if it is invariant under coordinatewise sign changes; both classes considered here have this property. Write

$$\Theta^2 = \{(\theta_i^2)_{i\ge1} : \theta \in \Theta\}.$$

We call Θ *quadratically convex* if Θ^2 is a convex subset of the nonnegative cone. Equivalently, whenever $\theta, \theta' \in \Theta$ and $0 \le \lambda \le 1$, there exists $\zeta \in \Theta$ such that

$$\zeta_i^2 = \lambda\theta_i^2 + (1-\lambda)(\theta_i')^2 \qquad \text{for all } i.$$

To form the *closed quadratic convex hull*, denoted $\mathrm{Q.Hull}(\Theta)$, take the convex hull of Θ^2 and close it in the ordinary ℓ_1 norm. Each resulting profile x represents all sequences with coordinates $\pm\sqrt{x_i}$. The closure includes limits of finite convex combinations. The next theorem shows that the restricted minimax risk is unchanged by this enlargement and identifies when it exceeds the unrestricted benchmark.

Theorem B.1 (Quadratic-Estimator Benchmark over L_p and Besov Balls) *In model* (1.1) *with* $\varepsilon = n^{-1/2}$, *let*

$$\Theta \in \left\{L_p(\alpha, M),\ B^{\alpha}_{p,q}(M)\right\}, \qquad p, q, \alpha, M > 0, \qquad s = \alpha + \frac{1}{2} - \frac{1}{p} > 0.$$

Keep the class parameters fixed.

With the closed quadratic convex hull defined above,

$$R_Q^*(n, \Theta) = R_Q^*\big(n, \mathrm{Q.Hull}(\Theta)\big).$$

For $0 < p < 2$, *as* $n \to \infty$,

$$R_Q^*(n, \Theta) \asymp n^{-r_Q(\alpha,p)}, \qquad r_Q(\alpha, p) = \min\left\{1, \frac{8s}{1+4s}\right\}.$$

Consequently, if $0 < p < 2$ *and* $\alpha < 1/p - 1/4$, *then*

$$r_Q(\alpha, p) < r(\alpha, p),$$

where $r(\alpha, p)$ *is the exponent in Theorem 4.7; no rule in* $\mathcal{Q}$ *is then minimax-rate optimal. At* $\alpha = 1/p - 1/4$, *both exponents equal* 1.

For $q < 2$, the Besov hull has fine index 2, correcting the formula in Cai and Low [32, equation (13)]; for $q \ge 2$, the index is unchanged.

The proof has two parts. First, identify the quadratic convex hulls and combine projection bounds with the main-text lower bounds to obtain the restricted rate. Second, prove the unrestricted upper bounds using projection for $p \ge 2$ and centered thresholding for $p < 2$. The sharp constant $4A(\Theta)$ follows when the remainder terms are $o(n^{-1})$; at the parametric boundary, only rates are asserted. We finish by comparing the two exponents.

Quadratic-convex-hull reduction. For a finite quadratic form, averaging over coordinate sign changes removes the off-diagonal terms without increasing maximum squared-error risk on an orthosymmetric class, by Jensen's inequality and sign invariance. Fan's author report [64, Proposition 1] records the reduction for rectangular classes. For the infinite-sequence class $\mathcal{Q}$, use the admissible diagonal rules (4.9). For such a rule, independence and $L_2(P_\theta)$ convergence give

$$\mathbb{E}_\theta \widehat{Q} = \sum_{i \ge 1} a_i\left(\theta_i^2 + \frac{1}{n}\right) + b, \qquad \mathrm{Var}_\theta(\widehat{Q}) = \sum_{i \ge 1} a_i^2 \,\mathrm{Var}_\theta(Y_i^2) = \sum_{i \ge 1} a_i^2\left(\frac{2}{n^2} + \frac{4\theta_i^2}{n}\right).$$

Hence the risk is

$$\mathbb{E}_\theta(\widehat{Q} - Q(\theta))^2 = \Big[\sum_{i\ge1}(a_i - 1)\theta_i^2 + \frac{1}{n}\sum_{i\ge1} a_i + b\Big]^2 + \sum_{i\ge1} a_i^2\Big(\frac{2}{n^2} + \frac{4\theta_i^2}{n}\Big).$$

Write $R_{a,b}(x)$ for this risk as a function of $x = (\theta_i^2)_{i\ge1} \in \Theta^2$. It is convex: the variance is affine in x, and the bias is affine before squaring. Orthosymmetry ensures that every profile has a nonnegative representative in Θ. For any convex combination

$$x = \sum_{j=1}^m \lambda_j x^{(j)}, \qquad \lambda_j \ge 0, \quad \sum_{j=1}^m \lambda_j = 1, \quad x^{(j)} \in \Theta^2,$$

convexity gives

$$R_{a,b}(x) \le \sum_{j=1}^m \lambda_j R_{a,b}(x^{(j)}) \le \sup_{y\in\Theta^2} R_{a,b}(y).$$

For passage to the closed hull, we check both admissibility and continuity. Because zero belongs to both parameter classes, admissibility gives convergence at zero, hence $\sum_i a_i^2 < \infty$ and convergence of $\sum_i a_i/n$. Thus (a_i) is bounded, and every nonnegative $x \in \ell_1$ satisfies $\sum_i |a_i| x_i < \infty$ and $\sum_i a_i^2 x_i < \infty$. The mean series and sum of variances therefore converge at every signal with squared profile x, so the same series defines an $L_2(P_\theta)$-convergent rule on the closed hull. Restriction gives the converse inclusion of admissible rule classes. Moreover, the displayed bias and variance are continuous in the ordinary ℓ_1 norm. Convexification and closure therefore leave the supremum risk unchanged:

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta(\widehat{Q} - Q(\theta))^2 = \sup_{\theta\in \mathrm{Q.Hull}(\Theta)} \mathbb{E}_\theta(\widehat{Q} - Q(\theta))^2.$$

Taking the infimum over the common admissible class gives

$$R_Q^*(n,\Theta) = R_Q^*\big(n, \mathrm{Q.\,Hull}(\Theta)\big).$$

This is the quadratic-convex-hull principle of Cai and Low [32, Theorem 1].

Hulls of sparse L_p and Besov balls. Let $0 < p < 2$, $s = \alpha + \frac{1}{2} - \frac{1}{p}$, and $x_i = \theta_i^2$. Membership in $L_p(\alpha, M)$ means

$$\sum_{i\ge1} i^{ps}|\theta_i|^p = \sum_{i\ge1} i^{ps} x_i^{p/2} \le M^p.$$

Put $r = p/2 < 1$ and $y_i = i^{2s}x_i/M^2$. The constraint is $\sum_i y_i^r \le 1$. Since every $y_i \le 1$ and $y_i \le y_i^r$ on $[0,1]$, the squared-coordinate image lies in the weighted simplex:

$$\sum_i y_i \le \sum_i y_i^r \le 1, \qquad \sum_{i\ge1} i^{2s} x_i \le M^2.$$

Conversely, every vertex $M^2 i^{-2s} e_i$ of this simplex belongs to the original squared-coordinate set, and finite truncations of every point in the simplex are convex combinations of these vertices and the origin. These truncations converge in weighted ℓ_1 for this L_p class and therefore in the ordinary ℓ_1 topology defining the hull. This proves

$$\mathrm{Q.\,Hull}(L_p(\alpha, M)) = L_2\left(\alpha + \frac{1}{2} - \frac{1}{p}, M\right).$$

For a Besov ball, write $x_{jk} = \theta_{jk}^2$ and set

$$E_j = \sum_k x_{jk}, \qquad z_j = 2^{2js} E_j, \qquad s = \alpha + \frac{1}{2} - \frac{1}{p}.$$

For every original squared profile, the levelwise bound gives

$$\sum_k x_{jk} \le \left(\sum_k x_{jk}^{p/2} \right)^{2/p}, \qquad \sum_j z_j^{q/2} \le M^q.$$

If $q \ge 2$, this is a convex constraint on the nonnegative level energies. Conversely, fix a squared profile satisfying this convex constraint on the level energies. On each level with $E_j > 0$, place squared amplitude E_j on coordinate k with probability x_{jk}/E_j, independently across levels; leave zero-energy levels unchanged. The mean squared coordinate vector is the prescribed one. Every realization has at most one nonzero coordinate per level, so its block ℓ_p (quasi-)norm and ℓ_2 norm are equal and it belongs to the original class $B_{p,q}^{\alpha}(M)$. Finite-level truncation and closure therefore give the fine index q in this case.

If $0 < q < 2$, the nonnegative $\ell_{q/2}$ ball in $z = (z_j)$ is nonconvex, and two levels show why the fine index must change. For $j = 0, 1$, let $\theta^{(j)}$ have one nonzero coefficient of magnitude $M2^{-js}$ on level j and zero coefficients elsewhere. Both vectors belong to $B_{p,q}^{\alpha}(M)$, and their squared-coordinate midpoint $x = \{(\theta^{(0)})^2 + (\theta^{(1)})^2\}/2$ has $z_0 = z_1 = M^2/2$. The corresponding vector with squared profile x has Besov defining expression

$$\left\{ 2 \left(\frac{M}{\sqrt{2}} \right)^q \right\}^{1/q} = M 2^{1/q - 1/2} > M.$$

Thus the quadratic convex hull is not contained in $B_{2,q}^{s}(M)$ when $q < 2$. To identify the hull, convexify the level energies. Set $u_j := z_j/M^2$. Then $\sum_j u_j^{q/2} \le 1$, so $0 \le u_j \le 1$ and $u_j \le u_j^{q/2}$. Hence $\sum_j z_j = M^2 \sum_j u_j \le M^2$. Every vertex of that simplex is obtained by putting all energy on one level and, within that level, on one coordinate. Finite convex combinations and closure fill the simplex, which is exactly the squared-coordinate image of $B_{2,2}^{s}(M)$. The weighted simplex and both proposed Besov profile sets are closed in ordinary ℓ_1: convergence in that norm gives coordinatewise convergence, and Fatou's lemma preserves the defining constraints. Thus the closure adds no profiles outside these sets. Combining the two cases proves

$$\mathrm{Q.Hull}(B_{p,q}^{\alpha}(M)) = B_{2,q\vee 2}^{\alpha+1/2-1/p}(M).$$

This establishes the fine-index correction noted above; the rate exponent is unchanged because it does not depend on the fine index.

The quadratic-estimator exponent. The convexified classes have smoothness $s = \alpha + \frac{1}{2} - \frac{1}{p}$. First consider $L_2(s, M) = \Theta_s(M^2)$, using (4.7) and (3.2). For the projection rule $\widehat{Q}_m = \sum_{i \le m} (Y_i^2 - n^{-1})$,

$$\sup_{\theta \in L_2(s,M)} \mathbb{E}_\theta (\widehat{Q}_m - Q(\theta))^2 \le M^4 m^{-4s} + \frac{4M^2}{n} + \frac{2m}{n^2}.$$

If $s \le 1/4$, take $m \asymp n^{2/(4s+1)}$; if $s > 1/4$, choose $n^{1/(4s)} \ll m \ll n$. These choices give the upper rate $n^{-\{1\wedge 8s/(1+4s)\}}$. For $s \le 1/4$, apply the dense lower-bound argument in Theorem 4.7, proved in Section 4.2, with $p = 2$ and smoothness s. It gives $cn^{-8s/(4s+1)}$ on $L_2(s, M)$ for all estimators, hence also for the restricted quadratic rules. For $s > 1/4$, the radial van Trees argument in the same proof gives the lower order n^{-1}. These lower arguments also apply to $B^s_{2,q\vee 2}(M)$, with its fine index fixed. For its matching upper bound, projection through level J has $m \asymp 2^J$ coefficients, tail energy $O(M^2 2^{-2Js})$, and total energy $O(M^2)$; the same choices of m therefore give the upper rate. Thus the dense exponent at smoothness s gives

$$r_Q(\alpha, p) = \min\Big\{1, \frac{8s}{1+4s}\Big\}.$$

The threshold $r_Q(\alpha, p) = 1$ is therefore equivalent to

$$\frac{8s}{1+4s} \ge 1 \quad \Longleftrightarrow \quad 4s \ge 1 \quad \Longleftrightarrow \quad \alpha \ge \frac{1}{p} - \frac{1}{4}.$$

This proves the displayed form of $r_Q(\alpha, p)$ in Theorem B.1.

The unrestricted dense range $p \ge 2$. We now prove the unrestricted upper bounds in Theorem 4.7 on the original classes. In the dense range, use the projection estimator $\widehat{Q}_m = \sum_{i\le m}(Y_i^2 - n^{-1})$. Independence gives

$$\mathbb{E}_\theta(\widehat{Q}_m - Q(\theta))^2 = \left(\sum_{i>m}\theta_i^2\right)^2 + \frac{4}{n}\sum_{i\le m}\theta_i^2 + \frac{2m}{n^2}.$$

For $L_p(\alpha, M)$, the tail bound is immediate when $p = 2$. If $p > 2$, Hölder's inequality and $s = \alpha + 1/2 - 1/p$ give

$$\begin{aligned}\sum_{i>m}\theta_i^2 &\le \left(\sum_{i>m} i^{ps}|\theta_i|^p\right)^{2/p}\left(\sum_{i>m} i^{-2sp/(p-2)}\right)^{1-2/p}\\ &\le C_{p,\alpha}M^2 m^{-2\alpha}.\end{aligned}$$

The series is summable because $2sp/(p-2) > 1$. For a Besov ball, on resolution level j,

$$\|\theta_j\|_2 \le 2^{j(1/2-1/p)}\|\theta_j\|_p \le M2^{-j\alpha},$$

so summation above level J, where $m \asymp 2^J$, gives the same bound. Consequently

$$\sup_{\theta\in\Theta}\mathbb{E}_\theta(\widehat{Q}_m - Q(\theta))^2 \le Cm^{-4\alpha} + \frac{4A(\Theta)}{n} + \frac{2m}{n^2}. \tag{B.1}$$

When $\alpha \le 1/4$, taking $m \asymp n^{2/(1+4\alpha)}$ proves the upper rate $n^{-8\alpha/(1+4\alpha)}$. When $\alpha > 1/4$, choose $n^{1/(4\alpha)} \ll m \ll n$; then the first and third terms in (B.1) are $o(n^{-1})$.

A centered thresholding bound. For the sparse upper bound, let $X \sim N(\theta, n^{-1})$, let $\tau \ge 1$, and set

$$U_\tau(X) = (X^2 - \tau/n)_+ - \mathbb{E}_0(X^2 - \tau/n)_+.$$

There is a universal constant C such that

$$\left|\mathbb{E}_\theta U_\tau(X) - \theta^2\right| \le C \min\{\theta^2, \tau/n\}, \tag{B.2}$$

$$\mathrm{Var}_\theta\{U_\tau(X)\} \le \frac{6\theta^2}{n} + \frac{Cw(\tau)}{n^2}, \qquad w(\tau) = (1+\tau^2)\exp\left\{-\frac{(\sqrt{\tau}-1)_+^2}{2}\right\}. \tag{B.3}$$

To verify the bias bound, write $u = \sqrt{n}\theta$, $Z \sim N(0,1)$, and $F_\tau(u) = \mathbb{E}\{((Z+u)^2 - \tau)_+\}$. The scaled bias is $F_\tau(u) - F_\tau(0) - u^2$. Since $(x-\tau)_+ \le x$ for $x \ge 0$ and $(x-\tau)_+ \ge x - \tau$, its absolute value is at most τ. To obtain the complementary bound Cu^2, use symmetry and split at $|u| = 1$. The scaled bias is even and vanishes together with its first derivative at zero. For $0 \le u \le 1$, differentiation under the Gaussian integral gives

$$F_\tau''(u) = 2\mathbb{P}\{|Z+u| > \sqrt{\tau}\} + 2\sqrt{\tau}\{\varphi(\sqrt{\tau}-u) + \varphi(\sqrt{\tau}+u)\}.$$

This expression is uniformly bounded for $\tau \ge 1$. For $u \ge 1$,

$$|F_\tau'(u) - 2u| \le 2\mathbb{E}|Z+u| + 2u \le Cu,$$

and integration from 1 gives the same quadratic bound. Rescaling proves (B.2).

For the variance, the map $x \mapsto (x - \tau/n)_+$ is 1-Lipschitz. If $u^2 \ge 1$, the independent-copy identity for variance therefore gives

$$\mathrm{Var}\{U_\tau(X)\} \le \mathrm{Var}(X^2) = \frac{4\theta^2}{n} + \frac{2}{n^2} \le \frac{6\theta^2}{n}.$$

If $u^2 < 1$, subtracting the null expectation leaves the variance unchanged, and the variance is bounded by the uncentered second moment:

$$\begin{aligned} n^2\,\mathrm{Var}\{U_\tau(X)\} &\le \mathbb{E}\{((Z+u)^2 - \tau)_+^2\} \\ &\le C\mathbb{E}\left[(1+Z^4)\mathbf{1}\{|Z| > (\sqrt{\tau}-1)_+\}\right] \le Cw(\tau), \end{aligned}$$

where the last inequality follows by two integrations by parts of the Gaussian tail. This proves (B.3). The same calculation also gives

$$0 \le \mathbb{E}_0(X^2 - \tau/n)_+ \le Cn^{-1}\sqrt{\tau}e^{-\tau/2},$$

so every centering term is finite.

Block bounds and the sparse estimator. For $0 < p < 2$, the estimator uses unthresholded squares on an initial block and centered thresholded squares thereafter. The class constraints control the thresholding bias within each block and the energy omitted beyond the last block. For $L_p(\alpha, M)$, let $D_0(m) = \{1, \dots, m\}$ and

$$D_j(m) = \{i : 2^{j-1}m < i \le 2^j m\}, \qquad j \ge 1.$$

For a Besov ball, write $m = 2^{J_m}$ with $J_m \ge 0$ an integer. Let $D_0(m)$ contain the coefficients on levels $0 \le \ell < J_m$, and, for $j \ge 1$, let $D_j(m)$ contain those on level $\ell = J_m + j - 1$.

Use the single-sequence reindexing of the Besov coefficients from Section 4.2. In either case, $|D_j(m)| \le C2^j m$. The block bound below holds for $j \ge 1$, and the tail bound for $J \ge 0$:

$$\sum_{i \in D_j(m)} |\theta_i|^p \le C M^p 2^{-jps} m^{-ps}, \tag{B.4}$$

$$\sum_{i \notin D_0(m) \cup \cdots \cup D_J(m)} \theta_i^2 \le C M^2 (2^J m)^{-2s}. \tag{B.5}$$

For the L_p ball, the first inequality follows by replacing every i^{ps} on $D_j(m)$ by its lower endpoint; the second follows by $\|x\|_2 \le \|x\|_p$ on each block and summing a geometric series. For the Besov ball, the normalized quantity on resolution level ℓ, $2^{\ell s}\|\theta_\ell\|_p$, is at most M, and the same two conclusions follow level by level. Thus no convexity or triangle inequality for the full quasi-norm is used.

For the Besov class, round each subsequent choice of m to a power of two of the same order. Let J_n be the largest nonnegative integer for which

$$2^{J_n} m \le n^{1/(4s)} \log n;$$

if the inequality already fails for $J_n = 0$, set $J_n = 0$. Define

$$\widehat{Q}_m = \sum_{i \in D_0(m)} (Y_i^2 - n^{-1}) + \sum_{j=1}^{J_n} \sum_{i \in D_j(m)} \left\{ (Y_i^2 - 2j/n)_+ - \mathbb{E}_0 (Y_i^2 - 2j/n)_+ \right\}.$$

Since $2^j w(2j) = (1+4j^2)\exp\{-(1-\log 2)j + \sqrt{2j} - 1/2\}$, the series $\sum_{j \ge 1} 2^j w(2j)$ converges. Hence the variance bound (B.3), the block sizes, and (B.5) imply

$$\mathrm{Var}_\theta(\widehat{Q}_m) \le \frac{4}{n} \sum_{i \in D_0(m)} \theta_i^2 + C \left\{ \frac{m}{n^2} + \frac{m^{-2s}}{n} \right\}. \tag{B.6}$$

For the bias, use $\min(1, x) \le x^{p/2}$ for $x \ge 0$, since $0 < p/2 < 1$. Equations (B.2) and (B.4) give

$$\begin{aligned} |\mathbb{E}_\theta \widehat{Q}_m - Q(\theta)| &\le C \sum_{j=1}^{J_n} (j/n)^{1-p/2} \sum_{i \in D_j(m)} |\theta_i|^p + C M^2 (2^{J_n} m)^{-2s} \\ &\le C m^{-ps} n^{p/2-1} + C M^2 (2^{J_n} m)^{-2s}. \end{aligned} \tag{B.7}$$

The series $\sum_{j \ge 1} j^{1-p/2} 2^{-jps}$ is finite because $s > 0$.

The sparse upper bound. To obtain the sparse rate, balance squared thresholding bias against the noise contribution to variance. Suppose first that $\alpha p \le 1/2$ and take $m \asymp n^{p/(1+2ps)}$. The conditions $s > 0$ and $\alpha p \le 1/2$ imply $p > 1$, $ps \le (p-1)/2 < 1/2$, and $m \gtrsim n$. Since $p/(1+2ps) < 1/(4s)$, the cutoff eventually exceeds $2m$, so $J_n \ge 1$. Maximality gives

$$(2^{J_n} m)^{-2s} \le C n^{-1/2} (\log n)^{-2s}.$$

Since $m^{1+2ps} \asymp n^p$,

$$\{m^{-ps} n^{p/2-1}\}^2 \asymp \frac{m}{n^2}.$$

Equations (B.6) and (B.7) therefore yield

$$\sup_{\theta\in\Theta}\mathbb{E}_\theta(\widehat{Q}_m - Q(\theta))^2 \le C\frac{m}{n^2} \asymp n^{-\{2-p/(1+2ps)\}}. \tag{B.8}$$

This includes the endpoint $\alpha p = 1/2$, where $m \asymp n$.

For the sharp constant in the strict parametric regime, recalibrate the initial block so that the squared bias and variance remainders are negligible: if $\alpha p > 1/2$, take $m \asymp n/\log n$. When $s \le 1/4$, (B.7) gives

$$m^{-ps}n^{p/2-1} \asymp n^{-p\alpha}(\log n)^{ps} = o(n^{-1/2}), \qquad (2^{J_n}m)^{-2s} = o(n^{-1/2}).$$

When $s > 1/4$, m eventually exceeds the cutoff, so $J_n = 0$ and the unestimated tail is at most $Cm^{-2s} = o(n^{-1/2})$. In both cases the squared bias is $o(n^{-1})$. The last two terms in (B.6) are also $o(n^{-1})$, while its leading term is at most $4A(\Theta)/n$. Hence

$$\sup_{\theta\in\Theta}\mathbb{E}_\theta(\widehat{Q}_m - Q(\theta))^2 \le \frac{4A(\Theta)}{n} + o(n^{-1}). \tag{B.9}$$

The sharp parametric constant. The radial lower bound in Section 4.2 identifies the sharp constant $4A(\Theta)$ when $\alpha p_* > 1/2$. To see when M^2 can replace the maximal energy, compare $A(\Theta)$ and M^2 across Besov fine indices. If $p < 2$ and $q \le 2$, put $t_j = 2^{js}\|\theta_j\|_p$. Then

$$Q(\theta) \le \sum_j 2^{-2js}t_j^2 \le \sum_j t_j^2 \le \left(\sum_j t_j^q\right)^{2/q} \le M^2,$$

and equality is attained by one coefficient on level zero. The same conclusion holds for $L_p(\alpha, M)$ with $p < 2$. It need not hold for Besov balls with $q > 2$. For example, take $p = 1$, $q = 4$, $\alpha = 1$, so $s = 1/2$, put $u = (5/4)^{-1/4}$, and set

$$\theta_{0,1} = Mu, \qquad \theta_{1,1} = Mu/2, \qquad \theta_{j,k} = 0 \quad \text{otherwise}.$$

The two nonzero normalized level quantities are Mu and $Mu/\sqrt{2}$, whose fourth powers sum to $M^4u^4(1+1/4) = M^4$. Thus $\theta \in B^1_{1,4}(M)$, while $Q(\theta) = M^2u^2(1+1/4) = M^2\sqrt{5/4} > M^2$. Here both $\alpha > 1/p - 1/4$ and $\alpha > 1/(2p)$ hold strictly, so the example lies in both cited source regimes. The radial lower-bound argument in Section 4.2, applied to interior rescalings of this θ, therefore shows that the $4M^2/n$ sharp-constant assertions for all Besov fine indices in Cai and Low [32, equations (16) and (22)] cannot hold under its displayed sequence-ball definition when $q = 4$; the class-dependent constant $4A(\Theta)/n$ in our theorem is necessary.

The dense upper bound (B.1) and the sparse upper bound (B.9) prove the matching upper constant. Together with the main-text lower bounds, these calculations establish

$$R^*(n,\Theta) \asymp n^{-r(\alpha,p)}, \qquad r(\alpha,p) = \begin{cases} 1, & \alpha p_* \ge 1/2, \\ 2 - \dfrac{p_*}{1+2p_*s_*}, & \alpha p_* < 1/2 \end{cases},$$

and, when $\alpha p_* > 1/2$,

$$R^*(n,\Theta) = 4A(\Theta)n^{-1}(1+o(1)).$$

This proves Theorem 4.7. Donoho and Nussbaum [61] develop the dense quadratic-estimation strategy; see also Fan's author report [64, Corollaries 4–5]. The centered

thresholding construction and the random-support lower bound are the mechanisms developed by Cai and Low [32].

The strict exponent gap. Assume now $0 < p < 2$, retaining the standing condition $s = \alpha + 1/2 - 1/p > 0$.

If $1/(2p) < \alpha < 1/p - 1/4$, then $\alpha p > 1/2$ and $s < 1/4$, so the preceding calculations give

$$r(\alpha, p) = 1, \qquad r_Q(\alpha, p) = \frac{8s}{1+4s} < 1,$$

and $R^*(n, \Theta)$ is parametric while $R^*_Q(n, \Theta)$ is strictly slower. At $\alpha = 1/p - 1/4$, the same calculation gives $r_Q(\alpha, p) = 1$, so the exponent gap closes.

If $\alpha = 1/(2p)$, then $r(\alpha, p) = 1$ by Theorem 4.7, whereas $s = 1/2 - 1/(2p) < 1/4$ for $p < 2$, and hence $r_Q(\alpha, p) < 1$. If $\alpha < 1/(2p)$, then

$$r(\alpha, p) = 2 - \frac{p}{1+2ps}, \qquad r_Q(\alpha, p) = \frac{8s}{1+4s}.$$

A direct subtraction gives

$$\left(2 - \frac{p}{1+2ps}\right) - \frac{8s}{1+4s} = \frac{2-p}{(1+2ps)(1+4s)} > 0.$$

Therefore $r_Q(\alpha, p) < r(\alpha, p)$ for every $0 < p < 2$ with $\alpha < 1/p - 1/4$, while the two exponents agree at the boundary $\alpha = 1/p - 1/4$. This gives Theorem B.1. □

B.1.3 Average absolute Gaussian mean: full-space extension

The bounded-cube argument in Section 4.3.2 proves Theorem 4.14(i). For part (ii), the moment-matching priors have growing support, so both their separation and their mixture distance must be recalibrated.

Proof of Theorem 4.14(ii) Retain the unit-variance Gaussian model, the average of the absolute coordinate means as the target, and the squared-error loss of Section 4.3.2, now with $\theta \in \mathbb{R}^d$. We use support radius $M_d \asymp \sqrt{\log d}$ and match moments through a degree of order $\log d$. First bound the mixture distance, then show that the target's fluctuations under each product prior are negligible relative to the prior mean gap. Finally, choose the constants in the support radius and matching degree to satisfy both requirements.

Mixture distance. Take $M_d = c\sqrt{\log d}$ with a constant $c > 0$ to be chosen and let $k_d = 2\lceil b \log d/2 \rceil$ with $b > 0$. This even degree satisfies the hypothesis of Lemma 4.13 for all sufficiently large d. For $j = 0, 1$, denote the scaled symmetric moment-matching measures on $[-M_d, M_d]$ by $\nu_{j,d}$, and put $\Pi_{j,d} := \nu_{j,d}^{\otimes d}$. Write $g_{j,d}(y) = \varphi(y) r_{j,d}(y)$ for the one-coordinate mixture density induced by $\nu_{j,d}$. Symmetry and the support bound give

$$r_{0,d}(y) = \int e^{yt - t^2/2}\, \nu_{0,d}(dt) \geq e^{-M_d^2/2} = d^{-c^2/2},$$

so the one-observation series bound in Lemma A.12, with $a = M_d$, $K = k_d$, $\pi_j = \nu_{j,d}$, and $c_0 = d^{-c^2/2}$, gives

$$\chi^2(g_{1,d}, g_{0,d}) \le 4d^{c^2/2} \sum_{r=k_d+1}^{\infty} \frac{M_d^{2r}}{r!}.$$

The tail terms have successive ratios at most $M_d^2/(k_d+2) \to c^2/b$. For fixed $b, c > 0$ satisfying $b > ec^2$, a geometric bound and Stirling's formula give

$$\sum_{r=k_d+1}^{\infty} \frac{M_d^{2r}}{r!} \le \exp\left\{ -\left[b \log\left(\frac{b}{ec^2}\right) + o(1) \right] \log d \right\}.$$

Consequently

$$\chi^2(g_{1,d}^{\otimes d}, g_{0,d}^{\otimes d}) \to 0 \quad \text{if} \quad 1 + \frac{c^2}{2} - b \log\left(\frac{b}{ec^2}\right) < 0.$$

Target separation and risk. Put $m_{j,d} := \mathbb{E}_{\Pi_{j,d}} T(\theta) = \int |t| \, \nu_{j,d}(dt)$ for $j = 0, 1$. The prior mean gap is

$$m_{1,d} - m_{0,d} = 2M_d \delta_{k_d} = 2\beta_* \frac{M_d}{k_d}(1 + o(1)) = 2\beta_* \frac{c}{b} (\log d)^{-1/2}(1 + o(1)).$$

Since the prior standard deviation is $O(M_d/\sqrt{d}) = o((\log d)^{-1/2})$, Theorem 2.14 yields

$$\inf_{\widehat{T}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta (\widehat{T} - T(\theta))^2 \ge \beta_*^2 \frac{c^2}{b^2} \frac{1}{\log d}(1 + o(1)).$$

Choice of constants. Take $c^2 = 4/(9e^2)$ and $b = 1$. Then $b > ec^2$ and

$$1 + \frac{c^2}{2} - b \log\left(\frac{b}{ec^2}\right) = \frac{2}{9e^2} - \log\left(\frac{9}{4}\right) < 0.$$

Thus the product-χ^2 condition is satisfied. Substitution into the preceding lower bound gives

$$\inf_{\widehat{T}} \sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta (\widehat{T} - T(\theta))^2 \ge \frac{4\beta_*^2}{9e^2 \log d}(1 + o(1)).$$

This proves part (ii), with the constant extracted in Cai and Low [29, 37]. □

B.1.4 Fourier bound for the average absolute mean

In the unit-noise Gaussian model of Section 4.3.2, the target is $T(\theta) = d^{-1} \sum_{i=1}^d |\theta_i|$, with $\theta \in \mathbb{R}^d$. The following Fourier construction verifies the stated full-space upper rate.

For a cutoff $\lambda > 0$, define

$$c_0 = \int_{\mathbb{R}} \frac{1 - \cos t}{t^2} \, dt > 0, \qquad h_\lambda(y) = |y| - \frac{1}{c_0} \int_{-\lambda}^{\lambda} \frac{e^{t^2/2} - 1}{t^2} \cos(ty) \, dt, \tag{B.10}$$

with both integrands defined at zero by continuity. The estimate $d^{-1} \sum_i h_\lambda(Y_i)$ satisfies

$$\sup_{\theta \in \mathbb{R}^d} \mathbb{E}_\theta \left[\left\{ \frac{1}{d} \sum_i h_\lambda(Y_i) - T(\theta) \right\}^2 \right] \le \frac{4}{c_0^2 \lambda^2} + \frac{(1 + c_0^{-1} \lambda^2 e^{\lambda^2/2})^2}{d}. \tag{B.11}$$

Taking $\lambda^2 = \frac{1}{2}\log(d+1)$ gives risk of order $1/\log(d+1)$.

Proof For $\lambda > 0$, use c_0 and h_λ from (B.10). We bound the bias and variance of each summand before averaging. The constant c_0 is finite and positive: its integrand is bounded near zero and at most $2/t^2$ away from zero. For a scalar observation $Y \sim N(\mu, 1)$, the identity

$$|x| = c_0^{-1} \int_{\mathbb{R}} \frac{1 - \cos(tx)}{t^2}\, dt$$

and $\mathbb{E}_\mu \cos(tY) = e^{-t^2/2}\cos(t\mu)$ show, after cancellation on $[-\lambda, \lambda]$, that

$$\mathbb{E}_\mu h_\lambda(Y) - |\mu| = c_0^{-1} \int_{|t|>\lambda} \frac{(1 - e^{-t^2/2})\cos(t\mu)}{t^2}\, dt.$$

Tonelli's theorem permits interchange of expectation and integration in the nonnegative absolute-value representation; Fubini's theorem applies to the bounded-cutoff correction. The bias is at most $2/(c_0\lambda)$ in absolute value, uniformly in μ, because $\int_{|t|>\lambda} t^{-2}\, dt = 2/\lambda$. A direct difference bound controls the variance without differentiating at the cusp. For $|t| \le \lambda$,

$$0 \le \frac{e^{t^2/2} - 1}{t^2} \le \frac{1}{2} e^{\lambda^2/2}, \qquad |\cos(tx) - \cos(ty)| \le |t|\,|x - y|.$$

The first inequality follows from $e^u - 1 \le u e^u$ for $u \ge 0$, with the value at zero obtained by continuity. Integrating the difference of the two correction terms and using $\big||x| - |y|\big| \le |x - y|$ gives

$$|h_\lambda(x) - h_\lambda(y)| \le L_\lambda |x - y|, \qquad L_\lambda = 1 + c_0^{-1}\lambda^2 e^{\lambda^2/2}.$$

Since $\mathbb{E}_\mu(Y - \mu)^2 = 1$,

$$\mathrm{Var}_\mu\{h_\lambda(Y)\} \le \mathbb{E}_\mu\{h_\lambda(Y) - h_\lambda(\mu)\}^2 \le L_\lambda^2.$$

Independence and the bias–variance decomposition give (B.11). Taking $\lambda^2 = \frac{1}{2}\log(d+1)$ makes the squared-bias term $O(1/\log(d+1))$ and the variance term smaller asymptotically; a larger constant covers the finite initial values of d. □

B.1.5 Projection bound for density estimation

Piecewise-polynomial projection gives an alternative to the kernel estimator in Remark 4.17. We record its risk bound and the Taylor approximation used in the confidence-band construction of Section 7.5.

Projection-estimator risk bound Let $X_1, \dots, X_n$ be independent observations with density $f \in \mathcal{D}_\beta(M)$, so $\|f\|_\infty \le M$, and put $r = \lceil \beta \rceil - 1$ and $\nu_\beta = \beta - r$. All function norms and inner products here use Lebesgue measure on $[0, 1]$. Partition $[0, 1]$ into N equal cells. Let S_N be the space of piecewise polynomials of degree at most r on these cells, with a fixed orthonormal basis $\varphi_1, \dots, \varphi_d$, $d = N(r+1)$, using bounded measurable representatives. Write Π_N for its L_2 projection and define

$$\widehat{f}_N = \sum_{\nu=1}^{d} \left\{ \frac{1}{n} \sum_{i=1}^{n} \varphi_\nu(X_i) \right\} \varphi_\nu.$$

On a cell beginning at a, the degree-r Taylor polynomial P_a has error at most $MN^{-\beta}/r!$. For $r = 0$ this is the Hölder condition; for $r \geq 1$, it follows from

$$f(x) - P_a(x) = \frac{1}{(r-1)!} \int_a^x (x-t)^{r-1} \{f^{(r)}(t) - f^{(r)}(a)\}\, dt,$$
$$|f(x) - P_a(x)| \leq \frac{M}{r!}(x-a)^{r+\nu_\beta}.$$

Thus $\|f - \Pi_N f\|_2^2 \leq M^2 N^{-2\beta}/(r!)^2$. Since $\mathbb{E}_f \widehat{f}_N = \Pi_N f$, independence of observations and orthonormality give

$$\begin{aligned} \mathbb{E}_f \|\widehat{f}_N - f\|_2^2 &= \|f - \Pi_N f\|_2^2 + \frac{1}{n} \sum_{\nu=1}^{d} \mathrm{Var}_f\{\varphi_\nu(X_1)\} \\ &\leq \frac{M^2}{(r!)^2} N^{-2\beta} + \frac{M(r+1)N}{n}. \end{aligned} \tag{B.12}$$

Orthonormality removes cross terms even though the estimated coefficients can be dependent. Balancing the approximation and variance terms with $N = \lceil n^{1/(2\beta+1)} \rceil$ proves the upper rate uniformly over $\mathcal{D}_\beta(M)$ for every $n \geq 1$, matching Theorem 4.16 and Example 3.3. The argument includes integer smoothness and uses no extension beyond $[0,1]$. Nonnegativity of the estimator is not required because the decision space is $L_2[0,1]$. □

B.2 Estimation comparisons for Chapter 5

In the Gaussian sequence experiment (3.1) on $\Theta_s(R^2)$, with known $s, R > 0$, we compare vector and quadratic-functional estimation with the zero-null testing benchmarks of Section 5.6. For the quadratic functional, we also examine the estimation obstruction away from zero.

For vector estimation, Example 3.2, with $\alpha = s$ and $M = R^2$, gives the squared-error risk lower rate $n^{-2s/(2s+1)}$. Keeping the first m observations and setting the remaining coefficients to zero gives risk at most $m/n + R^2(m+1)^{-2s}$; taking $m \asymp n^{1/(2s+1)}$ matches that rate. Thus the minimax root mean squared ℓ_2 error has order $n^{-s/(2s+1)}$, which is larger than the testing norm radius $n^{-2s/(4s+1)}$.

From quadratic-functional estimation to detection. An estimator of a quadratic functional induces a test when its error is small relative to the target gap. In model (3.1), let $\Theta \subset \ell_2$ contain 0, retain the quadratic functional (4.5), and test $\theta = 0$ against $\theta \in \Theta$ with $Q(\theta) \geq a$. Assume $a > 0$ and the alternative is nonempty. The reduction discussed by Cai and Low [32, Section 5] connects this testing problem to the quadratic-functional benchmarks of Section 4.2.

Write $\mathcal{E}_\Theta(a)$ for the minimax total error (5.1) in this testing problem, and retain the quadratic-functional risk $R^*(n, \Theta)$ from (4.6). Here a is an energy gap, corresponding to norm separation $\sqrt{a}$. Threshold any estimator $\widehat{Q}$ at $a/2$. Under the null, a false rejection implies $|\widehat{Q} - Q(0)| \geq a/2$; under every alternative, a false acceptance implies $|\widehat{Q} - Q(\theta)| \geq a/2$. Markov's inequality applied to squared error therefore gives

$$\mathcal{E}_\Theta(a) \leq \frac{8}{a^2} \sup_{\theta \in \Theta} \mathbb{E}_\theta \big(\widehat{Q} - Q(\theta)\big)^2, \qquad R^*(n, \Theta) \geq \frac{a^2}{8} \mathcal{E}_\Theta(a). \tag{B.13}$$

Thus a fixed positive lower bound on testing error at energy a forces estimation risk of order at least a^2; conversely, uniform squared-error risk for the quadratic functional small relative to a^2 supplies a test with small total error. The comparison uses the scale of the quadratic functional Q, and the hardest estimation parameters need not lie near zero.

Return now to the Sobolev ellipsoid $\Theta_s(R^2) = L_2(s, R)$ and recall the testing radius $r_n = n^{-2s/(4s+1)}$. Theorem 4.7 gives the minimax root mean squared error for estimating Q:

$$\left[\inf_{\widehat{Q}} \sup_{\theta\in\Theta_s(R^2)} \mathbb{E}_\theta\{\widehat{Q} - Q(\theta)\}^2\right]^{1/2} \asymp \max\{n^{-4s/(4s+1)}, n^{-1/2}\}. \tag{B.14}$$

The projection calculation (B.1), with $\alpha = s$ and $M = R$, supplies the upper order. For the matching lower bound on root mean squared error, testing against zero gives the term $n^{-4s/(4s+1)}$, while a two-point comparison away from zero gives the term $n^{-1/2}$.

Both terms are necessary. For the nonparametric term, apply (B.13) with energy gap $a = \rho^2$ and $\rho = cr_n$. Theorem 5.11 gives total error at least $3/4$, hence squared estimation risk at least $3\rho^4/32$. For the parametric term, now use $a \in (0, R)$ for a fixed first-coordinate amplitude and compare $\theta^{(0)} = (a, 0, \ldots)$ with $\theta^{(1)} = (a + h/\sqrt{n}, 0, \ldots)$ for fixed $0 < h \le 1/2$ and large n; both belong to the ellipsoid. Their Gaussian KL is $h^2/2$, so Pinsker gives total variation at most $h/2$. Their quadratic targets differ by $2ah/\sqrt{n} + O(n^{-1})$. Lemma 2.1 and its squared-error specialization in Remark 2.2 give squared-error risk at least one quarter of the squared target gap times the testing affinity. The gap is of order $n^{-1/2}$ and the affinity is bounded away from zero, giving the lower order n^{-1}. Thus the upper and lower rates match.

Measured by the quadratic functional Q, the testing gap $r_n^2 = n^{-4s/(4s+1)}$ matches the root mean squared error for estimating Q in order when $s \le 1/4$. For $s > 1/4$, detection needs a smaller gap: the energy statistic has no signal-dependent variance at zero, whereas comparing the two nonzero signals above gives the parametric estimation lower bound.

For the sparse L_p and Besov classes of Section 4.2, with $0 < p < 2$, $\alpha + 1/2 - 1/p > 0$, and $\alpha \le 1/(2p)$, Cai and Low [32, Section 5] identify an energy-separation rate $a_n \asymp n^{-r(\alpha,p)/2}$, where $r(\alpha, p)$ is the squared-error risk exponent in Theorem 4.7. Their estimation results alone do not determine a sharp testing boundary in the intervening range $1/(2p) < \alpha \le 1/p$.

B.3 Proofs for Chapter 6: honest confidence sets

This section proves the refinements of the Gaussian band-width bound that hold at every hypercube vertex and the miscoverage obstruction for relaxed bands. The worst-case Gaussian bound and its function-space reduction are proved in Section 6.5.1; the single-level confidence-ball proof appears in Section 6.6.1.

Proof of the assertions holding at every vertex in Proposition 6.16 We prove the two assertions that hold at every vertex, retaining the proposition's coordinate intervals and observation laws. If the procedure uses independent parameter-free randomization R, condition on (X, R) in the posterior calculations. Independence leaves the posterior unchanged; coverage is averaged over both X and R.

Equal lengths. Assume that the coordinate intervals have the same random length. We use a star prior centered at each vertex: coverage under the mixture forces a lower bound on the

common length, which we then transfer to the central vertex. Fix an arbitrary $\theta^{(0)}$ and put the uniform prior on the $(n+1)$-point star consisting of $\theta^{(0)}$ and its single-coordinate flips $\theta^{(i)}$, $1 \le i \le n$. If $L_i = dP_{\theta^{(i)}}/dP_{\theta^{(0)}}$, then under $P_{\theta^{(0)}}$ the L_i are independent and

$$\mathbb{E}_{\theta^{(0)}} L_i = 1, \qquad \mathbb{E}_{\theta^{(0)}} L_i^2 = e^{c_n^2}, \qquad \mathbb{E}_{\theta^{(0)}} L_i L_j = 1 \quad (i \neq j).$$

Let $\bar{P}$ be the observation law under the uniform star prior. Its likelihood ratio relative to $P_{\theta^{(0)}}$ is

$$\bar{L} = \frac{1 + \sum_{i=1}^n L_i}{n+1}, \qquad \chi^2(\bar{P}, P_{\theta^{(0)}}) = \frac{n(e^{c_n^2} - 1)}{(n+1)^2} = o(1),$$

both when $c_n^2 = c \log n$ with $c < 1$ and when c_n is fixed.

Let $p_{\max}(X)$ be the largest posterior mass of a star vertex. The uniform star prior gives

$$p_{\max}(X) = \frac{\max\{1, L_1, \dots, L_n\}}{1 + \sum_{i=1}^n L_i}.$$

Changing measure by $\bar{L}$ cancels the denominator. Since $\max\{1, L_1, \dots, L_n\} \le (1 + \sum_i L_i^2)^{1/2}$, Cauchy–Schwarz yields

$$\begin{aligned} \mathbb{E}_{\bar{P}} p_{\max}(X) &= \frac{1}{n+1} \mathbb{E}_{\theta^{(0)}} \max\{1, L_1, \dots, L_n\} \\ &\le \frac{\sqrt{1 + \sum_{i=1}^n \mathbb{E}_{\theta^{(0)}} L_i^2}}{n+1} = \frac{\sqrt{1 + n e^{c_n^2}}}{n+1} = o(1). \end{aligned}$$

Both limits are uniform over the central vertex $\theta^{(0)}$ in the stated regimes.

On the event that the common coordinate length is below 1, the product of coordinate intervals can contain at most one star vertex. Conditional on X and any auxiliary randomization, its posterior probability of covering the latent true vertex is therefore at most $p_{\max}(X)$. Let Q denote the joint law of the star vertex and the data, with data marginal $\bar{P}$. Averaging honesty over the star prior gives

$$1 - \alpha \le \mathbb{P}_Q\{\theta \in C(X)\} \le \mathbb{P}_{\bar{P}}\{L(C_1) \ge 1\} + \mathbb{E}_{\bar{P}} p_{\max}(X).$$

Since $\mathrm{TV}(\bar{P}, P_{\theta^{(0)}}) \to 0$,

$$\mathbb{P}_{\theta^{(0)}}\{L(C_1) \ge 1\} \ge 1 - \alpha - o(1).$$

The vertex $\theta^{(0)}$ was arbitrary. Therefore, for every $\delta > 0$ and all sufficiently large n, every $\theta \in \Theta_n$ satisfies

$$\mathbb{E}_{X|\theta} \sum_{i=1}^n L(C_i(X)) = n \mathbb{E}_{X|\theta} L(C_1(X)) \ge (1 - \alpha - \delta) n.$$

Fixed signal, arbitrary lengths. Assume $\alpha < 1/2$, as in the proposition's final assertion. Fix a vertex θ and let $\theta^{(i)}$ be obtained by flipping coordinate i. The two product-normal laws differ only in that coordinate, and Pinsker's inequality gives

$$\mathrm{TV}(P_\theta, P_{\theta^{(i)}}) \le \sqrt{\frac{1}{2} \mathrm{KL}(P_\theta, P_{\theta^{(i)}})} = \frac{c_n}{2}.$$

Simultaneous honesty implies coverage for coordinate i at both vertices. The coverage-transfer argument of Lemma 6.3, which also applies when $\alpha = 0$, gives

$$\mathbb{P}_\theta\{\theta_i, \theta_i^{(i)} \in C_i(X)\} \geq 1 - 2\alpha - \mathrm{TV}(P_\theta, P_{\theta^{(i)}}).$$

On this event the interval has length at least one. Hence

$$\mathbb{E}_\theta L(C_i(X)) \geq 1 - 2\alpha - \frac{c_n}{2}.$$

Choose $c_\alpha \leq 1 - 2\alpha$. Then the last display is at least $C_\alpha = (1 - 2\alpha)/2$ whenever $c_n \leq c_\alpha$. Summing over $i = 1, \ldots, n$ proves the final assertion, uniformly over every vertex θ and every n. □

Proof of Theorem 6.19 Fix $0 < M' < M$ and $h \in \Lambda^\beta(M')$, and put

$$a_n = \begin{cases} n^{-\beta}, & 0 < \beta \leq 1/4, \\ n^{-2\beta/(1+4\beta)}, & \beta > 1/4, \end{cases} \qquad W = \int_0^1 \{U(t) - L(t)\}\, dt.$$

Mixture construction. Let g be a nonnegative C^∞ function supported on $[0, 1]$, positive on $(0, 1)$, with all derivatives vanishing at the endpoints. If $\beta > 1/4$, set

$$m = \left\lfloor n^{2/(1+4\beta)} \right\rfloor, \qquad I_i = [(i-1)/m, i/m], \qquad x_i = (i-1)/m.$$

Choose $a_0 > 0$ sufficiently small, depending on the Hölder normalization and on $M - M'$, and set $A_n = a_0 m^{-\beta}$. For $B = (B_1, \ldots, B_m) \in \{-1, 1\}^m$, define

$$f_B(t) = h(t) + A_n \sum_{i=1}^m B_i g\{m(t - x_i)\}. \tag{B.15}$$

The bump interiors are disjoint, and all bumps vanish at shared endpoints. Put $\delta_H = (M - M')/2$ and choose a_0 small enough that every derivative bound and the top Hölder seminorm of $f_B - h$ are at most δ_H, uniformly over the signs and the resolution. For points in different cells, the vanishing boundary derivatives and the cross-cell argument in Section 4.4.1 give at most twice the within-cell seminorm bound; include this factor when choosing a_0. Since $h \in \Lambda^\beta(M')$, the triangle inequality bounds each corresponding quantity for f_B by $M' + \delta_H < M$. Thus every f_B belongs to $\Lambda^\beta(M)$. Moreover,

$$A_n \asymp a_n. \tag{B.16}$$

Let $P_0 = P_h$ and $P_\pi = 2^{-m} \sum_B P_{f_B}$. For the design points in I_i, define the standardized observations and mean-shift vector by

$$Z_i = \left(\frac{y_j - h(t_j)}{\sigma} : t_j \in I_i \right),$$
$$\gamma_i = \left(\frac{A_n}{\sigma} g\{m(t_j - x_i)\} : t_j \in I_i \right).$$

For one block, averaging the Gaussian likelihood ratios for the two signs gives

$$L_i(Z_i) = \frac{1}{2} \exp\left(\gamma_i^\top Z_i - \frac{1}{2}\|\gamma_i\|_2^2 \right) + \frac{1}{2} \exp\left(-\gamma_i^\top Z_i - \frac{1}{2}\|\gamma_i\|_2^2 \right).$$

Since $\gamma_i^\top Z_i \sim N(0, \|\gamma_i\|_2^2)$ under P_0, direct Gaussian integration gives

$$\mathbb{E}_0 L_i^2 = \cosh(\|\gamma_i\|_2^2).$$

Assign each shared cell endpoint to just one cell; its bump coefficient is zero. The resulting block likelihoods depend on disjoint Gaussian coordinates and are independent. Each block contains at most $n/m + 2$ design points, so $\|\gamma_i\|_2^2 \le (A_n^2 \|g\|_\infty^2/\sigma^2)(n/m + 2)$. Since $\beta > 1/4$, eventually $2m \le n$ and $n/m + 2 \le 2n/m$. Using $\log\cosh x \le x^2/2$ therefore yields

$$\begin{aligned} 1 + \chi^2(P_\pi, P_0) &= \prod_{i=1}^{m} \cosh(\|\gamma_i\|_2^2) \\ &\le \exp\left\{ \frac{1}{2} \sum_{i=1}^{m} \|\gamma_i\|_2^4 \right\} \\ &\le \exp\left\{ C_{g,\sigma} m A_n^4 (n/m)^2 \right\} \le \exp(C_{g,\sigma} a_0^4). \end{aligned} \tag{B.17}$$

Integer rounding changes only the constant because $n/m \to \infty$ when $\beta > 1/4$. Given any fixed $\delta > 0$, we may consequently choose a_0 still smaller so that

$$\mathrm{TV}(P_\pi, P_0) \le \frac{1}{2}\sqrt{\chi^2(P_\pi, P_0)} \le \delta. \tag{B.18}$$

For $0 < \beta \le 1/4$, switch from second-moment control to exact agreement at the design points: use $m = n$, $I_i = [(i-1)/n, i/n]$, $x_i = (i-1)/n$ for $1 \le i \le n$, and $A_n = a_0 n^{-\beta}$ in (B.15). Every design point is then an endpoint of every bump that meets it, so $f_B(t_j) = h(t_j)$ for all j and all B. Hence $P_\pi = P_0$ exactly. In this case

$$A_n = a_0 n^{-\beta} = a_0 a_n. \tag{B.19}$$

Thus, in both regimes, there is a fixed $c_A > 0$ such that $A_n \ge c_A a_n$, all alternatives lie in $\Lambda^\beta(M)$, and (B.18) holds, with $\delta = 0$ in the second regime.

With μ denoting Lebesgue measure on $[0,1]$, positivity of g on $(0,1)$ gives, for every $\delta_{\mathrm{cov}} > 0$, a $q = q(\delta_{\mathrm{cov}}) > 0$ such that $\mu\{u \in [0,1] : g(u) \ge q\} \ge 1 - \delta_{\mathrm{cov}}$. Consequently, for every sign vector B,

$$S_3(B) = \{t : |f_B(t) - h(t)| \ge q A_n\}, \qquad \mu\{S_3(B)\} \ge 1 - \delta_{\mathrm{cov}}.$$

Common coverage-to-width implication. The remaining argument uses four inputs: h and every f_B lie in the coverage class; $\mathrm{TV}(P_\pi, P_h) \le \delta$; $\mu\{S_3(B)\} \ge 1 - \delta_{\mathrm{cov}}$ for every B; and $A_n \ge c_A a_n$ with fixed $c_A > 0$. Retain the theorem's measurability assumptions and zero-width conventions for each relaxed guarantee.

Suppose $\mathbb{E}_h W < b a_n$ for sufficiently small fixed $b > 0$. A narrow band that meets the guarantee at h must fail at every f_B on a common data event. Transferring that event to the mixture yields a contradiction. This argument also applies to the white-noise construction in Appendix B.4.4, once its four inputs are checked.

Uncovered measure. Under the first guarantee, choose $\gamma_1, \gamma_2, \delta_{\mathrm{cov}}, \delta > 0$ so that

$$\gamma_1 + \delta < 1 - 2\alpha, \qquad \gamma_2 + \delta_{\mathrm{cov}} < \varepsilon, \tag{B.20}$$

and then choose $b > 0$ so small that $b/(\gamma_1\gamma_2) < q c_A$. The first inequality leaves room for probability losses from Markov's inequality and coverage transfer. The second leaves room

for spatial losses from locations where the band is wide or the perturbation is small. Suppose, to obtain a contradiction, that $w(CB, h) = \mathbb{E}_h W < ba_n$. Markov's inequality gives

$$\mathbb{P}_0(E_W) \geq 1 - \gamma_1, \qquad E_W = \{W \leq ba_n/\gamma_1\}.$$

On E_W, the set

$$S_1 = \left\{t : U(t) - L(t) \leq \frac{ba_n}{\gamma_1\gamma_2}\right\}$$

has measure at least $1 - \gamma_2$. Put $S_2 = \{t : h(t) \in [L(t), U(t)]\}$. The assumed guarantee at h gives $\mu(S_2) > 1/2 + \varepsilon$ with P_0-probability at least $1 - \alpha$. Thus the data event $E = E_W \cap \{\mu(S_2) > 1/2 + \varepsilon\}$ satisfies $\mathbb{P}_0(E) \geq 1 - \alpha - \gamma_1$ and, by (B.18),

$$\mathbb{P}_\pi(E) \geq 1 - \alpha - \gamma_1 - \delta > \alpha. \tag{B.21}$$

On E, put $S = S_1 \cap S_2$. Then $\mu(S) > 1/2 + \varepsilon - \gamma_2$. At every point of $S \cap S_3(B)$ the interval contains $h(t)$, but its width is strictly smaller than $qA_n \leq |f_B(t) - h(t)|$; it therefore cannot contain $f_B(t)$. Thus, for every B,

$$\mu\{N(CB, f_B)\} \geq \mu\{S \cap S_3(B)\} > \frac{1}{2} + \varepsilon - \gamma_2 - \delta_{\mathrm{cov}} > \frac{1}{2}. \tag{B.22}$$

Since (B.22) holds for every B on the same data event E, (B.21) implies that the uniform average of $\mathbb{P}_{f_B}\{\mu(N(CB, f_B)) \geq 1/2\}$ exceeds α. Some $f_B \in \Lambda^\beta(M)$ therefore violates the uncovered-measure guarantee.

Relative excess. For the second guarantee, retain its fixed relative-excess bound $r > 0$ and use integrated separation in place of the uncovered-measure argument. Choose $\gamma_1, \delta > 0$ and $0 < \delta_{\mathrm{cov}} < 1$ so that

$$\gamma_1 + \delta < 1 - 2\alpha. \tag{B.23}$$

With q and c_A supplied by the construction above, choose $b > 0$ so small that

$$\frac{(2r+1)b}{\gamma_1} < qc_A(1 - \delta_{\mathrm{cov}}). \tag{B.24}$$

Suppose that $\mathbb{E}_h W < ba_n$. Markov's inequality and the guarantee at h give

$$E' = \{W \leq ba_n/\gamma_1,\ RE(CB, h) \leq r\}, \qquad \mathbb{P}_0(E') \geq 1 - \alpha - \gamma_1.$$

The key observation is that projecting $f_B(t)$ and $h(t)$ onto $[L(t), U(t)]$ and applying the triangle inequality gives

$$|f_B(t) - h(t)| \leq e_{f_B}(t) + \{U(t) - L(t)\} + e_h(t). \tag{B.25}$$

Here the two excess terms measure the distances of the functions from the reported interval. If both relative excesses were at most r, their integrals would each be at most rW; the interval itself contributes W. This explains the factor $2r + 1$. If $RE(CB, f_B) \leq r$ also held on E', integration would imply

$$\begin{aligned} qc_A a_n(1 - \delta_{\mathrm{cov}}) \leq qA_n\mu\{S_3(B)\} &\leq \int_0^1 |f_B(t) - h(t)|\, dt \\ &\leq (2r+1)W \leq \frac{(2r+1)ba_n}{\gamma_1} < qc_A a_n(1 - \delta_{\mathrm{cov}}), \end{aligned} \tag{B.26}$$

a contradiction. This also covers $W = 0$: the zero-width convention forces both excess integrals to vanish if both relative excesses are at most r. Thus $E' \subseteq \{RE(CB, f_B) > r\}$ for every B. Transferring this common data event by (B.18) and averaging over the signs gives

$$2^{-m} \sum_{B} \mathbb{P}_{f_B}\{RE(CB, f_B) > r\} \geq \mathbb{P}_{\pi}(E') \geq 1 - \alpha - \gamma_1 - \delta > \alpha.$$

Some $f_B \in \Lambda^{\beta}(M)$ therefore violates the relative-excess guarantee.

Both contradictions give

$$w(CB, h) \geq b \begin{cases} n^{-\beta}, & 0 < \beta \leq 1/4, \\ n^{-2\beta/(1+4\beta)}, & \beta > 1/4, \end{cases},$$

uniformly over $h \in \Lambda^{\beta}(M')$ for all sufficiently large n: the amplitude and probability constants above depend on the fixed parameters and the slack $M - M'$, not on h. Taking $h = 0$ and any fixed $M' < M$ yields the worst-case bound. The two rates arise from the mixture argument above $\beta = 1/4$ and from perturbations that vanish at every design point at and below it. □

B.4 Proofs for Chapter 7: the cost of adaptation

The lower bounds and benchmark comparisons for fully honest bands and confidence balls are proved in Sections 7.5 and 7.6. This section gives the selector and interval constructions, the adaptive quadratic-functional upper bound, and the relaxed-band lower bound, each under the model and coverage or risk criterion of its main-text statement.

B.4.1 Two-class adaptive selector and fourth-moment bound

We construct one estimator that preserves the first-class risk while attaining the second-class bound in Theorem 7.1. There are two selection steps. The first produces a fallback estimator valid over the union of classes. The second retains the more accurate first-class estimator unless the data reject it. Affine estimators with directional bias bounds provide the comparisons for both steps. The ordered-modulus construction follows Cai and Low [31, 33]. Retain the theorem's setup and notation $a_1, a_2, a_{12}, \gamma_n^+$.

Ordered-modulus bias–variance tradeoff. Use the Gaussian measurements (1.3) with $\varepsilon = n^{-1/2}$: $\mathbb{E}_f Y(u) = \langle f, u\rangle_H$ and $\mathrm{Var}_f\, Y(u) = \|u\|_H^2/n$. As in Section 7.2, write the Hilbert norm as $\|\cdot\|_2$. For closed convex classes $\mathcal{F}$ and $\mathcal{H}$ with nonempty intersection and a variance budget $V \geq 0$, define

$$B(V; \mathcal{F}, \mathcal{H}) = \frac{1}{2} \sup_{\varepsilon > 0} \left\{ \omega(\varepsilon, \mathcal{F}, \mathcal{H}) - \sqrt{nV}\, \varepsilon \right\}.$$

The map $\varepsilon \mapsto \omega(\varepsilon, \mathcal{F}, \mathcal{H})$ is concave: if $(f_k, g_k) \in \mathcal{F} \times \mathcal{H}$ is admissible at radius ε_k, convexity makes $\big((1-t)f_0 + tf_1, (1-t)g_0 + tg_1\big)$ admissible at radius $(1-t)\varepsilon_0 + t\varepsilon_1$. Linearity of T gives the same convex combination of target differences; taking suprema proves concavity. For the scaling bound, choose $f_* \in \mathcal{F} \cap \mathcal{H}$ and contract both endpoints of a radius-$D\varepsilon$ pair

toward f_* by the factor $1/D$. Their distance and target difference shrink by the same factor, so

$$\omega(D\varepsilon, \mathcal{F}, \mathcal{H}) \le D\omega(\varepsilon, \mathcal{F}, \mathcal{H}), \qquad D \ge 1. \tag{B.27}$$

There is an affine Gaussian estimator $\widehat{T}_{\mathcal{F},\mathcal{H},V}$ satisfying

$$\operatorname{Var}(\widehat{T}_{\mathcal{F},\mathcal{H},V}) \le V, \tag{B.28}$$

$$\sup_{f\in\mathcal{F}}\{\mathbb{E}_f \widehat{T}_{\mathcal{F},\mathcal{H},V} - T(f)\} \le B(V;\mathcal{F},\mathcal{H}), \tag{B.29}$$

$$\inf_{g\in\mathcal{H}}\{\mathbb{E}_g \widehat{T}_{\mathcal{F},\mathcal{H},V} - T(g)\} \ge -B(V;\mathcal{F},\mathcal{H}). \tag{B.30}$$

Attained case. Suppose the supremum defining B is attained at $f_0 \in \mathcal{F}$, $g_0 \in \mathcal{H}$, with $\varepsilon_0 = \|g_0 - f_0\|_2 > 0$. Put $u = (g_0 - f_0)/\varepsilon_0$ and

$$c = T\left(\frac{f_0+g_0}{2}\right) - \sqrt{nV}\left\langle u, \frac{f_0+g_0}{2}\right\rangle, \qquad \widehat{T}_{\mathcal{F},\mathcal{H},V} = c + \sqrt{nV}\,Y(u).$$

The variance is V. Moving f_0 toward any $f \in \mathcal{F}$ with g_0 fixed, or g_0 toward any $g \in \mathcal{H}$ with f_0 fixed, optimality gives the respective one-sided derivative inequalities

$$-T(f - f_0) + \sqrt{nV}\langle u, f - f_0\rangle \le 0,$$
$$T(g - g_0) - \sqrt{nV}\langle u, g - g_0\rangle \le 0.$$

Thus the bias at f is no larger than the bias $B(V;\mathcal{F},\mathcal{H})$ at f_0, and the bias at g is no smaller than the bias $-B(V;\mathcal{F},\mathcal{H})$ at g_0.

Without attainment. If $B = +\infty$, the bias bounds are vacuous; otherwise put $C = \mathcal{H} - \mathcal{F}$, $M = \sqrt{nV}$, and $q = 2B$. Here M is the coefficient-norm budget. The definition of B gives $T(h) - M\|h\|_2 \le q$ for every $h \in C$. We first find a coefficient satisfying any finite collection of constraints $T(h_j) - \langle u, h_j\rangle \le q$, $1 \le j \le m$.

Let D be the compact convex set of vectors $(T(h_j) - \langle u, h_j\rangle)_{j=1}^m$ as u ranges over the radius-M ball in $\operatorname{span}\{h_1, \dots, h_m\}$. If D does not meet $Q = (-\infty, q]^m$, choose closest points $z_0 \in D$, $w_0 \in Q$. Then $w_{0j} = \min\{z_{0j}, q\}$ and $p = z_0 - w_0$ is nonzero with nonnegative coordinates. Minimality along each segment from z_0 to $z \in D$ gives $p^\top(z - z_0) \ge 0$, while $p^\top z_0 = q\sum_j p_j + \|p\|_2^2$. Set $\lambda_j = p_j/\sum_\ell p_\ell$ and $\bar{h} = \sum_j \lambda_j h_j \in C$. It follows that

$$q < \inf_{z\in D}\sum_j \lambda_j z_j = T(\bar{h}) - M\|\bar{h}\|_2 \le q,$$

a contradiction. Every finite collection of constraints therefore has a solution. The coefficient ball in the Hilbert space is weakly compact, and each constraint is weakly closed. The finite-intersection property gives one u satisfying all the constraints.

It remains to choose the intercept. Writing $R(x) = T(x) - \langle u, x\rangle$, fix f at a point of $\mathcal{F} \cap \mathcal{H}$ to see that $\sup_{g\in\mathcal{H}} R(g)$ is finite. Choose $c = \sup_{g\in\mathcal{H}} R(g) - B$ and $\widehat{T} = c + Y(u)$. Since $R(g) - R(f) \le 2B$ for every $f \in \mathcal{F}$ and $g \in \mathcal{H}$, both displayed bias inequalities follow; $\|u\|_2 \le \sqrt{nV}$ yields the variance bound. The argument uses linearity only on finite-dimensional spans and does not require a maximizing pair or ambient continuity of T.

For later use, set $r \ge n^{-1/2}$, $d = \omega(r, \mathcal{F}, \mathcal{H})$, and $V = d^2/(nr^2)$. This choice makes $\sqrt{nV} = d/r$, so the linear penalty in B equals the modulus at radius r. For $\varepsilon \ge r$, the scaling

bound (B.27) makes the penalized modulus nonpositive; for $\varepsilon \le r$, monotonicity bounds it by d. Hence

$$B(V;\mathcal{F},\mathcal{H}) \le \frac{d}{2}. \tag{B.31}$$

Union estimator and fourth moment. The fallback estimator must remain controlled even when it is used on an event selected from the same data. A fourth-moment bound lets Cauchy–Schwarz control its squared error on a rare rejection event; no independence between the estimator and that event is needed. We first construct this estimator and establish the bound. Let $G = \mathcal{F}_1 \cup \mathcal{F}_2$ and

$$W_n = \omega(n^{-1/2}, G) = \max\{a_1, a_2, \omega(n^{-1/2}, \mathcal{F}_1, \mathcal{F}_2), \omega(n^{-1/2}, \mathcal{F}_2, \mathcal{F}_1)\} = \max\{a_2, a_{12}\}.$$

Apply the ordered-modulus construction above with $\mathcal{F} = \mathcal{H} = \mathcal{F}_i$, $r = n^{-1/2}$, $d = a_i$, and $V = a_i^2$. Since the ordered modulus of one class against itself is its ordinary modulus, (B.28)–(B.31) give affine Gaussian estimators $\widetilde{T}_i$ with absolute bias at most $a_i/2$ and standard deviation at most a_i on $\mathcal{F}_i$. Consequently,

$$\sup_{f\in\mathcal{F}_i} \mathbb{E}_f(\widetilde{T}_i - T(f))^2 \le \frac{5}{4}a_i^2, \qquad i = 1, 2. \tag{B.32}$$

This argument requires only the assumed finiteness of the moduli on the parameter classes, not continuity of T on the ambient Hilbert space. For $i \ne j$, write $d_{ij}^{(0)} = \omega(n^{-1/2}, \mathcal{F}_i, \mathcal{F}_j)$ and use (B.28)–(B.31) with $V = (d_{ij}^{(0)})^2$ to construct $\widetilde{T}_{ij}$. Define

$$Z_{ij}^{+} = \frac{\widetilde{T}_{ij} - \widetilde{T}_i}{d_{ij}^{(0)} + a_i}, \qquad Z_{ij}^{-} = \frac{\widetilde{T}_i - \widetilde{T}_{ji}}{d_{ji}^{(0)} + a_i}, \qquad Z_{ij} = \max\{Z_{ij}^{+}, Z_{ij}^{-}\}.$$

The denominators are positive because $a_i > 0$. If $f \in \mathcal{F}_i$, both Gaussian variables Z_{ij}^{+} and Z_{ij}^{-} have mean at most one. Their variances are at most one because the standard deviation of a difference is at most the sum of the two standard deviations. Define the selector and union estimator by

$$\widehat{i} = \begin{cases} 1, & Z_{12} \le Z_{21}, \\ 2, & Z_{21} < Z_{12}, \end{cases} \qquad \widehat{T}_G = \widetilde{T}_{\widehat{i}}.$$

Fourth-moment bound. Fix $f \in \mathcal{F}_i$ and let $j \ne i$. If the wrong candidate has bias

$$B_j = |\mathbb{E}_f\widetilde{T}_j - T(f)| = \lambda W_n, \qquad \lambda \ge 3,$$

then one of the two Gaussian components of Z_{ji} has mean at least $(\lambda - 1)/2$: use Z_{ji}^{-} when the bias is positive and Z_{ji}^{+} when it is negative. The corresponding numerator has mean at least $B_j - d_{ij}^{(0)}/2$ in the first case and $B_j - d_{ji}^{(0)}/2$ in the second. Both ordered moduli are at most W_n, and either denominator is at most $2W_n$.

If $\widehat{i} = j$, at least one of the two components of Z_{ij} must be as large as the indicated component of Z_{ji}. Thus at least one of their two differences must be nonnegative. Each difference between a correct-class component and the indicated wrong-class component is Gaussian,

has variance at most four, and has mean at most $(3-\lambda)/2$. A union bound and the standard Gaussian tail inequality therefore give

$$\mathbb{P}_f(\widehat{i}=j) \le 2\exp\left\{-\frac{(\lambda-3)^2}{32}\right\}. \tag{B.33}$$

If $B_j < 3W_n$, the fourth moment of the normal error $\widetilde{T}_j - T(f)$ is at most CW_n^4. If $B_j \ge 3W_n$, Cauchy–Schwarz, the eighth-moment formula for a normal variable, and (B.33) yield

$$\mathbb{E}_f\left[|\widetilde{T}_j - T(f)|^4\,\mathbf{1}\{\widehat{i}=j\}\right] \le CW_n^4(1+\lambda^4)\exp\left\{-\frac{(\lambda-3)^2}{64}\right\} \le CW_n^4.$$

The correct candidate has fourth moment at most CW_n^4 by its Gaussian bias and variance bounds preceding (B.32). Combining this with the two wrong-selection cases gives

$$\sup_{f\in G}\mathbb{E}_f|\widehat{T}_G - T(f)|^4 \le CW_n^4. \tag{B.34}$$

Adaptive test and risk on $\mathcal{F}_1$. Let $\widehat{T}_1 = \widetilde{T}_1$. For $i \ne j$, define

$$\gamma_{ij} = \max\left\{e, \frac{d_{ij}^{(0)}}{a_1}\right\}, \qquad r_{ij} = \sqrt{\frac{\log\gamma_{ij}}{n}}, \qquad d_{ij} = \omega(r_{ij},\mathcal{F}_i,\mathcal{F}_j), \qquad s_{ij} = \frac{d_{ij}}{\sqrt{\log\gamma_{ij}}}.$$

Use the ordered-modulus construction with variance s_{ij}^2 and denote the resulting estimator by $\widehat{T}_{ij}$. Equations (B.28)–(B.31) show that its variance is at most s_{ij}^2, its upper bias on $\mathcal{F}_i$ is at most $d_{ij}/2$, and its lower bias on $\mathcal{F}_j$ is at least $-d_{ij}/2$. Put

$$b_{ij} = a_1 + d_{ij}, \qquad I_n = \mathbf{1}\left\{\widehat{T}_{12} - 12b_{12} - 12W_n \le \widehat{T}_1 \le \widehat{T}_{21} + 12b_{21} + 12W_n\right\},$$

and set

$$\widehat{T}^* = I_n\widehat{T}_1 + (1-I_n)\widehat{T}_G. \tag{B.35}$$

The event $I_n = 1$ retains the estimator for $\mathcal{F}_1$; rejection switches to the union estimator. To control risk on the first class, we next bound the rejection probability and combine it with the union estimator's fourth-moment bound.

For $f \in \mathcal{F}_1$, each rejection event is a one-sided tail of a Gaussian variable. Its variance is at most $(a_1+s_{ij})^2$, and the distance from its mean to the rejection threshold is at least $11b_{ij} + 12W_n$. Consequently

$$\mathbb{P}_f(I_n = 0) \le \sum_{i\ne j}\exp\left\{-\frac{(11b_{ij}+12W_n)^2}{2(a_1+s_{ij})^2}\right\}. \tag{B.36}$$

The union estimator's fourth moment is bounded by a constant times W_n^4. By Cauchy–Schwarz, a rejection probability of order at most $(a_1/W_n)^4$ therefore limits its contribution to squared risk to order a_1^2. We verify this by showing that each Gaussian-tail exponent is at least $4\log(W_n/a_1)$. Normalize one summand by

$$x = \frac{d_{ij}^{(0)}}{a_1}, \quad y = \frac{W_n}{a_1}, \quad L = \log\gamma_{ij}, \quad p = \frac{d_{ij}}{a_1}.$$

Then $1 \le y$, $x \le y$, and monotonicity plus (B.27) give $x \le p \le \sqrt{L}\,x$. The exponent in the corresponding summand of (B.36) is

$$E = \frac{\{11(1+p)+12y\}^2}{2(1+p/\sqrt{L})^2}.$$

There are three cases. When $x > e$ and $y \le x^2$, the logarithmic enlargement of the directional modulus suffices: $L = \log x$ and $p \ge x \ge \sqrt{L}$, so

$$E \ge \frac{1}{2}\left(\frac{11p}{1+p/\sqrt{L}}\right)^2 \ge \frac{121}{8}L \ge \frac{121}{16}\log y.$$

When $x > e$ and $y > x^2$, the union scale supplies the larger threshold: $p/\sqrt{L} \le x < \sqrt{y}$, giving $E \ge 18y \ge 4\log y$. Finally, when $x \le e$, no logarithmic enlargement is needed: $L = 1$ and $p = x$, giving $E \ge 72y^2/(1+e)^2 \ge 4\log y$. Thus both summands are at most y^{-4}, and (B.36) gives the uniform bound

$$\sup_{f\in\mathcal{F}_1} \mathbb{P}_f(I_n = 0) \le C\left(\frac{a_1}{W_n}\right)^4. \tag{B.37}$$

Using (B.32), (B.34), Cauchy–Schwarz, and (B.37),

$$\begin{aligned}\sup_{f\in\mathcal{F}_1} \mathbb{E}_f(\widehat{T}^* - T(f))^2 &\le \frac{5}{4}a_1^2 + \sup_{f\in\mathcal{F}_1}\{\mathbb{E}_f|\widehat{T}_G - T(f)|^4\}^{1/2}\mathbb{P}_f(I_n=0)^{1/2}\\ &\le Ca_1^2.\end{aligned}$$

Risk on $\mathcal{F}_2$. Set $D_n = b_{12} + b_{21} + W_n$, fix $f \in \mathcal{F}_2$, and write $\mu_f = \mathbb{E}_f\widehat{T}_1 - T(f)$. When the first-class estimator has large bias, we control its risk contribution through a bound on the acceptance probability; when its bias is bounded, the Gaussian risk bound applies directly.

Suppose that $|\mu_f| = \lambda D_n$ with $\lambda > 14$. If $\mu_f > 0$, acceptance and the upper-bias bound for $\widehat{T}_{21}$ on $\mathcal{F}_2$ give, respectively,

$$\widehat{T}_1 - \widehat{T}_{21} \le 12b_{21} + 12W_n,$$
$$\mathbb{E}_f(\widehat{T}_1 - \widehat{T}_{21}) \ge \mu_f - \frac{d_{21}}{2}.$$

The required downward deviation is therefore at least

$$\mu_f - \frac{d_{21}}{2} - 12b_{21} - 12W_n \ge (\lambda - 25/2)D_n.$$

If $\mu_f < 0$, use the other acceptance inequality and the lower-bias bound for $\widehat{T}_{12}$ on $\mathcal{F}_2$; the required upward deviation has the same lower bound. In either case the relevant difference is Gaussian with standard deviation at most $a_1 + s_{ij} \le D_n$. Therefore

$$\mathbb{P}_f(I_n = 1) \le \exp\left\{-\frac{(\lambda - 25/2)^2}{2}\right\}. \tag{B.38}$$

If $|\mu_f| \le 14D_n$, the mean squared error of $\widehat{T}_1$ is at most CD_n^2. If $|\mu_f| > 14D_n$, Cauchy–Schwarz, the fourth-moment formula for the Gaussian error $\widehat{T}_1 - T(f)$, and (B.38) again give

$$\mathbb{E}_f\left[(\widehat{T}_1 - T(f))^2 I_n\right] \le CD_n^2.$$

The accepted first-class contribution is now bounded in either bias regime. The rejection contribution is bounded by the union estimator's second moment, controlled by (B.34). Thus (B.35) implies

$$\sup_{f\in\mathcal{F}_2} \mathbb{E}_f(\widehat{T}^* - T(f))^2 \le C(D_n^2 + W_n^2). \tag{B.39}$$

Finally let

$$r_n^+ = \sqrt{\frac{\log \gamma_n^+}{n}}, \qquad \Omega_n = \omega_+(r_n^+, \mathcal{F}_1, \mathcal{F}_2).$$

Because $\gamma_{ij} \le \gamma_n^+$, one has $d_{ij} \le \Omega_n$. Together with $a_1 \le a_2$ and $W_n = \max\{a_2, a_{12}\} \le a_2 + \Omega_n$, this gives $D_n + W_n \le C(a_2 + \Omega_n)$. Substitution in (B.39) proves

$$\sup_{f\in\mathcal{F}_2} \mathbb{E}_f(\widehat{T}^* - T(f))^2 \le C\{a_2^2 + \Omega_n^2\},$$

which is the upper bound in Theorem 7.1.

Modulus calibration. We now compare the lower-bound and selector calibrations, then apply the main-text monotone-modulus calculation to the interval construction below.

Let $\mathcal{F}_1$ and $\mathcal{F}_2$ satisfy the closed-convex, nonempty-intersection, finiteness, and modulus-ordering conditions of Theorem 7.1. The theorem proves that the smallest possible risk order over $\mathcal{F}_2$ among estimators attaining the $\mathcal{F}_1$ benchmark with a suitable fixed constant is

$$\omega_+^2\left(\sqrt{\frac{\log \gamma_n^+}{n}}, \mathcal{F}_1, \mathcal{F}_2\right) + \omega^2\left(\frac{1}{\sqrt{n}}, \mathcal{F}_2\right). \tag{B.40}$$

For the fixed risk constant $C_0 > 0$ in the theorem, put $K = 1 + \frac{1}{2}|\log C_0|$. The definitions give

$$K^{-1} \log \gamma_n^+ \le \log \gamma_{n,0} \le K \log \gamma_n^+.$$

Monotonicity and (B.27), applied to both ordered moduli, therefore compare the two cross-class terms within constants depending only on C_0. This identifies the lower benchmark and the selector's upper benchmark up to constants, including the truncation at e; see Cai and Low [33, Theorems 1 and 3].

Monotone Hölder intervals. In the monotone Hölder setting, the one-sided persistence argument and opposing-cap construction in the proof of Corollary 7.12 give, for each fixed $0 < \beta \le 1$, $M > 0$, and all sufficiently small $\varepsilon > 0$,

$$\omega_+\big(\varepsilon, F_D(\beta, M), \mathcal{F}\big) \asymp \omega\big(\varepsilon, F_D(\beta, M)\big) \asymp M^{1/(2\beta+1)}\varepsilon^{2\beta/(2\beta+1)}, \tag{B.41}$$

where $\mathcal{F}$ is the union in Corollary 7.12 and the comparison constants may depend on β. This is a rate comparison, not an identity of moduli or leading constants. Scaling f and g by M^{-1} gives the stated radius dependence.

The small-radius qualification is necessary: constant shifts show that each modulus is at least ε, so this sublinear power cannot describe all radii. Only ε of order $n^{-1/2}$ is used below. Corollary 7.10 therefore gives, for every interval honest over $\mathcal{F}$,

$$L(CI, F_D(\beta, M)) \gtrsim n^{-\beta/(2\beta+1)}.$$

The common interval in Appendix B.4.2, honest over the entire nonincreasing class, attains this order on every fixed $F_D(\beta, M)$ and hence adapts over the union. Its local-modulus upper bound uses the same persistence argument, requiring only monotonicity of the alternative. See Cai and Low [30, 33] for the broader classwise context.

B.4.2 Functionwise shape-constrained intervals

The local-modulus calculations of Cai, Low, and Xia [39] illustrate Theorem 7.14 for point evaluation at the midpoint $x_0 = 1/2$ of $[0, 1]$ in Gaussian white noise. Honesty over the full monotone or convex class constrains expected interval length at each fixed function. The affine example shows how the choice of class changes this local difficulty.

Let $f_k(x) = -k(x - x_0)$ for $k \geq 0$, and let $\mathcal{F}_{\text{mon}}$ and $\mathcal{F}_{\text{conv}}$ be the full nonincreasing and convex classes of Theorem 7.14, with its right-continuous interior representatives for the former. Example 1 of Cai, Low, and Xia uses kt in the centered coordinate $t = x - x_0$ and the nondecreasing class. After translating the domain, the sign change $g \mapsto -g$ converts nondecreasing functions to nonincreasing functions while preserving right continuity, L_2 distance, and absolute target separation; no reflection or change of representative at a jump is used. For convexity, adding the same affine function to every curve is a bijection preserving L_2 distances and target differences; adding $-2k(x-x_0)$ transfers the local modulus at $k(x-x_0)$ to that at $-k(x - x_0)$.

For the monotone class, Cai, Low, and Xia [39, Example 1] give, for $k > 0$ and sufficiently small ε,

$$\omega(\varepsilon, f_k; \mathcal{F}_{\text{mon}}) = (3k)^{1/3}\varepsilon^{2/3}, \qquad \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f_k; \mathcal{F}_{\text{mon}}\right) \asymp n^{-1/3}.$$

For the convex class, the same example gives

$$\omega(\varepsilon, f_k; \mathcal{F}_{\text{conv}}) = 2\varepsilon, \qquad \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f_k; \mathcal{F}_{\text{conv}}\right) \asymp \frac{1}{\sqrt{n}}.$$

Thus, for each fixed $k > 0$, the honest-length benchmark at the same function is $n^{-1/3}$ in the monotone class and parametric in the convex class.

For $f(x) = (x - x_0)^2$ on $[0, 1]$, Cai, Low, and Xia [39, Example 4] give, for sufficiently small ε,

$$\omega(\varepsilon, f; \mathcal{F}_{\text{conv}}) = \frac{15^{2/5}}{2}\varepsilon^{4/5}, \qquad \omega\left(\frac{\zeta_\alpha}{\sqrt{n}}, f; \mathcal{F}_{\text{conv}}\right) \asymp n^{-2/5},$$

where the second expression substitutes $\varepsilon = \zeta_\alpha/\sqrt{n}$.

We now construct intervals attaining these local benchmarks in the same white-noise model, with noise level $\varepsilon = n^{-1/2}$ and target $T(f) = f(1/2)$. Here $Y(t)$ is the observed path. Write $\mathcal{F}_m = \mathcal{F}_{\text{mon}}$ and $\mathcal{F}_c = \mathcal{F}_{\text{conv}}$ for the full representative classes.

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Theorem 7.14 gives

$$L_\alpha^*(f, \mathcal{F}, T) \geq c_\alpha \omega(\zeta_\alpha/\sqrt{n}, f; \mathcal{F}). \tag{B.42}$$

For each full class with the specified representatives, we construct a single interval using observations on separated intervals and bounds based on second moments. This gives the functionwise rate comparison of Cai, Low, and Xia [39] without using their printed numerical constants.

Fix $0 < \alpha < 1/2$ and $n \geq 1$. For $j = 0, 1, \ldots$, set

$$h_j = 2^{-2j-4}, \qquad s_j = (nh_j)^{-1/2}, \qquad r_j = \frac{64}{\alpha} s_j.$$

Thus $h_{j+1} = h_j/4$ and $s_{j+1} = 2s_j$. All windows used below lie strictly inside $[0, 1]$. For a diagnostic sequence V_j, select $S = \min\{j \geq 0 : V_j \leq 64 s_j\}$; if this set is empty, put $S = 0$. We prove that the fallback event is null under every member of the class. All statistics are finite linear combinations of observed path increments, and the first-crossing rule is measurable. To order the endpoints, use $[a, \max\{a, b\}]$ in place of a proposed $[a, b]$. This interval has length $(b - a)_+$ and does not decrease coverage.

Convexity of a class $\mathcal{H}$ containing f and linearity of the target give

$$\omega(c\varepsilon, f; \mathcal{H}) \leq c\, \omega(\varepsilon, f; \mathcal{H}), \qquad c \geq 1. \tag{B.43}$$

Indeed, replace an alternative g by $f + (g - f)/c$. We work at an arbitrary fixed radius $z/\sqrt{n}$, $z > 0$, and finally take $z = \zeta_\alpha$.

Interval construction. Inner averages supply the endpoints; outer-shell diagnostics select the scale while preserving coverage by independence.

For the nonincreasing class, define

$$L_j = \frac{Y(x_0) - Y(x_0 - h_j)}{h_j}, \qquad R_j = \frac{Y(x_0 + h_j) - Y(x_0)}{h_j},$$
$$V_j^m = \frac{Y(x_0 - h_j) - Y(x_0 - 2h_j)}{h_j} - \frac{Y(x_0 + 2h_j) - Y(x_0 + h_j)}{h_j}.$$

Use V^m in the selection rule, denote the selected index by S_m, and put

$$CI_\alpha^m = [R_{S_m} - r_{S_m},\ \max\{R_{S_m} - r_{S_m}, L_{S_m} + r_{S_m}\}].$$

For the convex class, write

$$D(h) = \frac{Y(x_0 + h) - Y(x_0 - h)}{2h}, \qquad V_j^c = 2D(4h_j) - 3D(2h_j) + D(h_j).$$

Use V^c in the selection rule and denote the selected index by S_c. With $a_j = 2D(h_j/2) - D(h_j) - r_j$ and $b_j = D(h_j/2) + r_j$, put

$$CI_\alpha^c = [a_{S_c}, \max\{a_{S_c}, b_{S_c}\}].$$

Both rules depend only on the data and n, α; the reference indices below are used only in the analysis.

Bias, independence, and honesty. For a nonincreasing function, $\mathbb{E}_f R_j \le f(x_0) \le \mathbb{E}_f L_j$. Both centered averages have variance s_j^2. The diagnostic noise has variance $2s_j^2$, is supported on the two shells $[x_0 - 2h_j, x_0 - h_j]$ and $[x_0 + h_j, x_0 + 2h_j]$, and is independent across j. The inner pair (L_j, R_j) is independent of the diagnostic history through j. Chebyshev's inequality gives fixed-index noncoverage at most $2s_j^2/r_j^2 = 2\alpha^2/4096 \le \alpha$.

For the convex case, let $0 \le t \le 1/2$ denote an offset from x_0 and set

$$u(t) = \frac{f(x_0 + t) + f(x_0 - t)}{2} - f(x_0), \qquad b(h) = \frac{1}{h}\int_0^h u(t)\,dt.$$

Convexity gives $0 \le b(h/2) \le b(h)/2$. Hence $D(h_j/2)$ has nonnegative bias and $2D(h_j/2) - D(h_j)$ has nonpositive bias. The two centered inner averages $D(h_j/2), D(h_j)$ have variances at most $2s_j^2, s_j^2$, respectively. If their absolute noises are both at most $r_j/3$, the candidate interval covers $f(x_0)$. Thus its noncoverage is at most $27\alpha^2/4096 \le \alpha$.

The central increment cancels in V_j^c: its kernel is supported on $\{x \in [0,1] : h_j < |x - x_0| \le 4h_j\}$. These supports are disjoint across j, and are disjoint from the inner averages used for the interval at scale j. Consequently the diagnostics are independent and the current inner pair is independent of the entire diagnostic history through j. The diagnostic variance is at most $20s_j^2$. Define the first-crossing events

$$E_j = \{V_i > 64s_i \text{ for every } i < j,\ V_j \le 64s_j\}.$$

Each E_j belongs to that history. Almost-sure stopping, proved below, identifies E_j with $\{S = j\}$ up to null sets. Independence allows us to multiply each fixed-index noncoverage bound by $\mathbb{P}_f(E_j)$; summing proves honesty. Subsequent selected-event assertions are understood almost surely.

Almost-sure stopping. Write $\mu_j = \mathbb{E}_f V_j$ and $Z_j = V_j - \mu_j$. The deterministic geometry gives an index j_0 such that $\mu_j \le 32s_j$ for every $j \ge j_0$. For monotonicity, take the first index satisfying $f(x_0 - 2h_{j_0}) - f(x_0 + 2h_{j_0}) \le 32s_{j_0}$; the point gap decreases and s_j increases. For convexity an index with this property is constructed in the local-modulus argument below. Chebyshev's inequality and the diagnostic variance bounds give

$$\mathbb{P}_f\{V_j > 64s_j\} \le \frac{1}{16} \quad (j \ge j_0), \qquad \mathbb{P}_f\{S \ge j_0 + k\} \le 16^{-k}.$$

Independence of the diagnostic noises bounds the probability of no crossing through scale $j_0 + k - 1$ by 16^{-k}, giving the second inequality. Letting $k \to \infty$ proves almost-sure stopping, independently of the fallback definition of S.

Selected length moments. The stopping tail offsets the increasing noise at finer scales. To bound expected squared length, first bound the candidate length on $\{S = j\}$, then sum over j. Let A_j, B_j be the centered inner averages: $(L_j - \mathbb{E}_f L_j, R_j - \mathbb{E}_f R_j)$ in the monotone case and $(D(h_j/2) - \mathbb{E}_f D(h_j/2), D(h_j) - \mathbb{E}_f D(h_j))$ in the convex case. Put

$$F_j = s_j^2 + Z_j^2 + A_j^2 + B_j^2,$$

and let $K_\alpha > 0$ be a sufficiently large constant depending only on α. On $\{S = j\}$, the squared length is at most $K_\alpha F_j$. For monotonicity, the inner mean gap is nonnegative and at most

μ_j. For convexity, $b(h_j)$ is nonnegative and at most μ_j, and direct subtraction of the raw endpoints gives

$$b_j - a_j = D(h_j) - D(h_j/2) + 2r_j = b(h_j) - b(h_j/2) + B_j - A_j + 2r_j.$$

Since $b(h_j/2) \geq 0$, taking the positive part shows that, in either case, the length at a stopped scale is at most $\mu_j + |A_j| + |B_j| + 2r_j$, and $\mu_j \leq 64 s_j + |Z_j|$. Squaring these inequalities gives the claimed bound, retaining the positive part of the interval length.

Independence of the inner noises from $\{S = j\}$, and independence of the current diagnostic noise from the earlier survival event, yield

$$\mathbb{E}_f[F_j \mathbf{1}_{\{S=j\}}] \leq \begin{cases} 5 s_j^2 \mathbb{P}_f\{S \geq j\}, & \text{monotone,} \\ 24 s_j^2 \mathbb{P}_f\{S \geq j\}, & \text{convex.} \end{cases}$$

For the diagnostic term use $\{S = j\} \subseteq \{S \geq j\}$ before factoring: the diagnostic is not independent of its own stopping event. Since $s_j^2 = s_0^2 4^j$, summing the geometric series through the reference index and applying the stopping-tail bound afterward gives $\sum_{j \geq 0} s_j^2 \mathbb{P}_f\{S \geq j\} \leq 3 s_{j_0}^2$. Therefore

$$\mathbb{E}_f[\{L(CI_\alpha^m)\}^2] \leq 15 K_\alpha s_{j_0}^2, \qquad \mathbb{E}_f[\{L(CI_\alpha^c)\}^2] \leq 72 K_\alpha s_{j_0}^2. \tag{B.44}$$

Monotone local comparison. It remains to compare the reference scale with the local modulus. Fix $z > 0$, put $d_0 = \min\{z, 1\}$, and write $w = \omega(z/\sqrt{n}, f; \mathcal{F}_m)$. We claim $d_0 s_{j_0} \leq 16 w$. If $j_0 = 0$, use a constant shift and $s_0 = 4/\sqrt{n}$. Otherwise put $i = j_0 - 1$, $a = d_0 s_i / 8$, and let $t = 2h_i$ be a positive offset. Minimality gives $f(x_0 - t) - f(x_0 + t) > 32 s_i \geq 2a$. There is an admissible nonincreasing right-continuous alternative with target displacement a and squared norm distance at most $8a^2 t$. To see this, if $f(x_0 + t) \leq f(x_0) - a$, clip the function downward to $f(x_0) - a$ on $[x_0, 1]$. Otherwise $f(x_0 - t) \geq f(x_0) + a$. Put $c = f(x_0) + a$ and, by right continuity, choose $0 < \delta < t$ so that $f(x) \geq f(x_0) - a$ on $[x_0, x_0 + \delta]$. Set $g = \max\{f, c\}$ to the left of x_0, $g = c$ on $[x_0, x_0 + \delta)$, and $g = f$ thereafter. This nonincreasing, right-continuous function differs from f by at most $2a$ on a set of length at most $2t$. The alteration is supported in this interior neighborhood, remains in L_2, and changes the specified point value, including at a jump. Now $8a^2 t = d_0^2/(4n) \leq z^2/n$, so $w \geq a$ and $s_{j_0} = 2 s_i$ proves the claim.

Convex local comparison. Keep $d_0 = \min\{z, 1\}$, put $\varepsilon = d_0/\sqrt{n}$, $x = \varepsilon/4$, and set

$$H(t) = \sqrt{t}\, u(t), \quad r = \sup\{t \in [0, 1/4] : H(t) \leq x\}, \quad K = x/\sqrt{r}.$$

Continuity and monotonicity give $0 < r \leq 1/4$ and $H(r) \leq x$; if $r < 1/4$, equality holds. Choose a dyadic scale with $r/32 < h_{j_0} \leq r/8$. For all $j \geq j_0$,

$$0 \leq b(h_j) \leq \mu_j = \int_1^2 \{u(2h_j v) - u(h_j v)\}\, dv \leq u(4h_j), \qquad \sqrt{h_j}\mu_j \leq x/2.$$

Convexity and $u(0) = 0$ make each integrand at least $u(h_j v)$, which is at least $b(h_j)$ because u is nondecreasing. The last inequality follows from $4h_j < r$ and $H(4h_j) \leq x$. In particular $\mu_j \leq 32 s_j$, as used above.

We also have $K \leq \omega(\varepsilon, f; \mathcal{F}_c)$. If $r = 1/4$, then $K = \varepsilon/2$, and a constant shift suffices. Otherwise replace f on $[x_0 - r, x_0 + r]$ by its chord. The new function is convex, continuous

on the interior, and has target displacement $u(r) = K$. The chord gap is nonnegative and concave, vanishes at the endpoints, and has height K at x_0, so it is at most $2K$ everywhere. The squared norm distance is at most $8rK^2 = 8x^2 \le \varepsilon^2$. Monotonicity of the modulus and $h_{j_0} > r/32$ now give

$$s_{j_0}^2 \le \frac{512}{d_0^2}\omega(z/\sqrt{n}, f; \mathcal{F}_c)^2.$$

Length bounds and finiteness. The monotone comparison gives the analogous bound with 256 in place of 512. Inserting these inequalities into (B.44) bounds each squared length in expectation by $C_\alpha d_0^{-2}$ times the corresponding squared local modulus, for a constant C_α depending only on α. Cauchy–Schwarz then gives the expected-length bounds. Take $z = \zeta_\alpha$ and enlarge the constants if necessary to obtain

$$\inf_{g \in \mathcal{F}_m} \mathbb{P}_g\{g(x_0) \in CI_\alpha^m\} \ge 1 - \alpha, \tag{B.45}$$

$$\mathbb{E}_f L(CI_\alpha^m) \le C_\alpha^m \omega(\zeta_\alpha/\sqrt{n}, f; \mathcal{F}_m), \qquad f \in \mathcal{F}_m, \tag{B.46}$$

$$\inf_{g \in \mathcal{F}_c} \mathbb{P}_g\{g(x_0) \in CI_\alpha^c\} \ge 1 - \alpha, \tag{B.47}$$

$$\mathbb{E}_f L(CI_\alpha^c) \le C_\alpha^c \omega(\zeta_\alpha/\sqrt{n}, f; \mathcal{F}_c), \qquad f \in \mathcal{F}_c. \tag{B.48}$$

The constants depend only on α. The local modulus is finite at every fixed function and finite radius: at one fixed interior scale, left and right averages bracket the monotone target, and $2\mathbb{E}_g D(h/2) - \mathbb{E}_g D(h)$ and $\mathbb{E}_g D(h/2)$ bracket the convex target. These are continuous linear functionals on L_2, bounded on a fixed L_2 ball around f. Thus the expected lengths above are finite. Combining them with (B.42) proves the functionwise two-sided comparison. Each upper interval is a common nonrandomized rule over its full class; the lower theorem also allows independent randomization.

B.4.3 Adaptive quadratic-functional upper bound

The constrained-risk lower bound in Theorem 7.6(ii) is proved in Section 7.3. We establish the constructive assertion in part (i).

Proof of Theorem 7.6(i) We construct candidate estimates of the quadratic functional, identify one with the required risk for each class, and control the additional error from taking their penalized maximum. Retain Θ, s, p_*, s_*, $r(\alpha, p)$, and $A(\Theta)$ from the theorem and its setup; $\mathbb{E}_0$ denotes expectation at the zero sequence.

Write $\ell_n = \log n$ and take n sufficiently large that all integer blocks below are nonempty. Constants may depend on the fixed class parameters but not on n.

The universal estimator. Estimate the energy in three parts: use unthresholded squares for the low coordinates, candidate estimates indexed by cutoff for the middle coordinates, and increasing thresholds for the far tail. Selection applies only to the middle estimate.

Set

$$m_0 = \left\lfloor \frac{n}{\ell_n^2} \right\rfloor, \qquad m_k = 2^k m_0, \qquad J = \max\left\{k : m_k \le \frac{n^2}{\ell_n^2}\right\}.$$

Estimate the low-frequency energy without bias by $\widehat{\xi}_0$ below; for $1 \le k \le J$, let $d_k = m_k - m_0$ and define λ_k by

$$\widehat{\xi}_0 = \sum_{i=1}^{m_0} (Y_i^2 - n^{-1}), \qquad \lambda_k = \frac{d_k + 2\sqrt{d_k \ell_n} + 2\ell_n}{n}.$$

For $1 \le j \le J-k$ and $m_{k+j-1} < i \le m_{k+j}$, put $\tau_{k,i} = 2j$ and $\mu_{k,i} = \mathbb{E}_0(Y_i^2 - \tau_{k,i}/n)_+$. The kth hybrid estimate of $\xi_{\mathrm{mid}} = \sum_{i=m_0+1}^{m_J} \theta_i^2$ is

$$\widehat{\xi}_k = \left(\sum_{i=m_0+1}^{m_k} Y_i^2 - \lambda_k \right)_+ + \sum_{i=m_k+1}^{m_J} \left\{ (Y_i^2 - \tau_{k,i}/n)_+ - \mu_{k,i} \right\}. \tag{B.49}$$

For $j \ge 0$ and $m_{J+j} < i \le m_{J+j+1}$, where $m_{J+j} = 2^j m_J$, set $\gamma_i = 2(j+2)$ and

$$\widehat{\xi}_{\mathrm{tail}} = \sum_{i > m_J} \left\{ (Y_i^2 - \gamma_i \ell_n/n)_+ - \mathbb{E}_0(Y_i^2 - \gamma_i \ell_n/n)_+ \right\}.$$

Its target is the tail energy $\xi_{\mathrm{tail}} = \sum_{i>m_J} \theta_i^2$. Define the estimate as the partial-sum limit wherever it exists and as zero otherwise. For its summands T_i, the bounds below give $\sum_i |\mathbb{E}_\theta T_i| \le C \sum_i \theta_i^2 < \infty$ and $\sum_i \mathrm{Var}_\theta(T_i) < \infty$. Independence therefore gives almost-sure convergence; the class envelopes also give uniform L_2 convergence. This defines a single measurable rule without using the class parameters. Finally, for a sufficiently large universal constant K, set

$$\widehat{Q} = \widehat{\xi}_0 + \max_{1 \le k \le J} \left\{ \widehat{\xi}_k - K \frac{\sqrt{m_k \ell_n}}{n} \right\} + \widehat{\xi}_{\mathrm{tail}}. \tag{B.50}$$

This estimator contains no unknown smoothness, sparsity, or radius parameter.

Coordinate and block bounds. The centered-threshold calculation (B.2)–(B.3) gives, for $X \sim N(\vartheta, n^{-1})$ and $t \ge 1$,

$$\left| \mathbb{E}_\vartheta \{ (X^2 - t/n)_+ - \mathbb{E}_0 (X^2 - t/n)_+ \} - \vartheta^2 \right| \le C \min\{\vartheta^2, t/n\}, \tag{B.51}$$

$$\mathrm{Var}_\vartheta \{ (X^2 - t/n)_+ \} \le 6\vartheta^2/n + Cw(t)/n^2, \tag{B.52}$$

where w is the tail function in the cited variance bound and satisfies $\sum_{j \ge 1} 2^j w(2j) < \infty$. Also, $x \mapsto x_+$ is one-Lipschitz, so

$$\mathrm{Var} \left(\sum_{i=m_0+1}^{m_k} Y_i^2 - \lambda_k \right)_+ \le \frac{4}{n} \sum_{i=m_0+1}^{m_k} \theta_i^2 + \frac{2d_k}{n^2}.$$

Writing $\xi_{k,1} = \sum_{m_0 < i \le m_k} \theta_i^2$, combine the positive-part block's variance and Cauchy–Schwarz bias bounds with the coordinate bounds to obtain

$$\mathrm{Var}_\theta(\widehat{\xi}_k) \le C \frac{\xi_{\mathrm{mid}}}{n} + C \frac{m_k}{n^2}, \tag{B.53}$$

$$|\mathbb{E}_\theta \widehat{\xi}_k - \xi_{\mathrm{mid}}| \le C \left\{ \sqrt{\frac{\xi_{k,1}}{n}} + \frac{\sqrt{m_k \ell_n}}{n} + B_k(\theta) \right\}, \tag{B.54}$$

where

$$B_k(\theta) = \sum_{j=1}^{J-k} \sum_{i=m_{k+j-1}+1}^{m_{k+j}} \min\{\theta_i^2, 2j/n\}.$$

For the bias bound, subtract the expectation $d_k/n + \xi_{k,1}$ from the unthresholded block sum. Its standard deviation is at most $C\{\xi_{k,1}/n + d_k/n^2\}^{1/2}$, while $\lambda_k - d_k/n$ is at most $C\sqrt{m_k \ell_n}/n$. Equation (B.51) bounds every remaining coordinate. Independence and (B.52) prove (B.53).

Class envelopes and tail control. For $p \geq 2$, Hölder's inequality for $L_p(\alpha, M)$ and the levelwise ℓ_p–ℓ_2 inequality for $B^{\alpha}_{p,q}(M)$ give

$$\sum_{i>m} \theta_i^2 \leq C m^{-2\alpha}. \tag{B.55}$$

For $0 < p < 2$, the block envelope (B.4) and $\min(1, x) \leq x^{p/2}$ give

$$\sum_{j\geq 1} \sum_{i=m_{k+j-1}+1}^{m_{k+j}} \min\{\theta_i^2, 2j/n\} \leq C m_k^{-ps} n^{p/2-1}, \tag{B.56}$$

$$\sum_{i>m_J} \theta_i^2 \leq C m_J^{-2s}. \tag{B.57}$$

Indeed, the jth block's contribution to the left side of (B.56) is at most $(2j/n)^{1-p/2} \sum_i |\theta_i|^p$, and the resulting series $\sum_{j\geq 1} j^{1-p/2} 2^{-jps}$ is finite. For Besov sequences, a coordinate block $(a, 2a]$ intersects at most two resolution levels, each with $2^{\ell} \asymp a$. Summing their bounds $M^p 2^{-\ell ps}$ gives $C M^p a^{-ps}$, so the argument also covers cutoffs that do not coincide with level boundaries. The same envelopes together with (B.51)–(B.52) show that

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta(\widehat{\xi}_{\text{tail}} - \xi_{\text{tail}})^2 = \begin{cases} o(n^{-1}), & \alpha p_* > 1/2, \\ o(n^{-r(\alpha,p)}), & \alpha p_* \leq 1/2, \end{cases}. \tag{B.58}$$

For the variance, (B.51)– (B.52) give

$$\operatorname{Var}_\theta(\widehat{\xi}_{\text{tail}}) \leq \frac{6\xi_{\text{tail}}}{n} + \frac{C}{n^2} \sum_{j\geq 0} 2^j m_J w\{2(j+2)\ell_n\}.$$

The Gaussian-tail envelope $w(t) \leq Ce^{-3t/8}$ gives, for all sufficiently large n,

$$\sum_{j\geq 0} 2^j m_J w\{2(j+2)\ell_n\} \leq \frac{C m_J n^{-3/2}}{1 - 2n^{-3/4}} = O(n^{1/2}),$$

where $m_J \leq n^2$. The n^{-2} prefactor therefore makes this variance contribution $O(n^{-3/2})$. In the dense case,

$$|\mathbb{E}_\theta \widehat{\xi}_{\text{tail}} - \xi_{\text{tail}}| \leq C\xi_{\text{tail}} \leq C m_J^{-2\alpha}.$$

In the sparse case, the same calculation as in (B.56), now with threshold $2(j+2)\ell_n$, gives

$$|\mathbb{E}_\theta \widehat{\xi}_{\text{tail}} - \xi_{\text{tail}}| \leq C m_J^{-ps} n^{p/2-1} \ell_n^{1-p/2}.$$

Since $m_J \asymp n^2/\ell_n^2$, the dense squared-bias exponent is 8α, and the sparse squared-bias exponent is $2 + 4ps - p$. The first is larger than $r(\alpha, p)$ below the elbow and larger than one above it. For the sparse exponent,

$$2 + 4ps - p - r(\alpha, p) = p\left(4s - 1 + \frac{1}{1+2ps}\right) > 0,$$

when $\alpha p \leq 1/2$, while $2 + 4ps - p > 1$ when $\alpha p > 1/2$. This proves (B.58).

Rates at a class-dependent cutoff. For the proof, choose a class-dependent cutoff, called the oracle cutoff. Balance the squared uniform bound on thresholding bias over the class against the squared selection penalty, subject to the smallest available cutoff and dyadic rounding. The estimator still compares all candidates. Put $D = 1 + 2p_*s_*$ and choose the largest $k_* \in \{1, \dots, J\}$ for which

$$m_{k_*} \le \max\left\{2m_0, n^{p_*/D}\ell_n^{-1/D}\right\}. \tag{B.59}$$

Since $D > 1$ and $D - p_* = 2\alpha p_* - 1$, at or below the elbow we have $1 \le p_*/D < 2$. The proposed cutoff therefore lies between $2m_0$ and m_J eventually. When $\alpha p_* \le 1/2$, dyadic rounding and (B.55)–(B.56) give

$$\sup_{\theta\in\Theta} |\mathbb{E}_\theta \widehat{\xi}_{k_*} - \xi_{\mathrm{mid}}|^2 \le C R_{\mathrm{ad}}(n;\alpha,p), \tag{B.60}$$

$$\sup_{\theta\in\Theta} \mathrm{Var}_\theta(\widehat{\xi}_{k_*}) \le C\{R_{\mathrm{ad}}(n;\alpha,p)/\ell_n + n^{-1}\}, \tag{B.61}$$

where the rate and the penalty satisfy

$$R_{\mathrm{ad}}(n;\alpha,p) = n^{-r(\alpha,p)}\ell_n^{2p_*s_*/(1+2p_*s_*)},$$
$$\frac{m_{k_*}\ell_n}{n^2} \asymp n^{-2+p_*/D}\ell_n^{1-1/D} = R_{\mathrm{ad}}(n;\alpha,p).$$

When $p_* = 2$, the squared tail envelope is $m_{k_*}^{-4\alpha}$, which gives the same expression because $D = 1 + 4\alpha$. When $p_* = p < 2$, the square of $m_{k_*}^{-ps}n^{p/2-1}$ gives the same expression because $D = 1 + 2ps$. This proves both rate displays, including $\alpha p_* = 1/2$.

If $\alpha p_* > 1/2$, then $p_*/D < 1$, so $k_* = 1$ eventually. The dense and sparse tail bounds above, together with (B.54)–(B.58), give

$$\sup_{\theta\in\Theta} |\mathbb{E}_\theta(\widehat{\xi}_0 + \widehat{\xi}_1 + \widehat{\xi}_{\mathrm{tail}}) - Q(\theta)|^2 = o(n^{-1}).$$

Moreover,

$$\mathrm{Var}_\theta(\widehat{\xi}_0) = \frac{4}{n}\sum_{i\le m_0}\theta_i^2 + \frac{2m_0}{n^2},$$

and all variance terms outside the first m_0 coordinates are $o(n^{-1})$ uniformly on Θ. Since $\sup_{\theta\in\Theta}\sum_{i>m_0}\theta_i^2 = o(1)$,

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta(\widehat{\xi}_0 + \widehat{\xi}_1 + \widehat{\xi}_{\mathrm{tail}} - Q(\theta))^2 \le 4A(\Theta)n^{-1}(1 + o(1)). \tag{B.62}$$

Selector overshoot. A downward error of the penalized maximum is bounded by the penalized oracle candidate's error; an upward error is bounded using the sum of the candidates' squared positive deviations from the middle energy. We control this sum by pure-noise moment bounds and a blockwise stochastic comparison for nonzero means, which calibrate the common penalty.

Let $Z \sim N(0,1)$ and set $W_t = (Z^2 - t)_+ - \mathbb{E}(Z^2 - t)_+$. For every integer $r \ge 2$ and $t = 2j$, the Gaussian-square exponential moment at $3/8$ gives

$$\mathbb{E}|W_{2j}|^r \le 2r!(16/3)^r e^{-3j/4}. \tag{B.63}$$

Indeed, for $y \ge 0$ the exponential series gives $y^r \le r!(8/3)^r e^{3y/8}$. On $Z^2 > t$, apply this to $y = Z^2 - t$; on the complement the thresholded power is zero. Since $\mathbb{E}e^{3Z^2/8} = 2$,

$$\mathbb{E}(Z^2 - t)_+^r \le 2r!(8/3)^r e^{-3t/8}.$$

The inequality $|a - b|^r \le 2^{r-1}(|a|^r + |b|^r)$ and Jensen's inequality show that centering costs at most a factor 2^r. Put $b_* = 16/3$ and $C_* = 4b_*^2/(1 - 2e^{-3/4})$; the denominator is positive because $\log 2 < 3/4$. Expanding the centered exponential and using the moment bound yields

$$\log \mathbb{E}e^{sW_{2j}} \le \frac{s^2 v_j}{2(1 - b_* s)}, \qquad v_j = 4b_*^2 e^{-3j/4}, \quad 0 \le s < b_*^{-1}.$$

Here v_j is a variance proxy for the exponential-moment bound, not necessarily the variance itself. Independence adds these proxies while leaving b_* fixed. With at most $2^j m_k$ coordinates at level j, their sum is at most $C_* m_k$.

For nonzero means, concentrate each block's signal energy in one coordinate. This makes the thresholded-square sum stochastically larger and leaves the other coordinates as pure noise. We first prove the two-coordinate comparison by conditioning on the radius, then iterate within each equal-threshold block.

Let $X = \mu + Z \in \mathbb{R}^2$, where $Z \sim N_2(0, I_2)$, write $\|\mu\|_2 = a$, and let $\beta \in [0, \pi/4]$ be the angle of μ after reflections and interchange of the two coordinates. Conditional on $R = \|X\|_2 = r$, the angular density on the full circle is proportional to $\exp\{d\cos(\theta - \beta)\}$, where $d = ar$. Every strict upper level set of $(X_1^2 - t)_+ + (X_2^2 - t)_+$ is either empty, the full circle, or a union of four equal caps about the coordinate axes. Indeed, on $0 \le \theta \le \pi/4$ this statistic is nonincreasing in θ.

For the union of four caps with common half-width $b \in [0, \pi/4]$, the conditional probability is, apart from a normalizing constant independent of β,

$$F_b(\beta) = \int_{-b}^{b} K_d(u - \beta)\, du, \qquad K_d(v) = 2\{\cosh(d\cos v) + \cosh(d \sin v)\}.$$

The function K_d is even and $\pi/2$-periodic. Moreover, for $0 \le v \le \pi/4$,

$$K_d'(v) = 2d\{\cos v \sinh(d \sin v) - \sin v \sinh(d \cos v)\} \le 0,$$

because $x \mapsto \sinh(dx)/x$ is increasing on $(0, 1]$. Differentiating F_b and folding its arguments into $[0, \pi/4]$ gives $F_b'(\beta) \le 0$ for $0 \le \beta \le \pi/4$: the folded distance of $b + \beta$ from the nearest axis is at least $|b - \beta|$. Thus every conditional upper-tail probability is largest at $\beta = 0$. The law of R depends on μ only through a, so integration over R proves the same stochastic ordering unconditionally.

For fixed k and $1 \le j \le J - k$, put

$$\mathsf{e}_j = \sum_{i=m_{k+j-1}+1}^{m_{k+j}} \theta_i^2,$$

with the dependence on k suppressed. Apply the two-coordinate comparison repeatedly to the standardized observations $\sqrt{n}Y_i$ within each equal-threshold block. Their total squared mean is $n\mathsf{e}_j$, so the thresholded-square sum is stochastically largest when one standardized coordinate has mean $\sqrt{n\mathsf{e}_j}$ and the others have mean zero. For that concentrated vector,

$$\{(\sqrt{n\mathsf{e}_j} + Z)^2 - 2j\}_+ - n\mathsf{e}_j \le 2\sqrt{n\mathsf{e}_j}\, Z_+ + (Z^2 - 2j)_+.$$

The later blocks now have a stochastic upper bound. The first block still requires care: its positive part has two branches, and both must be retained when bounding squared overshoots. Write $\mathsf{e}_0 = \xi_{k,1}$, let U be the first block's centered error before its positive part, including its nonpositive deterministic offset, and let H be the sum of the later block errors. With $S_0 = (\mathsf{e}_0 + U)_+ - \mathsf{e}_0$, exactly

$$\widehat{\xi}_k - \xi_{\mathrm{mid}} = S_0 + H = \max\{U + H, H - \mathsf{e}_0\}.$$

Write the first-block observations as $Y_i = \theta_i + Z_i/\sqrt{n}$ and set

$$A_{k,0} = \frac{2}{\sqrt{n}} \left(\sum_{i=m_0+1}^{m_k} \theta_i Z_i \right)_+, \qquad U_k = \sum_{i=m_0+1}^{m_k} (Z_i^2 - 1).$$

Expanding the squares and discarding the nonpositive offset gives $U \le A_{k,0} + U_k/n$. Symmetry of the centered Gaussian linear form and its variance e_0 give $\mathbb{E}A_{k,0}^2 = 2\mathsf{e}_0/n$, also when $\mathsf{e}_0 = 0$.

On a separate product of standard Gaussian later blocks, let G_j be the first noise coordinate of block j, and let V_k be the sum of all centered pure-noise thresholded squares. Set

$$A_{k,+} = 2 \sum_{j=1}^{J-k} \sqrt{\mathsf{e}_j/n}\, G_{j,+}, \qquad H' = A_{k,+} + V_k/n,$$

where $G_{j,+} = (G_j)_+$ is the positive part of G_j. For $k = J$ these are empty sums. The blockwise stochastic comparison and scalar inequality above imply $H \le_{\mathrm{st}} H'$, meaning $\mathbb{P}(H > t) \le \mathbb{P}(H' > t)$ for every real t. To see that the comparison survives summation, condition on the other independent blocks, apply the scalar upper-tail comparison at the resulting threshold, and iterate. The first block is independent of both later-block products. For each fixed first-block observation and each penalty $p \ge 0$, the function $h \mapsto (S_0 + h - p)_+^2$ is nonnegative and nondecreasing. Integration of its upper level sets, followed by integration over the first block, therefore gives

$$\mathbb{E}_\theta (S_0 + H - p)_+^2 \le \mathbb{E}(S_0 + H' - p)_+^2.$$

On this comparison product space, without requiring a samplewise coupling, put

$$A_k = A_{k,0} + A_{k,+}, \qquad N_k = \max\{U_k + V_k, V_k\}.$$

Because $\mathsf{e}_0 \ge 0$ and $A_{k,0} \ge 0$, the exact two-branch identity implies $S_0 + H' \le A_k + N_k/n$. Consequently

$$\mathbb{E}_\theta \{\widehat{\xi}_k - \xi_{\mathrm{mid}} - p\}_+^2 \le 2\mathbb{E}A_k^2 + \frac{2}{n^2}\mathbb{E}(N_k - np)_+^2.$$

This separates the signal term from the noise term without assuming they are independent. The two branches of N_k are centered, although their maximum need not be, so we bound their exponential moments separately. First, Cauchy–Schwarz, the first-block moment, and $\mathbb{E}G_+^2 = 1/2$ give

$$\mathbb{E}A_k^2 \le \frac{2(J-k+1)}{n} \sum_{j=0}^{J-k} \mathsf{e}_j = \frac{2(J-k+1)\xi_{\mathrm{mid}}}{n} \le C\ell_n \xi_{\mathrm{mid}}/n.$$

The moment calculation above also applies at level zero, where $W_0 = Z^2 - 1$. Thus U_k has variance proxy $4b_*^2 d_k \le C_* m_k$, and V_k has variance proxy at most $C_* m_k$. They are independent, so the exponential moments of each of $U_k + V_k$ and V_k are bounded by

$$\exp\left\{\frac{s^2(2C_* m_k)}{2(1 - b_* s)}\right\}, \qquad 0 < s < b_*^{-1}.$$

For real x, a and $s > 0$, the exponential series gives $(x - a)_+^2 \le 4s^{-2} e^{s(x-a)}$. Hence

$$(N_k - a)_+^2 \le (U_k + V_k - a)_+^2 + (V_k - a)_+^2,$$
$$\mathbb{E}(N_k - a)_+^2 \le \frac{8}{s^2} \exp\left\{-sa + \frac{s^2(2C_* m_k)}{2(1 - b_* s)}\right\}.$$

The bound is finite. To make it summable over the candidate cutoffs, put $D_* = 2C_* + b_*$ and choose the universal multiplier $K = 2 + 24(C_* + b_*)$. This is at least $8D_* + 2$. For $L = \ell_n$, $u = \sqrt{L/m_k}$ and $s = u/(2D_*)$, large n ensures $1 \le L \le m_k$ and hence $b_* s \le 1/2$. Taking $a = K m_k u = K\sqrt{m_k L}$ gives

$$-sa + \frac{s^2(2C_* m_k)}{2(1 - b_* s)} \le -4L, \qquad \frac{8}{s^2} = \frac{32 D_*^2 m_k}{L}.$$

It follows that

$$\mathbb{E}\{N_k - K\sqrt{m_k \ell_n}\}_+^2 \le 32 D_*^2 m_k n^{-4}.$$

Using $\sum_{k=1}^{J} m_k \le 2m_J \le 2n^2/\ell_n^2$ in the second-moment comparison bounds the total noise contribution by $O(n^{-2})$ (in fact by a smaller order). The signal contribution is at most $CJ\ell_n \xi_{\mathrm{mid}}/n \le C\ell_n^2 \xi_{\mathrm{mid}}/n$. These estimates yield

$$\sum_{k=1}^{J} \mathbb{E}_\theta\left[\left\{\widehat{\xi}_k - K\frac{\sqrt{m_k \ell_n}}{n} - \xi_{\mathrm{mid}}\right\}_+^2\right] \le Cn^{-2} + C\frac{\ell_n^2 \xi_{\mathrm{mid}}}{n}. \tag{B.64}$$

The polynomial tail bounds in the class envelopes imply $\sup_{\theta \in \Theta} \xi_{\mathrm{mid}} \ell_n^2 = o(1)$ because $m_0 \asymp n/(\log n)^2$ and the tail-energy envelopes decay as $m_0^{-2\alpha}$ for $p \ge 2$ and m_0^{-2s} for $p < 2$, with $\alpha > 0$ and $s > 0$, respectively. Thus the right side of (B.64) is $o(n^{-1})$.

We now combine the oracle risk and the overshoot bound. For arbitrary real numbers $a_1, \dots, a_J$ and every k_*,

$$\left(\max_k a_k - \xi\right)^2 \le (a_{k_*} - \xi)^2 + \sum_k \{a_k - \xi\}_+^2.$$

Use this inequality with $a_k = \widehat{\xi}_k - K\sqrt{m_k \ell_n}/n$ and $\xi = \xi_{\mathrm{mid}}$. The penalty at the oracle index satisfies

$$\frac{m_{k_*} \ell_n}{n^2} \lesssim R_{\mathrm{ad}}(n; \alpha, p) \quad \text{if } \alpha p_* \le 1/2, \qquad \frac{m_1 \ell_n}{n^2} = o(n^{-1}) \quad \text{if } \alpha p_* > 1/2.$$

For $\alpha p_* \le 1/2$, the preceding bounds, together with (B.58) and the low-coordinate variance formula, prove the adaptive upper bound. For $\alpha p_* > 1/2$, preserve the leading constant by separating the unbiased low-coordinate error from the middle and tail errors. Write

$$M_n = \max_{1\le k\le J}\left\{\widehat{\xi}_k - K\frac{\sqrt{m_k \ell_n}}{n}\right\} - \xi_{\mathrm{mid}},$$
$$L_n = \widehat{\xi}_0 - \sum_{i\le m_0}\theta_i^2, \qquad R_n = M_n + \widehat{\xi}_{\mathrm{tail}} - \xi_{\mathrm{tail}}.$$

The maximum inequality with $k_* = 1$, the oracle bias and variance bounds, and (B.64) give $\sup_{\theta\in\Theta}\mathbb{E}_\theta M_n^2 = o(n^{-1})$. Together with (B.58), this implies $\sup_{\theta\in\Theta}\mathbb{E}_\theta R_n^2 = o(n^{-1})$. Now $\widehat{Q} - Q(\theta) = L_n + R_n$, where L_n has mean zero and is independent of R_n because they use disjoint coordinate sets. Consequently,

$$\mathbb{E}_\theta(\widehat{Q} - Q(\theta))^2 = \frac{4}{n}\sum_{i\le m_0}\theta_i^2 + \frac{2m_0}{n^2} + \mathbb{E}_\theta R_n^2.$$

Since $m_0 = o(n)$ and $A(\Theta) > 0$, taking the supremum proves the upper bound $4A(\Theta)n^{-1}(1 + o(1))$.

□

B.4.4 Relaxed-coverage band lower bound

We bound a band's width at a fixed function when its relaxed coverage guarantee must hold over a larger smoothness class in white noise. The proof uses a mixture of distributed perturbations and the common coverage-to-width implication proved earlier.

The transferred event depends only on the observed band and the reference function, not on the alternative's index; a narrow band on that event violates the relaxed guarantee for each sufficiently separated alternative.

Proof of Theorem 7.17 Perturbations and mixture distance. Write $\beta = \alpha_0$ and fix $0 < M_0' < M_0$ and $h \in \Lambda^\beta(M_0')$. A common mixture of perturbations forces a width lower bound at h under each relaxed coverage criterion.

For each criterion, first choose $\delta_{\mathrm{cov}} > 0$ and $\delta > 0$ satisfying the feasibility conditions in the common coverage-to-width implication. Then let $g = g_{\delta_{\mathrm{cov}}}$ be a nonnegative smooth bump supported in $(0, 1)$ and equal to 1 on an interval of length at least $1 - \delta_{\mathrm{cov}}$; put $E_g = \int_0^1 g^2 > 0$. Set

$$m = \lceil n^{2/(1+4\beta)}\rceil, \qquad x_i = (i-1)/m.$$

Choose $a_0 > 0$ sufficiently small, depending on g and the slack $M_0 - M_0'$, and put

$$A_n = a_0 m^{-\beta}, \qquad f_B(t) = h(t) + A_n\sum_{i=1}^m B_i g\{m(t - x_i)\}, \qquad B \in \{-1, 1\}^m.$$

Smoothness permits amplitude of order $m^{-\beta}$. The chosen resolution keeps $n^2 m^{-(1+4\beta)}$ bounded, which will control the sign mixture's second moment. Using every cell, rather

than one localized bump, separates each alternative from h over most of the domain; that spatial separation is needed to force integrated width.

The signed-bump version of (7.10) places $f_B - h$ in $\Lambda^\beta(a_0 C_g)$. Choose $a_0 C_g \le M_0 - M_0'$. Applying the triangle inequality separately to every derivative bound and the top Hölder seminorm then places every f_B in $\Lambda^\beta(M_0)$.

Let $P_0 = P_h$, $P_\pi = 2^{-m} \sum_B P_{f_B}$, and define the centered coordinates

$$\phi_i(t) = \sqrt{m/E_g}\, g\{m(t - x_i)\}, \qquad Z_i = \sqrt{n} \int_0^1 \phi_i(t)\{dY(t) - h(t)\, dt\}.$$

The disjoint normalized bumps are orthonormal, so the Z_i are independent standard normals under P_0 and have means $B_i \lambda_n$ under f_B, where $\lambda_n^2 = n A_n^2 E_g m^{-1}$. The drift difference $f_B - h$ lies in the bump span. The Gaussian measurements indexed by its orthogonal complement in $L_2[0,1]$ have the same joint law under all alternatives and P_0, and are independent of the bump coordinates. Thus the full likelihood ratio depends only on these coordinates. The block sign-mixture likelihood ratio and its second moment, by direct Gaussian integration, are

$$L_i(Z_i) = e^{-\lambda_n^2/2} \cosh(\lambda_n Z_i), \qquad \mathbb{E}_0 L_i^2 = \cosh(\lambda_n^2).$$

The blocks are independent, and $\log \cosh x \le x^2/2$ for every real x. Consequently,

$$\begin{aligned} 1 + \chi^2(P_\pi, P_0) &= \prod_{i=1}^m \cosh(\lambda_n^2) \\ &\le \exp\left(\frac{1}{2} m \lambda_n^4\right) = \exp\left(\frac{1}{2} n^2 A_n^4 E_g^2 m^{-1}\right) \\ &\le \exp(C a_0^4), \end{aligned}$$

because $n^2 m^{-(1+4\beta)}$ is bounded. Given any $\delta > 0$, reducing a_0 if necessary and using $\mathrm{TV}(P_\pi, P_0) \le \frac{1}{2}\sqrt{\chi^2(P_\pi, P_0)}$ gives

$$\mathrm{TV}(P_\pi, P_0) \le \delta. \tag{B.65}$$

With μ denoting Lebesgue measure, the plateau of $g = g_{\delta_{\mathrm{cov}}}$ gives $\mu\{u \in [0,1] : g(u) \ge q\} \ge 1 - \delta_{\mathrm{cov}}$ with $q = 1$. Thus, for every sign vector B,

$$S_3(B) = \{t : |f_B(t) - h(t)| \ge q A_n\} \quad \text{satisfies} \quad \mu\{S_3(B)\} \ge 1 - \delta_{\mathrm{cov}}. \tag{B.66}$$

Write $[L(t), U(t)]$ for the reported band and let $W = \int_0^1 \{U(t) - L(t)\}\, dt$. If $\mathbb{E}_h W = \infty$, the lower bound is immediate. Otherwise W is finite almost surely under P_h and, by equivalence of the white-noise laws, under every P_{f_B}; the relative-excess convention at infinite width therefore has no effect.

For this white-noise construction, apply the common coverage-to-width implication with its reference rate a_n and its perturbation amplitude both set to the present A_n, and with $c_A = q = 1$. Its class radii are $M = M_0$ and $M' = M_0'$. The relevant coverage guarantee holds at h and every f_B because they belong to the rough class, contained in the stated union. Class membership and (B.65)–(B.66) verify the remaining inputs. Thus either (7.13) or (7.14) implies $\mathbb{E}_h W \ge b A_n$ for a fixed $b > 0$, independent of n.

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Width and rate comparison. In either case, $\mathbb{E}_h W \geq bA_n$ and $A_n = a_0 m^{-\beta} \asymp n^{-2\beta/(1+4\beta)}$, proving the pointwise lower bound. Taking $h = 0$ gives the smoother-class bound because 0 lies in both Hölder balls.

For the final comparison, set

$$q_0 = \frac{2\alpha_0}{1+4\alpha_0}, \qquad q_1 = \frac{\alpha_1}{1+2\alpha_1},$$
$$q_1 - q_0 = \frac{\alpha_1 - 2\alpha_0}{(1+2\alpha_1)(1+4\alpha_0)}.$$

When $\alpha_1 > 2\alpha_0$, this difference is positive, and

$$\frac{n^{-q_0}}{(\log n/n)^{q_1}} = \frac{n^{q_1-q_0}}{(\log n)^{q_1}} \longrightarrow \infty.$$

The smoother-class benchmark is therefore unattainable beyond the stated factor-of-two range. □

B.5 The classification upper comparison

We verify (8.19) under the fixed smoothness, strict margin, and support assumptions of Section 8.2. This is a comparison for nearest-neighbor classification, not a new lower-bound argument.

Proof of (8.19) Let v_d be the volume of the Euclidean unit ball, put $a = \mu_{\min} c_0 v_d$, and choose $c_1 = \min\{1/4, a r_0^d/2\}$. Fix $x \in A$ and $1 \leq k \leq c_1 n$. The ball of radius $r_k = (2k/(na))^{1/d} \leq r_0$ around x contains a binomial number of training covariates with mean at least $2k$. If $R_k(x)$ is the distance to the kth nearest covariate, the binomial lower-tail bound gives

$$\mathbb{P}\{R_k(x) > r_k\} \leq e^{-k/4}.$$

Fix a measurable rule for distance ties. Conditional on the covariates, the selected labels are independent Bernoulli variables. On $\{R_k(x) \leq r_k\}$, their average conditional mean differs from $m_P(x)$ by at most $b = L r_k^\beta$. Hoeffding's inequality therefore gives, for $u > 0$,

$$\mathbb{P}\{|\widehat{m}_k(x) - m_P(x)| \geq b + u,\ R_k(x) \leq r_k\} \leq 2e^{-2ku^2},$$

where $\widehat{m}_k$ is the average of the k selected labels. Writing $\delta(x) = |m_P(x) - 1/2|$, a wrong decision at $\delta(x) > 2b$ consequently has probability at most $2e^{-k\delta(x)^2/2} + e^{-k/4}$.

To integrate this pointwise bound, set $h = \max\{b, k^{-1/2}\}$. By (8.16) and the strict margin condition, the region $0 < \delta \leq 2h$ contributes at most $4hC_{\mathrm{m}}(2h)^\alpha$ to expected excess risk. On each shell $2^j h < \delta \leq 2^{j+1}h$, $j \geq 1$, bound the loss and margin mass at the upper endpoint and the exponential at the lower endpoint. The sum of the concentration contributions is at most

$$\sum_{j \geq 1} 2^{j+3} h\, C_{\mathrm{m}} (2^{j+1}h)^\alpha e^{-2^{2j-1}} \leq C h^{1+\alpha}.$$

The series converges because its exponential factor dominates every geometric sequence. The radius-failure contribution over all shells is at most $e^{-k/4}$, since $2\delta \leq 1$ and the shells are

disjoint. Points with $\delta = 0$ contribute no excess risk. Finally, $h^{1+\alpha} \le b^{1+\alpha} + k^{-(1+\alpha)/2}$, which proves (8.19) with $c' = 1/4$. □

B.6 Convex cap geometry: reverse modulus comparisons

The tilted-cap construction in Section 4.5.1 proves $\rho_T(\varepsilon; f) \le \omega_T(\varepsilon; f)$ for $T \in \{Z, M\}$. Here we complete Theorem 4.21 by proving $\omega_T(\varepsilon; f) \le 3\rho_T(\varepsilon; f)$, following Cai, Chen, and Zhu [17, Proposition 2.2] and its proof in [18, Section C.3].

Use the normalization and continuous representatives of the main-text proof: $M(f) = 0$, $z = Z(f)$, $H = \rho_M(\varepsilon; f)$, and $D = \rho_Z(\varepsilon; f)$, with $\int_0^1 (H - f(t))_+^2 \, dt = \varepsilon^2$. In particular, $H, D > 0$. Let $I = \{t : f(t) \le H\} = [a, b]$ and let $v = b - a > 0$. Then

$$\varepsilon^2 \le H^2 v \le 2H^2 D. \tag{B.67}$$

The first inequality bounds the cap integrand by H^2; the second uses $I \subset [z - D, z + D]$.

Upper comparison for location. Let $g \in \mathcal{F}_{\mathrm{cvx}}$ satisfy $\|f - g\|_2 \le \varepsilon$, and put $y = Z(g)$. Suppose first that $y > z + 3D$. Then $b \le z + D < 1$, $f(b) = H$, and $[b, b + 2D] \subset [0, y]$. If $g(b) > H$, monotonicity to the left of y gives $g(t) > H$ throughout I, so integration yields the contradiction $\|f - g\|_2^2 > \varepsilon^2$. Otherwise $g(t) \le H$ for $b \le t \le b + 2D$, whereas convexity gives $f(t) \ge H + H(t - b)/(b - z)$. By (B.67),

$$\|f - g\|_2^2 \ge \frac{H^2}{(b - z)^2} \int_0^{2D} s^2 \, ds \ge \frac{8}{3} H^2 D \ge \frac{4}{3} \varepsilon^2,$$

again a contradiction. Reflection excludes $y < z - 3D$. Taking the supremum over g proves $\omega_Z(\varepsilon; f) \le 3D$.

Upper comparison: raising the minimum. If $M(g) > H$, then $g(t) > H$ on I, and the same integral comparison gives $\|f - g\|_2^2 > \varepsilon^2$. Thus $M(g) - M(f) \le H$.

Upper comparison: lowering the minimum. Write $M(g) = -A < 0$, $y = Z(g)$, and $K = \{t : g(t) \le 0\} = [c, d]$. Put $l = y - c$, $r = d - y$, and $w = l + r > 0$. A deep negative minimum costs squared distance either within K or through the steep rise outside it. First, convexity bounds g above by the chords from its minimum to the nonpositive endpoint values of K. Since $f \ge 0$, the two triangular integrals give

$$\int_K (f - g)^2 \, dt \ge \int_K g^2 \, dt \ge \frac{A^2 w}{3}. \tag{B.68}$$

For example, on $[c, y]$ one has $g(t) \le -A(t - c)/l$ when $l > 0$; integrating its square gives $A^2 l/3$. A side of zero length contributes zero.

Suppose $A > 3H$. If $w \ge v/3$, equation (B.68) gives $\|f - g\|_2^2 > H^2 v$, a contradiction. Hence $0 < w < v/3$.

Convexity forces a steep rise outside this narrow negative region. The part of I outside K has length at least $v - w$, so one of its two sides has length at least $(v - w)/2$. Reflect if necessary to make this the right side, of length

$$e = \big(b - \max\{d, a\}\big)_+ \ge \frac{v - w}{2}.$$

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In particular, $d < b \le 1$, so $g(d) = 0$ and $r = d - y > 0$. The secant from $(y, -A)$ to $(d, 0)$ gives $g(t) \ge A(t-d)/r$ for $t \ge d$. Set $s = \max\{a, d + Hw/A\}$ and $\delta = b - s$. Since $r \le w$ and $Hw/A < w/3$, we have

$$\delta \ge e - Hw/A > \frac{v-w}{2} - \frac{w}{3} = \frac{3v-5w}{6} > 0.$$

For $s < t < b$, the inequalities $f(t) \le H$ and $s \ge d + Hr/A$ imply

$$g(t) - f(t) \ge \frac{A}{r}(t-s) \ge \frac{A}{w}(t-s).$$

The interval is disjoint from K apart from a possible endpoint. Adding its triangular integral to (B.68), then writing $\gamma = w/v \in (0, 1/3)$ and using $A > 3H$, gives

$$\|f-g\|_2^2 \ge \frac{A^2 w}{3} + \frac{A^2\delta^3}{3w^2},$$
$$\frac{\|f-g\|_2^2}{H^2 v} > 3\gamma + \frac{(3-5\gamma)^3}{72\gamma^2} > 1.$$

For the last inequality, if $\gamma \le 1/4$, the second term is at least $343/288 > 1$. If $1/4 < \gamma < 1/3$, the two terms are respectively greater than $3/4$ and at least $8/27$, whose sum is $113/108 > 1$. This contradicts $\|f-g\|_2^2 \le \varepsilon^2 \le H^2 v$. Thus $M(f) - M(g) \le 3H$ and $\omega_M(\varepsilon; f) \le 3H$. With the cap-product bound proved in Chapter 4, the comparisons establish Theorem 4.21.

B.7 Gaussian discovery: the tail optimization

The remaining step in Theorem 9.5 is to find scores that provide both strong evidence of a signal and enough signal observations for discovery. Retain the fixed parameters $1/2 < \beta < 1$, $r > 0$, and $\sigma > 0$. Recall from (9.13) that $G_\sigma(u; r) = u^2 - (u - \sqrt{r})^2/\sigma^2$. The quantity $J_\sigma = J_\sigma(r, \beta)$ in (9.14) is the infimum of $(u - \sqrt{r})^2/\sigma^2$ over nonnegative scores satisfying $G_\sigma(u; r) \ge \beta$, with value $+\infty$ when the feasible set is empty.

Write $t = \sqrt{r}$ and $u_{\max} = t + \sigma\sqrt{1-\beta}$. On the score scale $u = x/\sqrt{2\log n}$, evidence requires $G_\sigma(u; r) > \beta$, whereas a diverging signal count requires $(u-t)^2/\sigma^2 < 1 - \beta$. The latter restricts scores to a neighborhood of the typical signal scale t, with right endpoint $u_{\max}$. We therefore seek an intersection of these two score ranges.

For $r \ge \beta$, $G_\sigma(t; r) = r$ and $\partial G_\sigma(t; r)/\partial u = 2t > 0$. Scores just to the right of t therefore have $G_\sigma > \beta$ and arbitrarily small positive signal-tail exponent when $r = \beta$; when $r > \beta$, a neighborhood of t already qualifies. Hence $J_\sigma = 0$ and discovery is attainable throughout this range.

We may assume $0 < r < \beta$ below. For $0 \le u \le t$, $G_\sigma(u; r) \le u^2 \le r < \beta$, so every feasible nonnegative score lies to the right of t. Denote the first feasible score by u_- when it exists. If the feasible set has interior, the two strict inequalities hold together exactly when $u_- < u_{\max}$: scores just beyond u_- then provide both strong evidence and a diverging signal count. Thus u_- is the evidence threshold, whereas $u_{\max}$ is the frequency limit. We first solve the equal-variance case, then examine how changing the variance alters the evidence threshold.

Equal signal variance: $\sigma = 1$. Here $G_1(u; r) = 2tu - r$, so $u_- = (\beta + r)/(2t) > t$. The inequality $J_1 < 1 - \beta$ is equivalent to $u_- < u_{\max}$, or $t + \sqrt{1-\beta} > 1$. This yields the third branch of (9.11).

Smaller signal variance: $0 < \sigma < 1$. Set $a = 1 - \sigma^2$. Completing the square gives

$$G_\sigma(u; r) = \frac{r}{a} - \frac{a}{\sigma^2}\left(u - \frac{t}{a}\right)^2.$$

The evidence first increases and then decreases with the score, with maximum r/a at $u_* = t/a$. An extremely large observation can favor the wider null distribution rather than the narrower signal distribution. If $r < a\beta$, no fixed posterior cutoff can eventually be exceeded. For $r > a\beta$, the feasible interval begins at

$$u_- = \frac{t - \sigma\sqrt{r - a\beta}}{a}, \qquad J_\sigma = \frac{(u_- - t)^2}{\sigma^2} = \left(\frac{\sigma t - \sqrt{r - a\beta}}{a}\right)^2.$$

For $a\beta < r < \beta$, the last expression decreases continuously from $\sigma^2\beta/a$ to zero: its positive square root has derivative $\{\sigma/(2\sqrt{r}) - 1/(2\sqrt{r - a\beta})\}/a < 0$. At the onset of feasibility, $\sigma^2\beta/a \le 1 - \beta$ exactly when $\beta \le a$. In this case, once convincing scores exist, they are also frequent enough: every $r > a\beta$ has $J_\sigma < 1 - \beta$, giving the first branch of (9.11).

If $\beta > a$, convincing scores are initially too rare: the limiting exponent $\sigma^2\beta/a$ exceeds $1 - \beta$. As r increases, J_σ decreases, and the unique transition occurs at $u_- = u_{\max}$. The identity $G_\sigma(u_{\max}; r) = u_{\max}^2 - (1 - \beta)$ then gives $u_{\max} = 1$, or $r = (1 - \sigma\sqrt{1 - \beta})^2$. At that value, $u_* = t/a > 1$ because $\beta > a$, so the crossing lies to the left of the vertex. It therefore occurs at the left endpoint of the feasible interval and gives the second branch.

One case requires a separate check. At $r = a\beta$, the closed feasible set defining J_σ is the singleton $\{u_*\}$, while the strict set $\{u \ge 0 : G_\sigma(u; r) > \beta\}$ is empty. A fixed posterior cutoff changes $G_\sigma \ge \beta$ by $O(1/\log n)$ and restricts u to a neighborhood of radius $O((\log n)^{-1/2})$ around u_*. Its signal probability is at most $n^{-\sigma^2\beta/a + o(1)}$. When $\beta > a$, this makes the expected qualifying signal count vanish, as required below the discovery boundary. When $\beta \le a$, the singleton occurs at the boundary itself, where the theorem makes no assertion.

Larger signal variance: $\sigma > 1$. Now larger positive scores always provide stronger evidence: $\partial G_\sigma(u; r)/\partial u = 2\{(\sigma^2 - 1)u + t\}/\sigma^2 > 0$ for $u \ge 0$. With $s = \sigma^2 - 1$, its feasible set begins at

$$u_- = \frac{\sigma\sqrt{r + s\beta} - t}{s} > t.$$

Again $J_\sigma < 1 - \beta$ exactly when $u_- < u_{\max}$. Monotonicity and the identity for $G_\sigma(u_{\max}; r)$ reduce this to $t + \sigma\sqrt{1 - \beta} > 1$. If $\beta \le 1 - \sigma^{-2}$, it holds for every $r > 0$, giving the zero boundary. Otherwise it gives $r > (1 - \sigma\sqrt{1 - \beta})^2$, the final branch.

Thus, strictly below the boundary, either the feasible set is empty or $J_\sigma > 1 - \beta$, including the singleton case. Strictly above it, a positive-width interval satisfies both strict inequalities in the main proof. The posterior-cutoff perturbations preserve these strict comparisons.

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